Intelligent Security API

Developer Guide

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1 Reading Guide

Chapter	Description
Overview	Includes the ISAPI overview, applicable products, terms and definitions, abbreviations, and update history.
ISAPI Framework	Read the chapter to take a quick look at the ISAPI framework and basic functions.
Quick Start Guide	Read the chapter to quickly understand the programming process of basic functions such as authentication, message parsing, real-time live view, playback, and event uploading.
API Reference	Start programming according to API definitions.
How-To Video Guidance	How-to videos demonstrate detailed steps of different integration tasks.

2 Overview

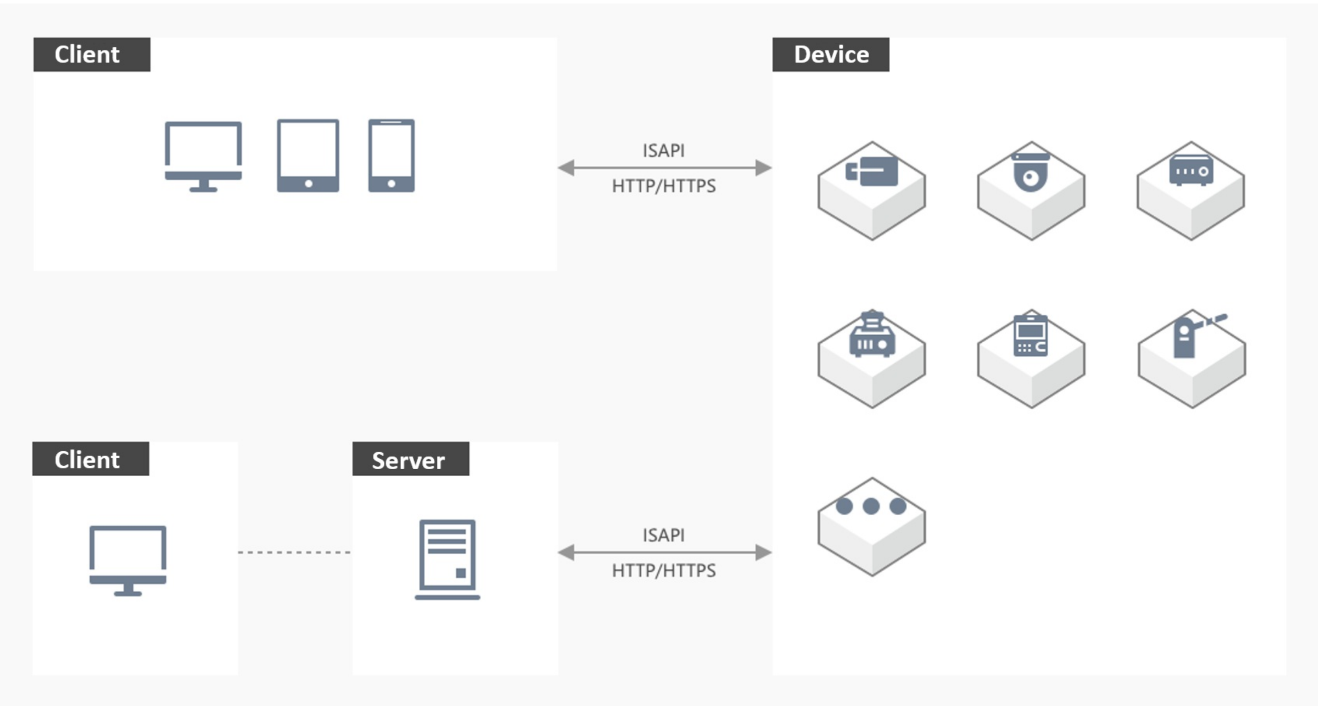
2.1 Introduction

Intelligent Security API (hereinafter referred to as ISAPI) is an application layer protocol based on HTTP (Hypertext Transfer Protocol) and adopts the REST (Representational State Transfer) architecture for communication between security devices (cameras, DVRs, NVRs, etc.) and the platform or client software.

Since established in 2013, ISAPI has included more than 11,000 APIs for different functions, including device management, vehicle recognition, parking lot management, intelligent facial application, access control management, interrogation management, and recording management. It is applicable to industries such as traffic, fire protection, education, and security inspection.

2.1.1 Application Scenario

When you integrate devices via ISAPI, the device acts as the server to listen on the fixed port and the user's application acts as the client to actively log in to the device for communication. To achieve the above goals, the device should be configured with a fixed IP address and the requests from the client can reach the server.



2.1.2 Layers in the Network Model

ISAPI is an application layer protocol based on HTTP, thereby it inherits all specifications and properties from HTTP.

Protocols frequently used along with ISAPI include SADP (Search Active Device Protocol) based on multicast for discovering and activating devices, RTSP (Real-Time Streaming Protocol) based on TCP/UDP for live view and video playback of devices, etc.

	Device Discovery	Signaling Interaction	Media Stream
Application Layer	SADP	ISAPI (HTTP)	RTSP/RTP
Transport Layer	MCAST	TCP	TCP/UDP
Network Layer	IP		
Network Interface	Hardware Drive Interface		

2.2 Product Scope

- Vehicle Access Control Management
    - ANPR Cameras

      DS-2CD9046-ESUL, DS-TCG204-DI, DS-TCG204-E1, DS-TCG205-B, DS-TCG205-B(Module), DS-TCG205-E, DS-TCG205-E(S), DS-TCG225-JIR, DS-TCG227, DS-TCG227-A, DS-TCG227-AIR, DS-TCG2A5-B, DS-TCG400-E, DS-TCG405-E, DS-TCG405-E(MOD), DS-TCG405-E(S), DS-TCG405-E/H, DS-TCG405-EP, DS-TCG406-E, DS-TCG406-E(ER), DS-TCG406-E(ES), DS-TCG406-E(S), DS-TMC200-E1, DS-TMC400-E, DS-TMC403-E, DS-TMC403-EL, iDS-2CD9545-ESU

2.3 Terms And Definitions

2.3.1 Event

It refers to the information uploaded by the device. The event is uploaded by the device in real time for the immediate response from the client or the platform. If the device is offline, the event will be stored in the cache first and then it will be uploaded again when the connection is restored.

2.3.2 Listen

After the platform starts listening service, when an event occurs, the event information will be sent to the listening port of the platform based on the IP address and port No., then the connection will be closed.

2.3.3 Listening Host

The listening host service can be enabled to receive the event information from devices.

2.3.4 Motion Detection

It can realize image changes recognition and moving objects detection of a specified area by camera. For example, if a person passes by or a camera is moved, the motion will be recognized and then an alarm will be uploaded for notifications, and the event can trigger recording and alarm output.

2.4 Symbols And Acronyms

admin: administrator

HC: Connect Mobile Client

HPP: Partner Pro

HTTP: Hypertext Transfer Protocol

NVR: Network Video Recorder

IPC: IP Camera

OSD: On-screen display

VMD: Motion Detection

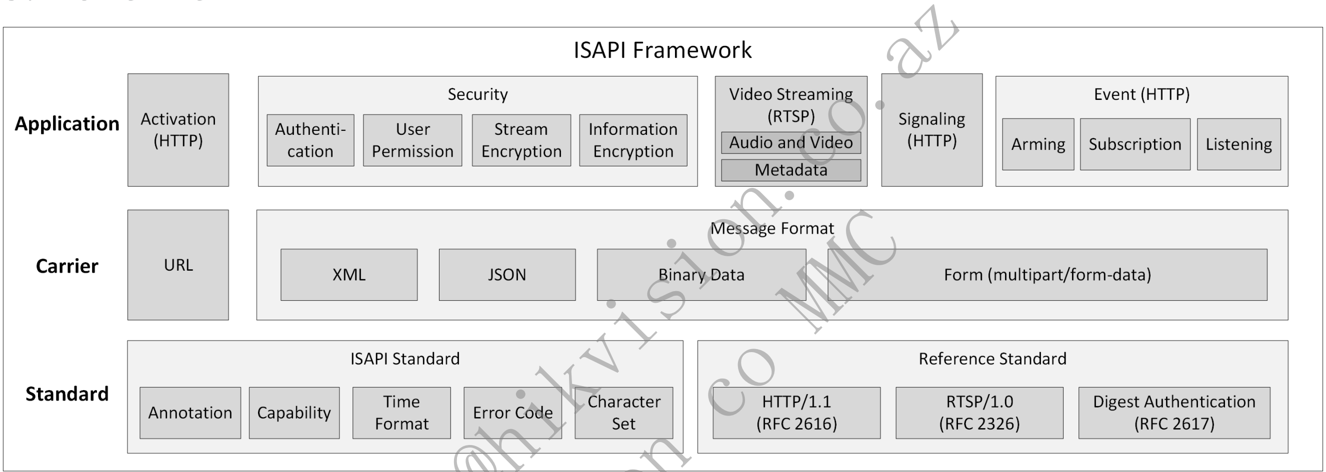
ANPR: Automatic Number Plate Recognition

2.5 Update History

No update record

3 ISAPI Framework

3.1 Overview



Notes:

In general, ISAPI refers to the communication protocol based on the HTTP standard. As ISAPI is usually used along with RTSP (Real-Time Streaming Protocol), the RTSP standard is brought into the ISAPI system.

The metadata scheme for transmitting additional information of the stream is extended based on the RTSP standard to transmit the video stream and the structured intelligent information of the stream simultaneously. It is compatible with the RTSP standard.

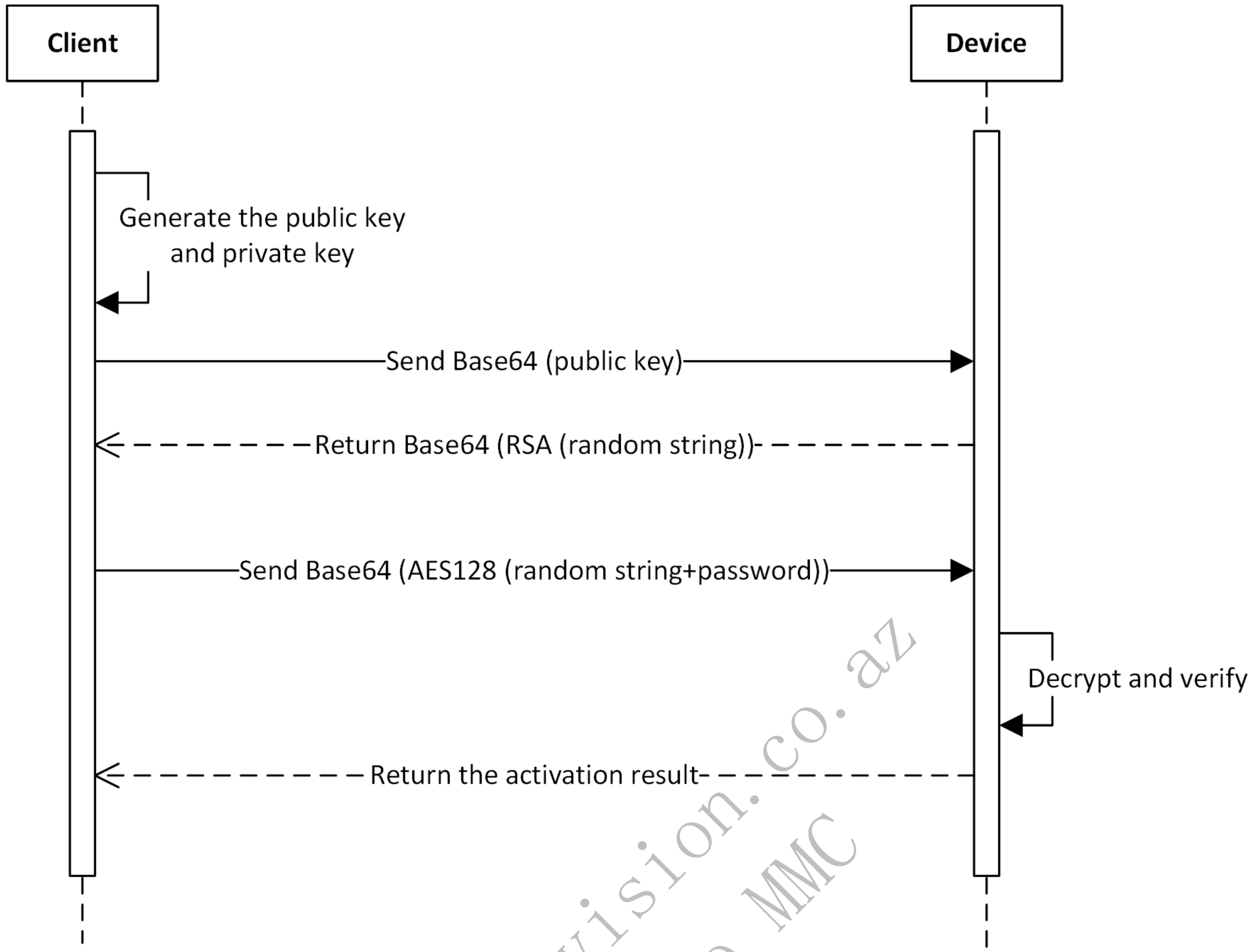
3.2 Activation

The purpose of activation is to ensure that the user can set the password for the device and the password meets the security requirement. After the device is activated, you can use the related functions.

ISAPI is a communication protocol running on the application layer. When activating the device via ISAPI, you should know the device's IP address and make sure that the device is connected to the client.

The web application built in the device supports activating the device via ISAPI. When you enter the device's IP address in the address bar of the web browser on the PC, you can activate the device according to the activation guide.

If you want to activate the device on your own application, you need to integrate the activation function via ISAPI. The API calling flow and related APIs are shown below.



Firstly, two operations are defined:

- bytesToHexstring: it is used to convert a byte array (the length is N) to a hexadecimal string (the length is 2N). For example, `127,10,23` -> `7f0a17`
- hexStringToBytes: it is used to convert a hexadecimal string (the length is 2N) to a byte array (the length is N). For example, `7f0a17` -> `127,10,23`

1. The client generates a public and private key pair (1024 bits), and gets the 128-byte modulus in the public key (hereinafter referred to as public key modulus). If the length is longer than 128, the leading 0 needs to be removed.
2. The client converts the public key modulus to a 256-byte public key string via bytesToHexstring and sends the public key string to the device in XML message (related URI: `POST /ISAPI/Security/challenge`) after being encoded by Base64.
3. The device parses the request to obtain a 256-byte public key string decoded by Base64 and converts it to a 128-byte public key modulus via hexStringToBytes. The complete public key is the combination of obtained public key modulus and public exponent (the default value is `'010001'`).
4. The device generates a 32-byte hexadecimal random string, calls the RSA API to encrypt the random string with the private key, converts the encrypted data to a string via bytesToHexstring, encodes the string by Base64, and then sends it to the client.
5. The client decodes the string from the device by Base64, converts it via hexStringToBytes to get the encrypted data, decrypts the encrypted data with the private key via RSA to obtain a 32-byte hexadecimal random string, converts the obtained string via hexStringToBytes to get a 16-byte AES key. Then the client uses the AES key to encrypt the `"string consisting of the first 16 characters of the random string and the real password"` by AES128 ECB mode (with zero-padding method) to get a ciphertext, and converts the ciphertext via bytesToHexstring, encodes it by Base64, and sends it to the device in XML message (related URI: `PUT /ISAPI/System/activate`). Note: If the first 16 characters of the random string are `aaaabbbbccccdddd` and the real password is `Abc12345`, the data before encryption is `aaaabbbbccccddddAbc12345`. This can ensure that the client uses the random string as the key for encryption.
6. The device decodes the string by Base64, converts it via hexStringToBytes to get the ciphertext, uses the AES key to decrypt the ciphertext by AES128 ECB mode, and gets the real password via removing the first 16 characters.

7. The device verifies the password and returns the activation result.

Notes:

- You can get the device's activation status by calling the URI `GET /SDK/activateStatus` which requires no authentication.
- Devices also support to be activated via SADP (Search Active Device Protocol) which is based on the communication protocol of the data link layer. With SADP, you do not have to know the IP address of the device but need to ensure that the device and the application running SADP are connected to the same router. SADP also supports discovering devices in the LAN, changing the password of the devices, and so on. The HCSadpSDK is provided for SADP integration, including the developer guide, plug-in, and sample demo which can be used as a simple SADP tool.

3.3 Security Mechanism

3.3.1 Authentication

When the client applications send requests to devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Currently, all mainstream request class libraries of HTTP have encapsulated digest authentication. See details in Authentication of Quick Start Guide.

3.3.2 User Permission

There are three kinds of users with different permissions for access control and management.

Administrator: Has the permission to access all supported resources and should keep activated all the time. It is also known as "admin".

Operator: Has the permission to access general resources and a part of advanced resources.

Normal User: Only has the permission to access general resources.

3.3.3 Information Encryption

During ISAPI integration, the HTTPS service of devices is enabled by default. When the client applications communicate with devices via HTTPS, the information can be transmitted securely.

3.4 Video Streaming

3.4.1 Audio and Video Stream

ISAPI supports getting and setting stream media parameters of the device, such as video resolution, encoding format, and stream.

Cameras support standard RTSP (Real-Time Streaming Protocol, see details in RFC 7826). Client applications can get the stream from devices via RTSP.

For details about real-time streaming and video playback, refer to **Real-Time Live View** and **Playback** in Quick Start Guide.

3.4.2 Metadata

The metadata is the structured intelligent information generated by intelligent devices. When the client applications get the audio and/or video stream from devices via RTSP, the metadata will be returned by the device at the same time. For example, to display the face target frame, face information, vehicle target frame, license plate number, vehicle information, and other information on the video stream, the client applications can overlay the above information on the video image.

Before using the metadata, you need to enable the metadata function of the device and then get the stream from the device via RTSP. Some devices support subscribing to the metadata by type. For details about the process of integrating the metadata function, refer to **Metadata Management**.

4 Quick Start Guide

4.1 Authentication

When the client applications send requests to the devices, they need to use digest authentication (see details in RFC 7616) for identity authentication.

Client applications only need to call APIs of the class library to implement the digest authentication. The sample code is shown below.

4.1.1 C/C++ (libcurl)

```
// #include <curl/curl.h>
// Callback Function
static size_t OnWriteData(void* buffer, size_t size, size_t nmemb, void* lpVoid)
{
    std::string* str = dynamic_cast<std::string*>((std::string *)lpVoid);
    if( NULL == str || NULL == buffer )
    {
        return -1;
    }

    char* pData = (char*)buffer;
    str->append(pData, size * nmemb);
    return nmemb;
}

std::string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
std::string strResponseData;
CURL *pCurlHandle = curl_easy_init();
curl_easy_setopt(pCurlHandle, CURLOPT_CUSTOMREQUEST, "GET");
curl_easy_setopt(pCurlHandle, CURLOPT_URL, strUrl.c_str());
// Set the user name and password
curl_easy_setopt(pCurlHandle, CURLOPT_USERPWD, "admin:admin12345");
// Set the authentication method to the digest authentication
curl_easy_setopt(pCurlHandle, CURLOPT_HTTPAUTH, CURLAUTH_DIGEST);
// Set the callback function
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEFUNCTION, OnWriteData);
// Set the parameters of the callback function to get the returned information
curl_easy_setopt(pCurlHandle, CURLOPT_WRITEDATA, &strResponseData);
// Timeout settings for receiving the data. If receiving data is not completed within 5 seconds, the application will exit directly
curl_easy_setopt(pCurlHandle, CURLOPT_TIMEOUT, 5);
// Set the redirection times to avoid too many redirections
curl_easy_setopt(pCurlHandle, CURLOPT_MAXREDIRS, 1);
// Connection timeout duration. If the duration is too short, the client application will be disconnected before the data request sent by the application reaches the device
curl_easy_setopt(pCurlHandle, CURLOPT_CONNECTTIMEOUT, 5);
CURLcode nRet = curl_easy_perform(pCurlHandle);
if (0 == nRet)
{
    // Output the received message
    std::cout << strResponseData << std::endl;
}
curl_easy_cleanup(pCurlHandle);
```

4.1.2 C# (WebClient)

```
// using System.Net;
// using System.Net.Security;
try
{
    string strUrl = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
    WebClient client = new WebClient();
    // Set the user name and password
    client.Credentials = new NetworkCredential("admin", "admin12345");
    byte[] responseData = client.DownloadData(strUrl);
    string strResponseData = Encoding.UTF8.GetString(responseData);
    // Output received information
    Console.WriteLine(strResponseData);
}
catch (Exception ex)
{
    Console.WriteLine(ex.Message);
}
```

4.1.3 Java (HttpClient)

```java
// import org.apache.commons.httpclient.HttpClient;
String url = "http://192.168.18.84:80/ISAPI/System/deviceInfo";
HttpClient client = new HttpClient();
// Set the user name and password
UsernamePasswordCredentials creds = new UsernamePasswordCredentials("admin", "admin12345");
client.getState().setCredentials(AuthScope.ANY, creds);
GetMethod method = new GetMethod(url);
method.setDoAuthentication(true);
int statusCode = client.executeMethod(method);
byte[] responseData = method.getResponseBodyAsString().getBytes(method.getResponseCharSet());
String strResponseData = new String(responseData, "utf-8");
method.releaseConnection();
// Output received information
System.out.println(strResponseData);
```

4.1.4 Python (requests)

```python
# - *- coding: utf-8 -*-

import requests
request_url = 'http://192.168.18.84:80/ISAPI/System/deviceInfo'
# Set the authentication information
auth = requests.auth.HTTPDigestAuth('admin', 'admin12345')
# Send the request and receive response
response = requests.get(request_url, auth=auth)
# Output response content
print(response.text)
```

4.2 Message Parsing

4.2.1 Message Format

During the process of communication and interaction via ISAPI, the request and response messages are often text data in XML or JSON format. Besides that, the data of firmware packages and configuration files is in binary format. A request can also be in form format with multiple formats of data (multipart/form-data).

4.2.1.1 XML

Generally, the `Content-Type` in the headers of the HTTP request is `application/xml; charset="UTF-8"`.

Request and response messages in XML format are all encoded with UTF-8 standards in ISAPI.

The namespace `http://www.isapi.org/ver20/XMLSchema` and ISAPI version number `2.0` of XML messages are configured by default, see the example below.

```xml
<?xml version="1.0" encoding="UTF-8"?>
<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <Node>
    <id>1</id>
    <enabled>true</enabled>
    <nodeName>nodeName</nodeName>
    <level>level1</level>
  </Node>
</NodeList>
```

4.2.1.2 JSON

The `Content-Type` in the headers of the HTTP request is often `application/json`.

To distinguish between APIs with XML messages and those with JSON messages, ISAPI adds the query parameter format=json to all request URLs with JSON messages, e.g., `http://192.168.1.1:80/ISAPI/System/Sensor/thermometrySensor?format=json`. Messages of request URLs without the query parameter format=json are usually in XML format. However, there may be some exceptions, and the message format is subject to the API definition.

Request and response messages in JSON format are all encoded by UTF-8 in ISAPI.

4.2.1.3 Binary Data

For the firmware and configuration files, the `Content-Type` in the header of an HTTP request is often `application/octet-stream`.

4.2.1.4 Form (multipart/form-data)

When multiple pieces of data are submitted at the same time in an ISAPI request (e.g., the person information and face picture need to be submitted at the same time when a face record is added to the face picture library), the `Content-Type` in the header of the corresponding HTTP request is usually `multipart/form-data, boundary=AaB03x`, where the boundary is a variable used to separate the entire HTTP body into multiple units and each unit is a piece of data with its own headers and body. In `Content-Disposition` of form unit headers, the `name` property refers to the form unit name, which is required for all form units; the `filename` property refers to the file name of form unit body, which is required only when the form unit body is a file. In headers of form units, `Content-Length` refers to the body length, which starts after CRLF(`\r\n`) and ends before two hyphens (`--`) of next form. There should be a CRLF used as the delimiter of two form units before two hyphens (`--`), and the `Content-Length` of previous form unit does not include the CRLF length. For the detailed format description, refer to RFC 1867 (Form-Based File Upload in HTML). Pay attention to two hyphens (`--`) before and after the boundary.

Notes

- In RFC specifications, it is strongly recommended to contain the field `Content-Length` in the entity header, and there is no requirement that the field `Content-Length` should be contained in the header of each form element. The absence of field `Content-Length` in the header should be considered when the client and device programs parse the form data.
- To avoid the conflict between message content and boundary value, it is recommended to use a longer and more complex string as the boundary value.

The example of ISAPI form data submitted by a client to a device is as follows.

```
POST /ISAPI/Intelligent/FDLib/pictureUpload
Content-Type: multipart/form-data; boundary=e5c2f8c5461142aea117791dade6414d
Content-Length: 56789

--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="PictureUploadData";
Content-Type: application/xml
Content-Length: 1234

<PictureUploadData/>
--e5c2f8c5461142aea117791dade6414d
Content-Disposition: form-data; name="face_picture"; filename="face_picture.jpg";
Content-Type: image/jpeg
Content-Length: 34567

Picture Data
--e5c2f8c5461142aea117791dade6414d--
```

The example of ISAPI form data responded from a device to a client is as follows.

In ISAPI messages, when there are multiple form units, three nodes (`pid`, `contentid`, and `filename`) are used for linking form units. The corresponding relations are as follows:

Node Name	Form Field	Description
pid	name	pid in XML/JSON messages corresponds to the name property of Content-Disposition in form headers.
contentid	Content-ID	contentid in XML/JSON messages corresponds to Content-ID in form headers.
filename	filename	filename in XML/JSON messages corresponds to filename property of Content-Disposition in form headers.

```
HTTP/1.1 200 OK
Content-Type: multipart/form-data; boundary=136a73438ecc4618834b999409d05bb9
Content-Length: 56789

--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="mixedTargetDetection";
Content-Type: application/json
Content-Length: 811

{
    "ipAddress": "172.6.64.7",
    "macAddress": "01:17:24:45:D9:F4",
    "channelID": 1,
    "dateTime": "2009-11-14T15:27+08:00",
    "eventType": "mixedTargetDetection",
    "eventDescription": "Mixed target detection",
    "deviceID": "123456789",
    "CaptureResult": [{
        "targetID": 1,
        "Human": {
            "Rect": {
                "height": 1.0,
                "width": 1.0,
                "x": 0.0,
                "y": 0.0
            },
            "contentID1": "humanImage", /*human body thumbnail*/
            "contentID2": "humanBackgroundImage", /*human body background picture*/
            "pId1": "9d48a26f7b8b4f2390c16808f93f3534", /*human body thumbnail ID */
            "pId2": "5EE7078E07BB47CF860DE8E4E9A85F28" /*ID of human body background picture*/
        }
    }]
}
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="9d48a26f7b8b4f2390c16808f93f3534"; filename="humanImage.jpg";
Content-Type: image/jpeg
Content-Length: 34567
Content-ID: humanImage

Picture Data
--136a73438ecc4618834b999409d05bb9
Content-Disposition: form-data; name="5EE7078E07BB47CF860DE8E4E9A85F28"; filename="humanBackgroundImage.jpg";
Content-Type: image/jpeg
Content-Length: 345678
Content-ID: humanBackgroundImage

Picture Data
--136a73438ecc4618834b999409d05bb9--
```


4.2.2 Annotation

The field descriptions of ISAPI request and response messages are marked as annotations in the example messages as shown below.

```
<?xml version="1.0" encoding="UTF-8"?>

<NodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, node list, attr:version{req, string, version No., range:[,]}-->
  <Node>
    <!--ro, opt, object, node information-->
    <id>
      <!--ro, req, int, node No., range:[,], step:, unit:, unitType:-->1
    </id>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <nodeName>
      <!--ro, req, string, node name, range:[1,32]-->test
    </nodeName>
    <level>
      <!--ro, opt, enum, level, subType:string,
      [level1#level 1,level2#level 2,level3#level 3]-->level1
    </level>
  </Node>
</NodeList>
```

```
{
    "name":  "test",
    /*ro, req, string, name, range:[1,32]*/
    "type":  "type1",
    /*ro, req, enum, type, subType:string, [type1#type 1,type2#type 2]*/
    "enabled":  true,
    /*ro, opt, bool, enable or not, desc:xxxxxxx*/
    "NodeList": {
    /*opt, object, node list, dep:and,{$.enabled,eq,true}*/
        "scene":  1,
        /*req, enum, scene, subType:int, [1#scene 1; 2#scene 2; 3#scene 3]*/
        "ID":  1
        /*req, int, No., range:[1,8], step:, unit:, unitType:*/
    }
}
```

Key annotations are shown in the table below.

Annotation	Description	Remark
ro	Attribute: Read-Only	This field can only be obtained and cannot be edited.
wo	Attribute: Write-Only	This field can only be edited and cannot be obtained.
req	Attribute: Required	This field is required for request messages sent to the device and response messages returned from the device.
opt	Attribute: Optional	This field is optional for request messages sent to the device and response messages returned from the device.
dep	Attribute: Dependent	This field is valid and required when specific conditions are satisfied.
object	Field Type: Object	The field of type object contains multiple sub-fields.
list	Field Type: List	The subType following it refers to the data type of sub-items in the list.
subType	Field Type: String	The range following it refers to the maximum and the minimum string size of the field.
int	Field Type: Int	The range following it refers to the maximum and the minimum value of the field.
float	Field Type: Float	The range following it refers to the maximum and the minimum value of the field.
bool	Field Type: Boolean	The value can be true or false.
enum	Field Type: Enumeration	The subType following it indicates that the enumerators are of type string or int. The [] following the subType contains all enumerators.
subType	Sub-Type of Field	When the type of field is list or enum, the value of subType is the data type of each sub-object.
desc	Field Description	The detailed description of the field.

4.2.3 Capability Set

ISAPI has designed capability sets for almost all functions, APIs, and fields. URLs for getting the capability set end with `/capabilities`. Some URLs may contain query parameters in the format: `/capabilities?format=json&type=xxx`.

There are two types of fields in the capability message of ISAPI: whether the device supports a function and the value range of a field in an API.

Whether the device supports a function: it is often in the format `isSupportXxxxxxxx`, which indicates that whether the device supports a function and a set of APIs for implementing this function.

The capability message example in JSON format is shown below:

```
{
    "isSupportMap":  true,
    /*ro, opt, bool, whether it supports the e-map function, desc:/ISAPI/SDT/Management/map/capabilities?format=json*/
    "isSupportAlgTrainResourceInfo":  true,
    /*ro, opt, bool, whether it supports only getting the resource information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/ResourceInfo?format=json*/
    "isSupportAlgTrainAuthInfo":  true,
    /*ro, opt, bool, whether it supports ony getting the authorization information of the algorithm training platform,
desc:/ISAPI/SDT/algorithmTraining/SoftLock/AuthInfo?format=json*/
    "isSupportAlgTrainNodeList":  true,
    /*ro, opt, bool, whether it supports only getting the node information of the algorithm training platform, desc:/ISAPI/SDT/algorithmTraining/NodeList?
format=json*/
    "isSupportNAS":  true
    /*ro, opt, bool, whether it supports mounting and unmounting NAS, desc:/ISAPI/SDT/Management/NAS/capabilities?format=json*/
}
```

The capability message example in XML format is shown below:

```
<isSupportNetworkStatus>
    <!--ro, opt, bool, whether it supports searching the network status, desc: related API (/ISAPI/System/Network/status?format=json)-->true
</isSupportNetworkStatus>
```

The value range of the field: the maximum value, minimum value, the maximum size, the minimum size, options, and so on of each field of the API.

The example of JSON format is shown below:

```
{
    "boolType": {
        /*req, object, example of the capability of type bool*/
        "@opt": [true, false]
        /*req, array, options, subType: bool*/
    },
    "integerType": {
        /*req, object, example of the capability of type integer*/
        "@min": 0,
        /*ro, req, int, the minimum value*/
        "@max": 100
        /*ro, req, int, the maximum value*/
    },
    "stringType": {
        /*req, object, example of the capability of type string*/
        "@min": 0,
        /*ro, req, int, the minimum string size*/
        "@max": 32
        /*ro, req, int, the maximum string size*/
    },
    "enumType": {
        /*req, object, capability example of type enum*/
        "@opt": ["enum1", "enum2", "enum3"]
        /*req, array, options, subType: string*/
    }
}
```

The example of XML format is shown below:

```
<boolType opt="true,false" def="true">
    <!--ro, opt, bool, example of the capability of type bool-->true
</boolType>
<integerType min="0" max="100">
    <!--ro, opt, int, example of the capability of type int-->50
</integerType>
<stringType min="0" max="64">
    <!--ro, opt, string, example of the capability of type string-->test
</stringType>
<enumType opt="red,white,black" def="red">
    <!--ro, opt, string, example of the capability of type enum-->white
</enumType>
```

Note: For the same capability set, devices of different models and versions may return different results. The values shown in this document are only examples for reference. The capability set actually returned by the device takes precedence.

4.2.4 Time Format

ISAPI adopts ISO 8601 Standard Time Format, which is the same as W3C Standard Date and Time Formats.

Format: `YYYY-MM-DDThh:mm:ss.sTZD`

```
YYYY = the year consisting of four decimal digits
MM = the month consisting of two decimal digits (01-January, 02-February, and so forth)
DD = the day consisting of two decimal digits (01 to 31)
hh = the hour consisting of two decimal digits (00 to 23, a.m. and p.m. are not allowed)
mm = the minute consisting of two decimal digits (00 to 59)
ss = the second consisting of two decimal digits (00 to 59)
s = one or more digits representing the fractional part of a second
TZD = time zone identifier (Z or +hh:mm or -hh:mm)
```

Example: 2017-08-16T20:17:06.123+08:00 refers to 20:17:06.123 on August 16, 2017 (local time which is 8 hours ahead of UTC). The plus sign (+) indicates that the local time is ahead of UTC, and the minus sign (-) means that the local time is behind UTC.

After the DST is enabled, the local time and time difference will change compared with UTC, and the values of related fields also need to be changed. Disabling the DST will bring into the opposite effect.

Example: In 1986, the DST was in effect from May 4 at 2:00 a.m. (GMT+8). During the DST period, the clocks were moved one hour ahead, which means that there was one less hour on that day. When the DST ends at 2:00 a.m. on September 14, 1986, the clocks were moved one hour back and there was an extra hour on that day. The changes of the time are as follows:

- DST Starts: 1986-05-04T02:00:00+08:00 --> 1986-05-04T03:00:00+09:00
- DST Ends: 1986-09-14T02:00:00+09:00 --> 1986-09-14T01:00:00+08:00

Notes:

- The time difference cannot be simply used to determine the time zone. Because when the DST starts, the time difference will change and it cannot represent the actual time zone.
- Both TZ (UTC time, e.g., 1986-05-03T18:00:00Z) and TD (local time and time difference, e.g., 1986-05-04T02:00:00+08:00) meet the time format standards of ISO 8601. In ISAPI, the TD format is recommended to be used in messages sent from the user applications and the devices.
- A few old-version devices will return the time in TZ format. For representing the time difference information and forward compatibility, an extra field `timeDiff` is added as shown in the example below. User applications need to support both TD format and TZ format when parsing the time in the messages returned by devices.

```
{
    "dateTime": "1986-05-03T18:00:00Z", /*device time. The value in TZ format is the UTC time and the value in TD format is the time difference between the device's local time and UTC*/
    "timeDiff": "+08:00" /*optional, time difference between the local time and UTC time. If this field does not exist, the user application will convert the dateTime into the local time for use*/
}
```


4.2.5 Character Set

To prevent characters not commonly used from resulting in exceptions in device programs and user applications, ISAPI limits the valid field values of type string to a specific range of characters. Character sets allowed to be used in the fields of type string in ISAPI are listed below.

- Single-byte character set: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and special characters (see details in the table below).
- Multi-byte character set: language characters based on Unicode and encoded by UTF-8 (UTF-8 encoding is a transformation format of Unicode character set. For details, refer to RFC 2044).

No.	Name	Special Character	No.	Name	Special Character
1	Open Parenthesis	(	18	Dollar Sign	$
2	Close Parenthesis	)	19	Percent Sign	%
3	Plus Sign	+	20	Ampersand	&
4	Comma	,	21	Close Single Quotation Mark	'
5	Minus Sign	-	22	Asterisk	*
6	Period	.	23	Slash	/
7	Semicolon	;	24	Smaller Than	<
8	Equal Sign	=	25	Greater Than	>
9	At Sign	@	26	Question Mark	?
10	Open Square Bracket	[	27	Caret	^
11	Close Square Bracket	]	28	Open Single Quotation Mark	'
12	Underscore	_	29	Vertical Bar	|
13	Open Brace	{	30	Tilde	~
14	Close Brace	}	31	Double Quotation Marks	"
15	Space		32	Colon	:
16	Exclamation Mark	!	33	Backslash	|
17	Octothorpe	#			

The valid characters that can be used in some special fields are listed below.

- User name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 30 in the special character table.
- Password: User Name: lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), and characters from No. 1 to No. 33 in the special character table.
- Names displayed on the UI (device name, person name, face picture library name, etc.): lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters.
- Normal fields of type string support lowercase letters (`a-z`), uppercase letters (`A-Z`), digits (`0-9`), characters from No. 1 to No. 15 in the special character table, and multi-byte characters by default.

4.2.6 Error Processing

When requesting via ISAPI failed (the HTTP status code is not 200), the device will return the HTTP status code and ISAPI error code. For HTTP status codes, refer to 10 Status Code Definitions in RFC 2616. For ISAPI error codes, refer to Error Code Dictionary.

Message Example:

```
HTTP/1.1 403 Forbidden
Content-Type: application/json; charset="UTF-8"
Date: Thu, 15 Jul 2021 20:43:30 GMT
Content-Length: 229
Connection: Keep-Alive

{
    "requestURL": "/ISAPI/Event/triggers/notifications/channels/whiteLightAlarm",
    "statusCode": 4,
    "statusString": "Invalid Operation",
    "subStatusCode": "notSupport",
    "errorCode": 1073741825,
    "errorMsg": "notSupport"
}
```

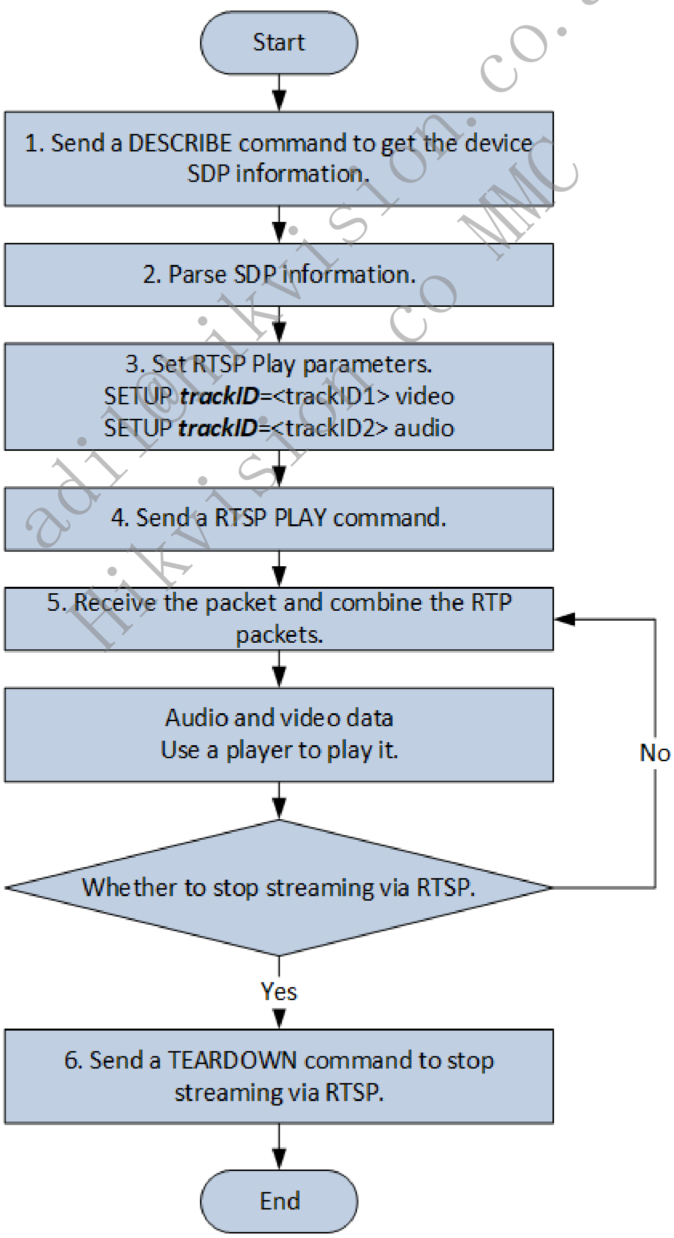
4.3 Real-Time Live View

4.3.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

4.3.2 API Calling Flow



1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`. Digest authentication with devices is required before this step. 2. The client parses the media SDP information returned by the device. 3. Set RTSP play parameters, that is to set the track ID parsed from SDP information via SETUP commands. For

example, trackID=1 indicates videos while trackID=2 indicates audios. 4. The client sends an RTSP PLAY command, and the device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`. 5. The client receives the RTP packet sent by the device. Divided RTP packets should be assembled on the client before being parsed. 6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address; `[:port]` is optional, and 554 by default; `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, the IP address of the target device is `172.7.203.11`, and the streaming address of main stream for channel 17 will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

4.3.3 Example

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.22.126
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B48020100000200000111710110401F000000FA000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J............X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

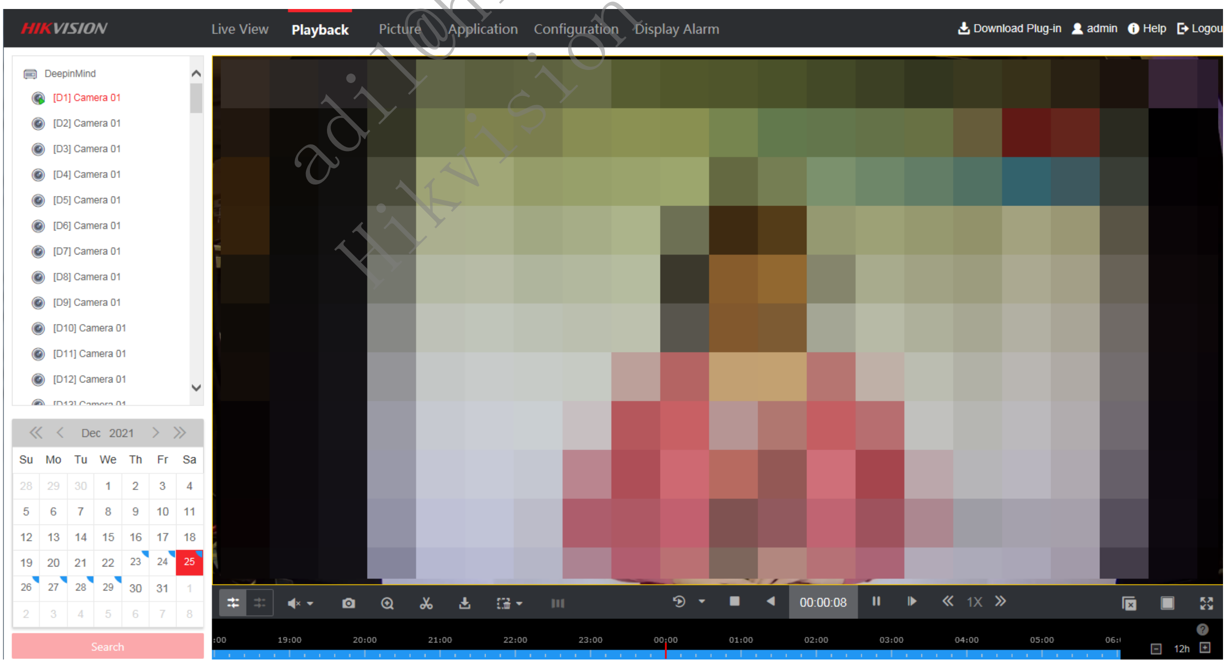
```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```

4.4 Playback

4.4.1 Introduction to the Function

Devices such as cameras and NVRs can store videos. Storage mediums such as SD card, TF card and HDDs are needed before recording videos. The process of playback starts with searching for footages and then getting video stream via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

Below is an example of playback on a web client. The calendar in the bottom-left corner shows the results of searching for videos. Dates with videos are shown on the calendar and you can click a date to play back searched videos of the corresponding day.



4.4.2 API Calling Flow

1. (Optional) Check the calendar to find dates with stored videos: `POST /ISAPI/ContentMgmt/record/tracks/<trackStreamID>/dailyDistribution`. `<trackStreamID>` = channel number * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream). For example, The rackStreamID of the main stream for channel 17 is 1701.
2. Searching for videos by parsing the `playbackURI` of the device to get the RTSP address `POST /ISAPI/ContentMgmt/search`.

3. Play videos back via RTSP. Steps of playback via RTSP is the same as that of real-time live view. Refer to streaming via RTSP in real-time streaming. See details in RFC 7826 (Real Time Streaming Protocol).

Notes:

- Playback via RTSP needs digest authentication. The method is the same as the one of ISAPI digest authentication.
- Send PAUSE command to pause playback. Send PLAY command to continue. If you want to perform fast forward and slow forward, you can send a PLAY command and set Scale in headers. See details in 10.6 PAUSE and 12.34 Scale in RFC 7826 (Real Time Streaming Protocol).

4.4.3 Example

4.4.3.1 Search for Videos According to Calendar

Request from Client

```
POST /ISAPI/ContentMgmt/record/tracks/101/dailyDistribution HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 119
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<trackDailyParam>
    <year>2021</year>
    <monthOfYear>08</monthOfYear>
</trackDailyParam>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:47:43 GMT
Content-Length: 2915
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<trackDailyDistribution version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <dayList>
        <day><id>1</id><dayOfMonth>1</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>2</id><dayOfMonth>2</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>3</id><dayOfMonth>3</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>4</id><dayOfMonth>4</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>5</id><dayOfMonth>5</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>6</id><dayOfMonth>6</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>7</id><dayOfMonth>7</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>8</id><dayOfMonth>8</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>9</id><dayOfMonth>9</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>10</id><dayOfMonth>10</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>11</id><dayOfMonth>11</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>12</id><dayOfMonth>12</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>13</id><dayOfMonth>13</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>14</id><dayOfMonth>14</dayOfMonth><record>false</record></day>
        <day><id>15</id><dayOfMonth>15</dayOfMonth><record>false</record></day>
        <day><id>16</id><dayOfMonth>16</dayOfMonth><record>false</record></day>
        <day><id>17</id><dayOfMonth>17</dayOfMonth><record>false</record></day>
        <day><id>18</id><dayOfMonth>18</dayOfMonth><record>true</record><recordType>time</recordType></day>
        <day><id>19</id><dayOfMonth>19</dayOfMonth><record>false</record></day>
        <day><id>20</id><dayOfMonth>20</dayOfMonth><record>false</record></day>
        <day><id>21</id><dayOfMonth>21</dayOfMonth><record>false</record></day>
        <day><id>22</id><dayOfMonth>22</dayOfMonth><record>false</record></day>
        <day><id>23</id><dayOfMonth>23</dayOfMonth><record>false</record></day>
        <day><id>24</id><dayOfMonth>24</dayOfMonth><record>false</record></day>
        <day><id>25</id><dayOfMonth>25</dayOfMonth><record>false</record></day>
        <day><id>26</id><dayOfMonth>26</dayOfMonth><record>false</record></day>
        <day><id>27</id><dayOfMonth>27</dayOfMonth><record>false</record></day>
        <day><id>28</id><dayOfMonth>28</dayOfMonth><record>false</record></day>
        <day><id>29</id><dayOfMonth>29</dayOfMonth><record>false</record></day>
        <day><id>30</id><dayOfMonth>30</dayOfMonth><record>false</record></day>
        <day><id>31</id><dayOfMonth>31</dayOfMonth><record>false</record></day>
    </dayList>
</trackDailyDistribution>
```

4.4.3.2 Search for Videos

Request from Client

```
POST /ISAPI/ContentMgmt/search HTTP/1.1
Host: 10.14.97.40
Connection: keep-alive
Content-Length: 486
Cache-Control: max-age=0
Accept: */*
X-Requested-With: XMLHttpRequest
If-Modified-Since: 0
User-Agent: Mozilla/5.0 (Windows NT 6.1; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/92.0.4515.131 Safari/537.36
Content-Type: application/x-www-form-urlencoded; charset=UTF-8
Origin: http://10.14.97.40
Referer: http://10.14.97.40/doc/index.html
Accept-Encoding: gzip, deflate
Accept-Language: zh-CN,zh;q=0.9
Cookie: WebSession_78aedfcc66=3b451c9d37cb637827da0815086e7ecfd9984b0095b3cb7198e8197a424a3279

<?xml version="1.0" encoding="utf-8"?>
<CMSearchDescription>
    <searchID>88C2CD4D-D3FA-4AD4-BD80-555C18205DCC</searchID>
    <trackList>
        <trackID>101</trackID>
    </trackList>
    <timeSpanList>
        <timeSpan>
            <startTime>2021-08-16T00:00:00Z</startTime>
            <endTime>2021-08-18T23:59:59Z</endTime>
        </timeSpan>
    </timeSpanList>
    <maxResults>100</maxResults>
    <searchResultPostion>0</searchResultPostion>
    <metadataList>
        <metadataDescriptor>//recordType.meta.std-cgi.com</metadataDescriptor>
    </metadataList>
</CMSearchDescription>
```

Response from Device

```
HTTP/1.1 200 OK
Vary: Accept-Encoding
X-Frame-Options: SAMEORIGIN
Content-Type: application/xml; charset="UTF-8"
X-Content-Type-Options: nosniff
Date: Wed, 18 Aug 2021 15:19:13 GMT
Content-Length: 1021
X-XSS-Protection: 1; mode=block
Connection: Keep-Alive
Accept-Ranges: bytes

<?xml version="1.0" encoding="UTF-8" ?>
<CMSearchResult version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <searchID>{88c2cd4d-d3fa-4ad4-bd80-555c18205dcc}</searchID>
    <responseStatus>true</responseStatus>
    <responseStatusStrg>OK</responseStatusStrg>
    <numOfMatches>1</numOfMatches>
    <matchList>
        <searchMatchItem>
            <sourceID>{00000000-0000-0000-0000-000000000000}</sourceID>
            <trackID>101</trackID>
            <timeSpan>
                <startTime>2021-08-18T15:18:15Z</startTime>
                <endTime>2021-08-18T15:19:08Z</endTime>
            </timeSpan>
            <mediaSegmentDescriptor>
            <contentType>video</contentType>
            <codecType>H.264-BP</codecType>
            <playbackURI>rtsp://10.14.97.40/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&amp;endtime=20210818T151908Z&amp;name=00000004667000100&amp;size=1400788</playbackURI>
            <lockStatus>unlock</lockStatus>
            <name>00000004667000100</name>
            </mediaSegmentDescriptor>
            <metadataMatches>
                <metadataDescriptor>recordType.meta..com/timing</metadataDescriptor>
            </metadataMatches>
        </searchMatchItem>
    </matchList>
</CMSearchResult>
```

4.4.3.3 Playback via RTSP

1. A client sends an RTSP DESCRIBE command.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 4
Authorization: Digest username="", realm="323852ae0234c718f2d4198b", nonce="34e30476d", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="883546f4c19dd156fb3a490266c99715"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
```

2. The server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 5
WWW-Authenticate: Digest realm="323852ae0234c718f2d4198b", nonce="55e5895b9", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 7
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="cf33e4dc6b86a2fdd2e5b26d25e7b99d"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Accept: application/sdp
```

4. The server responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 7
Content-Type: application/sdp
Content-Length: 546
Date: Wed, 18 Aug 2021 07:29:25 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s= Media Server V4.60.000
i= Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:4106
a=range:clock=20210818T151815Z-20210818T151908Z
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H265/90000
a=control:trackID=video
b=AS:4096
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=audio
b=AS:10
a=Media_header:MEDIAINFO=494D4B480201000004000500117101104 01F000000FA000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP commands two times, and the server responds to them.

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=video RTSP/1.0
CSeq: 8
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61048-61049
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61048-61049;server_port=62000-62001;ssrc=d6d3ed5
CSeq: 8
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

```
SETUP rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788/trackID=audio RTSP/1.0
CSeq: 9
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="44d97a3758b7f19f1a31fe042b1464b5"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Transport: RTP/AVP;unicast;client_port=61050-61051
Session: 225263317
```

```
RTSP/1.0 200 OK
Session: 225263317;timeout=60
Transport: RTP/AVP;unicast;client_port=61050-61051;server_port=62002-62003;ssrc=d6d3ed6
CSeq: 9
Accept-Ranges: UTC
Media-Properties: Random-Access=1.0s, Unlimited, Immutable,Scales="-1, 0.5, 0.25, 0.125,:1, 2, 4"
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

6. The client sends an RTSP PLAY request, and the server starts to push streams for playback.

```
PLAY rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 10
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="4d8b7ceb301c06b40a9ae411189a909f"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
Range: npt=0.000-
```

```
RTSP/1.0 200 OK
Range: clock=20210818T151815Z-20210818T151908Z
Session: 225263317
CSeq: 10
Date: Wed, 18 Aug 2021 07:29:25 GMT
```

7. The client sends an RTSP TEARDOWN request to stop playback.

```
TEARDOWN rtsp://10.14.97.40:554/ISAPI/Streaming/tracks/101/?starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788 RTSP/1.0
CSeq: 11
Authorization: Digest username="admin", realm="323852ae0234c718f2d4198b", nonce="55e5895b9", uri="rtsp://10.14.97.40:554/Streaming/tracks/101/?
starttime=20210818T151815Z&endtime=20210818T151908Z&name=00000004667000100&size=1400788", response="1d3f6f8d07d7087d341560b125445456"
User-Agent: LibVLC/3.0.3 (LIVE555 Streaming Media v2016.11.28)
Session: 225263317
```

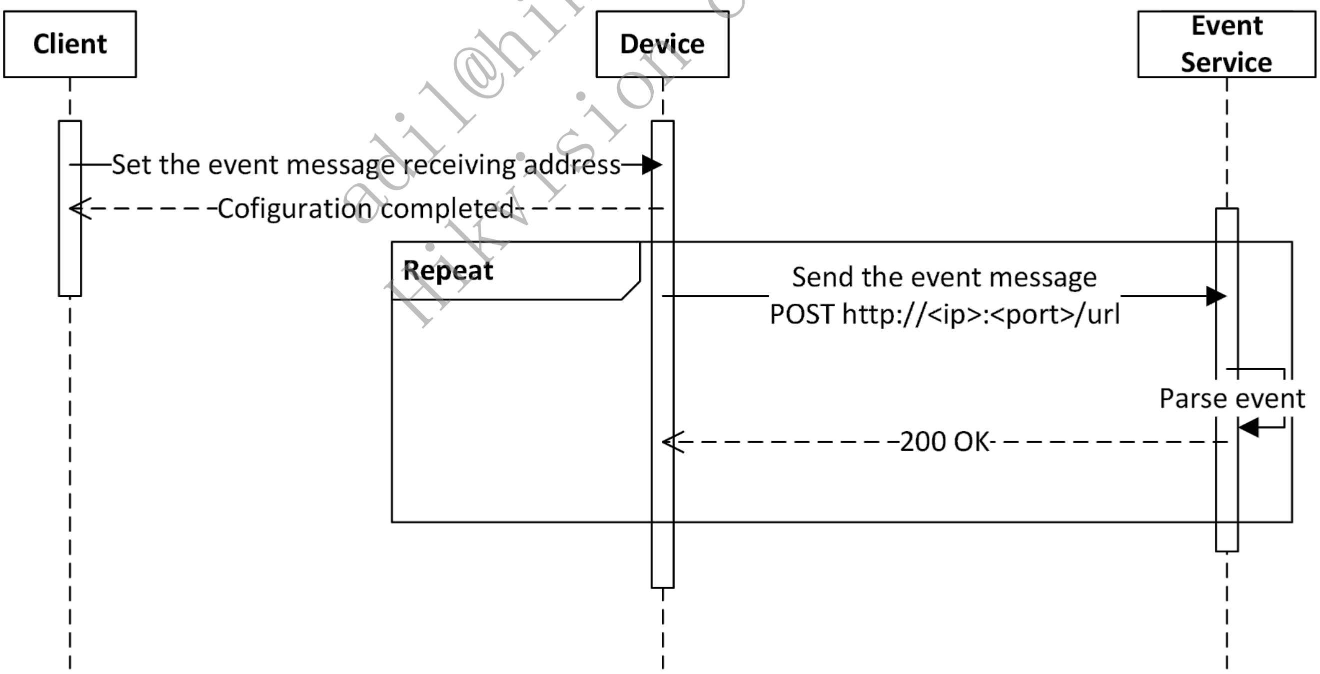
```
RTSP/1.0 200 OK
CSeq: 11
Date: Wed, 18 Aug 2021 07:29:38 GMT
Session: 225263317
Connection: close
```

4.5 Event Uploading

When the rules configured on the device are triggered, the device will generate event messages (e.g., motion detection, etc.) and actively upload them to the client. ISAPI supports three methods to receive event messages uploaded by the device, that is, in arming mode, in listening mode, and via subscription.

4.5.1 Listening

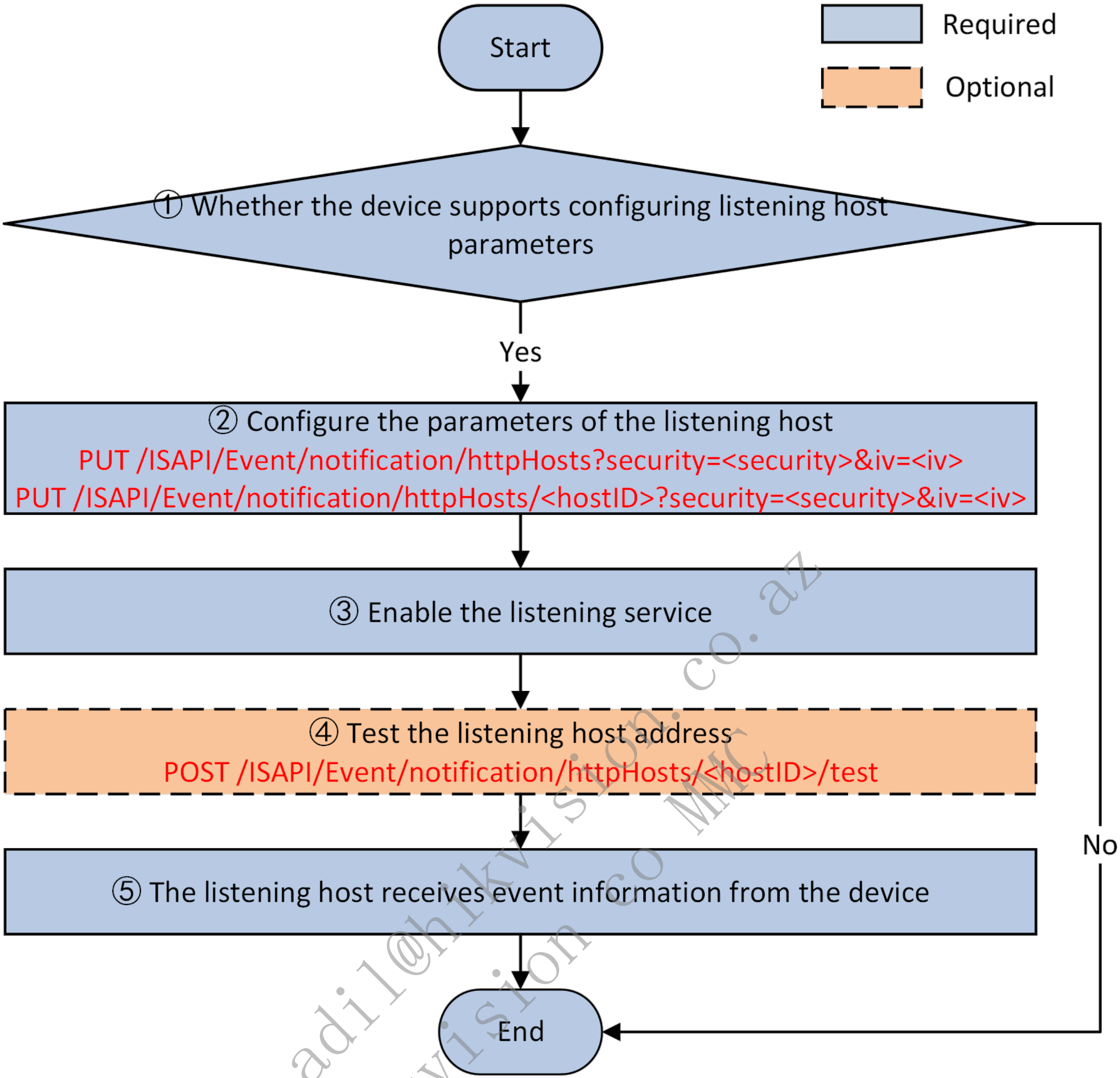
After a client enables the listening service, when an event occurs, the device will send the event information actively to the configured event receiving address. The event receiving address should be valid and configured on the device.



Notes:

- The client and event service can be the same program.
- In listening mode, no heartbeat information is generated on devices.

4.5.1.1 API Calling Flow



1. Check whether the device supports configuring listening host parameters.

   Get the configuration capability of the listening host: `GET /ISAPI/Event/notification/httpHosts/capabilities`. If the node `<HttpHostNotificationCap>` is returned and its value is true, it indicates that the device supports configuring listening host parameters.

2. Configure parameters of the listening host.

   Configure parameters of all listening hosts: `PUT /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Get parameters of all listening hosts: `GET /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Configure parameters of a listening host: `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

   Get parameters of a listening host: `GET /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`.

3. Enable the listening service.

   You need to enable the listening service of the listening host.

4. (Optional) Test the listening service.

   The platform applies the command to the device to test whether the listening host is available for the device: `POST /ISAPI/Event/notification/httpHosts/<hostID>/test`.

5. The listening host receives event information from the device.

   When an event occurs, the device creates an connection with the client and uploads alarm information actively. Meanwhile, the listening host receives data from the device. See details in **Event Messages**.

Note: You can also configure the listening parameters such as the time out via URL `/ISAPI/Event/notification/httpHosts/<hostID>/uploadCtrl`.

4.5.1.2 Event Messages

When an event occurs or an alarm is triggered in listening mode, the event/alarm information can be uploaded with binary data (such as pictures) and without binary data.

1. **Without Binary Data:**

The `Content-Type` in headers of the HTTP request sent by the device is usually `application/xml` or `application/json` as follows:

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: data_gateway_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: application/xml; <!--Content Type, which is used for the upper layer to distinguish different formats when parsing the message-->
Content-Length: text_length
Connection: keep-alive  <!--maintain the connection between the device and the server for better transmission performance-->

<EventNotificationAlert/>
```

Response by the Listening Host

```
HTTP/1.1 200 OK
Date: YourDate
Connection: close
```

2. **With Binary Data:**

The format of the data sent by the device is HTTP form (multipart/form-data). The `Content-Type` in headers of the HTTP request is usually `multipart/form-data, boundary=<frontier>`, of which boundary is a variable used to divide the HTTP body into multiple units, and each unit has its headers and body. See details in RFC 1867 (Form-based File Upload in HTML). An example is shown below. Please note two hyphens `--` before and after the boundary.

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, , related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: device_ip:port <!--HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: en-us
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml <!--some event messages are uploaded in JSON format, and the upper layer needs to distinguish the message format accroding to
Content-Type when parsing event messages-->

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[Picture Data]
--<frontier>--
```

Response by the Listening Host

```
HTTP/1.1 200 OK
Date: YourDate
Connection: close
```

The description of some keywords are as follows:

Keyword	Example	Description
Content-Type	multipart/form-data; boundary=frontier	Content type, multipart/form-data refers to data in form format.
boundary	frontier	Delimiter of the form message. A form message which starts with `--boundary` and ends with `--boundary--`.
Content-Disposition	form-data; name="Picture_Name";	Content description. `form-data` is a piece of form data.
filename	"Picture_Name"	File name. The file refers to the form message.
Content-Length	10	Content length, starting from the next `\r\n` to the next `--boundary`.

4.5.1.3 Exception Handling

Error Codes

statusCode	statusString	subStatusCode	errorCode	Description
6	Invalid Content	eventNotSupport	0x60001024	

5 Device Management (General)

5.1 Calling Flow of Device Packet Capture

5.1.1 Function Introduction

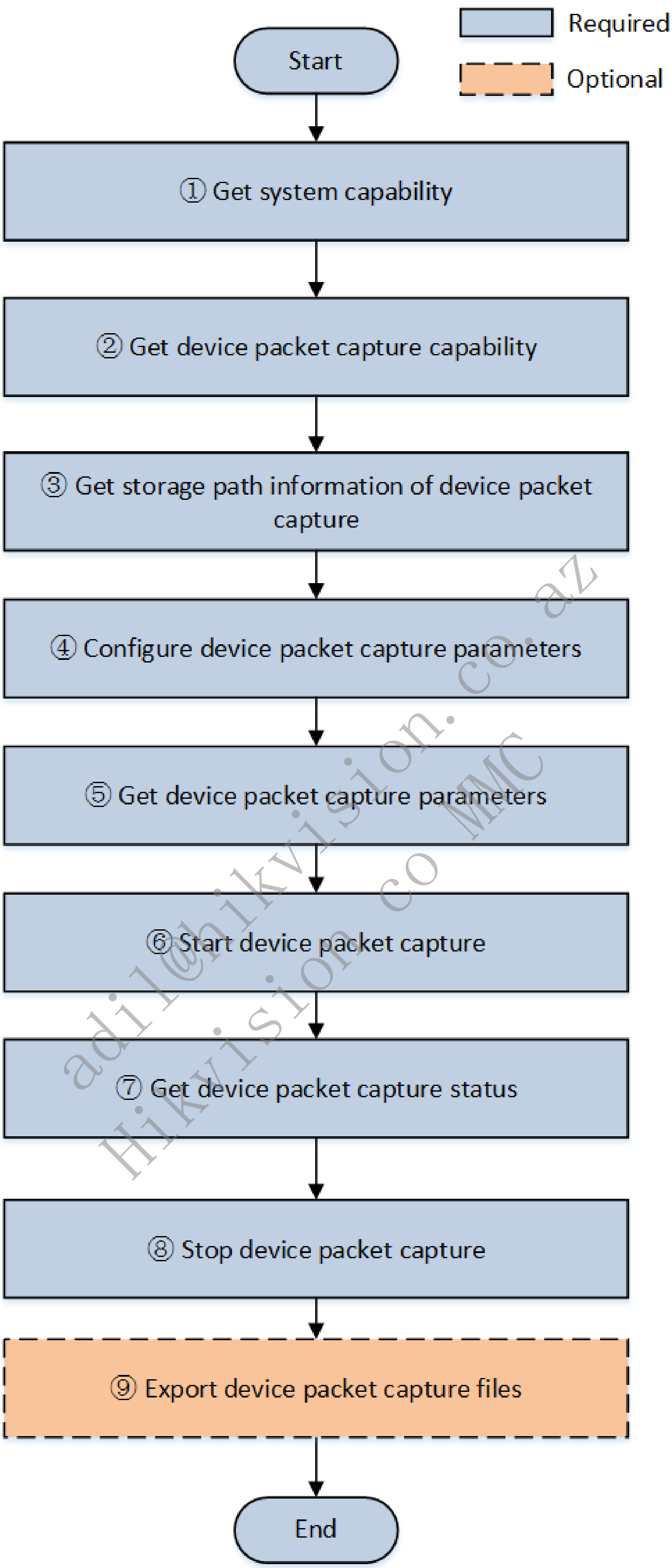
When problems arise after a device is deployed on site, interaction message between the device and the external network is necessary to help developers for troubleshooting. Packet capture can be stored on the local device, and packet capture files can be exported after capture is complete. Also, packet capture files can be uploaded to cloud storage, and packet capture data can be obtained in real-time even if the device does not have the storage space.

1. Device packet capture: save packet capture files on the local device, and export the files after capture is complete. Also, uploading packet capture files to cloud storage after capture is complete is supported. Then the client can obtain the storage URL and download packet capture files from the cloud storage.

2. Device real-time packet capture: after it is enabled, the device returns an URI for downloading packet capture data. The client can submit this URI to the browser to download the packet data. The device transmits packet data via HTTP Chunked, and users can store the packet data through the browser.

5.1.2 API Calling Flow

5.1.2.1 Device Packet Capture



1. Get device system capabilities: `GET /ISAPI/System/capabilities`. Get to know if the device supports packet capture by the field: `<isSupportNetworkCapture>true</isSupportNetworkCapture>`.

2. Check if the device supports packet captures: `GET /ISAPI/System/networkCapture/capabilities?format=json`. If `isSupportManualControl` is true, the device supports packet capture. If `isSupportManualControlAsyn` is true, the device supports asynchronous packet capture.

3. Get storage path information of device packet capture: `GET /ISAPI/System/networkCapture/StoragePathInfo?format=json`.

4. Configure device packet capture parameters such as capture duration, storage path, port, and address: `PUT /ISAPI/System/networkCapture/captureParams?format=json`.

5. Get device packet capture parameters such as capture duration, storage path, port, and address: `GET /ISAPI/System/networkCapture/captureParams?format=json`.

6. Start device packet capture: depending on the parameters, packet capture files can be saved on the local device for export after capture is complete, or packet capture files can be uploaded to cloud storage and downloaded via the storage URL, or the packet capture data can be returned in real-time.

Start device packet capture: `PUT /ISAPI/System/networkCapture/manualStart?format=json&asyn=<asyn>&realTime=<realTime>`.

Note:

- If the API does not contain URL parameters, the packet capture file is saved on the local device.
- If the API contains `asyn=true`, the packet capture file is automatically uploaded to cloud storage after capture is complete.

7. After starting capture, you can repeatedly get capture status, including whether the capture is ongoing, the size of the packet capture data, and the progress and storage URL for uploading the data to cloud storage.

Get status of device packet capture: `GET /ISAPI/System/networkCapture/manualStatus?format=json`.

8. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

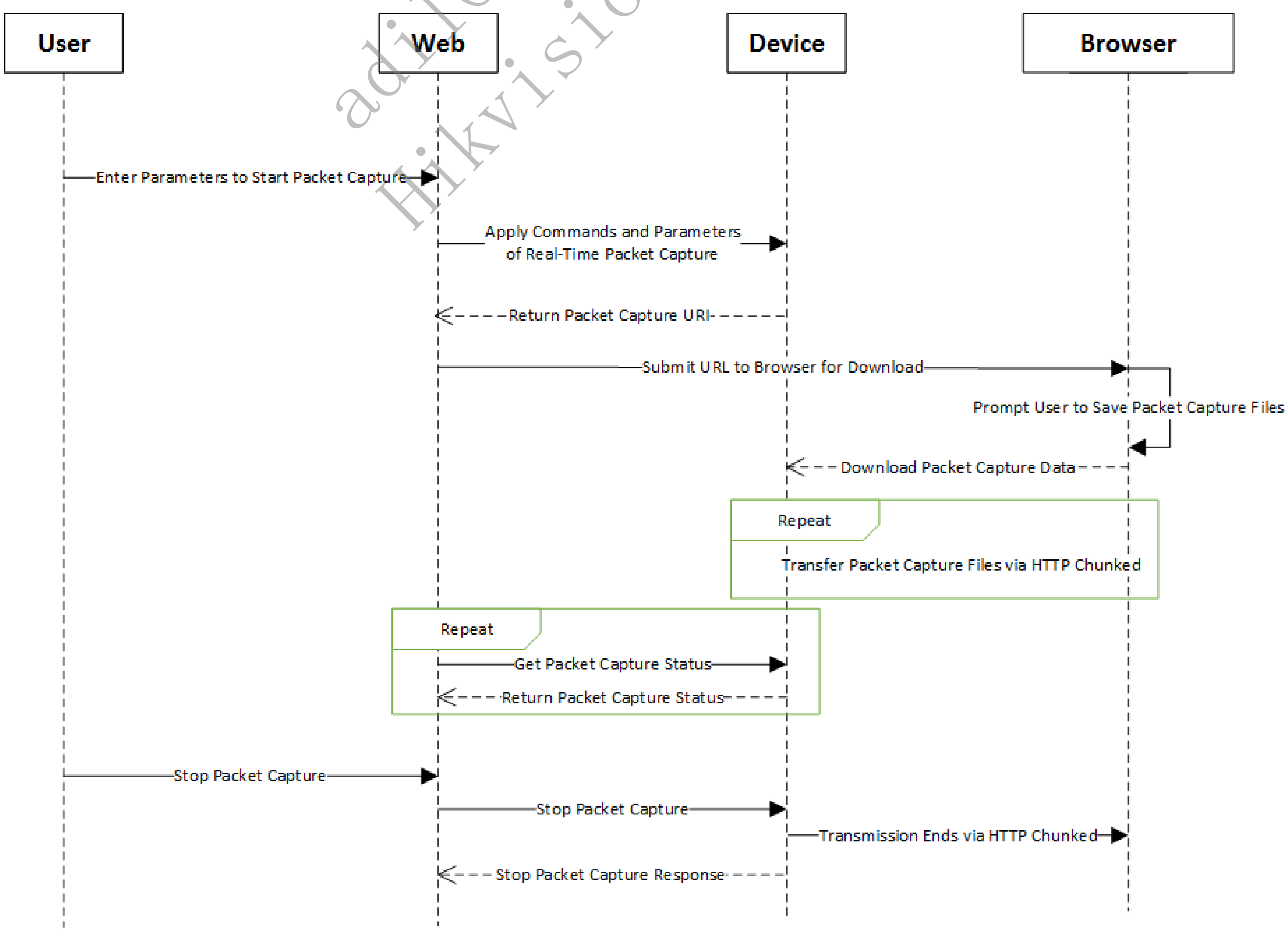
Stop device packet capture: `PUT /ISAPI/System/networkCapture/manualStop?format=json`.

9. (Optional) If packet capture data is stored on the local device, packet capture files need to be exported. If packet capture files are saved to cloud storage or captured in real-time, there is no need to export packet capture files.

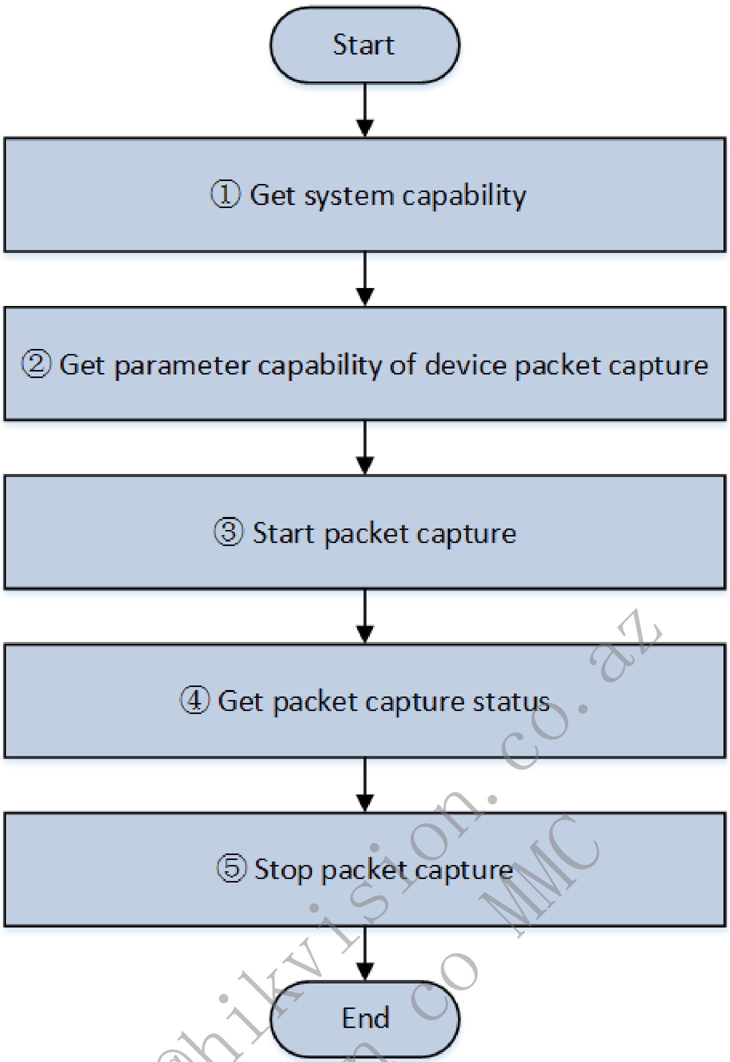
Export device packet capture files: `GET /ISAPI/System/networkCapture/exportFile?format=json`.

5.1.2.2 Device Real-Time Packet Capture

The process of real-time packet capture is shown as the following:



See the following figure for the calling flow:



1. Get device system capabilities: `GET /ISAPI/System/capabilities`. `<isSupportStartNetworkCapture>true</isSupportStartNetworkCapture>` indicates the device supports starting packet capture. `<isSupportStopNetworkCapture>true</isSupportStopNetworkCapture>` indicates the device supports stopping packet capture. `<isSupportGetNetworkCaptureStatus>true</isSupportGetNetworkCaptureStatus>` indicates the device supports getting packet capture status.

2. Get capabilities of packet capture parameters: `GET /ISAPI/System/NetworkCaptureParams/capabilities?format=json`. The `realTimeEnabled` field indicates whether the device supports real-time packet capture.

3. Set the field realTimeEnabled as true in the parameters applied to the device to start device packet capture. The device returns an URI, and the client can download the real-time packet capture data from the device through a browser.

Start packet capture: `POST /ISAPI/System/StartNetworkCapture?format=json&security=<security>&iv=<iv>`.

Note:

- If `realTimeEnabled=true` is contained when starting device packet capture, it indicates packet capture data is uploaded in real-time by HTTP Chunked. The URL for downloading the returned packet capture data is valid for 30 seconds by default. If the download is attempted after this time, the device should return an HTTP 404 status code.

4. After starting packet capture, you can repeatedly get packet capture status, including whether packet capture is ongoing and the size of the packet capture data.

Get packet capture status: `GET /ISAPI/System/GetNetworkCaptureStatus?format=json`.

5. Packet capture can be stopped at any time by calling the interface of stopping packet capture.

Stop device packet capture: `POST /ISAPI/System/StopNetworkCapture?format=json`.

5.2 Device Hardware Asset Management

5.2.1 Introduction to the Function

Hardware assets: the host assets (CPU, memory, and HDD) and peripheral assets (hardware connected to the host including camera, sensor, and USB flash drive).
Typical application: financial industry, where the head office needs to regularly count the security device assets of each branch to collect the number, operation status, and other basic information of deployed host/storage/peripheral cameras.
Extended application: management of industry-related service applications for assets customized by the industry platform (e.g., information and software assets, person assets, and service assets) based on the hardware asset integration.

5.2.2 API Calling Flow

1. Get the search capability of the device hardware asset data via `GET /ISAPI/System/deviceInfo/capabilities`: when `isSupportSearchHardwareAssets` is true, it indicates the device supports searching for asset data.
2. Search for device hardware asset data via `GET /ISAPI/System/deviceInfo/ExportDeviceAssets?format=json` to get the information of hardware assets on device including host assets, connected sub-device assets, HDD assets, etc.
3. Export the device hardware asset information in binary data in Excel format.

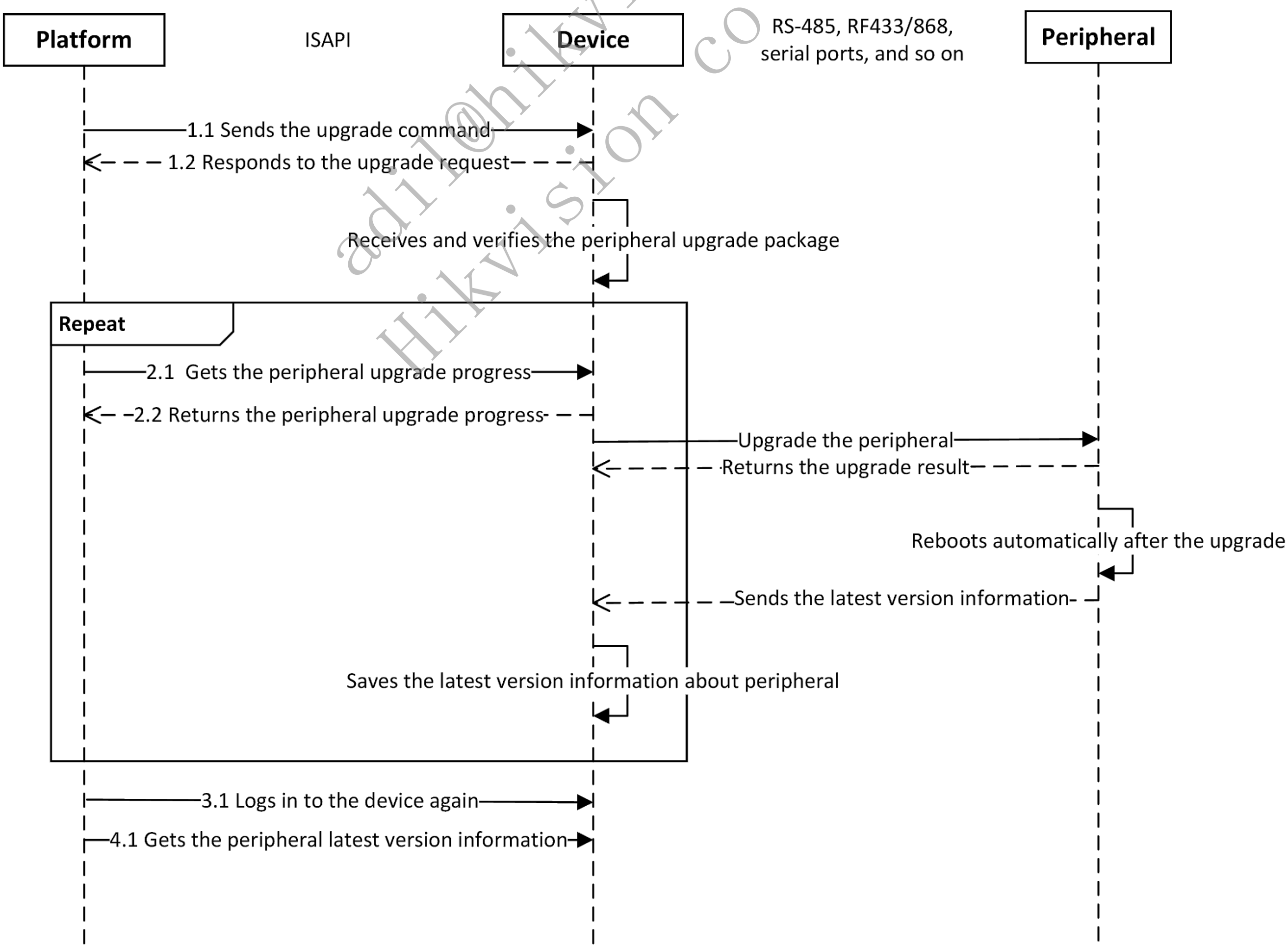
5.3 Device Peripherals Upgrade

5.3.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades device peripherals via ISAPI.

5.3.2 API Calling Flow

The sequence diagram of upgrading device peripherals by the platform is shown below.



1. Get the device system capability `GET /ISAPI/System/capabilities` and check whether the device supports upgrading peripherals. If the field `isSupportAcsUpdate` is returned and its value is true, it indicates that the device supports this function, otherwise, the device does not support this function.

2. Get the capability of upgrading the peripherals module `GET /ISAPI/System/AcsUpdate/capabilities`, and get the types and IDs of peripherals that support upgrading.
3. The platform sends the upgrade command `POST /ISAPI/System/updateFirmware?type=<type>&moduleAddress=<moduleAddress>&id=<indexID>`. In the URL `type` refers to the peripheral type, `moduleAddress` refers to the peripheral module address, and `indexID` refers to the ID of peripheral to be upgraded. The platform will apply the upgrade peripheral package to the device.
4. Get the peripheral upgrade progress `GET /ISAPI/System/upgradeStatus?type=<Type>`.
5. Log in to the device again.
6. Get the peripheral latest version information.

5.4 Device Time Sync

5.4.1 Introduction to the Function

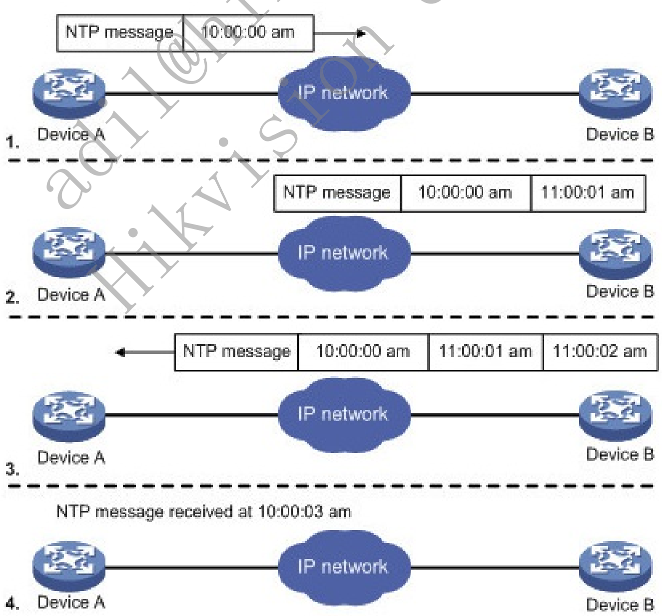
Time sync is a method to synchronize the time of all devices connecting to the NTP server, so that all devices can share the same clock time for providing related functions based on time. Supported time sync types: NTP time sync, manual sync, satellite time sync, platform time synchronization, etc. The following describes the method of NTP time sync.

5.4.1.1 NTP Time Sync

The local system of running NTP can receive sync from other clock sources (self as client), other clocks can sync from the local system (self as server), and sync with other devices.

The basic working principle of NTP is shown in the picture. Device A and Device B are connected via the network, and their systems follow their own independent system time. To auto sync their time, you can set device time auto sync via NTP. For example:

Before time sync between Device A and Device B, the time of Device A is 10:00:00 am, and that of Device B is 11:00:00 am. Device B is set as the server of NTP server, so that the time of Device A should be synchronized with that of Device B. The time of NTP message transmitted between Device A and Device B is 1 second.



The working process of system clock synchronization is as follows:

Device A sends an NTP message to Device B with a timestamp of 10:00:00 am (T1) that is when it leaves Device A.

When the NTP message reaches Device B. Device B will add its own timestamp, which is 11:00:01 am (T2).

Then the NTP message leaves Device B with Device B's timestamp, which is 11:00:02 am (T3).

Device A receives the response message, and the local time of Device A is 10:00:03 am (T4).

Above all, Device A can calculate two important parameters:

Round-trip delay of NTP message: Delay = (T4-T1) - (T3-T2) = 2 seconds.

Time difference between Device A and Device B: offset = ((T2-T1)+(T3-T4))/2=1 h.

Device A can sync its own time with that of Device B according to calculation results.

5.4.2 API Calling Flow

5.4.2.1 Time Sync Configuration

1. Get the Capability of Device Time synchronization Management

You can call this API to get the time sync types currently supported by the device, such as NTP time sync, manual time sync, satellite time sync, EZ platform time sync.

Get the capability: `GET /ISAPI/System/time/capabilities`.

2. Set device time synchronization management parameters

You can configure the time synchronization mode as follows：

Get device time synchronization management parameters: `GET /ISAPI/System/time`;

Set device time synchronization management parameters: `PUT /ISAPI/System/time`;

NTP time synchronization: See 4.2.2 NTP Time Sync (Client).

Manual time synchronization: Set the value of `timeMode` to `manual`, and set the device local time in nodes `localTime、timeZone`.

Satellite time synchronization: Set the value of `timeMode` to `satellite`, and set the device local time in nodes `satelliteInterval`.

Platform time synchronization: Set the value of `timeMode` to `platform`.

Note： For manual time synchronization (time offset including time zone offset): Set manual time synchronization: localTime refers to the local time on device (time offset excluded, in format like 2019-02-28T10:50:44); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00); Get manual time synchronization: localTime refers to the local time on device (time offset included, in format like 2019-02-28T10:50:44+8:30); timeZone refers to the time offset of local time on device (time offset format with DST disabled: CST-8:00:00; time offset format with DST enabled: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00).

5.4.2.2 Time Zone Configuration

1. Get device time zone configuration capability

Call `GET /ISAPI/System/capabilities` to get the system capability. When `isSupportTimeZone` is returned, the time zone configuration is supported by the device.

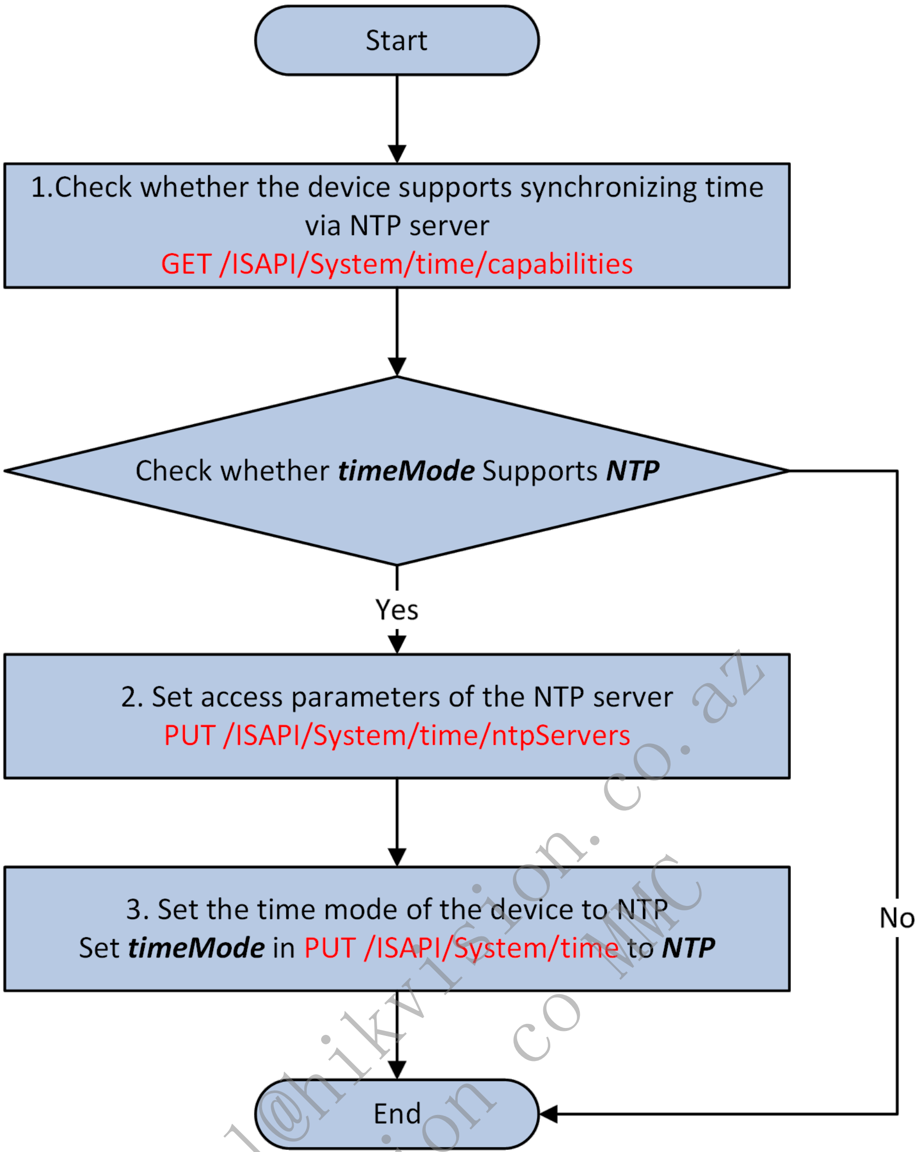
2. Configure time zone parameters

Get the device's time zone parameters: `GET /ISAPI/System/time/timeZone`. Set the device's time zone parameters: `PUT /ISAPI/System/time/timeZone`.

If DST (Daylight Saving Time) is disabled, the example of returned time zone parameters is: CST-8:00:00. It refers to UTC+8, and -8:00:00 is the UTC local time. If DST (Daylight Saving Time) is enabled, the example of returned time zone parameters is: CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.2/02:00:00. It refers to UTC+8, the DST time is 30 minutes ahead of local time, the DST starts at 02:00:00 on the first Sunday of April and ends at 02:00:00 on the fifth Tuesday of October. MX.Y.Z: X is the month, Y is the week number in the month, Z is the day of a week (0-Sunday, 1-Monday, 2-Tuesday, 3-Wednesday, 4-Thursday, 5-Friday, 6-Saturday).

5.4.2.3 NTP Time Sync (Client)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as client):



1. Check whether the device supports synchronizing time via NTP server Get the capability of the device: `GET /ISAPI/System/time/capabilities`, and check whether `timeMode` supports `NTP`.

2. Set access parameters of the NTP server

Supports accessing the NTP server by IP address to synchronize the device time.

Get the access parameter capability of the NTP server: `GET /ISAPI/System/time/ntpServers/capabilities`

Set access parameters of the NTP server: `PUT /ISAPI/System/time/ntpServers`

Get access parameters of the NTP server: `GET /ISAPI/System/time/ntpServers`

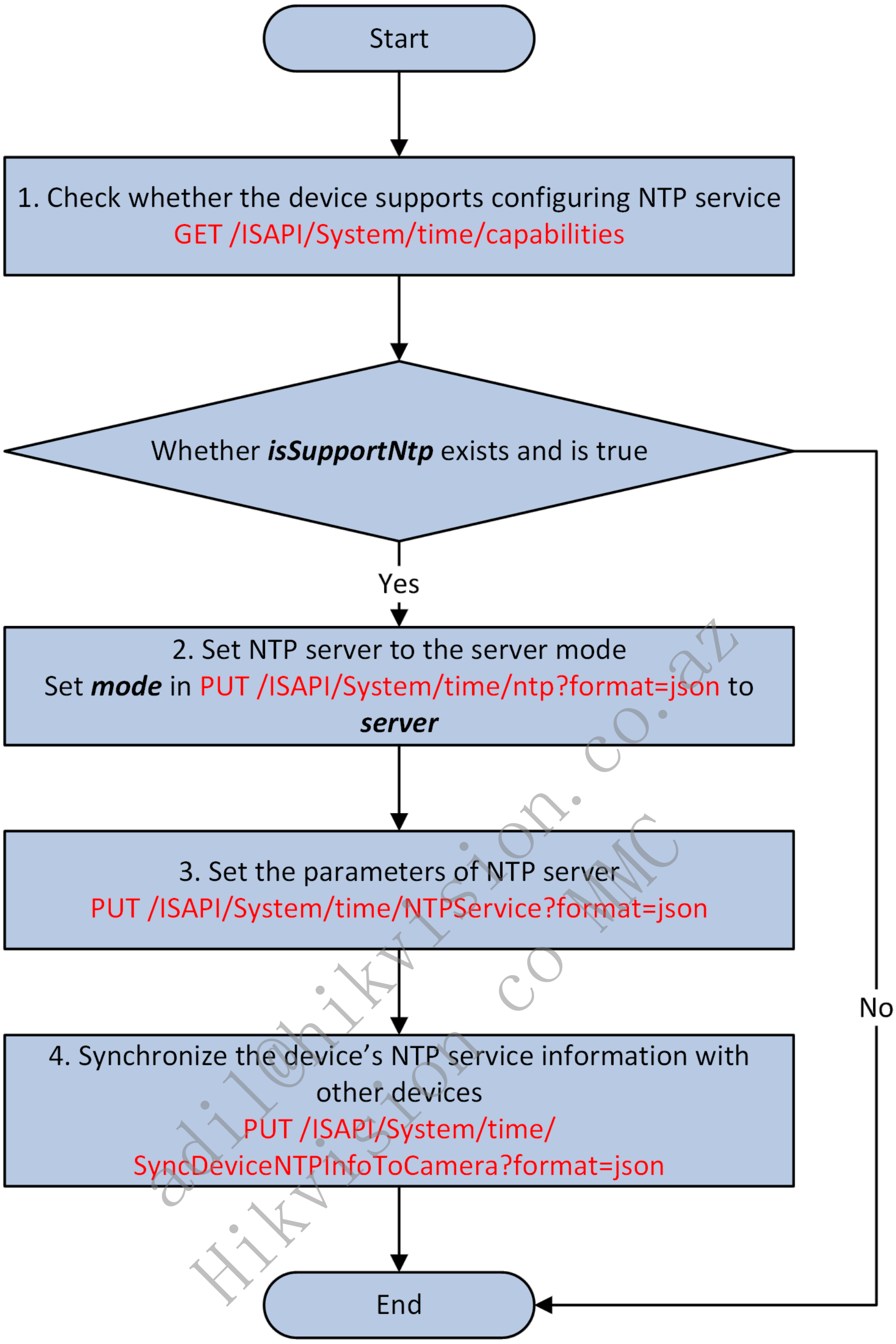
3. Set the time mode of the device to NTP

Supports setting the value of `timeMode` to `NTP`.

Get device time synchronization management parameters: `GET /ISAPI/System/time` Set device time synchronization management parameters: `PUT /ISAPI/System/time`

5.4.2.4 NTP Time Sync (Server Mode)

The local system running the NTP server can receive sync information from other clock sources (self as client), sync other clocks (self as server) as clock sources, and sync with other devices. Calling flow (self as server):



1. Check whether the device supports configuring NTP service Get the capability of device time synchronization management: `GET /ISAPI/System/time/capabilities`; If `isSupportNtp` is returned, it indicates that the device supports time synchronization management.

2. Set NTP server to the server mode

Supports setting the value of `mode` to `server`.

Get the capability of server mode: `GET /ISAPI/System/time/ntp/capabilities?format=json`

Set NTP to server mode: `PUT /ISAPI/System/time/ntp?format=json`

Get parameters of NTP server mode: `GET /ISAPI/System/time/ntp?format=json`

3. Set the parameters of NTP server

Supports setting the IP address of the NTP server.

Get the capability of NTP server: `GET /ISAPI/System/time/NTPService/capabilitis?format=json`

Set the NTP server parameters: `PUT /ISAPI/System/time/NTPService?format=json`

Get the parameters of the NTP server: `GET /ISAPI/System/time/NTPService?format=json`

4. Synchronize the device's NTP service information with other devices

Supports synchronizing the time information to the camera.

Get the capability set of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json`

Synchronize device's NTP service information with the camera: `PUT /ISAPI/System/time/SyncDeviceNTPInfoToCamera?format=json`

Get the progress of synchronizing device's NTP service information with the camera: `GET /ISAPI/System/time/SyncDeviceNTPInfoToCamera/Progress?format=json`

Search for the results of synchronizing device's NTP service information with the camera: `POST /ISAPI/System/time/SyncDeviceNTPInfoToCamera/SearchResult?format=json`

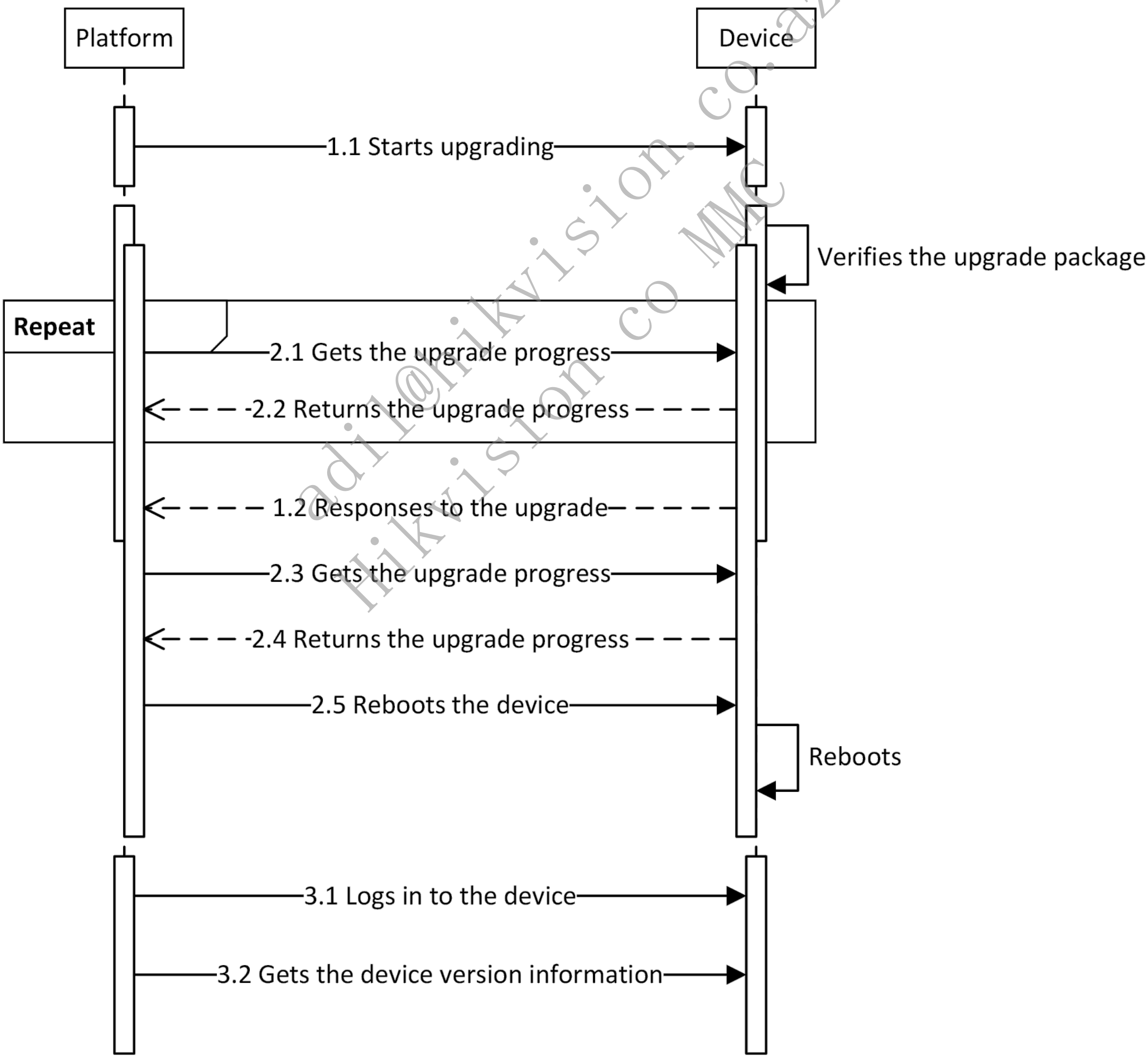
5.5 Device Upgrade

5.5.1 Introduction to the Function

The platform or client software or web client under the LAN upgrades devices via ISAPI.

5.5.2 API Calling Flow

The sequence diagram of upgrading devices by the platform is shown below.



1. **Upgrade devices.**

   Upgrade the device firmware: `POST /ISAPI/System/updateFirmware`.

2. **Get the device upgrade progress.**

Get the device upgrade progress: `GET /ISAPI/System/upgradeStatus`.

3. **Reboot devices.**

   Reboot devices: `PUT /ISAPI/System/reboot`.

5.6 Mutually Exclusive Functions

5.6.1 Introduction to the Function

Some functions are mutually exclusive due to the device performance (for example, function A and function B cannot run at the same time, i.e, only one of them is allowed at one time).

5.6.2 API Calling Flow

The following three APIs are available for the integration of mutually exclusive functions:

1. Get the information of mutually exclusive functions: `GET /ISAPI/System/mutexFunction/capabilities?format=json`. Call this URL to get the list of existing mutually exclusive functions supported by the device. Note: NVR devices only support setting exlusive function "perimeter" (perimeter protection), and do not support "linedetection" (line crossing detection), "fielddetection" (intrusion detection), "regionEntrance" (region entrance), or "regionExiting" (region exiting).

2. Search for the functions that are mutually exclusive with a specified function: `POST /ISAPI/System/mutexFunction?format=json`. Based on the list of mutually exclusive functions returned by `GET /ISAPI/System/mutexFunction/capabilities?format=json`, you can search for the mutual exclusion status of a specified function and see whether to change the settings and disbale the mutually exclusive function.

3. Get the mutual exclusion information when device function exception occurs: `GET /ISAPI/System/mutexFunctionErrorMsg`. After getting the error code, you can call this API to get the current mutually exclusive functions.

5.7 Sub-device Batch Upgrade

5.7.1 Introduction to the Function

Sub-device Batch Upgrade (Single Task): applicable to situations of upgrading multiple sub-devices using one upgrade package. Application scenarios include: when the UWB positioning anchor connects to the web, upgrading multiple tags (sub-devices) through the positioning engine (gateway), with the same upgrade package.

Management of Sub-device Batch Upgrade Tasks: There are many types of sub-devices, such as LoRa nodes, which have a slow upgrade process and different models of LoRa nodes in the field also require different upgrade packages. Therefore, it is necessary to support the creation of multiple upgrade tasks at once, entrusting the upgrade to the devices, to enhance the practicality of batch upgrading sub-devices.

5.7.2 API Calling Flow

Sub-device Batch Upgrade (Single Task)

Get the device capability: `/ISAPI/System/capabilities`.

If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that the device support the function.

If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that the device support getting the progress of batch upgraading sub-devices.

Sub-device batch upgrade: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.

Batch sub-device upgrade progress search: `POST /ISAPI/System/BulkUpgradeChildDeviceList/Search?format=json`.

Sub-device Batch Upgrade Task Management

Get device system capabilities `/ISAPI/System/capabilities`, If the node `isSupportBulkUpgradeChildDevice` exists and is true, it indicates that batch upgrade of sub devices is supported, If the node `isSupportSearchBulkUpgradeChildDeviceProgress` exists and is true, it indicates that progress of batch upgrading sub devices can be searched,

If the node `isSupportSearchBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading devices can be searched,

If the node `isSupportModifyBulkUpgradeChildDeviceTask` exists and is true, it indicates that batch upgrading sub devices can be edited,

If the node `isSupportDeleteBulkUpgradeChildDeviceTask` exists and is true, it indicates that task of batch upgrading sub devices can be deleted.

Get the capability of batch upgrading sub-device: `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`.

Batch upgrade sub devices: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`.****

Search for tasks of batch sub-device upgrade: `POST /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`.

Edit tasks of batch sub-device upgrade: **`PUT /ISAPI/System/ModifyBulkUpgradeChildDeviceTask?format=json`**.****

Delete tasks of batch sub-device upgrade: `PUT /ISAPI/System/DeleteBulkUpgradeChildDeviceTask?format=json`.

Note: 1. The two methods for batch upgrading child devices are the same API: `POST /ISAPI/System/BulkUpgradeChildDeviceList?format=json`. If the `BulkUpgradeChildDeviceListCap.taskID` is returned in `GET /ISAPI/System/BulkUpgradeChildDevice/capabilities?format=json`, it indicates that the task method is supported in the sub device upgrade. 2. To determine which sub devices can be upgraded, you can use the API `POST /ISAPI/IoTGateway/Childmanage/SearchChild?format=json`. Sub devices with the `tag` `isUpgradable` and `value` `true` indicate that the sub devices support upgrades.

5.8 User Management

5.8.1 Introduction to the Function

When the device is activated, you can log in to it via the `admin` account and corresponding password, and manage users as needed, including:

1. Change the password of the `admin` account. The user name cannot be edited.
2. Add, edit, and delete other users, including the user type, password, user name, and so on. A `Non-admin` user can log in to and operate the device after being created.

Remarks

1. Common user types:
    - Administrator (`admin`): has the permission of accessing all resources supported by the device, and can operate all functions of the device. The `admin` account cannot be deleted.
    - Operator (`operator`): has the permission to view. Their operation permissions are assigned by `admin`. Operator accounts are created by the administrator only.
    - User (`viewer`): has the permission to view only. They have no operation permission. User accounts are created by the administrator only.
2. User types related to the installer:
    - Local Administrator (`localAdmin`) : Activated by the device owner on the local software side (e.g., WEB) that accompanies the device. By default, a `localAdmin` has the permission to access the device and perform all functions supported by the device.
    - Local Installer (`localInstaller`) : Activated by `localAdmin`. The identity can be given to the installer for device installation and debugging on site. By default, a `localInstaller` has the permission to access and perform all functions supported by the device.
    - Local Operator (`localOperator`) : Created by `localAdmin` or`cloudAdmin`. A password is required to be set by `localAdmin`, `cloudAdmin`, `localInstaller`, `installerAdmin`, or `installerEmployee` before the local operator can log in via the local keypad. A local operator’s business functions are limited, and cannot configure device parameters. By default, only arming, disarming (including alarm clearing), and relay control operations are allowed. According to the actual business needs, there can be two types of local operator: one-time local

operator (can only log in once and the identity will expire after logging out), and temporary local operator (can log in within the validity period), both created by `localAdmin` or `cloudAdmin`.
    - Cloud Administrator (`cloudAdmin`) : Created by the device owner in the mobile software that accompanies the device (e.g. HC over the cloud) and then synchronized to the device. By default, a `CloudAdmin` has the permission to access and perform all functions supported by the device.
    - Cloud Operator (`cloudOperator`) : Created by `cloudAdmin` in the mobile software that accompanies the device (e.g. HC over the cloud) and then synchronized to the device for ordinary mobile users. A cloud operator's business functions are limited, and cannot configure device parameters, access and view the device, arm/disarm (including alarm clearing), or operate relays.
    - Installer Administrator (`installerAdmin`) : User on the installer business application, created on the business platform (e.g. HPC deployed in the cloud) and then synchronized to the device for remote device management by the business platform's admin. By default, an `installerAdmin` has the permission to access the device and perform all functions supported by the device. To facilitate installation management, when the business platform sets `installerAdmin` as the highest authority and synchronizes to the device, `installerAdmin` is allowed to modify user permissions of `localAdmin` and `cloudAdmin`.
    - Installer Employee (`installerEmployee`) : User on the installer business application, created on the business platform (e.g. HPC deployed in the cloud) and then synchronized to the device for the employees on the business platform to remotely install and debug the device. By default, an `installerEmployee` has the permission to access the device and perform all functions supported by the device.
3. User password:
    - To ensure the security of account information, it is recommended to create a password using eight to sixteen characters, including at least two kinds of the following categories: digits, lower case letters, upper case letters, and special characters, and the user name is not allowed in the password.
    - Risky passwords include the following categories: less than 8 characters, containing only one type of characters, same as the user name or reversed user name. To protect user data privacy and improve security, it is recommended to use a strong password.
    - The password strength rule is as follows:
        a. Strong password: including at least three kinds of the categories (digits, lower case letters, upper case letters, and special characters).
        b. Medium password: a combination of digits and special characters, lower case letters and special characters, upper case letters and special characters, or lower case letters and upper case letters.
        c. Weak password: a combination of digits and lower case letters or digits and upper case letters.

5.8.2 API Calling Flow

1. Get the user management capability of devices on the client: `GET /ISAPI/Security/users/<indexID>/capabilities`.

2. Add device users on the client: `POST /ISAPI/Security/users?security=<security>&iv=<iv>`.

    Remarks:

    - Only `admin` can create other types of users, and creating users requires login password verification (`<loginPassword>`) of `admin`.
    - If the user account is inactivated, it's required to log in to the account and change the user password (`PUT /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>`). The account is activated when the password is changed.
    - When the account is inactivated, it's not allowed to perform any operations except changing the user password. Otherwise, an error (`0x0020000f`) will be returned.

3. Edit the user information on the client: `PUT /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>`.

    Remarks:

    - It requires password verification of `admin` when `admin` changes the user password. The account turns

inactivated when the user password is changed by `admin`. Once logging out, the user needs to change the password first before the next login.
    - When `non-admin` users changed their passwords, the account remains activated.

4. Delete users on the client: `DELETE /ISAPI/Security/users?loginPassword=<loginPassword>&security=<security>&iv=<iv>`.

   Remarks:

   Only `admin` can delete users, and deleting users requires login password verification of `admin`.

5. Get the user information, including user name, activation status (`<userActivationStatus>`), and so on.

   Get a single user information: `GET /ISAPI/Security/users/<indexID>?security=<security>&iv=<iv>`.

   Get the information of all users: `GET /ISAPI/Security/users?security=<security>&iv=<iv>`.

   Get the information of online users: `GET /ISAPI/Security/onlineUser`. Online users refer to users who have logged in to the device. The information such as user name, user type, and IP address can be obtained.

Remarks:

- If multiple attempts of `admin` login password verification failed in the process of adding, editing, or deleting users, the `admin` will be locked. The remaining attempts are defined by the field `retryTimes` in the response message.
- The new password cannot be the same as the last password. Otherwise, an error (`0x400010E8`) will be returned.

5.8.3 Exception Handling

Error Code

statusCode	statusString	subStatusCode	errorCode	errorMsg	Descript
4	Invalid Operation	theAccountIsNotActivated	0x0020000f	The account is not activated.	
4	Invalid Operation	loginPasswordError	0x4000000C	Incorrect login password.	
4	Invalid Operation	theAnswerToTheUserSecurityQuestionIsDuplicate	0x4000A0B6	The answer to the user security question is duplicate.	Please se different answer.
4	Invalid Operation	theAnswerToTheUserSecurityQuestionIsTooShort	0x4000A0B7	The answer to the user security question is too short.	Please se longer answer.
4	Invalid Operation	cannotSameAsOldPassword	0x400010E8	New password cannot be the same as the old one.	Please se different passwor
6	Invalid Content	administratorPasswordError	0x60000042	Incorrect administrator password.	Please en the corre passwor you forg the passwor you can reset the passwor

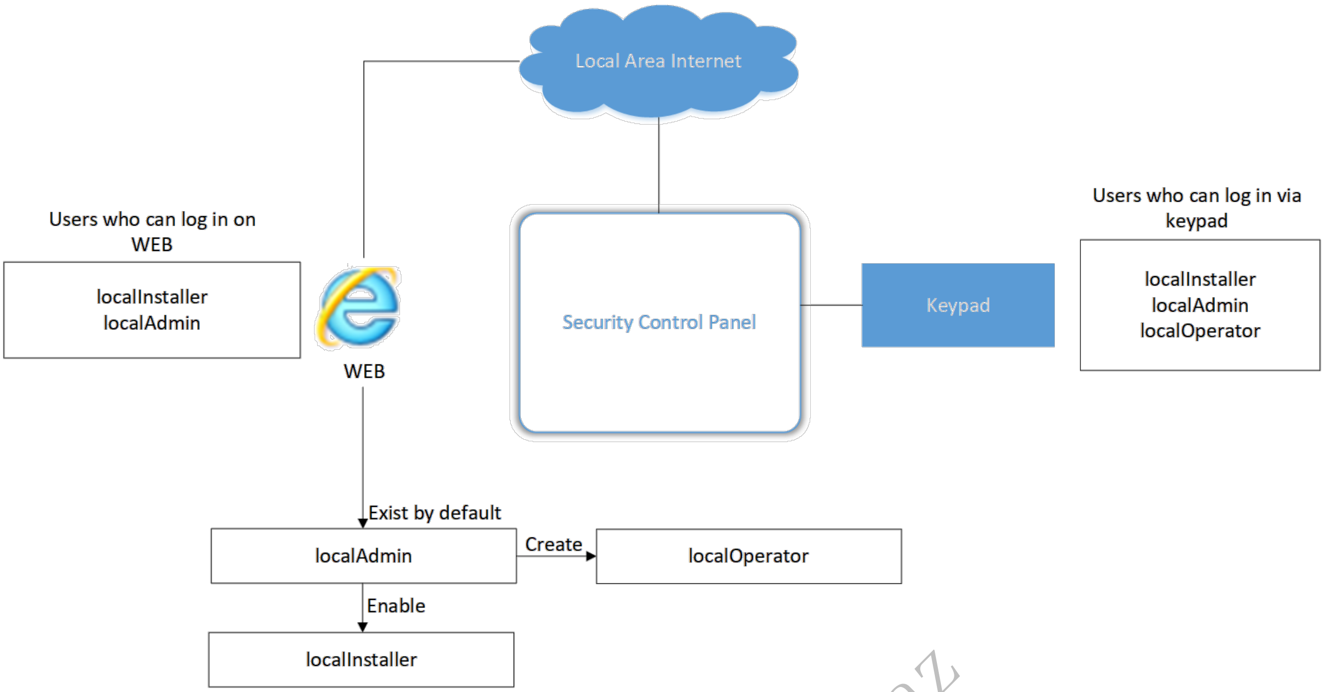
5.9 User Types Related to the Installer (supported by the security control panel)

For different application scenarios, e.g., local environment, cloud environment, and HPP installer environment, users related to the installer can be classified into seven types: `localAdmin`, `localInstaller`, `localOperator`, `cloudAdmin`, `installerAdmin`, `installEmployee`, and `cloudOperator`.

5.9.1 Create the User

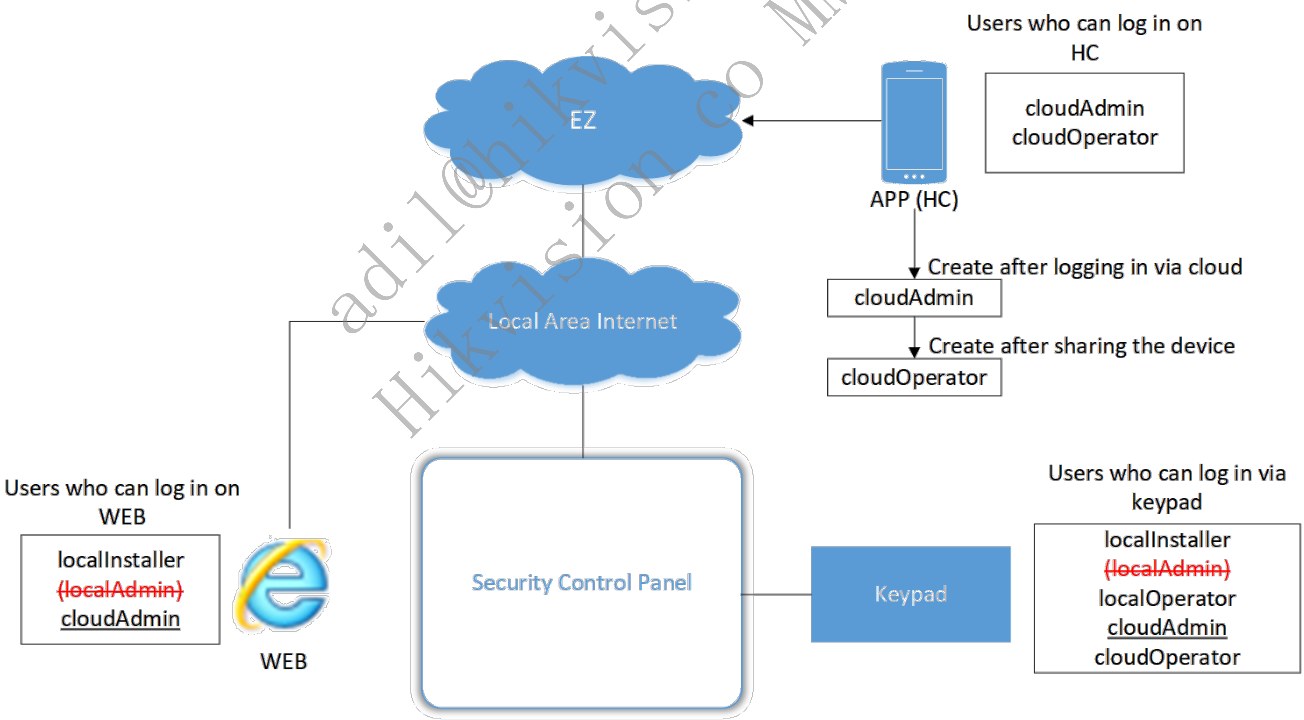
5.9.1.1 Local Environment

In a local environment (LAN), three types of users are involved: `localAdmin`, `localInstaller`, and `localOperator`. By default, two types of users exist on the device: `localAdmin` and `localInstaller`, but `localInstaller` is not enabled, which means that although a user ID is created, but the user cannot log in as `localInstaller` before the user type is enabled by `localAdmin`. `localOperator` is created by `localAdmin`, and can only log in via local keypad. Users who have been set with the keypad password can log in via keypad.



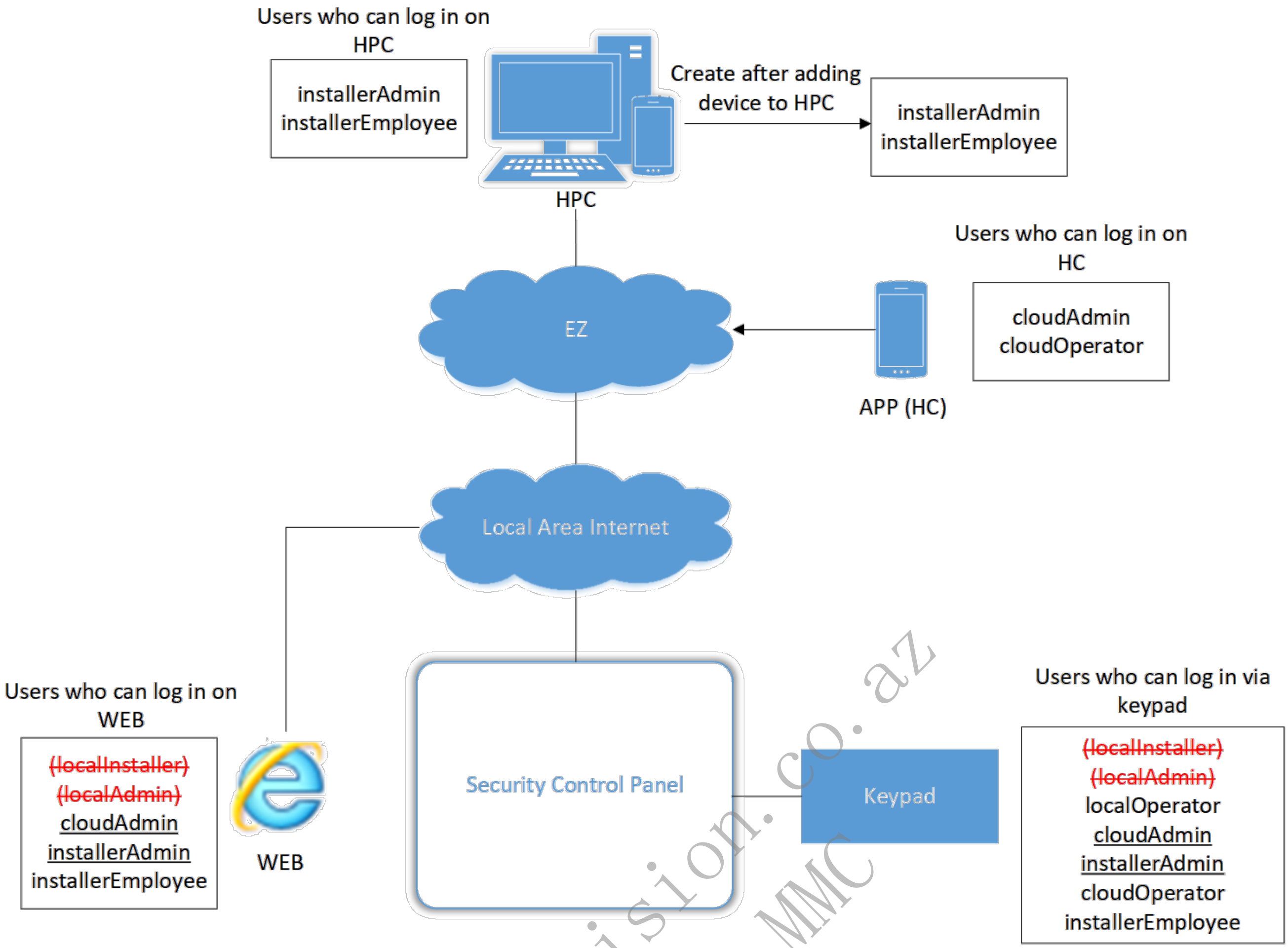
5.9.1.2 Cloud Environment

In a cloud environment, four types of users are involved: `cloudAdmin`, `cloudOperator`, `localInstaller`, and `localOperator`. After logging in to the device on HC application via cloud, a `cloudAdmin` can be created. Then, the `cloudAdmin` can share the device to create a `cloudOperator`. Note that after a `cloudAdmin` is created, the existing `localAdmin` will expire. Users who have been set with the keypad password can log via keypad.



5.9.1.3 HPP Installer Environment

In an HPP installer environment, five types of users are involved: `cloudAdmin`, `cloudOperator`, `localOperator`, `installerAdmin`, and `installerEmployee`. After a device is added to HPC, the user adding protocol will be applied, and `installerAdmin` and `installerEmployee` can be created (batch creating is supported). Note that after an `installerAdmin` is created, the existing `localInstaller` will expire. Users who have been set with the keypad password can log via keypad.



5.9.2 User Permissions

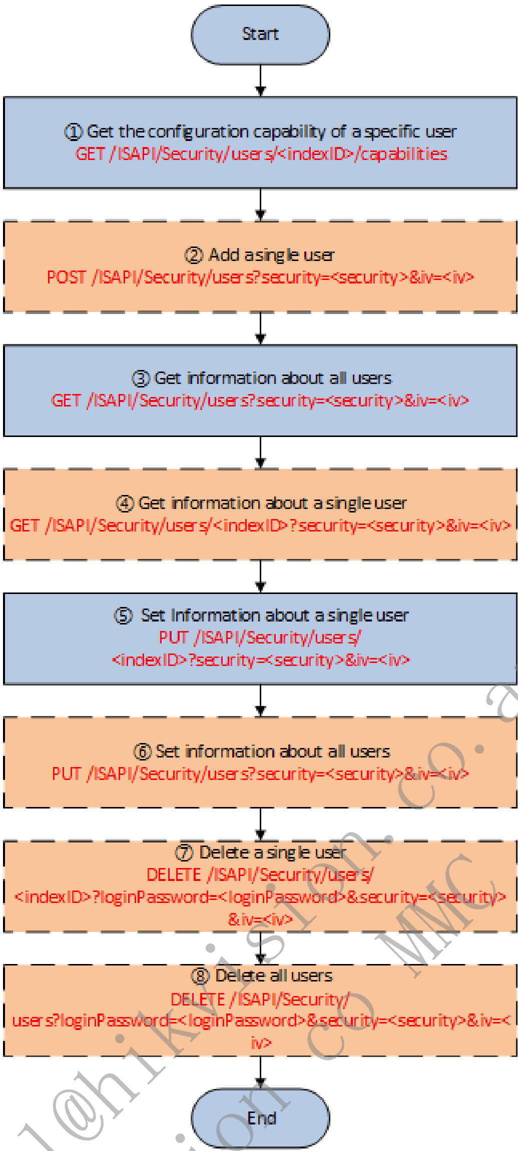
User Type / Permission	Local Admin	Local Installer	Local Operator	Cloud Admin	Cloud Operator	Installer Admin	Installer Employee
Arming	√	√	√	√	√	√	√
Disarming (Alarm Clearing)	√	√	√	√	√	√	√
Bypass	√	√	×	√	√	√	√
View logs and status	√	√	×	√	√	√	√
Configure parameters	√	√	×	√	×	√	√
Manage partitions	√	√	×	√	×	√	√
Operate relays	√	√	one-time local operator: × temporary local operator: √	√	√	√	√
Edit localAdmin's keypad password	√	√	×	×	×	√	√
Edit cloudAdmin's keypad password	×	√	×	√	×	√	√
Edit localInstaller's keypad password	×	√	×	×	×	×	×
Edit							

installerAdmin's keypad password	×	×	×	×	×	√	×
Edit installerEmployee's keypad password	×	×	×	×	×	√	√ (only the employee's own password)
Edit localOperator's keypad password	√	√	×	√	×	√	√
Edit cloudOperator's keypad password	×	√	×	√	√ (only the operator's own password)	√	√
Edit localAdmin's permission	×	×	×	×	×	×	×
Edit cloudAdmin's permission	×	×	×	×	×	×	×
Edit localInstaller's permission	√	×	×	√	×	×	×
Edit installerAdmin's permission	√	×	×	√	×	×	×
Edit installerEmployee's permission	√	×	×	√	×	×	×
Edit localOperator's permission	√	√	×	√	×	√	√
Edit cloudOperator's permission	×	√	×	√	×	√	√

Note: "Configure parameters" includes parameters of zones, sounders, keypads, card readers, keyfobs, cards, relays, repeaters, transmitters, network cameras, partitions, and so on.

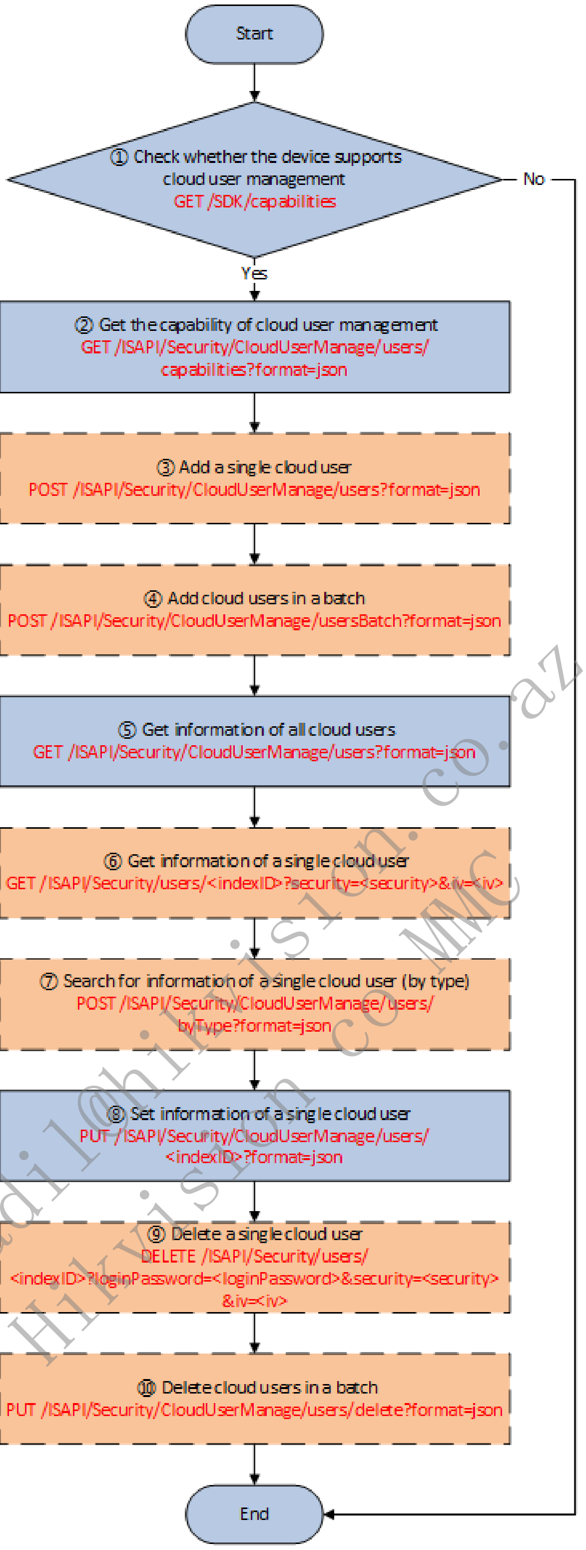
5.9.3 Manage User Information

5.9.3.1 Local User



1. Get the configuration capability of a specific user: GET /ISAPI/Security/users//capabilities
2. (Optional) Add a user: POST /ISAPI/Security/users?security=&iv=. The nodes userName, password, keypadPassword, and loginPassword in the message will be encrypted.
3. Get information of all users: GET /ISAPI/Security/users?security=&iv=. The nodes phoneNum, emailAddress, password, duressPassword, keypadPassword, and loginPassword in the message will be encrypted.
4. Get information of a single user: GET /ISAPI/Security/users/?security=&iv=. The index in the URL is the user ID.
5. Set permissions for a single user: PUT /ISAPI/Security/users/?security=&iv=. The index in the URL is the user ID.
6. (Optional) Set information of all users: PUT /ISAPI/Security/users?security=&iv=. This API can be called for batch configuring user information.
7. (Optional) Delete a single user: DELETE /ISAPI/Security/users/?loginPassword=&security=&iv=. Delete the user by indexID, and loginPassword is required for deleting the user. The loginPassword should be encrypted.
8. (Optional) Delete all users: DELETE /ISAPI/Security/users?loginPassword=&security=&iv=. loginPassword should be encrypted.

5.9.3.2 Cloud User



1. Check whether the device supports cloud user management: GET /SDK/capabilities. If the node value of isSupportCloudUserManage is true, cloud user management is supported in these URLs:

   /ISAPI/Security/CloudUserManage/users/capabilities?format=json

   /ISAPI/Security/CloudUserManage/users/?format=json

   /ISAPI/Security/CloudUserManage/users?format=json

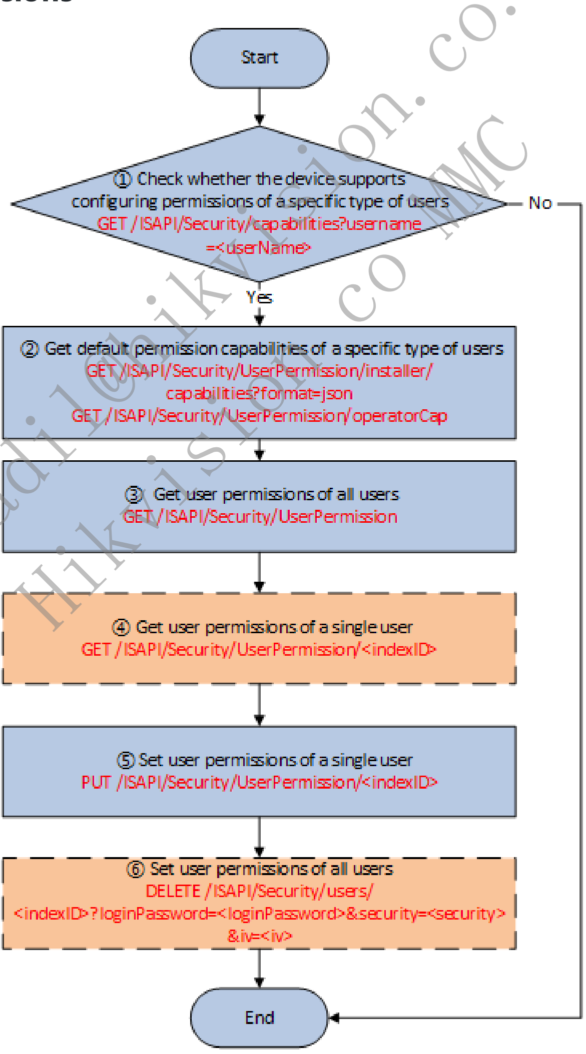
2. Get the capability of cloud user management: GET /ISAPI/Security/CloudUserManage/users/capabilities?format=json. If the node value of isSupportAddCloudUserList is true, it supports batch adding cloud users.

3. Add a single cloud user: POST /ISAPI/Security/CloudUserManage/users?format=json. The following cloud users need to create the user accounts on their own: coludAdmin, installerAdmin, installerEmployee, and cloudOperator, which is different from localOperator who needs to be created by admin users. Enter information such as user

name, password, e-mail, and phone number to create a user account and put into use.

4. (Optional) Add cloud users in a batch: POST /ISAPI/Security/CloudUserManage/usersBatch?format=json. For installerEmployee users that has been created on HPC, use this URL to apply them to the device and synchronize user information.
5. Get information of all cloud users: GET /ISAPI/Security/CloudUserManage/users?format=json
6. Get information of a single cloud user: GET /ISAPI/Security/CloudUserManage/users/?format=json
7. Search for information of a single cloud user (by type): /ISAPI/Security/CloudUserManage/users/byType?format=json. Support searching by e-mail, phone number, and user name.
8. Set information of a single cloud user: PUT /ISAPI/Security/CloudUserManage/users/?format=json
9. Delete a single cloud user: DELETE /ISAPI/Security/CloudUserManage/users/?format=json. Delete the user by specifying the user's indexID.
10. Delete cloud users in a batch: PUT /ISAPI/Security/CloudUserManage/users/delete?format=json. Support batch deleting by user names, user types, phone numbers, and e-mails.

5.9.4 Manage User Permissions



1. Check whether the device supports configuring permissions of a specific type of users: GET /ISAPI/Security/capabilities?username=. If the node value of isSupportInstallerCap is true, the installer users' permissions can be configured. There is no specific capability node for operator type users, and by default their permissions can be configured.
2. Get the default permission capabilities of a specific type of users.
   - For installer users: GET /ISAPI/Security/UserPermission/installer/capabilities?format=json

- For operator users: GET /ISAPI/Security/UserPermission/operatorCap

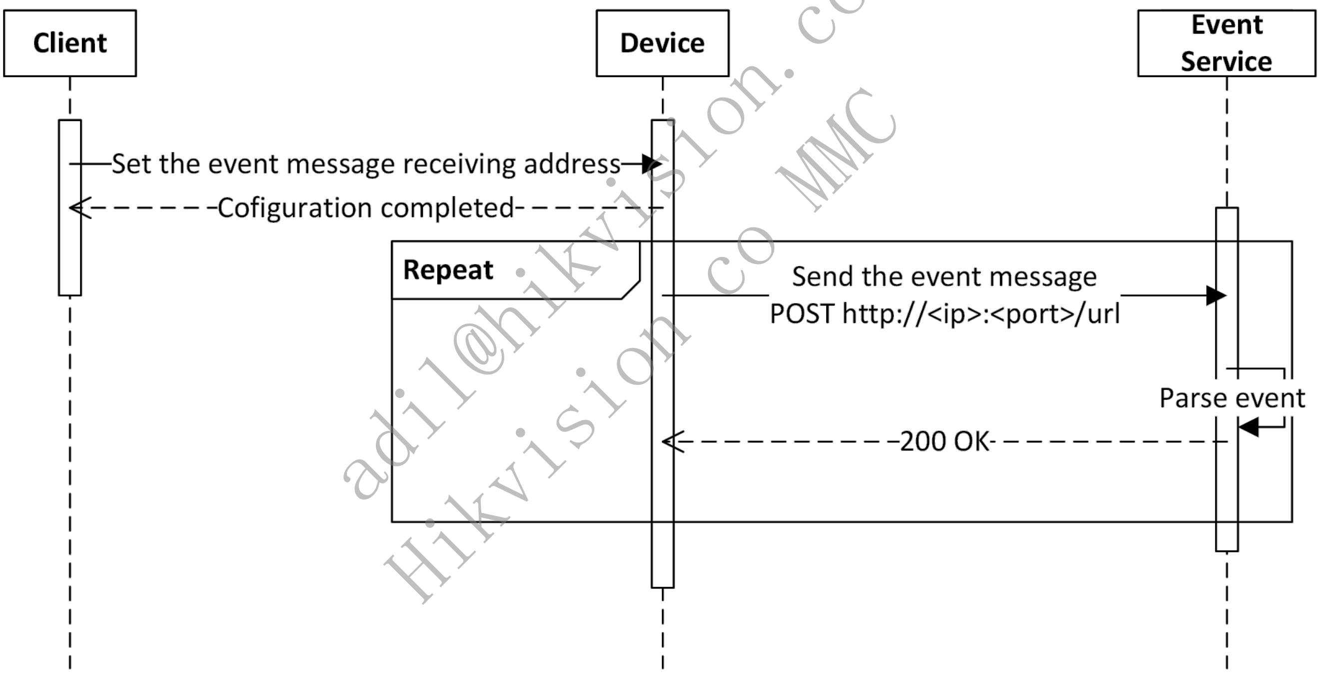
3. Get user permissions of all users: GET /ISAPI/Security/UserPermission
4. (Optional) Get user permissions of a single user: GET /ISAPI/Security/UserPermission/. The index in the URL is the user ID.
5. Set user permissions of a single user: PUT /ISAPI/Security/UserPermission/
6. (Optional) Set user permissions of all users: PUT /ISAPI/Security/UserPermission

6 Network Configuration

6.1 Listening Service

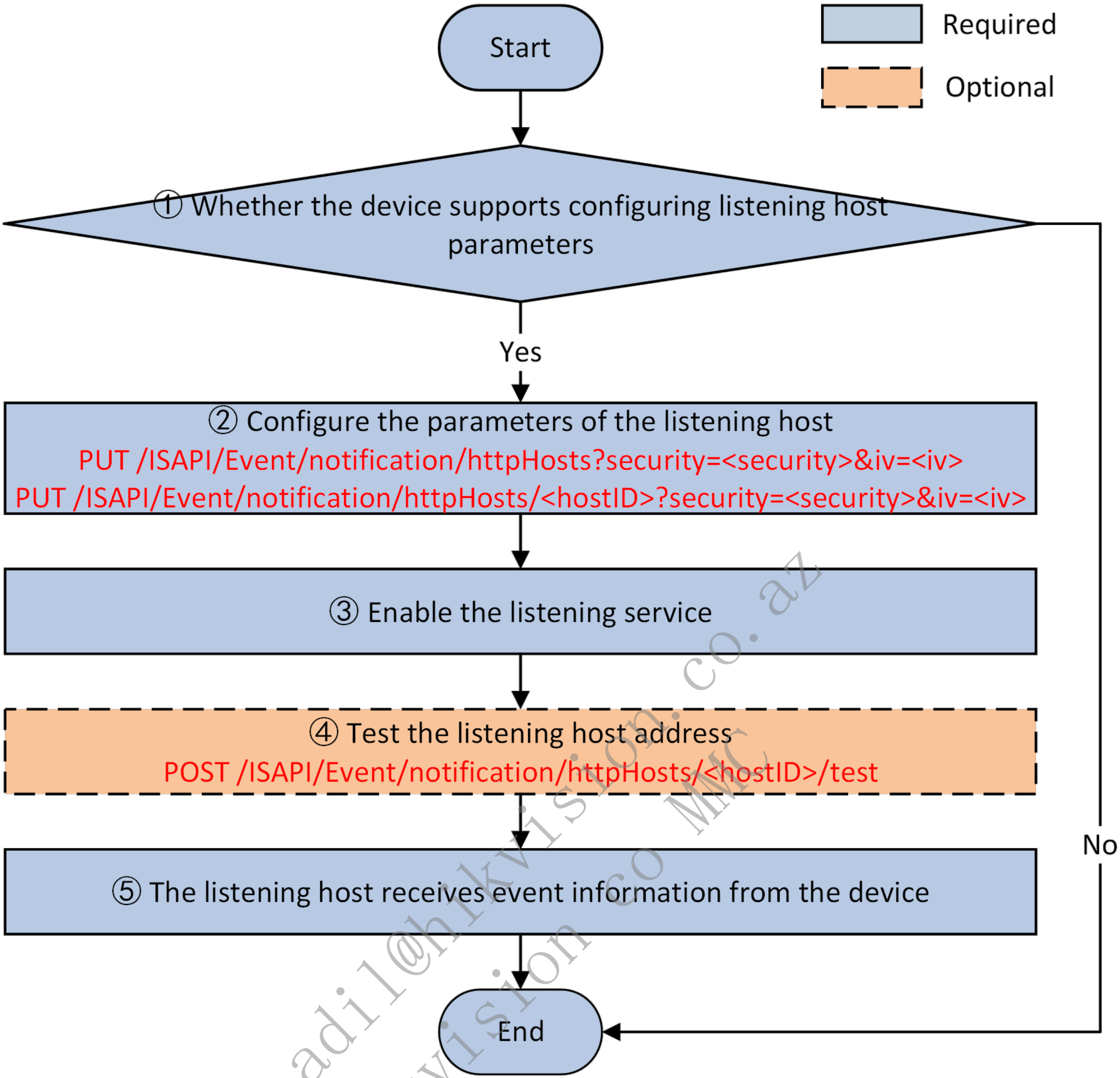
6.1.1 Introduction to the Function

When an event occurs, the device creates connection with the client and uploads alarm information. Meanwhile, the listening host receives data from the device. The IP address and the port No. of the listening host should be configured for the device. The HTTP listening service supports subscribing to specific events when adding or editing the listening host. Only the specified events will be uploaded by the device.(currently not available for the device)



6.1.2 API Calling Flow

6.1.2.1 Listening Service



1. **Check whether the device supports configuring listening host parameters:**

   Get the configuration capability of the listening host: `GET /ISAPI/Event/notification/httpHosts/capabilities`. If the node `<HttpHostNotificationCap>` exists in the returned message and its value is true, it indicates that the device supports configuring listening host parameters.

2. **Configure the parameters of the listening host:**

   Configure the parameters of all listening hosts: `PUT /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Get the parameters of all listening hosts: `GET /ISAPI/Event/notification/httpHosts?security=<security>&iv=<iv>`;

   Configure the parameters of a listening host: `PUT /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

   Get the parameters of a listening host: `GET /ISAPI/Event/notification/httpHosts/<hostID>?security=<security>&iv=<iv>`;

3. **Enable the listening service:**

   The user needs to enable the listening service of the listening host.

4. **(Optional) Test the listening service:**

   The platform applies the command to the device to test whether the listening host is available for the device: `POST /ISAPI/Event/notification/httpHosts/<hostID>/test`.

5. **The listening host receives event information from the device:**

   When an event occurs, the device creates connection with the client and uploads alarm information. Meanwhile, the listening host receives data from the device. See details in **Event Message Grammar**.

Remark: You can also configure the listening parameters such as the timeout.

6.1.2.2 Event Message Grammar

When an event occurs or an alarm is triggered, the event/alarm information can be with binary data (such as pictures) and without binary data.

1. **Without Binary Data:**

The `Content-Type` in the Headers of the HTTP request sent by the device is usually `application/xml` or `application/json` as follows:

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--/Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: data_gateway_ip:port <!--Host: HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: zh-cn
Date: YourDate
Content-Type: application/xml; <!--content type, which is used for the upper layer to distinguish different formats when parsing the message-->
Content-Length: text_length
Connection: keep-alive  <!--maintain the connection between the device and the server for better transmission performance-->

<EventNotificationAlert/>
```

Response by the Listening Host Notes:

1. `Content-Length` is required and its value is 0.
2. `Connection` is recommended to use keep-alive to reduce the port usage.

```
HTTP/1.1 200 OK
Content-Length: 0
Connection: keep-alive
```

2. **With Binary Data**

The format of the data sent by the device is HTTP form (multipart/form-data). The `Content-Type` in the Headers of the HTTP request sent by the device is usually `multipart/form-data, boundary=<frontier>`: boundary is a variable, which is used to divide the HTTP Body into multiple units and each unit has its Headers and Body. See details in RFC 1867 (Form-based File Upload in HTML). Please note the `--` before and after the boundary.

Alarm Message Sent by the Device

```
POST Request_URI HTTP/1.1 <!--Request_URI, related URI: POST /ISAPI/Event/notification/httpHosts-->
Host: device_ip:port <!--Host: HTTP server's domain name / IP address and port No., related URI: POST /ISAPI/Event/notification/httpHosts-->
Accept-Language: zh-cn
Date: YourDate
Content-Type: multipart/form-data;boundary=<frontier>
Content-Length: text_length
Connection: keep-alive <!--maintain the connection between the device and the server for better transmission performance-->

--<frontier>
Content-Disposition: form-data; name="Event_Type"
Content-Type: text/xml <!--Some alarms use JSON format, so the upper-layer application can distinguish based on the Content-Type during parsing-->

<EventNotificationAlert/>
--<frontier>
Content-Disposition: form-data; name="Picture_Name"
Content-Length: image_length
Content-Type: image/jpeg

[picture data]
--<frontier>--
```

Response by the Listening Host

Notes:

1. `Content-Length` is required and its value is 0.
2. `Connection` is recommended to use keep-alive to reduce the port usage.

```
HTTP/1.1 200 OK
Content-Length: 0
Connection: keep-alive
```

Here are the descriptions of the main keywords.

keyword	example	description
Content-Type	multipart/form-data; boundary=frontier	Content type, multipart/form-data refers to data in form format.
boundary	frontier	Separator of the form message. A form message starts with --boundary and ends with --boundary--.
Content-Disposition	form-data; name="Picture_Name";	Content description. form-data refers to data in the form format.
filename	"Picture_Name"	File name. The file refers to the form message.
Content-Length	10	Content length. The length of the content which starts from \r\n to the next --boundary.

6.1.3 Exception Handling

6.1.3.1 Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Description	Remarks
6	Invalid Content	eventNotSupport	0x60001024		Event subscription is not supported.	

7 Video (General)

7.1 . OSD

7.1.1 Introduction to the Function

On-screen display is applied in the display interface of cameras, and is used to display some characters or graphics on the video to provide some information to the audience. After OSD configurations on the client software, the configured information text will be overlaid on the video stream of cameras and displayed on the screen.

The typical application scenarios of dynamic OSD: elevator floor number overlay for elevator monitoring, characters overlay for toll stations on highway/road, notification information overlay during live video programs. The typical application scenarios of static OSD: fixed information (such as camera position, date, time) overlay on the video signal.

7.1.2 API Calling Flow

7.1.2.1 OSD

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`.

2. The client software configures OSD parameters, including whether to enable OSD, custom text overlay, time overlay, and overlay format, for a specified channel.

Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

Set: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

Remarks:

- The configuration URL supports various parameters, including custom text overlay, channel name overlay, time overlay, overlay format, and so on. Some of these parameters can also be configured by other URLs, as shown below: `TextOverlayList` can also be configured via URL of custom text overlay.

  `channelNameOverlay` can also be configured via URL of channel name overlay.

  `DateTimeOverlay` can also be configured via URL of channel time overlay.

- The overlaid information alignment mode (defined by `alignment`) refers to the alignment mode of custom text:

  `customize`: custom mode. The client software sends the overlay position coordinates of custom text to the device, and the device will overlay the characters on the stream for display according to the position coordinates.

  `alignRight`: right align. The device will display custom text on the right boundary of screen.

  `alignLeft`: left align. The device will display custom text on the left boundary of screen.

  `allRight`: all right align. The device will display all overlay contents, including custom text, the channel name, the time on the right boundary of screen.

  `allLeft`： all left align. The device will display all overlay contents, including custom text, the channel name, the time on the left boundary of screen.

7.1.2.2 Custom Text Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `TextOverlayList` is returned, it indicates that custom text overlay is supported and `size` defines the maximum number of custom texts.

2. The client software configures custom text OSD. `textID` refers to the custom text No., and the value range is: [1, `size`]:

   Create a custom text for a specified channel: `POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Get overlay parameters of a custom text for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Set overlay parameters of a custom text for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Delete a custom text: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>`.

   Get overlay parameters of all custom texts for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Set overlay parameters of all custom texts for a specified channel: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

   Delete all custom texts for a specified channel: `DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text`.

7.1.2.3 Channel Name Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `channelNameOverlay` is returned, it indicates that channel name overlay is supported.

2. The client software configures the channel name overlay, including whether to enable channel name overlay, overlay position, and so on.

   Get channel name overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

   Set channel name overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay`.

7.1.2.4 Channel Time Overlay

1. The client software gets the OSD capability of a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities`. If `DateTimeOverlay` is returned, it indicates that channel time overlay is supported.

2. The client software configures the channel time overlay, including whether to enable channel time overlay, overlay position, date format, and so on.

   Get channel time overlay parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

   Set channel time overlay parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay`.

7.2 Async Capturing Manually

7.2.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately responds by returning the URL of the storage server where the captured picture is stored. The client can request image data from the storage server based on the received URL. It is suitable for the limited network bandwidth between the client and the device or the client centrally manages a large number of devices. The client can choose to use the image URL address carried in the alarm content to request pictures from the storage server when the network conditions improve or there is no concurrency of a large number of requests, so as to reduce the probability of image download failure and the risk of network congestion.

7.2.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportSnapshotAsync` is returned and its value is `true`, async manual capturing shall be supported.

2. (Optional) The client gets the async manual capturing capability of a specified channel: `GET /ISAPI/Streaming/channels/<channelID>/picture/async/capabilities?format=json`.

3. The client controls the specified channel to manually and asynchronously capture pictures: `GET /ISAPI/Streaming/channels/<channelID>/picture/async?format=json&imageType=<imageType>&URLType=<URLType>`, and the device returns the URL of the storage server for capturing pictures.

Remarks:

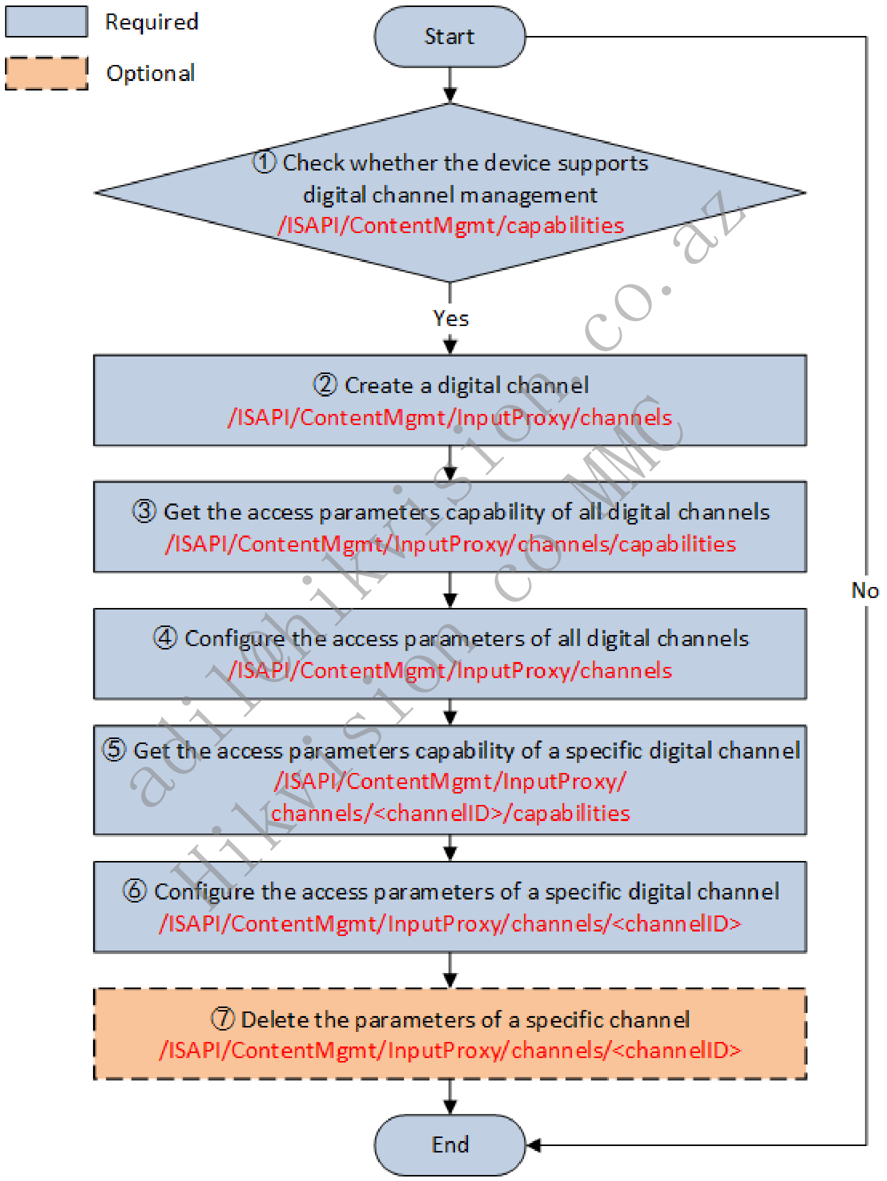
- If the step 2 returns a failure during the integration，it is recommended that you check whether the channel supports async manual capturing according to steps 1 and 3.

7.3 Digital Channel Access Management

7.3.1 Introduction to the Function

- Video channel includes analog channel and digital channel. The service involves channel access protocols, streaming protocols, address information, version No. of channel access firmware, and so on.
- Analog channel refers to the local video channels of the device.
- Digital channel refers to the video channels accessed by the device by IP address or POE port (e.g., NVR accessing the video channel of network camera).
- Analog channel No. and digital channel No. are unique. Devices that do not support digital channels will only return analog channel No. For example, analog channel No. ranges from 1 to 8 (determined by the number of analog channel ports of the device), digital channel No. ranges from 9 to 32 (within the maximum number of digital channels, determined by the number of added network cameras).

7.3.2 API Calling Flow



1. Get the capability node to check whether the device supports digital channel management: `GET /ISAPI/ContentMgmt/capabilities`. If the value of the node `iSptInputProxyChanCap` is true, it indicates that the device supports digital channel management.

2. Add digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

3. Get the access parameters capability of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities?security=<security>&iv=<iv>`.

4. Configure the access parameters of all digital channels:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.

5. Get the capability of configuring the access parameters of a specific digital channel: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/capabilities`.
6. Configure the access parameters of a specific digital channel:

   Get: `GET /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.

   Configure: `PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>?security=<security>&iv=<iv>`.
7. Delete the parameters of a specific digital channel: `DELETE /ISAPI/ContentMgmt/InputProxy/channels/<channelID>`.

7.3.3 Video Channel Application

1.Get the list of video channels, which is the sum of digital channels and analog channels. Digital channels and analog channels need to be obtained separately:

- 1) Call "Get all video input channel parameters" to get all analog channels via GET /ISAPI/System/Video/inputs/channels and VideoInputChannelList parameter is returned. The id represents the analog channel No.
- 2) Call the "Get all digital channel access parameters" to get all digital channels via GET /ISAPI/ContentMgmt/InputProxy/channels?security=&iv= and InputProxyChannelList parameter is returned. The id represents the digital channel No.

2. Video channel function configuration supports unified configuration of analog and digital channels. Both the analog channel interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/` and the digital channel interface starting with `/ISAPI/ContentMgmt/InputProxy/channels/<channelID>/` are supported. **It is recommended to use the channel configuration interface starting with `/ISAPI/System/Video/inputs/channels/<channelID>/`, where <channelID> represents all channel No.s supported by the device, including analog channels and digital channels.**

> For example, configure text overlay parameters for a channel: Both `GET/PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays` and `GET/PUT /ISAPI/ContentMgmt/InputProxy/channels/<channelID>/overlays` are supported, and it is recommended to use `/ISAPI/System/Video/inputs/channels/<channelID>/overlays`.

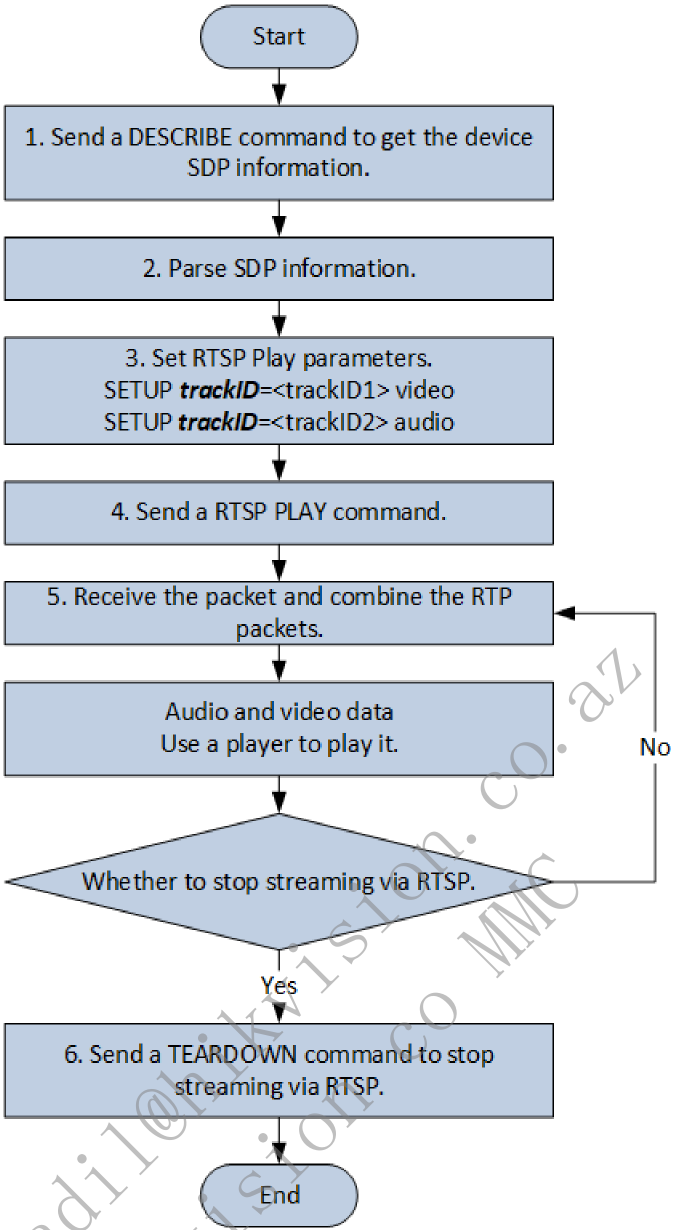
7.4 Live View

7.4.1 Introduction to the Function

Supports getting and setting stream media parameters of devices such as resolution, coding format, and stream type.

Supports streaming from products via RTSP (Real Time Streaming Protocol, see details in RFC 7826).

7.4.2 API Calling Flow



1. A client sends RTSP DESCRIBE commands such as `DESCRIBE /ISAPI/Streaming/channels/101 RTSP/1.0`; Digest authentication with devices is required.
2. The client parses the media SDP information returned by the device.
3. Set RTSP play parameters, set the track ID parsed from SDP information via SETUP commands. For example, trackID=1 indicates videos while trackID=2 indicates audios.
4. The client sends an RTSP PLAY command, and device will send audio stream, video stream, and metadata in the format of `PLAY /ISAPI/Streaming/channels/101 RTSP/1.0`.
5. The client receives the RTP package sent by device. Divided RTP packets should be assembled on the client before being parsed.
6. The client sends the command RTSP TEARDOWN to stop streaming.

Notes:

- Digest authentication is required in RTSP playback. The method is the same as that of ISAPI digest authentication.
- The address format for streaming from devices is `rtsp:// <host>[:port]/ISAPI/Streaming/channels/<ID>`, of which `<host>` is the device IP address, `[:port]` is optional, and 554 by default, `<ID>` is the device channel ID * 100 + stream type (1-main stream, 2-sub-stream, 3-third stream); for example, the IP address of the target device is `172.7.203.11`, and if the channel number is 172, the streaming address will be `rtsp://172.7.203.11:554/ISAPI/Streaming/channels/1701`.
- RTSP also supports containing user names and passwords in URL. The format is `rtsp://username:password@[address]:[port]/Streaming/Channels/[id](?parm1=value1&parm2-=value2…)`, such as `/Streaming/Channels/101?transportmode=unicast`.

7.4.3. Example

1. A client sends an RTSP DESCRIBE request.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:0
Accept:application/sdp
User-Agent:NKPlayer-1.00.00.081112
```

2. Server responds that authentication is required.

```
RTSP/1.0 401 Unauthorized
CSeq: 0
WWW-Authenticate: Digest realm="3521781c29acb312330dd668", nonce="026019333", algorithm="MD5"
```

3. The client sends an RTSP DESCRIBE request with authentication information again.

```
DESCRIBE rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:1
Accept:application/sdp
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="76a2c9c5b8edbd49838013cf1cf27941"
User-Agent:NKPlayer-1.00.00.081112
```

4. The device responds to SDP information.

```
RTSP/1.0 200 OK
CSeq: 1
Content-Type: application/sdp
Content-Length: 571
Date: Tue, 17 Nov 2020 02:09:45 GMT

v=0
o=- 1109162014219182 0 IN IP4 0.0.0.0
s=Media Server V4.22.126
i=Media Server Session Description : standard
e=NONE
c=IN IP4 0.0.0.0
t=0 0
a=control:*
b=AS:6154
a=range:npt=now-
m=video 0 RTP/AVP 96
i=Video Media
a=rtpmap:96 H264/90000
a=fmtp:96 profile-level-id=4D0014;packetization-mode=0
a=control:trackID=1
b=AS:6144
m=audio 0 RTP/AVP 8
i=Audio Media
a=rtpmap:8 PCMA/8000
a=control:trackID=2
b=AS:10
a=Media_header:MEDIAINFO=494D4B48020100000200000111710110401F000000FA000000000000000000000000000000000000;
a=appversion:1.0
```

5. The client sends RTSP SETUP requests, and the server responds to them.

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=1 RTSP/1.0
CSeq:2
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Transport:RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=0-1;ssrc=433122aa
CSeq: 2
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

```
SETUP rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101/trackID=2 RTSP/1.0
CSeq:3
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="ff343f5ff82deb028dd9b4932cc44201"
Session:1127293610;timeout=60
Transport:RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=0
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610;timeout=60
Transport: RTP/AVP/TCP;unicast;interleaved=2-3;ssrc=433122ab
CSeq: 3
Accept-Ranges: NPT
Media-Properties: No-Seeking, Time-Progressing, Time-Duration=0
Date: Tue, 17 Nov 2020 02:09:45 GMT
```

6. The client sends an RTSP PLAY request.

```
PLAY rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:4
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

7. The server sends audio and video stream data.

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 4
Date: Tue, 17 Nov 2020 02:09:45 GMT

$. .....d1.w....c....".T....g....).i.........a....7.S..~J...........X....X.
```

8. The client sends an RTSP TEARDOWN request, and the server responds to it.

```
TEARDOWN rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101 RTSP/1.0
CSeq:5
Authorization: Digest username="admin", realm="3521781c29acb312330dd668", nonce="026019333", uri="rtsp://10.21.84.147:554/ISAPI/Streaming/channels/101",
response="24edf8a6ff3ef767f7c49d1c847200bd"
Session:1127293610;timeout=60
Range:npt=0.000000-0.000000
User-Agent:NKPlayer-1.00.00.081112
```

```
RTSP/1.0 200 OK
Session: 1127293610
CSeq: 5
Date: Tue, 17 Nov 2020 02:09:50 GMT
```


7.5 Manual Capturing Events

7.5.1 Introduction to the Function

According to the preset information and capture interval configured by the client, the device manually captures images at the specified preset and reports the image information in the form of events. It is used for the client to identify emergency situations and perform an emergency patrol on some presets for analysis. It has a higher priority than the daily scheduled patrol capture. Common usage scenarios: according to the scheduled capturing event (eventType: imageCapture) rules, the store camera regularly audits in a number of key areas. If in the event of an emergency, and a worker wants to immediately re-inspect each area, you can set a manual capture rule to complete an independent audit. After the audit is completed, the camera will continue to audit according to the scheduled capturing rules.

7.5.2 API Calling Flow

1. The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualImageCapture` returns `true`, manual capturing shall be supported.
2. The client gets manual capturing capability for a specified channel: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture/capabilities?format=json`.
3. The client sets the parameters of manual capture events for a specified channel: Get parameters: `GET /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`. Set parameters: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/manualImageCapture?format=json`.
4. Receive the event uploaded by the device in arming or listening mode: `eventType:manualImageCapture`.

Remarks:

- Each time the client sets manual capture event parameters, the channel immediately takes a patrol capture of the relevant presets. When the last preset capture is finished, the manual capture event ends.

7.6 Motion Detection

7.6.1 Introduction to the Function

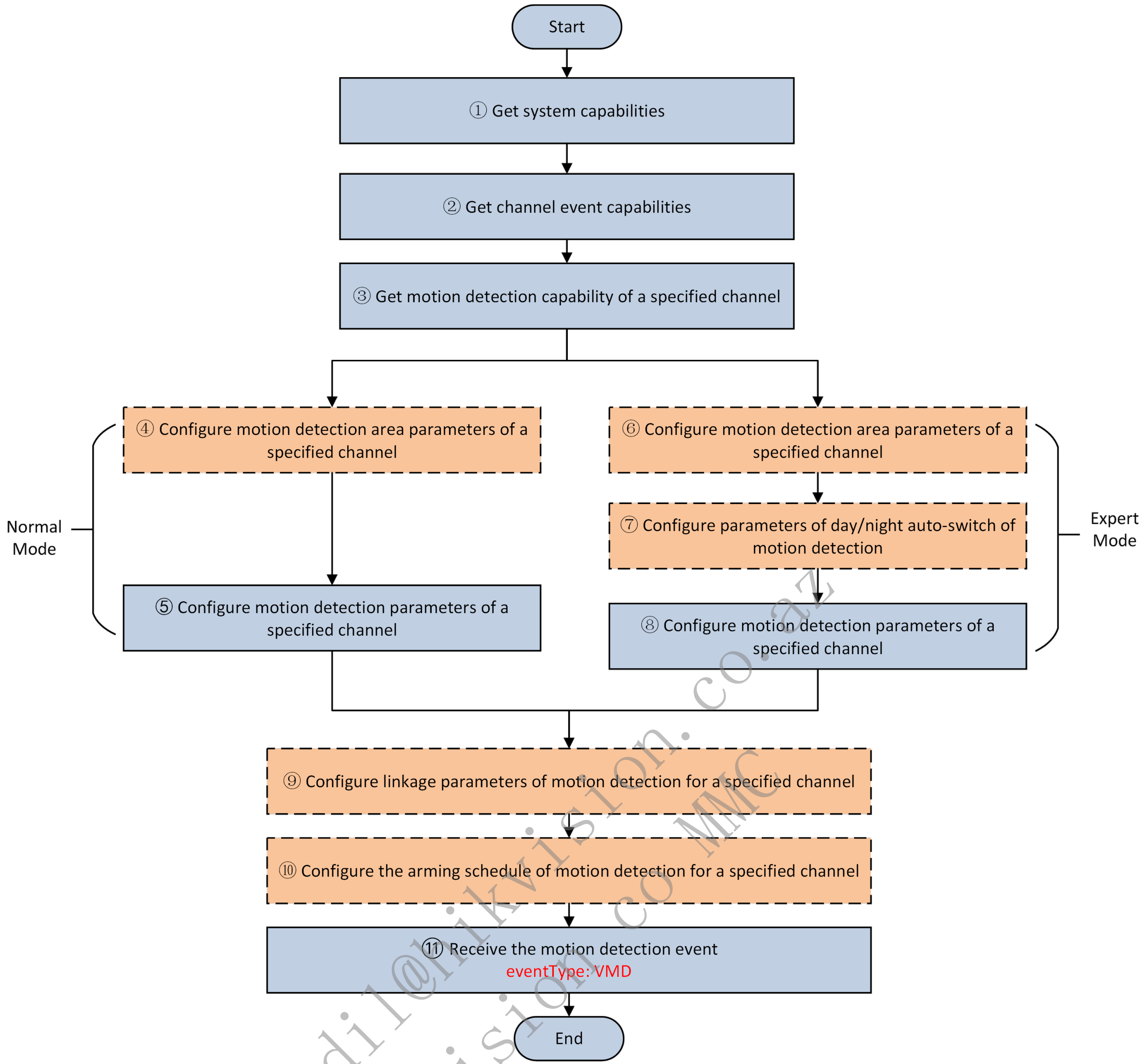
Commonly used in unattended camera recording and automatic alarm, motion detection can help to reduce the cost of manual monitoring for public institutions and companies. Besides, it can greatly improve monitoring efficiency and accuracy by avoiding the errors of manual monitoring due to fatigue after long-time work. For example, it can be applied in home and company theft proof, as well as monitoring and alarm in institutions like medical organizations and nursing homes.

Motion detection has two modes: normal mode and expert mode.

Normal mode: the default mode of motion detection. In this mode, all detected areas only support a same sensitivity.

Expert mode: in this mode, every detected area can have its own detection sensitivity and percentage of objects in area. It supports configuring different sensitivities according to area types. For example, for the motion detection in the company hall, you can set higher detection sensitivity for the gate area to focus on people's entering and exiting.

7.6.2 API Calling Flow



1. Get the system capability: `GET /ISAPI/System/capabilities`. If the node `isSupportMotionDetection` is returned and its value is `true`, it indicates that the device supports motion detection.

2. Get the device channel event capability. If the returned `eventType` contains `VMD`, it indicates that the corresponding channel supports motion detection.

   Get the event capability of all channels: `GET /ISAPI/Event/channels/capabilities`.

   Get the event capability of a specified channel: `GET /ISAPI/Event/channels/<channelID>/capabilities`.

3. Get the motion detection capability of a specified channel (normal mode): `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/capabilities`.

   Check the motion detection parameters (such as function enabling, detection sensitivity) supported by the device. You can check whether the device supports multiple areas via checking whether the node `regionType` contains `region`.

4. (Optional) Configure parameters of the motion detection area (normal mode).

   Capability: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout/capabilities`.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection/layout?regionType=<regionType>`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

5. Configure the motion detection parameters (such as motion detection enabling, detection sensitivity) (normal mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, for compatibility, both `<gridMap>` and `<RegionList>` will be returned regardless the `regionType` value in URL.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetection`. **Note:** If the device supports multiple areas of motion detection, to set multiple areas, you need to input `regionType=<region>` in URL.

6. (Optional) Configure the parameters of motion detection area (expert mode). You can configure different detection conditions such as sensitivity for different areas.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/regions`.

7. (Optional) Configure day/night auto switch of motion detection (expert mode). You can configure the time switch schedule of `day` and `night`.

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt/switch`.

8. Configure motion detection conditions (expert mode).

   Get: `GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

   Configure: `PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt`.

9. (Optional) Configure linkage parameters of the motion detection.

   Get: `GET /ISAPI/Event/triggers/VMD-<channelID>`.

   Configure: `PUT /ISAPI/Event/triggers/VMD-<channelID>`.

   Delete: `DELETE /ISAPI/Event/triggers/VMD-<channelID>`.

10. (Optional) Configure the arming schedule of motion detection.

    Get: `GET /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

    Configure: `PUT /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>`.

11. Receive events reported by device in arming or listening mode: `eventType:VMD`.

Remarks:

1. For motion detection (normal mode), the uploaded `VMD` event includes device basic information only, such as channel No. and event time. For details, see 4.3.1 Motion Detection Event (Normal Mode).

2. For motion detection (expert mode), in addition to the device basic information, the uploaded `VMD` event includes the detection area ID, area coordinates and area detection sensitivity. For details, see 4.3.2 Motion Detection Event (Expert Mode).

3. In practical employment, you can select one mode as needed. There are no logical relations between the two modes.

7.6.3 Message Example

7.6.3.1 Motion Detection Event (Normal Mode)


```xml
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T19:42:27+08:00</dateTime>
<activePostCount>4</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;&amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```


7.6.3.2 Motion Detection Event (Expert Mode)


```xml
<?xml version="1.0" encoding="UTF-8"?>
<EventNotificationAlert version="1.0" xmlns="urn:psialliance-org">
<ipAddress>10.17.114.252</ipAddress>
<protocolType>HTTP</protocolType>
<macAddress>64:db:8b:6e:4f:08</macAddress>
<channelID>1</channelID>
<dateTime>2018-03-13T20:36:34+08:00</dateTime>
<activePostCount>11</activePostCount>
<eventType>VMD</eventType>
<eventState>active</eventState>
<eventDescription>Motion alarm</eventDescription>
<DetectionRegionList>
<DetectionRegionEntry>
<regionID>1</regionID>
<sensitivityLevel>50</sensitivityLevel>
<RegionCoordinatesList>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>756</positionX>
<positionY>756</positionY>
</RegionCoordinates>
<RegionCoordinates>
<positionX>216</positionX>
<positionY>216</positionY>
</RegionCoordinates>
</RegionCoordinatesList>
</DetectionRegionEntry>
</DetectionRegionList>
<channelName>Camera 01</channelName>
<Extensions version="1.0" xmlns="urn:psialliance-org">
<serialNumber xmlns="urn:selfextension:psiaext-ver10-xsd">DS-2DF8236IX-AELW20170718CCWR799562898X</serialNumber>
<eventPush xmlns="urn:selfextension:psiaext-ver10-xsd">VMD&amp;&amp;DS-2DF8236IX-AELW20170718CCWR799562898X,2018-01-21T12:50:39+08:00,1,1.0</eventPush>
</Extensions>
</EventNotificationAlert>
```


7.7 Preset Dedicated Focusing

7.7.1 Function Introduction

The preset dedicated focusing algorithm is designed for specific scenarios (such as instrument and gauge images) where the image is not clear at a certain position, to achieve clearer and faster focusing. The dedicated focusing algorithm includes foreground focusing, background focusing, and area focusing.

7.7.2 API Calling Flow

1. The client gets the device's PTZ capabilities via URL `GET /ISAPI/PTZCtrl/capabilities`. If `isSupportPresetsProFocus` is returned with `true` value, the device supports preset dedicated focusing.

2. The client gets the capability of preset dedicated focusing for the specified channel via URL `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg/capabilities?format=json`.

3. The client batch configures presets' dedicated focusing parameters for the specified channel.
   - Get parameters: `GET /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/PresetsProFocusCfg?format=json`

4. The client configures the dedicated focusing parameters of a single preset for the specified channel. If a preset is not clear under the batch configuration in step 3, it can be configured separately.
   - Add dedicated focusing parameters of a preset for a specified channel: `POST /ISAPI/PTZCtrl/channels/<channelID>/AddPresetsProFocusCfg?format=json`
   - Get parameters: `POST /ISAPI/PTZCtrl/channels/<channelID>/GetPresetsProFocusCfg?format=json`
   - Configure: `PUT /ISAPI/PTZCtrl/channels/<channelID>/ModifyPresetsProFocusCfg?format=json`
   - Delete: `POST /ISAPI/PTZCtrl/channels/<channelID>/DeletePresetsProFocusCfg?format=json`
   - Get parameters of multiple presets: `GET /ISAPI/PTZCtrl/channels/<channelID>/GetMultiPresetsProFocusCfg?format=json`

7.7.3 Exception Handling

Error Codes

statusCode	statusString	subStatusCode	errorCode	errorMsg	Chinese Description
4	Invalid Operation	numberReachedLlimit	0x40001119	The number reached limit.	The configured items have reached the limit and cannot be added further. Please delete unnecessary ones and try again.

7.8 Sub-device's Transparent Transmission

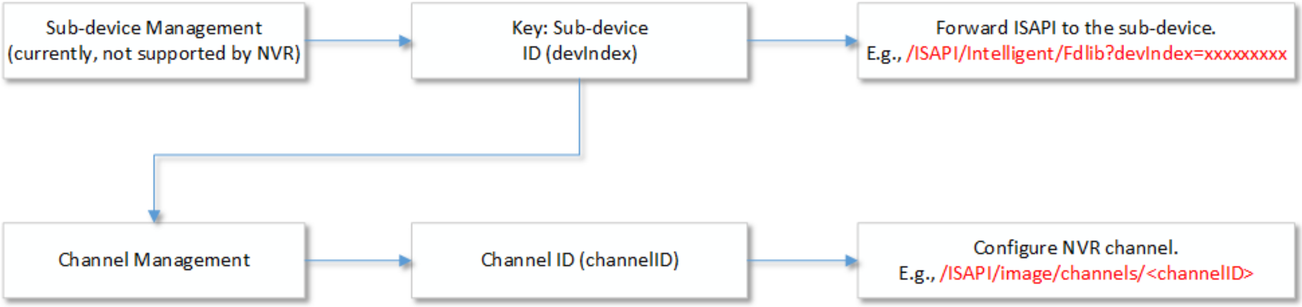
7.8.1 Introduction to the Function

In actual application scenarios, the NVR being used is generally not the latest version, if an IPC with latest functions is added to the system, the NVR will fail to recognize the IPC's new functions due to version incompatibility, so the platform (including device access platforms, client softwares, device's web plug-ins, etc, hereinafter referred to as the platform) will fail to operate new functions of the IPC.

Besides, in video channel resource management, the platform can only get information about digital channels (also known as IP channels) added to the NVR, but not the device that the channels belong to. For example, two video channels (channel 13 and channel 18) of a dual-lens people counting camera are added to the NVR. The platform can only get that digital channel 13 and 18 are added to the NVR, but not that the two channels belong to one camera. Such information loss may lead to repeated operations, which will further cause problems in actual practices. For example, users may fail to upgrade IPC via digital channel No. if not knowing the relationship between camera and digital channel.

To settle above problems, the sub-device's transmission function is introduced to the video channel resources management. In this way, users can directly access to the device of the digital channels via the devIndex (sub-device index) allocated to the device by NVR.

Notably, despite the introduction of the sub-device's transparent transmission function, the NVR does not function as an access gateway, and so it does not function with sub-device management on gateway. Refer to the Sub-device Management of Device Management for details.

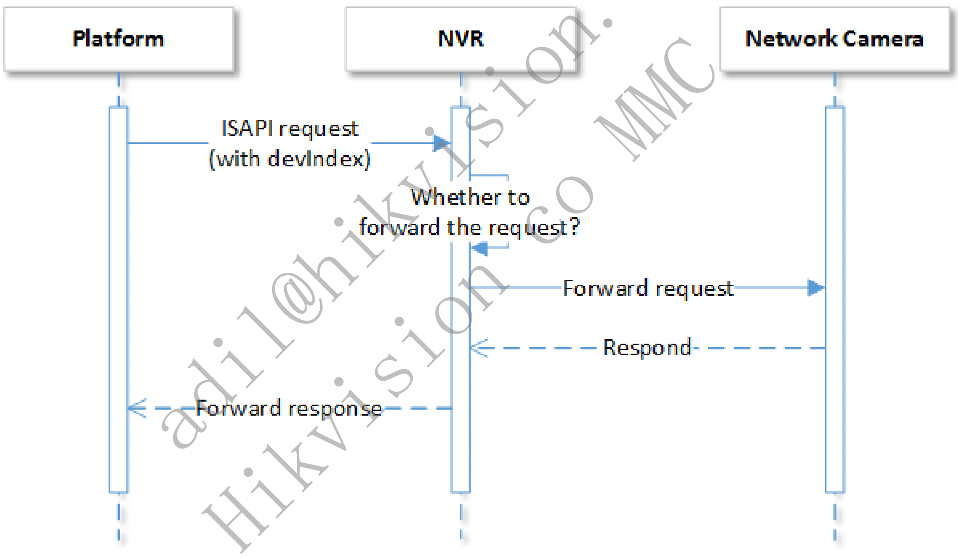


Steps are as follows:

1. First, the platform will check whether the NVR supports sub-device's transparent transmission. If it is supported, follow the next step. Second, add devIndex to the request URL.
2. For the ISAPI request sent to the NVR by the platform, if it contains devIndex, it will be forwarded to the specific sub-device; otherwise, the NVR will respond to the request.
3. When the request is forwarded to the sub-device by the NVR, delete the devIndex in the request URL so as to realize ISAPI interaction.

For example:

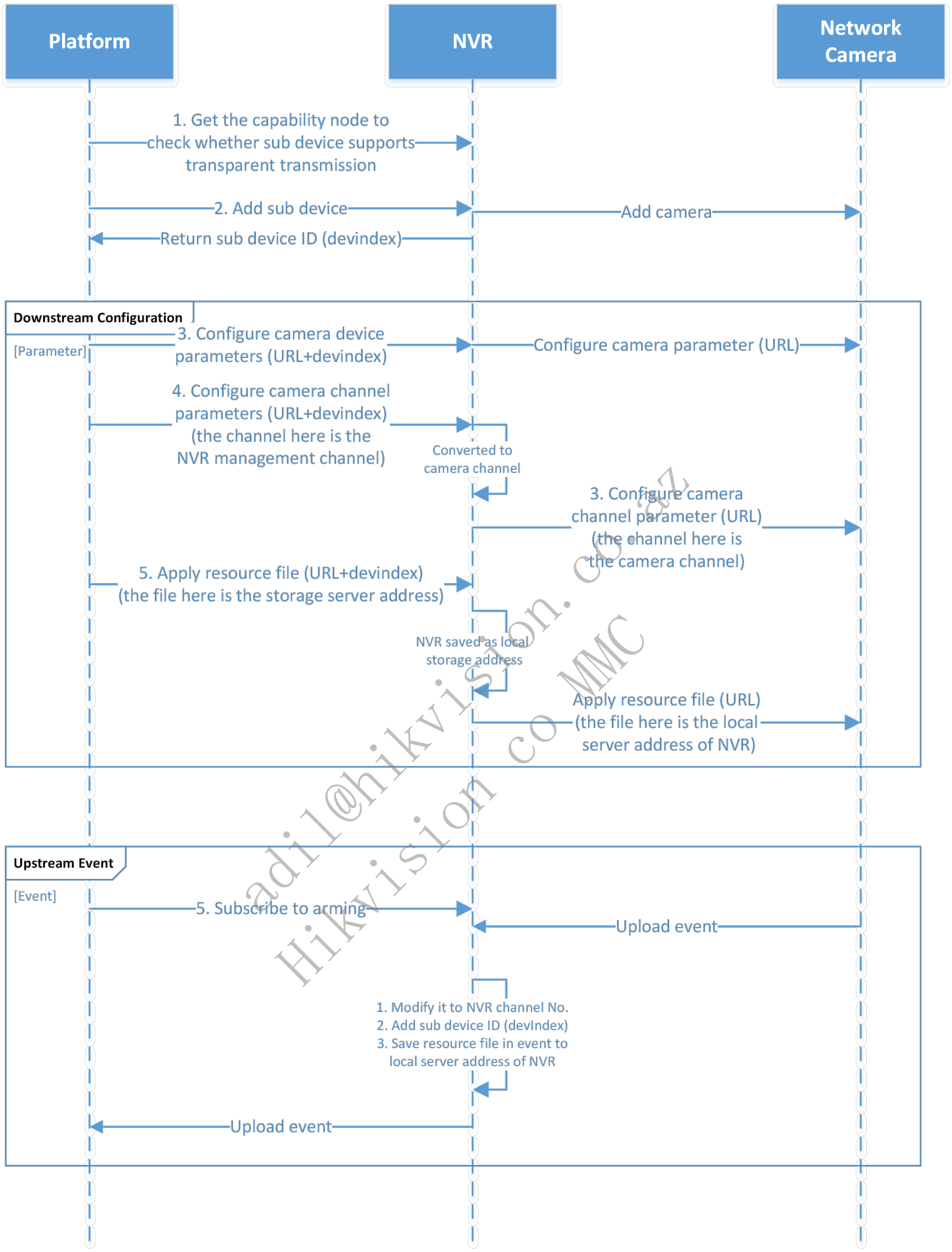
1. Send the request URL (`GET /ISAPI/System/deviceInfo`) to get the information of the platform's NVR device.
2. Send the request URL (`GET /ISAPI/System/deviceInfo?devIndex=<devIndex>`) to get the information of a specific sub-device of the NVR.



Note:

1. This function is applicable to transitory connections, while inapplicable to persistent connections (e.g., arming subscription, two-way audio).
2. It is required to delete the devIndex in the URL before NVR forwards the ISAPI request.
3. For the VCA function of NVR, the transmission solution of sub-devices is not adopted. You can switch the analysis unit by the following URI: /ISAPI/Smart/analysisUnitSwitch/channels/`<channelID>`/event/`<EventType>`.

7.8.2 API Calling Flow



1. Get the capability node to check whether the device supports transparent transmission. If the value of the node `isSupportDevIndex` is true, it indicates that the device supports transparent transmission.
2. Create digital channels: `POST /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`. The sub device ID`devindex` will be returned.
3. Get access parameters of all digital channels: `GET /ISAPI/ContentMgmt/InputProxy/channels?security=<security>&iv=<iv>`.
   Each digital channel information contains one parameter `devIndex`, which is the same for digital channels of a same physical device.

4. When splicing ISAPI request, if the URL contains `devIndex=<devIndex>`, it represents a request for the sub-device. For example, you can send `12345678` (the request URL) to get the information of the sub-device `GET /ISAPI/System/deviceInfo?devIndex=12345678` of the target device. Note: 1) For the downstream configuration of a video channel, the platform will apply the channel managed by NVR, and before applying, the NVR will transform channel into the video channel. 2) When the platform applies the storage address of resource files, the camera cannot access the storage service to get resource files because of the network isolation between the camera and platform, so the storage address must be saved as a local storage address by the NVR before being applied.

7.9 Sync Capturing Manually

7.9.1 Introduction to the Function

The client sends a capture command to the device, and the device immediately returns the captured picture data in respond. It is often used in scenarios with high network bandwidth between clients and devices to facilitate the rapid transmission of image data, such as a LAN built on a company campus.

7.9.2 API Calling Flow

1. (Optional) The client gets the system capability: `GET /ISAPI/System/capabilities`, and if `isSupportManualSnapPicture` returns `true`, sync manual capturing shall be supported.

2. (Optional) The client gets sync manual capturing capability for a specified channel: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture/capabilities?format=json`.

3. The client controls the specified channel to perform sync manual capturing: `GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>`, and the device returns the captured binary picture data.

Remarks:

- If you only need to capture a JEPG picture, you can jump to step 3.
- If you need to capture pictures other than JEPG or capture in the designated area of the video, you need to perform steps 1 and 2 to determine whether the device supports it. If it does, go to step 3.
- In step 3, the value ranges of `videoResolutionWidth` and `videoResolutionHeight` are the same as `<videoResolutionWidth> and <videoResolutionHeight>` returned by `GET /ISAPI/Streaming/channels/<trackStreamID>/capabilities`.

8 Video Management

8.1 Search for Event Video

8.1.1 Introduction to the Function

This function supports searching for the videos of linked cameras when an event occurs.

8.1.2 API Calling Flow

1. Check whether the device supports searching for event videos.

Get the device sysgem capability: `GET /ISAPI/System/capabilities`. If the node `<SerialCap>` is returned and its value is true, the device supports searching for event videos.

2. Search for event videos: `POST /ISAPI/ContentMgmt/eventRecordSearch?format=json`.

8.2 Search for Videos and Pictures

8.2.1 Introduction to the Function

This function supports searching for the video/picture data stored in the device. Video data source includes scheduled recording, manual recording and event recording, and picture data source includes scheduled capture and event linkage capture.

8.2.2 API Calling Flow

1. Check the device's supported time format. Get device's system capabilities: `GET /ISAPI/System/capabilities`.
   - If node `isSupportTimeTypeSTD` is returned with value `true`, it indicates that the device supports the ISO 8601 standard time, and URLs of functions related to ISO 8601 time parameters support additional query parameter `timeType=STD`.
   - If node `isSupportTimeTypeSTD` is not returned. Call `GET /ISAPI/System/time/timeType?format=json` to get the time type. If the `type` field returns `local`, it indicates that the device video search and playback use local time. If the `type` field returns `UTC`, it indicates that the device video search and playback use ISO 8601 time format.
   - If the device does not support API `GET /ISAPI/System/time/timeType?format=json`. Call `GET /ISAPI/System/deviceInfo` and check the time type of device via field `deviceType`. If `deviceType` (device type) is CVR, NVR, DVR, or ITCCAM, the device video search and playback use local time. Otherwise, follow the protocol specifications and use the TZ format, for example, `2024-05-13T16:00:00Z`.

2. Check whether the device supports video/picture search. Get the device's video search capability via `GET /ISAPI/ContentMgmt/capabilities`. If `<isSupportRecordSearch>` is returned, it indicates that the device supports video/picture search.

3. Search for videos and pictures:
   i. Get the capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/capabilities`. Get the picture search types that the device supports searching via the node `<pictureSearchType>` returned in the message, including pictures of event capture, scheduled capture and manual capture. Get the video types that the device supports searching for via the node `<recordSearchType>` returned in the message.
   ii. Get the parameters capability of searching for videos and pictures: `GET /ISAPI/ContentMgmt/search/capabilities`. Get the periods when the device supports searching for videos and pictures via the node `<timeSpanList>` returned in the message. Get the file tags that the device supports searching via the node `<labelSearch>` returned in the message.
   iii. Search for videos and pictures: `POST /ISAPI/ContentMgmt/search?security=<security>&iv=<iv>`.
   - The URL containing `timeType=STD` indicates that the time in the input and output messages are according to the ISO 8601 time format. The URL without `timeType=STD` indicates to use device's original time.
   - Check whether it is picture or video to be searched via the node `<trackID>` in the request message `<CMSearchDescription>`. "101" (search for the main stream videos of channel 1), "103" (search for the pictures of channel 1), and it is channel 1 by default .

   Note:

   - The value of `CMSearchDescription`.`trackIDList`.`trackID` is obtained from the `id` value returned in the response message of `GET /ISAPI/ContentMgmt/record/tracks`. The meaning of `id`: `101` is the main stream of channel 1, `102` is the sub-stream of channel 1, `103` is the capture schedule of channel 1, `201` is the main stream of channel 2, and so on. When the device saves only one copy of channel videos, it does not distinguish between the main and sub stream, and only supports setting `trackID` to `xx1`.
   - If `<trackIDList>` is returned in response message of `GET /ISAPI/ContentMgmt/search/capabilities`, it indicates that the device supports accurate search, and the returned value in `<trackIDList>` indicates the channels that support independent video search.

8.3 Search for Videos by Target Type

8.3.1 Introduction to the Function

This function realizes persons, vehicles or mixed targets distinguishing when searching for motion detection event videos. Searching for videos by target type is newly added. The search conditions include target type information, and the returned result of the search contains the start and end time as well as the file name of the video.

8.3.2 API Calling Flow

1. Check whether the device supports searching by target type:

Get the device capability: `GET /ISAPI/System/capabilities`. If the node is returned and its value is true, it indicates that the device supports searching by target type. Currently, the device supports searching by person and by vehicle.

2. Get the capability of searching by target type: `GET /ISAPI/ContentMgmt/SearchByTargetType/capabilities?format=json`.

3. Search for videos by target type: `POST /ISAPI/ContentMgmt/SearchByTargetType?format=json`.

9 Vehicle Recognition

9.1 Motor Vehicle Recognition

9.1.1 Introduction to the Function

This function is to detect whether there are vehicles appearing in the image for certain time periods. If a vehicle appears and the set recognition conditions are met, it will be captured, and its related information will be analyzed and uploaded as alarms. The integrator receives the event and analyzes the specific content as needed.

9.1.2. Device Event Upload

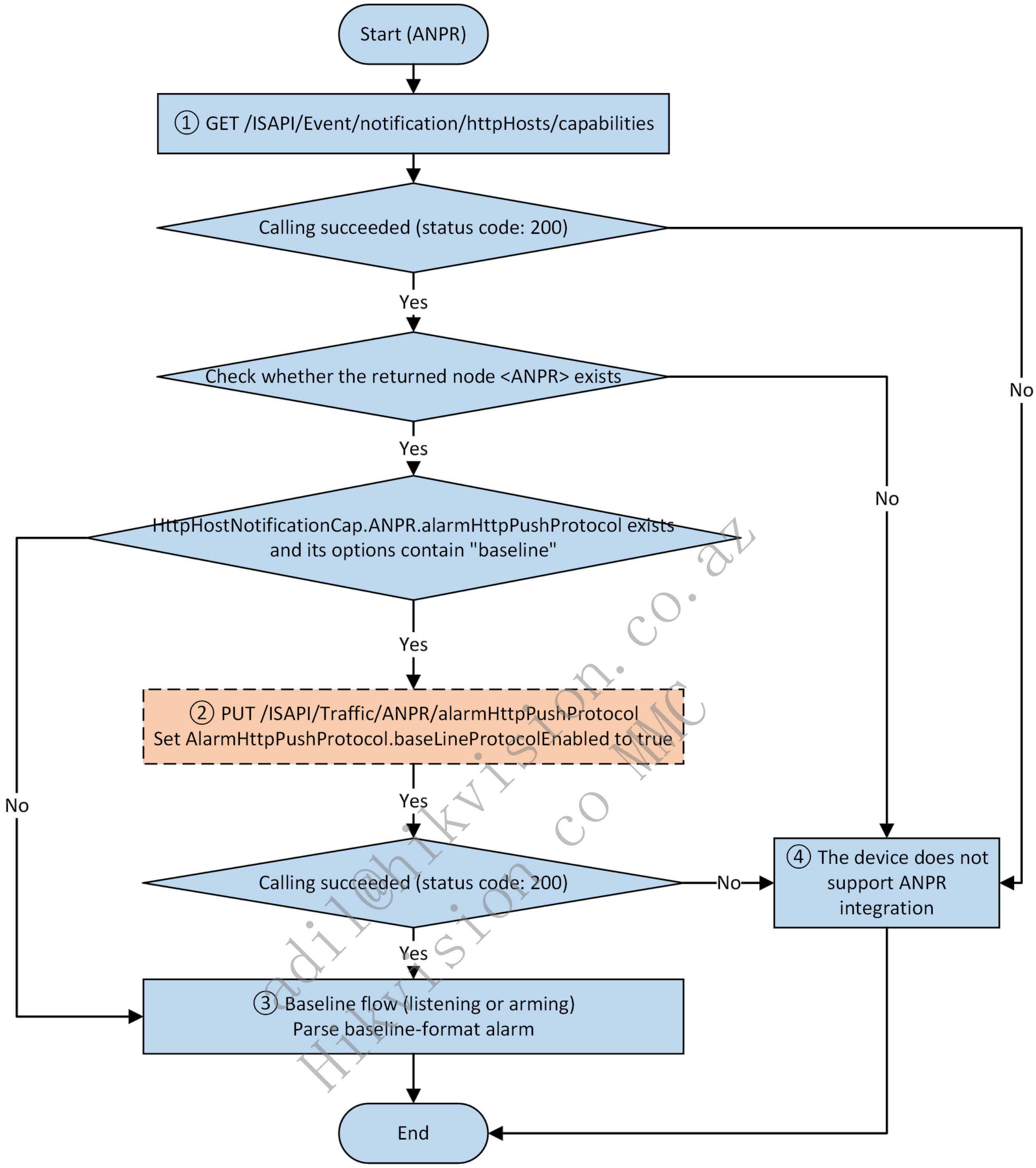
9.1.2.1 Check Whether Device Supports Baseline Vehicle Recognition Event

The integrator get the capability from device.

1. Get the system capability set via `GET /ISAPI/System/capabilities`: `true` is returned for `<ITCCap>.<isSupportVehicleDetection>` by device or `<ITCCap>.<isSupportHVTVehicleDetection>` is `true`.
2. Get the vehicle capture recognition capability via `GET /ISAPI/ITC/capabilities`: `true` is returned for `<isSupportVehicleDetection>` by device or `<isSupportHVTVehicleDetection>` is `true`.
3. Get the traffic service capability via `/ISAPI/Traffic/capabilities`: `<plateCap>` is returned by device.
4. Get the trigger mode capability set via `GET /ISAPI/ITC/TriggerMode/capabilities`: `<TriggerMode>` is returned.

When any one of the 4 conditions are met, it indicates the vehicle recognition event detection is supported. If none are met, the vehicle recognition event is not supported and please do not proceed to the following steps.

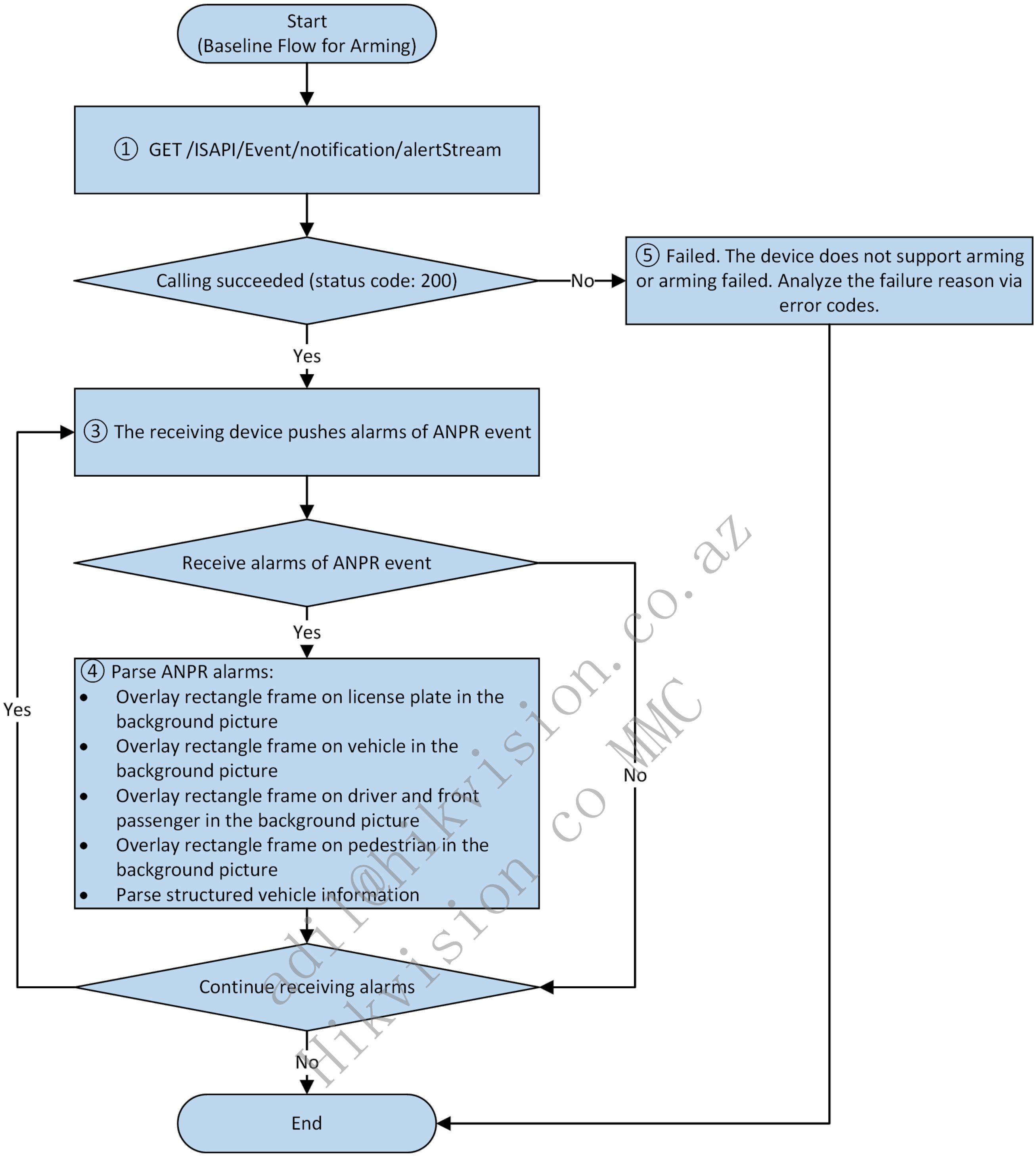
9.1.2.2 Check Whether Device Supports Baseline Flow



1. (Client) Get the configuration capability of listening host via `GET /ISAPI/Event/notification/httpHosts/capabilities`.
2. (Client) Configure the alarm mode switch for device via `PUT /ISAPI/Traffic/ANPR/alarmHttpPushProtocol` and set `AlarmHttpPushProtocol.baseLineProtocolEnabled` to `true`.
3. Receive and parse alarms according to the baseline flow. See details in `4.3 Baseline Flow Event Upload` and `4.4 Baseline Flow Message Format and Example`.
4. The ANPR function integration is not supported by device.

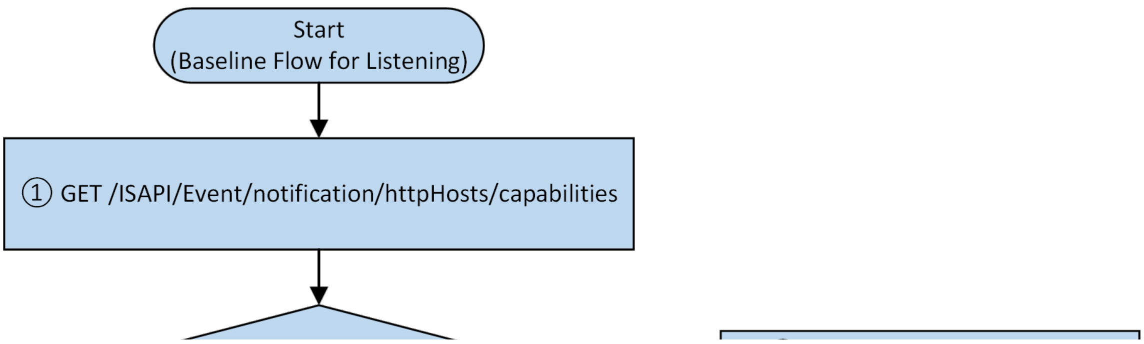
9.1.3 Baseline Flow Event Upload

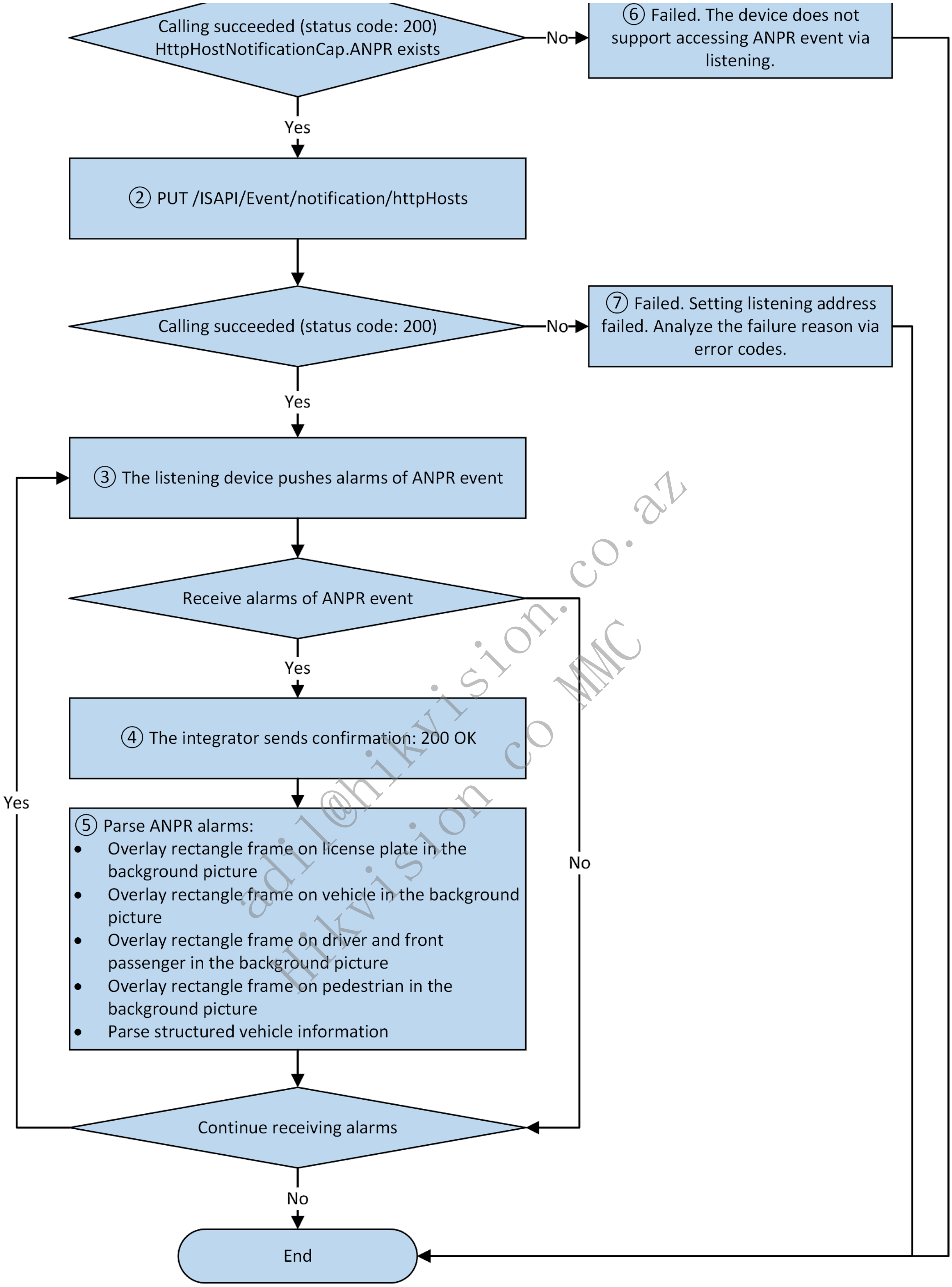
9.1.3.1 Arming Flow



1. Arm the device.
2. Check whether arming succeeded.
3. Receive alarm notifications and check whether `eventType==ANPR`.
4. Parse ANPR alarm content.
5. Arming failed. Analyze the failure reason via error codes.

9.1.3.2 Listening Flow





1. Get the capability set of device HTTP listening host location.
2. Change the HTTP listening host location, IP address, port configuration to the IP address of the server for receiving alarm notifications.
3. Receive alarm notifications and check whether `eventType==ANPR`.
4. The integrator sends confirmation (200 OK) to the device and notify the device of the successful alarm receiving; if the integrator does not reply, the device will consider the alarm notification lost and upload it again.
5. Parse ANPR alarm content.

6. The device does not support accessing ANPR event via listening.
7. Setting listening address failed. Analyze the failure reason via error codes.

9.1.4 Baseline Flow Message Format and Example

9.1.4.1 Baseline Flow Event Upload Method Configuration

Set `baseLineProtocolEnabled` to `true`,

URI: `/ISAPI/Traffic/ANPR/alarmHttpPushProtocol`

Method: `GET, PUT`

Body:

```
<AlarmHttpPushProtocol version="2.0" xmlns="http://www.isapi.org/ver20/XMLSchema">
    <baseLineProtocolEnabled>true</baseLineProtocolEnabled>
</AlarmHttpPushProtocol>
```

9.1.2.2 Baseline Flow Event Upload Example

Reported by device.

```
POST /test HTTP/1.1
Accept: text/html, application/xhtml+xml, */*
Accept-Language: zh-CN
Content-Type: multipart/form-data; boundary=7e13971310878
User-Agent: Mozilla/5.0 (compatible; MSIE 9.0; Windows NT 6.1; WOW64; Trident/5.0)
Accept-Encoding: gzip, deflate
Host: 10.10.36.29:8080
Content-Length: 9907
Connection: Keep-Alive
Cache-Control: no-cache

--7e13971310878
Content-Disposition: form-data; name="anpr.xml";filename="anpr.xml";
Content-Type: text/xml
Content-Length: 150

<EventNotificationAlert/>
-----------------------------7e13971310878
Content-Disposition: form-data; name="licensePlatePicture.jpg";filename="licensePlatePicture.jpg";
Content-Type: image/jpeg
Content-Length: 200

......JFIF.....`.`.....C........... .
..
................ $.' ",#..(7),01444.'9=82<.342...C. ....

.2!.!222222222222222222222222222222222222222222222222222......q.<..".....................................
.....................}........!1A..Qa."q.2....#B...R..$3br.
.....%&'()*456789:CDEFGHIJSTUVWXYZcdefghijstuvwxyz...............................................................................................
.....................w.......!1..AQ.aq."2...B.... #3R..br.
.$4.
--7e13971310878
Content-Disposition: form-data; name="detectionPicture.jpg";filename="detectionPicture.jpg";
Content-Type: image/jpeg
Content-Length: 9907

......JFIF.....`.`.....C........... .
..
................ $.' ",#..(7),01444.'9=82<.342...C. ....

.2!.!222222222222222222222222222222222222222222222222222......q.<..".....................................
.....................}........!1A..Qa."q.2....#B...R..$3br.
.....%&'()*456789:CDEFGHIJSTUVWXYZcdefghijstuvwxyz...............................................................................................
.....................w.......!1..AQ.aq."2...B.... #3R..br.
.$4.
--7e13971310878--
```

The receiver responds.

```
HTTP/1.1 200 OK
Connection: close
```

Note: When there are multiple pictures of the same type, the following naming rule for will be applied: add suffix (e.g., "_1", "_2") after the extended name. For example, the names of three detection pictures can be detectionPicture_1.jpg, detectionPicture_2.jpg, and detectionPicture_3.jpg.

Attribute Value	Attribute Type Character String
ANPR Alarm XML	anpr.xml
Detection Picture (Background Picture)	detectionPicture.jpg
License Plate Thumbnail	licensePlatePicture.jpg
Driver's Face Matting	pilotPicture.jpg
Passenger’s Face Matting	copilotPicture.jpg
Composite Picture	compositePicture.jpg
License Plate Binary Picture	plateBinaryPicture.jpg
Non-Motor Vehicle Matting	nonMotorPicture.jpg
Pedestrian Picture	pedestrianDetectionPicture.jpg
Pedestrian Matting	pedestrianPicture.jpg

10 Entrance & Exit Management and Control

10.1 Entrance & Exit Station Management

10.1.1 Function Introduction

In the parking fee management scenarios (e.g., parking lots), the entrance & exit station is used to configure charging rules, and to charge via cards/tickets.

10.1.2 API Calling Flow

1. Check if the device supports vehicle card data management

Get the total capabilities of entrance & exit station: `GET /ISAPI/Parking/TME/capabilities`. A success return indicates that the device supports card data management.

2. Set entrance & exit station parameters

Configure the basic rule parameters.

- Get the capabilities of entrance & exit station parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEConfiguration/capabilities`.
- Configure entrance & exit station parameters: `PUT /ISAPI/Parking/channels/<channelID>/TMEConfiguration`.
- Get entrance & exit station parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEConfiguration`.

(Optional) Configure the ticket printing format & content.

- Get the capabilities of ticket printing format & content: `GET /ISAPI/Parking/channels/<channelID>/paperPrintFormatInfo/capabilities`.
- Configure the ticket printing format & content: `PUT /ISAPI/Parking/channels/<channelID>/paperPrintFormatInfo`.
- Get the ticket printing format & content: `GET /ISAPI/Parking/channels/<channelID>/paperPrintFormatInfo`.

3. Search for card and vehicle information of entrance/exit

- Search for card information of entrance/exit: `POST /ISAPI/Parking/channels/<channelID>/search/TMECard`.

- Search for vehicle information of entrance/exit: `POST /ISAPI/Parking/channels/<channelID>/search/TMEVehicle`.

4. Manage vehicle card information of entrance/exit

Configure the vehicle card information of entrance/exit.

- Batch set multiple vehicle card information of entrance/exit: `PUT /ISAPI/Parking/channels/<channelID>/TMECardVehicle`.
- Batch get multiple vehicle card information of entrance/exit: `GET /ISAPI/Parking/channels/<channelID>/TMECardVehicle`.
- Set a single vehicle card information of entrance/exit: `PUT /ISAPI/Parking/channels/<channelID>/TMECardVehicle/<cardID>`. The `cardID` in URL specifies the vehicle card ID.
- Get a single vehicle card information of entrance/exit: `GET /ISAPI/Parking/channels/<channelID>/TMECardVehicle/<cardID>`. The `cardID` in URL specifies the vehicle card ID.
- Export the vehicle card information as binary data: `GET /ISAPI/Parking/channels/<channelID>/TMECardChargeData`.

5. Set audio parameters of entrance & exit station

- Set audio parameters: `PUT /ISAPI/Parking/channels/<channelID>/TMEVoiceConfiguration`.
- Get audio parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEVoiceConfiguration`.

6. Top-up and payment

The client sends account top-up information to the entrance & exit station to top up the account.

- Get capabilities of account top-up: `GET /ISAPI/Parking/channels/<channelID>/chargeAmount/capabilities`.
- Set account top-up parameters: `PUT /ISAPI/Parking/channels/<channelID>/chargeAmount`.

The client sends payment information to the entrance & exit station, which informs the vehicle owner of payment amount.

- Get capabilities of payment information: `GET /ISAPI/Parking/channels/<channelID>/paperChargeInfo/capabilities`.
- Set the payment information: `PUT /ISAPI/Parking/channels/<channelID>/paperChargeInfo`.

7. Entrance & exit station status

Get the entrance & exit station status: `GET /ISAPI/Parking/channels/<channelID>/TMESystemInfo`.

8. Offline mode

Configure offline mode parameters and charging rules in offline mode.

Offline mode parameters include whether to enable offline mode, whether to charge, whether to trigger an alarm, etc.

- Get the capability of offline mode parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEOfflineConfiguration/capabilities`.
- Set offline mode parameters: `PUT /ISAPI/Parking/channels/<channelID>/TMEOfflineConfiguration`.
- Get offline mode parameters: `GET /ISAPI/Parking/channels/<channelID>/TMEOfflineConfiguration`.

When the offline mode is enabled and charging is required, set the charging rule in offline mode.

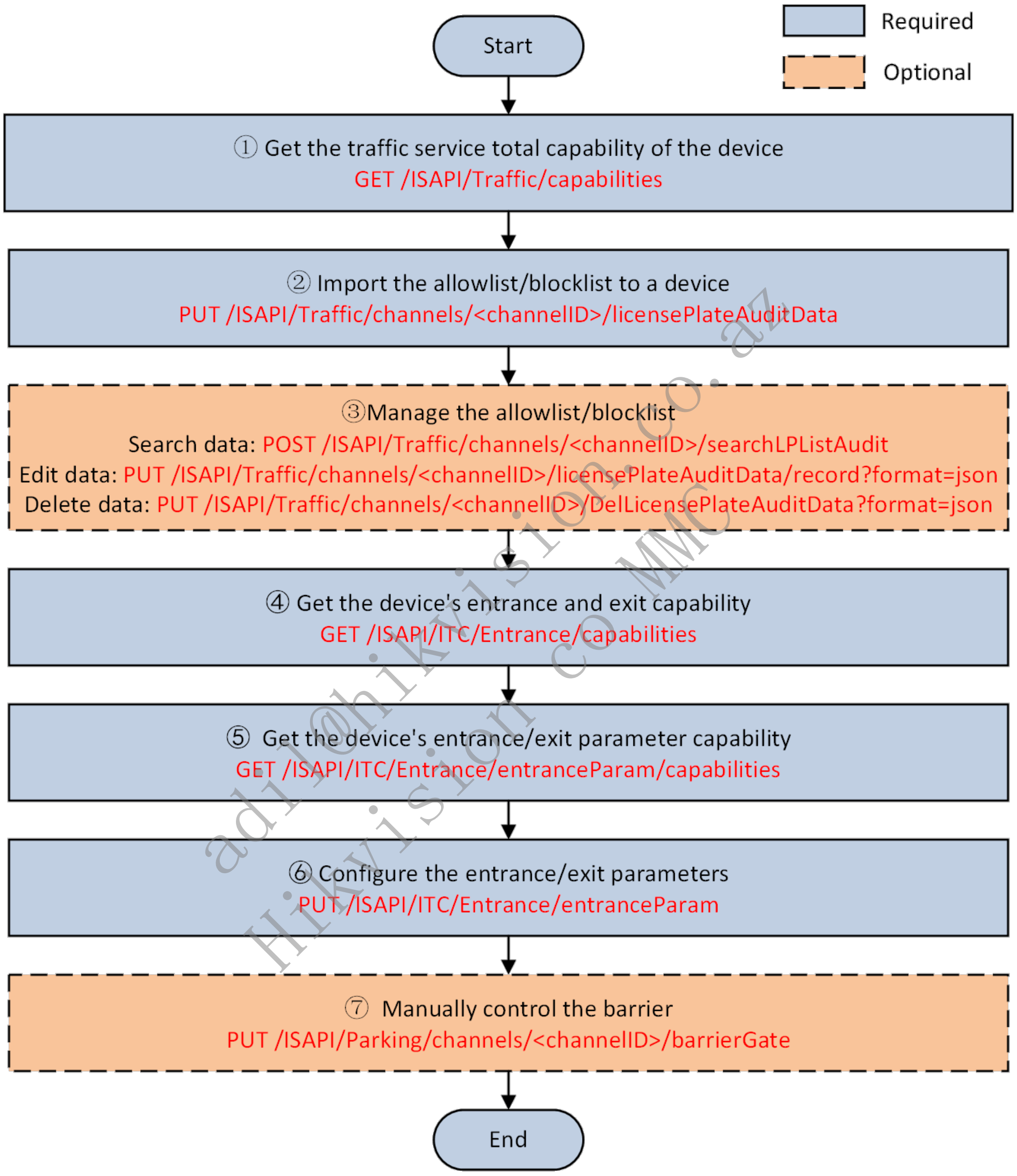
- Get the capability of charging rules in offline mode: `GET /ISAPI/Parking/channels/<channelID>/TMEChargeRule/capabilities`.
- Batch set charging rules in offline mode: `PUT /ISAPI/Parking/channels/<channelID>/TMEChargeRule`.
- Batch get charging rules in offline mode: `GET /ISAPI/Parking/channels/<channelID>/TMEChargeRule`.
- Set a single charging rule in offline mode: `PUT /ISAPI/Parking/channels/<channelID>/TMEChargeRule/<ruleID>`. The `ruleID` specifies the charging rule ID.
- Get a single charging rule in offline mode: `GET /ISAPI/Parking/channels/<channelID>/TMEChargeRule/<ruleID>`. The `ruleID` specifies the charging rule ID.

10.2 Entrance and Exit Barrier Control

10.2.1 Function Introduction

Devices use license plate recognition to compare captured license plate with the vehicle allowlist/blocklist, then automatically control the barrier based on the predefined rules. Manual control is also supported. Typical applications include entrances/exits, toll stations, parking lots, etc.

10.2.2 API Calling Flow



Steps

1. Get the device's traffic service total capability: `GET /ISAPI/Traffic/capabilities`.

   Check the license plate recognition capabilities. Key capability fields are:

   - `$.TrafficCap.plateCap.isSupportPlateList` (whether the device supports importing/exporting the license plate blocklist and allowlist),
   - `$.TrafficCap.plateCap.isSupportLicenseModify` (whether the device supports adding or editing the license plate of imported allowlist/blocklist),
   - `$.TrafficCap.plateCap.isSupportLPAuditDataDelete` (whether the device supports deleting the imported allowlist/blocklist).

2. Import a vehicle allowlist/blocklist to the device: `PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData`.

   The default supported format is Excel. The sheet should include the following information: serial number, license plate number, group (0 for blocklist, 1 for allowlist), card number, license plate color, license plate type.

3. (Optional) Manage the vehicle allowlist/blocklist.
   - Search the data: `POST /ISAPI/Traffic/channels/<channelID>/searchLPListAudit`.
   - Add or edit the data: `PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData/record?format=json`.
   - Delete the data: `PUT /ISAPI/Traffic/channels/<channelID>/DelLicensePlateAuditData?format=json`.

4. Get the device's entrance and exit capability: `GET /ISAPI/ITC/Entrance/capabilities`.

   Related capability fields are:
   - `$.EntranceCap.supportEntrance` (whether it supports entrance and exit control),
   - `$.EntranceCap.supportBarrierGateNum` (number of barriers).

5. Get the entrance and exit parameter capability of the device: `GET /ISAPI/ITC/Entrance/entranceParam/capabilities`.

6. Set the device's entrance and exit parameters: `PUT /ISAPI/ITC/Entrance/entranceParam`.

   The `$.EntranceParamList.EntranceParam.ctrlMod` field sets the barrier control mode:
   - 0-camera control (automatic),
   - 1-platform control (manual),
   - 2-both camera and platform are available.

7. (Optional) Manually control the barrier: `PUT /ISAPI/Parking/channels/<channelID>/barrierGate`.

11 API Reference

11.1 设备通用

11.1.1 监控点信息

11.1.1.1 Get camera information

Request URL

GET /ISAPI/Traffic/channels/<channelID>/basic

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<BasicInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, use CRIndex; otherwise, use countryIndex; if there is only countryIndex in the message and countryIndex is "253", additional logical processing is needed to enable CRIndex, attr:version{req, string, protocolVersion}-->
  <channelID>
    <!--ro, req, string, channel No.-->test
  </channelID>
  <directionNo>
    <!--ro, req, enum, "Upward,downward,bidirectional,westward,northward,eastward,southward", subType:string, desc:"Upward, downward, bidirectional, westward, northward, eastward, southward"-->Upward
  </directionNo>
  <monitoringSiteID>
    <!--ro, req, string, camera No.-->test
  </monitoringSiteID>
  <deviceID>
    <!--ro, req, string, device ID-->test
  </deviceID>
  <monitorDescription>
    <!--ro, req, string-->test
  </monitorDescription>
  <MonitorInfoList>
    <!--ro, opt, array, subType:object-->
    <MonitorInfo>
      <!--ro, opt, object, camera information-->
      <information>
        <!--ro, req, string-->test
      </information>
    </MonitorInfo>
  </MonitorInfoList>
  <defaultCHN>
    <!--ro, req, string-->test
  </defaultCHN>
  <region>
    <!--ro, req, enum, region, subType:string, desc:"AFandAM" (Africa and America), "APAC" (Asia-Pacific Region), "All" (all regions), "ER" (Russian Region), "EU" (Europe Region), "EUandCIS" (Europe and Russia), "HKandMO" (Hong Kong China and Macao China), "ME" (Middle East), "THAandLA" (Thailand and Laos), "other" (other)-->AFandAM
  </region>
  <countryIndex>
    <!--ro, opt, int, country/region index, desc:country/region index-->1
  </countryIndex>
  <CRIndex>
    <!--ro, req, enum, country/region, subType:int, desc:0 (Unsupported. It indicates that the algorithm library does not support recognizing license plates of this country), 1 (Czech Republic), 10 (Moldova), 100 (Reserved), 101 (Reserved), 102 (Reserved), 103 (Reserved), 104 (Egypt), 105 (Libya), 106 (Sudan), 107 (Tunisia), 108 (Algeria), 109 (Morocco), 11 (Russia), 110 (Ethiopia), 111 (Eritrea), 112 (Somalia Democratic), 113 (Djibouti), 114 (Kenya), 115 (Tanzania), 116 (Uganda), 117 (Rwanda), 118 (Burundi), 119 (Seychelles), 12 (Ukraine), 120 (Chad), 121 (Central African), 122 (Cameroon), 123 (Equatorial Guinea), 124 (Gabon), 125 (Republic of the Congo (Congo-Brazzaville)), 126 (Democratic Republic of the Congo (Congo-Kinshasa)), 127 (São Tomé and Príncipe), 128 (Mauritania), 129 (Western Sahara (Saharaui)), 13 (Belgium), 130 (Senegal), 131 (Gambia), 132 (Mali), 133 (Burkina Faso), 134 (Guinea), 135 (Guinea-Bissau), 136 (Cape Verde), 137 (Sierra Leone), 138 (Liberia), 139 (Ivory Coast), 14 (Bulgaria), 140 (Ghana), 141 (Togo), 142 (Benin), 143 (Niger), 144 (Zambia), 145 (Angola), 146 (Zimbabwe), 147 (Malawi), 148 (Mozambique), 149 (Botswana), 15 (Denmark), 150 (Namibia), 151 (South Africa), 152 (Swaziland), 153 (Lesotho), 154 (Madagascar), 155 (Comoros), 156 (Mauritius), 157 (Nigeria), 158 (South Sudan), 159 (Saint Helena), 16 (Finland), 160 (Mayotte), 161 (Reunion), 162 (Canary Islands), 163 (AZORES), 164 (Madeira), 165 (Reserved), 166 (Reserved), 167 (Reserved), 168 (Reserved), 169 (Canada), 17 (United Kingdom Great Britain), 170 (Greenland), 171 (St. Pierre and Miquelon), 172 (United State), 173 (Bermuda), 174 (Mexico), 175 (Guatemala), 176 (Belize), 177 (El Salvador), 178 (Honduras), 179 (Nicaragua), 18 (Greece), 180 (Costa Rica), 181 (Panama), 182 (Bahamas), 183 (Turks and  Caicos Islands), 184 (Cuba), 185 (Jamaica), 186 (Cayman Islands), 187 (Haiti), 188 (Dominican), 189 (Puerto Rico), 19 (Croatia), 190 (United States Virgin Islands), 191 (British Virgin Islands), 192 (Anguilla), 193 (Antigua and Barbuda), 194 (Collectivité de Saint-Martin), 195 (Autonomous country), 196 (Saint-Barthélemy), 197 (Saint Kitts and Nevis), 198 (Montserrat), 199 (Guadeloupe), 2 (France), 20 (Hungary), 200 (Dominica), 201 (Martinique), 202 (St. Lucia), 203 (Saint Vincent and the Grenadines), 204 (Grenada), 205 (Barbados), 206 (Trinidad and Tobago), 207 (Curaçao), 208 (Aruba), 209 (Netherlands Antilles), 21 (Israel), 210 (Colombia), 211 (Venezuela), 212 (Guyana), 213 (Suriname), 214 (French Guiana), 215 (Ecuador), 216 (Peru), 217 (Bolivia), 218 (Paraguay), 219 (Chile), 22 (Luxembourg), 220 (Brazil), 221 (Uruguay), 222 (Argentina), 223 (Reserved), 224 (Reserved), 225 (Reserved), 226 (Reserved), 227 (Australia), 228 (New Zealand ), 229 (Papua New Guinea), 23 (Macedonia, North Macedonia after 2018), 230 (Solomon Islands), 231 (Vanuatu), 232 (New Caledonia), 233 (Palau), 234 (Federated States of Micronesia), 235 (Marshall Island), 236 (Northern Mariana Islands), 237 (Guam), 238 (Nauru), 239 (Kiribati), 24 (Norway), 240 (Fiji Islands), 241 (Tonga), 242 (Tuvalu), 243 (Wallis et Futuna), 244 (Samoa), 245 (Eastern Samoa), 246 (Tokelau), 247 (Niue), 248 (Cook Islands), 249 (French Polynesia), 25 (Portugal), 250 (Pitcairn Islands), 251 (Hawaii State), 252 (Reserved), 253 (invalid), 254 (Unrecognized), 255 (All), 26 (Romania), 27 (Serbia), 28 (Azerbaijan), 29 (Georgia), 3 (Germany), 30 (Kazakhstan), 31 (Lithuania), 32 (Turkmenistan), 33 (Uzbekistan), 34 (Latvia), 35 (Estonia), 36 (Albania), 37 (Austria), 38 (Bosnia and Herzegovina), 39 (Ireland), 4 (Spain), 40 (Iceland), 41 (Vatican), 42 (Malta), 43 (Sweden), 44 (Switzerland), 45 (Cyprus), 46 (Turkey), 47 (Slovenia), 48 (Montenegro), 49 (Kosovo), 5 (Italy), 50 (Andorra), 51 (Armenia), 52 (Monaco), 53 (Liechtenstein), 54 (San Marino), 55 (Reserved), 56 (Reserved), 57 (Reserved), 58 (Reserved), 59 (China), 6 (Netherlands), 60 (Bahrain), 61 (South Korea), 62 (Lebanon), 63 (Nepal), 64 (Thailand), 65 (Pakistan), 66 (United Arab Emirates), 67 (Bhutan), 68 (Oman), 69 (North Korea), 7 (Poland), 70 (Philippines), 71 (Cambodia), 72 (Qatar), 73 (Kyrgyzstan), 74 (Maldives), 75 (Malaysia), 76 (Mongolia), 77 (Saudi Arabia), 78 (Brunei), 79 (Laos), 8 (Slovakia), 80 (Japan), 81 (Turkey), 82 (Palestinian), 83 (Tajikistan), 84 (Kuwait), 85 (Syria), 86 (India), 87 (Indonesia), 88 (Afghanistan), 89 (Sri Lanka), 9 (Belarus), 90 (Iraq), 91 (Vietnam), 92 (Iran), 93 (Yemen), 94 (Jordan), 95 (Burma/Myanmar), 96 (Sikkim), 97 (Bangladesh), 98 (Singapore), 99 (Democratic Republic of Timor-Leste)-->0
  </CRIndex>
  <radarTrack>
    <!--ro, opt, bool, whether to enable the radar pattern-->true
  </radarTrack>
  <closeDistanceCapture>
    <!--ro, opt, bool, whether to enable close recapture-->true
  </closeDistanceCapture>
  <continuousSnapType>
    <!--ro, opt, enum, burst type, subType:string, desc:"time"-->time
  </continuousSnapType>
  <intervalTime>
    <!--ro, opt, int, unit: ms. This node is valid only when <continuousSnapType> is "time", unit:ms, dep:and,{$.BasicInfo.continuousSnapType,eq,time}, desc:This node is valid only when <continuousSnapType> is "time"-->1
  </intervalTime>
  <applicationScene>
    <!--ro, opt, enum, subType:int-->1
  </applicationScene>
</BasicInfo>
```

11.1.1.2 Set basic parameters for the camera of a specific channel

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/basic

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BasicInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, basic parameters of a camera, attr:version{req, string, protocolVersion}-->
  <channelID>
    <!--req, string, channel No.-->test
  </channelID>
  <directionNo>
    <!--req, enum, "Upward,downward,bidirectional,westward,northward,eastward,southward", subType:string, desc:"Upward", "downward", "bidirectional",
"westward", "northward", "eastward", "southward"-->Upward
  </directionNo>
  <monitoringSiteID>
    <!--req, string, camera No.-->test
  </monitoringSiteID>
  <deviceID>
    <!--req, string, device ID-->test
  </deviceID>
  <monitorDescription>
    <!--req, string-->test
  </monitorDescription>
  <MonitorInfoList>
    <!--opt, array, subType:object-->
    <MonitorInfo>
      <!--opt, object, list-->
      <information>
        <!--req, string-->test
      </information>
    </MonitorInfo>
  </MonitorInfoList>
  <defaultCHN>
    <!--req, string-->test
  </defaultCHN>
  <region>
    <!--req, enum, region, subType:string, desc:"AFandAM" (Africa and America), "APAC" (Asia-Pacific Region), "All" (all regions), "ER" (Russian Region),
"EU" (Europe Region), "EUandCIS" (Europe and Russia), "HKandMO" (Hong Kong China and Macao China), "ME" (Middle East), "THAandLA" (Thailand and Laos),
"other" (other)-->AFandAM
  </region>
  <countryIndex>
    <!--opt, int, country index, desc:"0"-generic-->1
  </countryIndex>
  <CRIndex>
    <!--req, enum, country/region, subType:int, desc:country or region index,which contains countryIndex; when CRIndex is "0",it represents generic-->0
  </CRIndex>
  <radarTrack>
    <!--opt, bool, whether to enable the radar pattern-->true
  </radarTrack>
  <closeDistanceCapture>
    <!--opt, bool, whether to enable close recapture-->true
  </closeDistanceCapture>
  <continuousSnapType>
    <!--opt, enum, burst type, subType:string, desc:"time"-->time
  </continuousSnapType>
  <intervalTime>
    <!--opt, int, unit: ms. This node is valid only when <continuousSnapType> is "time", unit:ms, dep:and,{$.BasicInfo.continuousSnapType,eq,time},
desc:this node is valid only when <continuousSnapType> is "time"-->1
  </intervalTime>
  <applicationScene>
    <!--opt, enum, subType:int-->1
  </applicationScene>
</BasicInfo>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
</ResponseStatus>
```

11.1.1.3 Get the basic configuration capability

Request URL

GET /ISAPI/Traffic/channels/<channelID>/basic/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<BasicInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <channelID>
    <!--ro, req, string-->test
  </channelID>
  <directionNo opt="Upward,downward,bidirectional,westward,northward,eastward,southward">
    <!--ro, req, enum, subType:string, attr:opt{req, string}-->Upward
  </directionNo>
  <monitoringSiteID min="1" max="10">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test
  </monitoringSiteID>
  <deviceID min="1" max="10">
    <!--ro, req, string, device ID, attr:min{req, int},max{req, int}-->test
  </deviceID>
  <monitorDescription min="1" max="10">
    <!--ro, req, string, attr:min{req, int},max{req, int}-->test
  </monitorDescription>
  <MonitorInfoList size="5">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <MonitorInfo>
      <!--ro, opt, object-->
      <information>
        <!--ro, req, string-->test
      </information>
    </MonitorInfo>
  </MonitorInfoList>
  <defaultCHN min="1" max="10">
    <!--ro, req, string, abbreviation of the province supported by license plate recognition, attr:min{req, int},max{req, int}-->test
  </defaultCHN>
  <region opt="default,EU,CIS,EU_CIS,ME">
    <!--ro, req, enum, region, subType:string, attr:opt{req, string}, desc:region: "default"-general,"EU"-Europe,"CIS"-Russia,"EU_CIS"-Europe and
Russia,"ME"-the Middle East-->AFandAM
  </region>
  <countryIndex
opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,25,26,28,29,30,31,33,34,35,36,37,38,39,40,42,43,44,45,46,47,49,51,53,55,59,60,62,65,68,70
,71,72,73,76,77,84,87,89,90,94,95,91,104,107,108,110,114,115,130,137,139,140,144,145,146,151,156,157,169,174,175,177,180,181,188,206,210,215,216,217,218,219
,220,221,222,227,228">
    <!--ro, opt, int, country/region index, attr:opt{req, string}, desc:country or region index: 0-general,1-Czech Republic,2-France,3-Germany,4-Spain,5-
Italy,6-Netherlands,7-Poland,8-Slovakia,11-Russia,12-Ukraine,14-Bulgaria,17-United Kingdom,18-Greece,19-Croatia,20-Hungary,21-Israel (Asia),23-Republic of
Macedonia,30-Kazakhstan (Asia),33-Uzbekistan (Asia),37-Austria,39-Ireland,44-Switzerland,46-Turkey,55-Oran,73-Kyrgyzstan,76-Mongolia-->1
  </countryIndex>
  <CRIndex
opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,25,26,28,29,30,31,33,34,35,36,37,38,39,40,42,43,44,45,46,47,49,51,53,55,59,60,62,65,68,70
,71,72,73,76,77,84,87,89,90,94,95,91,104,107,108,110,114,115,130,137,139,140,144,145,146,151,156,157,169,174,175,177,180,181,188,206,210,215,216,217,218,219
,220,221,222,227,228,256">
    <!--ro, req, enum, country/region, subType:int, attr:opt{req, string}, desc:country or region index,which contains <countryIndex> and adds regions; when
<CRIndex> is "0,it indicates generic; <CRIndex> is preferred over <countryIndex>-->0
  </CRIndex>
  <defaultCHN>
    <!--ro, req, string, abbreviation of the province supported by license plate recognition-->test
  </defaultCHN>
  <radarTrack>
    <!--ro, opt, bool, whether to enable the radar pattern-->true
  </radarTrack>
  <closeDistanceCapture>
    <!--ro, opt, bool, whether to enable close recapture-->true
  </closeDistanceCapture>
  <continuousSnapType opt="time">
    <!--ro, opt, enum, burst type, subType:string, attr:opt{req, string}, desc:"time"-->time
  </continuousSnapType>
  <intervalTime min="1" max="10">
    <!--ro, opt, int, unit: ms. This node is valid only when <continuousSnapType> is "time", unit:ms, dep:and,{$.BasicInfo.continuousSnapType,eq,time},
attr:min{req, int},max{req, int}, desc:unit: ms. This node is valid only when <continuousSnapType> is "time"-->1
  </intervalTime>
  <applicationScene opt="0,1" def="0">
    <!--ro, opt, enum, subType:int, attr:opt{req, string},def{req, string}-->1
  </applicationScene>
</BasicInfo>
```


11.1.2 信息管理

11.1.2.1 Get configuration capability of the device information parameters

Request URL

GET /ISAPI/System/deviceInfo/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName min="0" max="132">
    <!--ro, req, string, device name, range:[1,132], attr:min{opt, string},max{opt, string}-->test
  </deviceName>
  <deviceID min="1" max="128">
    <!--ro, opt, string, device ID, range:[1,128], attr:min{opt, string},max{opt, string}-->123456
  </deviceID>
  <deviceDescription min="1" max="128">
    <!--ro, opt, string, device description, range:[1,128], attr:min{opt, string},max{opt, string}-->test
  </deviceDescription>
  <deviceLocation opt="STD-CGI">
    <!--ro, opt, string, device location, range:[1,32], attr:opt{opt, string}-->hangzhou
  </deviceLocation>
  <systemContact opt="STD-CGI">
    <!--ro, opt, string, manufacturer, range:[1,32], attr:opt{opt, string}-->test
  </systemContact>
  <model min="1" max="64">
    <!--ro, opt, string, device model, range:[1,64], attr:min{opt, string},max{opt, string}-->iDS-9632NX-I8/X
  </model>
  <serialNumber min="1" max="48">
    <!--ro, opt, string, device serial No., range:[1,48], attr:min{opt, string},max{opt, string}-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress min="1" max="64">
    <!--ro, opt, string, MAC address, range:[1,64], attr:min{opt, string},max{opt, string}-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion min="1" max="64">
    <!--ro, opt, string, device firmware version, range:[1,64], attr:min{opt, string},max{opt, string}-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate min="1" max="64">
    <!--ro, opt, string, release date of the device firmware version, range:[1,64], attr:min{opt, string},max{opt, string}-->build 181129
  </firmwareReleasedDate>
  <encoderVersion min="1" max="64">
    <!--ro, opt, string, device description, range:[1,64], attr:min{opt, string},max{opt, string}-->V7.3
  </encoderVersion>
  <encoderReleasedDate min="1" max="64">
    <!--ro, opt, string, release date, range:[1,64], attr:min{opt, string},max{opt, string}-->build 20161223
  </encoderReleasedDate>
  <bootVersion min="1" max="16">
    <!--ro, opt, string, boot version, range:[1,16], attr:min{opt, string},max{opt, string}-->V1.3.4
  </bootVersion>
  <bootReleasedDate min="1" max="16">
    <!--ro, opt, string, release date of boot, range:[1,16], attr:min{opt, string},max{opt, string}-->100316
  </bootReleasedDate>
  <hardwareVersion min="1" max="24">
    <!--ro, opt, string, hardware version, range:[1,24], attr:min{opt, string},max{opt, string}-->0x0
  </hardwareVersion>
  <telecontrolID min="1" max="255">
    <!--ro, opt, int, remote control ID, range:[1,255], attr:min{opt, string},max{opt, string}-->8
  </telecontrolID>
  <subChannelEnabled>
    <!--ro, opt, bool, whether to enable two channels-->true
  </subChannelEnabled>
  <thrChannelEnabled>
    <!--ro, opt, bool, whether to enable three channels-->true
  </thrChannelEnabled>
  <isSupportPipeID>
    <!--ro, opt, bool, whether building pipeline (PIPEiD), desc:GET /ISAPI/System/deviceInfo/PipeID?format=json-->true
  </isSupportPipeID>
</DeviceInfo>
```


11.1.2.2 Get device information parameters

Request URL

GET /ISAPI/System/deviceInfo?serialNumber=<serialNumber>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	--
controllerID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--ro, req, string, device name, range:[0,132]-->test
  </deviceName>
  <deviceID>
    <!--ro, req, string, device No., range:[1,128]-->test
  </deviceID>
  <deviceDescription>
    <!--ro, opt, string, device description, range:[1,128]-->test
  </deviceDescription>
  <deviceLocation>
    <!--ro, opt, string, device location, range:[1,128]-->hangzhou
  </deviceLocation>
  <systemContact>
    <!--ro, opt, string, manufacturer, range:[1,32]-->STD-CGI
  </systemContact>
  <model>
    <!--ro, req, string, device model, range:[1,64]-->iDS-9632NX-I8/X
  </model>
  <serialNumber>
    <!--ro, req, string, device serial No., range:[1,48]-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress>
    <!--ro, req, string, MAC address, range:[1,64]-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion>
    <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate>
    <!--ro, opt, string, release date of the device firmware version-->2019-11-01
  </firmwareReleasedDate>
  <encoderVersion>
    <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
  </encoderVersion>
  <encoderReleasedDate>
    <!--ro, opt, string, release date of the device encoder version-->2019-11-02
  </encoderReleasedDate>
  <bootVersion>
    <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
  </bootVersion>
  <bootReleasedDate>
    <!--ro, opt, string, release date of boot-->2019-11-03
  </bootReleasedDate>
  <hardwareVersion>
    <!--ro, opt, string, hardware version, range:[1,24]-->0x0
  </hardwareVersion>
  <subDeviceType>
    <!--ro, opt, enum, device sub type, subType:string, desc:"accessControlTerminal" (access control terminal; it is valid when the value of deviceType is
"ACS"), "attendanceCheckDevice" (attendance check devices; it is valid when the value of deviceType is "ACS"), "multiChannelAccessController" (multi-channel
access controller; it is valid when the value of deviceType is "ACS"), "personAndIdCardDevice" (person and ID card device; it is valid when the value of
deviceType is "ACS"), "doorStation" (door station; it is valid when the value of deviceType is "VIS"), "indoor" (indoor station; it is valid when the value
of deviceType is "VIS"), "mainStation" (main station; it is valid when the value of deviceType is "VIS"), "dropGate" (swing barrier; it is valid when the
value of deviceType is "PersonnelChannel"), "wingGate" (flap barrier; it is valid when the value of deviceType is "PersonnelChannel"), "threeRollerGate"
(tripod turnstile; it is valid when the value of deviceType is "PersonnelChannel")-->accessControlTerminal
  </subDeviceType>
  <telecontrolID>
    <!--ro, opt, int, remote control ID, range:[1,255]-->1
  </telecontrolID>
  <supportBeep>
    <!--ro, opt, bool, whether it supports buzzer-->true
  </supportBeep>
  <supportVideoLoss>
    <!--ro, opt, bool, whether it supports video loss-->true
  </supportVideoLoss>
  <customizedInfo>
    <!--ro, opt, string, customized project No., range:[1,32], desc:no value will be returned if it is baseline device. Custom project No. will be returned
if is custom device-->test
  </customizedInfo>
  <subSerialNumber>
    <!--ro, opt, string, sub serial No.-->test
  </subSerialNumber>
</DeviceInfo>
```


Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
DeviceInfo	[Message content]	application/xml	--	--	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

11.1.2.3 Set device information parameters

Request URL

PUT /ISAPI/System/deviceInfo?serialNumber=<serialNumber>

Query Parameter

Parameter Name	Parameter Type	Description
serialNumber	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<DeviceInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device information, attr:version{opt, string, protocolVersion}-->
  <deviceName>
    <!--req, string, device name, range:[1,132]-->test
  </deviceName>
  <deviceID>
    <!--opt, string, device No., range:[1,128]-->test
  </deviceID>
  <deviceDescription>
    <!--opt, string, device description, range:[1,128]-->test
  </deviceDescription>
  <deviceLocation>
    <!--opt, string, device location, range:[1,128]-->hangzhou
  </deviceLocation>
  <systemContact>
    <!--ro, opt, string, manufacturer, range:[1,32]-->STD-CGI
  </systemContact>
  <model>
    <!--ro, req, string, device model, range:[1,64]-->iDS-9632NX-I8/X
  </model>
  <serialNumber>
    <!--ro, req, string, device serial No., range:[1,48]-->iDS-9632NX-I8/X1620181209CCRRC77605411WCVU
  </serialNumber>
  <macAddress>
    <!--ro, req, string, MAC Address, range:[1,64]-->44:47:cc:c8:d9:e4
  </macAddress>
  <firmwareVersion>
    <!--ro, req, string, device firmware version, range:[1,64]-->V4.1.40
  </firmwareVersion>
  <firmwareReleasedDate>
    <!--ro, opt, string, release date of the device firmware version-->2019-11-01
  </firmwareReleasedDate>
  <encoderVersion>
    <!--ro, opt, string, device encoder version No., range:[1,32]-->V7.3
  </encoderVersion>
  <encoderReleasedDate>
    <!--ro, opt, string, release date of the device encoder version-->2019-11-02
  </encoderReleasedDate>
  <bootVersion>
    <!--ro, opt, string, boot version, range:[1,16]-->V1.3.4
  </bootVersion>
  <bootReleasedDate>
    <!--ro, opt, string, release date of boot-->2019-11-03
  </bootReleasedDate>
  <hardwareVersion>
    <!--ro, opt, string, hardware version, range:[1,16]-->0x0
  </hardwareVersion>
  <telecontrolID>
    <!--opt, int, remote control ID, range:[1,255]-->1
  </telecontrolID>
  <supportBeep>
    <!--ro, opt, bool, whether to support buzzer-->true
  </supportBeep>
  <supportVideoLoss>
    <!--ro, opt, bool, whether the device supports video loss-->true
  </supportVideoLoss>
</DeviceInfo>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```


11.1.2.4 Get firmware code list

Request URL

GET /ISAPI/System/firmwareCodeV2?startIndex=<startIndex>&maxNumber=<maxNumber>

Query Parameter

Parameter Name	Parameter Type	Description
startIndex	string	To control the message size, the quantity of returned firmware IDs returned by device depends on the client request. For example, if there are 100 firmware IDs of a device, and up to 32 IDs are allowed in every request, then the device returns only 32 IDs according to the start index (startIndex). Examples: GET /ISAPI/System/firmwareCodeV2?startIndex=1&maxNumber=32
maxNumber	string	To control the message size, the quantity of returned firmware IDs returned by device depends on the client request. For example, if there are 100 firmware IDs of a device, and up to 32 IDs are allowed in every request, then the device returns only 32 IDs according to the start index (startIndex). Examples: GET /ISAPI/System/firmwareCodeV2?startIndex=1&maxNumber=32

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<FirmwareCodeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, firmware ID list, subType:object, attr:version{req, string, protocolVersion}-->
  <FirmwareCode>
    <!--ro, opt, object, firmware ID-->
    <index>
      <!--ro, req, int, index, desc:it starts from 1-->1
    </index>
    <code>
      <!--ro, req, string, firmware ID, desc:N/A-->0000000200000100000000010000000100000001000000000000000001ffffffff0001020307e1060100001000
    </code>
    <version>
      <!--ro, req, string, version-->V5.5.160 build 210416
    </version>
    <channel>
      <!--ro, opt, int, channel No., desc:this node is for IP channel-->1
    </channel>
    <devTypeName>
      <!--ro, opt, string, device model name which IP channel corresponds to, desc:device model name which IP channel corresponds to-->test
    </devTypeName>
    <moduleType>
      <!--ro, opt, enum, device component type, subType:string, desc:"cardReader" (card reader), "FPModule" (fingerprint module), "securityModule" (security
module), "extendModule" (I/O extended module), "channelController" (lane controller), "IRModule" (infrared module), "lampModule" (indicator module),
"elevatorController" (sub elevator controller), "FPAlgorithmProgram" (fingerprint algorithm programma of card reader), "dockStationHost" (dock station
host), "dockStationScreen" (dock station screen)-->cardReader
    </moduleType>
    <moduleID>
      <!--ro, opt, int, device component No., desc:it depends on the value of moduleType-->1
    </moduleID>
  </FirmwareCode>
</FirmwareCodeList>
```

11.1.3 IO输出配置

11.1.3.1 Set the parameters of synchronous signal output

Request URL

PUT /ISAPI/System/syncSignalOutput/<alarmOutID>

Query Parameter

Parameter Name	Parameter Type	Description
alarmOutID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<SyncSignalOutputList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SyncSignalOutput>
    <!--opt, object, the parameters of synchronous signal output-->
    <IOWorkMode>
      <!--req, enum, IO output mode, subType:string, desc:"flashLight" (strobe light control mode), "polarizer" (polarizer control mode), "continuousLight"
(solid light control mode), "flashLightSwitch" (strobe light switching mode (used to switch between the strobe supplement light mode and IR mode)). When
this node is set to "polarizer", only <outputStatus>, <detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>, <startMinute>,
<endHour>, and <endMinute> are valid, and the <outputStatus> can only be set to "high" or "low". When this node is set to "continuousLight", only
<detectBrightnessEnable>, <brightnessThreshold>, <flashEnabled>, <startHour>, <startMinute>, <endHour>, and <endMinute> are valid-->flashLight
    </IOWorkMode>
    <id>
      <!--req, int, index, range:[1,8]-->0
    </id>
    <defaultStatus>
      <!--req, enum, IO default status, subType:string, desc:"high" (high electrical level), "low" (low electrical level)-->high
    </defaultStatus>
    <outputStatus>
      <!--req, enum, IO effective status, subType:string, desc:"high", "low", "pulse"-->high
    </outputStatus>
    <aheadTime>
      <!--req, int, IO pre-output time, unit: microsecond, range:[0,10]-->0
    </aheadTime>
    <timeDelay>
      <!--req, int, IO effective duration, unit: microsecond, range:[0,10]-->0
    </timeDelay>
    <freqMultiplyulti>
      <!--req, int, frequency multiplication, range:[1,15]-->0
    </freqMultiplyulti>
    <dutyRate>
      <!--req, int, duty ratio, range:[0,40]-->0
    </dutyRate>
    <postFlashEnable>
      <!--req, bool, checkpoint output-->true
    </postFlashEnable>
```

```
    </postFlashEnable>
    <illegalFlashEnable>
      <!--req, bool, violation output-->true
    </illegalFlashEnable>
    <videoFlashEnable>
      <!--req, bool, video output-->true
    </videoFlashEnable>
    <detectBrightnessEnable>
      <!--req, bool, whether to enable flash light for automatic brightness detection-->true
    </detectBrightnessEnable>
    <brightnessThreshold>
      <!--opt, int, brightness threshold of the enabled flash light, range:[0,100]-->0
    </brightnessThreshold>
    <flashEnabled>
      <!--req, bool, whether to enable flash light-->true
    </flashEnabled>
    <startHour>
      <!--opt, int, start time in hour, range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--opt, int, start time in minute, range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--opt, int, end time in hour, range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--opt, int, end time in minute, range:[0,59]-->0
    </endMinute>
    <plateBrightness>
      <!--req, bool, whether to enable flash light by license plate brightness-->true
    </plateBrightness>
    <incrBrightEnable>
      <!--opt, bool, whether to enable brightness enhancement mode (for solid light mode)-->true
    </incrBrightEnable>
    <incrBrightTime>
      <!--req, int, brightness enhancement duration. This node is valid only when <incrBrightEnable> is "true", range:[0,10000], unit:ms-->0
    </incrBrightTime>
    <incrBrightPercent>
      <!--req, int, percentage of brightness enhancement, range:[0,100]-->0
    </incrBrightPercent>
    <brightness>
      <!--opt, int, solid light brightness (for solid light mode), range:[0,100]-->0
    </brightness>
    <delayCaptureTime>
      <!--opt, int, delayed capture time, range:[1,1000], unit:ms-->1
    </delayCaptureTime>
    <manualBrightnessEnable>
      <!--req, bool, whether to enable adjusting brightness manually-->true
    </manualBrightnessEnable>
    <irLightBrightness>
      <!--opt, int, brightness of IR supplement light, range:[0,100]-->0
    </irLightBrightness>
    <whiteLightBrightness>
      <!--opt, int, brightness limit of white supplement light, range:[0,100]-->0
    </whiteLightBrightness>
    <fanModeEnable>
      <!--req, bool-->true
    </fanModeEnable>
  </SyncSignalOutput>
</SyncSignalOutputList>
```


Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.1.3.2 Get the parameters of synchronous signal output

Request URL

GET /ISAPI/System/syncSignalOutput/<alarmOutID>

Query Parameter

Parameter Name	Parameter Type	Description
alarmOutID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<SyncSignalOutputList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <SyncSignalOutput>
    <!--ro, opt, object-->
    <IOWorkMode>
      <!--ro, req, enum, IO output mode, subType:string, desc:IO output mode-->flashLight
    </IOWorkMode>
    <id>
      <!--ro, req, int, ID, range:[1,8]-->0
    </id>
    <defaultStatus>
      <!--ro, req, enum, default I/O status, subType:string, desc:"high" (high level), "low" (low level)-->high
    </defaultStatus>
    <outputStatus>
      <!--ro, req, enum, IO effective status, subType:string, desc:"high, low, pulse"-->high
    </outputStatus>
    <aheadTime>
      <!--ro, req, int, IO pre-output time, range:[0,10]-->0
    </aheadTime>
    <timeDelay>
      <!--ro, req, int, IO effective duration, range:[0,10]-->0
    </timeDelay>
    <freqMultiplyulti>
      <!--ro, req, int, frequency multiplication, range:[1,15]-->0
    </freqMultiplyulti>
    <dutyRate>
      <!--ro, req, int, duty ratio, range:[0,40]-->0
    </dutyRate>
    <postFlashEnable>
      <!--ro, req, bool, checkpoint output-->true
    </postFlashEnable>
    <illegalFlashEnable>
      <!--ro, req, bool, violation output-->true
    </illegalFlashEnable>
    <videoFlashEnable>
      <!--ro, req, bool, video output-->true
    </videoFlashEnable>
    <detectBrightnessEnable>
      <!--ro, req, bool, whether to enable flash light for automatic brightness detection-->true
    </detectBrightnessEnable>
    <brightnessThreshold>
      <!--ro, opt, int, brightness threshold of the enabled flash light, range:[0,100]-->0
    </brightnessThreshold>
    <flashEnabled>
      <!--ro, req, bool, whether to enable flash light-->true
    </flashEnabled>
    <startHour>
      <!--ro, opt, int, start time in hour, range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--ro, opt, int, start time in minute, range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--ro, opt, int, end time in hour, range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--ro, opt, int, end time in minute, range:[0,59]-->0
    </endMinute>
    <plateBrightness>
      <!--ro, req, bool, whether to enable flash light by license plate brightness-->true
    </plateBrightness>
    <incrBrightEnable>
      <!--ro, opt, bool, whether to enable brightness enhancement mode (for solid light mode)-->true
    </incrBrightEnable>
    <incrBrightTime>
      <!--ro, req, int, brightness enhancement duration, range:[0,10000], unit:ms-->0
    </incrBrightTime>
    <incrBrightPercent>
      <!--ro, req, int, percentage of brightness enhancement, range:[0,100]-->0
    </incrBrightPercent>
    <brightness>
      <!--ro, opt, int, solid light brightness (for solid light mode), range:[0,100]-->0
    </brightness>
    <delayCaptureTime>
      <!--ro, opt, int, delayed capture time, range:[1,1000], unit:ms-->1
    </delayCaptureTime>
```

```xml
    <manualBrightnessEnable>
      <!--ro, req, bool, whether to enable adjusting brightness manually-->true
    </manualBrightnessEnable>
    <irLightBrightness>
      <!--ro, opt, int, brightness of IR supplement light, range:[0,100]-->0
    </irLightBrightness>
    <whiteLightBrightness>
      <!--ro, opt, int, range:[0,100]-->0
    </whiteLightBrightness>
    <fanModeEnable>
      <!--ro, req, bool-->true
    </fanModeEnable>
  </SyncSignalOutput>
</SyncSignalOutputList>
```

11.1.3.3 Get the configuration capability of synchronous signal output

Request URL

GET /ISAPI/System/syncSignalOutput/<alarmOutID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
alarmOutID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<SyncSignalOutputList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <IOOutNum min="3" max="8">
    <!--ro, req, string, number of IO outputs, attr:min{req, int},max{req, int}-->test
  </IOOutNum>
  <SyncSignalOutput>
    <!--ro, opt, object-->
    <id min="1" max="8">
      <!--ro, req, int, ID, attr:min{req, int},max{req, int}-->0
    </id>
    <IOWorkMode opt="flashLight,polarizer,continuousLight,flashLightSwitch,stvobeLight,burstFlashLight,fan">
      <!--ro, req, enum, IO output mode, subType:string, attr:opt{req, string}, desc:IO output mode-->fan
    </IOWorkMode>
    <defaultStatus opt="hige,low">
      <!--ro, req, enum, default I/O status, subType:string, attr:opt{req, string}, desc:"high” (high level), "low” (low level)-->high
    </defaultStatus>
    <outputStatus opt="high,low,pulse">
      <!--ro, req, enum, default status, subType:string, attr:opt{req, string}, desc:default status-->high
    </outputStatus>
    <aheadTime min="0" max="10">
      <!--ro, req, int, pre-output time, attr:min{req, int},max{req, int}-->0
    </aheadTime>
    <timeDelay min="0" max="10">
      <!--ro, req, int, output duration, attr:min{req, int},max{req, int}-->0
    </timeDelay>
    <freqMultiplyulti min="1" max="15">
      <!--ro, req, int, frequency multiplication, attr:min{req, int},max{req, int}-->0
    </freqMultiplyulti>
    <dutyRate min="0" max="40">
      <!--ro, req, int, duty ratio, attr:min{req, int},max{req, int}-->0
    </dutyRate>
    <postFlashEnable>
      <!--ro, req, bool-->true
    </postFlashEnable>
    <illegalFlashEnable>
      <!--ro, req, bool-->true
    </illegalFlashEnable>
    <videoFlashEnable>
      <!--ro, req, bool-->true
    </videoFlashEnable>
    <detectBrightnessEnable>
      <!--ro, req, bool-->true
    </detectBrightnessEnable>
    <brightnessThreshold min="0" max="100">
      <!--ro, opt, int, brightness threshold, attr:min{req, int},max{req, int}-->0
    </brightnessThreshold>
    <flashEnabled>
      <!--ro, req, bool-->true
    </flashEnabled>
    <startHour min="0" max="10">
      <!--ro, opt, int, start time in hour, attr:min{req, int},max{req, int}-->0
    </startHour>
    <startMinute min="0" max="10">
      <!--ro, opt, int, start time in minute, attr:min{req, int},max{req, int}-->0
    </startMinute>
    <endHour min="0" max="10">
      <!--ro, opt, int, end time in hour, attr:min{req, int},max{req, int}-->0
    </endHour>
    <endMinute min="0" max="10">
      <!--ro, opt, int, end time in minute, attr:min{req, int},max{req, int}-->0
    </endMinute>
    <plateBrightness>
      <!--ro, req, bool-->true
    </plateBrightness>
    <incrBrightEnable>
      <!--ro, opt, bool, whether to enable brightness enhancement mode (for solid light mode)-->true
    </incrBrightEnable>
    <incrBrightTime min="0" max="10000">
      <!--ro, req, int, brightness enhancement duration, attr:min{req, int},max{req, int}-->0
    </incrBrightTime>
    <incrBrightPercent min="0" max="100">
      <!--ro, req, int, percentage of brightness enhancement, attr:min{req, int},max{req, int}-->0
    </incrBrightPercent>
    <brightness min="0" max="100">
      <!--ro, opt, int, solid light brightness (for solid light mode), attr:min{req, int},max{req, int}-->0
    </brightness>
    <delayCaptureTime min="1" max="1000">
      <!--ro, opt, int, delayed capture time, attr:min{req, int},max{req, int}-->0
    </delayCaptureTime>
    <manualBrightnessEnable>
      <!--ro, req, bool, whether to enable adjusting brightness manually-->true
    </manualBrightnessEnable>
  </SyncSignalOutput>
</SyncSignalOutputList>
```


11.1.4 系统维护

11.1.4.1 Get the storage capability of the device

Request URL

GET /ISAPI/ContentMgmt/capabilities

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<RacmCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device storage capability, attr:version{req, string, protocolVersion}-->
  <isSupportSMARTTest>
    <!--ro, opt, bool, whether the device supports HDD detection, desc:related URI: /ISAPI/ContentMgmt/Storage/hdd/SMARTTest/config-->true
  </isSupportSMARTTest>
  <isSupportMainAndSubRecord>
    <!--ro, opt, bool, whether the device supports the main stream and sub-stream recording-->true
  </isSupportMainAndSubRecord>
  <isSupportLogConfig>
    <!--ro, opt, bool, whether the device supports log configuration, desc:related URI: /ISAPI/ContentMgmt/logConfig/capabilities-->true
  </isSupportLogConfig>
  <isSupportPictureInfo>
    <!--ro, opt, bool, whether it supports picture search on the SD card of the capture camera-->true
  </isSupportPictureInfo>
  <isSupportLogSearch>
    <!--ro, opt, bool, whether it supports log search-->true
  </isSupportLogSearch>
  <isSupportRecordSearch>
    <!--ro, opt, bool, whether it supports video search-->true
  </isSupportRecordSearch>
</RacmCap>
```


11.1.4.2 Get the device status capability

Request URL

GET /ISAPI/ITC/ITCStatus/capabilities

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ITCStatusCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isSupportfortifyLinkStatus>
    <!--ro, req, bool-->true
  </isSupportfortifyLinkStatus>
  <isSupportDetectorStatus>
    <!--ro, req, bool-->true
  </isSupportDetectorStatus>
  <relatedTriggerInputs>
    <!--ro, req, int, number of linked triggering inputs-->1
  </relatedTriggerInputs>
  <relatedSyncOutputs>
    <!--ro, req, int, number of linked synchronous outputs-->1
  </relatedSyncOutputs>
</ITCStatusCap>
```


11.1.4.3 Get the live view status

Request URL

GET /ISAPI/ITC/ITCStatus/previewStatus

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<PreviewStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, live view status, attr:version{req, string, protocolVersion}-->
  <previewCount>
    <!--ro, req, int, number of channels for live view-->1
  </previewCount>
  <previewCfgList>
    <!--ro, opt, array, subType:object-->
    <previewCfg>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, int-->1
      </id>
      <previewIP>
        <!--ro, req, string, live view connection IP-->test
      </previewIP>
      <channelType>
        <!--ro, req, enum, channel type, subType:string, desc:"mainChannel", "subChannel", "noChannel"-->mainChannel
      </channelType>
      <frameRate>
        <!--ro, req, int, frame rate, desc:the value equals to the maximum frame rate *100. This node is set to 0 if there is no live view-->1
      </frameRate>
      <resolutionWidth>
        <!--ro, req, int, width of the video resolution-->1
      </resolutionWidth>
      <resolutionHeight>
        <!--ro, req, int, height of the video resolution-->1
      </resolutionHeight>
    </previewCfg>
  </previewCfgList>
  <snapResolutionWidth>
    <!--ro, req, int, width of the capture resolution-->1
  </snapResolutionWidth>
  <snapResolutionHeight>
    <!--ro, req, int, height of the capture resolution-->1
  </snapResolutionHeight>
  <triggerType>
    <!--ro, req, enum, trigger mode, subType:string, desc:trigger mode-->normal
  </triggerType>
</PreviewStatus>
```


11.1.4.4 Get the system security capability

Request URL

GET /ISAPI/Security/capabilities?username=<userName>&deviceIndex=<deviceIndex>

Query Parameter

Parameter Name	Parameter Type	Description
userName	string	user name
deviceIndex	string	For centralized splicing controllers connecting to LED sending cards, the URL needs to include the parameter deviceIndex= to specify LED sending card The value of deviceIndex is the value of "id" when "portType" is LED and can be obtained via GET /ISAPI/DisplayDev/Video/outputs/channels.

Request Message

None

Response Message

```xml
&lt;?xml version="1.0" encoding="UTF-8"?&gt;
&lt;SecurityCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0"&gt;
  &lt;!--ro, req, object, system security capability, attr:version{req, string, protocolVersion}--&gt;
  &lt;supportUserNums&gt;
    &lt;!--ro, opt, int, supported maximum number of users--&gt;0
  &lt;/supportUserNums&gt;
  &lt;userBondIpNums&gt;
    &lt;!--ro, opt, int, supported maximum number of IP addresses that can be bound--&gt;0
  &lt;/userBondIpNums&gt;
  &lt;userBondMacNums&gt;
    &lt;!--ro, opt, int, supported maximum number of MAC addresses that can be bound--&gt;0
  &lt;/userBondMacNums&gt;
  &lt;securityVersion opt="1,2,3,4"&gt;
    <!--ro, opt, string, encryption capability set, attr:opt{req, string}, desc:the encryption capability of each version consists of two parts: encryption algorithm and the range of encrypted nodes currently 1 refers to AES128 encryption and 2 refers to AES256 encryption, the range of encrypted nodes is described in each protocol-->1
  &lt;/securityVersion&gt;
  &lt;keyIterateNum&gt;
    <!--ro, opt, int, secret key iteration times, dep:or,{$.SecurityCap.securityVersion,eq,1},{$.SecurityCap.securityVersion,eq,2}, desc:this node depends on the node securityVersion, the range is between 100 and 1000-->100
  &lt;/keyIterateNum&gt;
  &lt;isSupportUserCheck&gt;
    <!--ro, opt, bool, whether the device supports verifying the login password when editing (editing/adding/deleting) user parameters, dep:or, {$.SecurityCap.securityVersion,eq,0},{$.SecurityCap.securityVersion,eq,1}, desc:it is an added capability, which indicates that whether supporting the login password verification for editing/adding/deleting user parameters, this node depends on the node securityVersion, which means that it is only valid for the versions that support encrypting the sensitive information-->true
  &lt;/isSupportUserCheck&gt;
  &lt;isSupportONVIFUserManagement&gt;
    &lt;!--ro, opt, bool, whether the device supports user management of Open Network Video Interface Protocol--&gt;true
  &lt;/isSupportONVIFUserManagement&gt;
  &lt;isSupportConfigFileImport&gt;
    &lt;!--ro, opt, bool, whether the device supports importing the configuration file, desc:true (support), this node is not returned (not support)--&gt;true
  &lt;/isSupportConfigFileImport&gt;
  &lt;isSupportConfigFileExport&gt;
    &lt;!--ro, opt, bool, whether the device supports exporting the configuration file, desc:true (support), this node is not returned (not support)--&gt;true
  &lt;/isSupportConfigFileExport&gt;
  &lt;cfgFileSecretKeyLenLimit min="0" max="16"&gt;
    &lt;!--ro, opt, string, length limit of the configuration file's verification key, attr:min{req, int},max{req, int}--&gt;1
  &lt;/cfgFileSecretKeyLenLimit&gt;
  &lt;isIrreversible&gt;
    <!--ro, opt, bool, whether the device supports irreversible password storage, desc:If this function is not supported, the plaintext password of the user information will be stored in the device; otherwise, the password will be hashed for storage in the device.-->true
  &lt;/isIrreversible&gt;
  &lt;salt&gt;
    &lt;!--ro, opt, string, the specific salt used by the user to log in--&gt;test
  &lt;/salt&gt;
  &lt;isSupportGB35114Certificate&gt;
    &lt;!--ro, opt, bool, whether supports GB35114 protocol and CA/SIP authentication certificate--&gt;true
  &lt;/isSupportGB35114Certificate&gt;
  &lt;isSupportAuthenticationMode opt="uKey,private"&gt;
    &lt;!--ro, opt, enum, verification mode, subType:string, attr:opt{req, string}, desc:“uKey”, “private”--&gt;private
  &lt;/isSupportAuthenticationMode&gt;
  &lt;isSupportUserWithIssuerDN&gt;
    &lt;!--ro, opt, bool--&gt;true
  &lt;/isSupportUserWithIssuerDN&gt;
  &lt;isSupportDeviceCertificatesManagement&gt;
    &lt;!--ro, opt, bool, whether the device supports device certificate management, desc:if this node is not returned, it indicates that device does not support this function--&gt;true
  &lt;/isSupportDeviceCertificatesManagement&gt;
  &lt;SecurityAdvanced&gt;
    &lt;!--ro, opt, object, security advanced parameters--&gt;
    &lt;noOperationEnabled&gt;
      &lt;!--ro, req, bool, whether to enable the control timeout when there is no operation--&gt;true
    &lt;/noOperationEnabled&gt;
    &lt;noOperationTime min="1" max="60" def="15"&gt;
      &lt;!--ro, req, int, control timeout when there is no operation, range:[1,60], unit:min, attr:min{req, int},max{req, int},def{req, int}--&gt;15
    &lt;/noOperationTime&gt;
    &lt;passwordLimitLevel opt="middling,high"&gt;
      <!--ro, opt, enum, password strength limit level, subType:string, attr:opt{req, string},
desc:the configured password is required by the following limit _x000D_
“middling” (middle level), “high” (high level), middle level password contains 8 to 16 characters, including at least 2 types of the following characters: digits, lower case letters, upper case letters, and special characters; high level password contains 8 to 16 characters, including at least 3 types of the following characters: digits, lower case letters, upper case letters, and special characters-->middling
    &lt;/passwordLimitLevel&gt;
  &lt;/SecurityAdvanced&gt;
  &lt;isSupportSoftwareLicense&gt;
    <!--ro, opt, bool, whether support configuring the product open source statement, desc:if this function is supported, this node will be returned and its value is true; otherwise, this node will not be returned. Related URI: /ISAPI/Security/softwareLicense?format=json-->true
  &lt;/isSupportSoftwareLicense&gt;
  &lt;isSupportPasswordProtection&gt;
    &lt;!--ro, opt, bool, whether it supports configuring SDK protocol authentication mode parameters, desc:PUT/GET /ISAPI/Security/adminAccesses/SDKPasswordProtection?format=json&security=&lt;security&gt;&iv=&lt;iv&gt;--&gt;true
  &lt;/isSupportPasswordProtection&gt;
&lt;/SecurityCap&gt;
```

11.1.4.5 Get device system capability

Request URL

GET /ISAPI/System/capabilities?type=<all>

Query Parameter

Parameter Name	Parameter Type	Description
all	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<DeviceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, device system capability, attr:version{opt, string, protocolVersion}-->
  <isSupportPreview>
    <!--ro, opt, bool, whether the device supports live view-->true
  </isSupportPreview>
  <isSupportSystemMaintain>
    <!--ro, opt, bool, whether the device supports system maintenance, desc:for traffic devices, this node is required and must be true-->true
  </isSupportSystemMaintain>
  <isSupportReboot>
    <!--ro, opt, bool, whether the device supports rebooting, desc:for traffic devices, this node is required and must be true.-->true
  </isSupportReboot>
  <isSupportFactoryReset>
    <!--ro, opt, bool, whether the device supports restoring to default settings, desc:for traffic devices, this node is required and must be true-->true
  </isSupportFactoryReset>
  <isSupportConfigurationData>
    <!--ro, opt, bool, whether the device supports importing and exporting the configuration files safely, desc:for traffic devices, this node is required
and must be true-->true
  </isSupportConfigurationData>
  <isSupportUpdatefirmware>
    <!--ro, opt, bool, whether the device supports upgrading, desc:for traffic devices, this node is required and must be true-->true
  </isSupportUpdatefirmware>
  <isSupportDeviceInfo>
    <!--ro, opt, bool, whether the device supports getting the device information, desc:for traffic devices, this node is required and must be true-->true
  </isSupportDeviceInfo>
  <isSupportTime>
    <!--ro, opt, bool, whether the device supports time configuration, desc:for traffic device, this node is required and must be true. For other devices,
the isSupportTimeCap is used to configure time-->true
  </isSupportTime>
  <SysCap>
    <!--ro, opt, object, system capability-->
    <isSupportDst>
      <!--ro, opt, bool, whether the device supports DST (Daylight Saving Time)-->true
    </isSupportDst>
    <NetworkCap>
      <!--ro, opt, object, network capability, desc:related URI: /ISAPI/System/Network/capabilities-->
      <isSupportWireless>
        <!--ro, req, bool, whether the device supports wireless network-->true
      </isSupportWireless>
      <isSupportPPPoE>
        <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
      </isSupportPPPoE>
      <isSupportBond>
        <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
      </isSupportBond>
      <isSupport802_1x>
        <!--ro, req, bool, whether the device supports 802.1x protocol-->true
      </isSupport802_1x>
      <isSupportNtp>
        <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
      </isSupportNtp>
      <isSupportFtp>
        <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
      </isSupportFtp>
      <isSupportUpnp>
        <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play ) protocol-->true
      </isSupportUpnp>
      <isSupportDdns>
        <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
      </isSupportDdns>
      <isSupportHttps>
        <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
      </isSupportHttps>
      <isSupport28181>
        <!--ro, opt, bool, N/A-->true
      </isSupport28181>
      <SnmpCap>
        <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
        <isSupport>
          <!--ro, req, bool, whether the device supports SNMP-->true
        </isSupport>
      </SnmpCap>
      <isSupportIPFilter>
        <!--ro, opt, bool, whether the device supports IP filtering-->true
      </isSupportIPFilter>
```

```xml
      <isSupportEhome>
        <!--ro, opt, bool, whether the device supports ISUP server configuration-->true
      </isSupportEhome>
      <isSupportWirelessDial>
        <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
      </isSupportWirelessDial>
      <isSupportWifiProbe>
        <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
      </isSupportWifiProbe>
      <VerificationCodeModification>
        <!--ro, opt, object, device verification code configuration-->
        <verificationCodeType opt="normal,empty">
          <!--ro, opt, string, verification code type, attr:opt{opt, string}-->empty
        </verificationCodeType>
      </VerificationCodeModification>
      <isSupportIntegrate>
        <!--ro, opt, bool, whether the device supports access protocol configuration-->true
      </isSupportIntegrate>
      <isSupportGetLinkSocketIP>
        <!--ro, opt, bool, whether the device supports getting the SocketIP of the current link-->true
      </isSupportGetLinkSocketIP>
      <isSupportWebSocket>
        <!--ro, opt, bool, whether the device supports WebSocket-->true
      </isSupportWebSocket>
      <isSupportWebSocketS>
        <!--ro, opt, bool, whether the device supports WebSocketS-->true
      </isSupportWebSocketS>
      <isSupportEventDataOverWebSocket opt="true,false">
        <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{opt, string}-->true
      </isSupportEventDataOverWebSocket>
      <isSupportPingDeny>
        <!--ro, opt, bool, whether the device supports configuring parameters to prevent the device from being tested using ping command-->true
      </isSupportPingDeny>
      <isSupportClusterIntercom>
        <!--ro, opt, bool, whether the 7200 intercom protocol is transmitted, desc:if this function is supported, this node will be returned and its value
is true; otherwise, this node will be returned and its value is true or not be returned.-->true
      </isSupportClusterIntercom>
      <isSupportTSOverWebSocket>
        <!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
      </isSupportTSOverWebSocket>
    </NetworkCap>
    <SerialCap>
      <!--ro, opt, object, range of RS-485 serial port numbers supported by the device-->
      <rs485PortNums>
        <!--ro, opt, int, the maximum number of RS-485 serial ports supported by the device-->true
      </rs485PortNums>
      <supportRS232Config>
        <!--ro, opt, bool, whether the device supports configuring parameters of RS-232 serial ports-->true
      </supportRS232Config>
      <rs422PortNums>
        <!--ro, opt, int, the maximum number of RS-422 serial ports supported by the device-->true
      </rs422PortNums>
      <rs232PortNums>
        <!--ro, opt, int, the maximum number of RS-232 serial ports supported by the device-->true
      </rs232PortNums>
      <rs485WorkMode opt="Led,CaptureTrigger">
        <!--ro, opt, string, RS-485 working modes supported by the device, attr:opt{opt, string}, desc:Led (external LED screen, which is used for LED
display), CaptureTrigger (external device, which is used to trigger the transmission of the captured data)-->test
      </rs485WorkMode>
    </SerialCap>
    <isSupportSubscribeEvent>
      <!--ro, opt, bool, whether the device supports subscribing to events, desc:related URI: /ISAPI/Event/notification/subscribeEventCap-->true
    </isSupportSubscribeEvent>
    <isSupportSubscribeIOTInfo>
      <!--ro, opt, bool, whether the device supports subscribing to the IoT information-->true
    </isSupportSubscribeIOTInfo>
  </SysCap>
  <ITCCap>
    <!--ro, opt, object, related node and URIs: the node <isSupportVehicleDetection> in /ISAPI/ITC/capability and /ISAPI/ITC/capabilities-->
    <isSupportVehicleDetection>
      <!--ro, opt, bool, whether the device supports vehicle detection-->true
    </isSupportVehicleDetection>
    <isSupportlicencePlateAuditData>
      <!--ro, opt, bool, whether the device supports importing or exporting the license plate blocklist and allowlist data-->true
    </isSupportlicencePlateAuditData>
    <isSupportSyncSignalOutput>
      <!--ro, opt, bool, whether the device supports configuring output parameters-->true
    </isSupportSyncSignalOutput>
  </ITCCap>
  <RacmCap>
    <!--ro, opt, object, UI before picture search-->
    <inputProxyNums>
      <!--ro, opt, int, supported number of digital channels, desc:related URI: /ISAPI/ContentMgmt/InputProxy/channels/<ID>-->1
    </inputProxyNums>
  </RacmCap>
  <TestCap>
    <!--ro, opt, object, test capability-->
    <isSupportEmailTest>
      <!--ro, opt, bool, whether the device supports email test-->true
    </isSupportEmailTest>
  </TestCap>
  <isSupportRTMP>
    <!--ro, opt, bool, whether the device supports RTMP configuration, desc:related URI: /ISAPI/Streaming/channels/<ID>/RTMPCfg/capabilities-->true
  </isSupportRTMP>
```

```xml
  <VideoIntercomCap>
    <!--ro, opt, object, video intercom capability, desc:related URI: /ISAPI/VideoIntercom/capabilities-->
    <platform opt="R0,R1,S29,N10,N8,N4">
      <!--ro, opt, enum, platform type, subType:string, attr:opt{req, string}, desc:"R0", "R1"; it will be returned if the device supports-->R0
    </platform>
  </VideoIntercomCap>
  <isSupportDumpData>
    <!--ro, opt, bool, whether the device supports exporting the Dump file-->true
  </isSupportDumpData>
  <AIDEventSupport
opt="abandonedObject,pedestrian,congestion,roadBlock,construction,trafficAccident,fogDetection,wrongDirection,illegalParking,SSharpDriving,lowSpeed,dragRacing,obstacle,vehNoYieldPedest,illegalMannedVeh,illegalMannedNonMotorVeh,umbrellaTentInstall,nonMotorVehOnVehLane,wearingNoHelmet,pedestRedLightRunning,pedestOnNonMotorVehLane,pedestOnVehLane,conflagration,smoke,VehicleSpeedDrop,lowVisibility,animalsOccupy,targetSpeedDetection,emergencyLane,gasser,changeLaneContinuously,RegionalVehicleStatistics">
    <!--ro, opt, string, supported AID event, attr:opt{opt, string}, desc:supported AID event-->test
  </AIDEventSupport>
  <TFSEventSupport opt="illegalParking
,wrongDirection,crossLane,laneChange,vehicleExist,turnRound,parallelParking,notKeepDistance,notSlowZebraCrossing,overtakeRightSide,lowSpeed,dragRacing,changeLaneContinuously,SSharpDriving,largeVehicleOccupyLine,jamCrossLine,obstacle,blackSmokeVehicle,turnRightStop,occupyDedicatedLane,notDriveInDedicatedLane,TFS_nonZipperMerge,overSpeed,emergencyLane,gasser">
    <!--ro, opt, string, supported TFS, attr:opt{opt, string}-->test
  </TFSEventSupport>
  <isSupportVehicleFaceRecognition>
    <!--ro, opt, bool, whether the device supports checkpoint vehicle recognition, desc:background: the traffic devices support "facial recognition" (driver), however, the corresponding node is not added to the device to notify the integration of this capability-->true
  </isSupportVehicleFaceRecognition>
  <isSupportLCDScreen>
    <!--ro, opt, bool, whether the device supports LCD display screen-->true
  </isSupportLCDScreen>
  <isSupportEmmcInfo>
    <!--ro, opt, bool, the capability of supporting the EMMC configuration, desc:related URI: /ISAPI/System/emmcInfo/capabilities?format=json-->true
  </isSupportEmmcInfo>
  <isSupportVehicleIllegalType opt="1,2...29">
    <!--ro, opt, string, the vehicle violation types supported by the device, attr:opt{opt, string}, desc:the content value is the same as the index value, refer to NET_ITS_PLATE_RESULT.NET_DVR_VEHICLE_INFO.byIllegalType the capture camera is connected to the new platform, and the device supports "traffic violation alarm"; however, the corresponding node is not added to the device to notify the integration of this capability; that is, the traffic violation alarm capability lacks details. If later the platform needs to develop capability with details, the current device will be incompatible with the capability; and detailed capability is hidden in the configuration of all trigger modes, which will influence overall access-->test
  </isSupportVehicleIllegalType>
  <isSupportRadar>
    <!--ro, opt, bool, whether the device supports radar functions, desc:related URI: /ISAPI/Radar/capabilities?format=json-->true
  </isSupportRadar>
  <isSupportISUPHttpPassthrough>
    <!--ro, opt, bool, whether the device supports HTTP transparent transmission via ISUP V5.0-->true
  </isSupportISUPHttpPassthrough>
  <isSupportDetectorMotionAlarm>
    <!--ro, opt, bool, whether the device supports the detector moving alarm, desc:eventType: detectorMotionAlarm-->true
  </isSupportDetectorMotionAlarm>
  <isSupportRadarVideoDetection>
    <!--ro, opt, bool, related URI: /ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json, desc:related URI: /ISAPI/Intelligent/channels/<ID>/radarVideoDetection/capabilities?format=json-->true
  </isSupportRadarVideoDetection>
  <isSupportOnlineUpgradeTask>
    <!--ro, opt, bool, whether device online upgrading is supported-->true
  </isSupportOnlineUpgradeTask>
  <isSupportObjectServer>
    <!--ro, opt, bool, whether the device supports the object storage service, desc:related URI: /ISAPI/System/objectServer/capabilities?format=json-->true
  </isSupportObjectServer>
  <isSupportDrivingDirectionAtIntersection>
    <!--ro, opt, bool, whether the device supports intersection driving direction detection, desc:eventType: drivingDirectionAtIntersection; this node is not returned if not supported-->true
  </isSupportDrivingDirectionAtIntersection>
  <supportQuickConfigType opt="icr,dayNightGate,syncSignalOutput">
    <!--ro, opt, string, the list of supported functions of the quick configuration pop-up on the web, attr:opt{opt, string},
desc:the web quick configuration does not distinguish the configuration order, and it displays all supported function lists on the pop-up on the tab page
    icr refers to ICR switch configuration (related URL: /ISAPI/Image/channels/<channelID>/); icr
   dayNightGate refers to day and night mode switching (related URL: /ISAPI/Image/channels/<channelID>/dayNightGate);
    IOOutputs refers to synchronous signal output (related URL: /ISAPI/System/syncSignalOutput)-->syncSignalOutput
  </supportQuickConfigType>
  <isSupportUsersLoginNotification>
    <!--ro, opt, bool, whether the device supports parameters of notifications on user login, desc:/ISAPI/System/UsersLoginNotification?format=json-->true
  </isSupportUsersLoginNotification>
  <isSupportChildmanage>
    <!--ro, opt, bool, whether the device supports gateway sub-device topology management, desc:GET /ISAPI/IoTGateway/Childmanage/capabilities?format=json-->true
  </isSupportChildmanage>
  <isSupportPowerIndicatorLightCtrlParams>
    <!--ro, opt, bool, whether the device supports parameters of device battery indicator control, desc:GET/PUT /ISAPI/System/PowerIndicatorLightCtrlParams?format=json-->true
  </isSupportPowerIndicatorLightCtrlParams>
  <isSupportMaintenanceServiceAuthorization>
    <!--ro, opt, bool, whether it supports device maintenance service authorization information, desc:GET/PUT /ISAPI/System/MaintenanceServiceAuthorization?format=json-->true
  </isSupportMaintenanceServiceAuthorization>
  <isSupportVehicleTurnBackDetectionEvent>
    <!--ro, opt, bool-->true
  </isSupportVehicleTurnBackDetectionEvent>
</DeviceCap>
```


11.1.4.6 Export device configuration file

Request URL

GET /ISAPI/System/configurationData?secretkey=<secretkey>&mode=<mode>

Query Parameter

Parameter Name	Parameter Type	Description
secretkey	string	The verification key, it is provided by the upper-layer. It should be encrypted for exporting and recorded for importing.
mode	enum	--

Request Message

None

Response Message

Binary Data

11.1.4.7 Get the languages supported by the device

Request URL

GET /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--ro, req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian",
"Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish",
"Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian"-Slovenian,
"Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian", "Uzbek", "Kazak",
"Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
</DeviceLanguage>
```

11.1.4.8 Set device language parameters

Request URL

PUT /ISAPI/System/DeviceLanguage

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, languages supported by the device, attr:version{req, string, protocolVersion}-->
  <language>
    <!--req, enum, language, subType:string, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese), "English", "Russian", "Bulgarian",
"Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish", "Romanian", "Turkish", "Japanese",
"Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic", "Sovenian" (Slovenian), "Croatian",
"Lithuanian", "Serbian", "BrazilianPortuguese" (Brazilian Portuguese), "Indonesian", "Ukrainian", "EURSpanish"-->SimChinese
  </language>
</DeviceLanguage>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.1.4.9 Get the capability of configuring the device language

Request URL

GET /ISAPI/System/DeviceLanguage/capabilities

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<DeviceLanguage xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, device language configuration, attr:version{req, string, protocolVersion}-->
  <language
opt="SimChinese,TraChinese,English,Russian,Bulgarian,Hungarian,Greek,German,Italian,Czech,Slovakia,French,Polish,Dutch,Portuguese,Spanish,Romanian,Turkish,J
apanese,Danish,Swedish,Norwegian,Finnish,Korean,Thai,Estonia,Vietnamese,Hebrew,Latvian,Arabic,Sovenian,Croatian,Lithuanian,Serbian,BrazilianPortuguese,Indon
esian,Ukrainian,EURSpanish,Uzbek,Kazak,Kirghiz,Farsi,Azerbaidzhan,Burmese,Mongolian,Anglicism,Estonian">
    <!--ro, req, enum, language, subType:string, attr:opt{req, string}, desc:"SimChinese" (simplified Chinese), "TraChinese" (traditional Chinese),
"English", "Russian", "Bulgarian", "Hungarian", "Greek", "German", "Italian", "Czech", "Slovakia", "French", "Polish", "Dutch", "Portuguese", "Spanish",
"Romanian", "Turkish", "Japanese", "Danish", "Swedish", "Norwegian", "Finnish", "Korean", "Thai", "Estonia", "Vietnamese", "Hebrew", "Latvian", "Arabic",
"Sovenian"-Slovenian, "Croatian", "Lithuanian", "Serbian", "BrazilianPortuguese"-Brazilian Portuguese, "Indonesian", "Ukrainian", "EURSpanish", "Sovenian",
"Uzbek", "Kazak", "Kirghiz", "Farsi", "Azerbaidzhan", "Burmese", "Mongolian"-->SimChinese
  </language>
  <upgradeFirmWareEnabled>
    <!--ro, opt, bool, whether to enable upgrading the firmware-->true
  </upgradeFirmWareEnabled>
</DeviceLanguage>
```


11.1.4.10 Set restoring devices' default parameters

Request URL

PUT /ISAPI/System/factoryReset?mode=<mode>&childDevID=<devIndex>&loginPassword=<loginPassword>&module=<module>

Query Parameter

Parameter Name	Parameter Type	Description
mode	enum	--
devIndex	string	--
loginPassword	string	--
module	enum	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```


11.1.4.11 Get the network service capability

Request URL

GET /ISAPI/System/Network/capabilities

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<NetworkCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, get the network service capability, attr:version{opt, string, protocolVersion}-->
  <isSupportWireless>
    <!--ro, req, bool, whether the device supports wireless network-->true
  </isSupportWireless>
  <isSupportPPPoE>
    <!--ro, req, bool, whether the device supports PPPoE (Point to Point Protocol over Ethernet)-->true
  </isSupportPPPoE>
  <isSupportBond>
    <!--ro, req, bool, whether the device supports NIC (Network Interface Card) bonding-->true
  </isSupportBond>
  <isSupport802_1x>
    <!--ro, req, bool, whether the device supports 802.1x protocol-->true
  </isSupport802_1x>
  <isSupportNtp>
    <!--ro, opt, bool, whether the device supports NTP (Network Time Protocol)-->true
  </isSupportNtp>
  <isSupportFtp>
    <!--ro, opt, bool, whether the device supports FTP (File Transfer Protocol)-->true
  </isSupportFtp>
  <isSupportUpnp>
    <!--ro, opt, bool, whether the device supports UPnP (Universal Plug and Play) protocol-->true
  </isSupportUpnp>
  <isSupportDdns>
    <!--ro, opt, bool, whether the device supports DDNS (Dynamic Domain Name System) service-->true
  </isSupportDdns>
  <isSupportHttps>
    <!--ro, opt, bool, whether the device supports HTTPS (Hypertext Transfer Protocol Secure)-->true
  </isSupportHttps>
  <isSupport28181>
    <!--ro, opt, bool, N/A-->true
  </isSupport28181>
  <SnmpCap>
    <!--ro, opt, object, SNMP (Simple Network Management Protocol) capability-->
    <isSupport>
      <!--ro, req, bool, whether the device supports SNMP-->true
    </isSupport>
  </SnmpCap>
  <isSupportIPFilter>
    <!--ro, opt, bool, whether the device supports IP filtering-->true
  </isSupportIPFilter>
  <isSupportEZVIZ>
    <!--ro, opt, bool, whether the device supports EZ protocol-->true
  </isSupportEZVIZ>
  <isSupportEhome>
    <!--ro, opt, bool, whether the device supports EHome (ISUP) protocol-->true
  </isSupportEhome>
  <isSupportWirelessDial>
    <!--ro, opt, bool, whether the device supports wireless dial-up protocol-->true
  </isSupportWirelessDial>
```

```
  <isSupportWifiProbe>
    <!--ro, opt, bool, whether the device supports Wi-Fi signal detection-->true
  </isSupportWifiProbe>
  <VerificationCodeModification>
    <!--ro, opt, object, verification code editing-->
    <verificationCodeType opt="normal,empty">
      <!--ro, opt, string, verification code type, attr:opt{req, string}-->test
    </verificationCodeType>
  </VerificationCodeModification>
  <isSupportIntegrate>
    <!--ro, opt, bool, whether the device supports access protocol configuration-->true
  </isSupportIntegrate>
  <isSupportWebSocket>
    <!--ro, opt, bool, whether the device supports WebSocket-->true
  </isSupportWebSocket>
  <isSupportWebSocketS>
    <!--ro, opt, bool, whether the device supports WebSocketS-->true
  </isSupportWebSocketS>
  <isSupportVideoImgDB>
    <!--ro, opt, bool, whether the device supports image and video library configuration-->true
  </isSupportVideoImgDB>
  <isSupportEventDataOverWebSocket opt="true,false">
    <!--ro, opt, bool, whether the device supports uploading the event data via WebSocket, attr:opt{req, string}-->true
  </isSupportEventDataOverWebSocket>
  <isSupportSip>
    <!--ro, opt, bool, whether the device supports private SIP (VOIP)-->true
  </isSupportSip>
  <isSupportGantryETCServer>
    <!--ro, opt, bool, N/A, desc:N/A-->true
  </isSupportGantryETCServer>
  <isSupportDSTPServerCfg>
    <!--ro, opt, bool, whether the device supports DTSP server address configuration, desc:GET/PUT /ISAPI/System/Network/DSTPAccess/DSTPServerCfg?
format=json-->true
  </isSupportDSTPServerCfg>
  <isSupportDSTPProtocolsCap>
    <!--ro, opt, bool, whether the device supports DTSP capability, desc:GET /ISAPI/System/Network/DSTPAccess/DSTPProtocolsCap?format=json-->true
  </isSupportDSTPProtocolsCap>
  <isSupportTSOverWebSocket>
    <!--ro, opt, bool, whether the device supports transparent transmission via the WebSocket serial port, desc:(TS:TransportSerial)-->true
  </isSupportTSOverWebSocket>
  <isSupportMDNSParams>
    <!--ro, opt, bool-->true
  </isSupportMDNSParams>
</NetworkCap>
```

11.1.4.12 Set SSH parameters

Request URL

PUT /ISAPI/System/Network/ssh?readerID=<readerID>&type=<type>&SOCChipID=<SOCChipID>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
readerID	string	--
type	enum	--
SOCChipID	string	--
controllerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SSH xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--wo, opt, object, attr:version{opt, string, protocolVersion}-->
  <enabled>
    <!--wo, req, bool, whether to enable the function-->true
  </enabled>
  <port>
    <!--wo, opt, int-->22
  </port>
</SSH>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, read-only,status description: OK,Device Busy,Device Error,Invalid Operation,Invalid XML Format,Invalid XML Content,Reboot,Additional 
Error, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" 
(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.1.4.13 Reboot device

Request URL

PUT /ISAPI/System/reboot?childDevID=<devIndex>&module=<module>&loginPassword=<loginPassword>&controllerID=<controllerID>

Query Parameter

Parameter Name	Parameter Type	Description
devIndex	string	Reboot a sub-device via specifying sub-device ID. The ID of the sub-device connected to the current device can be searched via /ISAPI/IoTGateway/Childmanage/SearchChild?format=json.
module	enum	The "module" specifies a module to restart, without requiring a device reboot, corresponding to the capability (/ISAPI/System/capabilities/deviceReboot->module). "algorithmProgram (traffic terminal radar-assisted algorithm program, supports independent algorithm program startup), "radar" (restart radar module of the radar-assisted camera).
loginPassword	string	--
controllerID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, 
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to 
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

11.1.4.14 Get the video analysis task capability

Request URL

GET /ISAPI/System/Video/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<VideoCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <videoInputPortNums>
    <!--ro, opt, int, number of video input ports-->0
  </videoInputPortNums>
  <videoOutputPortNums>
    <!--ro, opt, int, number of video output ports-->0
  </videoOutputPortNums>
  <menuNums>
    <!--ro, opt, int, number of local menus that can be displayed-->0
  </menuNums>
  <externalChannelNum>
    <!--ro, opt, bool, number of extended analog channels-->true
  </externalChannelNum>
  <isSupportHeatmap>
    <!--ro, opt, bool, whether the device supports heat map-->true
  </isSupportHeatmap>
  <isSupportCounting>
    <!--ro, opt, bool, whether the device supports people counting-->true
  </isSupportCounting>
  <countingType>
    <!--ro, opt, enum, statistics type, subType:string, desc:statistics type-->human
  </countingType>
  <isSupportPicture>
    <!--ro, opt, bool, whether the device supports OSD picture overlay, desc:related URI: /ISAPI/System/Video/inputs/channels/<channelID>/image/picture--
>true
  </isSupportPicture>
  <isSupportPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view-->true
  </isSupportPreviewSwitch>
  <isSupportRecodStatus>
    <!--ro, opt, bool, whether the device supports searching for the recording status-->true
  </isSupportRecodStatus>
  <isSupportPrivacyMask>
    <!--ro, opt, bool, whether the device supports privacy mask-->true
  </isSupportPrivacyMask>
  <isSupportBinocularPreviewSwitch>
    <!--ro, opt, bool, whether the device supports switching live view of the dual-lens camera-->true
  </isSupportBinocularPreviewSwitch>
```

```
  <isSupportCalibCheck>
    <!--ro, opt, bool, whether the device supports calibration verification-->true
  </isSupportCalibCheck>
  <isSupportPIP>
    <!--ro, opt, bool-->true
  </isSupportPIP>
  <isSupportFocusVideoMode>
    <!--ro, opt, bool-->true
  </isSupportFocusVideoMode>
  <isSupportExternalChannel>
    <!--ro, opt, bool, whether the device supports extended analog channels-->true
  </isSupportExternalChannel>
  <isSupportMultiChannelCounting>
    <!--ro, opt, bool-->true
  </isSupportMultiChannelCounting>
  <isSupportCountingCollection>
    <!--ro, opt, bool, whether the device supports people counting ANR-->true
  </isSupportCountingCollection>
  <isSupportHeatmapCollection>
    <!--ro, opt, bool, whether the device supports heat map data ANR-->true
  </isSupportHeatmapCollection>
  <channelFlexible opt="name,enable,online,linknum">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->name
  </channelFlexible>
  <isSupportOutputsResource>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/resource?format=json, desc:related URI: /ISAPI/System/Video/outputs/resource?format=json--
>true
  </isSupportOutputsResource>
  <OSDLanguage opt="GBK,EUC-KR,Hebrew" def="GBK">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->GBK
  </OSDLanguage>
  <isSupportMixedChannel>
    <!--ro, opt, bool-->true
  </isSupportMixedChannel>
  <isSupportMixedChannelStatus>
    <!--ro, opt, bool-->true
  </isSupportMixedChannelStatus>
  <isSupportOutputCourseware>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/courseware/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/courseware/capabilities?format=json-->true
  </isSupportOutputCourseware>
  <isSupportVideoInputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/inputs/mode/capabilities, desc:related URI: /ISAPI/System/Video/inputs/mode/capabilities-->true
  </isSupportVideoInputMode>
  <isSupportVideoOutputMode>
    <!--ro, opt, bool, related URI: /ISAPI/System/Video/outputs/mode/capabilities?format=json, desc:related URI:
/ISAPI/System/Video/outputs/mode/capabilities?format=json-->true
  </isSupportVideoOutputMode>
  <PreviewMode>
    <!--ro, opt, object-->
    <PIPType>
      <!--ro, opt, object-->
      <mainScreenChannelID>
        <!--ro, opt, int-->0
      </mainScreenChannelID>
      <subScreenChannelID>
        <!--ro, opt, int-->0
      </subScreenChannelID>
    </PIPType>
  </PreviewMode>
  <isSupportViewTag>
    <!--ro, opt, bool-->true
  </isSupportViewTag>
  <isSupportMenuStatus>
    <!--ro, opt, bool-->true
  </isSupportMenuStatus>
</VideoCap>
```

11.1.5 时间管理

11.1.5.1 Get device time synchronization management parameters

Request URL

GET /ISAPI/System/time

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--ro, req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--ro, opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}-->2019-02-28T10:50:44+08:30
  </localTime>
  <timeZone>
    <!--ro, opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:daylight saving time configuration in time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the device time returned currently is in DST (Daylight Saving Time) system, desc:whether the device time returned currently is in DST (Daylight Saving Time) system-->true
  </isSummerTime>
  <platformType>
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform-->EZVIZ
  </platformType>
  <platformNo>
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--ro, opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--ro, opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--ro, opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```

11.1.5.2 Set device time synchronization management parameters

Request URL

PUT /ISAPI/System/time

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, time management, attr:version{opt, string, protocolVersion}-->
  <timeMode>
    <!--req, enum, time synchronization mode, subType:string, desc:“NTP” (NTP time synchronization), “manual” (manual time synchronization), “satellite” (satellite time synchronization), “platform” (platform time synchronization), “NONE” (time synchronization is not allowed or no time synchronization source), “GB28181” (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime>
    <!--opt, string, local time, range:[0,256], dep:and,{$.Time.timeMode,eq,manual}, desc:the time difference between local time and coordinated universal time (UTC) is configured via the timeZone field-->2019-02-28T10:50:44
  </localTime>
  <timeZone>
    <!--opt, string, time zone, range:[0,256], dep:and,{$.Time.timeMode,eq,manual},{$.Time.timeMode,eq,NTP}, desc:time zone-->CST-8:00:00DST00:30:00,M4.1.0/02:00:00,M10.5.0/02:00:00
  </timeZone>
  <satelliteInterval>
    <!--opt, int, satellite time synchronization interval, step:1, unit:min, dep:and,{$.Time.timeMode,eq,satellite}, desc:unit: minute-->60
  </satelliteInterval>
  <isSummerTime>
    <!--ro, opt, bool, whether the time returned by the current device is that in the DST (daylight saving time), desc:whether the time returned by the current device is that in the DST (daylight saving time)-->true
  </isSummerTime>
  <platformType>
    <!--opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, desc:exists only when the timeMode is selected as platform, related URI: /ISAPI/System/Network/EZVIZ-->EZVIZ
  </platformType>
  <platformNo>
    <!--opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <ethernetPort>
    <!--opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}-->0
  </ethernetPort>
  <IANA>
    <!--opt, enum, subType:string-->Africa/Abidjan
  </IANA>
  <windowsZone>
    <!--opt, enum, subType:string-->Dateline Standard Time
  </windowsZone>
  <standard>
    <!--opt, enum, format, subType:string, desc:"PAL", "NTSC"-->PAL
  </standard>
</Time>
```


Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <FailedNodeInfoList>
    <!--ro, opt, object, information list of failed nodes, desc:for the manual time synchronization of central analysis cluster, this field is returned if time synchronization failed-->
      <FailedNodeInfo>
        <!--ro, opt, object, information of failed nodes-->
        <nodeID>
          <!--ro, req, string, node ID, range:[0,64]-->test
        </nodeID>
        <nodeIP>
          <!--ro, req, string, node IP, range:[0,20]-->test
        </nodeIP>
        <reason>
          <!--ro, opt, string, reason why the node failed to synchronize time, range:[0,128]-->test
        </reason>
      </FailedNodeInfo>
  </FailedNodeInfoList>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to quickly identify and evaluate issues-->badXmlFormat
  </description>
</ResponseStatus>
```

11.1.5.3 Get the capability of device time synchronization management

Request URL

GET /ISAPI/System/time/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Time xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, time management capability set, attr:version{opt, string, protocolVersion}-->
  <timeMode opt="NTP,manual,satellite,SDK,28181,ONVIF,ALL(任何支持的校时方式都允许校时),NONE(不允校时或无校时源),platform,PTP">
    <!--ro, req, enum, time synchronization mode, subType:string, attr:opt{opt, string}, desc:"NTP" (NTP time synchronization), "manual" (manual time synchronization), "satellite" (satellite time synchronization), "platform" (platform time synchronization), "NONE" (time synchronization is not allowed or no time synchronization source), "GB28181" (GB28181 time synchronization)-->NTP
  </timeMode>
  <localTime min="0" max="256">
    <!--ro, opt, string, local time, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTime>
  <timeZone min="0" max="256">
    <!--ro, opt, string, time zone, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </timeZone>
  <satelliteInterval min="0" max="3600">
    <!--ro, opt, int, satellite time synchronization interval, step:1, unit:min, attr:min{opt, string},max{opt, string}, desc:unit: minute-->60
  </satelliteInterval>
  <timeType opt="local,UTC">
    <!--ro, opt, enum, time type, subType:string, attr:opt{opt, string}, desc:"local" (local time), "UTC" (UTC time)-->local
  </timeType>
  <platformType opt="EZVIZ">
    <!--ro, opt, enum, platform type, subType:string, dep:and,{$.Time.timeMode,eq,platform}, attr:opt{opt, string}, desc:platform type-->EZVIZ
  </platformType>
  <platformNo min="1" max="2">
    <!--ro, opt, int, platform No., range:[1,2], dep:and,{$.Time.timeMode,eq,GB28181}, attr:min{req, int},max{req, int}, desc:it is the only ID, which is configured via platformNo in GB28181List, related URI: /ISAPI/System/Network/SIP/<SIPServerID>-->1
  </platformNo>
  <isSupportHistoryTime>
    <!--ro, opt, bool, supported capability of the historical time synchronization list, desc:related URI: /ISAPI/System/time/historyInfo?format=json-->true
  </isSupportHistoryTime>
  <isSupportTimeFilter>
    <!--ro, opt, bool, supported capability of filtering time synchronization, desc:related URI: /ISAPI/System/time/filter/capabilities?format=json-->true
  </isSupportTimeFilter>
  <displayFormat opt="MM/dd/yyyy hh:mm,mm,dd-MM-yyyy hh,MM-dd-yyyy hh:mm,yyyy-MM-dd hh:mm,MM/dd/yyyy hh:mm:ss,dd-MM-yyyy hh:mm:ss,MM-dd-yyyy hh:mm:ss,yyyy-MM-dd hh:mm:ss,yyyy-MM-dd,dd-MM-yyyy,MM-dd-yyyy">
    <!--ro, opt, enum, time display format, subType:string, attr:opt{req, string}, desc:if this node is returned, it indicates that the device supports configuring time display format, related URI: /ISAPI/System/time/timeType?format=json-->MM/dd/yyyy hh:mm
  </displayFormat>
  <isSupportSyncDeviceNTPInfoToCamera>
    <!--ro, opt, bool, the capability of synchronizing device's NTP service information with the camera, desc:related URI: /ISAPI/System/time/SyncDeviceNTPInfoToCamera/capabilities?format=json-->true
  </isSupportSyncDeviceNTPInfoToCamera>
  <isSupportNTPService>
    <!--ro, opt, bool-->true
  </isSupportNTPService>
  <isSupportTimeZoneListInfo>
    <!--ro, opt, bool-->true
  </isSupportTimeZoneListInfo>
  <isSupportAutoEnterSummerTime>
    <!--ro, opt, bool-->true
  </isSupportAutoEnterSummerTime>
  <ethernetPort min="0" max="255">
    <!--ro, opt, int, range:[0,255], dep:and,{$.Time.timeMode,eq,PTP}, attr:min{req, int},max{req, int}-->1
  </ethernetPort>
  <IANA
opt="Africa/Abidjan,Africa/Accra,Africa/Addis_Ababa,Africa/Algiers,Africa/Blantyre,Africa/Brazzaville,Africa/Cairo,Africa/Casablanca,Africa/Conakry,Africa/Dakar,Africa/Dar_es_Salaam,Africa/Djibouti,Africa/Freetown,Africa/Gaborone,Africa/Harare,Africa/Johannesburg,Africa/Kampala,Africa/Khartoum,Africa/Kigali,Africa/Kinshasa,Africa/Lagos,Africa/Maseru,Africa/Mogadishu,Africa/Nairobi,Africa/Sao_Tome,Africa/Timbuktu,Africa/Tripoli,Africa/Tunis,America/Anchorage,America/Aruba,America/Asuncion,America/Barbados,America/Belize,America/Bogota,America/Buenos_Aires,America/Cancun,America/Caracas,America/Cayman,America/Chicago,America/Curacao,America/Dawson_Creek,America/Denver,America/Detroit,America/Dominica,America/Edmonton,America/El_Salvador,America/Fortaleza,America/Grand_Turk,America/Grenada,America/Guatemala,America/Guyana,America/Halifax,America/Havana,America/Indiana/Indianapolis,America/Indiana/Knox,America/Indiana/Marengo,America/Indiana/Petersburg,America/Indiana/Tell_City,America/Indiana/Vevay,America/Indiana/Vincennes,America/Indiana/Winamac,America/Jamaica,America/La_Paz,America/Lima,America/Los_Angeles,America/Louisville,America/Managua,America/Martinique,America/Mendoza,America/Metlakatla,America/Mexico_City,America/Monterrey,America/Montevideo,America/Montreal,America/Nassau,America/New_York,America/North_Dakota/Beulah,America/North_Dakota/Center,America/North_Dakota/New_Salem,America/Panama,America/Phoenix,America/Port_of_Spain,America/Port-au-
Prince,America/Puerto_Rico,America/Santo_Domingo,America/Sao_Paulo,America/St_Johns,America/St_Kitts,America/St_Lucia,America/St_Thomas,America/Tijuana,America/Toronto,America/Vancouver,America/Winnipeg,Antarctica/South_Pole,Arctic/Longyearbyen,Asia/Almaty,Asia/Amman,Asia/Anadyr,Asia/Aqtau,Asia/Baghdad,Asia/Bahrain,Asia/Baku,Asia/Bangkok,Asia/Beirut,Asia/Calcutta,Asia/Damascus,Asia/Dhaka,Asia/Dubai,Asia/Gaza,Asia/Hebron,Asia/Ho_Chi_Minh,Asia/Hong_Kong,Asia/Jakarta,Asia/Jerusalem,Asia/Kabul,Asia/Karachi,Asia/Kathmandu,Asia/Kuala_Lumpur,Asia/Kuwait,Asia/Macau(China),Asia/Manila,Asia/Muscat,Asia/Phnom_Penh,Asia/Pyongyang,Asia/Rangoon,Asia/Riyadh,Asia/Seoul,Asia/Shanghai,Asia/Singapore,Asia/Taipei,Asia/Tehran,Asia/Tel_Aviv,Asia/Tokyo,Asia/Ulaanbaatar,Atlantic/Bermuda,Atlantic/Canary,Atlantic/Cape_Verde,Atlantic/Reykjavik,Atlantic/Stanley,Australia/Adelaide,Australia/Brisbane,Australia/Canberra,Australia/Darwin,Australia/Melbourne,Australia/NSW,Australia/Perth,Australia/Queensland,Australia/Sydney,Australia/Victoria,Canada/Newfoundland,Canada/Saskatchewan,Chile/EasterIsland,Europe/Amsterdam,Europe/Andorra,Europe/Athens,Europe/Belfast,Europe/Belgrade,Europe/Berlin,Europe/Bratislava,Europe/Brussels,Europe/Bucharest,Europe/Budapest,Europe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
```

```
pe/Chisinau,Europe/Copenhagen,Europe/Dublin,Europe/Gibraltar,Europe/Helsinki,Europe/Isle_of_Man,Europe/Istanbul,Europe/Kiev,Europe/Lisbon,Europe/London,Euro
pe/Luxembourg,Europe/Madrid,Europe/Malta,Europe/Minsk,Europe/Monaco,Europe/Moscow,Europe/Oslo,Europe/Paris,Europe/Prague,Europe/Riga,Europe/Rome,Europe/San_
Marino,Europe/Sarajevo,Europe/Simferopol,Europe/Skopje,Europe/Sofia,Europe/Stockholm,Europe/Tallinn,Europe/Vatican,Europe/Vienna,Europe/Vilnius,Europe/Warsa
w,Europe/Zagreb,Europe/Zurich,Indian/Antananarivo,Indian/Maldives,Indian/Mauritius,Pacific/Auckland,Pacific/Fiji,Pacific/Guam,Pacific/Honolulu,Pacific/Kirit
imati,Pacific/Noumea,America/Antigua,MST,Asia/Kolkata,America/Santiago,Africa/Monrovia,Asia/Colombo,America/Chihuahua,America/Tegucigalpa,America/Paramaribo
,Europe/Ljubljana,Asia/Ashgabat,Africa/Asmara,Asia/Brunei,Africa/Bangui,Africa/Banjul,Asia/Bishkek,Africa/Bissau,Africa/Bujumbura,Asia/Dili,Asia/Dushanbe,Pa
cific/Funafuti,Pacific/Guadalcanal,Africa/Juba,America/St_Vincent,Africa/Libreville,Africa/Lome,Africa/Luanda,Africa/Lusaka,Pacific/Majuro,Africa/Malabo,Afr
ica/Maputo,Africa/Mbabane,Indian/Comoro,Africa/Ndjamena,Pacific/Palau,Africa/Niamey,Europe/Nicosia,Africa/Nouakchott,Pacific/Tongatapu,Africa/Ouagadougou,Am
erica/Lower_Princes,Europe/Podgorica,Pacific/Efate,Africa/Porto-
Novo,Asia/Tashkent,Asia/Tbilisi,Asia/Thimphu,Europe/Tirane,Europe/Vaduz,Asia/Vientiane,Africa/Windhoek,Africa/Douala,Pacific/Nauru,Asia/Yerevan,Etc/GMT+12,E
tc/GMT+11,America/Adak,Pacific/Marquesas,Etc/GMT+9,Etc/GMT+8,America/Whitehorse,Pacific/Easter,America/Regina,America/Indianapolis,America/Cuiaba,America/Ar
aguaina,America/Cayenne,America/Godthab,America/Punta_Arenas,America/Miquelon,America/Bahia,Etc/GMT+2,Atlantic/Azores,Etc/UTC,Europe/Kaliningrad,Europe/Astr
akhan,Europe/Samara,Europe/Saratov,Europe/Volgograd,Asia/Yekaterinburg,Asia/Qyzylorda,Asia/Katmandu,Asia/Omsk,Asia/Barnaul,Asia/Hovd,Asia/Krasnoyarsk,Asia/N
ovosibirsk,Asia/Tomsk,Asia/Irkutsk,Australia/Eucla,Asia/Chita,Asia/Yakutsk,Pacific/Port_Moresby,Australia/Hobart,Asia/Vladivostok,Australia/Lord_Howe,Pacifi
c/Bougainville,Asia/Srednekolymsk,Asia/Magadan,Pacific/Norfolk,Asia/Sakhalin,Asia/Kamchatka,Etc/GMT-12,Pacific/Chatham,Etc/GMT-13,Pacific/Apia">
    <!--ro, opt, enum, IANA synchronization, subType:string, attr:opt{opt, string}-->Africa/Abidjan
  </IANA>
  <isSupportSearchDeviceIANAList>
    <!--ro, opt, bool, whether the device supports searching for supported IANA list, desc:POST /ISAPI/System/time/SearchDeviceIANAList?format=json-->true
  </isSupportSearchDeviceIANAList>
  <windowsZone opt="Dateline Standard Time,UTC-11,Aleutian Standard Time,Hawaiian Standard Time,Marquesas Standard Time,Alaskan Standard Time,UTC-09,Pacific
Standard Time (Mexico),UTC-08,Pacific Standard Time,US Mountain Standard Time,Mountain Standard Time (Mexico),Mountain Standard Time,Yukon Standard
Time,Central America Standard Time,Central Standard Time,Easter Island Standard Time,Central Standard Time (Mexico),Canada Central Standard Time,SA Pacific
Standard Time,Eastern Standard Time (Mexico),Eastern Standard Time,Haiti Standard Time,Cuba Standard Time,US Eastern Standard Time,Turks And Caicos Standard
Time,Paraguay Standard Time,Atlantic Standard Time,Venezuela Standard Time,Central Brazilian Standard Time,SA Western Standard Time,Pacific SA Standard
Time,Newfoundland Standard Time,Tocantins Standard Time,E. South America Standard Time,SA Eastern Standard Time,Argentina Standard Time,Greenland Standard
Time,Montevideo Standard Time,Magallanes Standard Time,Saint Pierre Standard Time,Bahia Standard Time,UTC-02,Azores Standard Time,Cape Verde Standard
Time,UTC,GMT Standard Time,Greenwich Standard Time,Sao Tome Standard Time,Morocco Standard Time,W. Europe Standard Time,Central Europe Standard Time,Romance
Standard Time,Central European Standard Time,W. Central Africa Standard Time,Jordan Standard Time,GTB Standard Time,Middle East Standard Time,Egypt Standard
Time,E. Europe Standard Time,Syria Standard Time,West Bank Standard Time,South Africa Standard Time,FLE Standard Time,Israel Standard Time,South Sudan
Standard Time,Kaliningrad Standard Time,Sudan Standard Time,Libya Standard Time,Namibia Standard Time,Arabic Standard Time,Turkey Standard Time,Arab
Standard Time,Belarus Standard Time,Russian Standard Time,E. Africa Standard Time,Iran Standard Time,Arabian Standard Time,Astrakhan Standard
Time,Azerbaijan Standard Time,Russia Time Zone 3,Mauritius Standard Time,Saratov Standard Time,Georgian Standard Time,Volgograd Standard Time,Caucasus
Standard Time,Afghanistan Standard Time,West Asia Standard Time,Ekaterinburg Standard Time,Pakistan Standard Time,Qyzylorda Standard Time,India Standard
Time,Sri Lanka Standard Time,Nepal Standard Time,Central Asia Standard Time,Bangladesh Standard Time,Omsk Standard Time,Myanmar Standard Time,SE Asia
Standard Time,Altai Standard Time,W. Mongolia Standard Time,North Asia Standard Time,N. Central Asia Standard Time,Tomsk Standard Time,China Standard
Time,North Asia East Standard Time,Singapore Standard Time,W. Australia Standard Time,Taipei Standard Time,Ulaanbaatar Standard Time,Aus Central W. Standard
Time,Transbaikal Standard Time,Tokyo Standard Time,North Korea Standard Time,Korea Standard Time,Yakutsk Standard Time,Cen. Australia Standard Time,AUS
Central Standard Time,E. Australia Standard Time,AUS Eastern Standard Time,West Pacific Standard Time,Tasmania Standard Time,Vladivostok Standard Time,Lord
Howe Standard Time,Bougainville Standard Time,Russia Time Zone 10,Magadan Standard Time,Norfolk Standard Time,Sakhalin Standard Time,Central Pacific
Standard Time,Russia Time Zone 11,New Zealand Standard Time,UTC+12,Fiji Standard Time,Chatham Islands Standard Time,UTC+13,Tonga Standard Time,Samoa
Standard Time,Line Islands Standard Time">
    <!--ro, opt, enum, subType:string, attr:opt{opt, string}-->Dateline Standard Time
  </windowsZone>
  <isSupportTimeStateParam>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeStateParam?format=json-->true
  </isSupportTimeStateParam>
  <localTimeDST min="0" max="256">
    <!--ro, opt, string, range:[0,256], attr:min{opt, string},max{opt, string}-->test
  </localTimeDST>
  <isSupportStandard>
    <!--ro, opt, bool-->true
  </isSupportStandard>
  <isSupportPTPParams>
    <!--ro, opt, bool, desc:GET/PUT /ISAPI/System/time/PTPParams?format=json-->true
  </isSupportPTPParams>
  <isSupportTimeZoneLock>
    <!--ro, opt, bool, desc:GET /ISAPI/System/time/TimeZoneLock?format=json-->true
  </isSupportTimeZoneLock>
</Time>
```


11.1.5.4 Get device local time

Request URL

GET /ISAPI/System/time/localTime

Query Parameter

None

Request Message

None

Response Message

Binary Data

11.1.5.5 Set device local time

Request URL

PUT /ISAPI/System/time/localTime

Query Parameter

None

Request Message

Binary Data

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to 
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```


11.1.5.6 Set parameters of all NTP servers

Request URL

PUT /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--opt, object, NTP server information-->
    <id>
      <!--req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```


11.1.5.7 Get the parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{req, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType>
      <!--ro, req, enum, NTP server address type, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, NTP server domain name, range:[1,64]-->12345
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IPv4 address, range:[1,32]-->192.168.1.112
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port No., range:[1,65535], desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval>
      <!--ro, opt, int, time synchronization interval, range:[1,10800], unit:min-->1440
    </synchronizeInterval>
    <enabled>
      <!--ro, opt, bool, whether to enable, desc:disabled (by default)-->false
    </enabled>
  </NTPServer>
</NTPServerList>
```


11.1.5.8 Set the parameters of a NTP server

Request URL

PUT /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, NTP server domain name地址去掉, range:[1,64]-->12345
  </hostName>
  <ipAddress>
    <!--opt, string, IPv4 address, range:[1,32], desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, range:[1,128], desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--opt, bool-->false
  </enabled>
</NTPServer>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```


11.1.5.9 Get the parameters of a NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	NTP server No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <addressingFormatType>
    <!--ro, req, enum, IP address type of NTP server, subType:string, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, NTP server domain name, range:[1,64], dep:and,{$.NTPServer.addressingFormatType,eq,hostname}-->xxx12345
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IPv4 address, range:[1,32], dep:and,{$.NTPServer.addressingFormatType,eq,ipAddress}-->192.168.1.112
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, range:[1,128]-->1030:C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port No., range:[1,65535], step:1, desc:port No.-->1
  </portNo>
  <synchronizeInterval>
    <!--ro, opt, int, time synchronization interval, range:[1,10800], step:1, unit:min, desc:NTP time synchronization interval, unit: minute-->1440
  </synchronizeInterval>
  <enabled>
    <!--ro, opt, bool-->false
  </enabled>
  <devicePortNoValid>
    <!--ro, opt, enum, subType:string-->yes
  </devicePortNoValid>
  <portType>
    <!--ro, opt, enum, subType:string-->auto
  </portType>
  <customPortNo>
    <!--ro, opt, int, range:[1,65535], step:1, dep:and,{$.NTPServer.portType,eq,custom}-->1
  </customPortNo>
  <hostNameExampleList>
    <!--ro, opt, array, subType:object-->
    <hostNameExample>
      <!--ro, opt, object-->
      <hostName>
        <!--ro, opt, string-->xxx12345
      </hostName>
    </hostNameExample>
  </hostNameExampleList>
</NTPServer>
```


11.1.5.10 Get the configuration capability of a specific NTP server

Request URL

GET /ISAPI/System/time/ntpServers/<NTPServerID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
NTPServerID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<NTPServer xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, NTP server information, attr:version{req, string, protocolVersion}-->
  <id min="1" max="1">
    <!--ro, req, string, ID, range:[0,1], attr:min{req, int},max{req, int}-->1
  </id>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, IP address type of NTP server, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)--
>hostname
  </addressingFormatType>
  <hostName min="0" max="68">
    <!--ro, opt, string, NTP server domain name, range:[0,68], attr:min{req, int},max{req, int}-->time.windows.com
  </hostName>
  <ipAddress min="0" max="32">
    <!--ro, opt, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
  </ipAddress>
  <ipv6Address min="0" max="128">
    <!--ro, opt, string, IPv6 address, range:[0,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030::C9B4:FF12:48AA:1A2B
  </ipv6Address>
  <portNo min="0" max="65535">
    <!--ro, opt, int, port No., range:[0,65535], attr:min{req, int},max{req, int}, desc:port No.-->123
  </portNo>
  <synchronizeInterval min="0" max="10800">
    <!--ro, opt, int, time synchronization interval, range:[0,10800], unit:min, attr:min{req, int},max{req, int}, desc:NTP time synchronization interval,
unit: minute-->1440
  </synchronizeInterval>
  <enabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </enabled>
  <portType opt="auto,custom" def="auto">
    <!--ro, opt, enum, subType:string, attr:opt{req, string},def{req, string}-->auto
  </portType>
  <customPortNo min="1" max="65535">
    <!--ro, opt, int, range:[0,65535], dep:and,{$.NTPServer.portType,eq,custom}, attr:min{req, int},max{req, int}-->123
  </customPortNo>
</NTPServer>
```

11.1.5.11 Get the configuration capability of parameters of a specific NTP (Network Time Protocol) server

Request URL

GET /ISAPI/System/time/ntpServers/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<NTPServerList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, NTP server information list, subType:object, attr:version{opt, string, protocolVersion}-->
  <NTPServer>
    <!--ro, opt, object, NTP server information-->
    <id>
      <!--ro, req, string, ID-->1
    </id>
    <addressingFormatType opt="ipaddress,hostname">
      <!--ro, req, enum, NTP server address type, subType:string, attr:opt{req, string}, desc:"ipaddress" (IP address), "hostname" (domain name)-->hostname
    </addressingFormatType>
    <hostName min="1" max="64">
      <!--ro, opt, string, NTP server domain name, range:[1,64], attr:min{req, int},max{req, int}-->12345
    </hostName>
    <ipAddress min="1" max="32">
      <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:IPv4 address-->192.168.1.112
    </ipAddress>
    <ipv6Address min="1" max="128">
      <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:IPv6 address-->1030:C9B4:FF12:48AA:1A2B
    </ipv6Address>
    <portNo min="1" max="65535">
      <!--ro, opt, int, port No., range:[1,65535], attr:min{req, int},max{req, int}, desc:the default port No. is 123-->123
    </portNo>
    <synchronizeInterval min="1" max="10800">
      <!--ro, opt, int, time synchronization interval, unit:min, attr:min{req, int},max{req, int}-->1440
    </synchronizeInterval>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable, attr:opt{req, string}, desc:true (enable), false (disable)-->true
    </enabled>
  </NTPServer>
</NTPServerList>
```


11.1.6 设备本地升级

11.1.6.1 Get the device upgrading status and progress

Request URL

GET /ISAPI/System/upgradeStatus?type=<type>&childDevID=<childDevID>&id=<id>

Query Parameter

Parameter Name	Parameter Type	Description
type	string	"usbmic" (array microphone module)
childDevID	string	--
id	string	The id indicates the peripheral No. to be upgraded.

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<upgradeStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, upgrade status and result, attr:version{req, string, protocolVersion}-->
  <upgrading>
    <!--ro, req, bool, upgrade status-->true
  </upgrading>
  <percent>
    <!--ro, req, int, upgrade progress (% complete), range:[0,100]-->1
  </percent>
  <reboot>
    <!--ro, opt, bool, dep:and,{$.upgradeStatus.percent,eq,100}-->false
  </reboot>
  <upgradeStage>
    <!--ro, opt, enum, subType:string-->software
  </upgradeStage>
</upgradeStatus>
```


11.1.7 用户管理

11.1.7.1 Log in to the device by digest

Request URL

GET /ISAPI/Security/userCheck

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<userCheck xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <statusValue>
    <!--ro, req, enum, status code, subType:int, desc:"200"(succeeded), "401"(failed)-->200
  </statusValue>
  <statusString>
    <!--ro, opt, enum, status, subType:string, desc:"OK", "Unauthorized"-->OK
  </statusString>
  <isDefaultPassword>
    <!--ro, opt, bool-->true
  </isDefaultPassword>
  <isRiskPassword>
    <!--ro, opt, bool-->true
  </isRiskPassword>
  <isActivated>
    <!--ro, opt, bool-->true
  </isActivated>
  <residualValidity>
    <!--ro, opt, int-->1
  </residualValidity>
  <lockStatus>
    <!--ro, opt, enum, locking status, subType:string, desc:"unlock", "locked"-->unlock
  </lockStatus>
  <unlockTime>
    <!--ro, opt, int, unlocking remaining time, unit:s-->1
  </unlockTime>
  <retryLoginTime>
    <!--ro, opt, int, remaining login attempts-->1
  </retryLoginTime>
</userCheck>
```

11.1.7.2 Get all users' permissions

Request URL

GET /ISAPI/Security/UserPermission

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<UserPermissionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, list of user permissions, subType:object, attr:version{req, string, protocolVersion}-->
  <UserPermission>
    <!--ro, req, object, user permissions-->
    <id>
      <!--ro, req, string, index, range:[1,32]-->test
    </id>
    <userID>
      <!--ro, req, string, user ID, range:[1,32]-->test
    </userID>
    <userType>
      <!--ro, req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission),
"operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)--
>admin
    </userType>
    <remotePermission>
      <!--ro, opt, object, remote permission node-->
      <record>
        <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
      </record>
      <playBack>
        <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
      </playBack>
      <preview>
        <!--ro, opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
      </preview>
      <ptzControl>
        <!--ro, opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
      </ptzControl>
      <logOrStateCheck>
        <!--ro, opt, bool, whether to enable permission for checking logs and device status-->true
      </logOrStateCheck>
      <parameterConfig>
        <!--ro, opt, bool, whether to enable the permission for remote parameters setting-->true
      </parameterConfig>
      <restartOrShutdown>
        <!--ro, opt, bool, whether to enable the permission to reboot or power off-->true
      </restartOrShutdown>
      <upgrade>
        <!--ro, opt, bool, whether to enable the upgrade permission-->true
      </upgrade>
      <voiceTalk>
        <!--ro, opt, bool, whether to enable the two-way audio permission (of all channels)-->true
      </voiceTalk>
      <transParentChannel>
        <!--ro, opt, bool, whether to enable the transparent channel permission (of all channels), desc:namely, the serial port control-->true
      </transParentChannel>
      <contorlLocalOut>
        <!--ro, opt, bool, whether to enable the local output control permission-->true
      </contorlLocalOut>
      <alarmOutOrUpload>
        <!--ro, opt, bool, whether to enable permission for alarm uploading and output-->true
      </alarmOutOrUpload>
    </remotePermission>
  </UserPermission>
</UserPermissionList>
```

11.1.7.3 Set the user permission of the device

Request URL

PUT /ISAPI/Security/UserPermission

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<UserPermissionList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, list of user permissions, attr:version{req, string, protocolVersion}-->
  <UserPermission>
    <!--req, object, user permissions-->
    <id>
      <!--req, string, index, range:[1,32]-->test
    </id>
    <userID>
      <!--req, string, user ID, range:[1,32]-->test
    </userID>
    <userType>
      <!--req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission), "operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)-->admin
    </userType>
    <remotePermission>
      <!--opt, object, remote permission node-->
      <record>
        <!--opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
      </record>
      <playBack>
        <!--opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
      </playBack>
      <preview>
        <!--opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
      </preview>
      <ptzControl>
        <!--opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
      </ptzControl>
      <logOrStateCheck>
        <!--opt, bool, whether to enable permission for checking logs and device status-->true
      </logOrStateCheck>
      <parameterConfig>
        <!--opt, bool, whether to enable the permission for remote parameters setting-->true
      </parameterConfig>
      <restartOrShutdown>
        <!--opt, bool, whether to enable the permission to reboot or power off-->true
      </restartOrShutdown>
      <upgrade>
        <!--opt, bool, whether to enable the upgrade permission-->true
      </upgrade>
      <voiceTalk>
        <!--opt, bool, whether to enable the two-way audio permission (of all channels)-->true
      </voiceTalk>
      <transParentChannel>
        <!--opt, bool, whether to enable the transparent channel, desc:namely, the serial port control-->true
      </transParentChannel>
      <contorlLocalOut>
        <!--opt, bool, whether to enable the local output control permission-->true
      </contorlLocalOut>
      <alarmOutOrUpload>
        <!--opt, bool, whether to enable permission for alarm uploading and output-->true
      </alarmOutOrUpload>
    </remotePermission>
  </UserPermission>
</UserPermissionList>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <AttachInfo>
    <!--ro, opt, object, additional information about the status of batch operations, desc:this node is returned when the statusCode is 8 (Batch Operation)-->
    <StatusList>
      <!--ro, req, object, status information list-->
      <Status>
        <!--ro, req, object, status information-->
        <index>
          <!--ro, req, int, ID, desc:to distinguish each resource in the batch operation-->0
        </index>
        <statusCode>
          <!--ro, req, enum, status code, subType:int, desc:it only indicates the status of the partition (area) operation, and its return mode is the same with the above nodes; 0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
        </statusCode>
        <statusString>
          <!--ro, req, enum, status information, subType:string, desc:it only indicates the status of the partition (area) operation, and its return mode is the same with the above nodes; "OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
        </statusString>
        <subStatusCode>
          <!--ro, req, string, sub status code, which describes the error in details, range:[1,64], desc:it only indicates the status of the subsystem operation, and its return mode is the same with the above nodes-->ok
        </subStatusCode>
      </Status>
    </StatusList>
  </AttachInfo>
</ResponseStatus>
```


11.1.7.4 Get the permission of a single user

Request URL

GET /ISAPI/Security/UserPermission/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<UserPermission xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user permissions, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, index, range:[1,32]-->test
  </id>
  <userID>
    <!--ro, req, string, user ID, range:[1,32]-->test
  </userID>
  <userType>
    <!--ro, req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission),
"operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)--
>admin
  </userType>
  <remotePermission>
    <!--ro, opt, object, remote permission node-->
    <record>
      <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
    </record>
    <playBack>
      <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--ro, opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
    </preview>
    <ptzControl>
      <!--ro, opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
    </ptzControl>
    <logOrStateCheck>
      <!--ro, opt, bool, whether to enable permission for checking logs and device status-->true
    </logOrStateCheck>
    <parameterConfig>
      <!--ro, opt, bool, whether to enable the configuration permission for remote parameters-->true
    </parameterConfig>
    <restartOrShutdown>
      <!--ro, opt, bool, whether to enable the permission to reboot or power off-->true
    </restartOrShutdown>
    <upgrade>
      <!--ro, opt, bool, whether to enable the upgrade permission-->true
    </upgrade>
    <voiceTalk>
      <!--ro, opt, bool, whether to enable the two-way audio permission (of all channels)-->true
    </voiceTalk>
    <transParentChannel>
      <!--ro, opt, bool, whether to enable the transparent channel permission (of all channels), desc:the serial port control-->true
    </transParentChannel>
    <contorlLocalOut>
      <!--ro, opt, bool, whether to enable the local output control permission-->true
    </contorlLocalOut>
    <alarmOutOrUpload>
      <!--ro, opt, bool, whether to enable permission for alarm uploading and output-->true
    </alarmOutOrUpload>
  </remotePermission>
</UserPermission>
```


11.1.7.5 Set permissions for a single user

Request URL

PUT /ISAPI/Security/UserPermission/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<UserPermission xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user permissions, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, index, range:[1,32]-->test
  </id>
  <userID>
    <!--req, string, user ID, range:[1,32]-->test
  </userID>
  <userType>
    <!--req, enum, user type, subType:string, desc:"admin" (administrator, who has all permissions and can review and edit user's permission), "operator"
(operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user)-->admin
  </userType>
  <remotePermission>
    <!--opt, object, remote permission node-->
    <record>
      <!--opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
    </record>
    <playBack>
      <!--opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
    </preview>
    <ptzControl>
      <!--opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
    </ptzControl>
    <logOrStateCheck>
      <!--opt, bool, whether to enable permission for checking logs and device status-->true
    </logOrStateCheck>
    <parameterConfig>
      <!--opt, bool, whether to enable the permission for remote parameters setting-->true
    </parameterConfig>
    <restartOrShutdown>
      <!--opt, bool, whether to enable the permission to reboot or power off-->true
    </restartOrShutdown>
    <upgrade>
      <!--opt, bool, whether to enable the upgrade permission-->true
    </upgrade>
    <voiceTalk>
      <!--opt, bool, whether to enable the two-way audio permission (of all channels)-->true
    </voiceTalk>
    <transParentChannel>
      <!--opt, bool, whether to enable the transparent channel, desc:namely, the serial port control-->true
    </transParentChannel>
    <contorlLocalOut>
      <!--opt, bool, whether to enable the local output control permission-->true
    </contorlLocalOut>
    <alarmOutOrUpload>
      <!--opt, bool, whether to enable permission for alarm uploading and output-->true
    </alarmOutOrUpload>
  </remotePermission>
</UserPermission>
```


Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.1.7.6 Get the user permission capability of the operator

Request URL

GET /ISAPI/Security/UserPermission/operatorCap

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<UserPermissionCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the capability of user permission, attr:version{req, string, protocolVersion}-->
  <userType opt="admin,operator,viewer,installer,manufacturer,single">
    <!--ro, req, enum, user type, subType:string, attr:opt{req, string}, desc:"admin" (administrator, who has all permissions and can review and edit user's permission), "operator" (operator, who has default permissions), "viewer" (viewer, who has default permissions), "installer", "manufacturer", "single" (single user), "installerAdmin" (installation admin), "installEmployee" (installation employee)-->admin
  </userType>
  <remotePermissionCap>
    <!--ro, req, object, see details in the message of XML_remotePermissionCap-->
    <record>
      <!--ro, opt, bool, whether to enable the recording permission (of all channels), desc:true (enable), false (disable)-->true
    </record>
    <playBack>
      <!--ro, opt, bool, whether to enable the playback permission (of all channels), desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--ro, opt, bool, whether to enable the live view permission (of all channels), desc:true (enable), false (disable)-->true
    </preview>
    <ptzControl>
      <!--ro, opt, bool, whether to enable the PTZ control permission (of all channels), desc:true (enable), false (disable)-->true
    </ptzControl>
    <logOrStateCheck>
      <!--ro, opt, bool, whether to enable permission for checking logs and device status-->true
    </logOrStateCheck>
    <parameterConfig>
      <!--ro, opt, bool, whether to enable the remote parameters setting permission of the device-->true
    </parameterConfig>
    <restartOrShutdown>
      <!--ro, opt, bool, whether to enable the permission to reboot or power off-->true
    </restartOrShutdown>
    <upgrade>
      <!--ro, opt, bool, whether to enable the upgrade permission-->true
    </upgrade>
    <voiceTalk>
      <!--ro, opt, bool, whether to enable the two-way audio permission (of all channels)-->true
    </voiceTalk>
    <transParentChannel>
      <!--ro, opt, bool, whether to enable the transparent channel, desc:namely, the serial port control-->true
    </transParentChannel>
    <contorlLocalOut>
      <!--ro, opt, bool, whether to enable the local output control permission-->true
    </contorlLocalOut>
    <alarmOutOrUpload>
      <!--ro, opt, bool, whether to enable permission for alarm uploading and output-->true
    </alarmOutOrUpload>
  </remotePermissionCap>
</UserPermissionCap>
```


11.1.7.7 Get the user permission capability of the viewer

Request URL

GET /ISAPI/Security/UserPermission/viewerCap

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<UserPermissionCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <userType opt="admin,operator,viewer,installer,manufacturer,single,installerAdmin,installEmployee">
    <!--ro, req, enum, user type, subType:string, attr:opt{req, string}, desc:"admin"-administrator,"operator","viewer","installer","manufacturer"-->admin
  </userType>
  <remotePermissionCap>
    <!--ro, req, object-->
    <record>
      <!--ro, opt, bool, desc:true (enable), false (disable)-->true
    </record>
    <playBack>
      <!--ro, opt, bool, desc:true (enable), false (disable)-->true
    </playBack>
    <preview>
      <!--ro, opt, bool, desc:true (enable), false (disable)-->true
    </preview>
    <ptzControl>
      <!--ro, opt, bool, desc:true (enable), false (disable)-->true
    </ptzControl>
    <logOrStateCheck>
      <!--ro, opt, bool-->true
    </logOrStateCheck>
    <parameterConfig>
      <!--ro, opt, bool-->true
    </parameterConfig>
    <restartOrShutdown>
      <!--ro, opt, bool-->true
    </restartOrShutdown>
    <upgrade>
      <!--ro, opt, bool-->true
    </upgrade>
    <voiceTalk>
      <!--ro, opt, bool-->true
    </voiceTalk>
    <transParentChannel>
      <!--ro, opt, bool-->true
    </transParentChannel>
    <contorlLocalOut>
      <!--ro, opt, bool-->true
    </contorlLocalOut>
    <alarmOutOrUpload>
      <!--ro, opt, bool-->true
    </alarmOutOrUpload>
  </remotePermissionCap>
</UserPermissionCap>
```


11.1.7.8 Get the configuration capability of a specific user

Request URL

GET /ISAPI/Security/users/<indexID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	user (index) ID (it indicates the single ID in the array)

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user, attr:version{req, string, protocolVersion}-->
  <id min="1" max="200">
    <!--ro, req, int, user parameter No., range:[1,200], attr:min{req, int},max{req, int}-->1
  </id>
  <userName min="1" max="32">
    <!--ro, req, string, user name, range:[1,32], attr:min{req, int},max{req, int}-->test
  </userName>
  <password min="1" max="16">
    <!--ro, req, string, current password, range:[1,16], attr:min{req, int},max{req, int}, desc:the password will not be returned-->null
  </password>
  <bondIpAddressList>
    <!--ro, opt, array, list of socket IP address that bound to the user, subType:object-->
    <bondIpAddress>
      <!--ro, opt, object, socket IP address that bound to the user-->
      <id>
        <!--ro, req, int, bounded IP address No., range:[1,16]-->1
      </id>
    </bondIpAddress>
  </bondIpAddressList>
  <bondMacAddressList>
    <!--ro, opt, array, list of MAC address that bound to the user, subType:object-->
    <bondMacAddress>
      <!--ro, opt, object, MAC address that bound to the user-->
      <id>
        <!--ro, req, int, bounded MAC address No., range:[1,16]-->1
      </id>
    </bondMacAddress>
  </bondMacAddressList>
  <userLevel opt="Administrator,Operator,Viewer,installer,manufacturer,single,installerAdmin,installEmployee,Employee,DepartmentManager">
    <!--ro, opt, enum, user type, subType:string, attr:opt{req, string}, desc:"Administrator", "Operator", "Viewer" (viewer, which is not applied on devices currently), "installer", "manufacturer", "single" (single user, which is not applied on devices currently), "installerAdmin" (installation admin), "installEmployee" (installation employee), "Employee" (for attendance), "DepartmentManager" (department manager, for attendance)-->Administrator
  </userLevel>
  <PasswordPeriodPrompt>
    <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password expired shall be prompted during logging-in. password changing is not required in using-->
    <passwordPeriodType opt="permanent,byDate" def="permanent">
      <!--ro, req, enum, notification type of the password validity period, subType:string, attr:opt{req, string},def{req, string}, desc:"permanent" (take effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
    </passwordPeriodType>
    <passwordPeriodbyDate min="1" max="1200" def="90">
      <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and, {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}, attr:min{req, int},max{req, int},def{req, int}-->90
    </passwordPeriodbyDate>
  </PasswordPeriodPrompt>
  <IssuerDN opt="true,false">
    <!--ro, opt, bool, whether it has a bound DN, attr:opt{req, string}, desc:a bound DN refers to a bound server directory.The process of the client sending a request to establish session with the LDAP server is called binding.When binding, the client needs to specify the access account so as to access the directory information on the server.The user searches for contents according to the bound directory.This node will be used only when logging in to the device (GB35114) via pin (UKey)-->true
  </IssuerDN>
</User>
```


11.1.7.9 Get information about a single user

Request URL

GET /ISAPI/Security/users/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, user, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, int, user parameter No.-->1
  </id>
  <enabled>
    <!--ro, opt, bool, whether to enable user-->true
  </enabled>
  <userName>
```

```xml
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
    <phoneNum>
      <!--ro, opt, string, phone number, range:[1,11], desc:the sensitive information should be encrypted-->test
    </phoneNum>
    <emailAddress>
      <!--ro, opt, string, Email Address, range:[1,32], desc:the sensitive information should be encrypted-->test
    </emailAddress>
    <duressPassword>
      <!--ro, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </duressPassword>
    <keypadPassword>
      <!--ro, opt, string, Keypad Password, range:[1,16], desc:the sensitive information should be encrypted. entering password is required when the security
control panel is armed or disarmed via keypad-->test
    </keypadPassword>
    <singleKeypadEnable>
      <!--ro, opt, bool, whether to enable the keypad password for one time-->true
    </singleKeypadEnable>
    <invalidkeypadPassword>
      <!--ro, opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
    </invalidkeypadPassword>
    <userOperateType>
      <!--ro, opt, enum, user type, subType:int,
desc:1 (if userOperateType is 1 or none, password is valid. if method is POST, password is required. otherwise this node is optional)
2 (if userOperateType is 2, keypadPassword is valid. if method is POST, keypadPassword is required. otherwise this node is optional)
3 (if userOperateType is 3,both  passwordkeypad and Password is valid. if method is POST, password and keypadPassword is required. otherwise this node is
optional) 1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
    </userOperateType>
    <bondIpAddressList>
      <!--ro, opt, object, list of socket IP address that bound to the user-->
      <bondIpAddress>
        <!--ro, opt, object, socket IP address that bound to the user-->
        <id>
          <!--ro, req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <ipAddress>
          <!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, bounded IPv6 address, range:[1,128]-->test
        </ipv6Address>
      </bondIpAddress>
    </bondIpAddressList>
    <bondMacAddressList>
      <!--ro, opt, object, list of mac address that bound to the user-->
      <bondMacAddress>
        <!--ro, opt, object, mac address that bound to the user-->
        <id>
          <!--ro, req, int, bounded mac address No., range:[1,16], desc:It starts from 1-->1
        </id>
        <macAddress>
          <!--ro, opt, string, bounded mac address, range:[1,32]-->test
        </macAddress>
      </bondMacAddress>
    </bondMacAddressList>
    <userLevel>
      <!--ro, opt, enum, user level, subType:string, desc:"Administrator", "Operator", "Viewer" (viewer, which is not applied on devices currently),
"installer", "manufacturer", "single" (single user, which is not applied on devices currently), "installerAdmin" (installation admin), "installEmployee"
(installation employee), "Employee" (for attendance), "DepartmentManager" (department manager, for attendance)-->Administrator
    </userLevel>
    <attribute>
      <!--ro, opt, object, user's special attributes-->
      <inherent>
        <!--ro, opt, bool, whether it can be deleted, desc:true: the user cannot be deleted. false / the node is invalid: the user can be deleted-->true
      </inherent>
    </attribute>
    <cardList>
      <!--ro, opt, object, list of cards that belong to the user-->
      <card>
        <!--ro, opt, object, card-->
        <id>
          <!--ro, opt, int, No., range:[1,16]-->1
        </id>
        <name>
          <!--ro, opt, string, name, range:[1,32]-->test
        </name>
      </card>
    </cardList>
    <remoteCtrlList>
      <!--ro, opt, object, list of keyfobs that belong to the user-->
      <remoteCtrl>
        <!--ro, opt, object, keyfob-->
        <id>
          <!--ro, opt, int, No., range:[1,16]-->1
        </id>
        <name>
          <!--ro, opt, string, name, range:[1,32]-->test
        </name>
      </remoteCtrl>
    </remoteCtrlList>
    <userNo>
      <!--ro, opt, int, displayed No. corresponding to the user ID, desc:the No. is used on the App and the Web Client and reporting alarms to ARC. it is
valid to all user types. the format of the No. is 500 + id-->501
    </userNo>
```

```
  <adminType>
    <!--ro, opt, enum, admin type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the administrator account is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN" (user account created on LAN)-->Cloud
  </adminType>
  <installerType>
    <!--ro, opt, enum, installer type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the installer account is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN" (user account created on LAN)-->Cloud
  </installerType>
  <PasswordPeriodPrompt>
    <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password expired shall be prompted during logging-in, but the expired password does not affect the usage-->
    <passwordPeriodType>
      <!--ro, req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired shall be prompted in time, but password changing is not required in using-->permanent
    </passwordPeriodType>
    <passwordPeriodbyDate>
      <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and, {$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
    </passwordPeriodbyDate>
  </PasswordPeriodPrompt>
  <userActivationStatus>
    <!--ro, opt, bool, user activation status, desc:true (activated; default value), false (inactivated)-->true
  </userActivationStatus>
  <invalidDuressPassword>
    <!--ro, opt, bool, whether current duress password is invalid, desc:if the duress password is not configured, the value is true-->true
  </invalidDuressPassword>
  <passwordType>
    <!--ro, opt, enum, subType:string-->simple
  </passwordType>
  <temporaryLocalOperator>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <scheduledOperationList>
      <!--ro, opt, array, subType:object, dep:or,{$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
      <scheduledOperation>
        <!--ro, opt, object-->
        <beginTime>
          <!--ro, req, datetime-->1970-01-01T00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, datetime-->1970-01-01T00:00:00+08:00
        </endTime>
      </scheduledOperation>
    </scheduledOperationList>
    <validPeriodStatus>
      <!--ro, opt, enum, subType:string-->notReached
    </validPeriodStatus>
    <remainingValidTime>
      <!--ro, opt, int, range:[0,720], step:1, unit:h-->1
    </remainingValidTime>
  </temporaryLocalOperator>
  <IssuerDN>
    <!--ro, opt, bool-->true
  </IssuerDN>
</User>
```

11.1.7.10 Set information for a single user

Request URL

PUT /ISAPI/Security/users/<indexID>

Query Parameter

Parameter Name	Parameter Type	Description
indexID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, user parameter No., range:[1,16], desc:supported range: [1,16]-->1
  </id>
  <enabled>
    <!--opt, bool, whether to enable user-->true
  </enabled>
  <userName>
    <!--req, string, user name, range:[1,32]-->test
```

```xml
  </userName>
  <phoneNum>
    <!--wo, opt, string, phone number, range:[1,11], desc:the sensitive information should be encrypted-->test
  </phoneNum>
  <emailAddress>
    <!--wo, opt, string, email address, range:[1,32], desc:the sensitive information should be encrypted-->test
  </emailAddress>
  <password>
    <!--wo, opt, string, password, range:[1,16], desc:the sensitive information should be encrypted-->test
  </password>
  <duressPasswordMode>
    <!--wo, opt, enum, subType:object-->auto
  </duressPasswordMode>
  <duressPassword>
    <!--wo, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
  </duressPassword>
  <keypadPassword>
    <!--wo, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the
security control panel is armed or disarmed via keypad-->test
  </keypadPassword>
  <invalidkeypadPassword>
    <!--opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
  </invalidkeypadPassword>
  <bondIpAddressList>
    <!--opt, object, list of socket IP address that bound to the user-->
    <bondIpAddress>
      <!--opt, object, socket IP address that bound to the user-->
      <id>
        <!--req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
      </id>
      <ipAddress>
        <!--opt, string, bounded IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, bounded IPv6 address, range:[1,32]-->test
      </ipv6Address>
    </bondIpAddress>
  </bondIpAddressList>
  <bondMacAddressList>
    <!--opt, object, list of MAC address that bound to the user-->
    <bondMacAddress>
      <!--opt, object, MAC address that bound to the user-->
      <id>
        <!--req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
      </id>
      <macAddress>
        <!--opt, string, bounded MAC address, range:[1,32]-->test
      </macAddress>
    </bondMacAddress>
  </bondMacAddressList>
  <attribute>
    <!--opt, object, user's special attributes-->
    <inherent>
      <!--opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
    </inherent>
  </attribute>
  <loginPassword>
    <!--req, string, confirm password, range:[1,16], desc:confirming password is required when adding and editing user information. Encrypting the sensitive
information is required when the confirm password is transmitted via HTTP-->King726014
  </loginPassword>
  <cardList>
    <!--opt, object, list of cards that belong to the user-->
    <card>
      <!--opt, object, card-->
      <id>
        <!--opt, int, No., range:[1,16]-->1
      </id>
      <name>
        <!--opt, string, name, range:[1,32]-->test
      </name>
    </card>
  </cardList>
  <remoteCtrlList>
    <!--opt, object, list of keyfobs that belong to the user-->
    <remoteCtrl>
      <!--opt, object, keyfob-->
      <id>
        <!--opt, int, No., range:[1,16]-->1
      </id>
      <name>
        <!--opt, string, name, range:[1,32]-->test
      </name>
    </remoteCtrl>
  </remoteCtrlList>
  <PasswordPeriodPrompt>
    <!--opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
expired shall be prompted during logging-in, but the expired password does not affect the usage-->
    <passwordPeriodType>
      <!--req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by
day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired
shall be prompted in time, but password changing is not required in using-->permanent
    </passwordPeriodType>
    <passwordPeriodbyDate>
```

```xml
        <!--opt, int, validity period of the password taking effect by day, range:[1,1200], unit:大, dep:and,
{$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
      </passwordPeriodbyDate>
    </PasswordPeriodPrompt>
    <temporaryLocalOperator>
      <!--opt, object, information of temporary local operator, desc:the temporary local operator can only log in on local keypad-->
      <scheduledOperationList>
        <!--opt, array, validity period list of the permission of temporary local operator, subType:object, dep:and,
{$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
        <scheduledOperation>
          <!--opt, object, validity period of the permission of temporary local operator-->
          <beginTime>
            <!--req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
          </beginTime>
          <endTime>
            <!--req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
          </endTime>
        </scheduledOperation>
      </scheduledOperationList>
    </temporaryLocalOperator>
    <IssuerDN>
      <!--opt, bool, whether it has a bound DN, desc:a bound DN refers to a bound server directory. The process of the client sending a request to establish
session with the LDAP server is called binding. When binding, the client needs to specify the access account so as to access the directory information on
the server. The user searches for contents according to the bound directory. N/A-->true
    </IssuerDN>
    <userGroupID>
      <!--opt, int, user group ID, range:[0,3], desc:if this node does not exist or its value is 0, it indicates that there are no groups. As for the security
inspection, the upper-level software supports dividing users into several groups and delivering the group IDs to them, which is managed by the device.-->0
    </userGroupID>
    <facePicture>
      <!--opt, object-->
      <filePathType>
        <!--req, enum, subType:string-->multipart
      </filePathType>
      <filePath>
        <!--opt, string, range:[0,262000]-->test
      </filePath>
    </facePicture>
  </User>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
User	[Message content]	application/xml	--	--	--
facePicture	[Binary picture data]	image/jpeg	facePicture	facePicture.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, 
used for fast debugging-->badXmlFormat
  </description>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"lock", "unlock". It is returned when verifying loginPassword failed-->lock
  </lockStatus>
  <resLockTime>
    <!--ro, opt, int, range:[0,1800], unit:s, dep:or,{$.ResponseStatus.lockStatus,eq,lock}-->1
  </resLockTime>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying loginPassword failed-->1
  </retryTimes>
  <lockType>
    <!--ro, opt, enum, subType:string, dep:or,{$.ResponseStatus.lockStatus,eq,lock}-->lockIP
  </lockType>
</ResponseStatus>
```

11.1.7.11 Delete all users

Request URL

DELETE /ISAPI/Security/users?loginPassword=<loginPassword>

Query Parameter

Parameter Name	Parameter Type	Description
loginPassword	string	The password length cannot exceed 16.Confirming password is required when deleting user information. It is required to encrypt the sensitive information when the confirmed password is transmitted via HTTP. It is also required due to the security upgrade.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"lock", "unlock". It is returned when verifying loginPassword failed-->lock
  </lockStatus>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying loginPassword failed-->1
  </retryTimes>
  <resLockTime>
    <!--ro, opt, int, range:[0,1800], unit:s, dep:and,{$.ResponseStatus.lockStatus,eq,lock}-->1
  </resLockTime>
  <lockType>
    <!--ro, opt, enum, subType:string, dep:or,{$.ResponseStatus.lockStatus,eq,lock}-->lockIP
  </lockType>
</ResponseStatus>
```

11.1.7.12 Add a user

Request URL

POST /ISAPI/Security/users

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<User xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, user ID, range:[1,64]-->1
  </id>
  <userName>
    <!--req, string, user name, range:[1,32]-->test
  </userName>
  <password>
    <!--wo, req, string, password, range:[1,16], desc:the sensitive information should be encrypted-->test
  </password>
  <duressPasswordMode>
    <!--wo, opt, enum, subType:object-->auto
  </duressPasswordMode>
  <duressPassword>
    <!--wo, opt, string, range:[1,16]-->1
  </duressPassword>
  <keypadPassword>
    <!--wo, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the 
security control panel is armed or disarmed via keypad-->test
  </keypadPassword>
  <singleKeypadEnable>
    <!--opt, bool, whether to enable single keypad operation, desc:true (the password applied to the device is valid only once, and will become invalid when 
the user logs out)-->true
  </singleKeypadEnable>
  <bondIpAddressList>
    <!--opt, object, list of socket IP address that bound to the user-->
    <bondIpAddress>
      <!--opt, object, socket IP address that bound to the user-->
      <id>
        <!--req, int, bounded IP address No., range:[1,16]-->1
      </id>
      <ipAddress>
        <!--opt, string, bounded IPv4 address, range:[1,32]-->test
      </ipAddress>
      <ipv6Address>
        <!--opt, string, bounded IPv6 address, range:[1,32]-->test
      </ipv6Address>
    </bondIpAddress>
```

```
      </bondIpAddress>
    </bondIpAddressList>
    <bondMacAddressList>
      <!--opt, object, list of MAC address that bound to the user-->
      <bondMacAddress>
        <!--opt, object, MAC address that bound to the user-->
        <id>
          <!--req, int, bounded MAC address No., range:[1,16]-->1
        </id>
        <macAddress>
          <!--opt, string, bounded MAC address, range:[1,32]-->test
        </macAddress>
      </bondMacAddress>
    </bondMacAddressList>
    <userLevel>
      <!--opt, enum, user level, subType:string, desc:"Administrator", "Operator", "Viewer", "installer", "manufacturer", "single" (single user),
"installerAdmin" (installation admin), "installEmployee" (installation employee)-->Administrator
    </userLevel>
    <attribute>
      <!--opt, object, user's special attributes-->
      <inherent>
        <!--opt, bool, whether the user cannot be deleted, desc:true (the user cannot be deleted), false / the node is not supported (the user can be
deleted)-->true
      </inherent>
    </attribute>
    <loginPassword>
      <!--req, string, confirm password, range:[1,16], desc:confirming password is required when adding and editing user information. It is required to
encrypt the sensitive information when the confirmed password is transmitted via HTTP-->King726014
    </loginPassword>
    <PasswordPeriodPrompt>
      <!--opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
expired shall be prompted during logging-in, but the expired password does not affect the usage-->
      <passwordPeriodType>
        <!--req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect by
day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired
shall be prompted in time, but password changing is not required in using-->permanent
      </passwordPeriodType>
      <passwordPeriodbyDate>
        <!--opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
{$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
      </passwordPeriodbyDate>
    </PasswordPeriodPrompt>
    <userID>
      <!--opt, string, user No., range:[1,64], desc:UUID (unique ID of the user)-->test
    </userID>
    <temporaryLocalOperator>
      <!--opt, object, information of temporary local operator, desc:the temporary local operator can only log in on local keypad-->
      <enabled>
        <!--req, bool, whether to enable the permission of temporary local operator-->true
      </enabled>
      <scheduledOperationList>
        <!--opt, array, validity period list of the permission of temporary local operator, subType:object-->
        <scheduledOperation>
          <!--opt, object, validity period of the permission of temporary local operator-->
          <beginTime>
            <!--req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
          </beginTime>
          <endTime>
            <!--req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
          </endTime>
        </scheduledOperation>
      </scheduledOperationList>
    </temporaryLocalOperator>
    <userGroupID>
      <!--opt, int, user group ID, range:[0,3], desc:no group if this node does not exist or its value is 0. As for the security inspection, the upper-level
software supports dividing users into several groups and delivering the group IDs to them, which is managed by the device-->0
    </userGroupID>
    <facePicture>
      <!--opt, object-->
      <filePathType>
        <!--req, enum, subType:string-->multipart
      </filePathType>
      <filePath>
        <!--opt, string, range:[0,262000]-->test
      </filePath>
    </facePicture>
  </User>
```


Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
User	[Message content]	application/xml	--	--	--
facePicture	[Binary picture data]	image/jpeg	facePicture	facePicture.jpg	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot Required" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, 
used for fast debugging-->badXmlFormat
  </description>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"lock", "unlock". It is returned when verifying loginPassword failed-->lock
  </lockStatus>
  <unlockTime>
    <!--ro, opt, int, range:[0,3600], unit:s, dep:or,{$.ResponseStatus.lockStatus,eq,lock}-->1
  </unlockTime>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying loginPassword failed. Each time the password login fails, it will 
minus 1, until it reaches 0 and locks the device-->1
  </retryTimes>
</ResponseStatus>
```

11.1.7.13 Get information about all users

Request URL

GET /ISAPI/Security/users

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<UserList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, user list, subType:object, attr:version{req, string, protocolVersion}-->
  <User>
    <!--ro, opt, object, user-->
    <id>
      <!--ro, req, int, user parameter No., range:[1,200]-->1
    </id>
    <enabled>
```

```
      <!--ro, opt, bool, whether to enable user-->true
    </enabled>
    <userName>
      <!--ro, req, string, user name, range:[1,32]-->test
    </userName>
    <userNickName>
      <!--ro, opt, string, user's nickname, range:[1,32], desc:userNickName is the user's nickname on the Hik-Connect client, and the user can modify it.
The web client can get userNickName via the device-->test
    </userNickName>
    <phoneNum>
      <!--ro, opt, string, phone number, range:[1,13], desc:the sensitive information should be encrypted-->test
    </phoneNum>
    <emailAddress>
      <!--ro, opt, string, email address, range:[1,32], desc:the sensitive information should be encrypted-->test
    </emailAddress>
    <duressPassword>
      <!--ro, opt, string, duress password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </duressPassword>
    <keypadPassword>
      <!--ro, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the
security control panel is armed or disarmed via keypad-->test
    </keypadPassword>
    <singleKeypadEnable>
      <!--ro, opt, bool, whether to enable single keypad operation-->true
    </singleKeypadEnable>
    <invalidkeypadPassword>
      <!--ro, opt, bool, whether the current keypad password is invalid, desc:it is valid after setting singleKeypadEnable-->true
    </invalidkeypadPassword>
    <userOperateType>
      <!--ro, opt, enum, user operation type, subType:int,
desc:1 (if userOperateType is 1 or none, password is valid. if method is POST, password is required. otherwise this node is optional)
2 (if userOperateType is 2, keypadPassword is valid. if method is POST, keypadPassword is required. otherwise this node is optional)
3 (if userOperateType is 3,both  passwordkeypad and Password is valid. if method is POST, password and keypadPassword is required. otherwise this node is
optional) 1 (network user), 2 (keypad user), 3 (network user and keypad user)-->1
    </userOperateType>
    <bondIpAddressList>
      <!--ro, opt, object, list of socket IP address that bound to the user-->
      <bondIpAddress>
        <!--ro, opt, object, socket IP address that bound to the user-->
        <id>
          <!--ro, req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <ipAddress>
          <!--ro, opt, string, bounded IPv4 address, range:[1,32]-->test
        </ipAddress>
        <ipv6Address>
          <!--ro, opt, string, bounded IPv6 address, range:[1,128]-->test
        </ipv6Address>
      </bondIpAddress>
    </bondIpAddressList>
    <bondMacAddressList>
      <!--ro, opt, object, list of MAC address that bound to the user-->
      <bondMacAddress>
        <!--ro, opt, object, MAC address that bound to the user-->
        <id>
          <!--ro, req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <macAddress>
          <!--ro, opt, string, bounded MAC address, range:[1,32]-->test
        </macAddress>
      </bondMacAddress>
    </bondMacAddressList>
    <userLevel>
      <!--ro, opt, enum, user level, subType:string, desc:"Administrator", "Operator", "Viewer", "installer", "manufacturer", "single" (single user),
"installerAdmin" (installation admin), "installEmployee" (installation employee), "Employee", "DepartmentManager"-->Administrator
    </userLevel>
    <attribute>
      <!--ro, opt, object, user's special attributes-->
      <inherent>
        <!--ro, opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false / the node is invalid (the user can be deleted)-->true
      </inherent>
    </attribute>
    <cardList>
      <!--ro, opt, object, list of cards that belong to the user-->
      <card>
        <!--ro, opt, object, card-->
        <id>
          <!--ro, opt, int, No., range:[1,16]-->1
        </id>
        <name>
          <!--ro, opt, string, name, range:[1,32]-->test
        </name>
      </card>
    </cardList>
    <remoteCtrlList>
      <!--ro, opt, object, list of keyfobs that belong to the user-->
      <remoteCtrl>
        <!--ro, opt, object, keyfob-->
        <id>
          <!--ro, opt, int, No., range:[1,16]-->1
        </id>
        <name>
          <!--ro, opt, string, name, range:[1,32]-->test
```

```xml
        </name>
      </remoteCtrl>
    </remoteCtrlList>
    <userNo>
      <!--ro, opt, int, displayed No. corresponding to the user ID, desc:the No. is used on the App and the Web Client and reporting alarms to ARC. it is
valid to all user types. the format of the No. is 500 + id-->501
    </userNo>
    <adminType>
      <!--ro, opt, enum, admin type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the administrator account
is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
(user account created on LAN)-->Cloud
    </adminType>
    <installerType>
      <!--ro, opt, enum, installer type, subType:string, desc:for hybrid security control panel, it is used for distinguishing whether the installer account
is created on the cloud or in the local area network. It is displayed both on the App and the Web Client. "Cloud" (user account created on the cloud), "LAN"
(user account created on LAN)-->Cloud
    </installerType>
    <PasswordPeriodPrompt>
      <!--ro, opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
expired shall be prompted during logging-in, but the expired password does not affect the usage-->
      <passwordPeriodType>
        <!--ro, req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take
effect by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password
expired shall be prompted in time, but password changing is not required in using-->permanent
      </passwordPeriodType>
      <passwordPeriodbyDate>
        <!--ro, opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
{$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
      </passwordPeriodbyDate>
    </PasswordPeriodPrompt>
    <userActivationStatus>
      <!--ro, opt, bool, user activation status, desc:true (activated; default value), false (inactivated)-->true
    </userActivationStatus>
    <invalidDuressPassword>
      <!--ro, opt, bool, whether current duress password is invalid, desc:if the duress password is not configured, the value is true-->true
    </invalidDuressPassword>
    <userID>
      <!--ro, opt, string, user No., range:[1,64], desc:UUID (unique ID of the user)-->test
    </userID>
    <passwordType>
      <!--ro, opt, enum, password type, subType:string,
desc:1. Attendance check devices support setting simple password and complex password. The password is used for device activation and login.
2. The password should contain 6 digits. "simple", "complex"-->simple
    </passwordType>
    <temporaryLocalOperator>
      <!--ro, opt, object, information of temporary local operator, desc:the temporary local operator can only log in on local keypad-->
      <enabled>
        <!--ro, req, bool, whether to enable the permission of temporary local operator-->true
      </enabled>
      <scheduledOperationList>
        <!--ro, opt, array, validity period list of the permission of temporary local operator, subType:object, dep:or,
{$.UserList.User.temporaryLocalOperator.enabled,eq,true}-->
        <scheduledOperation>
          <!--ro, opt, object, validity period of the permission of temporary local operator-->
          <beginTime>
            <!--ro, req, datetime, start time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
          </beginTime>
          <endTime>
            <!--ro, req, datetime, end time of the validity period (UTC time)-->1970-01-01T00:00:00+08:00
          </endTime>
        </scheduledOperation>
      </scheduledOperationList>
      <validPeriodStatus>
        <!--ro, opt, enum, validity period status, subType:string, desc:"notReached" (the validity period has not started), "available" (within the validity
period), "expired"-->notReached
      </validPeriodStatus>
      <remainingValidTime>
        <!--ro, opt, int, remaining validity period, range:[0,720], step:1, unit:h-->1
      </remainingValidTime>
    </temporaryLocalOperator>
    <userGroupID>
      <!--ro, opt, int, user group ID, range:[0,3], desc:if this node does not exist or its value is 0, it indicates that there are no groups. As for the
security inspection, the upper-level software supports dividing users into several groups and delivering the group IDs to them, which is managed by the
device.-->0
    </userGroupID>
  </User>
</UserList>
```


11.1.7.14 Set information about all users

Request URL

PUT /ISAPI/Security/users

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<UserList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, user list, attr:version{req, string, protocolVersion}-->
  <User>
    <!--opt, object, user-->
    <id>
      <!--req, int, user parameter No., range:[1,64]-->1
    </id>
    <enabled>
      <!--opt, bool, whether to enable user-->true
    </enabled>
    <userName>
      <!--req, string, user name, range:[1,32]-->test
    </userName>
    <password>
      <!--wo, req, string, password, range:[1,16], desc:the sensitive information should be encrypted-->test
    </password>
    <keypadPassword>
      <!--wo, opt, string, keypad password, range:[1,16], desc:the sensitive information should be encrypted. The keypad password is required when the
security control panel is armed or disarmed via keypad-->test
    </keypadPassword>
    <userOperateType>
      <!--opt, enum, user type, subType:int, desc:1 (network user), 2 (keypad user), 3 (network and keypad user). When userOperateType is 1 or none, the
node password is valid and it is required when the method is POST. When userOperateType is 2, the node keypadPassword is valid and it is required when the
method is POST. When userOperateType is 3, both nodes password and keypadPassword are valid and they are required when the method is POST.-->1
    </userOperateType>
    <bondIpAddressList>
      <!--opt, object, list of socket IP address that bound to the user-->
      <bondIpAddress>
        <!--opt, object, socket IP address that bound to the user-->
        <id>
          <!--req, int, bounded IP address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <ipAddress>
          <!--opt, string, bounded IPv4 address, range:[1,32]-->test
        </ipAddress>
        <ipv6Address>
          <!--opt, string, bounded IPv6 address, range:[1,32]-->test
        </ipv6Address>
      </bondIpAddress>
    </bondIpAddressList>
    <bondMacAddressList>
      <!--opt, object, list of MAC address that bound to the user-->
      <bondMacAddress>
        <!--opt, object, MAC address that bound to the user-->
        <id>
          <!--req, int, bounded MAC address No., range:[1,16], desc:it starts from 1-->1
        </id>
        <macAddress>
          <!--opt, string, bounded MAC address, range:[1,32]-->test
        </macAddress>
      </bondMacAddress>
    </bondMacAddressList>
    <userLevel>
      <!--opt, enum, user level, subType:string, desc:"Administrator", "Operator", "Viewer", "installer", "manufacturer", "single" (single user),
"installerAdmin" (installation admin), "installEmployee" (installation employee)-->Administrator
    </userLevel>
    <attribute>
      <!--opt, object, user's special attributes-->
      <inherent>
        <!--opt, bool, whether it can be deleted, desc:true (the user cannot be deleted), false/the node is invalid (the user can be deleted)-->true
      </inherent>
    </attribute>
    <loginPassword>
      <!--req, string, confirm password, range:[1,16], desc:confirming password is required when adding and editing user information. Encrypting the
sensitive information is required when the confirm password is transmitted via HTTP-->King726014
    </loginPassword>
    <PasswordPeriodPrompt>
      <!--opt, object, configure notifications of the password validity period, desc:if the password validity period expired, a notification of password
expired shall be prompted during logging-in. password changing is not required in using-->
      <passwordPeriodType>
        <!--req, enum, notification type of the password validity period, subType:string, desc:"permanent" (take effect permanently), "byDate" (take effect
by day). Take effect permanently: notifications of the password expired shall not prompted permanently. Take effect by day: notification of password expired
shall be prompted in time, but password changing is not required in using-->permanent
      </passwordPeriodType>
      <passwordPeriodbyDate>
        <!--opt, int, validity period of the password taking effect by day, range:[1,1200], unit:天, dep:and,
{$.User.PasswordPeriodPrompt.passwordPeriodType,eq,byDate}-->90
      </passwordPeriodbyDate>
    </PasswordPeriodPrompt>
    <userID>
      <!--opt, string, user No., range:[1,64], desc:UUID (unique ID of the user)-->test
    </userID>
  </User>
</UserList>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <lockStatus>
    <!--ro, opt, enum, lock status, subType:string, desc:"lock", "unlock". It is returned when verifying loginPassword failed-->lock
  </lockStatus>
  <retryTimes>
    <!--ro, opt, int, remaining attempts, range:[0,6], desc:it is returned when verifying loginPassword failed-->1
  </retryTimes>
</ResponseStatus>
```

11.2 网络配置

11.2.1 设备监听管理

11.2.1.1 Check whether the HTTP listening server is working normally

Request URL

POST /ISAPI/Event/notification/httpHosts/<hostID>/test

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, event subscription parameters, see details in the message of XML_SubscribeEvent, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, listening host ID-->test
  </id>
  <url>
    <!--req, string, URL-->test
  </url>
  <protocolType>
    <!--req, enum, protocol type, subType:string, desc:"HTTP,HTTPS,EHome". When the value of this node is EHome,this message can be used to configure the information (such as IP address and port No.) for the AMS (Alarm Management Server) of ISUP 5.0 to listen on. For communication via ISUP 5.0 (MQTT),the listening parameters can be configured and obtained; For other communication methods (such as communication via HTTP,HTTPS,EZVIZ Open Platform,etc.),the listening parameters can only be obtained and cannot be configured-->HTTPS
  </protocolType>
  <parameterFormatType>
    <!--req, enum, alarm/event information format, subType:string, desc:"XML,JSON"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"ipaddress, hostname"-->ipaddress
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostname}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IP address, dep:and,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}-->test
  </ipAddress>
  <ipv6Address>
    <!--opt, string, IPv6 address, dep:and,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}-->test
  </ipv6Address>
  <portNo>
    <!--opt, int, port-->1
  </portNo>
  <userName>
    <!--opt, string, user name-->test
  </userName>
  <password>
    <!--opt, string, password-->test
```

```xml
    </password>
    <httpAuthenticationMethod>
      <!--req, enum, HTTP authentication method, subType:string, desc:"MD5digest,none"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--opt, object-->
      <detectionUpLoadPicturesType>
        <!--opt, enum, subType:string-->all
      </detectionUpLoadPicturesType>
      <videoUploadEnabled>
        <!--opt, bool-->false
      </videoUploadEnabled>
    </ANPR>
    <Extensions>
      <!--opt, object-->
      <intervalBetweenEvents>
        <!--opt, int-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--opt, enum, subType:string-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--opt, bool-->true
    </httpBroken>
    <SubscribeEvent>
      <!--opt, object-->
      <heartbeat>
        <!--opt, int, heartbeat interval time, unit:s, desc:heartbeat interval time-->1
      </heartbeat>
      <eventMode>
        <!--req, enum, "all, list", subType:string, desc:"all,list"-->all
      </eventMode>
      <EventList>
        <!--opt, array, event list, it is valid only when eventMode is "list", subType:object-->
        <Event>
          <!--opt, object-->
          <type>
            <!--req, string, type, desc:type-->test
          </type>
          <minorAlarm>
            <!--opt, string, alarm sub type, desc:alarm sub type-->0x400,0x401,0x402,0x403
          </minorAlarm>
          <minorException>
            <!--opt, string, exception sub type, desc:exception sub type-->0x400,0x401,0x402,0x403
          </minorException>
          <minorOperation>
            <!--opt, string, operation sub type, desc:operation sub type-->0x400,0x401,0x402,0x403
          </minorOperation>
          <minorEvent>
            <!--opt, string-->0x01,0x02,0x03,0x04
          </minorEvent>
          <pictureURLType>
            <!--opt, enum, alarm picture format, subType:string, desc:"binary" (binary), "localURL" (device local URL), "cloudStorageURL" (cloud storage URL)-
->binary
          </pictureURLType>
          <channels>
            <!--opt, string, channel No.-->test
          </channels>
        </Event>
      </EventList>
      <channels>
        <!--opt, string, "1,2,3,4..."-->test
      </channels>
      <pictureURLType>
        <!--opt, enum, alarm picture format, subType:string, desc:alarm picture format-->binary
      </pictureURLType>
      <ChangedUploadSub>
        <!--opt, object, subscribe to messages-->
        <interval>
          <!--opt, int, the lifecycle of arm GUID, unit:s, desc:the lifecycle of arm GUID-->1
        </interval>
        <StatusSub>
          <!--opt, object, status-->
          <all>
            <!--opt, bool, whether to subscribe to all-->true
          </all>
          <channel>
            <!--opt, bool, subscribe to channel status, desc:reporting is not required if all is true-->true
          </channel>
          <hd>
            <!--opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
          </hd>
          <capability>
            <!--opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
          </capability>
        </StatusSub>
      </ChangedUploadSub>
    </SubscribeEvent>
  </HttpHostNotification>
```

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostTestResult xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <errorDescription>
    <!--ro, req, enum, error description, subType:string, desc:error description-->ok
  </errorDescription>
</HttpHostTestResult>
```


11.2.1.2 Set the parameters of a listening host

Request URL

PUT /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	Security control panel ID, which is returned by the URL (POST /ISAPI/Event/notification/httpHosts?security=&iv=).

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>
<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, listening host, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, listening host ID, range:[1,10]-->test
  </id>
  <url>
    <!--req, string, URL-->test
  </url>
  <protocolType>
    <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
  </protocolType>
  <parameterFormatType>
    <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
  </addressingFormatType>
  <hostName>
    <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
  </hostName>
  <ipAddress>
    <!--opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
  </ipAddress>
  <portNo>
    <!--opt, int, port number-->1
  </portNo>
  <ANPR>
    <!--opt, object, ANPR-->
    <detectionUpLoadPicturesType>
      <!--opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture", "detectionPicture"(detected picture)-->all
    </detectionUpLoadPicturesType>
  </ANPR>
</HttpHostNotification>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK"(succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot"(reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:the custom error message description returned by the application is used to 
quickly identify and evaluate issues-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:N/A-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

11.2.1.3 Get the parameters of a listening host

Request URL

GET /ISAPI/Event/notification/httpHosts/<hostID>

Query Parameter

Parameter Name	Parameter Type	Description
hostID	string	Security control panel ID, which is returned by the URL (POST /ISAPI/Event/notification/httpHosts?security=&iv=).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotification xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, listening host, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string, listening host ID, range:[1,10]-->test
  </id>
  <url>
    <!--ro, req, string, URL-->test
  </url>
  <protocolType>
    <!--ro, req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocolType>
  <parameterFormatType>
    <!--ro, req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
  </parameterFormatType>
  <addressingFormatType>
    <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
  </addressingFormatType>
  <hostName>
    <!--ro, opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
  </hostName>
  <ipAddress>
    <!--ro, opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IP address-->test
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:IPv6 address-->test
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, port number-->1
  </portNo>
  <userName>
    <!--ro, opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
  </userName>
```

```xml
  <httpAuthenticationMethod>
    <!--ro, req, enum, authentication method, subType:string, desc:"MD5digest" (MD5), "none", "base64"-->MD5digest
  </httpAuthenticationMethod>
  <ANPR>
    <!--ro, opt, object, ANPR-->
    <detectionUpLoadPicturesType>
      <!--ro, opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture" (license plate picture), "detectionPicture" (detected picture)-->all
    </detectionUpLoadPicturesType>
    <videoUploadEnabled>
      <!--ro, opt, bool, whether to upload violation pre-records, desc:when it is enabled, the device will upload the stored video via the "relationVideo" event-->false
    </videoUploadEnabled>
  </ANPR>
  <Extensions>
    <!--ro, opt, object, range-->
    <intervalBetweenEvents>
      <!--ro, opt, int, event interval-->1
    </intervalBetweenEvents>
  </Extensions>
  <uploadImagesDataType>
    <!--ro, opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
  </uploadImagesDataType>
  <httpBroken>
    <!--ro, opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
  </httpBroken>
  <SubscribeEvent>
    <!--ro, opt, object, picture uploading modes of all events which contain pictures-->
    <heartbeat>
      <!--ro, opt, int, heartbeat interval time-->30
    </heartbeat>
    <eventMode>
      <!--ro, req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
    </eventMode>
    <EventList>
      <!--ro, opt, array, event list, subType:object-->
      <Event>
        <!--ro, opt, object, channel information linked to event-->
        <type>
          <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system), "ADASAlarm" (advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent" (access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection), "PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving", "advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection), "blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis", "databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access), "ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection), "linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection), "peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting", "personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection", "radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection), "raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance", "regionExiting", "retention" (people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection), "scenechangedetection", "sensorAlarm" (angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm), "shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control), "CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent", "TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison" (human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
        </type>
        <minorAlarm>
          <!--ro, opt, string, alarm sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
        </minorAlarm>
        <minorException>
          <!--ro, opt, string, minor exception type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
        </minorException>
        <minorOperation>
          <!--ro, opt, string, minor operation type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
        </minorOperation>
        <minorEvent>
          <!--ro, opt, string, minor event type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
        </minorEvent>
        <pictureURLType>
          <!--ro, opt, enum, alarm picture format of the specified event, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL)-->binary
        </pictureURLType>
```

```xml
        <channels>
          <!--ro, opt, string, listen to the events on the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
        </channels>
      </Event>
    </EventList>
    <channels>
      <!--ro, opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. If some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
    </channels>
    <pictureURLType>
      <!--ro, opt, enum, alarm picture format, subType:string, desc:"binary", "localURL" (local URL), "cloudStorageURL" (cloud storage URL), "EZVIZURL" (EZ URL). The node indicates the upload mode of all event pictures. If the node is applied, <pictureURLType> of <Event> will be invalid. If the node is not applied, the pictures are uploaded in the default mode. The default data type of uploaded pictures for front-end devices is binary, and for back-end devices is local URL of the device-->binary
    </pictureURLType>
    <ChangedUploadSub>
      <!--ro, opt, object, subscribe to messages-->
      <interval>
        <!--ro, opt, int, the lifecycle of arming GUID, desc:within the interval, if the client software does not reconnect to the device, a new GUID will be generated by the device-->5
      </interval>
      <StatusSub>
        <!--ro, opt, object, sub status-->
        <all>
          <!--ro, opt, bool, whether to subscribe to all-->true
        </all>
        <channel>
          <!--ro, opt, bool, channel subscription status (whether the channel is subscribed), desc:it is not required if the value of <all> is true-->true
        </channel>
        <hd>
          <!--ro, opt, bool, HDD subscription status (whether the HDD is subscribed), desc:it is not required if the value of <all> is true-->true
        </hd>
        <capability>
          <!--ro, opt, bool, subscription status of capability set change (whether the capability set change is subscribed), desc:it is not required if the value of <all> is true-->true
        </capability>
      </StatusSub>
    </ChangedUploadSub>
  </SubscribeEvent>
  <PackingSpaceRecognition>
    <!--ro, opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event: PackingSpaceRecognition-->
    <upLoadPicturesType>
      <!--ro, req, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes"(upload specified types of pictures), "notUpload"(not upload pictures)-->all
    </upLoadPicturesType>
    <PicturesTypes>
      <!--ro, opt, array, specified list of uploaded picture types, subType:object, dep:and,{$.HttpHostNotification.PackingSpaceRecognition.upLoadPicturesType,eq,picturesTypes}-->
      <picturesType>
        <!--ro, opt, enum, uploaded picture type, subType:string, desc:"backgroundImage"(captured background picture), "plateImage"(license plate thumbnail)-->backgroundImage
      </picturesType>
    </PicturesTypes>
  </PackingSpaceRecognition>
  <enabled>
    <!--ro, opt, bool, whether to enable-->true
  </enabled>
  <netWork>
    <!--ro, opt, enum, network, subType:int, desc:1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
  </netWork>
  <method>
    <!--ro, opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
  </method>
  <pictureAttachEnabled>
    <!--ro, opt, bool, whether to enable attaching pictures, desc:true by default-->true
  </pictureAttachEnabled>
  <contentType>
    <!--ro, opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not limited to pictures only-->multipart
  </contentType>
  <eventTemplateID>
    <!--ro, opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-->1
  </eventTemplateID>
  <customTriggersEnabled>
    <!--ro, opt, bool, whether to enable custom linkage, desc:true (listening service's linkage method is unaffected by upload center's linkage method, that is, independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->false
  </customTriggersEnabled>
  <checkResponseEnabled>
    <!--ro, opt, bool, whether to enable verifying listening host response, desc:when a device uploads an event to the listening host, the host should respond with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms' listening services fail to respond. This parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response from the listening host. When enabled, failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the event-->false
  </checkResponseEnabled>
</HttpHostNotification>
```

11.2.1.4 Get the capabilities of listening hosts parameters

Request URL

GET /ISAPI/Event/notification/httpHosts/capabilities?type=<type>

Query Parameter

Parameter Name	Parameter Type	Description
type	enum	custom refers to that the listening service is independent and unaffected by the upload center linkage method. default refers to that the listening service is controlled by the linkage method of the uploading center. When the type is empty or default, the listening service is the default.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<HttpHostNotificationCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, capabilities of subscribing to event types and event channels, attr:version{req, string, protocolVersion}-->
  <hostNumber>
    <!--ro, req, int, the number of listening hosts-->0
  </hostNumber>
  <urlLen min="1" max="10">
    <!--ro, req, string, URL length, attr:min{opt, int},max{req, int}-->test
  </urlLen>
  <url>
    <!--ro, opt, string, URL-->test
  </url>
  <protocolType opt="HTTP,HTTPS,EHome">
    <!--ro, req, enum, protocol type, subType:string, attr:opt{req, string}, desc:"HTTP", "HTTPS", "EHome"-->HTTP
  </protocolType>
  <parameterFormatType opt="XML,querystring,JSON">
    <!--ro, req, enum, alarm parameter format type, subType:string, attr:opt{req, string}, desc:"xml" (XML format), "querystring", "json" (JSON format)--
>XML
  </parameterFormatType>
  <addressingFormatType opt="ipaddress,hostname">
    <!--ro, req, enum, address format type, subType:string, attr:opt{req, string}, desc:"ipaddress", "hostname"-->ipaddress
  </addressingFormatType>
  <hostName min="1" max="64">
    <!--ro, opt, string, domain name, attr:min{req, int},max{req, int}, desc:it is valid when addressingFormatType is "hostname"-->test
  </hostName>
  <ipAddress opt="ipv4,ipv6">
    <!--ro, opt, string, IP address, attr:opt{req, string}, desc:it is valid when addressingFormatType is "ipaddress"-->test
  </ipAddress>
  <portNo min="0" max="10">
    <!--ro, opt, string, port number, range:[0,65535], attr:min{req, int},max{req, int}-->1
  </portNo>
  <userNameLen min="0" max="10">
    <!--ro, opt, string, user name length, attr:min{req, int},max{req, int}-->1
  </userNameLen>
  <passwordLen min="0" max="10">
    <!--ro, opt, string, password length, attr:min{req, int},max{req, int}-->1
  </passwordLen>
  <httpAuthenticationMethod opt="MD5digest,none,base64,basic" def="MD5digest">
    <!--ro, req, enum, HTTP authentication method, subType:string, attr:opt{req, string},def{opt, string}, desc:"MD5digest"(MD5), "none", "base64"--
>MD5digest
  </httpAuthenticationMethod>
  <ANPR>
    <!--ro, opt, object, ANPR-->
    <detectionUpLoadPicturesType opt="all,licensePlatePicture,detectionPicture,notUpload">
      <!--ro, opt, enum, uploaded pictures type, subType:string, attr:opt{req, string}, desc:"all", "licensePlatePicture" (license plate picture),
"detectionPicture" (detected picture), "notUpload" (no picture upload)-->all
    </detectionUpLoadPicturesType>
    <alarmHttpPushProtocol opt="baseline,custom">
      <!--ro, opt, string, alarm HTTP push protocol, attr:opt{req, string}-->baseline
    </alarmHttpPushProtocol>
  </ANPR>
  <uploadImagesDataType opt="URL,binary">
    <!--ro, opt, enum, uploaded picture type, subType:string, attr:opt{req, string}, desc:"URL", "binary" (default value). For cloud storage, only "URL" is
supported-->URL
  </uploadImagesDataType>
  <SubscribeEventCap>
    <!--ro, opt, object, subscribe to changing status of capability set-->
    <heartbeat min="0" max="180" def="10">
      <!--ro, opt, string, heartbeat interval time, attr:min{req, int},max{req, int},def{opt, int}-->1
    </heartbeat>
  </SubscribeEventCap>
  <enabled>
    <!--ro, opt, bool, whether to enable-->true
  </enabled>
  <checkResponseEnabled opt="true,false" def="true">
    <!--ro, opt, bool, whether to enable verifying listening host response, attr:opt{req, string},def{req, bool}, desc:when a device uploads an event to the
listening host, the host should respond with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms'
listening services fail to respond. This parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response
from the listening host. When enabled, failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the
event-->true
  </checkResponseEnabled>
</HttpHostNotificationCap>
```

11.2.1.5 Set IP address of receiving server(s)

Request URL

PUT /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<HttpHostNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, listening host list, subType:object, attr:version{req, string, protocolVersion}-->
  <HttpHostNotification>
    <!--opt, object, listening host-->
    <id>
      <!--req, string, ID-->test
    </id>
    <url>
      <!--req, string, URL-->test
    </url>
    <protocolType>
      <!--req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome"-->HTTP
    </protocolType>
    <parameterFormatType>
      <!--req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
    </parameterFormatType>
    <addressingFormatType>
      <!--req, enum, address types, subType:string, desc:"hostname" (host name), "ipaddress" (ip address)-->hostname
    </addressingFormatType>
    <hostName>
      <!--opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipAddress}, desc:only one of IPv4 and IPv6 address exists when
the address type is "ipaddress"-->test
    </ipAddress>
    <portNo>
      <!--opt, int, port No.-->1
    </portNo>
    <httpAuthenticationMethod>
      <!--req, enum, authentication method, subType:string, desc:"MD5digest" (MD5), "none"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--opt, object, ANPR-->
      <detectionUpLoadPicturesType>
        <!--opt, enum, uploaded pictures type, subType:string, desc:uploaded pictures type-->all
      </detectionUpLoadPicturesType>
    </ANPR>
  </HttpHostNotification>
</HttpHostNotificationList>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error message description, range:[0,1024], desc:detailed information of custom error returned by device applications, used
for fast debugging-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string, error codes categorized by functional modules, desc:none-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

11.2.1.6 Get parameters of all listening hosts

Request URL

GET /ISAPI/Event/notification/httpHosts

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<HttpHostNotificationList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, listening host list, subType:object, attr:version{req, string, protocolVersion}-->
  <HttpHostNotification>
    <!--ro, opt, object, listening host-->
    <id>
      <!--ro, req, string, subscribe to ID, range:[1,10]-->test
    </id>
    <url>
      <!--ro, req, string, URL-->test
    </url>
    <protocolType>
      <!--ro, req, enum, protocol type, subType:string, desc:"HTTP", "HTTPS", "EHome" (ISUP)-->HTTP
    </protocolType>
    <parameterFormatType>
      <!--ro, req, enum, parameter format type, subType:string, desc:"JSON", "XML"-->JSON
    </parameterFormatType>
    <addressingFormatType>
      <!--ro, req, enum, address type, subType:string, desc:"hostname", "ipaddress"-->hostname
    </addressingFormatType>
    <hostName>
      <!--ro, opt, string, host name, dep:and,{$.HttpHostNotification.addressingFormatType,eq,hostName}-->test
    </hostName>
    <ipAddress>
      <!--ro, opt, string, IP address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}, desc:only one of IPv4 and IPv6 address exists when
the address type is "ipaddress"-->test
    </ipAddress>
    <ipv6Address>
      <!--ro, opt, string, IPv6 Address, dep:or,{$.HttpHostNotification.addressingFormatType,eq,ipaddress}, desc:only one of IPv4 and IPv6 address exists
when the address type is "ipaddress"-->test
    </ipv6Address>
    <portNo>
      <!--ro, opt, int, port number-->1
    </portNo>
    <userName>
      <!--ro, opt, string, user name, dep:and,{$.HttpHostNotification.httpAuthenticationMethod,eq,MD5digest}-->test
    </userName>
    <httpAuthenticationMethod>
      <!--ro, req, enum, authentication method, subType:string, desc:"MD5digest"(MD5), "none", "base64"-->MD5digest
    </httpAuthenticationMethod>
    <ANPR>
      <!--ro, opt, object, ANPR-->
      <detectionUpLoadPicturesType>
        <!--ro, opt, enum, uploaded pictures type, subType:string, desc:"all", "licensePlatePicture", "detectionPicture"(detected picture)-->all
      </detectionUpLoadPicturesType>
      <videoUploadEnabled>
        <!--ro, opt, bool, whether to upload violation pre-records, desc:when it is enabled, the device will upload the stored video via the "relationVideo"
event-->false
      </videoUploadEnabled>
    </ANPR>
    <Extensions>
      <!--ro, opt, object, range-->
      <intervalBetweenEvents>
        <!--ro, opt, int, event interval-->1
      </intervalBetweenEvents>
    </Extensions>
    <uploadImagesDataType>
      <!--ro, opt, enum, picture data type, subType:string, desc:"URL", "binary" (default value). For cloud storage, only "URL" is supported-->URL
    </uploadImagesDataType>
    <httpBroken>
      <!--ro, opt, bool, whether to enable the automatic network replenishment, desc:if the ANR function is enabled, it will be applied to all events-->true
    </httpBroken>
    <SubscribeEvent>
      <!--ro, opt, object, picture uploading modes of all events which contain pictures-->
      <heartbeat>
        <!--ro, opt, int, heartbeat interval time-->30
      </heartbeat>
      <eventMode>
        <!--ro, req, enum, event mode, subType:string, desc:"all" (all alarms need to be reported), "list" (only listed alarms need to be reported)-->all
      </eventMode>
      <EventList>
        <!--ro, opt, array, event list, subType:object-->
        <Event>
          <!--ro, opt, object, channel information linked to event-->
          <type>
            <!--ro, req, enum, event type, subType:string, desc:refer to event type list (eventType): "ADAS"(advanced driving assistance system),
"ADASAlarm"(advanced driving assistance alarm), "AID"(traffic incident detection), "ANPR"(automatic number plate recognition), "AccessControllerEvent"
(access controller event), "CDsStatus" (CD burning status), "DBD"(driving behavior detection) "GPSUpload" (GPS information upload), "HFPD"(frequently
appeared person detection), "IO"(I/O alarm), "IOTD" (IoT device detection), "LES" (logistics scanning event), "LFPD"(rarely appeared person detection),
"PALMismatch" (video standard mismatch), "PIR", "PeopleCounting" (people counting), "PeopleNumChange" (people number change detection), "Standup"(standing
up detection), "TMA"(thermometry alarm), "TMPA"(temperature measurement pre-alarm), "VMD"(motion detection), "abnormalAcceleration", "abnormalDriving",
"advReachHeight", "alarmResult", "attendance", "attendedBaggage", "audioAbnormal", "audioexception", "behaviorResult"(abnormal event detection),
"blindSpotDetection"(blind spot detection alarm), "cardMatch", "changedStatus", "collision", "containerDetection", "crowdSituationAnalysis",
"databaseException", "defocus"(defocus detection), "diskUnformat"(disk unformatted), "diskerror", "diskfull", "driverConditionMonitor"(driver status
monitoring alarm); "emergencyAlarm", "faceCapture", "faceSnapModeling", "facedetection", "failDown"(People Falling Down), "faultAlarm", "fielddetection"
(intrusion detection), "fireDetection", "fireEscapeDetection", "flowOverrun", "framesPeopleCounting", "getUp"(getting up detection), "group" (people
gathering), "hdBadBlock"(HDD bad sector detection event), "hdImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high
```

```
gathering)", "hddBadBlock"(HDD bad sector detection event), "hddImpact"(HDD impact detection event), "heatmap"(heat map alarm), "highHDTemperature"(HDD high
temperature detection event), "highTempAlarm"(HDD high temperature alarm), "hotSpare"(hot spare exception), "illaccess"(invalid access),
"ipcTransferAbnormal", "ipconflict"(IP address conflicts), "keyPersonGetUp"(key person getting up detection), "leavePosition"(absence detection),
"linedetection"(line crossing detection), "listSyncException"(list synchronization exception), "loitering"(loitering detection), "lowHDTemperature"(HDD low
temperature detection event), "mixedTargetDetection"(multi-target-type detection), "modelError", "nicbroken"(network disconnected), "nodeOffline"(node
disconnected), "nonPoliceIntrusion", "overSpeed"(overspeed alarm), "overtimeTarry"(staying overtime detection), "parking"(parking detection),
"peopleNumChange", "peopleNumCounting", "personAbnormalAlarm"(person ID exception alarm), "personDensityDetection", "personQueueCounting",
"personQueueDetection", "personQueueRealTime"(real-time data of people queuing-up detection), "personQueueTime"(waiting time detection), "playCellphone"
(playing mobile phone detection), "pocException"(video exception), "poe"(POE power exception), "policeAbsent", "radarAlarm", "radarFieldDetection",
"radarLineDetection", "radarPerimeterRule"(radar rule data), "radarTargetDetection", "radarVideoDetection"(radar-assisted target detection),
"raidException", "rapidMove", "reachHeight"(climbing detection), "recordCycleAbnormal"(insufficient recording period), "recordException", "regionEntrance",
"regionExiting", "retention"(people overstay detection), "rollover", "running"(people running), "safetyHelmetDetection"(hard hat detection),
"scenechangedetection", "sensorAlarm"(angular acceleration alarm), "severeHDFailure"(HDD major fault detection), "shelteralarm"(video tampering alarm),
"shipsDetection", "sitQuietly"(sitting detection), "smokeAndFireDetection", "smokeDetection", "softIO", "spacingChange"(distance exception), "sysStorFull"
(storaging full alarm of cluster system), "takingElevatorDetection"(elevator electric moped detection), "targetCapture", "temperature"(temperature
difference alarm), "thermometry"(temperature alarm), "thirdPartyException", "toiletTarry"(in-toilet overtime detection), "tollCodeInfo"(QR code information
report), "tossing"(thrown object detection), "unattendedBaggage", "vehicleMatchResult"(uploading list alarms), "vehicleRcogResult", "versionAbnormal"
(cluster version exception), "videoException", "videoloss", "violationAlarm", "violentMotion"(violent motion detection), "yardTarry"(playground overstay
detection), "AccessControllerEvent", "IDCardInfoEvent", "FaceTemperatureMeasurementEvent", "QRCodeEvent"(QR code event of access control),
"CertificateCaptureEvent"(person ID capture comparison event), "UncertificateCompareEvent", "ConsumptionAndTransactionRecordEvent", "ConsumptionEvent",
"TransactionRecordEvent", "SetMealQuery"(searching consumption set meals), "ConsumptionStatusQuery"(searching the consumption status), "humanBodyComparison"
(human body comparison), "regionTargetNumberCounting" (regional target statistics)-->mixedTargetDetection
          </type>
          <minorAlarm>
            <!--ro, opt, string, alarm sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorAlarm>
          <minorException>
            <!--ro, opt, string, exception sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorException>
          <minorOperation>
            <!--ro, opt, string, operation sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of
event is "AccessControllerEvent"-->0x400,0x401,0x402,0x403
          </minorOperation>
          <minorEvent>
            <!--ro, opt, string, event sub type, desc:refer to the macro definition of uploaded events. "IDCardInfoEvent" is required when the type of event
is "AccessControllerEvent"-->0x01,0x02,0x03,0x04
          </minorEvent>
          <pictureURLType>
            <!--ro, opt, enum, alarm picture format of the specified event, subType:string, desc:alarm picture format of the specified event-->binary
          </pictureURLType>
          <channels>
            <!--ro, opt, string, listen to the events on the specified channel No. list, desc:if all channels are being listened to, the node shall not be
applied. if some channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
          </channels>
        </Event>
      </EventList>
      <channels>
        <!--ro, opt, string, listen to the specified channel No. list, desc:if all channels are being listened to, the node shall not be applied. if some
channels are being listened to, the channel No. shall be listed and separated by commas-->1,2,3,4
      </channels>
      <pictureURLType>
        <!--ro, opt, enum, alarm picture format, subType:string, desc:the node indicates the picture types of all events which contain pictures to be
uploaded. if the node is applied, the pictureURLType of the Event will not take effect. if the node is not applied, the pictures reported from the device by
default mode: the default data type of uploaded pictures captured from front-end devices is binary. the default data type of uploaded pictures reported from
storage devices is local URL of the device.-->binary
      </pictureURLType>
      <ChangedUploadSub>
        <!--ro, opt, object, subscribe to messages-->
        <interval>
          <!--ro, opt, int, the lifecycle of arming GUID, desc:if GUID is not reconnected in the internal, a new GUID will be generated since the device
start a new arm period-->5
        </interval>
        <StatusSub>
          <!--ro, opt, object, sub status-->
          <all>
            <!--ro, opt, bool, subscribe to all?-->true
          </all>
          <channel>
            <!--ro, opt, bool, subscribe to channel status, desc:reporting is not required if all is true-->true
          </channel>
          <hd>
            <!--ro, opt, bool, subscribe to HDD status, desc:reporting is not required if all is true-->true
          </hd>
          <capability>
            <!--ro, opt, bool, subscribe to changing status of capability set, desc:reporting is not required if all is true-->true
          </capability>
        </StatusSub>
      </ChangedUploadSub>
    </SubscribeEvent>
    <PackingSpaceRecognition>
      <!--ro, opt, object, current control parameters of event listened by parking space detector on the security control panel, desc:related event:
PackingSpaceRecognition-->
      <upLoadPicturesType>
        <!--ro, opt, enum, uploaded picture type, subType:string, desc:"all", "picturesTypes" (upload specified types of pictures), "notUpload" (not upload
pictures)-->all
      </upLoadPicturesType>
      <PicturesTypes>
        <!--ro, opt, array, specified list of uploaded picture type, subType:object, dep:and,
{$.HttpHostNotificationList[*].HttpHostNotification.PackingSpaceRecognition.upLoadPicturesType,eq,picturesTypes}-->
        <picturesType>
          <!--ro, opt, enum, uploaded picture type, subType:string, desc:"backgroundImage" (captured background picture), "plateImage" (license plate
thumbnail)-->backgroundImage
```

```
        </picturesType>
      </PicturesTypes>
    </PackingSpaceRecognition>
    <fileUploadType>
      <!--ro, opt, enum, subType:string-->cloudStrage
    </fileUploadType>
    <enabled>
      <!--ro, opt, bool, whether to enable-->true
    </enabled>
    <netWork>
      <!--ro, opt, enum, network, subType:int, desc:1 (wired network), 2 (mobile network: 3G/4G/5G/GPRS), 3 (wireless network)-->1
    </netWork>
    <method>
      <!--ro, opt, enum, request method, subType:string, desc:"POST", "PUT", "GET"-->POST
    </method>
    <pictureAttachEnabled>
      <!--ro, opt, bool, whether to enable attaching pictures, desc:true by default-->true
    </pictureAttachEnabled>
    <contentType>
      <!--ro, opt, enum, HTTP content format, subType:string, desc:"multipart" (form format) by default. It is the format for all uploaded events, not limited to pictures only-->multipart
    </contentType>
    <eventTemplateID>
      <!--ro, opt, int, event message template ID, desc:related URL (GET /ISAPI/Event/notification/EventTemplateParams?format=json) is to get template list-->1
    </eventTemplateID>
    <customTriggersEnabled>
      <!--ro, opt, bool, whether to enable custom linkage, desc:true (listening service's linkage method is unaffected by upload center's linkage method, that is, independent), false or node doesn't exist (use the previous logic, that upload center sends data to listening service)-->false
    </customTriggersEnabled>
    <checkResponseEnabled>
      <!--ro, opt, bool, whether to enable verifying listening host response, desc:when a device uploads an event to the listening host, the host should respond with HTTP 200 OK status. Otherwise, the device assumes that the event was not received. However, some platforms' listening services fail to respond. This parameter is used to address the actual cause. When disabled, the device skips the verification of 200 OK response from the listening host. When enabled, failure to receive 200 OK indicates the event upload failure, and the device will log the failure or retransmit the event-->true
    </checkResponseEnabled>
  </HttpHostNotification>
</HttpHostNotificationList>
```


11.3 视频通用

11.3.1 抓拍图像

11.3.1.1 Set the captured picture parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/capture

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ITCImageSnap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <SnapColor>
    <!--opt, object-->
    <brightnessLevel>
      <!--req, int, brightness, range:[0,100]-->1
    </brightnessLevel>
    <contrastLevel>
      <!--req, int, contrast, range:[0,100]-->1
    </contrastLevel>
    <redBrightEnhanceLevel>
      <!--opt, int, range:[0,100]-->1
    </redBrightEnhanceLevel>
  </SnapColor>
  <SnapShutter>
    <!--opt, object-->
    <snapShutterLevel>
      <!--req, int, range:[120,14887]-->120
    </snapShutterLevel>
    <redSnapShutterLevel>
      <!--opt, int, range:[120,14887]-->120
    </redSnapShutterLevel>
  </SnapShutter>
  <SnapGain>
    <!--opt, object-->
    <snapGainLevel>
```

```
      <snapGainLevel>
        <!--req, int, range:[0,100]-->1
      </snapGainLevel>
      <redSnapGainLevel>
        <!--opt, int, range:[0,100]-->1
      </redSnapGainLevel>
    </SnapGain>
    <SnapNoiseReduce2D>
      <!--opt, object-->
      <noiseReduce2DEnable>
        <!--req, bool-->true
      </noiseReduce2DEnable>
      <noiseReduce2DLevel>
        <!--opt, int, 2D noise reduction level, range:[0,100]-->1
      </noiseReduce2DLevel>
    </SnapNoiseReduce2D>
    <ShadowHighlight>
      <!--opt, object-->
      <mode>
        <!--req, enum, subType:string-->close
      </mode>
      <level>
        <!--opt, int, range:[0,100]-->50
      </level>
    </ShadowHighlight>
    <SnapLPDE>
      <!--opt, object-->
      <enable>
        <!--req, bool-->true
      </enable>
      <level>
        <!--opt, int, range:[0,100]-->50
      </level>
    </SnapLPDE>
    <SnapPlateContrast>
      <!--opt, object-->
      <plateContrastLevel>
        <!--opt, int, range:[0,100]-->1
      </plateContrastLevel>
    </SnapPlateContrast>
    <SnapPlateSaturation>
      <!--opt, object-->
      <plateSaturationLevel>
        <!--opt, int, range:[0,100]-->1
      </plateSaturationLevel>
    </SnapPlateSaturation>
    <SnapCCM>
      <!--opt, object-->
      <level>
        <!--opt, int, range:[0,100]-->80
      </level>
    </SnapCCM>
    <carWindowEnhancement>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <brightenLevel>
        <!--opt, int, range:[0,100]-->1
      </brightenLevel>
      <defogLevel>
        <!--opt, int, range:[0,100]-->1
      </defogLevel>
    </carWindowEnhancement>
    <CapLightMode>
      <!--opt, object-->
      <capLightWorkMode>
        <!--req, enum, subType:string-->flash
      </capLightWorkMode>
    </CapLightMode>
    <CarLightColor>
      <!--opt, object-->
      <carLightColorLevel>
        <!--req, int, range:[0,100]-->1
      </carLightColorLevel>
    </CarLightColor>
    <SnapWhiteBalance>
      <!--opt, object-->
      <whiteBalanceLevel>
        <!--opt, int, range:[0,100]-->1
      </whiteBalanceLevel>
    </SnapWhiteBalance>
  </ITCImageSnap>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.1.2 Get the capture parameters capability of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capture/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ITCImageSnapCaps xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <SnapColor>
    <!--ro, opt, object-->
    <brightnessLevel min="0" max="100">
      <!--ro, req, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </brightnessLevel>
    <contrastLevel min="0" max="100">
      <!--ro, req, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </contrastLevel>
    <redBrightEnhanceLevel min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </redBrightEnhanceLevel>
  </SnapColor>
  <SnapShutter>
    <!--ro, opt, object-->
    <snapShutterLevel min="50" max="9000">
      <!--ro, req, int, range:[120,14887], attr:min{req, int},max{req, int}-->120
    </snapShutterLevel>
    <redSnapShutterLevel min="50" max="9000">
      <!--ro, opt, int, range:[120,14887], attr:min{req, int},max{req, int}-->120
    </redSnapShutterLevel>
  </SnapShutter>
  <SnapGain>
    <!--ro, opt, object-->
    <snapGainLevel min="0" max="100">
      <!--ro, req, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </snapGainLevel>
    <redSnapGainLevel min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </redSnapGainLevel>
  </SnapGain>
  <SnapNoiseReduce2D>
    <!--ro, opt, object-->
    <noiseReduce2DEnable>
      <!--ro, req, bool-->true
    </noiseReduce2DEnable>
    <noiseReduce2DLevel min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
    </noiseReduce2DLevel>
  </SnapNoiseReduce2D>
  <ShadowHighlight>
    <!--ro, opt, object-->
    <mode opt="close,SAE,HLC,SAE_HLC,DCE">
      <!--ro, req, enum, subType:string, attr:opt{req, string}-->close
```

```
      </mode>
      <level min="0" max="100" def="0">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->50
      </level>
    </ShadowHighlight>
    <SnapLPDE>
      <!--ro, opt, object-->
      <enable>
        <!--ro, req, bool-->true
      </enable>
      <level min="0" max="100" def="0">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->50
      </level>
    </SnapLPDE>
    <SnapPlateContrast>
      <!--ro, opt, object-->
      <plateContrastLevel min="0" max="100" def="0">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->1
      </plateContrastLevel>
    </SnapPlateContrast>
    <SnapPlateSaturation>
      <!--ro, opt, object-->
      <plateSaturationLevel min="0" max="100" def="0">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->1
      </plateSaturationLevel>
    </SnapPlateSaturation>
    <SnapCCM>
      <!--ro, opt, object-->
      <level min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->80
      </level>
    </SnapCCM>
    <carWindowEnhancement>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <brightenLevel min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
      </brightenLevel>
      <defogLevel min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
      </defogLevel>
    </carWindowEnhancement>
    <CapLightMode>
      <!--ro, opt, object-->
      <capLightWorkMode opt="flash,noflash">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->flash
      </capLightWorkMode>
    </CapLightMode>
    <CarLightColor>
      <!--ro, opt, object-->
      <carLightColorLevel min="0" max="100">
        <!--ro, req, int, range:[0,100], attr:min{req, int},max{req, int}-->1
      </carLightColorLevel>
    </CarLightColor>
    <SnapWhiteBalance>
      <!--ro, opt, object-->
      <whiteBalanceLevel min="0" max="100">
        <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->1
      </whiteBalanceLevel>
    </SnapWhiteBalance>
    <isSupportSnapClip>
      <!--ro, opt, bool-->true
    </isSupportSnapClip>
  </ITCImageSnapCaps>
```


11.3.1.3 Set the parameters of the license plate background enhancement

Request URL

PUT /ISAPI/Image/channels/<channelID>/capture/CCM

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapCCM xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of license plate background enhancement, attr:version{req, string, protocolVersion}-->
  <level>
    <!--opt, int, enhancement level, range:[0,100]-->80
  </level>
</SnapCCM>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML 
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, 
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.1.4 Get the parameters of the license plate background enhancement

Request URL

GET /ISAPI/Image/channels/<channelID>/capture/CCM?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapCCM xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of license plate background enhancement, attr:version{req, string, protocolVersion}-->
  <level>
    <!--ro, opt, int, enhancement level, range:[0,100]-->80
  </level>
</SnapCCM>
```

11.3.1.5 Set the license plate enhancement parameters of the captured pictures of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/capture/LPDE

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapLPDE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--req, bool, whether to enable license plate enhancement-->true
  </enable>
  <level min="0" max="100" def="50">
    <!--opt, int, license plate enhancement level, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->50
  </level>
</SnapLPDE>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, read-only,describe the error reason in detail, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.1.6 Get the license plate enhancement parameters of the captured pictures of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capture/LPDE?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapLPDE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, license plate enhancement parameters of the captured picture, attr:version{req, string, protocolVersion}-->
  <enable>
    <!--ro, req, bool, whether to enable license plate enhancement-->true
  </enable>
  <level min="0" max="100" def="50">
    <!--ro, opt, int, license plate enhancement level, range:[0,100], attr:min{req, int},max{req, int},def{req, int}-->50
  </level>
</SnapLPDE>
```

11.3.1.7 Set the license plate contrast parameters of the captured images of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/capture/plateContrast

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapPlateContrast xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <plateContrastLevel>
    <!--opt, int, license plate contrast level, range:[0,100]-->1
  </plateContrastLevel>
</SnapPlateContrast>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.1.8 Get the license plate contrast parameters of the captured pictures of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capture/plateContrast?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapPlateContrast xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, license plate contrast parameters of the captured picture, attr:version{req, string, protocolVersion}-->
  <plateContrastLevel>
    <!--ro, opt, int, license plate contrast level, range:[0,100]-->1
  </plateContrastLevel>
</SnapPlateContrast>
```

11.3.1.9 Set the license plate saturation parameters of the captured pictures of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/capture/plateSaturation

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapPlateSaturation xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <plateSaturationLevel>
    <!--opt, int, license plate saturation level, range:[0,100]-->1
  </plateSaturationLevel>
</SnapPlateSaturation>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.1.10 Get the license plate saturation parameters of the captured pictures of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capture/plateSaturation?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapPlateSaturation xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, license plate saturation parameters of the captured picture, attr:version{req, string, protocolVersion}-->
  <plateSaturationLevel>
    <!--ro, opt, int, license plate saturation level, range:[0,100]-->1
  </plateSaturationLevel>
</SnapPlateSaturation>
```

11.3.1.11 Get the capture parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/capture?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ITCImageSnap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <SnapColor>
    <!--ro, opt, object-->
    <brightnessLevel>
      <!--ro, req, int, range:[0,100]-->1
    </brightnessLevel>
    <contrastLevel>
      <!--ro, req, int, range:[0,100]-->1
    </contrastLevel>
    <redBrightEnhanceLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </redBrightEnhanceLevel>
  </SnapColor>
  <SnapShutter>
    <!--ro, opt, object-->
    <snapShutterLevel>
      <!--ro, req, int, range:[120,14887]-->120
    </snapShutterLevel>
    <redSnapShutterLevel>
      <!--ro, opt, int, range:[120,14887]-->120
    </redSnapShutterLevel>
  </SnapShutter>
  <SnapGain>
    <!--ro, opt, object-->
    <snapGainLevel>
      <!--ro, req, int, range:[0,100]-->1
    </snapGainLevel>
    <redSnapGainLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </redSnapGainLevel>
  </SnapGain>
  <SnapNoiseReduce2D>
    <!--ro, opt, object-->
    <noiseReduce2DEnable>
      <!--ro, req, bool-->true
    </noiseReduce2DEnable>
    <noiseReduce2DLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </noiseReduce2DLevel>
  </SnapNoiseReduce2D>
  <ShadowHighlight>
    <!--ro, opt, object-->
    <mode>
      <!--ro, req, enum, subType:string-->close
    </mode>
    <level>
      <!--ro, opt, int, range:[0,100]-->50
    </level>
  </ShadowHighlight>
  <SnapLPDE>
    <!--ro, opt, object-->
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <level>
      <!--ro, opt, int, range:[0,100]-->50
    </level>
  </SnapLPDE>
  <SnapPlateContrast>
    <!--ro, opt, object-->
    <plateContrastLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </plateContrastLevel>
  </SnapPlateContrast>
  <SnapPlateSaturation>
    <!--ro, opt, object-->
    <plateSaturationLevel>
      <!--ro, opt, int, range:[0,100]-->1
    </plateSaturationLevel>
  </SnapPlateSaturation>
  <SnapCCM>
    <!--ro, opt, object-->
    <level>
      <!--ro, opt, int, range:[0,100]-->80
```

```
      </level>
    </SnapCCM>
    <carWindowEnhancement>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <brightenLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </brightenLevel>
      <defogLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </defogLevel>
    </carWindowEnhancement>
    <CapLightMode>
      <!--ro, opt, object-->
      <capLightWorkMode>
        <!--ro, req, enum, subType:string-->flash
      </capLightWorkMode>
    </CapLightMode>
    <CarLightColor>
      <!--ro, opt, object-->
      <carLightColorLevel>
        <!--ro, req, int, range:[0,100]-->1
      </carLightColorLevel>
    </CarLightColor>
    <SnapWhiteBalance>
      <!--ro, opt, object-->
      <whiteBalanceLevel>
        <!--ro, opt, int, range:[0,100]-->1
      </whiteBalanceLevel>
    </SnapWhiteBalance>
  </ITCImageSnap>
```


11.3.1.12 Get parameters of JPEG picture capturing of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/JPEGParam

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<JPEGParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, JPEG picture parameters, attr:version{req, string, protocolVersion}-->
  <JPEGSize>
    <!--ro, opt, int, JPEG picture size, range:[64,8196], unit:KB-->64
  </JPEGSize>
  <MergeJPEGSize>
    <!--ro, opt, int, JPEG picture size of the composite picture, range:[64,8196], unit:KB-->64
  </MergeJPEGSize>
  <EXIFInformationEnabled>
    <!--ro, opt, bool-->true
  </EXIFInformationEnabled>
</JPEGParam>
```


11.3.1.13 Set the JPEG picture size parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/JPEGParam

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<JPEGParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, JPEG picture parameters, attr:version{req, string, protocolVersion}-->
  <JPEGSize>
    <!--opt, int, JPEG picture size, range:[64,8196], unit:KB-->64
  </JPEGSize>
  <MergeJPEGSize>
    <!--opt, int, range:[64,8196], unit:KB-->64
  </MergeJPEGSize>
  <EXIFInformationEnabled>
    <!--opt, bool-->true
  </EXIFInformationEnabled>
</JPEGParam>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.1.14 Get the capability of JPEG picture capturing of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/JPEGParam/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<JPEGParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, JPEG picture parameters, attr:version{req, string, protocolVersion}-->
  <JPEGSize min="64" max="8196">
    <!--ro, opt, int, JPEG picture size, range:[64,8196], unit:KB, attr:min{req, int},max{req, int}-->64
  </JPEGSize>
  <MergeJPEGSize min="64" max="8196">
    <!--ro, opt, int, JPEG picture size of the composite picture, range:[64,8196], unit:KB, attr:min{req, int},max{req, int}-->64
  </MergeJPEGSize>
  <EXIFInformationEnabled opt="true,false" def="false">
    <!--ro, opt, bool, attr:opt{req, string},def{req, bool}-->true
  </EXIFInformationEnabled>
</JPEGParam>
```

11.3.1.15 Get the capability of captured picture encoding of a specified channel

Request URL

GET /ISAPI/ITC/Snapshot/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<CapResInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, captured picture encoding, attr:version{req, string, protocolVersion}-->
  <capResolutionWidth>
    <!--ro, req, int, width of the captured picture resolution-->1
  </capResolutionWidth>
  <capResolutionHeight>
    <!--ro, req, int, height of the captured picture resolution-->1
  </capResolutionHeight>
</CapResInfo>
```


11.3.1.16 Get parameters of captured picture encoding of a specified channel

Request URL

GET /ISAPI/ITC/Snapshot/channels/<channelID>/capResInfo

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<CapResolution xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, captured picture encoding, attr:version{req, string, protocolVersion}-->
  <capResolutionWidth>
    <!--ro, opt, int, width of the captured picture resolution-->1
  </capResolutionWidth>
  <capResolutionHeight>
    <!--ro, opt, int, height of the captured picture resolution-->1
  </capResolutionHeight>
</CapResolution>
```


11.3.1.17 Set the captured picture encoding parameters for a specified channel

Request URL

PUT /ISAPI/ITC/Snapshot/channels/<channelID>/capResInfo

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<CapResolution xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, captured picture encoding parameters, attr:version{req, string, protocolVersion}-->
  <capResolutionWidth>
    <!--opt, int, width of the captured picture resolution-->1
  </capResolutionWidth>
  <capResolutionHeight>
    <!--opt, int, height of the captured picture resolution-->1
  </capResolutionHeight>
</CapResolution>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.1.18 Get the parameters of all channels for capturing pictures by schedule

Request URL

GET /ISAPI/Snapshot/channels

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapshotChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{opt, string, protocolVersion}-->
  <SnapshotChannel>
    <!--ro, opt, object-->
    <id>
      <!--ro, req, string, ID-->test
    </id>
    <videoInputChannelID>
      <!--ro, req, string-->test
    </videoInputChannelID>
    <timingCapture>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <supportSchedule>
        <!--ro, opt, bool-->true
      </supportSchedule>
      <compress>
        <!--ro, opt, object-->
        <pictureCodecType>
          <!--ro, req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->JPEG
        </pictureCodecType>
        <pictureWidth>
          <!--ro, req, int-->1
        </pictureWidth>
        <pictureHeight>
          <!--ro, req, int-->1
        </pictureHeight>
        <quality>
          <!--ro, opt, int, picture quality, range:[0,100]-->1
        </quality>
        <captureInterval>
          <!--ro, opt, int, unit: milliseconds, desc:unit: milliseconds-->1
        </captureInterval>
        <captureNumber>
          <!--ro, opt, int, number of captured pictures-->1
        </captureNumber>
      </compress>
    </timingCapture>
    <eventCapture>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <supportSchedule>
        <!--ro, opt, bool-->true
      </supportSchedule>
      <compress>
        <!--ro, opt, object-->
        <pictureCodecType>
          <!--ro, req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->JPEG
        </pictureCodecType>
        <pictureWidth>
          <!--ro, req, int-->1
        </pictureWidth>
        <pictureHeight>
          <!--ro, req, int-->1
        </pictureHeight>
        <quality>
          <!--ro, opt, int, picture quality, range:[0,100]-->1
        </quality>
        <captureInterval>
          <!--ro, opt, int, unit: milliseconds, desc:unit: milliseconds-->1
        </captureInterval>
        <captureNumber>
          <!--ro, opt, int, number of captured pictures-->1
        </captureNumber>
      </compress>
    </eventCapture>
  </SnapshotChannel>
</SnapshotChannelList>
```


11.3.1.19 Set picture capturing parameters of all channels

Request URL

PUT /ISAPI/Snapshot/channels

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapshotChannelList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, picture capturing parameters of all channels, subType:object, attr:version{opt, string, protocolVersion}-->
  <SnapshotChannel>
    <!--opt, object, picture capturing parameters of a channel-->
    <id>
      <!--req, string, ID-->test
    </id>
    <videoInputChannelID>
      <!--req, string-->test
    </videoInputChannelID>
    <timingCapture>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <supportSchedule>
        <!--opt, bool-->true
      </supportSchedule>
      <compress>
        <!--opt, object-->
        <pictureCodecType>
          <!--req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->JPEG
        </pictureCodecType>
        <pictureWidth>
          <!--req, int, picture resolution (width)-->1
        </pictureWidth>
        <pictureHeight>
          <!--req, int, picture resolution (height)-->1
        </pictureHeight>
        <quality>
          <!--opt, int, picture quality, range:[0,100]-->1
        </quality>
        <captureInterval>
          <!--opt, int, capture interval, desc:unit: millisecond-->1
        </captureInterval>
        <captureNumber>
          <!--opt, int, number of captured pictures-->1
        </captureNumber>
      </compress>
    </timingCapture>
    <eventCapture>
      <!--opt, object-->
      <enabled>
        <!--req, bool-->true
      </enabled>
      <supportSchedule>
        <!--opt, bool-->true
      </supportSchedule>
      <compress>
        <!--opt, object-->
        <pictureCodecType>
          <!--req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->JPEG
        </pictureCodecType>
        <pictureWidth>
          <!--req, int, picture resolution (width)-->1
        </pictureWidth>
        <pictureHeight>
          <!--req, int, picture resolution (height)-->1
        </pictureHeight>
        <quality>
          <!--opt, int, picture quality, range:[0,100]-->1
        </quality>
        <captureInterval>
          <!--opt, int, Capture Interval, desc:unit: millisecond-->1
        </captureInterval>
        <captureNumber>
          <!--opt, int, number of captured pictures-->1
        </captureNumber>
      </compress>
    </eventCapture>
  </SnapshotChannel>
</SnapshotChannelList>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.2 图像曝光

11.3.2.1 Get the configuration parameters of the low illumination electronic shutter of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/DSS

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<DSS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, low illumination electronic shutter, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <DSSLevel>
    <!--ro, opt, enum, low illumination electronic shutter level in exposure, subType:string,
desc:"*1.25,*1.5,*2,*3*4,*6,*8,*12,*16,*24,*32,*48,*64,*96,*128,*256,*512,auto". "*2" indicates that the speed of the low illumination electronic shutter is
2 times of the normal shutter speed-->*1.25
  </DSSLevel>
  <mode>
    <!--ro, opt, enum, mode, subType:string, desc:"frame" (frame mode, it indicates that node DSSLevel is valid), "level" (level mode, it indicates that
slowShutterLevel is valid). If this node does not exist, the default mode is frame mode-->frame
  </mode>
  <slowShutterLevel>
    <!--ro, req, int, shutter level, range:[1,6]-->1
  </slowShutterLevel>
</DSS>
```


11.3.2.2 Set the configuration parameters of the low illumination electronic shutter for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/DSS

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<DSS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, low illumination electronic shutter, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <DSSLevel>
    <!--opt, enum, low illumination level, subType:string, desc:low illumination level-->*1.25
  </DSSLevel>
  <mode>
    <!--opt, enum, mode, subType:string, desc:"frame" (frame mode, it indicates that node DSSLevel is valid), "level" (level mode, it indicates that
slowShutterLevel is valid). If this node does not exist, the default mode is frame mode-->frame
  </mode>
  <slowShutterLevel>
    <!--req, int, slow shutter level in exposure, range:[1,6]-->1
  </slowShutterLevel>
</DSS>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.2.3 Set the exposure mode for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/exposure

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--req, enum, exposure type, subType:string, desc:exposure type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--opt, int, automatic aperture level-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--opt, object-->
    <enabled>
      <!--req, bool-->true
    </enabled>
    <Type>
      <!--opt, enum, "AUTO", "MANUAL", subType:string, desc:"AUTO", "MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--opt, int, focal length-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
  </OverexposeSuppress>
  <pIris>
    <!--opt, object, aperture-->
    <pIrisType>
      <!--opt, enum, aperture type, subType:string, desc:aperture type-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--opt, int, aperture level-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--opt, object, face exposure-->
    <enabled>
      <!--opt, bool, whether to enable face exposure-->true
    </enabled>
    <sensitivity>
      <!--opt, int, sensitivity of face exposure-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is available when ExposureType is set to "pIris-General"-->
    <irisLevel>
      <!--opt, int, aperture level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
</Exposure>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.2.4 Get the exposure mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/exposure?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Exposure xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, exposure mode, attr:version{req, string, protocolVersion}-->
  <ExposureType>
    <!--ro, req, enum, lens type, subType:string, desc:lens type-->auto
  </ExposureType>
  <autoIrisLevel>
    <!--ro, opt, int-->1
  </autoIrisLevel>
  <OverexposeSuppress>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <Type>
      <!--ro, opt, enum, this node depends on <enabled>,"AUTO,MANUAL", subType:string, desc:this node depends on <enabled>,"AUTO,MANUAL"-->AUTO
    </Type>
    <DistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </DistanceLevel>
    <shortIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </shortIRDistanceLevel>
    <longIRDistanceLevel>
      <!--ro, opt, int, this node depends on  <Type>-->1
    </longIRDistanceLevel>
    <supplementLightIntensity1>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity1>
    <supplementLightIntensity2>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity2>
    <supplementLightIntensity3>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity3>
    <supplementLightIntensity4>
      <!--ro, opt, int, range:[1,100], dep:and,{$.Exposure.OverexposeSuppress.Type,eq,MANUAL}-->1
    </supplementLightIntensity4>
    <hightLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </hightLightDistanceLevel>
    <lowLightDistanceLevel>
      <!--ro, opt, int, range:[1,100]-->50
    </lowLightDistanceLevel>
  </OverexposeSuppress>
  <pIris>
    <!--ro, opt, object, this node depends on <ExposureType>,"AUTO,MANUAL"-->
    <pIrisType>
      <!--ro, opt, enum, subType:string-->AUTO
    </pIrisType>
    <IrisLevel>
      <!--ro, opt, int, this node depends on <pIrisType>-->1
    </IrisLevel>
  </pIris>
  <faceExposure>
    <!--ro, opt, object, face exposure-->
    <enabled>
      <!--ro, opt, bool-->true
    </enabled>
    <sensitivity>
      <!--ro, opt, int, sensitivity-->1
    </sensitivity>
  </faceExposure>
  <PIrisGeneral>
    <!--ro, opt, object, it is available when <ExposureType> is set to "pIris-General", desc:it is valid when <ExposureType> is set to "pIris-General"-->
    <irisLevel>
      <!--ro, opt, int, iris level of general lens, range:[1,100]-->1
    </irisLevel>
  </PIrisGeneral>
  <combinationParams>
    <!--ro, opt, int, range:[0,100]-->50
  </combinationParams>
  <opticalFilter>
    <!--ro, opt, enum, subType:string-->none
  </opticalFilter>
  <lowGhostingExposure>
    <!--ro, opt, object-->
    <lowGhostingShutterLevel>
      <!--ro, opt, enum, subType:string-->1/100
    </lowGhostingShutterLevel>
  </lowGhostingExposure>
  <exposureCompensateLevel>
    <!--ro, opt, int, range:[-100,100]-->0
  </exposureCompensateLevel>
  <exposureControl>
    <!--ro, opt, enum, subType:string-->center
  </exposureControl>
</Exposure>
```

11.3.2.5 Set the iris parameters of a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/iris

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Iris xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <IrisLevel>
    <!--req, int-->1
  </IrisLevel>
  <irisSpeed>
    <!--opt, int-->1
  </irisSpeed>
</Iris>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.2.6 Get the iris parameters of a specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/iris

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Iris xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <IrisLevel>
    <!--ro, req, int-->1
  </IrisLevel>
  <irisSpeed>
    <!--ro, opt, int-->1
  </irisSpeed>
</Iris>
```

11.3.2.7 Set the exposure time parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/shutter

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Shutter xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <ShutterLevel>
    <!--opt, enum, exposure level, subType:string, desc:"1/1", "1/2", "1/3", "1/6", "1/12", "1/25", "1/50", "1/75", "1/100", "1/120", "1/125", "1/150",
"1/175", this node is valid when exposure type is "shutter first"-->1/1
  </ShutterLevel>
  <scheduleEnabled>
    <!--opt, bool-->true
  </scheduleEnabled>
  <scheduleShutterLevel>
    <!--opt, int-->1
  </scheduleShutterLevel>
  <ScheduleList>
    <!--opt, array, subType:object-->
    <Schedule>
      <!--opt, object-->
      <startHour>
        <!--opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--opt, int, end time (hour)-->1
      </endHour>
      <endMinute>
        <!--opt, int, end time (minute)-->1
      </endMinute>
    </Schedule>
  </ScheduleList>
  <redShutterLevel>
    <!--opt, int, range:[0,20000]-->1
  </redShutterLevel>
</Shutter>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.2.8 Set the lens aperture parameters for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/iris

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<IrisData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, aperture, attr:version{req, string, protocolVersion}-->
  <iris>
    <!--opt, int, aperture adjusting speed, range:[-100,100], desc:aperture adjusting speed-->1
  </iris>
</IrisData>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.3 图像聚焦

11.3.3.1 Set the regional focus parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/regionalFocus

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<RegionalFocus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, regional focus parameters, attr:version{req, string, protocolVersion}-->
  <StartPoint>
    <!--opt, object, start point, desc:the origin is the upper-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
  </StartPoint>
  <EndPoint>
    <!--opt, object, end point, desc:the origin is the upper-left corner of the screen-->
    <positionX>
      <!--req, int, X-coordinate, range:[0,1000]-->0
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate, range:[0,1000]-->0
    </positionY>
  </EndPoint>
</RegionalFocus>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.3.2 Set the focus parameters for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/focus

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<FocusData xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of focus control, attr:version{req, string, protocolVersion}-->
  <focus>
    <!--req, int, focus control, range:[-100,100], desc:focus control-->0
  </focus>
</FocusData>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.4 图像调节

11.3.4.1 Set the brightness enhancement parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/brightEnhance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, brightness enhancement parameters, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--req, bool, whether to enable-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.4.2 Get the brightness enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/brightEnhance?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--ro, req, bool, whether to enable brightness enhancement-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--ro, req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

11.3.4.3 Set the image adjustment parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--opt, object, gray scale-->
    <grayScaleMode>
      <!--opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, bool, whether to enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted--
>true
  </nightMode>
  <redBrightnessLevel>
    <!--opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--opt, int, range:[0,100]-->0
  </sharpnessLevel>
</Color>
```


Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.4.4 Get the image adjustment parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/color?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool, whether to enable night mode-->true
  </nightMode>
  <redBrightnessLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
</Color>
```


11.3.4.5 Get the contrast enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lse

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--ro, opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```


11.3.4.6 Set the contrast enhancement parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/lse

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--opt, enum, LSE switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.4.7 Get the contrast enhancement capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lse/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, req, int, contrast enhancement level, range:[0,100]-->50
  </lseLevel>
  <lseHaloLevel>
    <!--ro, req, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, opt, int, brightness level, range:[0,100], dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```


11.3.4.8 Set the mounting scenario mode of a camera

Request URL

PUT /ISAPI/Image/channels/<channelID>/mountingScenario

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor“, “outdoor”, “day”, “night”, “morning”, “nightfall”, “mode1”, “mode2”, “mode3”, “mode4”, “street”, “lowIllumination”-low illumination, “custom1”, “custom2”, “normal”, “road”, “faceSnap”-face picture capture, “highway”, “road2”, “backlight”-back light, “frontlight”-front light, “facenight”, “fastRoadAndLight”, “fastRoadNoLihgt”, “normalRoadAndLight”, “normalRoadNoLight”, “cityRoadAndBright”-->indoor
  </mode>
</MountingScenario>
```


Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, “Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.4.9 Get the capability of configurations of mounting scenario mode

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR,sportSnap,dustMoisture,rampWay">
    <!--ro, req, enum, Mode, subType:string, attr:opt{req, string}, desc:"lowIllumination"-low illumination, "backlight"-back light, "frontlight"-front
light, "faceSnap"-face picture capture-->indoor
  </mode>
  <isSupportRecommendation>
    <!--ro, opt, bool-->true
  </isSupportRecommendation>
  <modeNameList size="2">
    <!--ro, opt, object, attr:size{req, int}-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

11.3.4.10 Get the mounting scenario mode of a camera

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	Parameter type. When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured, it will be the value of the parameter configured on the device.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-
low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:"indoor", "outdoor", "day", "night",
"morning", "nightfall", "mode1", "mode2"," mode3", "mode4", "highway", "road2", "faceSnap", "backlight", "frontlight", "lowIllumination"-low illumination,
"backlight"-back light, "frontlight"-front light, "faceSnap"-face picture capture-->indoor
  </mode>
  <modeNameList>
    <!--ro, opt, array, subType:object-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode>
        <!--ro, req, enum, subType:string-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

11.3.4.11 Restore the image parameters of a specific channel to default settings

Request URL

PUT /ISAPI/Image/channels/<channelID>/restore

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```


11.3.4.12 Set the sharpness control parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/sharpness

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.4.13 Get the sharpness control parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/sharpness?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--ro, opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--ro, req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```


11.3.4.14 Get the image mode parameters of all channels

Request URL

GET /ISAPI/Image/channels/imageModes

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ImageModeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, image mode configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <ImageMode>
    <!--ro, opt, object, image mode-->
    <type>
      <!--ro, req, enum, type, subType:string, desc:"standard", "indoor", "outdoor", "dimLight", "bright", "soft", "vivid"-->standard
    </type>
    <recommendation>
      <!--ro, req, object, recommended configurations-->
      <brightnessLevel>
        <!--ro, opt, int, brightness, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, contrast, range:[0,100]-->1
      </contrastLevel>
      <sharpnessLevel>
        <!--ro, opt, int, sharpness, range:[0,100]-->1
      </sharpnessLevel>
      <saturationLevel>
        <!--ro, opt, int, saturation, range:[0,100]-->1
      </saturationLevel>
      <hueLevel>
        <!--ro, opt, int, hue, range:[0,100]-->1
      </hueLevel>
      <deNoiseLevel>
        <!--ro, opt, int, noise reduction, range:[0,100]-->1
      </deNoiseLevel>
    </recommendation>
  </ImageMode>
</ImageModeList>
```

11.3.5 图像调节

11.3.5.1 Set the brightness enhancement parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/brightEnhance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, brightness enhancement parameters, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--req, bool, whether to enable-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.5.2 Get the brightness enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/brightEnhance?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<BrightEnhance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <brightEnhanceEnabled>
    <!--ro, req, bool, whether to enable brightness enhancement-->true
  </brightEnhanceEnabled>
  <brightEnhanceLevel>
    <!--ro, req, int, brightness enhancement level, range:[0,100]-->1
  </brightEnhanceLevel>
</BrightEnhance>
```

11.3.5.3 Set the image adjustment parameters in auto mode of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/color

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--opt, object, gray scale-->
    <grayScaleMode>
      <!--opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--opt, bool, whether to enable night mode, when its value is "true", the saturation can be adjusted, otherwise, the saturation cannot be adjusted--
>true
  </nightMode>
  <redBrightnessLevel>
    <!--opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--opt, int, range:[0,100]-->0
  </sharpnessLevel>
</Color>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.5.4 Get the image adjustment parameters in auto mode of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/color?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Color xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <brightnessLevel>
    <!--ro, opt, int, brightness-->24
  </brightnessLevel>
  <contrastLevel>
    <!--ro, opt, int, contrast-->22
  </contrastLevel>
  <saturationLevel>
    <!--ro, opt, int, saturation, dep:and,{$.Color.nightMode,eq,true}-->33
  </saturationLevel>
  <hueLevel>
    <!--ro, opt, int, hue-->55
  </hueLevel>
  <grayScale>
    <!--ro, opt, object-->
    <grayScaleMode>
      <!--ro, opt, enum, gray scale mode, subType:string, desc:"indoor", "outdoor"-->outdoor
    </grayScaleMode>
  </grayScale>
  <nightMode>
    <!--ro, opt, bool, whether to enable night mode-->true
  </nightMode>
  <redBrightnessLevel>
    <!--ro, opt, int, range:[0,100]-->0
  </redBrightnessLevel>
  <sharpnessLevel>
    <!--ro, opt, int, sharpness, range:[0,100]-->0
  </sharpnessLevel>
  <hueMode>
    <!--ro, opt, enum, subType:string-->warm
  </hueMode>
</Color>
```

11.3.5.5 Get the contrast enhancement parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lse

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--ro, opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

11.3.5.6 Set the contrast enhancement parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/lse

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--opt, int, contrast enhancement level, range:[0,100]-->1
  </lseLevel>
  <lseHaloLevel>
    <!--opt, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--opt, enum, LSE switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--req, int, brightness level, dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.5.7 Get the contrast enhancement capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/lse/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LSE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <lseLevel>
    <!--ro, req, int, contrast enhancement level, range:[0,100]-->50
  </lseLevel>
  <lseHaloLevel>
    <!--ro, req, int, halo inhibition intensity, range:[0,100]-->1
  </lseHaloLevel>
  <switchMode>
    <!--ro, opt, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness-->open
  </switchMode>
  <timeSwitch>
    <!--ro, req, object, switch time, dep:and,{$.LSE.switchMode,eq,timeCtrl}-->
    <startTime>
      <!--ro, opt, datetime, start time-->1970-01-01T00:00:00+08:00
    </startTime>
    <endTime>
      <!--ro, opt, datetime, end time-->1970-01-01T00:00:00+08:00
    </endTime>
  </timeSwitch>
  <lightLevel>
    <!--ro, opt, int, brightness level, range:[0,100], dep:and,{$.LSE.switchMode,eq,lightCtrl}-->1
  </lightLevel>
</LSE>
```

11.3.5.8 Set the mounting scenario mode of a camera

Request URL

PUT /ISAPI/Image/channels/<channelID>/mountingScenario

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:"indoor", "outdoor", "day", "night", "morning", "nightfall", "mode1", "mode2", "mode3", "mode4", "street", "lowIllumination"-low illumination, "custom1", "custom2", "normal", "road", "faceSnap"-face picture capture, "highway", "road2", "backlight"-back light, "frontlight"-front light, "facenight", "fastRoadAndLight", "fastRoadNoLihgt", "normalRoadAndLight", "normalRoadNoLight", "cityRoadAndBright"-->indoor
  </mode>
</MountingScenario>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.5.9 Get the capability of configurations of mounting scenario mode

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR,sportSnap,dustMoisture,rampWay">
    <!--ro, req, enum, Mode, subType:string, attr:opt{req, string}, desc:"lowIllumination"-low illumination, "backlight"-back light, "frontlight"-front
light, "faceSnap"-face picture capture-->indoor
  </mode>
  <isSupportRecommendation>
    <!--ro, opt, bool-->true
  </isSupportRecommendation>
  <modeNameList size="2">
    <!--ro, opt, object, attr:size{req, int}-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode
opt="indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,street,lowIllumination,custom1,custom2,normal,road,faceSnap,highway,road2,backlight,
frontlight,facenight,fastRoadAndLight,fastRoadNoLight,normalRoadAndLight,normalRoadNolight,riverSeaProtection,perimeterProtection,forestFirePrevention,foggy
,faceFirst,plateFirst,moveFirst,autoWDR">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```


11.3.5.10 Get the mounting scenario mode of a camera

Request URL

GET /ISAPI/Image/channels/<channelID>/mountingScenario?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	Parameter type. When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured, it will be the value of the parameter configured on the device.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<MountingScenario xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <mode>
    <!--ro, req, enum, "indoor,outdoor,day,night,morning,nightfall,mode1,mode2,mode3,mode4,highway,road2,faceSnap,backlight,frontlight","lowIllumination"-low illumination,"backlight"-back light,"frontlight"-front light,"faceSnap"-face picture capture, subType:string, desc:“indoor”, “outdoor”, “day”, “night”, “morning”, “nightfall”, “mode1”, “mode2”,“ mode3”, “mode4”, “highway”, “road2”, “faceSnap”, “backlight”, “frontlight”, “lowIllumination”-low illumination, “backlight”-back light, “frontlight”-front light, “faceSnap”-face picture capture-->indoor
  </mode>
  <modeNameList>
    <!--ro, opt, array, subType:object-->
    <modeNameItem>
      <!--ro, opt, object-->
      <mode>
        <!--ro, req, enum, subType:string-->indoor
      </mode>
      <modeName>
        <!--ro, req, string-->test
      </modeName>
    </modeNameItem>
  </modeNameList>
</MountingScenario>
```

11.3.5.11 Restore the image parameters of a specific channel to default settings

Request URL

PUT /ISAPI/Image/channels/<channelID>/restore

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel ID

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

11.3.5.12 Set the sharpness control parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/sharpness

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->/ISAPI/xxxx
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.5.13 Get the sharpness control parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/sharpness?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Sharpness xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, sharpness, attr:version{opt, string, protocolVersion}-->
  <SharpnessMode>
    <!--ro, opt, enum, mode, subType:string, desc:"manual", "auto"-->manual
  </SharpnessMode>
  <SharpnessLevel>
    <!--ro, req, int, sharpness level-->1
  </SharpnessLevel>
</Sharpness>
```

11.3.5.14 Get the image mode parameters of all channels

Request URL

GET /ISAPI/Image/channels/imageModes

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ImageModeList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, image mode configurations, subType:object, attr:version{req, string, protocolVersion}-->
  <ImageMode>
    <!--ro, opt, object, image mode-->
    <type>
      <!--ro, req, enum, type, subType:string, desc:"standard", "indoor", "outdoor", "dimLight", "bright", "soft", "vivid"-->standard
    </type>
    <recommendation>
      <!--ro, req, object, recommended configurations-->
      <brightnessLevel>
        <!--ro, opt, int, brightness, range:[0,100]-->1
      </brightnessLevel>
      <contrastLevel>
        <!--ro, opt, int, contrast, range:[0,100]-->1
      </contrastLevel>
      <sharpnessLevel>
        <!--ro, opt, int, sharpness, range:[0,100]-->1
      </sharpnessLevel>
      <saturationLevel>
        <!--ro, opt, int, saturation, range:[0,100]-->1
      </saturationLevel>
      <hueLevel>
        <!--ro, opt, int, hue, range:[0,100]-->1
      </hueLevel>
      <deNoiseLevel>
        <!--ro, opt, int, noise reduction, range:[0,100]-->1
      </deNoiseLevel>
    </recommendation>
  </ImageMode>
</ImageModeList>
```


11.3.6 图像补光

11.3.6.1 Set the day/night mode switching parameters for a specified channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/dayNightGate

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<DayNightGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, day/night mode switching parameters, attr:version{req, string, protocolVersion}-->
  <brightnessThreld>
    <!--opt, int, brightness-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--opt, object, day/night mode control-->
    <nightModeCtrlMode>
      <!--opt, enum, day/night switching mode, subType:string, desc:"manual" (manual switch), "time" (timed switch), "auto" (auto switch)-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--opt, object, manual switch-->
      <manualModeVal>
        <!--opt, enum, manual switching mode, subType:string, desc:"day", "night"-->day
      </manualModeVal>
    </manualMode>
    <timeMode>
      <!--opt, object, timed switch-->
      <switchList>
        <!--opt, array, subType:object-->
        <switch>
          <!--opt, object-->
          <nightModeTmeId>
            <!--opt, int-->1
          </nightModeTmeId>
          <nightModeStartHour>
            <!--opt, int, range:[0,23]-->1
          </nightModeStartHour>
          <nightModeStartMinute>
            <!--opt, int, range:[0,59]-->1
          </nightModeStartMinute>
          <nightModeEndHour>
            <!--opt, int, range:[0,23]-->1
          </nightModeEndHour>
          <nightModeEndMinute>
            <!--opt, int, range:[0,59]-->1
          </nightModeEndMinute>
        </switch>
      </switchList>
    </timeMode>
    <autoMode>
      <!--opt, object, auto switch-->
      <autoSwtichThresholdVal>
        <!--opt, int, threshold for switching, range:[1,100]-->1
      </autoSwtichThresholdVal>
    </autoMode>
  </nightModeCtrl>
</DayNightGate>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.6.2 Get the day/night switch parameters of the specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/dayNightGate

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<DayNightGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, day/night switch, attr:version{req, string, protocolVersion}-->
  <brightnessThreld>
    <!--ro, opt, int, brightness-->1
  </brightnessThreld>
  <nightModeCtrl>
    <!--ro, opt, object, day/night mode switching control-->
    <nightModeCtrlMode>
      <!--ro, opt, enum, day/night switch mode, subType:string, desc:"manual" (manual switch mode), "time" (timed switch mode), "auto" (automatic switch
mode)-->manual
    </nightModeCtrlMode>
    <manualMode>
      <!--ro, opt, object, manual switch-->
      <manualModeVal>
        <!--ro, opt, enum, "day", "night", subType:string, desc:"day", "night"-->day
      </manualModeVal>
    </manualMode>
    <timeMode>
      <!--ro, opt, object, timed switch-->
      <scheduleType>
        <!--ro, opt, enum, subType:string-->day
      </scheduleType>
      <switchList>
        <!--ro, opt, array, subType:object-->
        <switch>
          <!--ro, opt, object, timed switch settings-->
          <nightModeTmeId>
            <!--ro, opt, int, night mode ID-->1
          </nightModeTmeId>
          <nightModeStartHour>
            <!--ro, opt, int, night mode starting hour, range:[0,23]-->1
          </nightModeStartHour>
          <nightModeStartMinute>
            <!--ro, opt, int, night mode starting minute, range:[0,59]-->1
          </nightModeStartMinute>
          <nightModeEndHour>
            <!--ro, opt, int, night mode ending hour, range:[0,23]-->1
          </nightModeEndHour>
          <nightModeEndMinute>
            <!--ro, opt, int, night mode ending minute, range:[0,59]-->1
          </nightModeEndMinute>
        </switch>
      </switchList>
      <timeScheduleList>
        <!--ro, opt, array, subType:object, range:[0,2]-->
        <timeSchedule>
          <!--ro, opt, object-->
          <triggerData>
            <!--ro, opt, object-->
            <startTime>
              <!--ro, opt, object-->
              <monthOfYear>
                <!--ro, opt, int, range:[1,12]-->0
              </monthOfYear>
            </startTime>
            <endTime>
              <!--ro, opt, object-->
              <monthOfYear>
                <!--ro, opt, int, range:[1,12]-->0
              </monthOfYear>
            </endTime>
          </triggerData>
          <timeSwitchList>
            <!--ro, opt, array, subType:object, range:[0,1]-->
            <timeSwitch>
              <!--ro, opt, object-->
              <startHour>
                <!--ro, opt, int, range:[0,23]-->1
              </startHour>
              <startMinute>
                <!--ro, opt, int, range:[0,59]-->1
              </startMinute>
              <endHour>
                <!--ro, opt, int, range:[0,23]-->1
              </endHour>
              <endMinute>
                <!--ro, opt, int, range:[0,59]-->1
              </endMinute>
            </timeSwitch>
          </timeSwitchList>
        </timeSchedule>
      </timeScheduleList>
      <dayNightStatus>
        <!--ro, opt, enum, subType:string-->day
```

```
        <!--ro, opt, enum, subType:string-->day
      </dayNightStatus>
    </timeMode>
    <autoMode>
      <!--ro, opt, object, automatic switch-->
      <autoSwtichThresholdVal>
        <!--ro, opt, int, threshold for automatic switch, range:[0,100]-->1
      </autoSwtichThresholdVal>
      <autoNightSwtichThresholdVal>
        <!--ro, opt, int, range:[0,100]-->1
      </autoNightSwtichThresholdVal>
      <autoDayToNightDelayTime>
        <!--ro, opt, int, range:[0,120]-->1
      </autoDayToNightDelayTime>
      <autoNightToDayDelayTime>
        <!--ro, opt, int, range:[0,120]-->1
      </autoNightToDayDelayTime>
      <ICRChange>
        <!--ro, opt, bool-->false
      </ICRChange>
      <lightChange>
        <!--ro, opt, bool-->false
      </lightChange>
      <WDRChange>
        <!--ro, opt, bool-->false
      </WDRChange>
      <dayNightStatus>
        <!--ro, opt, enum, subType:string-->day
      </dayNightStatus>
    </autoMode>
  </nightModeCtrl>
</DayNightGate>
```

11.3.6.3 Set the IR-cut filter parameters for cameras of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/icr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ICR configuration parameters, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--opt, object, IR-cut filter-->
    <ICRCtrlMode>
      <!--req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-manually
switch, "time"-switch by schedule, "auto"-automatically switch-->close
    </ICRCtrlMode>
  </ICRCtrl>
</ImageIcrE>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.6.4 Get the IR-cut filter parameters for cameras of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/icr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ICR parameters configuration, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--ro, opt, object, IR-cut filter-->
    <ICRCtrlMode>
      <!--ro, req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-
manually switch, "time"-switch by schedule, "auto"-automatically switch-->close
    </ICRCtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,manual}-->
      <ManualPresetVal>
        <!--ro, req, enum, manually switch, subType:string, desc:"day", "night"-->day
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,time}-->
      <scheduleType>
        <!--ro, opt, enum, subType:string-->day
      </scheduleType>
      <SwitchList>
        <!--ro, opt, array, schedule list for switching, subType:object, range:[0,2], dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,day}, desc:up to
two time period control can be supported. For example: time period is from 6:00  to 18:00,the preset ID is 1-->
        <Preset>
          <!--ro, opt, object, preset-->
          <PresetId>
            <!--ro, req, int, preset ID-->1
          </PresetId>
          <PresetVal>
            <!--ro, opt, enum, "day","night", subType:string, desc:"day", "night"-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId>
            <!--ro, req, int, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--ro, opt, int, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--ro, opt, int, range:[0,23]-->1
          </endHour>
          <endMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </endMinute>
          <PresetVal>
            <!--ro, opt, enum, subType:string-->day
          </PresetVal>
        </TimeSwitch>
      </SwitchList>
      <dayNightStatus>
        <!--ro, opt, enum, subType:string-->day
      </dayNightStatus>
      <timeScheduleList>
        <!--ro, opt, array, subType:object, range:[0,2], dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,range}-->
        <timeSchedule>
          <!--ro, opt, object-->
          <triggerData>
            <!--ro, opt, object-->
            <startTime>
              <!--ro, opt, object-->
              <monthOfYear>
                <!--ro, opt, int, range:[1,12]-->1
              </monthOfYear>
            </startTime>
            <endTime>
              <!--ro, opt, object-->
              <monthOfYear>
```

```
                <!--ro, opt, int, range:[1,12]-->1
              </monthOfYear>
            </endTime>
          </triggerData>
          <timeSwitchList>
            <!--ro, opt, array, subType:object, range:[0,2]-->
            <timeSwitch>
              <!--ro, opt, object-->
              <timeID>
                <!--ro, req, int, range:[0,1]-->1
              </timeID>
              <startHour>
                <!--ro, opt, int, range:[0,23]-->1
              </startHour>
              <startMinute>
                <!--ro, opt, int, range:[0,59]-->1
              </startMinute>
              <endHour>
                <!--ro, opt, int, range:[0,23]-->1
              </endHour>
              <endMinute>
                <!--ro, opt, int, range:[0,59]-->1
              </endMinute>
              <dayNightVal>
                <!--ro, opt, enum, subType:string-->day
              </dayNightVal>
            </timeSwitch>
          </timeSwitchList>
        </timeSchedule>
      </timeScheduleList>
    </TimeMode>
    <AutoMode>
      <!--ro, opt, object, automatically switch according to the light change, dep:and,{$.ImageIcrE.ICRCtrl.ICRCtrlMode,eq,auto}-->
      <DayNightFilterTh>
        <!--ro, req, int, day/night auto switch sensitivity, range:[0,100]-->1
      </DayNightFilterTh>
      <ICRAutoSwitch min="0" max="100">
        <!--ro, opt, int, ICR auto-switch and threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
      </ICRAutoSwitch>
      <dayNightStatus>
        <!--ro, opt, enum, subType:string-->day
      </dayNightStatus>
    </AutoMode>
  </ICRCtrl>
  <ECtrl>
    <!--ro, opt, object, -E device,halo inhibition filter,which can be used to inhibit the halo-->
    <ECtrlMode>
      <!--ro, req, enum, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, subType:string, desc:"close", "manual"-
manually switch, "time"-switch by schedule, "auto"-automatically switch-->close
    </ECtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,manual}-->
      <ManualPresetVal>
        <!--ro, req, int, manually switch, range:[0,100]-->1
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,time}-->
      <SwitchList>
        <!--ro, opt, array, schedule list for switching, subType:object, range:[0,2], desc:up to two time period control can be supported. For example: time
period is from 6:00  to 18:00,the preset ID is 1-->
        <Preset>
          <!--ro, opt, object, preset-->
          <PresetId>
            <!--ro, req, int, preset ID-->1
          </PresetId>
          <PresetVal>
            <!--ro, opt, enum, "day", "night", subType:string, desc:"day", "night”-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId>
            <!--ro, req, int, range:[0,2]-->1
          </timeId>
          <startHour>
            <!--ro, opt, int, range:[0,23]-->1
          </startHour>
          <startMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </startMinute>
          <endHour>
            <!--ro, opt, int, range:[0,23]-->1
          </endHour>
          <endMinute>
            <!--ro, opt, int, range:[0,59]-->1
          </endMinute>
          <PresetVal>
            <!--ro, opt, enum, subType:string-->day
          </PresetVal>
        </TimeSwitch>
      </SwitchList>
    </TimeMode>
    <AutoMode>
```

```
    <AutoMode>
      <!--ro, opt, object, automatically switch according to the light change, dep:and,{$.ImageIcrE.ECtrl.ECtrlMode,eq,auto}-->
      <DayNightFilterTh>
        <!--ro, req, int, day/night auto switch sensitivity, range:[0,100]-->1
      </DayNightFilterTh>
      <ExceptionCatchTh>
        <!--ro, req, int, exception detection sensitivity, range:[0,100]-->1
      </ExceptionCatchTh>
    </AutoMode>
  </ECtrl>
</ImageIcrE>
```

11.3.6.5 Get the IR-cut filter capability for cameras of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/icr/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ImageIcrE xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ICR parameters configuration, attr:version{req, string, protocolVersion}-->
  <ICRCtrl>
    <!--ro, opt, object, IR-cut filter-->
    <ICRCtrlMode opt="close,manual,time,auto">
      <!--ro, req, string, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, attr:opt{req, string}-->test
    </ICRCtrlMode>
    <ManualMode>
      <!--ro, opt, object, manually switch-->
      <ManualPresetVal opt="day,night">
        <!--ro, req, string, manually switch, attr:opt{req, string}-->test
      </ManualPresetVal>
    </ManualMode>
    <TimeMode>
      <!--ro, opt, object, switch by schedule-->
      <scheduleType opt="day,range">
        <!--ro, opt, string, attr:opt{req, string}-->day
      </scheduleType>
      <SwitchList size="2">
        <!--ro, opt, array, subType:object, dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,day}, attr:size{req, int}-->
        <Preset>
          <!--ro, req, object-->
          <PresetId min="0" max="1">
            <!--ro, req, int, attr:min{req, int},max{req, int}-->1
          </PresetId>
          <PresetVal opt="day,night">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
          </PresetVal>
        </Preset>
        <TimeSwitch>
          <!--ro, opt, object-->
          <timeId min="0" max="1">
            <!--ro, req, int, attr:min{req, int},max{req, int}-->1
          </timeId>
          <startHour min="0" max="23">
            <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
          </startHour>
          <startMinute min="0" max="59">
            <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
          </startMinute>
          <endHour min="0" max="23">
            <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
          </endHour>
          <endMinute min="0" max="59">
            <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
          </endMinute>
          <PresetVal opt="day,night">
            <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
          </PresetVal>
        </TimeSwitch>
      </SwitchList>
      <timeScheduleList size="2">
        <!--ro, opt, object, dep:and,{$.ImageIcrE.ICRCtrl.TimeMode.scheduleType,eq,range}, attr:size{req, int}-->
        <timeSchedule>
          <!--ro, opt, object-->
```

```
        <triggerData>
          <!--ro, opt, object-->
          <startTime>
            <!--ro, opt, object-->
            <monthOfYear min="1" max="12">
              <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->0
            </monthOfYear>
          </startTime>
          <endTime>
            <!--ro, opt, object-->
            <monthOfYear min="1" max="12">
              <!--ro, opt, int, range:[1,12], attr:min{req, int},max{req, int}-->0
            </monthOfYear>
          </endTime>
        </triggerData>
        <timeSwitchList size="2">
          <!--ro, opt, object, attr:size{req, int}-->
          <timeSwitch>
            <!--ro, opt, object-->
            <timeID min="0" max="1">
              <!--ro, req, int, range:[0,1], attr:min{req, int},max{req, int}-->1
            </timeID>
            <startHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </startHour>
            <startMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </startMinute>
            <endHour min="0" max="23">
              <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
            </endHour>
            <endMinute min="0" max="59">
              <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
            </endMinute>
            <dayNightVal opt="day,night">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
            </dayNightVal>
          </timeSwitch>
        </timeSwitchList>
      </timeSchedule>
    </timeScheduleList>
  </TimeMode>
  <AutoMode>
    <!--ro, opt, object, automatically switch according to the light change-->
    <DayNightFilterTh min="0" max="100">
      <!--ro, req, int, day/night auto switch sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </DayNightFilterTh>
    <ICRAutoSwitch min="0" max="100">
      <!--ro, opt, int, ICR auto-switch and threshold, range:[0,100], attr:min{req, int},max{req, int}-->1
    </ICRAutoSwitch>
  </AutoMode>
</ICRCtrl>
<ECtrl>
  <!--ro, opt, object, -E device,halo inhibition filter,which can be used to inhibit the halo-->
  <ECtrlMode opt="close,manual,time,auto">
    <!--ro, req, string, "close","manual"-manually switch,"time"-switch by schedule,"auto"-automatically switch, attr:opt{req, string}-->test
  </ECtrlMode>
  <ManualMode>
    <!--ro, opt, object, manually switch-->
    <ManualPresetVal min="0" max="100">
      <!--ro, req, int, manually switch, range:[0,100], attr:min{req, int},max{req, int}-->1
    </ManualPresetVal>
  </ManualMode>
  <TimeMode>
    <!--ro, opt, object, switch by schedule-->
    <SwitchList size="2">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <Preset>
        <!--ro, req, object-->
        <PresetId min="0" max="1">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </PresetId>
        <PresetVal opt="day,night">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
        </PresetVal>
      </Preset>
      <TimeSwitch>
        <!--ro, opt, object-->
        <timeId min="0" max="1">
          <!--ro, req, int, attr:min{req, int},max{req, int}-->1
        </timeId>
        <startHour min="0" max="23">
          <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
        </startHour>
        <startMinute min="0" max="59">
          <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
        </startMinute>
        <endHour min="0" max="23">
          <!--ro, opt, int, range:[0,23], attr:min{req, int},max{req, int}-->1
        </endHour>
        <endMinute min="0" max="59">
          <!--ro, opt, int, range:[0,59], attr:min{req, int},max{req, int}-->1
        </endMinute>
        <PresetVal opt="day,night">
```

```
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->day
      </PresetVal>
    </TimeSwitch>
  </SwitchList>
</TimeMode>
<AutoMode>
  <!--ro, opt, object, automatically switch according to the light change-->
  <DayNightFilterTh min="0" max="100">
    <!--ro, req, int, day/night auto switch sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
  </DayNightFilterTh>
  <ExceptionCatchTh min="0" max="100">
    <!--ro, req, int, "0-100", range:[0,100], attr:min{req, int},max{req, int}-->1
  </ExceptionCatchTh>
</AutoMode>
</ECtrl>
</ImageIcrE>
```

11.3.7 图像白平衡

11.3.7.1 Set the white balance parameters of s specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/whiteBalance

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--opt, enum, white balance type, subType:string, desc:"auto", "atuo1", "auto2" "manual", "indoor", "outdoor", "autotrace", "onece", "sodiumlight",
"mercurylight"-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--req, int, this node depends on <WhiteBalanceStyle>, desc:this node depends on <WhiteBalanceStyle>-->1
  </WhiteBalanceBlue>
</WhiteBalance>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0-OK, 1-OK, 2-Device Busy, 3-Device Error, 4-Invalid Operation, 5-Invalid XML Format, 6-Invalid XML
Content, 7-Reboot Required-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.7.2 Get the white balance parameters of s specified channel

Request URL

GET /ISAPI/Image/channels/<channelID>/whiteBalance?parameterType=<parameterType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
parameterType	string	When parameterType is recommendation, the obtained value is recommended by device. If parameterType is not configured as an input parameter, it wil be the value of the parameter in the current PUT URL.

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<WhiteBalance xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, white balance, attr:version{req, string, protocolVersion}-->
  <WhiteBalanceStyle>
    <!--ro, opt, enum, "auto,manual,indoor,outdoor,autotrace,onece,sodiumlight,mercurylight,auto0,atuo1,fluorescent,natural,warm,incandescent",
subType:string, desc:"auto", "auto1", "atuo2", "manual", "indoor", "outdoor", "autotrace", "onece", "sodiumlight", "mercurylight"-->auto
  </WhiteBalanceStyle>
  <WhiteBalanceRed>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceRed>
  <WhiteBalanceBlue>
    <!--ro, req, int, white balance, range:[0,100], desc:white balance-->1
  </WhiteBalanceBlue>
  <whiteBalanceLevel>
    <!--ro, opt, int, range:[0,100], dep:or,{$.WhiteBalance.WhiteBalanceStyle,eq,auto},{$.WhiteBalance.WhiteBalanceStyle,eq,auto1},
{$.WhiteBalance.WhiteBalanceStyle,eq,auto2}-->1
  </whiteBalanceLevel>
  <customColorTemperatureLevel>
    <!--ro, opt, int, range:[1800,10000]-->1
  </customColorTemperatureLevel>
  <WhiteBalanceRedFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceRedFineTuning>
  <WhiteBalanceBlueFineTuning>
    <!--ro, opt, int, range:[0,100]-->1
  </WhiteBalanceBlueFineTuning>
  <autoMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,auto}-->video
  </autoMode>
  <manualMode>
    <!--ro, opt, enum, subType:string, dep:and,{$.WhiteBalance.WhiteBalanceStyle,eq,manual}-->colorFineTuning
  </manualMode>
</WhiteBalance>
```


11.3.8 手动抓拍

11.3.8.1 Get the manual capture result information of a specified channel

Request URL

GET /ISAPI/Streaming/channels/<trackStreamID>/picture?snapShotImageType=<snapShotImageType>&videoResolutionWidth=<videoResolutionWidth>&videoResolutionHeight=<videoResolutionHeight>&imageQuality=<imageQuality>&x=<x>&y=<y>&hight=<hight>&width=<width>&maxSnapResolution=<maxSnapResolution>&livingBodyDetect=<livingBodyDetect>

Query Parameter

Parameter Name	Parameter Type	Description
trackStreamID	string	Channel No. * 100 + stream type (-1: main stream, -2: sub-stream).
snapShotImageType	string	Image formats: "bayer" (image not processed by ISP, which can be used to test whether the image clarity of the factory equipment meets the standard), "JPEGWITHRULE" (this type of image is specially for HIKMICRO devices, indicating a JPEG image captured with temperature rule information overlaid), "JPEGWITHOUTRULE" (when this type is used by HIKMICRO devices, it indicates a JPEG image captured without temperature rule information overlaid), "JPEG" (when this type is used by HIKMICRO devices, whether the captured JPEG image includes rule information depends on the enabling status of overlay temperature information on stream function. If it is enabled, the image returned will have rule information overlaid; otherwise, it will not. If the HIKMICRO device calls the URL without using this parameter, the logic for the returned JPEG image is consistent with the above JPEG type, depending on the enabling status of overlay temperature information on stream function).
videoResolutionWidth	string	Resolution width, and if not configured, it is 704 by default (D1 resolution width).
videoResolutionHeight	string	Resolution height, and if not configured, it is 576 by default (D1 resolution height).
imageQuality	string	Image quality: "best", "better", "general"
x	string	x-coordinate of the specified area, the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen
y	string	y-coordinate of the specified area, the value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen
hight	string	Height of the specified area. The value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen.
width	string	Width of the specified area. The value ranges from 0.000 to 1.000. The origin is the upper-left corner of the screen.
maxSnapResolution	enum	Capture images using the maximum image capture resolution. For certain devices, such as low-power devices, the actual maximum image capture resolution differs from the video encoding resolution. During certain intelligent detections, images are captured at the maximum image capture resolution. If not configured, the default value is 1.
livingBodyDetect	enum	Whether to perform face anti-spoofing detection when capturing images.

Request Message

None

Response Message

Binary Data

11.3.9 移动侦测

11.3.9.1 Get the arming schedule of motion detection for a specified channel

Request URL

GET /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->VMD_video1
  </id>
  <eventType>
    <!--ro, opt, string-->VMD
  </eventType>
  <inputIOPortID>
    <!--ro, opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--ro, opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--ro, req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--ro, opt, object-->
      <dayOfWeek>
        <!--ro, opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--ro, opt, array, subType:object-->
    <TimeBlock>
      <!--ro, opt, object-->
      <TimeRange>
        <!--ro, req, object-->
        <beginTime>
          <!--ro, req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--ro, req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```


11.3.9.2 Set the arming schedule of motion detection for a specified channel

Request URL

PUT /ISAPI/Event/schedules/motionDetections/VMD_video<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Schedule xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->VMD_video1
  </id>
  <eventType>
    <!--opt, string-->VMD
  </eventType>
  <inputIOPortID>
    <!--opt, string-->test
  </inputIOPortID>
  <outputIOPortID>
    <!--opt, string-->test
  </outputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <TimeBlockList size="8">
    <!--req, array, subType:object, attr:size{opt, int}-->
    <TimeBlock>
      <!--opt, object-->
      <dayOfWeek>
        <!--opt, enum, subType:int-->1
      </dayOfWeek>
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </TimeBlockList>
  <HolidayBlockList>
    <!--opt, array, subType:object-->
    <TimeBlock>
      <!--opt, object-->
      <TimeRange>
        <!--req, object-->
        <beginTime>
          <!--req, time-->10:00:00
        </beginTime>
        <endTime>
          <!--req, time-->10:00:00
        </endTime>
      </TimeRange>
    </TimeBlock>
  </HolidayBlockList>
</Schedule>
```


Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.9.3 Get the linkage parameters of motion detection for a specified channel

Request URL

GET /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
triggersEnabledStatus	string	Linkage enabling status (if this node is included in the URL, the enabling and disabling of the linkage mode should be determined based on the triggerEnabled status in the linkage list). This node is supported if the value of isSupporttriggersEnabled in /ISAPI/Event/triggersCap contains is true.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, req, string-->1
  </id>
  <eventType>
    <!--ro, req, string-->VMD
  </eventType>
  <eventDescription>
    <!--ro, opt, string-->test
  </eventDescription>
  <videoInputChannelID>
    <!--ro, opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--ro, opt, string-->1
  </dynVideoInputChannelID>
  <EventTriggerNotificationList>
    <!--ro, opt, array, subType:object-->
  </EventTriggerNotificationList>
</EventTrigger>
```

11.3.9.4 Set the linkage parameters of motion detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/VMD-<channelID>?triggersEnabledStatus=<triggersEnabledStatus>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
triggersEnabledStatus	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->1
  </id>
  <eventType>
    <!--req, enum, subType:string-->VMD
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
```

```xml
    </dynVideoInputChannelID>
    <intervalBetweenEvents>
      <!--opt, int, unit:s-->1
    </intervalBetweenEvents>
    <WLSensorID>
      <!--opt, string-->1
    </WLSensorID>
    <EventTriggerNotificationList>
      <!--opt, array, subType:object-->
      <EventTriggerNotification>
        <!--opt, object-->
        <id>
          <!--req, string-->test
        </id>
        <notificationMethod>
          <!--req, enum, subType:string-->email
        </notificationMethod>
        <notificationRecurrence>
          <!--opt, enum, subType:string-->beginning
        </notificationRecurrence>
        <notificationInterval>
          <!--opt, int, unit:ms-->1
        </notificationInterval>
        <outputIOPortID>
          <!--opt, string--> 1
        </outputIOPortID>
        <dynOutputIOPortID>
          <!--opt, string--> 1
        </dynOutputIOPortID>
        <videoInputID>
          <!--opt, string-->1
        </videoInputID>
        <dynVideoInputID>
          <!--opt, string-->1
        </dynVideoInputID>
        <ptzAction>
          <!--opt, object-->
          <ptzChannelID>
            <!--req, string-->1
          </ptzChannelID>
          <actionName>
            <!--req, enum, subType:string-->preset
          </actionName>
          <actionNum>
            <!--opt, int-->1
          </actionNum>
          <presetDurationTime>
            <!--opt, int, unit:s-->1
          </presetDurationTime>
        </ptzAction>
        <cellphoneNumber>
          <!--opt, string, range:[1,11]-->test
        </cellphoneNumber>
        <LightAudioByIPAction>
          <!--opt, object-->
          <notificationType>
            <!--req, enum, subType:string-->audio
          </notificationType>
          <notificationName>
            <!--opt, string-->test
          </notificationName>
        </LightAudioByIPAction>
        <SupplementLightAlarm>
          <!--opt, object-->
          <durationTime>
            <!--opt, int, range:[0,90], unit:s-->5
          </durationTime>
        </SupplementLightAlarm>
      </EventTriggerNotification>
    </EventTriggerNotificationList>
  </EventTrigger>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.9.5 Get motion detection parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, motion detection parameters of a specified video input channel, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
  <minObjectSize>
    <!--ro, opt, int, minimum target resolution-->0
  </minObjectSize>
  <maxObjectSize>
    <!--ro, opt, int, maximum target resolution-->0
  </maxObjectSize>
  <ROI>
    <!--ro, opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->0
    </normalizedScreenHeight>
  </ROI>
  <enableHighlight>
    <!--ro, opt, bool, whether to enable highlight-->true
  </enableHighlight>
  <MotionDetectionSwitch>
    <!--ro, opt, object, day/night settings switch-->
    <type>
      <!--ro, opt, enum, type, subType:string, desc:"off", "auto", "schedule" (scheduled)-->off
    </type>
    <Schedule>
      <!--ro, opt, object, schedule-->
      <scheduleType>
        <!--ro, req, enum, schedule type, subType:string, desc:"day" (daytime), "night" (night)-->day
      </scheduleType>
      <TimeRange>
        <!--ro, req, object, time period-->
        <beginTime>
          <!--ro, req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--ro, req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
  </MotionDetectionSwitch>
  <activeMode>
    <!--ro, opt, enum, effective mode, subType:string, desc:"normal", "expert"-->normal
  </activeMode>
```

```
    <MotionDetectionRegionList size="10">
      <!--ro, opt, array, detection area list, subType:object, attr:size{req, int}-->
      <MotionDetectionRegion>
        <!--ro, opt, object, detection area-->
        <id>
          <!--ro, req, string, ID-->test
        </id>
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <sensitivityLevel>
          <!--ro, opt, int, sensitivity, range:[0,100]-->0
        </sensitivityLevel>
        <daySensitivityLevel>
          <!--ro, opt, int, sensitivity (during day time), range:[0,100]-->0
        </daySensitivityLevel>
        <nightSensitivityLevel>
          <!--ro, opt, int, sensitivity (during night), range:[0,100]-->0
        </nightSensitivityLevel>
        <objectSize>
          <!--ro, opt, int, percentage, range:[0,100]-->0
        </objectSize>
        <dayObjectSize>
          <!--ro, opt, int, percentage (during day time), range:[0,100]-->0
        </dayObjectSize>
        <nightObjectSize>
          <!--ro, opt, int, percentage (during night), range:[0,100]-->0
        </nightObjectSize>
        <RegionCoordinatesList>
          <!--ro, req, array, rule region list, subType:object, range:[0,4], desc:rectangle-->
          <RegionCoordinates>
            <!--ro, opt, object, area, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--ro, req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </MotionDetectionRegion>
    </MotionDetectionRegionList>
    <enableWithMoving>
      <!--ro, opt, bool, whether to enable dynamic analysis-->true
    </enableWithMoving>
  </MotionDetectionExt>
```

11.3.9.6 Set motion detection parameters of a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/motionDetectionExt

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<MotionDetectionExt xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, motion detection parameters of a specified video input channel, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <minObjectSize>
    <!--opt, int, minimum target resolution-->0
  </minObjectSize>
  <maxObjectSize>
    <!--opt, int, maximum target resolution-->0
  </maxObjectSize>
  <ROI>
    <!--opt, object, region of interest (ROI)-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->0
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->0
    </normalizedScreenHeight>
  </ROI>
  <enableHighlight>
    <!--opt, bool, whether to enable highlight-->true
  </enableHighlight>
  <MotionDetectionSwitch>
```

```
    <!--opt, object, day/night settings switch-->
    <type>
      <!--opt, enum, type, subType:string, desc:"off", "auto", "schedule" (scheduled)-->off
    </type>
    <Schedule>
      <!--opt, object, schedule-->
      <scheduleType>
        <!--req, enum, schedule type, subType:string, desc:"day" (daytime), "night" (night)-->day
      </scheduleType>
      <TimeRange>
        <!--req, object, time period-->
        <beginTime>
          <!--req, time, start time-->00:00:00+08:00
        </beginTime>
        <endTime>
          <!--req, time, end time-->00:00:00+08:00
        </endTime>
      </TimeRange>
    </Schedule>
  </MotionDetectionSwitch>
  <activeMode>
    <!--opt, enum, effective mode, subType:string, desc:"normal", "expert"-->normal
  </activeMode>
  <MotionDetectionRegionList size="10">
    <!--opt, array, detection area list, subType:object, attr:size{req, int}-->
    <MotionDetectionRegion>
      <!--opt, object, detection area-->
      <id>
        <!--req, string, ID-->test
      </id>
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <sensitivityLevel>
        <!--opt, int, sensitivity, range:[0,100]-->0
      </sensitivityLevel>
      <daySensitivityLevel>
        <!--opt, int, sensitivity (during day time), range:[0,100]-->0
      </daySensitivityLevel>
      <nightSensitivityLevel>
        <!--opt, int, sensitivity (during night), range:[0,100]-->0
      </nightSensitivityLevel>
      <objectSize>
        <!--opt, int, percentage, range:[0,100]-->0
      </objectSize>
      <dayObjectSize>
        <!--opt, int, percentage (during day time), range:[0,100]-->0
      </dayObjectSize>
      <nightObjectSize>
        <!--opt, int, percentage (during night), range:[0,100]-->0
      </nightObjectSize>
      <RegionCoordinatesList>
        <!--req, array, rule region list, subType:object, range:[0,4], desc:rectangle-->
        <RegionCoordinates>
          <!--opt, object, area, desc:the origin is the lower-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </MotionDetectionRegion>
  </MotionDetectionRegionList>
  <enableWithMoving>
    <!--opt, bool, whether to enable dynamic analysis-->true
  </enableWithMoving>
</MotionDetectionExt>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device 
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.3.10 定时抓拍

11.3.10.1 Set parameters of capturing pictures by schedule for a specified channel

Request URL

PUT /ISAPI/Snapshot/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapshotChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, No.-->1
  </id>
  <videoInputChannelID>
    <!--req, string, video input channel ID-->1
  </videoInputChannelID>
  <timingCapture>
    <!--opt, object, configurations of capturing pictures by schedule-->
    <enabled>
      <!--req, bool, whether to enable capturing pictures by schedule-->true
    </enabled>
    <supportSchedule>
      <!--opt, bool, whether to enable configuring the schedule for picture capturing-->true
    </supportSchedule>
    <compress>
      <!--opt, object, compression configurations-->
      <pictureCodecType>
        <!--req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->PNG
      </pictureCodecType>
      <pictureWidth>
        <!--req, int, width of the picture resolution-->1024
      </pictureWidth>
      <pictureHeight>
        <!--req, int, height of the picture resolution-->1024
      </pictureHeight>
      <quality>
        <!--opt, int, picture quality, range:[0,100], desc:picture quality-->50
      </quality>
      <captureInterval>
        <!--opt, int, capture interval, unit:ms-->100
      </captureInterval>
      <captureNumber>
        <!--opt, int, number of captured pictures-->1
      </captureNumber>
    </compress>
  </timingCapture>
  <eventCapture>
    <!--opt, object-->
    <enabled>
      <!--req, bool, whether to enable capturing events-->true
    </enabled>
    <supportSchedule>
      <!--opt, bool, whether to enable configuring the schedule for event capturing-->true
    </supportSchedule>
    <compress>
      <!--opt, object-->
      <pictureCodecType>
        <!--req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->PNG
      </pictureCodecType>
      <pictureWidth>
        <!--req, int, width of the picture resolution-->1024
      </pictureWidth>
      <pictureHeight>
        <!--req, int, height of the picture resolution-->1024
      </pictureHeight>
      <quality>
        <!--opt, int, picture quality, range:[0,100], desc:picture quality-->15
      </quality>
      <captureInterval>
        <!--opt, int, capture interval, unit:ms-->100
      </captureInterval>
      <captureNumber>
        <!--opt, int, number of captured pictures-->1
      </captureNumber>
    </compress>
  </eventCapture>
</SnapshotChannel>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.10.2 Get the parameters of a specific channel for capturing pictures by schedule

Request URL

GET /ISAPI/Snapshot/channels/<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<SnapshotChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--ro, req, string, ID-->1
  </id>
  <videoInputChannelID>
    <!--ro, req, string, video input channel ID-->1
  </videoInputChannelID>
  <timingCapture>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable capturing pictures by schedule-->true
    </enabled>
    <supportSchedule>
      <!--ro, opt, bool-->true
    </supportSchedule>
    <compress>
      <!--ro, opt, object-->
      <pictureCodecType>
        <!--ro, req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->PNG
      </pictureCodecType>
      <pictureWidth>
        <!--ro, req, int, picture resolution (width)-->1024
      </pictureWidth>
      <pictureHeight>
        <!--ro, req, int, picture resolution (height)-->1024
      </pictureHeight>
      <quality>
        <!--ro, opt, int, picture quality, range:[0,100], desc:picture quality (in percentage),it is between 1 and 100-->50
      </quality>
      <captureInterval>
        <!--ro, opt, int, capture interval, unit:ms-->100
      </captureInterval>
      <captureNumber>
        <!--ro, opt, int, number of captured pictures-->1
      </captureNumber>
    </compress>
  </timingCapture>
  <eventCapture>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable capturing pictures by schedule-->true
    </enabled>
    <supportSchedule>
      <!--ro, opt, bool-->true
    </supportSchedule>
    <compress>
      <!--ro, opt, object-->
      <pictureCodecType>
        <!--ro, req, enum, picture format, subType:string, desc:"JPEG", "BMP", "GIF", "PNG"-->PNG
      </pictureCodecType>
      <pictureWidth>
        <!--ro, req, int, picture resolution (width)-->1024
      </pictureWidth>
      <pictureHeight>
        <!--ro, req, int, picture resolution (height)-->1024
      </pictureHeight>
      <quality>
        <!--ro, opt, int, picture quality, range:[0,100], desc:picture quality (in percentage),it is between 1 and 100-->15
      </quality>
      <captureInterval>
        <!--ro, opt, int, capture interval, unit:ms-->100
      </captureInterval>
      <captureNumber>
        <!--ro, opt, int, number of captured pictures-->1
      </captureNumber>
    </compress>
  </eventCapture>
</SnapshotChannel>
```


11.3.10.3 Get the configuration capability of capturing pictures by schedule

Request URL

GET /ISAPI/Snapshot/channels/<channelID>/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<SnapshotChannel xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <id opt="1">
    <!--ro, req, string, ID, attr:opt{opt, string}-->1
  </id>
  <videoInputChannelID opt="1">
    <!--ro, req, string, video input channel ID, attr:opt{opt, string}-->1
  </videoInputChannelID>
  <timingCapture>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{opt, string}-->true
    </enabled>
    <supportSchedule opt="true,false">
      <!--ro, req, bool, attr:opt{opt, string}-->true
    </supportSchedule>
    <compress>
      <!--ro, opt, object-->
      <pictureCodecType opt="JPEG">
        <!--ro, req, enum, picture format, subType:string, attr:opt{opt, string}, desc:picture format: "JPEG,BMP,GIF,PNG"-->PNG
      </pictureCodecType>
      <pictureWidth opt="1280">
        <!--ro, req, int, attr:opt{opt, string}-->1024
      </pictureWidth>
      <pictureHeight opt="720">
        <!--ro, req, int, attr:opt{opt, string}-->1024
      </pictureHeight>
      <quality opt="40,60,80">
        <!--ro, opt, int, picture quality, range:[0,100], attr:opt{opt, string}, desc:quality in percentage,the value is between 0 and 100-->50
      </quality>
      <captureInterval min="1000" max="604800000">
        <!--ro, opt, int, capture interval, unit: milliseconds, unit:ms, attr:min{opt, string},max{opt, string}-->100
      </captureInterval>
      <captureNumber min="1" max="120" def="4">
        <!--ro, opt, int, attr:min{opt, string},max{opt, string},def{opt, string}-->1
      </captureNumber>
    </compress>
  </timingCapture>
  <eventCapture>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{opt, string}-->true
    </enabled>
    <supportSchedule opt="false">
      <!--ro, opt, bool, attr:opt{opt, string}-->bool
    </supportSchedule>
    <compress>
      <!--ro, opt, object-->
      <pictureCodecType opt="JPEG">
        <!--ro, req, string, picture format, attr:opt{opt, string}-->JPEG
      </pictureCodecType>
      <pictureWidth opt="1280">
        <!--ro, req, int, attr:opt{opt, string}-->1024
      </pictureWidth>
      <pictureHeight opt="720">
        <!--ro, req, int, attr:opt{opt, string}-->1024
      </pictureHeight>
      <quality opt="40,60,80">
        <!--ro, opt, int, picture quality, range:[0,100], attr:opt{opt, string}, desc:quality in percentage,the value is between 0 and 100-->50
      </quality>
      <captureInterval min="1000" max="604800000">
        <!--ro, opt, int, capture interval, unit: milliseconds, unit:ms, attr:min{opt, string},max{opt, string}-->100
      </captureInterval>
      <captureNumber min="1" max="120" def="4">
        <!--ro, opt, int, attr:min{opt, string},max{opt, string},def{opt, string}-->1
      </captureNumber>
    </compress>
  </eventCapture>
  <PromptDescription>
    <!--ro, opt, object-->
    <prompt1>
      <!--ro, opt, string, description-->test
    </prompt1>
  </PromptDescription>
  <StreamInformationList>
    <!--ro, opt, array, subType:object-->
    <StreamInformation>
      <!--ro, opt, object, list-->
      <streamType opt="mainstream,substream,stream3,stream4,stream5">
        <!--ro, opt, string, stream type, attr:opt{opt, string}-->mainstream
      </streamType>
      <timingCapture>
        <!--ro, opt, object-->
        <enabled opt="true,false">
          <!--ro, req, bool, attr:opt{opt, string}-->true
        </enabled>
```

```xml
        </enabled>
        <supportSchedule opt="true,false">
          <!--ro, req, bool, attr:opt{opt, string}-->true
        </supportSchedule>
        <compress>
          <!--ro, opt, object-->
          <pictureCodecType opt="JPEG">
            <!--ro, req, string, picture format, attr:opt{opt, string}-->JPEG
          </pictureCodecType>
          <pictureWidth opt="1280">
            <!--ro, req, int, attr:opt{opt, string}-->1024
          </pictureWidth>
          <pictureHeight opt="720">
            <!--ro, req, int, attr:opt{opt, string}-->1024
          </pictureHeight>
          <quality opt="40,60,80">
            <!--ro, opt, int, picture quality, range:[0,100], attr:opt{opt, string}, desc:quality in percentage,the value is between 0 and 100-->50
          </quality>
          <captureInterval min="1000" max="604800000">
            <!--ro, opt, int, capture interval, unit: milliseconds, unit:ms, attr:min{opt, string},max{opt, string}-->100
          </captureInterval>
          <captureNumber min="1" max="120" def="4">
            <!--ro, opt, int, attr:min{opt, string},max{opt, string},def{opt, string}-->1
          </captureNumber>
        </compress>
      </timingCapture>
      <eventCapture>
        <!--ro, opt, object-->
        <enabled opt="true,false">
          <!--ro, req, bool, attr:opt{opt, string}-->true
        </enabled>
        <supportSchedule opt="true,false">
          <!--ro, req, bool, attr:opt{opt, string}-->true
        </supportSchedule>
        <compress>
          <!--ro, opt, object-->
          <pictureCodecType opt="JPEG">
            <!--ro, req, string, picture format, attr:opt{opt, string}-->JPEG
          </pictureCodecType>
          <pictureWidth opt="1280">
            <!--ro, req, int, attr:opt{opt, string}-->1280
          </pictureWidth>
          <pictureHeight opt="720">
            <!--ro, req, int, attr:opt{opt, string}-->720
          </pictureHeight>
          <quality opt="40,60,80">
            <!--ro, opt, int, quality in percentage, the value is between 0 and 100, attr:opt{opt, string}-->100
          </quality>
          <captureInterval min="1000" max="604800000">
            <!--ro, opt, int, capture interval, unit:ms, attr:min{opt, string},max{opt, string}-->1000
          </captureInterval>
          <captureNumber min="1" max="120" def="4">
            <!--ro, opt, int, attr:min{opt, string},max{opt, string},def{opt, string}-->1
          </captureNumber>
        </compress>
      </eventCapture>
      <PromptDescription>
        <!--ro, opt, object-->
        <prompt1>
          <!--ro, opt, string, this node will be returned only when it is supported. If this node is not supported, the upper node will still exist-->test
        </prompt1>
      </PromptDescription>
    </StreamInformation>
  </StreamInformationList>
</SnapshotChannel>
```


11.3.11 视频通道资源管理

11.3.11.1 Get the digital channel access capability

Request URL

GET /ISAPI/ContentMgmt/InputProxy/channels/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<InputProxyChannelListCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, digital channel access management capability, attr:version{req, string, protocolVersion}-->
  <InputProxyChannel>
```

```xml
    <!--ro, opt, object, input channel-->
    <id>
      <!--ro, req, string, index-->test
    </id>
    <name>
      <!--ro, opt, string, name, range:[1,32]-->test
    </name>
    <sourceInputPortDescriptor>
      <!--ro, req, object, input channel description-->
      <addressingFormatType opt="ipaddress,hostname">
        <!--ro, req, enum, address type, subType:string, attr:opt{req, string}, desc:"hostname", "ipaddress". It is invalid when the protocol type is 
GB28181-->hostname
      </addressingFormatType>
      <hostName min="1" max="64">
        <!--ro, opt, string, domain name, range:[1,64], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </hostName>
      <ipAddress min="1" max="32">
        <!--ro, opt, string, IPv4 address, range:[1,32], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </ipAddress>
      <ipv6Address min="1" max="128">
        <!--ro, opt, string, IPv6 address, range:[1,128], attr:min{req, int},max{req, int}, desc:it is invalid when the protocol type is GB28181-->test
      </ipv6Address>
      <managePortNo min="1" max="65535">
        <!--ro, req, int, port No. under management, attr:min{req, int},max{req, int}-->1
      </managePortNo>
      <srcInputPort min="1" max="256">
        <!--ro, req, string, device channel No., range:[1,256], attr:min{req, int},max{req, int}-->test
      </srcInputPort>
      <userName min="1" max="32">
        <!--ro, req, string, user name, range:[1,32], attr:min{req, int},max{req, int}-->test
      </userName>
      <password min="1" max="16">
        <!--ro, req, string, password, range:[1,16], attr:min{req, int},max{req, int}-->test
      </password>
      <streamType opt="auto,tcp,udp">
        <!--ro, opt, enum, streaming protocol type, subType:string, attr:opt{req, string}, desc:"auto", "TCP", "UDP"-->auto
      </streamType>
      <deviceID min="1" max="32">
        <!--ro, opt, string, device ID, range:[1,32], attr:min{req, int},max{req, int}-->test
      </deviceID>
      <activateEnabled opt="true,false">
        <!--ro, opt, bool, whether the device needs to be activated, attr:opt{req, string}, desc:if this node does not exist, it indicates that the device 
has already been activated. Application scenario: supports adding an unactivated device to the NVR channel. If this node is true, the device is unactivated 
and needs to be activated by the NVR. The NVR will auto-activate the device (MAC address is needed because IP access is unavailable for unactivated 
devices), and will add the device to NVR-->true
      </activateEnabled>
      <macAddress min="0" max="48">
        <!--ro, opt, string, MAC address, range:[0,48], attr:min{req, int},max{req, int}, desc:it is required when the value of activateEnabled is true--
>test
      </macAddress>
      <cameraType>
        <!--ro, opt, enum, camera type, subType:string, desc:"", "", "teacherLocation" (teacher locating), "studentLocation" (student locating), 
"teacherFull" (teacher full view), "studentFull" (student full view), "media", "blackboard",  "blackboardLocation", "IPCamera", "IPDome", "none"-->none
      </cameraType>
      <streamMode opt="main,sub">
        <!--ro, opt, enum, stream type, subType:string, attr:opt{req, string}, desc:"main", "sub"-->main
      </streamMode>
      <getSubStream opt="true,false">
        <!--ro, opt, bool, whether to get sub-stream, attr:opt{req, string}, desc:when this node is not configured, its value is true by default, which 
means the sub-streaming will start after the device is connected-->true
      </getSubStream>
    </sourceInputPortDescriptor>
    <online opt="true,false">
      <!--ro, opt, bool, whether the channel is online, attr:opt{req, string}-->true
    </online>
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the channel, attr:opt{req, string}, desc:the default value is true-->true
    </enabled>
    <anrInterval min="1" max="24">
      <!--ro, opt, int, time interval for automatic network replenishment, attr:min{req, int},max{req, int}-->1
    </anrInterval>
  </InputProxyChannel>
  <FilterchannelsByIp>
    <!--ro, opt, bool, whether filtering channel information by IP address is supported-->true
  </FilterchannelsByIp>
  <FilterchannelsByStreamID>
    <!--ro, opt, bool, whether filtering channel information by stream ID is supported-->true
  </FilterchannelsByStreamID>
  <isSupportSmartConfig>
    <!--ro, opt, bool, -ro, 
desc:search for connected wireless network camera automatically and then activate and add the network camera. The automatic match period is 120 seconds. The 
match process will end as soon as the number of channels reaches the upper limit.
    If its value is true, it indicates that the following URLs are supported:
PUT /ISAPI/ContentMgmt/InputProxy/channels/smartConfig?format=json;
GET ISAPI/ContentMgmt/InputProxy/channels/smartConfig/status?format=json;
PUT /ISAPI/ContentMgmt/InputProxy/channels/smartConfig/stop?format=json-->true
  </isSupportSmartConfig>
  <isSupportDevIndex>
    <!--ro, opt, bool, whether transparent transmission of sub-devices ID is supported, desc:related URLs: /ISAPI/ContentMgmt/InputProxy/channels. Each 
channel corresponds to a devIndex node. The sub-devices do not include the gateway sub-devices in /ISAPI/ContentMgmt/DeviceMgmt/addDevice?format=json-->true
  </isSupportDevIndex>
  <displayGetStream>
    <!--ro, opt, bool, whether the node configuration is displayed on the web channel configuration page, desc:currently, the configuration is not displayed 
on the NVR channel page, and is displayed on the AI DeepinMind channel page-->true
```

```
    </displayGetStream>
  </InputProxyChannelListCap>
```


11.3.12 OSD

11.3.12.1 Get the OSD parameters of a specified channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--ro, req, object, screen size-->
    <normalizedScreenWidth>
      <!--ro, req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--ro, req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent>
      <!--ro, req, bool, whether it is transparent-->true
    </transparent>
    <flashing>
      <!--ro, req, bool, whether it is flickering-->true
    </flashing>
  </attribute>
  <TextOverlayList>
    <!--ro, opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--ro, opt, object, text content-->
      <id>
        <!--ro, req, string, text No.-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--ro, req, string, overlay text-->test
      </displayText>
      <isPersistentText>
        <!--ro, opt, bool, whether the overlaid text remains the previous settings after device rebooting, desc:if the value is true or the node is not
transmitted, it indicates to write OSD to flash, and it is not suitable for frequent or fast callings. If the value is false, it indicates not to writer OSD
to flash and the OSD settings will not restore-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object, channel time information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
```

```
    <displayWeek>
      <!--ro, opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--ro, opt, object, channel name overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--ro, opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
  <frontColorMode>
    <!--ro, opt, enum, font color mode, subType:string, desc:"auto", "customize", "outline"-->auto
  </frontColorMode>
  <frontColor>
    <!--ro, opt, string, font color, desc:hexadecimal digit-->000000
  </frontColor>
  <alignment>
    <!--ro, opt, enum, alignment mode, subType:string, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (left align), "GB", "adsorption"
(adsorption mode)-->alignRight
  </alignment>
  <publicSecurity>
    <!--ro, opt, bool-->true
  </publicSecurity>
  <boundary>
    <!--ro, opt, int, the minimum boundary, range:[0,2], dep:and,{$.VideoOverlay.alignment,eq,GB}, desc:unit: number of characters-->1
  </boundary>
  <upDownboundary>
    <!--ro, opt, int, range:[0,2]-->1
  </upDownboundary>
  <PackingSpaceRecognitionOverlay>
    <!--ro, opt, object, display of parking space detection result-->
    <isParkedEnabled>
      <!--ro, opt, bool, whether to display the parking space status (occupied or free)-->true
    </isParkedEnabled>
  </PackingSpaceRecognitionOverlay>
  <VisibilityDisplay>
    <!--ro, opt, object, visibility information overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable. If it is enabled, the visibility information will be overlaid-->true
    </enabled>
    <positionX>
      <!--ro, req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--ro, req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </VisibilityDisplay>
</VideoOverlay>
```


11.3.12.2 Set OSD parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay, attr:version{req, string, protocolVersion}-->
  <normalizedScreenSize>
    <!--req, object, screen size-->
    <normalizedScreenWidth>
      <!--req, int, normalized width-->1
    </normalizedScreenWidth>
    <normalizedScreenHeight>
      <!--req, int, normalized height-->1
    </normalizedScreenHeight>
  </normalizedScreenSize>
```

```
  </normalizedScreenSize>
  <attribute>
    <!--opt, object, attributes-->
  </attribute>
  <TextOverlayList>
    <!--opt, array, text content list, subType:object-->
    <TextOverlay>
      <!--opt, object, text content-->
      <id>
        <!--req, string, text No.-->1
      </id>
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
      </positionY>
      <displayText>
        <!--req, string, overlay text-->test
      </displayText>
      <isPersistentText>
        <!--opt, bool, whether the overlaid text remains the previous settings after device rebooting, desc:if the value is true or the node is not
transmitted, it indicates to write OSD to flash, and it is not suitable for frequent or fast callings. If the value is false, it indicates not to writer OSD
to flash and the OSD settings will not restore-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--opt, object, channel time information overlay-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
    <displayWeek>
      <!--opt, bool, whether to display week information-->true
    </displayWeek>
  </DateTimeOverlay>
  <channelNameOverlay>
    <!--opt, object, channel name overlay-->
    <enabled>
      <!--req, bool, whether to enable channel name overlay-->true
    </enabled>
    <positionX>
      <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionX>
    <positionY>
      <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen. The range is the same with that of
normalizedScreenSize. Default value: x[0,704] y[0,576]-->1
    </positionY>
  </channelNameOverlay>
  <fontSize>
    <!--opt, enum, font size, subType:string, desc:"adaptive", "16*16", "32*32", "48*48", "64*64", "96*96", "128*128"-->96*96
  </fontSize>
  <frontColorMode>
    <!--opt, enum, font color mode, subType:string, desc:"auto", "customize", "outline"-->auto
  </frontColorMode>
  <frontColor>
    <!--opt, string, font color, desc:hexadecimal digit-->000000
  </frontColor>
  <alignment>
    <!--opt, enum, alignment mode, subType:string, desc:"customize" (custom), "alignRight" (right align), "alignLeft" (left align), "GB", "adsorption"
(adsorption mode)-->alignRight
  </alignment>
  <publicSecurity>
    <!--opt, bool-->true
  </publicSecurity>
  <boundary>
    <!--opt, int, range:[0,2], dep:and,{$.VideoOverlay.alignment,eq,GB}, desc:unit: number of characters-->000000
  </boundary>
  <upDownboundary>
    <!--opt, int, range:[0,2]-->1
  </upDownboundary>
</VideoOverlay>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, it is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

11.3.12.3 Get the OSD capability of a specific video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VideoOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_BatteryPowerOverlay, attr:version{req, string, protocolVersion}-->
  <attribute>
    <!--ro, opt, object, attributes-->
    <transparent opt="true,false">
      <!--ro, req, bool, whether it is transparent, attr:opt{req, string}-->true
    </transparent>
    <flashing opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </flashing>
  </attribute>
  <TextOverlayList size="8">
    <!--ro, opt, array, subType:object, attr:size{req, int}-->
    <TextOverlay>
      <!--ro, opt, object-->
      <id>
        <!--ro, req, string, listening host ID-->1
      </id>
      <enabled>
        <!--ro, req, bool, whether to enable-->true
      </enabled>
      <positionX>
        <!--ro, req, int, range:[0,704]-->1
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayText min="0" max="64">
        <!--ro, req, string, attr:min{req, int},max{req, int}-->test
      </displayText>
      <isPersistentText>
        <!--ro, opt, bool-->true
      </isPersistentText>
    </TextOverlay>
  </TextOverlayList>
  <DateTimeOverlay>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
    </enabled>
    <positionX>
      <!--ro, req, int, range:[0,704]-->1
```

```
      </positionX>
      <positionY>
        <!--ro, req, int, range:[0,576]-->1
      </positionY>
      <displayWeek opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </displayWeek>
    </DateTimeOverlay>
    <channelNameOverlay>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <positionX>
        <!--ro, opt, int, range:[0,1920]-->1
      </positionX>
      <positionY>
        <!--ro, opt, int, range:[0,1920]-->1
      </positionY>
      <name min="0" max="128">
        <!--ro, opt, string, range:[0,128], attr:min{req, int},max{req, int}-->Camera 01
      </name>
    </channelNameOverlay>
    <fontSize opt="16*16,32*32,48*48,64*64,80*80,96*96,128*128,adaptive">
      <!--ro, opt, string, font size, attr:opt{req, string}-->adaptive
    </fontSize>
    <frontColorMode opt="auto,customize,outline" def="auto">
      <!--ro, opt, string, font color mode, attr:opt{req, string},def{opt, string}-->outline
    </frontColorMode>
    <frontColor>
      <!--ro, opt, string, font color, desc:font color-->000000
    </frontColor>
    <alignment opt="customize,alignRight,alignLeft,GB,GB2025,allRight,allLeft,adsorption">
      <!--ro, opt, string, alignment mode, attr:opt{req, string}-->customize
    </alignment>
    <lowRightTextNumber min="1" max="6">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </lowRightTextNumber>
    <lowLeftTextNumber min="7" max="8">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->7
    </lowLeftTextNumber>
    <boundary min="0" max="2">
      <!--ro, opt, int, dep:and,{$.VideoOverlay.alignment,eq,GB}, attr:min{req, int},max{req, int}-->1
    </boundary>
    <upDownboundary min="0" max="5">
      <!--ro, opt, int, attr:min{opt, int},max{opt, int}-->1
    </upDownboundary>
    <videoCharacterOverlayLevel min="0" max="4">
      <!--ro, opt, int, range:[0,4], attr:min{req, int},max{req, int}-->0
    </videoCharacterOverlayLevel>
  </VideoOverlay>
```

11.3.12.4 Get the channel name overlay parameters of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, channel name overlay parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--ro, opt, int, ID-->1
  </id>
  <enabled>
    <!--ro, req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--ro, opt, string, channel name, range:[0,128]-->Camera 01
  </name>
  <OverlayRegion>
    <!--ro, opt, object, coordinates of overlaid string frame (the origin is the is the upper-left corner of the screen)-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

11.3.12.5 Set the channel name overlay parameters for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/channelNameOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<channelNameOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, ID-->1
  </id>
  <enabled>
    <!--req, bool, whether to enable channel name overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate-->1
  </positionY>
  <name>
    <!--opt, string, channel name, range:[0,32]-->Camera 01
  </name>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</channelNameOverlay>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.12.6 Set the parameters of date and time information overlay for a specified video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, configurations of date and time information overlay of video input channels, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable date and time information overlay-->true
  </enabled>
  <positionX>
    <!--req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY", "CHR-
YYYY/MM/DD", "CHR-MM/DD/YYYY", "CHR-DD/MM/YYYY"-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--opt, enum, time format, subType:string, desc:"12hour" (12-hour system), "24hour" (24-hour system)-->12hour
  </timeStyle>
  <displayWeek>
    <!--opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--opt, bool, whether to display date-->true
  </displayDate>
  <displayTime>
    <!--opt, bool, whether to display time-->true
  </displayTime>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.12.7 Get the parameters of date and time information overlay of a specific channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/dateTimeOverlay

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<DateTimeOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable date and time overlay-->true
  </enabled>
  <positionX>
    <!--ro, req, int, X-coordinate of the upper-left corner of the overlay area, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--ro, req, int, Y-coordinate of the upper-left corner of the overlay area, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <dateStyle>
    <!--ro, opt, enum, date format, subType:string, desc:"YYYY-MM-DD", "MM-DD-YYYY", "DD-MM-YYYY", "CHR-YYYY-MM-DD", "CHR-MM-DD-YYYY", "CHR-DD-MM-YYYY"--
>YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, time format, subType:string, desc:time format-->12hour
  </timeStyle>
  <displayWeek>
    <!--ro, opt, bool, whether to display week information-->true
  </displayWeek>
  <displayDate>
    <!--ro, opt, bool-->true
  </displayDate>
  <displayTime>
    <!--ro, opt, bool-->true
  </displayTime>
  <OverlayRegion>
    <!--ro, opt, object-->
    <width>
      <!--ro, req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--ro, req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</DateTimeOverlay>
```

11.3.12.8 Delete the text overlay parameters of a specific video input channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

11.3.12.9 Set the text overlay parameters of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_TextOverlay, subType:object, attr:version{opt, string, protocolVersion}-->
  <TextOverlay>
    <!--opt, object-->
    <id>
      <!--req, string, text No.-->test
    </id>
    <enabled>
      <!--req, bool-->true
    </enabled>
    <positionX>
      <!--req, int, X-coordinate-->1
    </positionX>
    <positionY>
      <!--req, int, Y-coordinate-->1
    </positionY>
    <displayText>
      <!--req, string, overlay character-->test
    </displayText>
    <OverlayRegion>
      <!--opt, object-->
      <width>
        <!--req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.3.12.10 Get parameters of text information overlay of a specified video input channel

Request URL

GET /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlayList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, text information overlay, subType:object, attr:version{req, string, protocolVersion}-->
  <TextOverlay>
    <!--ro, opt, object, text overlay configuration-->
    <id>
      <!--ro, req, string, text No.-->test
    </id>
    <enabled>
      <!--ro, req, bool, whether to enable-->true
    </enabled>
    <positionX>
      <!--ro, req, int, X-coordinate of the upper-left corner of the overlaid position, range:[0,704], desc:the origin is the lower-left corner of the
screen-->1
    </positionX>
    <positionY>
      <!--ro, req, int, Y-coordinate of the upper-left corner of the overlaid position, range:[0,576], desc:the origin is the lower-left corner of the
screen-->1
    </positionY>
    <displayText>
      <!--ro, req, string, overlaid text-->test
    </displayText>
    <OverlayRegion>
      <!--ro, opt, object-->
      <width>
        <!--ro, req, int, width, range:[0,1000]-->0
      </width>
      <height>
        <!--ro, req, int, height, range:[0,1000]-->0
      </height>
    </OverlayRegion>
  </TextOverlay>
</TextOverlayList>
```

11.3.12.11 Add a text to overlay on the video for a specific video input channel

Request URL

POST /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, text overlay settings, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string, ID-->test
  </id>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <positionX>
    <!--req, int, range:[0,704]-->1
  </positionX>
  <positionY>
    <!--req, int, the origin is the lower-left corner of the screen, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, text overlay-->test
  </displayText>
</TextOverlay>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:error reason description in detail-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

11.3.12.12 Set the parameters of a specific piece of text overlay information of a specific video input channel

Request URL

PUT /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TextOverlay xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{opt, string, protocolVersion}-->
  <id>
    <!--req, string, text No.-->test
  </id>
  <enabled>
    <!--req, bool-->true
  </enabled>
  <positionX>
    <!--req, int, x-coordinate, range:[0,704], desc:the origin is the lower-left corner of the screen-->1
  </positionX>
  <positionY>
    <!--req, int, y-coordinate, range:[0,576], desc:the origin is the lower-left corner of the screen-->1
  </positionY>
  <displayText>
    <!--req, string, overlay character-->test
  </displayText>
  <OverlayRegion>
    <!--opt, object-->
    <width>
      <!--req, int, width, range:[0,1000]-->0
    </width>
    <height>
      <!--req, int, height, range:[0,1000]-->0
    </height>
  </OverlayRegion>
</TextOverlay>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.12.13 Delete a specified piece of text overlay of a video input channel

Request URL

DELETE /ISAPI/System/Video/inputs/channels/<channelID>/overlays/text/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
textID	string	Text overlay ID

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```


11.3.13 图像光照均衡

11.3.13.1 Set the WDR (Wide Dynamic Range) parameters of a specific channel

Request URL

PUT /ISAPI/Image/channels/<channelID>/hdr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode>
    <!--req, enum, WDR (Wide Dynamic Range) mode, subType:string, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto"-->Sdr
  </hdrMode>
  <HdrSwitch>
    <!--opt, object-->
    <hdrSwitchMode>
      <!--req, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness. <timeSwitch> nad <lightSwitch>
are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--req, object, enable WDR by time-->
      <startHour>
        <!--opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--opt, int, start time (hour)-->1
      </endHour>
      <endMinute>
        <!--opt, int, start time (minute)-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel>
    <!--opt, enum, WDR level, subType:string, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel>
    <!--opt, int-->1
  </SdrLevel>
</Hdr>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.3.13.2 Get the WDR (Wide Dynamic Range) parameters of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/hdr

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode>
    <!--ro, req, enum, WDR mode, subType:string, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto".-->Sdr
  </hdrMode>
  <HdrSwitch>
    <!--ro, opt, object-->
    <hdrSwitchMode>
      <!--ro, req, enum, switch mode, subType:string, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness. <timeSwitch> nad
<lightSwitch> are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--ro, req, object, enable WDR by time-->
      <startHour>
        <!--ro, opt, int, start time (hour)-->1
      </startHour>
      <startMinute>
        <!--ro, opt, int, start time (minute)-->1
      </startMinute>
      <endHour>
        <!--ro, opt, int, start time (hour)-->1
      </endHour>
      <endMinute>
        <!--ro, opt, int, start time (minute)-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel>
    <!--ro, opt, enum, WDR level, subType:string, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel>
    <!--ro, opt, int-->1
  </SdrLevel>
</Hdr>
```


11.3.13.3 Get the WDR (Wide Dynamic Range) capability of a specific channel

Request URL

GET /ISAPI/Image/channels/<channelID>/hdr/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Hdr xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <hdrMode opt="close,realHdr,digitalHdr,Sdr,BLC">
    <!--ro, req, enum, WDR mode, subType:string, attr:opt{req, string}, desc:"close", "realHdr"-WDR, "digitalHdr"-digital WDR, "Sdr"-super WDR, "auto".--
>Sdr
  </hdrMode>
  <HdrSwitch>
    <!--ro, opt, object-->
    <hdrSwitchMode opt="open,timeCtrl,lightCtrl">
      <!--ro, req, enum, switch mode, subType:string, attr:opt{req, string}, desc:"open", "timeCtrl"-enabled by time, "lightCtrl"-enabled by brightness.
<timeSwitch> nad <lightSwitch> are invalid when this node is set to "open"-->lightCtrl
    </hdrSwitchMode>
    <timeSwitch>
      <!--ro, req, object, enable WDR by time-->
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->1
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->1
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->1
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->1
      </endMinute>
    </timeSwitch>
  </HdrSwitch>
  <hdrLevel opt="level1,level2,level3,level4,level5,level6,level7">
    <!--ro, opt, enum, WDR level, subType:string, attr:opt{req, string}, desc:"level1", "level2", "level3", "level4", "level5", "level6", "level7"-->level7
  </hdrLevel>
  <SdrLevel min="0" max="100" def="50">
    <!--ro, opt, int, dep:and,{$.Hdr.hdrMode,eq,Sdr}, attr:min{req, int},max{req, int},def{req, int}-->1
  </SdrLevel>
</Hdr>
```


11.4 录像管理

11.4.1 录像查询

11.4.1.1 Get the capability of searching for video and picture files

Request URL

GET /ISAPI/ContentMgmt/search/capabilities

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<CMSearchDescription xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, search conditions, attr:version{opt, string, protocolVersion}-->
  <timeSpanList>
    <!--ro, opt, array, list of time periods, subType:object-->
    <timeSpan>
      <!--ro, opt, object, time period-->
      <laneNumber min="1" max="99">
        <!--ro, opt, int, lane No., range:[1,99], attr:min{req, int},max{req, int}-->1
      </laneNumber>
      <carType opt="all,unknow,coach,truck,car,minibus,smalltruck,bigcar,smallcar,human,tumbrel,trike,suv,midcar">
        <!--ro, opt, enum, vehicle type, subType:string, attr:opt{req, string}, desc:"coach", "truck", "car", "minibus", "smalltruck", "bigcar", "smallcar",
"human", "tumbrel", "trike"; "suv"; "midcar", "all", "unknown"-->coach
      </carType>
      <startTime>
        <!--ro, req, datetime, start time-->1970-01-01T00:00:00+08:00
      </startTime>
      <endTime>
        <!--ro, req, datetime, end time-->1970-01-01T00:00:00+08:00
      </endTime>
    </timeSpan>
  </timeSpanList>
</CMSearchDescription>
```

11.5 车辆识别

11.5.1 机动车识别

11.5.1.1 Set ANPR arming schedules of all channels

Request URL

PUT /ISAPI/Event/schedules/vehicledetects

Query Parameter

None

Request Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<VehicledetectScheduleList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, arming time list, subType:object, attr:version{req, string, protocolVersion}-->
  <Schedule>
    <!--opt, object, arming schedule-->
    <id>
      <!--req, string, index, desc:the format is event type-channel No.-->vehicledetect-1
    </id>
    <eventType>
      <!--opt, enum, event type, subType:string-->vehicledetect
    </eventType>
    <videoInputChannelID>
      <!--opt, string, video input channel ID-->1
    </videoInputChannelID>
    <TimeBlockList size="8">
      <!--req, array, arming time list, subType:object, attr:size{opt, int}-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <dayOfWeek>
          <!--opt, enum, days of the week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)-->1
        </dayOfWeek>
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </TimeBlockList>
    <HolidayBlockList>
      <!--opt, array, holiday arming time, subType:object-->
      <TimeBlock>
        <!--opt, object, arming time-->
        <TimeRange>
          <!--req, object, time range-->
          <beginTime>
            <!--req, time, start time-->10:00:00
          </beginTime>
          <endTime>
            <!--req, time, end time-->10:00:00
          </endTime>
        </TimeRange>
      </TimeBlock>
    </HolidayBlockList>
  </Schedule>
</VehicledetectScheduleList>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.5.1.2 Set the linkage parameters of vehicle detection for a specified channel

Request URL

PUT /ISAPI/Event/triggers/vehicledetection-<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventTrigger xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, string-->vehicledetection-1
  </id>
  <eventType>
    <!--req, string-->vehicledetection
  </eventType>
  <eventDescription>
    <!--opt, string-->test
  </eventDescription>
  <inputIOPortID>
    <!--opt, string-->1
  </inputIOPortID>
  <dynInputIOPortID>
    <!--opt, string-->1
  </dynInputIOPortID>
  <direction>
    <!--opt, enum, subType:string-->both
  </direction>
  <videoInputChannelID>
    <!--opt, string-->1
  </videoInputChannelID>
  <dynVideoInputChannelID>
    <!--opt, string-->1
  </dynVideoInputChannelID>
  <intervalBetweenEvents>
    <!--opt, int, unit:s-->1
  </intervalBetweenEvents>
  <WLSensorID>
    <!--opt, string-->1
  </WLSensorID>
  <EventTriggerNotificationList>
    <!--opt, array, subType:object-->
    <EventTriggerNotification>
      <!--opt, object-->
      <id>
        <!--req, string-->test
      </id>
      <notificationMethod>
        <!--req, enum, subType:string-->email
      </notificationMethod>
      <notificationRecurrence>
        <!--opt, enum, subType:string-->beginning
      </notificationRecurrence>
      <notificationInterval>
        <!--opt, int, unit:ms-->1
      </notificationInterval>
      <outputIOPortID>
        <!--opt, string, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,IO}-->1
      </outputIOPortID>
      <videoInputID>
        <!--opt, string-->1
      </videoInputID>
      <dynVideoInputID>
        <!--opt, string-->1
      </dynVideoInputID>
      <ptzAction>
        <!--opt, object-->
        <ptzChannelID>
          <!--req, string-->1
        </ptzChannelID>
        <actionName>
          <!--req, enum, subType:string-->preset
        </actionName>
        <actionNum>
          <!--opt, int-->1
        </actionNum>
      </ptzAction>
      <WhiteLightAction>
        <!--opt, object-->
        <whiteLightDurationTime>
          <!--opt, int, range:[1,60], unit:s-->1
        </whiteLightDurationTime>
      </WhiteLightAction>
      <wiegandOutputID>
        <!--opt, int, dep:and,{$.EventTrigger.EventTriggerNotificationList[*].EventTriggerNotification.notificationMethod,eq,wiegand}-->1
      </wiegandOutputID>
    </EventTriggerNotification>
  </EventTriggerNotificationList>
</EventTrigger>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```


11.5.1.3 Get the total capability of vehicle capture and recognition

Request URL

GET /ISAPI/ITC/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<ITCCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <isSupportITCStatus>
    <!--ro, opt, bool-->true
  </isSupportITCStatus>
  <isSupportITCSetUp>
    <!--ro, opt, bool-->true
  </isSupportITCSetUp>
  <isSupportManualCap>
    <!--ro, opt, bool-->true
  </isSupportManualCap>
  <isSupportIllegalUploadPic>
    <!--ro, opt, bool-->true
  </isSupportIllegalUploadPic>
  <isSupportContinueCap>
    <!--ro, opt, bool-->true
  </isSupportContinueCap>
  <isSupportWiper>
    <!--ro, opt, bool-->true
  </isSupportWiper>
  <isSupportEntranceCap>
    <!--ro, opt, bool-->true
  </isSupportEntranceCap>
  <isSupportPlateRecognitionParam>
    <!--ro, opt, bool-->true
  </isSupportPlateRecognitionParam>
  <isSupportSyncSignalOutput>
    <!--ro, opt, bool-->true
  </isSupportSyncSignalOutput>
  <isSupportImageMerge>
    <!--ro, opt, bool-->true
  </isSupportImageMerge>
  <isSupportCarFeatureParam>
    <!--ro, opt, bool-->true
  </isSupportCarFeatureParam>
  <isSupportSnapshot>
    <!--ro, opt, bool-->true
  </isSupportSnapshot>
  <isSupportNetStorage>
    <!--ro, opt, bool-->true
  </isSupportNetStorage>
  <isSupportEntranceShowCap>
    <!--ro, opt, bool-->true
  </isSupportEntranceShowCap>
  <isSupportAlgorithmsState>
    <!--ro, opt, bool-->true
  </isSupportAlgorithmsState>
  <isSupportAlgorithmsVersion>
    <!--ro, opt, bool-->true
  </isSupportAlgorithmsVersion>
  <isSupportVehicleDetection>
    <!--ro, opt, bool-->true
  </isSupportVehicleDetection>
  <isSupportRealtimeView>
    <!--ro, opt, bool-->true
  </isSupportRealtimeView>
  <isSupportDifferIllegalPic>
    <!--ro, opt, bool-->true
  </isSupportDifferIllegalPic>
  <isSupportEntranceShow>
    <!--ro, opt, bool-->true
  </isSupportEntranceShow>
  <LSTSCap>
    <!--ro, opt, object-->
    <isSupportLiveReceivePicture>
      <!--ro, opt, bool-->true
    </isSupportLiveReceivePicture>
  </LSTSCap>
</ITCCap>
```


11.5.1.4 Get the vehicle feature parameters capability

Request URL

GET /ISAPI/ITC/carFeatureParam/capabilities?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	optional, for specifying channel No. for multi-channel devices; when this node is not returned, it is channel 1 by default

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<CarFeatureParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle feature parameters, attr:version{req, string, protocolVersion}-->
  <isSupportCarColor>
    <!--ro, opt, bool, whether to support vehicle color recognition-->true
  </isSupportCarColor>
  <isSupportCarLogo>
    <!--ro, opt, bool, whether to support vehicle logo recognition-->true
  </isSupportCarLogo>
  <isSupportSafeBelt>
    <!--ro, opt, bool, whether to support safety belt detection, desc:whether to support safety belt detection-->true
  </isSupportSafeBelt>
  <isSupportSunVisor>
    <!--ro, opt, bool, whether to support sun visor detection-->true
  </isSupportSunVisor>
  <isSupportHighPollutVehicle>
    <!--ro, opt, bool, whether to support yellow-label vehicle detection-->true
  </isSupportHighPollutVehicle>
  <isSupportDangerousVehicle>
    <!--ro, opt, bool, whether to support dangerous goods transport vehicle detection 1-->true
  </isSupportDangerousVehicle>
  <isSupportVehicleSubBrand>
    <!--ro, opt, bool, whether to support  vehicle sub-brand detection-->true
  </isSupportVehicleSubBrand>
  <isSupportEnvprosign>
    <!--ro, opt, bool, whether to support dangerous goods transport vehicle detection 2-->true
  </isSupportEnvprosign>
  <isSupportFaceDetect>
    <!--ro, req, bool, whether to enable face matting-->true
  </isSupportFaceDetect>
  <isSupportUphone>
    <!--ro, opt, bool, whether to support phone call detection-->true
  </isSupportUphone>
  <isSupportPendant>
    <!--ro, opt, bool, whether to support window hangings detection-->true
  </isSupportPendant>
  <isSupportPilotFace>
    <!--ro, req, bool, whether to support driver's face matting-->true
  </isSupportPilotFace>
  <isSupportCopilotFace>
    <!--ro, req, bool, whether to support front passenger's face picture matting-->true
  </isSupportCopilotFace>
  <isSupportNonmotorface>
    <!--ro, opt, bool, whether to support non-motor vehicle driver's face matting-->true
  </isSupportNonmotorface>
  <faceImgScale min="1" max="10">
    <!--ro, req, int, close-up face picture zooming ratio, range:[1,10], attr:min{req, int},max{req, int}-->1
  </faceImgScale>
  <faceImgRatio min="1" max="3">
    <!--ro, req, int, close-up face picture matting ratio, range:[1,3], attr:min{req, int},max{req, int}, desc:close-up face picture matting ratio-->1
  </faceImgRatio>
  <FaceImgInfoEx>
    <!--ro, opt, object, extended information of face matting-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable face matting types of close-up face picture, attr:opt{req, string}, desc:the node faceImgScale will be invalid
when this function is enabled-->false
    </enabled>
    <faceImgType opt="custom,IDCard">
      <!--ro, opt, enum, close-up face picture type, subType:string, attr:opt{req, string}, desc:"custom", "IDCard" (ID picture)-->custom
    </faceImgType>
    <faceCloseupRatio>
      <!--ro, opt, object, target capture size, desc:this node is valid when faceImgType is "custom"-->
      <upRatio min="0" max="100">
        <!--ro, req, int, enlarge from top, attr:min{req, int},max{req, int}-->20
      </upRatio>
      <downRatio min="0" max="100">
        <!--ro, req, int, enlarge from bottom, attr:min{req, int},max{req, int}-->20
      </downRatio>
      <leftRatio min="0" max="100">
        <!--ro, req, int, enlarge from left, attr:min{req, int},max{req, int}-->20
      </leftRatio>
      <rightRatio min="0" max="100">
        <!--ro, req, int, enlarge from right, attr:min{req, int},max{req, int}-->20
      </rightRatio>
    </faceCloseupRatio>
  </FaceImgInfoEx>
  <faceImgOutput opt="none,overPic,alarmUpload,overPicAndAlarmUpload">
    <!--ro, opt, enum, close-up face picture output mode, subType:string, attr:opt{req, string}, desc:"none" (no output), "overPic" (overlay on scene
picture), "alarmUpload" (upload arm), "overPicAndAlarmUpload" (overlay on scene picture and upload arm)-->none
  </faceImgOutput>
  <isSupportDrawFaceRect>
    <!--ro, opt, bool, whether to overlay frame on face matting-->true
  </isSupportDrawFaceRect>
  <isSupportFaceImgBrightenEnhence>
    <!--ro, opt, bool, whether to support close-up face picture contrast enhancement-->true
  </isSupportFaceImgBrightenEnhence>
```

```xml
    </isSupportFaceImgBrightenEnhence>
    <faceImgBrightenEnhenceLevel min="0" max="100">
      <!--ro, req, int, contrast enhancement level, range:[0,100], attr:min{req, int},max{req, int}-->0
    </faceImgBrightenEnhenceLevel>
    <tissueBoxEnable>
      <!--ro, opt, bool, whether to support enabling tissue box detection-->true
    </tissueBoxEnable>
    <frontChildEnable>
      <!--ro, opt, bool, whether to support enabling baby in arm detection-->true
    </frontChildEnable>
    <labelEnable>
      <!--ro, opt, bool, whether to support enabling label detection-->true
    </labelEnable>
    <decorationEnable>
      <!--ro, opt, bool, whether to support enabling decoration detection-->true
    </decorationEnable>
    <safeBeltSensitivity min="1" max="100">
      <!--ro, opt, int, safety belt detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:this node is valid when safeBeltEnabled is
"true"-->1
    </safeBeltSensitivity>
    <uphoneSensitivity min="1" max="100">
      <!--ro, opt, int, phone call detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:this node is valid when uphoneEnabled is
"true"-->1
    </uphoneSensitivity>
    <pendantSensitivity min="1" max="100">
      <!--ro, opt, int, window hangings detection sensitivity, range:[1,100], attr:min{req, int},max{req, int}, desc:this node is valid when pendantEnabled is
"true"-->1
    </pendantSensitivity>
    <isSupportSmoking>
      <!--ro, opt, bool, whether to support smoking detection-->true
    </isSupportSmoking>
    <isSupportHelmet>
      <!--ro, opt, bool, whether to support helmet detection-->true
    </isSupportHelmet>
    <isSupportTwoWheelVehicle>
      <!--ro, opt, bool, whether to support manned two wheeler detection-->true
    </isSupportTwoWheelVehicle>
    <targetCutout>
      <!--ro, opt, object, target face matting-->
      <enabled opt="true,false">
        <!--ro, opt, bool, whether to enable target face matting, attr:opt{req, string}-->true
      </enabled>
      <size>
        <!--ro, req, enum, target matting ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
      </size>
      <zoom>
        <!--ro, req, enum, zooming in/out ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
      </zoom>
    </targetCutout>
    <playMobilePhone>
      <!--ro, opt, bool, whether to enable driver using mobile phone detection-->true
    </playMobilePhone>
    <isSupportShed>
      <!--ro, opt, bool, whether to enable non-motor vehicle umbrella tent detection-->true
    </isSupportShed>
    <separatData>
      <!--ro, opt, bool, upload license plate No. and picture information separately and for multiple times-->true
    </separatData>
    <helmetDaySensitivity min="1" max="100">
      <!--ro, req, int, Sensitivity for Not Wearing Helmet at Daytime, attr:min{req, int},max{req, int}-->0
    </helmetDaySensitivity>
    <helmetNightSensitivity min="1" max="100">
      <!--ro, req, int, Sensitivity for Not Wearing Helmet at Night, attr:min{req, int},max{req, int}-->0
    </helmetNightSensitivity>
    <isSupportFocedCopilotFace>
      <!--ro, opt, bool, whether to mandate face matting for front passenger-->true
    </isSupportFocedCopilotFace>
    <smokingSensitivity min="0" max="100">
      <!--ro, opt, int, smoking detection sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </smokingSensitivity>
    <helmetCheckMode opt="checkOutFirst,checkAccurateFirst,custorm">
      <!--ro, opt, enum, helmet detection mode, subType:string, dep:and,{$.CarFeatureParam.isSupportHelmet,eq,true}, attr:opt{req, string},
desc:1. "checkOutFirst", "checkAccurateFirst"; it is "checkAccurateFirst" by default when this node is not returned
2. during detection of not wearing helmet: "checkOutFirst" is for higher detection rate but comes with higher false alarm rate which requires manual
verification; "checkAccurateFirst" is for high-quality detection but comes with higher under-detection rate-->checkOutFirst
    </helmetCheckMode>
    <reflectiveStripeEnabled opt="true,false">
      <!--ro, opt, bool, whether to enable reflective stripe detection, attr:opt{req, string}-->true
    </reflectiveStripeEnabled>
  </CarFeatureParam>
```


Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
CarFeatureParam	[Message content]	application/xml	--	--	--

Notes:

newly added reflectiveStripeEnabled as reflective stripe detection for vehicle feature parameters

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

11.5.1.5 Get the vehicle feature configuration parameters

Request URL

GET /ISAPI/ITC/carFeatureParam?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	optional, for specifying channel No. for multi-channel devices; when this node is not returned, it is channel 1 by default

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<CarFeatureParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle feature parameters, attr:version{req, string, protocolVersion}-->
  <colorEnabled>
    <!--ro, req, bool, vehicle color recognition-->true
  </colorEnabled>
  <carLogoEnabled>
    <!--ro, req, bool, vehicle log recognition-->true
  </carLogoEnabled>
  <sunVisorEnabled>
    <!--ro, req, bool, sun visor detection-->true
  </sunVisorEnabled>
  <faceDetectEnabled>
    <!--ro, req, bool, face picture matting-->true
  </faceDetectEnabled>
  <uphoneEnabled>
    <!--ro, opt, bool, phone call detection-->true
  </uphoneEnabled>
  <pendantEnabled>
    <!--ro, opt, bool, window hangings detection-->true
  </pendantEnabled>
  <tissueBoxEnable>
    <!--ro, opt, bool, tissue box detection-->true
  </tissueBoxEnable>
  <frontChildEnable>
    <!--ro, opt, bool, baby in arm detection-->true
  </frontChildEnable>
  <labelEnable>
    <!--ro, opt, bool, label detection-->true
  </labelEnable>
  <uphoneSensitivity>
    <!--ro, opt, int, phone call detection sensitivity, range:[1,100]-->1
  </uphoneSensitivity>
  <pendantSensitivity>
    <!--ro, opt, int, window hangings detection sensitivity, range:[1,100]-->1
  </pendantSensitivity>
  <decorationEnable>
    <!--ro, opt, bool, decoration detection-->true
  </decorationEnable>
  <OutputCtrl>
    <!--ro, opt, object, face matting-->
```

```
    <hostFaceEnabled>
      <!--ro, req, bool, driver's face matting-->true
    </hostFaceEnabled>
    <viceFaceEnabled>
      <!--ro, req, bool, front passenger's face matting-->true
    </viceFaceEnabled>
    <nonMotorFaceEnabled>
      <!--ro, req, bool, non-motor vehicle driver's face matting-->true
    </nonMotorFaceEnabled>
    <faceImgScale>
      <!--ro, req, int, close-up face picture zooming ratio, range:[1,10]-->1
    </faceImgScale>
    <faceImgRatio>
      <!--ro, req, enum, close-up face picture cutting ratio, subType:int, desc:1 (small), 2 (medium), 3 (large)-->1
    </faceImgRatio>
    <FaceImgInfoEx>
      <!--ro, opt, object, extended information of face matting-->
      <enabled>
        <!--ro, req, bool, whether to enable face matting types of close-up face picture, desc:the node faceImgScale will be invalid when this function is
enabled-->false
      </enabled>
      <faceImgType>
        <!--ro, req, enum, close-up face picture type, subType:string, desc:"custom", "IDCard" (ID picture)-->IDCard
      </faceImgType>
      <faceCloseupRatio>
        <!--ro, opt, object, target capture size, desc:this node is valid when faceImgType is "custom"-->
        <upRatio>
          <!--ro, req, int, enlarge from top, range:[0,100]-->20
        </upRatio>
        <downRatio>
          <!--ro, req, int, enlarge from bottom, range:[0,100]-->20
        </downRatio>
        <leftRatio>
          <!--ro, req, int, enlarge from left, range:[0,100]-->20
        </leftRatio>
        <rightRatio>
          <!--ro, req, int, enlarge from right, range:[0,100]-->20
        </rightRatio>
      </faceCloseupRatio>
    </FaceImgInfoEx>
    <faceImgOutputMode>
      <!--ro, req, enum, close-up face picture output mode, subType:string, desc:"none" (no output), "overPic" (overlay on scene picture), "alarmUpload"
(upload arm), "overPicAndAlarmUpload" (overlay on scene picture and upload arm)-->none
    </faceImgOutputMode>
    <DrawFaceRectEnabled>
      <!--ro, req, bool, whether to overlay frame on face matting-->true
    </DrawFaceRectEnabled>
    <faceImgBrightenEnhenceEnabled>
      <!--ro, req, bool, close-up face picture contrast enhancement-->true
    </faceImgBrightenEnhenceEnabled>
    <faceImgBrightenEnhenceLevel>
      <!--ro, req, int, contrast enhancement level, range:[0,100]-->0
    </faceImgBrightenEnhenceLevel>
    <faceCopyIllegal>
      <!--ro, opt, bool, whether need enable face matting for violation picture-->true
    </faceCopyIllegal>
    <forcedViceFaceEnabled>
      <!--ro, opt, bool, whether to mandate face matting for front passenger-->true
    </forcedViceFaceEnabled>
  </OutputCtrl>
  <smokingEnabled>
    <!--ro, req, bool, smoking detection-->true
  </smokingEnabled>
  <helmetEnabled>
    <!--ro, opt, bool, helmet detection-->true
  </helmetEnabled>
  <twoWheelVehicleEnabled>
    <!--ro, opt, bool, manned two wheeler detection-->true
  </twoWheelVehicleEnabled>
  <playMobilePhone>
    <!--ro, opt, bool, whether to enable driver using mobile phone detection-->true
  </playMobilePhone>
  <isSupportShed>
    <!--ro, opt, bool, whether to enable non-motor vehicle umbrella tent detection-->true
  </isSupportShed>
  <separatData>
    <!--ro, opt, bool, upload license plate No. and picture information separately and for multiple times-->true
  </separatData>
  <helmetDaySensitivity>
    <!--ro, req, int, sensitivity for detecting not wearing helmet at daytime-->0
  </helmetDaySensitivity>
  <helmetNightSensitivity>
    <!--ro, req, int, sensitivity for detecting not wearing helmet at night-->0
  </helmetNightSensitivity>
  <shedSensitivity>
    <!--ro, opt, int, non-motor vehicle umbrella tent detection sensitivity, range:[1,100]-->0
  </shedSensitivity>
  <twoWheelVehicleSensitivity>
    <!--ro, opt, int, manned two wheeler detection sensitivity, range:[1,100]-->0
  </twoWheelVehicleSensitivity>
  <smokingSensitivity>
    <!--ro, opt, int, smoking detection sensitivity, range:[0,100]-->0
  </smokingSensitivity>
  <CarWindowFeature>
```

```xml
<CarWindowFeature>
    <!--ro, opt, object, vehicle window feature detection-->
    <tempPlateEnable>
      <!--ro, opt, bool, whether to enable detection of temporary license plate on the window, desc:detect whether there is temporary license plate on the window-->true
    </tempPlateEnable>
    <tempPlateSensitivity>
      <!--ro, opt, int, detection sensitivity of temporary license plate on the window, range:[0,100]-->1
    </tempPlateSensitivity>
    <passCardEnable>
      <!--ro, opt, bool, whether to enable detection of entry & exit pass on the window, desc:whether there is entry & exit pass on the window-->true
    </passCardEnable>
    <passCardSensitivity>
      <!--ro, opt, int, detection sensitivity of entry & exit pass on the window, range:[0,100]-->1
    </passCardSensitivity>
    <carCardEnable>
      <!--ro, opt, bool, whether to enable detection of cards attached to the window, desc:whether there is any card (business cards, flyers, etc) attached to the window-->true
    </carCardEnable>
    <carCardSensitivity>
      <!--ro, opt, int, detection sensitivity of cards attached to the window, range:[0,100]-->1
    </carCardSensitivity>
  </CarWindowFeature>
  <CarBodyFeature>
    <!--ro, opt, object, bodywork feature detection-->
    <enabled>
      <!--ro, opt, bool, whether to enable bodywork feature detection, desc:whether the vehicle is with spare tire, with sunroof, with roof rack, painted, people sticking out of the sunroof, covered (for dump truck)-->true
    </enabled>
  </CarBodyFeature>
  <vehicleUseEnable>
    <!--ro, opt, bool, whether to enable vehicle purpose detection, desc:recognize bus, school bus, taxi, and ambulance-->true
  </vehicleUseEnable>
  <helmetCheckMode>
    <!--ro, opt, enum, helmet detection mode, subType:string, dep:and,{$.CarFeatureParam.helmetEnabled,eq,true},
desc:1. "checkOutFirst", "checkAccurateFirst"; it is "checkAccurateFirst" by default when this node is not returned
2. during detection of not wearing helmet: "checkOutFirst" is for higher detection rate but comes with higher false alarm rate which requires manual verification; "checkAccurateFirst" is for high-quality detection but comes with higher under-detection rate-->checkOutFirst
  </helmetCheckMode>
  <reflectiveStripeEnabled>
    <!--ro, opt, bool, whether to enable reflective stripe detection-->true
  </reflectiveStripeEnabled>
</CarFeatureParam>
```


11.5.1.6 Set the vehicle feature configuration parameters

Request URL

PUT /ISAPI/ITC/carFeatureParam?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	optional, for specifying channel No. for multi-channel devices; when this node is not returned, it is channel 1 by default

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<CarFeatureParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, vehicle feature parameters, attr:version{req, string, protocolVersion}-->
  <colorEnabled>
    <!--req, bool, whether to enable vehicle color recognition-->true
  </colorEnabled>
  <carLogoEnabled>
    <!--req, bool, whether to enable vehicle log recognition-->true
  </carLogoEnabled>
  <faceDetectEnabled>
    <!--req, bool, Face Picture Matting-->true
  </faceDetectEnabled>
  <pendantEnabled>
    <!--opt, bool, whether to enable window hangings detection-->true
  </pendantEnabled>
  <tissueBoxEnable>
    <!--opt, bool, Tissue Box Detection-->true
  </tissueBoxEnable>
  <frontChildEnable>
    <!--opt, bool, Baby in Arm Detection-->true
  </frontChildEnable>
  <labelEnable>
    <!--opt, bool, Label Detection-->true
  </labelEnable>
  <uphoneSensitivity>
    <!--opt, int, Phone Call Detection, range:[1,100]-->1
  </uphoneSensitivity>
  <pendantSensitivity>
    <!--opt, int, window hangings detection sensitivity, range:[1,100]-->1
  </pendantSensitivity>
  <decorationEnable>
    <!--opt, bool, Decoration Detection-->true
  </decorationEnable>
</CarFeatureParam>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.7 Get the parameters of traffic rule violation dictionary

Request URL

GET /ISAPI/ITC/illegalDictionary

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<IllegalDictionary xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, traffic rule violation dictionary, attr:version{req, string, protocolVersion}-->
  <IllegalCodeList>
    <!--ro, req, array, violation code list, subType:object-->
    <IllegalCodeItem>
      <!--ro, req, object, violation code-->
      <idx>
        <!--ro, req, int, ID, range:[0,65535]-->0
      </idx>
      <illegalCode>
        <!--ro, req, string, violation code, range:[0,64]-->test
      </illegalCode>
      <illegalName>
        <!--ro, req, string, violation name, range:[0,256]-->test
      </illegalName>
      <illegalStringCode>
        <!--ro, opt, string, violation code in string format, range:[0,128]-->test
      </illegalStringCode>
      <illegalDescription>
        <!--ro, opt, string, violation description, range:[0,128]-->test
      </illegalDescription>
    </IllegalCodeItem>
  </IllegalCodeList>
</IllegalDictionary>
```


11.5.1.8 Set parameters of traffic violation dictionary

Request URL

PUT /ISAPI/ITC/illegalDictionary

Query Parameter

None

Request Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<IllegalDictionary xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, traffic rule violation dictionary, attr:version{req, string, protocolVersion}-->
  <IllegalCodeList>
    <!--req, array, violation code list, subType:object-->
    <IllegalCodeItem>
      <!--req, object, violation code-->
      <idx>
        <!--req, int, index, range:[0,65535]-->0
      </idx>
      <illegalCode>
        <!--req, string, illegal action code, range:[0,64]-->test
      </illegalCode>
      <illegalName>
        <!--req, string, illegal action name, range:[0,256]-->test
      </illegalName>
      <illegalStringCode>
        <!--opt, string, illegal action code (string), range:[0,128]-->test
      </illegalStringCode>
      <illegalDescription>
        <!--opt, string, illegal action description, range:[0,128]-->test
      </illegalDescription>
    </IllegalCodeItem>
  </IllegalCodeList>
</IllegalDictionary>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device 
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.9 Get the capability of traffic rule violation dictionary

Request URL

GET /ISAPI/ITC/illegalDictionary/capabilities

Query Parameter

None

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<IllegalDictionary xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, traffic rule violation dictionary, attr:version{req, string, protocolVersion}-->
  <IllegalCodeList>
    <!--ro, req, array, violation code list, subType:object-->
    <IllegalCodeItem>
      <!--ro, req, object, violation code-->
      <idx min="0" max="64">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
      </idx>
      <illegalCode min="0" max="64">
        <!--ro, req, string, violation code, attr:min{req, int},max{req, int}-->test
      </illegalCode>
      <illegalName min="0" max="256">
        <!--ro, req, string, violation name, attr:min{req, int},max{req, int}-->test
      </illegalName>
      <illegalStringCode min="0" max="10">
        <!--ro, opt, string, violation code (string), attr:min{req, int},max{req, int}-->test
      </illegalStringCode>
      <illegalCodeLetterNum min="0" max="10">
        <!--ro, opt, int, supported number of letters in violation code, attr:min{req, int},max{req, int}-->0
      </illegalCodeLetterNum>
      <illegalDescription size="256">
        <!--ro, opt, string, violation description, attr:size{req, int}-->test
      </illegalDescription>
    </IllegalCodeItem>
  </IllegalCodeList>
  <isSupportReset>
    <!--ro, opt, bool, whether it supports restoring to default setting, desc:/ISAPI/ITC/illegalDictionary/reset-->true
  </isSupportReset>
  <isSupportOverspeedAdvancedParam>
    <!--ro, opt, bool, whether it supports overspeed advanced configuration of traffic rule violation dictionary-->true
  </isSupportOverspeedAdvancedParam>
</IllegalDictionary>
```


11.5.1.10 Get the capability of arming parameters for traffic violations capture

Request URL

GET /ISAPI/ITC/illegalSchedules/capabilities?format=json&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "eventPriorityEnable": {
    /*ro, opt, object, whether to enable violation priority*/
        "@opt": [true, false]
        /*ro, opt, array, options, subType:bool*/
    },
    "IllegalScheduleListCap": {
    /*ro, req, object, list of arming parameters for traffic violations capture, desc:this node is only available for ANPR violation information, related node is illegalInfo*/
        "@size":  1,
        /*ro, req, int, the maximum number supported in a single capture*/
        "Schedule": {
        /*ro, opt, object, arming parameters*/
            "eventType":  "post,driveLine",
            /*ro, req, string, event type, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way driving), "banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change), "safeBelt", "uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger" (manned non-motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against direction guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion), "leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock", "construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing), "changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm), "notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop", "animalsOccupy", "suonaDet" (whistle), "roarDet"*/
            "noPlateCaptureEnable": {
            /*ro, opt, object, whether to enable no plate vehicle capture*/
                "@opt": [true, false]
                /*ro, opt, array, options, subType:bool*/
            },
            "eventPriority": {
            /*ro, opt, object, event capture priority, desc:value range 0 to 10; the higher the value, the higher the priority; when the value is 0, this node will not have priority*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  10
                /*ro, req, int, the maximum value*/
            },
            "TimeBlockList": {
            /*ro, opt, object, list of time periods*/
                "@size":  7,
                /*ro, req, int, the maximum number supported in a single capture*/
                "TimeBlock": {
                /*ro, opt, object, time period*/
                    "dayOfWeek": {
                    /*ro, req, object, day of a week, desc:Monday, Tuesday, Wednesday, Thursday, Friday, Saturday, Sunday*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  7
                        /*ro, req, int, the maximum value*/
                    },
                    "TimeRangeList": {
                    /*ro, opt, object, list of upload intervals*/
                        "@size":  4,
                        /*ro, req, int, the maximum number supported in a single capture*/
                        "TimeRange": {
                        /*ro, opt, object, time interval*/
                            "beginTime":  "00:00:00+08:00",
                            /*ro, req, time, start time*/
                            "endTime":  "00:00:00+08:00"
                            /*ro, req, time, end time*/
                        }
                    }
                }
            },
            "RelatedLanList": {
            /*ro, opt, object, list of linked lane information*/
                "@size":  6,
                /*ro, opt, int, size*/
                "RelatedLan": {
                /*ro, opt, object, linked lane information*/
                    "lanID": {
                    /*ro, opt, object, whether to enable linked lane No.*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  6
                        /*ro, req, int, the maximum value*/
                    }
                }
            }
        }
    }
}
```

11.5.1.11 Get the arming parameters for traffic violation capture

Request URL

GET /ISAPI/ITC/illegalSchedules?format=json&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "eventPriorityEnable":  true,
    /*ro, req, bool, whether to enable violation priority*/
    "IllegalScheduleList": [
    /*ro, req, array, arming time of traffic violation capture, subType:object, desc:this node is only available for ANPR violation information, which is
related to the node illegalInfo*/
        {
            "Schedule": {
            /*ro, opt, object, arming schedules*/
                "eventType":  "post",
                /*ro, req, enum, event type, subType:string, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way
driving), "banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change),
"safeBelt", "uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger"
(manned non-motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay"
(motor vehicle on non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against
direction guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion),
"leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight",
"leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red
light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock",
"construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing),
"changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm),
"notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield
at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to
vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when
turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop",
"animalsOccupy", "suonaDet" (whistle), "roarDet"*/
                "noPlateCaptureEnable":  true,
                /*ro, opt, bool, whether to enable no plate vehicle capture*/
                "eventPriority":  0,
                /*ro, opt, int, event capture priority, range:[0,10], desc:value ranges from 0 to 10; the higher the value, the higher the priority; when
the value is 0, this event will not have priority*/
                "TimeBlockList": [
                /*ro, opt, array, list of time periods, subType:object, range:[0,7]*/
                    {
                        "TimeBlock": {
                        /*ro, opt, object, time period*/
                            "dayOfWeek":  1,
                            /*ro, req, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6
(Saturday), 7 (Sunday)*/
                            "TimeRangeList": [
                            /*ro, opt, array, list of time intervals, subType:object, range:[0,4], desc:4 time periods can be configured for a day*/
                                {
                                    "TimeRange": {
                                    /*ro, opt, object, time interval*/
                                        "beginTime":  "00:00:00+08:00",
                                        /*ro, req, time, start time*/
                                        "endTime":  "00:00:00+08:00"
                                        /*ro, req, time, end time*/
                                    }
                                }
                            ]
                        }
                    }
                ],
                "RelatedLanList": [
                /*ro, opt, array, list of linked lane information, subType:object, range:[0,6]*/
                    {
                        "RelatedLan": {
                        /*ro, opt, object, linked lane information*/
                            "lanID":  1
                            /*ro, opt, int, whether to enable linked lane No.*/
                        }
                    }
                ]
            }
        }
    ]
}
```

11.5.1.12 Set the arming parameters for traffic violations capture

Request URL

PUT /ISAPI/ITC/illegalSchedules?format=json&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "eventPriorityEnable":  true,
    /*req, bool, whether to enable violation priority*/
    "IllegalScheduleList": [
    /*req, array, arming time of traffic violations capture, subType:object, desc:this node is only available for ANPR violation information, related node is illegalInfo*/
        {
            "Schedule": {
            /*opt, object, arming schedules*/
                "eventType":  "post",
                /*req, enum, event type, subType:string, desc:"post" (checkpoint), "driveLine" (driving on the lane line), "reverse" (wrong-way driving), "banSign" (prohibition violation), "overSpeed" (driving in overspeed), "lowSpeed" (driving in low speed), "changeLane" (illegal lane change), "safeBelt", "uphone" (making a phone call when driving), "gasser" (queue jumping), "congestion", "helmet" (driver is wearing a helmet), "twoWheelPassenger" (manned non-motor vehicle), "nonMotorExist" (non-motor vehicle on motor vehicle lane), "intersectionCongest" (overstay at intersection), "nonDriveWay" (motor vehicle on non-motor vehicle lane), "redLight" (red light running), "intersectionStop" (stop vehicle at intersection), "direction" (driving against direction guidance), "greenLightStop (stop vehicle when green light), "illegalTurn", "accident", "runGreeLight" (running green light in congestion), "leftNotYieldStraight" (left turn not yield to straight), "rightNotYieldLeft" (right turn not yield to left turn), "uTurnNotYieldGoingStaight", "leftSideDrivingAfterLeftTurn", "failYieldPed" (not yield to pedestrian), "illegalParking" (illegal parking), "nonMotorRedLight" (non-motor vehicle red light running), "nonMotorCrossing" (non-motor vehicle crossing line), "nonMotorOccupyZebraCrossing" (non-motor vehicle crossing zebra line), "roadBlock", "construction", "notKeepDistance" (not keeping a safe distance), "overtakeRightSide" (overtaking on the right), "dragRacing" (street racing), "changeLaneContinuously" (continuous lane change), "SSharpDriving" (slalom driving), "abandonedObject" (thrown object), "pedestrian" (pedestrian pre-alarm), "notSlowZebraCrossing", "overThermal", "turnRightStop" (large-sized vehicle not slowing down when turning right), "enterNotYieldToInRoundabout" (not yield at roundabout intersection), "rampNotYieldToMainRoad" (vehicle from ramp not yielding to vehicle on main road), "notYieldToRightRoad" (not yielding to vehicle from right side), "TruckOccupyOvertakingLane" (large truck occupying overtaking lane), "TruckTurnRightStop" (large truck not slowing down when turning right), "banStraight" (going straight prohibited), "emergencyLane" (occupy emergency lane), "smoke", "conflagration", "VehicleSpeedDrop", "animalsOccupy", "suonaDet" (whistle), "roarDet"*/
                "noPlateCaptureEnable":  true,
                /*opt, bool, whether to enable no plate vehicle capture*/
                "eventPriority":  0,
                /*opt, int, event capture priority, range:[0,10], desc:value range 0 to 10; the higher the value, the higher the priority; when the value is 0, this node will not have priority*/
                "TimeBlockList": [
                /*opt, array, list of time periods, subType:object, range:[0,7]*/
                    {
                        "TimeBlock": {
                        /*opt, object, time period*/
                            "dayOfWeek":  1,
                            /*req, enum, day of a week, subType:int, desc:1 (Monday), 2 (Tuesday), 3 (Wednesday), 4 (Thursday), 5 (Friday), 6 (Saturday), 7 (Sunday)*/
                            "TimeRangeList": [
                            /*opt, array, list of time intervals, subType:object, range:[0,4], desc:4 time periods can be configured for a day*/
                                {
                                    "TimeRange": {
                                    /*opt, object, time interval*/
                                        "beginTime":  "00:00:00+08:00",
                                        /*req, time, start time*/
                                        "endTime":  "00:00:00+08:00"
                                        /*req, time, end time*/
                                    }
                                }
                            ]
                        }
                    }
                ],
                "RelatedLanList": [
                /*opt, array, list of linked lane information, subType:object, range:[0,6]*/
                    {
                        "RelatedLan": {
                        /*opt, object, linked lane information*/
                            "lanID":  1
                            /*opt, int, whether to enable linked lane No.*/
                        }
                    }
                ]
            }
        }
    ]
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

11.5.1.13 Get ANPR configuration capability

Request URL

GET /ISAPI/ITC/plateRecognitionParam/capabilities?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID (this optional parameter is supported when isSupportChannelPlateRecognitionParam is "true" returned from GET /ISAPI/ITC/capabilities).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PlateRecognitionParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ANPR parameters, attr:version{req, string, protocolVersion}-->
  <isSupportdefaultCHN>
    <!--ro, opt, string, province name-->test
  </isSupportdefaultCHN>
  <isSupportfrontPlateReco>
    <!--ro, opt, bool, whether to support front license plate recognition-->true
  </isSupportfrontPlateReco>
  <isSupportrearPlateReco>
    <!--ro, opt, bool, whether to support rear license plate recognition-->true
  </isSupportrearPlateReco>
  <isSupportsmallPlateReco>
    <!--ro, opt, bool, whether to support small license plate recognition-->true
  </isSupportsmallPlateReco>
  <isSupportlargePlateReco>
    <!--ro, opt, bool, whether to support large license plate recognition-->true
  </isSupportlargePlateReco>
  <isSupportfrontAndRearReg>
    <!--ro, opt, bool, whether to enable front license plate recognition and rear license plate recognition-->true
  </isSupportfrontAndRearReg>
  <nation opt="EU,ER,EUandCIS,ME,APAC,AFandAM,All">
    <!--ro, opt, enum, country/region, subType:string, attr:opt{req, string}, desc:"ER" (Russian-speaking region), "EU" (Europe), "ME" (Middle East), "APAC"
(Asia-Pacific region), "AFandAM" (Africa and America), "All" (all regions), "EUandCIS" (Europe and Russia)-->ER
  </nation>
  <countryIndex opt="1,2,3,4,5,6,7,8,12,14,17,18,19,20,23,39,44,55">
    <!--ro, opt, int, country/region index, attr:opt{req, string}-->0
  </countryIndex>
  <fakePlateFilterEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->false
  </fakePlateFilterEnabled>
  <citySeparatorEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </citySeparatorEnabled>
  <fakePlateDetectionUploadEnabled opt="true,false">
    <!--ro, opt, bool, dep:and,{$.PlateRecognitionParam.fakePlateFilterEnabled,eq,true}, attr:opt{req, string}-->false
  </fakePlateDetectionUploadEnabled>
  <provinceThresholds min="0" max="100">
    <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
  </provinceThresholds>
  <yellowGreenPlateRecoRule opt="yellow,green,yellowGreen,yellowGreenBicolor">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->yellow
  </yellowGreenPlateRecoRule>
  <nonStandardPlateRegEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </nonStandardPlateRegEnabled>
  <threePlateRecoEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </threePlateRecoEnabled>
```

```
    <newEnergyGreenPlateRecoRule opt="green,newEnergyGreen,gradationGreen">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->green
    </newEnergyGreenPlateRecoRule>
    <customPlateRule>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <customPlateColorEnabled opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </customPlateColorEnabled>
      <customPlateColor opt="black,blue,green,white,yellow">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->black
      </customPlateColor>
      <wildcardRuleList size="10">
        <!--ro, opt, object, attr:size{req, int}-->
        <wildcardRule>
          <!--ro, opt, object-->
          <enabled opt="true,false">
            <!--ro, req, bool, attr:opt{req, string}-->true
          </enabled>
          <wildcard min="0" max="16">
            <!--ro, opt, string, range:[0,16], attr:min{req, int},max{req, int}-->蓝园A?!!**
          </wildcard>
        </wildcardRule>
      </wildcardRuleList>
    </customPlateRule>
    <plateMappingRule>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <plateMappingList size="64">
        <!--ro, opt, array, subType:object, range:[0,64], attr:size{req, int}-->
        <plateMapping>
          <!--ro, opt, object-->
          <detectPlateNo min="0" max="16">
            <!--ro, opt, string, range:[0,16], attr:min{req, int},max{req, int}-->浙A12348
          </detectPlateNo>
          <replacePlateNo min="0" max="16">
            <!--ro, opt, string, range:[0,16], attr:min{req, int},max{req, int}-->浙A1234B
          </replacePlateNo>
        </plateMapping>
      </plateMappingList>
    </plateMappingRule>
    <primaryAndSecondaryPlate>
      <!--ro, opt, array, subType:object-->
      <enabled>
        <!--ro, opt, array, subType:object-->
        <mainlandPlateEnable opt="true,false">
          <!--ro, opt, bool, attr:opt{req, string}-->true
        </mainlandPlateEnable>
        <hongKongPlateEnable opt="true,false">
          <!--ro, opt, bool, attr:opt{req, string}-->true
        </hongKongPlateEnable>
        <maCaoPlateEnable opt="true,false">
          <!--ro, opt, bool, attr:opt{req, string}-->true
        </maCaoPlateEnable>
      </enabled>
      <plateRegionType>
        <!--ro, opt, array, subType:object-->
        <mainAreaOption opt="mainland,hongKong,maCao">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->mainland
        </mainAreaOption>
        <secondAreaOption opt="mainland,hongKong,maCao">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->hongKong
        </secondAreaOption>
      </plateRegionType>
    </primaryAndSecondaryPlate>
  </PlateRecognitionParam>
```


11.5.1.14 Set ANPR parameters

Request URL

PUT /ISAPI/ITC/plateRecognitionParam?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<PlateRecognitionParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ANPR parameters, attr:version{req, string, protocolVersion}-->
  <defaultCHN>
    <!--req, string, province/state abbreviation-->test
  </defaultCHN>
  <frontPlateRecoEnabled>
    <!--opt, bool, front license plate recognition, desc:front license plate recognition-->true
  </frontPlateRecoEnabled>
  <rearPlateRecoEnabled>
    <!--opt, bool, rear license plate recognition-->true
  </rearPlateRecoEnabled>
  <smallPlateRecoEnabled>
    <!--opt, bool, small license plate recognition-->true
  </smallPlateRecoEnabled>
  <largePlateRecoEnabled>
    <!--opt, bool, large license plate recognition-->true
  </largePlateRecoEnabled>
  <farmVehicleEnabled>
    <!--opt, bool, agricultural vehicle recognition-->true
  </farmVehicleEnabled>
  <motorEnabled>
    <!--opt, bool, motorcycle recognition-->true
  </motorEnabled>
  <fuzzyDiscEnabled>
    <!--opt, bool, fuzzy recognition-->true
  </fuzzyDiscEnabled>
  <microPlateRegEnabled>
    <!--opt, bool, tiny license plate recognition-->true
  </microPlateRegEnabled>
  <cAPlateRegEnabled>
    <!--opt, bool, civil aviation license plate recognition-->true
  </cAPlateRegEnabled>
  <tiltPlateRegEnabled>
    <!--opt, bool, tilted license plate recognition-->true
  </tiltPlateRegEnabled>
  <superPlateRecoEnabled>
    <!--opt, bool, oversized license plate recognition-->true
  </superPlateRecoEnabled>
  <embassyPlateRecoEnabled>
    <!--opt, bool, embassy license plate recognition-->true
  </embassyPlateRecoEnabled>
</PlateRecognitionParam>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, 
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.15 Get ANPR parameters

Request URL

GET /ISAPI/ITC/plateRecognitionParam?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID (this optional parameter is supported when isSupportChannelPlateRecognitionParam is "true" returned from GET /ISAPI/ITC/capabilities).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognitionParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, ANPR parameters, attr:version{req, string, protocolVersion}-->
  <defaultCHN>
    <!--ro, opt, string, province name-->test
  </defaultCHN>
  <frontPlateRecoEnabled>
    <!--ro, opt, bool, front license plate recognition, desc:license plate recognition principles: when <isSupportfrontAndRearReg> is true and both front
and rear license plate recognition are supported, "true" is returned and the front license plate recognition parameters will be modified by default; when
<isSupportfrontAndRearReg> is false and either front or rear license plate recognition is supported, "false" is returned-->true
  </frontPlateRecoEnabled>
  <rearPlateRecoEnabled>
    <!--ro, opt, bool, rear license plate recognition-->true
  </rearPlateRecoEnabled>
  <smallPlateRecoEnabled>
    <!--ro, opt, bool, small license plate recognition-->true
  </smallPlateRecoEnabled>
  <largePlateRecoEnabled>
    <!--ro, opt, bool, large license plate recognition-->true
  </largePlateRecoEnabled>
  <farmVehicleEnabled>
    <!--ro, opt, bool, agricultural vehicle recognition-->true
  </farmVehicleEnabled>
  <motorEnabled>
    <!--ro, opt, bool, motorcycle recognition-->true
  </motorEnabled>
  <fuzzyDiscEnabled>
    <!--ro, opt, bool, fuzzy recognition-->true
  </fuzzyDiscEnabled>
  <microPlateRegEnabled>
    <!--ro, opt, bool, tiny license plate recognition-->true
  </microPlateRegEnabled>
  <cAPlateRegEnabled>
    <!--ro, opt, bool, civil aviation license plate recognition-->true
  </cAPlateRegEnabled>
  <tiltPlateRegEnabled>
    <!--ro, opt, bool, tilted license plate recognition-->true
  </tiltPlateRegEnabled>
  <superPlateRecoEnabled>
    <!--ro, opt, bool, large license plate recognition-->true
  </superPlateRecoEnabled>
  <embassyPlateRecoEnabled>
    <!--ro, opt, bool, embassy license plate recognition-->true
  </embassyPlateRecoEnabled>
  <nation>
    <!--ro, opt, enum, country/region, subType:string, desc:"ER" (Russian-speaking region), "EU" (Europe), , "EUandCIS" (Europe and Russia), "ME" (Middle
East), "APAC" (Asia-Pacific region), "AFandAM" (Africa and America), "All" (all regions)-->ER
  </nation>
  <countryIndex>
    <!--ro, opt, int, country/region index, desc:refer to the <CRIndex> data field-->0
  </countryIndex>
  <nonMotorPlateCityRecoEnabled>
    <!--ro, opt, bool, whether to enable belonged city recognition of non-motor vehicle license plate-->true
  </nonMotorPlateCityRecoEnabled>
  <city>
    <!--ro, opt, string, license plate belonged city-->test
  </city>
  <fakePlateFilterEnabled>
    <!--ro, opt, bool, filter fake license plate, desc:filter fake license plate-->false
  </fakePlateFilterEnabled>
  <citySeparatorEnabled>
    <!--ro, opt, bool-->true
  </citySeparatorEnabled>
  <fakePlateDetectionUploadEnabled>
    <!--ro, opt, bool, dep:and,{$.PlateRecognitionParam.fakePlateFilterEnabled,eq,true}-->false
  </fakePlateDetectionUploadEnabled>
  <provinceThresholds>
    <!--ro, opt, int, range:[0,100]-->0
  </provinceThresholds>
  <secondLicensePlateRegEnabled>
    <!--ro, opt, bool-->true
  </secondLicensePlateRegEnabled>
  <yellowGreenPlateRecoRule>
```

```xml
    <!--ro, opt, enum, subType:string-->yellow
  </yellowGreenPlateRecoRule>
  <nonStandardPlateIllegalHandleRules>
    <!--ro, opt, enum, subType:string-->none
  </nonStandardPlateIllegalHandleRules>
  <nonStandardPlatePostHandleRules>
    <!--ro, opt, enum, subType:string-->none
  </nonStandardPlatePostHandleRules>
  <nonStandardPlateThresholds>
    <!--ro, opt, int, range:[0,100]-->0
  </nonStandardPlateThresholds>
  <nonStandardPlateMinCorrectFactor>
    <!--ro, opt, int, range:[0,100]-->0
  </nonStandardPlateMinCorrectFactor>
  <nonStandardPlateMaxCorrectFactor>
    <!--ro, opt, int, range:[0,100]-->0
  </nonStandardPlateMaxCorrectFactor>
  <nonStandardVehicleCorrectFactor>
    <!--ro, opt, int, range:[0,100]-->0
  </nonStandardVehicleCorrectFactor>
  <nonStandardPlateRegEnabled>
    <!--ro, opt, bool-->true
  </nonStandardPlateRegEnabled>
  <threePlateRecoEnabled>
    <!--ro, opt, bool-->true
  </threePlateRecoEnabled>
  <newEnergyGreenPlateRecoRule>
    <!--ro, opt, enum, subType:string-->green
  </newEnergyGreenPlateRecoRule>
  <customPlateRule>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <customPlateColorEnabled>
      <!--ro, opt, bool-->true
    </customPlateColorEnabled>
    <customPlateColor>
      <!--ro, opt, enum, subType:string-->black
    </customPlateColor>
    <wildcardRuleList>
      <!--ro, opt, array, subType:object, range:[0,10]-->
      <wildcardRule>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, req, bool-->true
        </enabled>
        <wildcard>
          <!--ro, opt, string, range:[0,16]-->蓝园A?!!**
        </wildcard>
      </wildcardRule>
    </wildcardRuleList>
  </customPlateRule>
  <plateMappingRule>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <plateMappingList>
      <!--ro, opt, array, subType:object, range:[0,64]-->
      <plateMapping>
        <!--ro, opt, object-->
        <detectPlateNo>
          <!--ro, opt, string, range:[0,16]-->浙A12348
        </detectPlateNo>
        <replacePlateNo>
          <!--ro, opt, string, range:[0,16]-->浙A1234B
        </replacePlateNo>
      </plateMapping>
    </plateMappingList>
  </plateMappingRule>
  <primaryAndSecondaryPlate>
    <!--ro, opt, array, subType:object-->
    <enabled>
      <!--ro, opt, array, subType:object-->
      <mainlandPlateEnable>
        <!--ro, opt, bool-->true
      </mainlandPlateEnable>
      <hongKongPlateEnable>
        <!--ro, opt, bool-->true
      </hongKongPlateEnable>
      <maCaoPlateEnable>
        <!--ro, opt, bool-->true
      </maCaoPlateEnable>
    </enabled>
    <plateRegionType>
      <!--ro, opt, array, subType:object-->
      <mainAreaOption>
        <!--ro, opt, enum, subType:string-->mainland
      </mainAreaOption>
      <secondAreaOption>
        <!--ro, opt, enum, subType:string-->hongKong
      </secondAreaOption>
    </plateRegionType>
```

```
    </primaryAndSecondaryPlate>
</PlateRecognitionParam>
```

11.5.1.16 Get trigger mode capability of traffic cameras

Request URL

GET /ISAPI/ITC/TriggerMode/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TriggerMode xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, trigger mode, attr:version{req, string, protocolVersion}-->
  <usedTriggerMode
opt="postIOSpeed,postSingleIO,postRS485,postRadar,postVTCoil,postHVT,postMPR,epoliceRS485,postEpoliceRS485,videoEpolice,TPS,videoMonitor,postNoComityPed,pos
tRedLightPed,postSideRadarSpeed,parkingIncident">
    <!--ro, req, enum, trigger mode supported by the device, different devices support different trigger modes, and a device may support multiple trigger
modes, subType:string, attr:opt{req, string}, desc:"postMobile", "postDetect", "TPS", "trafficIncident", "trafficIncidentTPS", "roadWarning", "postIOSpeed",
"postSingleIO", "postRS485", "postRadar", "postVTCoil", "postHVT", "postMPR", "epoliceRS485", "postEpoliceRS485", "videoEpolice", "videoMonitor",
"postNoComityPed", "postRedLightPed", "postSideRadarSpeed", "parkingIncident", "vehicleDetect", "boxNumDetection", "parkSpeedDetection"-->postMobile
  </usedTriggerMode>
  <isSupportChannelConfig>
    <!--ro, opt, bool, whether it supports channel configuration, desc:related URL: /ISAPI/ITC/channels/<ID>/TriggerMode/capabilities-->true
  </isSupportChannelConfig>
  <isSupportChannelSwitchStatus>
    <!--ro, opt, bool, whether it supports switching trigger mode status by channel No., desc:true (the optional node <channelID> in
/ISAPI/ITC/TriggerMode/switchStatus?format=json&channelID=<channelID> is supported)-->true
  </isSupportChannelSwitchStatus>
</TriggerMode>
```

11.5.1.17 Get parameters of intersection violation system vehicle detector

Request URL

GET /ISAPI/ITC/TriggerMode/epoliceRS485

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <trafficLightSource>
    <!--ro, req, enum, traffic light source, subType:int, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <EpoliceRS485LaneList>
    <!--ro, opt, array, lane list, subType:object-->
    <EpoliceRS485Lane>
      <!--ro, opt, object, lane-->
      <laneId>
        <!--ro, req, int, lane ID, range:[1,4]-->0
      </laneId>
      <laneDirectionType>
        <!--ro, req, enum, lane direction, subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--ro, req, int, linked lane No., range:[1,16]-->0
      </relatedDriveWay>
      <redLightDriveWay>
        <!--ro, req, int, red light lane No., range:[1,16]-->0
      </redLightDriveWay>
      <yellowLightDriveWay>
        <!--ro, req, int, yellow light lane No., range:[1,16]-->0
      </yellowLightDriveWay>
      <OSDDriveWay>
        <!--ro, req, int, overlay lane No., range:[1,99]-->0
      </OSDDriveWay>
      <distance>
        <!--ro, req, int, coil distance, range:[0,2000], unit:cm-->0
      </distance>
      <recordEnable>
        <!--ro, req, bool, whether to enable recording-->true
      </recordEnable>
      <recordType>
        <!--ro, req, enum, recording type, subType:string, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
      </recordType>
      <preRecordTime>
        <!--ro, req, int, pre-record duration, range:[0,100]-->0
      </preRecordTime>
      <recordDelayTime>
        <!--ro, req, int, post-record duration, range:[0,100]-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--ro, req, int, timeout duration, range:[0,100]-->0
      </recordTimeOut>
      <serialProtocol>
        <!--ro, req, enum, serial port protocol, subType:int, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
      </serialProtocol>
      <runRedLightCaptureLogic>
        <!--ro, req, enum, red light running capture logic, subType:string, desc:"singelPro" (single_enter 1_exit 1_delay 1_1), "singlePro2" (singelPro), "doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit 1_delay 2_1), "doublePro4" (double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter 1_exit 2_delay 2_2), "doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
      </runRedLightCaptureLogic>
      <oppositeDirectionCaptureLogic>
        <!--ro, req, enum, Wrong-Way Driving Capture Logic, subType:string, desc:"noCap" (do not capture), "reversePro1" (double/three_enter_enter_delay), "reversePro2" (double/three_enter_delay)-->noCap
      </oppositeDirectionCaptureLogic>
      <intervalType>
        <!--ro, req, enum, Burst Interval Type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
      </intervalType>
      <interval>
        <!--ro, req, int, interval, range:[0,2000]-->0
      </interval>
      <IOOutList>
        <!--ro, opt, array, list of I/O output ports, subType:object-->
        <IOOut>
          <!--ro, req, object, I/O output port-->
          <id>
            <!--ro, req, int, linked I/O output No., range:[1,10]-->0
          </id>
          <enabled>
            <!--ro, req, bool, whether to link the I/O output-->true
          </enabled>
        </IOOut>
      </IOOutList>
      <flashMode>
        <!--ro, req, enum, flashing mode, subType:string, desc:"together" (simultaneously), "alternation" (alternatively)-->together
      </flashMode>
    </EpoliceRS485Lane>
  </EpoliceRS485LaneList>
</EpoliceRS485>
```

11.5.1.18 Set parameters of intersection violation system vehicle detector

Request URL

PUT /ISAPI/ITC/TriggerMode/epoliceRS485

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <trafficLightSource>
    <!--req, enum, traffic light source, subType:int, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <EpoliceRS485LaneList>
    <!--opt, array, lane list, subType:object-->
    <EpoliceRS485Lane>
      <!--opt, object, lane-->
      <laneId>
        <!--req, int, lane ID, range:[1,4]-->0
      </laneId>
      <laneDirectionType>
        <!--req, enum, lane direction, subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--req, int, linked lane No., range:[1,16]-->0
      </relatedDriveWay>
      <redLightDriveWay>
        <!--req, int, red light lane No., range:[1,16]-->0
      </redLightDriveWay>
      <yellowLightDriveWay>
        <!--req, int, yellow light lane No., range:[1,16]-->0
      </yellowLightDriveWay>
      <OSDDriveWay>
        <!--req, int, Overlay Lane No., range:[1,99]-->0
      </OSDDriveWay>
      <distance>
        <!--req, int, coil distance, range:[0,2000], unit:cm-->0
      </distance>
      <recordEnable>
        <!--req, bool, whether to enable recording-->true
      </recordEnable>
      <recordType>
        <!--req, enum, recording type, subType:string, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
      </recordType>
      <preRecordTime>
        <!--req, int, pre-record duration, range:[0,100]-->0
      </preRecordTime>
      <recordDelayTime>
        <!--req, int, post-record, range:[0,100]-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--req, int, timeout duration, range:[0,100]-->0
      </recordTimeOut>
      <serialProtocol>
        <!--req, enum, serial port protocol, subType:int, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
      </serialProtocol>
      <runRedLightCaptureLogic>
        <!--req, enum, Red Light Running Capture Logic, subType:string, desc:"singelPro" (single_enter 1_exit 1_delay 1_1), "singlePro2" (singelPro), "doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit 1_delay 1_1), "doublePro4" (double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter 1_exit 2_delay 2_2), "doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
      </runRedLightCaptureLogic>
      <oppositeDirectionCaptureLogic>
        <!--req, enum, Wrong-Way Driving Capture Logic, subType:string, desc:"noCap" (do not capture), "reversePro1" (double/three_enter_enter_delay), "reversePro2" (double/three_enter_delay)-->noCap
      </oppositeDirectionCaptureLogic>
      <intervalType>
        <!--req, enum, Burst Interval Type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
      </intervalType>
      <interval>
        <!--req, int, interval, range:[0,2000]-->0
      </interval>
      <IOOutList>
        <!--opt, array, list of I/O output ports, subType:object-->
        <IOOut>
          <!--req, object, I/O output port-->
          <id>
            <!--req, int, linked I/O output No., range:[1,10]-->0
          </id>
          <enabled>
            <!--req, bool, whether to link the I/O output-->true
          </enabled>
        </IOOut>
      </IOOutList>
      <flashMode>
        <!--req, enum, flashing mode, subType:string, desc:"together" (simultaneously), "alternation" (alternatively)-->together
      </flashMode>
    </EpoliceRS485Lane>
  </EpoliceRS485LaneList>
</EpoliceRS485>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, 
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.19 Get the parameters capability of intersection violation system vehicle detector

Request URL

GET /ISAPI/ITC/TriggerMode/epoliceRS485/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="6">
    <!--ro, req, int, number of supported lanes, range:[1,6], attr:min{req, int},max{req, int}-->0
  </relatedLaneCount>
  <trafficLightSource opt="0,1">
    <!--ro, req, enum, traffic light source, subType:int, attr:opt{req, string}, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <relatedDriveWay min="1" max="16">
    <!--ro, req, int, linked lane No., range:[1,16], attr:min{req, int},max{req, int}-->0
  </relatedDriveWay>
  <redLightDriveWay min="1" max="16">
    <!--ro, req, int, red light lane No., range:[1,16], attr:min{req, int},max{req, int}-->0
  </redLightDriveWay>
  <yellowLightDriveWay min="1" max="16">
    <!--ro, req, int, yellow light lane No., range:[1,16], attr:min{req, int},max{req, int}-->0
  </yellowLightDriveWay>
  <OSDDriveWay min="1" max="99">
    <!--ro, req, int, overlay lane No., range:[1,99], attr:min{req, int},max{req, int}-->0
  </OSDDriveWay>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, lane direction, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to
north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to
northeast)-->0
  </laneDirectionType>
  <distance min="0" max="2000">
    <!--ro, req, int, coil distance, range:[0,200], attr:min{req, int},max{req, int}-->0
  </distance>
  <recordEnable>
    <!--ro, req, bool, whether to enable recording-->true
  </recordEnable>
  <recordType opt="delayRecord,preRecord">
    <!--ro, req, enum, recording type, subType:string, attr:opt{req, string}, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
  </recordType>
  <preRecordTime min="0" max="100">
    <!--ro, req, int, pre-record, range:[0,100], attr:min{req, int},max{req, int}-->0
  </preRecordTime>
  <recordDelayTime min="0" max="100">
    <!--ro, req, int, post-record, range:[0,100], attr:min{req, int},max{req, int}-->0
  </recordDelayTime>
  <recordTimeOut min="0" max="100">
    <!--ro, req, int, timeout duration, range:[0,100], attr:min{req, int},max{req, int}-->0
  </recordTimeOut>
  <serialProtocol opt="0,1,2">
    <!--ro, req, enum, serial port protocol, subType:int, attr:opt{req, string}, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
  </serialProtocol>
  <runRedLightCaptureLogic opt="singelPro,doublePro1,doublePro2,doublePro3,doublePro4,doublePro5,doublePro6,doublePro7,doublePro8,singlePro2,other">
    <!--ro, req, enum, red light running capture logic, subType:string, attr:opt{req, string}, desc:"singelPro" (single_enter 1_exit 1_delay 1_1),
"singlePro2" (singelPro), "doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit
1_delay 2_1), "doublePro4" (double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter
1_exit 2_delay 2_2), "doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
  </runRedLightCaptureLogic>
  <oppositeDirectionCaptureLogic opt="noCap,reversePro1,reversePro2">
    <!--ro, req, enum, Wrong-Way Driving Capture Logic, subType:string, attr:opt{req, string}, desc:"noCap" (do not capture), "reversePro1"
(double/three_enter_enter_delay), "reversePro2" (double/three_enter_delay)-->noCap
  </oppositeDirectionCaptureLogic>
  <intervalType opt="time,distance">
    <!--ro, req, enum, Burst Interval Type, subType:string, attr:opt{req, string}, desc:"time" (by time), "distance" (by distance)-->time
  </intervalType>
  <interval>
    <!--ro, req, int, interval, range:[0,2000]-->0
  </interval>
  <IOOut min="1" max="7">
    <!--ro, req, int, number of I/O outputs, range:[1,7], attr:min{req, int},max{req, int}-->0
  </IOOut>
  <flashMode opt="together,alternation">
    <!--ro, req, enum, flashing mode, subType:string, attr:opt{req, string}, desc:"together" (simultaneously), "alternation" (alternatively)-->together
  </flashMode>
</EpoliceRS485>
```

11.5.1.20 Get the recommended configurations of intersection violation system vehicle detector

Request URL

GET /ISAPI/ITC/TriggerMode/epoliceRS485/recommendation

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EpoliceRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of intersection violation system vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <trafficLightSource>
    <!--ro, req, enum, traffic light source, subType:int, desc:0-vehicle detector, 1-traffic light detector.-->0
  </trafficLightSource>
  <EpoliceRS485LaneList>
    <!--ro, opt, array, lane list, subType:object-->
    <EpoliceRS485Lane>
      <!--ro, opt, object, lane-->
      <laneId>
        <!--ro, req, int, lane ID, range:[1,4]-->0
      </laneId>
      <laneDirectionType>
        <!--ro, req, enum, lane direction, subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--ro, req, int, linked lane No., range:[1,16]-->0
      </relatedDriveWay>
      <redLightDriveWay>
        <!--ro, req, int, red light lane No., range:[1,16]-->0
      </redLightDriveWay>
      <yellowLightDriveWay>
        <!--ro, req, int, yellow light lane No., range:[1,16]-->0
      </yellowLightDriveWay>
      <OSDDriveWay>
        <!--ro, req, int, overlay lane No., range:[1,99]-->0
      </OSDDriveWay>
      <distance>
        <!--ro, req, int, coil distance, range:[0,2000], unit:cm-->0
      </distance>
      <recordEnable>
        <!--ro, req, bool, whether to enable recording-->true
      </recordEnable>
      <recordType>
        <!--ro, req, enum, recording type, subType:string, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
      </recordType>
      <preRecordTime>
        <!--ro, req, int, pre-record duration, range:[0,100]-->0
      </preRecordTime>
      <recordDelayTime>
        <!--ro, req, int, post-record, range:[0,100]-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--ro, req, int, timeout duration, range:[0,100]-->0
      </recordTimeOut>
      <serialProtocol>
        <!--ro, req, enum, serial port protocol, subType:int, desc:0-vehicle detector, 1-OEM, 2-other vehicle detector-->0
      </serialProtocol>
      <runRedLightCaptureLogic>
        <!--ro, req, enum, Red Light Running Capture Logic, subType:string, desc:"singelPro" (single_enter 1_exit 1_delay 1_1), "singlePro2" (singelPro), "doublePro1" (double_enter 1_exit 1_exit 2_1), "doublePro2" (double_enter 1_exit 2_delay 2_1), "doublePro3" (double_enter 2_exit 1_delay 2_1), "doublePro4" (double/three_enter 2_exit 2_delay 2_1), "doublePro5" (double/three_enter 2_exit 2_delay 2_2), "doublePro6" (double/three_enter 1_exit 2_delay 2_2), "doublePro7" (double_enter 1_exit 1_delay 2_1), "doublePro8" (double_enter 1_exit 1_delay 2_2), "other"-->singelPro
      </runRedLightCaptureLogic>
      <oppositeDirectionCaptureLogic>
        <!--ro, req, enum, Wrong-Way Driving Capture Logic, subType:string, desc:"noCap" (do not capture), "reversePro1" (double/three_enter_enter_delay), "reversePro2" (double/three_enter_delay)-->noCap
      </oppositeDirectionCaptureLogic>
      <intervalType>
        <!--ro, req, enum, burst interval type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
      </intervalType>
      <interval>
        <!--ro, req, int, interval, range:[0,2000]-->0
      </interval>
      <IOOutList>
        <!--ro, opt, array, list of I/O output ports, subType:object-->
        <IOOut>
          <!--ro, req, object, I/O output port-->
          <id>
            <!--ro, req, int, linked I/O output No., range:[1,10]-->0
          </id>
          <enabled>
            <!--ro, req, bool, whether to link the I/O output-->true
          </enabled>
        </IOOut>
      </IOOutList>
      <flashMode>
        <!--ro, req, enum, flashing mode, subType:string, desc:"together" (simultaneously), "alternation" (alternatively)-->together
      </flashMode>
    </EpoliceRS485Lane>
  </EpoliceRS485LaneList>
</EpoliceRS485>
```

11.5.1.21 Get the status of video detection traffic lights

Request URL

GET /ISAPI/ITC/TriggerMode/lightStatus?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID (this optional parameter is supported when isSupportChannelLightStatus is "true" returned from /ISAPI/ITC/TriggerMode/capabilities).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VepLightStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, status of video detection traffic lights, attr:version{req, string, protocolVersion}-->
  <lightStatus opt="normal,abnomal">
    <!--ro, req, enum, traffic light status, subType:string, attr:opt{req, string}, desc:"normal", "abnormal"-->normal
  </lightStatus>
</VepLightStatus>
```

11.5.1.22 Get the license plate recognition configuration capability of a specific trigger mode

Request URL

GET /ISAPI/ITC/TriggerMode/plateRecognition/<triggerMode>/capabilities?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
triggerMode	enum	--
channelID	string	Camera ID (this optional parameter is supported when isSupportChannelPlateRecognition is "true" returned from /ISAPI/ITC/TriggerMode/capabilities).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the license plate recognition parameters of a specific trigger mode, attr:version{req, string, protocolVersion}-->
  <triggerMode opt="postIOSpeed,postSingleIO,postRS485,postRadar,postVTCoil,postHVT,postMpr,epoliceRS485,postEpoliceRS485,videoEpolice,postBOT">
    <!--ro, req, enum, trigger modes supported by device, subType:string, attr:opt{req, string}, desc:"postiospeed" (checkpoint i/o speed detection),
"postsingleio" (checkpoint single i/o trigger), "postrs485" (checkpoint vehicle detector), "postradar" (checkpoint radar), "postvtcoil" (checkpoint virtual
coil), "posthvt" (checkpoint multi-target-type detection), "postmpr" (entrance and exit video trigger), "epolicers485" (intersection violation system
vehicle detector), "postepolicers485" (checkpoint intersection violation system vehicle detector), "videoepolice" (video intersection violation system),
"postbot" (bus lane occupation)-->postIOSpeed
  </triggerMode>
  <backupType opt="0,1">
    <!--ro, req, enum, backup mode type, subType:int, attr:opt{req, string}, desc:0-backup mode, 1-normal configuration mode. When this node is 0, the
current trigger mode is checkpoint vehicle detector; when this node is 1, the current trigger mode is checkpoint virtual coil-->0
  </backupType>
  <plateRecogEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable license plate recognition function of this region, attr:opt{req, string}-->true
  </plateRecogEnabled>
  <regionMode opt="rectangle,polygon">
    <!--ro, req, enum, area type, subType:string, attr:opt{req, string}, desc:"rectangle", "polygon"-->rectangle
  </regionMode>
  <plateRegionCount min="1" max="6">
    <!--ro, req, int, total number of linked license plate recognition regions, range:[1,6], attr:min{req, int},max{req, int}-->0
  </plateRegionCount>
</PlateRecognition>
```

11.5.1.23 Set the license plate recognition parameters of a specific trigger mode

Request URL

PUT /ISAPI/ITC/TriggerMode/plateRecognition/<triggerMode>?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
triggerMode	enum	--
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<PlateRecognition xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <triggerMode>
    <!--req, enum, "postIOSpeed,postSingleIO,postRS485,postRadar,postVTCoil,postMixedLane,postMpr,epoliceRS485,postEpoliceRS485,videoEpolice",
subType:string, desc:"postIOSpeed,postSingleIO,postRS485,postRadar,postVTCoil,postMixedLane,postMpr,epoliceRS485,postEpoliceRS485,videoEpolice"--
>postIOSpeed
  </triggerMode>
  <backupType>
    <!--req, enum, backup mode type, subType:int, desc:backup mode type-->0
  </backupType>
  <plateRecogEnabled>
    <!--opt, bool, whether to enable license plate recognition function of this region-->true
  </plateRecogEnabled>
  <regionMode>
    <!--req, enum, region type, subType:string, desc:region type-->rectangle
  </regionMode>
  <plateRegionCount>
    <!--req, int, total number of linked license plate recognition regions, range:[1,6]-->0
  </plateRegionCount>
  <PlateRegionList>
    <!--req, array, subType:object-->
    <PlateRegion>
      <!--req, object-->
      <regionId>
        <!--req, int, region ID, range:[1,6]-->0
      </regionId>
      <RegionCoordinatesList>
        <!--req, array, region coordinate list, subType:object, range:[3,10]-->
        <RegionCoordinates>
          <!--req, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </PlateRegion>
  </PlateRegionList>
</PlateRecognition>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.24 Get parameters of checkpoint intersection violation system vehicle detector

Request URL

GET /ISAPI/ITC/TriggerMode/postEpoliceRS485

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
</PostRS485>
```

11.5.1.25 Set parameters of checkpoint intersection violation system vehicle detector

Request URL

PUT /ISAPI/ITC/TriggerMode/postEpoliceRS485

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <backupMode>
    <!--req, enum, backup mode, subType:int, desc:"0" (none), "1" (trigger of checkpoint virtual coil), "2" (mixed trigger mode)-->0
  </backupMode>
  <maxSwitchTime>
    <!--req, int, fault-tolerant time, range:[5,255]-->0
  </maxSwitchTime>
  <RS485LaneList>
    <!--opt, array, lane list, subType:object-->
    <RS485Lane>
      <!--opt, object, lane-->
      <laneId>
        <!--req, int, line No., range:[1,4]-->0
      </laneId>
      <relatedDriveWay>
        <!--req, int, related lane No., range:[1,99]-->0
      </relatedDriveWay>
      <OSDDriveWay>
        <!--req, int, OSD lane No., range:[1,99]-->0
      </OSDDriveWay>
      <laneType>
        <!--req, enum, lane type, subType:string, desc:"none"-not configured, "highway"-high way, "urbanFastWay"-urban expressway, "other"-->none
      </laneType>
      <distance>
        <!--req, int, coil distance, range:[0,20000], unit:cm-->0
      </distance>
      <delayTime>
        <!--req, int, trigger delay time, range:[0,2000]-->0
      </delayTime>
      <delayDistance>
        <!--req, int, trigger delay distance, range:[0,255]-->0
      </delayDistance>
      <speedCapEnabled>
        <!--req, bool, whether to enable low speed trigger mode-->true
      </speedCapEnabled>
      <lowSpeedCapEnable>
        <!--req, bool, whether to enable capture when low speed is detected-->true
      </lowSpeedCapEnable>
      <emergencyCapEnable>
        <!--req, bool, whether to enable capture when emergency lane occupancy is detected-->true
      </emergencyCapEnable>
      <signSpeed>
        <!--req, int, marked speed limit, range:[0,255], unit:km/h-->0
      </signSpeed>
      <speedLimit>
        <!--req, int, speed limit, range:[0,255], unit:km/h-->0
      </speedLimit>
      <bigCarSignSpeed>
        <!--req, int, speed limit for large-sized vehicle, range:[0,255], unit:km/h-->0
      </bigCarSignSpeed>
```

```
    </bigCarSignSpeed>
    <bigCarSpeedLimit>
      <!--req, int, marked speed limit for large-sized vehicle, range:[0,255], unit:km/h-->0
    </bigCarSpeedLimit>
    <lowSpeedLimit>
      <!--req, int, min. speed limit for small-sized vehicle, range:[0,255], unit:km/h-->0
    </lowSpeedLimit>
    <bigCarLowSpeedLimit>
      <!--req, int, min. speed limit for large-sized vehicle, range:[0,255], unit:km/h-->0
    </bigCarLowSpeedLimit>
    <carHighSpeed>
      <!--req, int, abnormal speeding, range:[0,255], unit:km/h, desc:this node is valid for lane 1 only-->0
    </carHighSpeed>
    <carLowSpeed>
      <!--req, int, abnormal low speed, range:[0,255], unit:km/h, desc:this node is valid for lane 1 only-->0
    </carLowSpeed>
    <bigCarHighSpeed>
      <!--req, int, abnormal speeding of large-sized vehicle, range:[0,255], unit:km/h, desc:this node is valid for lane 1 only-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed>
      <!--req, int, abnormal low speed of large-sized vehicle, range:[0,255], unit:km/h, desc:this node is valid for lane 1 only-->0
    </bigCarLowSpeed>
    <snapTimes>
      <!--req, int, capture times, range:[0,5]-->0
    </snapTimes>
    <intervalType>
      <!--req, enum, burst interval type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
    </intervalType>
    <IntervalList>
      <!--opt, array, list of time intervals, subType:object-->
      <Interval>
        <!--req, object, interval information-->
        <value>
          <!--req, int, burst interval, desc:unit: ms/dm; capture interval by time is between 70 and 2000; capture interval by distance is between 1 and 
20-->0
        </value>
      </Interval>
    </IntervalList>
    <IOOutList>
      <!--opt, array, list of I/O output ports, subType:object-->
      <IOOut>
        <!--req, object, I/O output port-->
        <id>
          <!--req, int, linked I/O output No., range:[1,10]-->0
        </id>
        <enabled>
          <!--req, bool, whether to link the I/O output-->true
        </enabled>
      </IOOut>
    </IOOutList>
    <flashMode>
      <!--req, enum, flashing mode, subType:string, desc:"together", "alternation"-->together
    </flashMode>
    <laneUsage>
      <!--req, enum, lane purpose, subType:string, desc:"carriageWay" (normal lane), "banTrucks" (lane banning large-sized vehicle), "emergency" 
(emergency lane)-->carriageWay
    </laneUsage>
    <laneDirectionType>
      <!--req, enum, lane direction, subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to 
south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
    </laneDirectionType>
    <emergencyTimeSwitchList>
      <!--opt, array, time period for emergency lanes, subType:object-->
      <emergencyTimeSwitch>
        <!--req, object, emergency lane time-->
        <timeId>
          <!--req, int, index, range:[1,4]-->0
        </timeId>
        <startHour>
          <!--opt, int, start time (hour), range:[0,23]-->0
        </startHour>
        <startMinute>
          <!--opt, int, start time (minute), range:[0,59]-->0
        </startMinute>
        <endHour>
          <!--opt, int, end time (hour), range:[0,23]-->0
        </endHour>
        <endMinute>
          <!--opt, int, end time (minute), range:[0,59]-->0
        </endMinute>
      </emergencyTimeSwitch>
    </emergencyTimeSwitchList>
    <LaneLine>
      <!--opt, object, lane line-->
      <lineName min="0" max="64">
        <!--req, string, line name, range:[0,64], attr:min{req, int},max{req, int}-->test
      </lineName>
      <RegionCoordinatesList num="2">
        <!--opt, array, region coordinate list, subType:object, range:[0,2], attr:num{opt, object}-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->0
          </positionX>
```

```
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </LaneLine>
    </RS485Lane>
  </RS485LaneList>
  <LaneRightBoundaryLine>
    <!--req, object, right border line of lane-->
    <lineName>
      <!--req, enum, line name, subType:string, desc:"laneRightBoundaryLine" (right border line of lane)-->laneRightBoundaryLine
    </lineName>
    <lineType>
      <!--req, enum, line type, subType:string, desc:"unknown" (white line), "white" (white solid line), "singleYellow" (single yellow line), "doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
    </lineType>
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--req, int, x-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--req, int, y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </LaneRightBoundaryLine>
  <safeBelt>
    <!--opt, bool, whether to enable seatbelt detection-->true
  </safeBelt>
  <safeBeltCapNo>
    <!--opt, int, number of captured pictures in which the driver does not wear the seatbelt, range:[1,3]-->0
  </safeBeltCapNo>
  <uphone>
    <!--opt, bool, whether to enable phone call detection-->true
  </uphone>
  <uphoneCapNo>
    <!--opt, int, number of captured pictures in which the driver is making call, range:[1,3]-->0
  </uphoneCapNo>
</PostRS485>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

11.5.1.26 Get the parameters capability of checkpoint intersection violation system vehicle detector

Request URL

GET /ISAPI/ITC/TriggerMode/postEpoliceRS485/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
```

```xml
  <!--ro, opt, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="6">
    <!--ro, req, int, number of supported lanes, attr:min{req, int},max{req, int}-->0
  </relatedLaneCount>
  <maxSwitchTime min="5" max="255">
    <!--ro, req, int, fault-tolerant time, attr:min{req, int},max{req, int}-->0
  </maxSwitchTime>
  <backupMode opt="0,1,2">
    <!--ro, req, enum, Backup Mode, subType:int, attr:opt{req, string}, desc:"0" (none), "1" (trigger of checkpoint virtual coil) Pattern-->0
  </backupMode>
  <relatedDriveWay min="1" max="99">
    <!--ro, req, int, linked lane No., attr:min{req, int},max{req, int}-->0
  </relatedDriveWay>
  <OSDDriveWay min="1" max="99">
    <!--ro, req, int, lane No. to be linked, attr:min{req, int},max{req, int}-->0
  </OSDDriveWay>
  <laneType opt="none,highway,urbanFastWay,other">
    <!--ro, req, enum, lane attribute, subType:string, attr:opt{req, string}, desc:"none"-not configured, "highway"-high way, "urbanFastWay"-urban expressway, "other"-->none
  </laneType>
  <laneUsage opt="carriageWay,emergency">
    <!--ro, req, enum, lane purpose, subType:string, attr:opt{req, string}, desc:"carriageWay" (normal lane), "emergency" (emergency lane)-->carriageWay
  </laneUsage>
  <distance min="0" max="20000">
    <!--ro, req, int, coil distance, attr:min{req, int},max{req, int}-->0
  </distance>
  <delayTime min="0" max="2000">
    <!--ro, req, int, trigger delay time, attr:min{req, int},max{req, int}-->0
  </delayTime>
  <delayDistance min="0" max="255">
    <!--ro, req, int, trigger delay distance, attr:min{req, int},max{req, int}-->0
  </delayDistance>
  <speedCapEnabled>
    <!--ro, req, bool, overspeed capture-->true
  </speedCapEnabled>
  <lowSpeedCapEnable>
    <!--ro, req, bool, low speed capture-->true
  </lowSpeedCapEnable>
  <emergencyCapEnable>
    <!--ro, req, bool, occupying emergency lane capture-->true
  </emergencyCapEnable>
  <signSpeed min="0" max="255">
    <!--ro, req, int, marked speed limit, attr:min{req, int},max{req, int}-->0
  </signSpeed>
  <speedLimit min="0" max="255">
    <!--ro, req, int, speed limit, attr:min{req, int},max{req, int}-->0
  </speedLimit>
  <bigCarSignSpeed min="0" max="255">
    <!--ro, req, int, marked speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSignSpeed>
  <bigCarLowSpeedLimit min="0" max="255">
    <!--ro, req, int, low speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarLowSpeedLimit>
  <lowSpeedLimit min="0" max="60">
    <!--ro, req, int, low speed limit for small-sized vehicle, attr:min{req, int},max{req, int}-->0
  </lowSpeedLimit>
  <bigCarSpeedLimit min="0" max="60">
    <!--ro, req, int, speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSpeedLimit>
  <carHighSpeed min="0" max="255">
    <!--ro, req, int, abnormal overspeed, attr:min{req, int},max{req, int}-->0
  </carHighSpeed>
  <carLowSpeed min="0" max="255">
    <!--ro, req, int, abnormal low speed, attr:min{req, int},max{req, int}-->0
  </carLowSpeed>
  <bigCarHighSpeed min="0" max="255">
    <!--ro, req, int, abnormal overspeed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarHighSpeed>
  <bigCarLowSpeed min="0" max="255">
    <!--ro, req, int, abnormal low speed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarLowSpeed>
  <snapTimes min="0" max="5">
    <!--ro, req, int, continuous capture times, attr:min{req, int},max{req, int}-->0
  </snapTimes>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, types of lane directions, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->0
  </laneDirectionType>
  <intervalType opt="time,distance">
    <!--ro, req, enum, Burst Interval Type, subType:string, attr:opt{req, string}, desc:"time" (by time), "distance" (by distance)-->time
  </intervalType>
  <IOOut min="1" max="7">
    <!--ro, req, int, I/O outputs, attr:min{req, int},max{req, int}-->0
  </IOOut>
  <flashMode opt="together,alternation">
    <!--ro, req, enum, flashing mode, subType:string, attr:opt{req, string}, desc:"together" (simultaneously), "alternation" (alternatively)-->together
  </flashMode>
  <emergencyTimeSwitchList size="4">
    <!--ro, req, array, time period for emergency lanes, subType:object, attr:size{req, int}-->
    <emergencyTimeSwitch>
      <!--ro, req, object, emergency lane time-->
      <timeId min="1" max="4">
```

```
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
      </timeId>
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->0
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, end time (hour), attr:min{req, int},max{req, int}-->0
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, end time (minute), attr:min{req, int},max{req, int}-->0
      </endMinute>
    </emergencyTimeSwitch>
  </emergencyTimeSwitchList>
  <noSupportPlateRegion>
    <!--ro, opt, bool, areas where ANPR is not supported-->true
  </noSupportPlateRegion>
  <LaneLine>
    <!--ro, opt, object, lane line-->
    <lineName min="0" max="64">
      <!--ro, req, string, line name, attr:min{req, int},max{req, int}-->test
    </lineName>
    <RegionCoordinatesList num="2">
      <!--ro, opt, array, region coordinate list, subType:object, range:[0,2], attr:num{opt, object}-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, x-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </LaneLine>
  <LaneRightBoundaryLine>
    <!--ro, req, object, right border line of lane-->
    <lineName>
      <!--ro, req, enum, line name, subType:string, desc:"laneRightBoundaryLine" (right border line of lane)-->laneRightBoundaryLine
    </lineName>
    <lineType>
      <!--ro, req, enum, line type, subType:string, desc:"unknown" (white line), "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
    </lineType>
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--ro, req, int, x-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </LaneRightBoundaryLine>
  <safeBelt>
    <!--ro, opt, bool, whether to enable seatbelt detection-->true
  </safeBelt>
  <safeBeltCapNo min="1" max="3">
    <!--ro, opt, int, number of captured pictures in which the driver does not wear the seatbelt, attr:min{req, int},max{req, int}-->0
  </safeBeltCapNo>
  <uphone>
    <!--ro, opt, bool, whether to enable phone call detection-->true
  </uphone>
  <uphoneCapNo min="1" max="3">
    <!--ro, opt, int, number of captured pictures in which the driver is making call, attr:min{req, int},max{req, int}-->0
  </uphoneCapNo>
</PostRS485>
```

11.5.1.27 Get the recommended configurations of checkpoint intersection violation system vehicle detector

Request URL

GET /ISAPI/ITC/TriggerMode/postEpoliceRS485/recommendation

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<PostRS485 xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, parameters of checkpoint vehicle detector, attr:version{req, string, protocolVersion}-->
</PostRS485>
```

11.5.1.28 Set the parameters of ANPR system

Request URL

PUT /ISAPI/ITC/TriggerMode/postPRS

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<PostPRS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, all detection lines, attr:version{req, string, protocolVersion}-->
  <triggerType>
    <!--req, enum, condition for triggering alarm output, subType:string, desc:"videoDetection", “IO”, “RS485”, “radarDetection”-->videoDetection
  </triggerType>
  <captureMode>
    <!--req, enum, capture mode, subType:string, dep:and,{$.PostPRS.triggerType,eq,videoDetection}, desc:“strobe”, “flashing”-->strobe
  </captureMode>
  <totalLaneNum>
    <!--req, int, number of total lanes, range:[0,10]-->0
  </totalLaneNum>
  <LaneParamList size="5">
    <!--opt, array, lane parameter list, subType:object, attr:size{req, int}-->
    <LaneParam>
      <!--opt, object, lane parameters-->
      <id>
        <!--req, int, index-->0
      </id>
      <laneNo>
        <!--req, int, lane No., range:[0,10]-->0
      </laneNo>
      <relatedDriveWay>
        <!--req, int, linked lane No., range:[1,99]-->0
      </relatedDriveWay>
      <enterExitDirectionType>
        <!--ro, req, enum, direction type of entrance and exit, subType:int, desc:this node is valid when enterExitDetectEnable is "true"; 0 (unknown), 1
(enter up to down), 2 (exit up to down)-->0
      </enterExitDirectionType>
      <captureMode>
        <!--req, enum, capture mode, subType:string, dep:and,{$.PostPRS.triggerType,eq,videoDetection}, desc:“strobe”, “flashing”-->strobe
      </captureMode>
      <RS485No>
        <!--opt, int, 485 No., range:[0,10], dep:and,{$.PostPRS.triggerType,eq,RS485}-->0
      </RS485No>
      <IONo>
        <!--opt, int, IO port No., range:[0,10], dep:and,{$.PostPRS.triggerType,eq,IO}-->0
      </IONo>
      <IODefStatus>
        <!--opt, enum, IO default status, subType:string, dep:and,{$.PostPRS.triggerType,eq,IO}, desc:"risingEdge", "fallingEdge"-->risingEdge
      </IODefStatus>
      <ViolationDetectLine>
        <!--req, object, draw violation detection line on the single lane-->
        <lineName>
          <!--req, enum, line name, subType:string, desc:"laneLine"-->laneLine
        </lineName>
        <NormalizedScreenSize>
          <!--req, object, normalized coordinates-->
          <normalizedScreenWidth>
            <!--req, int, normalized width, range:[0,1000]-->1
          </normalizedScreenWidth>
          <normalizedScreenHeight>
            <!--req, int, normalized height, range:[0,1000]-->1
          </normalizedScreenHeight>
        </NormalizedScreenSize>
        <RegionCoordinatesList>
          <!--opt, array, area coordinates list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ViolationDetectLine>
      <TriggerLine>
        <!--req, object, Triggering Line-->
```

```
        <!--req, object, triggering line-->
        <lineName>
          <!--req, enum, line name, subType:string, desc:"triggerLine" (triggering line)-->triggerLine
        </lineName>
        <NormalizedScreenSize>
          <!--req, object, normalized coordinates-->
          <normalizedScreenWidth>
            <!--req, int, normalized width, range:[0,1000]-->1
          </normalizedScreenWidth>
          <normalizedScreenHeight>
            <!--req, int, normalized height, range:[0,1000]-->1
          </normalizedScreenHeight>
        </NormalizedScreenSize>
        <RegionCoordinatesList>
          <!--opt, array, area coordinates list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </TriggerLine>
      <forwardRadarIO opt="IO1,IO2,IO3,IO4">
        <!--opt, enum, linked I/O output No. of forward radar, subType:string, dep:and,{$.PostPRS.triggerType,eq,HRT}, attr:opt{req, string}, desc:"IO1",
"IO2", "IO3", "IO4"-->IO1
      </forwardRadarIO>
      <backwardRadarIO opt="IO1,IO2,IO3,IO4">
        <!--opt, enum, linked I/O output No. of backward radar, subType:string, dep:and,{$.PostPRS.triggerType,eq,HRT}, attr:opt{req, string}, desc:"IO1",
"IO2", "IO3", "IO4"-->IO1
      </backwardRadarIO>
    </LaneParam>
  </LaneParamList>
  <VirtualLane>
    <!--req, object, virtual line-->
    <lineName>
      <!--req, enum, line name, subType:string, desc:"laneRightBoundaryLine" (right border line of lane)-->laneRightBoundaryLine
    </lineName>
    <NormalizedScreenSize>
      <!--req, object, normalized coordinates-->
      <normalizedScreenWidth>
        <!--req, int, normalized width, range:[0,1000]-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--req, int, normalized height, range:[0,1000]-->1
      </normalizedScreenHeight>
    </NormalizedScreenSize>
    <RegionCoordinates>
      <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </VirtualLane>
  <noPlatCarCap>
    <!--opt, bool, whether to enable no plate vehicle capture-->true
  </noPlatCarCap>
  <sceneMode>
    <!--opt, enum, scene mode, subType:string, desc:"commonEntrance" (common entrance and exit scene), "tollGate" (toll station scene (vehicles will stay
for longer time)), "ParkingEntrance" (entrance and exit of underground parking lot scene (there will be dark day and night))-->commonEntrance
  </sceneMode>
  <capPicMode>
    <!--opt, enum, capture mode, subType:string, desc:captured picture type: "scene"-scene picture,"sceneCloseup"-scene picture and close-up picture-->scene
  </capPicMode>
  <radarDetection>
    <!--opt, object, radar detection mode-->
    <detectDistance1>
      <!--req, int, radar detection distance, range:[0,2000], unit:cm-->0
    </detectDistance1>
    <detectDistance2>
      <!--req, int, radar detection distance, range:[0,2000], unit:cm-->0
    </detectDistance2>
    <alarmDistance>
      <!--req, int, radar alarm distance, range:[0,600], unit:cm-->0
    </alarmDistance>
  </radarDetection>
  <detectPosition>
    <!--ro, opt, enum, detection position, subType:string, desc:"bodyStock" (vehicle body), "headStock" (vehicle head)-->headStock
  </detectPosition>
  <recordEnable>
    <!--opt, bool, whether to enable recording-->true
  </recordEnable>
  <recordType>
    <!--opt, enum, recording type, subType:string, desc:"preRecord", "delayRecord"-->preRecord
  </recordType>
  <beginTime>
    <!--opt, enum, start time of pre-record when capturing pictures, subType:int, desc:0 (default: the 2nd picture), 1 (the 1st picture), 2 (the 2nd
picture), 3 (the 3rd picture)-->0
```

```
  </beginTime>
  <preRecordTime>
    <!--opt, int, pre-record, range:[0,100], unit:s-->0
  </preRecordTime>
  <recordDelayTime>
    <!--opt, int, post-record, range:[0,100], unit:s-->0
  </recordDelayTime>
  <recordTimeOut>
    <!--opt, int, timeout threshold, range:[0,100], unit:s-->0
  </recordTimeOut>
  <enterExitDetectEnable>
    <!--ro, opt, bool, whether to enable entrance and exit detection-->true
  </enterExitDetectEnable>
  <twoWheelCaptureEnable>
    <!--ro, opt, bool, whether to enable capturing two wheelers-->true
  </twoWheelCaptureEnable>
  <humanNonMotorCaptureEnable>
    <!--ro, opt, bool, whether to enable capturing vehicles, non-motor vehicles, and pedestrian-->true
  </humanNonMotorCaptureEnable>
</PostPRS>
```

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.29 Get the parameters of ANPR system

Request URL

GET /ISAPI/ITC/TriggerMode/postPRS

Query Parameter

None

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<PostPRS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, all detection lines, attr:version{req, string, protocolVersion}-->
  <triggerType>
    <!--ro, req, enum, triggering mode, subType:string, desc:"videoDetection", "IO", "RS485", "radarDetection", "HRT" (double radars mixed traffic)--
>videoDetection
  </triggerType>
  <captureMode>
    <!--ro, req, enum, capture mode, subType:string, dep:and,{$.PostPRS.triggerType,eq,videoDetection}, desc:“strobe”, “flashing”-->strobe
  </captureMode>
  <totalLaneNum>
    <!--ro, req, int, number of total lanes, range:[0,10]-->0
  </totalLaneNum>
  <LaneParamList size="5">
    <!--ro, opt, array, lane parameter list, subType:object, attr:size{req, int}-->
    <LaneParam>
      <!--ro, opt, object, lane parameters-->
      <id>
        <!--ro, req, int, ID-->0
      </id>
      <laneNo>
        <!--ro, req, int, lane No., range:[0,10]-->0
```

```
      </laneNo>
      <relatedDriveWay>
        <!--ro, req, int, linked lane No., range:[1,99]-->0
      </relatedDriveWay>
      <enterExitDirectionType>
        <!--ro, req, enum, direction type of entrance and exit, subType:int, desc:this node is valid when enterExitDetectEnable is "true"; 0 (unknown), 1
(enter up to down), 2 (exit up to down)-->0
      </enterExitDirectionType>
      <captureMode>
        <!--ro, req, enum, capture mode, subType:string, dep:and,{$.PostPRS.triggerType,eq,videoDetection}, desc:"strobe", "flashing"-->strobe
      </captureMode>
      <RS485No>
        <!--ro, opt, int, 485 No., range:[0,10], dep:and,{$.PostPRS.triggerType,eq,RS485}-->0
      </RS485No>
      <IONo>
        <!--ro, opt, int, IO port No., range:[0,10], dep:and,{$.PostPRS.triggerType,eq,IO}-->0
      </IONo>
      <IODefStatus>
        <!--ro, opt, enum, IO default status, subType:string, dep:and,{$.PostPRS.triggerType,eq,IO}, desc:"risingEdge", "fallingEdge"-->risingEdge
      </IODefStatus>
      <ViolationDetectLine>
        <!--ro, req, object, draw violation detection line on the single lane-->
        <lineName>
          <!--ro, req, enum, line name, subType:string, desc:"laneLine"-->laneLine
        </lineName>
        <NormalizedScreenSize>
          <!--ro, req, object, normalized coordinates-->
          <normalizedScreenWidth>
            <!--ro, req, int, normalized width, range:[0,1000]-->1
          </normalizedScreenWidth>
          <normalizedScreenHeight>
            <!--ro, req, int, normalized height, range:[0,1000]-->1
          </normalizedScreenHeight>
        </NormalizedScreenSize>
        <RegionCoordinatesList>
          <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ViolationDetectLine>
      <TriggerLine>
        <!--ro, req, object, Triggering Line-->
        <lineName>
          <!--ro, req, enum, line name, subType:string, desc:"triggerLine" (triggering line)-->triggerLine
        </lineName>
        <NormalizedScreenSize>
          <!--ro, req, object, normalized coordinates-->
          <normalizedScreenWidth>
            <!--ro, req, int, normalized width, range:[0,1000]-->1
          </normalizedScreenWidth>
          <normalizedScreenHeight>
            <!--ro, req, int, normalized height, range:[0,1000]-->1
          </normalizedScreenHeight>
        </NormalizedScreenSize>
        <RegionCoordinatesList>
          <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </TriggerLine>
      <forwardRadarIO opt="IO1,IO2,IO3,IO4">
        <!--ro, opt, enum, linked I/O output No. of forward radar, subType:string, dep:and,{$.PostPRS.triggerType,eq,HRT}, attr:opt{req, string},
desc:"IO1", "IO2", "IO3", "IO4"-->IO1
      </forwardRadarIO>
      <backwardRadarIO opt="IO1,IO2,IO3,IO4">
        <!--ro, opt, enum, linked I/O output No. of backward radar, subType:string, dep:and,{$.PostPRS.triggerType,eq,HRT}, attr:opt{req, string},
desc:"IO1", "IO2", "IO3", "IO4"-->IO1
      </backwardRadarIO>
    </LaneParam>
  </LaneParamList>
  <VirtualLane>
    <!--ro, req, object, virtual line-->
    <lineName>
      <!--ro, req, enum, line name, subType:string, desc:"laneRightBoundaryLine" (right border line of lane)-->laneRightBoundaryLine
    </lineName>
    <NormalizedScreenSize>
      <!--ro, req, object, normalized coordinates-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width, range:[0,1000]-->1
```

```
        </normalizedScreenWidth>
        <normalizedScreenHeight>
          <!--ro, req, int, normalized height, range:[0,1000]-->1
        </normalizedScreenHeight>
      </NormalizedScreenSize>
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </VirtualLane>
    <noPlatCarCap>
      <!--ro, opt, bool, no plate vehicle capture-->true
    </noPlatCarCap>
    <sceneMode>
      <!--ro, opt, enum, scene mode, subType:string, desc:"commonEntrance" (common entrance and exit scene), "tollGate" (toll station scene),
"ParkingEntrance" (entrance and exit of parking lot scene)-->commonEntrance
    </sceneMode>
    <capPicMode>
      <!--ro, opt, enum, capture mode, subType:string, desc:captured picture type: "scene"-scene picture,"sceneCloseup"-scene picture and close-up picture--
>scene
    </capPicMode>
    <radarDetection>
      <!--ro, opt, object, radar detection mode-->
      <detectDistance1>
        <!--ro, req, int, radar detection distance 1, range:[0,2000], unit:cm-->0
      </detectDistance1>
      <detectDistance2>
        <!--ro, req, int, radar detection distance 2, range:[0,2000], unit:cm-->0
      </detectDistance2>
      <alarmDistance>
        <!--ro, req, int, radar alarm distance, range:[0,600], unit:cm-->0
      </alarmDistance>
    </radarDetection>
    <detectPosition>
      <!--ro, opt, enum, detection position, subType:string, desc:"bodyStock" (vehicle body), "headStock" (vehicle head)-->headStock
    </detectPosition>
    <recordEnable>
      <!--ro, opt, bool, whether to enable recording-->true
    </recordEnable>
    <recordType>
      <!--ro, opt, enum, recording type, subType:string, desc:"preRecord", "delayRecord"-->preRecord
    </recordType>
    <beginTime>
      <!--ro, opt, enum, start time of pre-record when capturing pictures, subType:int, desc:0 (default: the 2nd picture), 1 (the 1st picture), 2 (the 2nd
picture), 3 (the 3rd picture)-->0
    </beginTime>
    <preRecordTime>
      <!--ro, opt, int, pre-record, range:[0,100], unit:s-->0
    </preRecordTime>
    <recordDelayTime>
      <!--ro, opt, int, post-record, range:[0,100], unit:s-->0
    </recordDelayTime>
    <recordTimeOut>
      <!--ro, opt, int, timeout threshold, range:[0,100], unit:s-->0
    </recordTimeOut>
    <enterExitDetectEnable>
      <!--ro, opt, bool, whether to enable entrance and exit detection-->true
    </enterExitDetectEnable>
    <twoWheelCaptureEnable>
      <!--ro, opt, bool, whether to enable capturing two wheelers-->true
    </twoWheelCaptureEnable>
    <humanNonMotorCaptureEnable>
      <!--ro, opt, bool, whether to enable capturing vehicles, non-motor vehicles, and pedestrian-->true
    </humanNonMotorCaptureEnable>
  </PostPRS>
```


11.5.1.30 Get the trigger mode configuration capability of ANPR system

Request URL

GET /ISAPI/ITC/TriggerMode/postPRS/capabilities

Query Parameter

None

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>
```

```xml
<PostPRS xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, road parameters, attr:version{req, string, protocolVersion}-->
  <triggerType opt="videoDetection,IO,RS485,radarDetection,HRT">
    <!--ro, req, enum, condition for triggering alarm output, subType:string, attr:opt{req, string}, desc:"videoDetection", "IO", "RS485", "radarDetection", "HRT" (double radars mixed traffic)-->videoDetection
  </triggerType>
  <captureMode opt="strobe,flashing">
    <!--ro, req, enum, capture mode, subType:string, attr:opt{req, string}, desc:"strobe", "flashing", it is valid when triggerType is "videoDetection"-->strobe
  </captureMode>
  <totalLaneNum min="0" max="10">
    <!--ro, req, int, number of total lanes, range:[0,10], attr:min{req, int},max{req, int}-->0
  </totalLaneNum>
  <LaneParamList size="5">
    <!--ro, opt, object, list of lane parameters, attr:size{req, int}-->
    <LaneParam>
      <!--ro, opt, object, lane parameters-->
      <id>
        <!--ro, req, int, ID, range:[1,5]-->0
      </id>
      <laneNo min="0" max="10">
        <!--ro, req, int, lane No., range:[0,10], attr:min{req, int},max{req, int}-->0
      </laneNo>
      <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
        <!--ro, req, enum, lane direction type, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->0
      </laneDirectionType>
      <enterExitDirectionType opt="0,1,2">
        <!--ro, req, enum, direction type of entrance and exit, subType:int, attr:opt{req, string}, desc:this node is valid when enterExitDetectEnable is "true"; 0 (unknown), 1 (enter up to down), 2 (exit up to down)-->0
      </enterExitDirectionType>
      <relatedDriveWay min="1" max="99">
        <!--ro, req, int, linked lane No., range:[1,99], attr:min{req, int},max{req, int}-->0
      </relatedDriveWay>
      <captureMode opt="strobe,flashing">
        <!--ro, req, enum, Capture Mode, subType:string, attr:opt{req, string}, desc:"strobe,flashing",it is valid when triggerType is "videoDetection"-->strobe
      </captureMode>
      <RS485No min="0" max="10">
        <!--ro, opt, int, 485 No., range:[0,10], attr:min{req, int},max{req, int}, desc:it is valid when triggerType is "RS485"-->0
      </RS485No>
      <IONo min="0" max="10">
        <!--ro, opt, int, IO port No., range:[0,10], attr:min{req, int},max{req, int}, desc:it is valid when triggerType is "IO”-->0
      </IONo>
      <IODefStatus opt="risingEdge,fallingEdge">
        <!--ro, opt, enum, IO default status, subType:string, attr:opt{req, string}, desc:it is valid when triggerType is "IO"; "risingEdge", "fallingEdge"-->risingEdge
      </IODefStatus>
      <ViolationDetectLine>
        <!--ro, req, object, draw violation detection line on the single lane-->
        <lineName min="0" max="64">
          <!--ro, req, enum, "laneLine", subType:string, attr:min{req, int},max{req, int}, desc:"laneLine"-->laneLine
        </lineName>
        <NormalizedScreenSize>
          <!--ro, req, object, normalized coordinates-->
          <normalizedScreenWidth>
            <!--ro, req, int, normalized width, range:[0,1000]-->1
          </normalizedScreenWidth>
          <normalizedScreenHeight>
            <!--ro, req, int, normalized height, range:[0,1000]-->1
          </normalizedScreenHeight>
        </NormalizedScreenSize>
        <RegionCoordinatesList size="5">
          <!--ro, opt, array, area coordinates list, subType:object, range:[0,2], attr:size{req, int}-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ViolationDetectLine>
      <TriggerLine>
        <!--ro, req, object, Triggering Line-->
        <lineName min="0" max="64">
          <!--ro, req, enum, "laneLine", subType:string, attr:min{req, int},max{req, int}, desc:"laneLine"-->triggerLine
        </lineName>
        <NormalizedScreenSize>
          <!--ro, req, object, normalized coordinates-->
          <normalizedScreenWidth>
            <!--ro, req, int, normalized width, range:[0,1000]-->1
          </normalizedScreenWidth>
          <normalizedScreenHeight>
            <!--ro, req, int, normalized height, range:[0,1000]-->1
          </normalizedScreenHeight>
        </NormalizedScreenSize>
        <RegionCoordinatesList size="5">
          <!--ro, opt, array, area coordinates list, subType:object, range:[0,2], attr:size{req, int}-->
          <RegionCoordinates>
            <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
```

```xml
            <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </TriggerLine>
      <forwardRadarIO opt="IO1,IO2,IO3,IO4">
        <!--ro, opt, enum, linked I/O output No. of forward radar, subType:string, dep:and,{$.PostPRS.triggerType,eq,HRT}, attr:opt{req, string},
desc:"IO1", "IO2", "IO3", "IO4"-->IO1
      </forwardRadarIO>
      <backwardRadarIO opt="IO1,IO2,IO3,IO4">
        <!--ro, opt, enum, linked I/O output No. of backward radar, subType:string, dep:and,{$.PostPRS.triggerType,eq,HRT}, attr:opt{req, string},
desc:"IO1", "IO2", "IO3", "IO4"-->IO1
      </backwardRadarIO>
    </LaneParam>
  </LaneParamList>
  <VirtualLane>
    <!--ro, req, object, virtual line-->
    <lineName min="0" max="64">
      <!--ro, req, enum, "laneLine", subType:string, attr:min{req, int},max{req, int}, desc:"laneRightBoundaryLine" (right border line of lane)--
>laneRightBoundaryLine
    </lineName>
    <NormalizedScreenSize>
      <!--ro, req, object, normalized coordinates-->
      <normalizedScreenWidth>
        <!--ro, req, int, normalized width, range:[0,1000]-->1
      </normalizedScreenWidth>
      <normalizedScreenHeight>
        <!--ro, req, int, normalized height, range:[0,1000]-->1
      </normalizedScreenHeight>
    </NormalizedScreenSize>
    <RegionCoordinates>
      <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </VirtualLane>
  <noPlatCarCap>
    <!--ro, opt, bool, No Plate Vehicle Capture-->true
  </noPlatCarCap>
  <sceneMode opt="commonEntrance,tollGate,ParkingEntrance">
    <!--ro, opt, enum, scene mode, subType:string, attr:opt{req, string}, desc:"commonEntrance" (common entrance and exit), "tollGate" (toll station
(vehicles will stay for longer time)), "ParkingEntrance"-entrance and exit of underground parking lot-->commonEntrance
  </sceneMode>
  <capPicMode opt="scene,sceneCloseup">
    <!--ro, opt, enum, Capture Mode, subType:string, attr:opt{req, string}, desc:"scene" (scene picture), "sceneCloseup" (close-up picture)-->scene
  </capPicMode>
  <radarDetection>
    <!--ro, opt, object, radar detection mode-->
    <detectDistance1 min="0" max="2000">
      <!--ro, req, int, radar detection distance, range:[0,2000], unit:cm, attr:min{req, int},max{req, int}-->0
    </detectDistance1>
    <detectDistance2 min="0" max="2000">
      <!--ro, req, int, radar detection distance, range:[0,2000], unit:cm, attr:min{req, int},max{req, int}-->0
    </detectDistance2>
    <alarmDistance min="0" max="600" def="300">
      <!--ro, req, int, radar alarm distance, range:[0,600], unit:cm, attr:min{req, int},max{req, int},def{req, int}-->0
    </alarmDistance>
  </radarDetection>
  <detectPosition opt="tailStock,headStock">
    <!--ro, opt, enum, detection position, subType:string, attr:opt{req, string}, desc:"bodyStock" (vehicle body), "headStock" (vehicle head)-->headStock
  </detectPosition>
  <recordEnable opt="false,true">
    <!--ro, opt, bool, whether to enable recording, attr:opt{req, string}-->true
  </recordEnable>
  <recordType opt="delayRecord,preRecord">
    <!--ro, opt, enum, recording type, subType:string, attr:opt{req, string}, desc:"preRecord", "delayRecord"-->preRecord
  </recordType>
  <preRecordTime min="0" max="100">
    <!--ro, opt, int, pre-record, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
  </preRecordTime>
  <recordDelayTime min="0" max="100">
    <!--ro, opt, int, post-record, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
  </recordDelayTime>
  <recordTimeOut min="0" max="100">
    <!--ro, opt, int, timeout threshold, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
  </recordTimeOut>
  <enterExitDetectEnable opt="false,true">
    <!--ro, opt, bool, whether to enable entrance and exit detection, attr:opt{req, string}-->true
  </enterExitDetectEnable>
  <twoWheelCaptureEnable opt="false,true">
    <!--ro, opt, bool, whether to enable capturing two wheelers, attr:opt{req, string}-->true
  </twoWheelCaptureEnable>
  <humanNonMotorCaptureEnable opt="false,true">
    <!--ro, opt, bool, whether to enable capturing vehicles, non-motor vehicles, and pedestrian, attr:opt{req, string}-->true
  </humanNonMotorCaptureEnable>
```

```
</PostPRS>
```

11.5.1.31 Set radar parameters

Request URL

PUT /ISAPI/ITC/TriggerMode/postRadar/param

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<PostRadarParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, radar parameters, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int, serial port number, range:[1,5]-->0
  </id>
  <softWareVersion min="0" max="32">
    <!--opt, string, software version No., range:[0,32], attr:min{req, int},max{req, int}-->test
  </softWareVersion>
  <workMode opt="continue,headTrig,tailTrig,moveTrig,doubleTrig,other">
    <!--opt, enum, radar working mode, subType:string, attr:opt{req, string}, desc:radar working mode-->continue
  </workMode>
  <speedType>
    <!--opt, enum, speed type, subType:string, desc:speed type-->single
  </speedType>
  <directionFilter>
    <!--opt, enum, "nonefilter"-unfiltered,"outputCome"-output the direction that the target comes from,"outputGo"-output the direction that the target goes
to, subType:string, desc:"nonefilter" (unfiltered), "outputCome" (output the direction that the target comes from), "outputGo" (output the direction that
the target goes to)-->nonefilter
  </directionFilter>
  <angleCorrect>
    <!--opt, int, corrected angle, range:[0,70]-->0
  </angleCorrect>
  <sensitivity>
    <!--opt, int, sensitivity, range:[11,240]-->0
  </sensitivity>
  <speedLowLimit>
    <!--opt, int, speed lower-limit, range:[1,150], unit:km/h-->1
  </speedLowLimit>
  <trigDistance>
    <!--opt, int, trigger distance, range:[0,40], unit:m-->0
  </trigDistance>
</PostRadarParam>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.32 Get statuses of switching trigger mode

Request URL

GET /ISAPI/ITC/TriggerMode/switchStatus?format=json&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No. (this optional parameter is supported when isSupportChannelSwitchStatus is "true" returned from GET /ISAPI/ITC/TriggerMode/capabilities)

Request Message

None

Response Message

```
{
    "switchStatus":  "completed"
    /*ro, req, enum, status, subType:string, desc:"completed", "executing"*/
}
```

11.5.1.33 Get the trigger mode configuration capability of traffic incidents

Request URL

GET /ISAPI/ITC/TriggerMode/trafficIncident/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message

```
{
    "TrafficIncidentCap": {
    /*ro, opt, object, configuration capability of traffic incident detection*/
        "enabled": [true, false],
        /*ro, req, array, whether to enable traffic incident detection, subType:bool*/
        "sceneMode":  "highway,urban,tunnel",
        /*ro, opt, string, scene mode, desc:"highway", "urban" (urban road), "tunnel"*/
        "isSupportChannelTrafficIncident":  false
        /*ro, opt, bool, whether it support configuring parameters of traffic incident detection, desc:if this node is true, it indicates that channelID can
be set in /ISAPI/ITC/TriggerMode/trafficIncident?format=json&channelID=<channelID> and /ISAPI/ITC/TriggerMode/trafficIncident/defaultParam?
format=json&channelID=<channelID>*/
    }
}
```

11.5.1.34 Get the trigger mode parameters of traffic incidents

Request URL

GET /ISAPI/ITC/TriggerMode/trafficIncident?format=json&channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No. (this optional parameter is supported when isSupportChannelTrafficIncident is "true" returned from GET /ISAPI/ITC/TriggerMode/trafficIncident/capabilities?format=json)

Request Message

None

Response Message

```
{
    "TrafficIncident": {
    /*ro, opt, object, parameters of traffic incident detection*/
        "enabled":  true,
        /*ro, req, bool, whether to enable the function*/
        "sceneMode":  "highway"
        /*ro, opt, enum, scene mode, subType:string, desc:"highway", "urban" (urban road), "tunnel"*/
    }
}
```

11.5.1.35 Get the capability of video intersection violation parameters

Request URL

GET /ISAPI/ITC/TriggerMode/videoEpolice/capabilities?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID (this optional parameter is supported when isSupportChannelVideoEpolice is "true" returned from /ISAPI/ITC/TriggerMode/capabilities).

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoEpolice xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, video intersection violation parameters, attr:version{req, string, protocolVersion}-->
  <intervalType opt="time,distance">
    <!--ro, req, enum, burst interval type, subType:string, attr:opt{req, string}, desc:"time" (by time), "distance" (by distance)-->time
  </intervalType>
  <interval min="1" max="1000">
    <!--ro, req, int, time interval, range:[1,1000], attr:min{req, int},max{req, int}-->0
  </interval>
  <post>
    <!--ro, req, bool, whether to enable checkpoint detection-->true
  </post>
  <postCapNo opt="1,3">
    <!--ro, req, int, captured pictures of checkpoint, range:[1,3], attr:opt{req, string}-->1
  </postCapNo>
  <checkpointQuickTriggerEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </checkpointQuickTriggerEnabled>
  <driveLine>
    <!--ro, req, bool, whether to enable driving on lane line detection-->true
  </driveLine>
  <drivelineCapNo opt="2,3">
    <!--ro, req, int, capture times, range:[2,3], attr:opt{req, string}-->0
  </drivelineCapNo>
  <driveLineSensitivity min="0" max="100">
    <!--ro, req, int, over line sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
  </driveLineSensitivity>
  <reverse>
    <!--ro, req, bool, whether to enable wrong-way driving detection-->true
  </reverse>
  <reverseCapNo opt="2,3">
    <!--ro, req, int, number of captured pictures, range:[2,3], attr:opt{req, string}-->0
  </reverseCapNo>
  <redLight>
    <!--ro, req, bool, whether to enable red light running detection-->true
  </redLight>
  <nonMotorRedLightEnabled opt="true,false">
    <!--ro, opt, bool, attr:opt{req, string}-->true
  </nonMotorRedLightEnabled>
  <direction>
    <!--ro, req, bool, whether to enable detection of driving in wrong lane at intersection-->true
  </direction>
  <intersectionCongest>
    <!--ro, req, bool, whether to enable intersection congestion detection-->true
  </intersectionCongest>
  <nonDriveWay>
    <!--ro, opt, bool, whether to enable detection of motor vehicle on non-motor vehicle lane-->true
  </nonDriveWay>
  <nonDriveWayCapNo opt="2,3">
    <!--ro, opt, int, capture times, range:[2,3], attr:opt{req, string}-->0
  </nonDriveWayCapNo>
  <changeLane>
    <!--ro, req, bool, whether to enable illegal lane change detection-->true
```

```xml
    </changeLane>
    <changeLaneCapNo opt="2,3">
      <!--ro, req, int, number of captured pictures, range:[2,3], attr:opt{req, string}-->0
    </changeLaneCapNo>
    <changeLaneFirstImageReserveEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </changeLaneFirstImageReserveEnabled>
    <banSign>
      <!--ro, opt, bool, whether to enable ban breaking detection-->true
    </banSign>
    <banSignCapNo opt="2,3">
      <!--ro, opt, int, number of captured pictures, range:[2,3], attr:opt{req, string}-->0
    </banSignCapNo>
    <notCatchingReverseVehiclesEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </notCatchingReverseVehiclesEnabled>
    <intersectionStop>
      <!--ro, opt, object, stop vehicle on intersection-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
      </enable>
      <lastTime min="10" max="180">
        <!--ro, req, int, congestion duration, unit: second, range:[0,180], attr:min{req, int},max{req, int}-->0
      </lastTime>
      <sensitivity min="1" max="100">
        <!--ro, opt, int, triggering sensitivity, which is between 0 to 100, range:[1,100], attr:min{req, int},max{req, int}-->0
      </sensitivity>
      <bigCarRatio min="1" max="100">
        <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->0
      </bigCarRatio>
      <smallCarRatio min="1" max="100">
        <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->0
      </smallCarRatio>
      <capNo opt="3">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->3
      </capNo>
    </intersectionStop>
    <greenLightStop>
      <!--ro, opt, bool, whether to enable detection of parking at green light-->true
    </greenLightStop>
    <overSpeed>
      <!--ro, req, bool, whether to enable overspeed detection-->true
    </overSpeed>
    <overSpeedCapNo opt="2,3">
      <!--ro, req, int, number of captured pictures in which the overspeed is detected, range:[2,3], attr:opt{req, string}-->0
    </overSpeedCapNo>
    <detectMode>
      <!--ro, opt, enum, detection mode, subType:string, desc:"video", "coilLoop" (video and coil)-->video
    </detectMode>
    <illegalTurn>
      <!--ro, req, bool, whether to enable illegal U-turning detection-->true
    </illegalTurn>
    <reverseLaneCheckpointFilteringEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </reverseLaneCheckpointFilteringEnabled>
    <TrafficLight opt="IO,RS485,videoDetect,network,autoDetectTrafficLight">
      <!--ro, req, object, traffic lights, attr:opt{req, string}-->
      <source>
        <!--ro, req, enum, source, subType:string, desc:"IO", "RS485", "videoDetect" (video detection), "network"-->IO
      </source>
      <IOLightList>
        <!--ro, opt, array, list of I/O light, subType:object-->
        <IOLight>
          <!--ro, opt, object, I/O light-->
          <lightId>
            <!--ro, req, int, light ID, range:[1,6]-->0
          </lightId>
          <lightType>
            <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->left
          </lightType>
          <relatedIO>
            <!--ro, req, enum, NON_IO, subType:string, desc:“NON_IO” (none), “IO1”, “IO2”, “IO3”, “IO4”, “IO5”, “IO6”, “IO7”, “IO8”-->IO8
          </relatedIO>
          <redLightStatus>
            <!--ro, req, enum, red light level status, subType:string, desc:"low,high"-->low
          </redLightStatus>
        </IOLight>
      </IOLightList>
      <RS485LightList>
        <!--ro, opt, array, list of 485 indicator lights, subType:object-->
        <RS485Light>
          <!--ro, opt, object, 485 indicator light-->
          <lightId>
            <!--ro, req, int, "1-3", range:[1,6]-->0
          </lightId>
          <lightType>
            <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:guiding direction of traffic light: "left,straight,right"-->left
          </lightType>
          <relatedLightChan>
            <!--ro, req, int, linked channel of red light, range:[0,16]-->0
          </relatedLightChan>
          <yellowLightChan>
            <!--ro, req, int, linked channel of yellow light, range:[0,16]-->0
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      </yellowLightChan>
    </RS485Light>
  </RS485LightList>
  <NetworkLightList>
    <!--ro, opt, array, list of network indicator lights, subType:object-->
    <NetworkLight>
      <!--ro, opt, object, network indicator light-->
      <lightId>
        <!--ro, req, int, light ID, range:[1,6]-->0
      </lightId>
      <lightType>
        <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:guiding direction of traffic light: "left,straight,right"-->left
      </lightType>
      <relatedLightChan>
        <!--ro, req, int, linked channel of red light, range:[0,16]-->0
      </relatedLightChan>
      <yellowLightChan>
        <!--ro, req, int, linked channel of yellow light, range:[0,16]-->0
      </yellowLightChan>
    </NetworkLight>
  </NetworkLightList>
  <videoDetectLightList>
    <!--ro, opt, array, list of video detection traffic lights, subType:object-->
    <videoDetectLight>
      <!--ro, opt, object, list-->
      <videoDetectLightId>
        <!--ro, req, int, area ID, range:[1,12]-->0
      </videoDetectLightId>
      <lightNum>
        <!--ro, req, int, number of lights, range:[0,3]-->0
      </lightNum>
      <straightLight>
        <!--ro, req, bool, whether to enable the straight signal-->true
      </straightLight>
      <leftLight>
        <!--ro, req, bool, whether to enable the left turn signal-->true
      </leftLight>
      <rightLight>
        <!--ro, req, bool, whether to enable the right turn signal-->true
      </rightLight>
      <redLight>
        <!--ro, req, bool, whether to enable the red light-->true
      </redLight>
      <greenLight>
        <!--ro, req, bool, whether to enable the green light-->true
      </greenLight>
      <yellowLight>
        <!--ro, req, bool, whether to enable the yellow light-->true
      </yellowLight>
      <yellowLightTime>
        <!--ro, req, int, yellow light duration, range:[0,10], unit:s-->0
      </yellowLightTime>
      <rectRegion>
        <!--ro, req, object, detection area-->
        <positionLeft>
          <!--ro, req, int, X-coordinate of upper left corner, range:[0,65535]-->0
        </positionLeft>
        <positionTop>
          <!--ro, req, int, Y-coordinate of upper left corner, range:[0,65535]-->0
        </positionTop>
        <positionRight>
          <!--ro, req, int, X-coordinate of lower right corner, range:[0,65535]-->0
        </positionRight>
        <positionBottom>
          <!--ro, req, int, Y-coordinate of lower right corner, range:[0,65535]-->0
        </positionBottom>
      </rectRegion>
    </videoDetectLight>
  </videoDetectLightList>
  <autoDetectTrafficLight>
    <!--ro, opt, object-->
    <detectRegionList size="8">
      <!--ro, opt, object, attr:size{req, int}-->
      <detectRegion>
        <!--ro, opt, object-->
        <height>
          <!--ro, req, float, range:[0,1]-->0.001
        </height>
        <width>
          <!--ro, req, float, range:[0,1]-->0.001
        </width>
        <x>
          <!--ro, req, float, range:[0,1]-->0.001
        </x>
        <y>
          <!--ro, req, float, range:[0,1]-->0.001
        </y>
      </detectRegion>
    </detectRegionList>
    <trafficLightPanelList size="8">
      <!--ro, opt, object, attr:size{req, int}-->
      <trafficLightPanel>
        <!--ro, opt, object-->
        <ruleID min="1" max="8">
```

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          <!--ro, req, int, range:[1,8], attr:min{req, int},max{req, int}-->1
        </ruleID>
        <lampPanelDirection opt="level,vertical">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->level
        </lampPanelDirection>
        <lampPanelType opt="combination,integrated">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->combination
        </lampPanelType>
        <lampSequence opt="redYellowGreen,greenYellowRed">
          <!--ro, opt, enum, subType:string, attr:opt{req, string}-->redYellowGreen
        </lampSequence>
        <lampPanelRegion>
          <!--ro, opt, object-->
          <height>
            <!--ro, req, float, range:[0,1]-->0.001
          </height>
          <width>
            <!--ro, req, float, range:[0,1]-->0.001
          </width>
          <x>
            <!--ro, req, float, range:[0,1]-->0.001
          </x>
          <y>
            <!--ro, req, float, range:[0,1]-->0.001
          </y>
        </lampPanelRegion>
        <trafficLightList size="8">
          <!--ro, opt, object, attr:size{req, int}-->
          <trafficLight>
            <!--ro, opt, object-->
            <trafficLightType
opt="roundLight,leftTurnArrowLight,straightArrowLight,rightTurnArrowLight,turnSignal,countdownTimer,nonMotorVehicleLight,pedestrianSignal,leftAndTurn,nonMot
orLeft">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->roundLight
            </trafficLightType>
            <lampPanelRegion>
              <!--ro, opt, object-->
              <height>
                <!--ro, req, float, range:[0,1]-->0.001
              </height>
              <width>
                <!--ro, req, float, range:[0,1]-->0.001
              </width>
              <x>
                <!--ro, req, float, range:[0,1]-->0.001
              </x>
              <y>
                <!--ro, req, float, range:[0,1]-->0.001
              </y>
            </lampPanelRegion>
          </trafficLight>
        </trafficLightList>
      </trafficLightPanel>
    </trafficLightPanelList>
  </autoDetectTrafficLight>
</TrafficLight>
<relatedLaneCount min="1" max="6">
  <!--ro, req, int, number of supported lanes, range:[1,6], attr:min{req, int},max{req, int}-->0
</relatedLaneCount>
<relatedDriveWay min="1" max="99">
  <!--ro, req, int, linked lane No., range:[1,99], attr:min{req, int},max{req, int}-->0
</relatedDriveWay>
<recordEnable>
  <!--ro, req, bool, whether to enable recording when traffic violation happened-->true
</recordEnable>
<recordType opt="delayRecord,preRecord">
  <!--ro, req, enum, recording type, subType:string, attr:opt{req, string}, desc:"preRecord" (pre-record), "delayRecord" (post-record)-->preRecord
</recordType>
<preRecordTime min="0" max="100">
  <!--ro, req, int, pre-record time, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
</preRecordTime>
<recordDelayTime min="0" max="100">
  <!--ro, req, int, post-record time, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
</recordDelayTime>
<recordTimeOut min="0" max="100">
  <!--ro, req, int, recording timeout, range:[0,100], unit:s, attr:min{req, int},max{req, int}-->0
</recordTimeOut>
<signSpeed min="0" max="255">
  <!--ro, opt, int, marked speed limit, range:[0,255], unit:km/h, attr:min{req, int},max{req, int}-->0
</signSpeed>
<speedLimit min="0" max="255">
  <!--ro, opt, int, speed limit, range:[0,255], unit:km/h, attr:min{req, int},max{req, int}-->0
</speedLimit>
<usageType opt="carriageWay,bus,slow,nonmotor,reverse,emergency,banNonMotor,ramp,fast,special">
  <!--ro, req, enum, lane usage, subType:string, attr:opt{req, string}, desc:"carriageWay" (roadway), "bus" (bus lane), "slow" (slow lane), "nonmotor"
(non-motor lane), "reverse" (contraflow lane), "emergency" (emergency lane), "banNonMotor" (non-motor vehicle vehicles prohibited lane), "ramp"--
>carriageWay
</usageType>
<LaneUsageList size="5">
  <!--ro, opt, object, list of lane purposes, attr:size{req, int}, desc:list of lane purposes-->
  <laneUsage opt="banTrucks,bus,banLeft,banRight,motor,banStraight,variableLane">
    <!--ro, req, enum, lane purpose, subType:string, attr:opt{req, string}, desc:lane usage type: "bus"-bus lane,"banTrucks"-truck prohibited
lane,"banLeft"-left turn prohibited lane,"banRight"-right turn prohibited lane,"motor"-motorcycle lane-->banTrucks
```

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    </laneUsage>
  </LaneUsageList>
  <banLeftTimeSwitchList size="4">
    <!--ro, opt, array, left-turn-prohibited time period, subType:object, attr:size{req, int}-->
    <banLeftTimeSwitch>
      <!--ro, req, object, left-turn-prohibited time period-->
      <timeId min="1" max="4">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
      </timeId>
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->0
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, end time (hour), attr:min{req, int},max{req, int}-->0
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, end time (minute), attr:min{req, int},max{req, int}-->0
      </endMinute>
    </banLeftTimeSwitch>
  </banLeftTimeSwitchList>
  <banRightTimeSwitchList size="4">
    <!--ro, opt, object, right-turn prohibited time period, attr:size{req, int}-->
    <banRightTimeSwitch>
      <!--ro, req, object, right-turn prohibited time period-->
      <timeId min="1" max="4">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
      </timeId>
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->0
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, end time (hour), attr:min{req, int},max{req, int}-->0
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, end time (minute), attr:min{req, int},max{req, int}-->0
      </endMinute>
    </banRightTimeSwitch>
  </banRightTimeSwitchList>
  <banStraightTimeSwitchList size="4">
    <!--ro, opt, object, straight prohibited time period, attr:size{req, int}-->
    <banStraightTimeSwitch>
      <!--ro, req, object, straight prohibited time period-->
      <timeId min="1" max="4">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
      </timeId>
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->0
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, end time (hour), attr:min{req, int},max{req, int}-->0
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, end time (minute), attr:min{req, int},max{req, int}-->0
      </endMinute>
    </banStraightTimeSwitch>
  </banStraightTimeSwitchList>
  <variableLaneTimeSwitchList size="4">
    <!--ro, opt, object, time period for changeable lanes, attr:size{req, int}-->
    <variableLaneTimeSwitch>
      <!--ro, req, object, time for changeable lanes-->
      <timeId min="1" max="4">
        <!--ro, req, int, index, attr:min{req, int},max{req, int}-->0
      </timeId>
      <startHour min="0" max="23">
        <!--ro, opt, int, start time (hour), attr:min{req, int},max{req, int}-->0
      </startHour>
      <startMinute min="0" max="59">
        <!--ro, opt, int, start time (minute), attr:min{req, int},max{req, int}-->0
      </startMinute>
      <endHour min="0" max="23">
        <!--ro, opt, int, end time (hour), attr:min{req, int},max{req, int}-->0
      </endHour>
      <endMinute min="0" max="59">
        <!--ro, opt, int, end time (minute), attr:min{req, int},max{req, int}-->0
      </endMinute>
      <laneDirection opt="unkown,left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight,leftWait,straightWait,strightWaitRight">
        <!--ro, req, enum, driving direction, subType:string, attr:opt{req, string}, desc:"unknown", "left", "straight", "leftStraight", "right",
"leftRight", "rightStraight", "leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->leftRightStraight
      </laneDirection>
    </variableLaneTimeSwitch>
  </variableLaneTimeSwitchList>
  <variableLaneTimeSwitchType opt="day,week">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->week
  </variableLaneTimeSwitchType>
  <variableLaneWeekSwitchList size="7">
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    <variableLaneWeekSwitchList size="7">
      <!--ro, opt, object, attr:size{req, int}-->
      <variableLaneWeekSwitch>
        <!--ro, req, object-->
        <dayOfWeek opt="1,2,3,4,5,6,7">
          <!--ro, req, enum, subType:int, attr:opt{req, string}-->1
        </dayOfWeek>
        <timeRangeList>
          <!--ro, req, object-->
          <timeRange>
            <!--ro, req, object-->
            <ID min="1" max="4">
              <!--ro, req, int, range:[1,4], attr:min{req, int},max{req, int}-->1
            </ID>
            <startHour min="0" max="23">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </startHour>
            <startMinute min="0" max="59">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </startMinute>
            <endHour min="0" max="23">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </endHour>
            <endMinute min="0" max="59">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </endMinute>
            <laneDirection opt="unkown,left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight,leftWait,straightWait,strightWaitRight">
              <!--ro, opt, enum, subType:string, attr:opt{req, string}-->unknown
            </laneDirection>
          </timeRange>
        </timeRangeList>
      </variableLaneWeekSwitch>
    </variableLaneWeekSwitchList>
    <directionType opt="unkown,left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight,leftWait,straightWait,strightWaitRight">
      <!--ro, req, enum, driving direction, subType:string, attr:opt{req, string}, desc:direction type,multiple options can be supported at one time--
>leftRightStraight
    </directionType>
    <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
      <!--ro, req, enum, lane direction type, subType:int, attr:opt{req, string}, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south
to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to
northeast)-->0
    </laneDirectionType>
    <stopLineDistance min="0" max="300">
      <!--ro, req, int, distance between the vehicle and the stop line in the second captured picture of red light running violation, range:[0,300], unit:m,
attr:min{req, int},max{req, int}, desc:distance between the vehicle and the stop line in the second captured picture of red light running violation-->0
    </stopLineDistance>
    <loopSensitivity min="1" max="100">
      <!--ro, opt, int, loop sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
    </loopSensitivity>
    <capPositionOffset min="0" max="300">
      <!--ro, req, int, second captured picture deviation, range:[0,300], attr:min{req, int},max{req, int}, desc:unit: pixel-->50
    </capPositionOffset>
    <videoDetectLightId min="1" max="12">
      <!--ro, req, int, area ID of video detection traffic light, range:[1,12], attr:min{req, int},max{req, int}-->1
    </videoDetectLightId>
    <lightNum min="0" max="12">
      <!--ro, req, int, number of lights, range:[1,3], attr:min{req, int},max{req, int}-->1
    </lightNum>
    <straightLight>
      <!--ro, req, bool, straight signal-->true
    </straightLight>
    <leftLight>
      <!--ro, req, bool, left turn signal-->true
    </leftLight>
    <rightLight>
      <!--ro, req, bool, right turn signal-->true
    </rightLight>
    <greenLight>
      <!--ro, req, bool, green light-->true
    </greenLight>
    <yellowLight>
      <!--ro, req, bool, yellow light-->true
    </yellowLight>
    <yellowLightTime min="0" max="10000">
      <!--ro, req, int, yellow time, range:[0,10000], unit:ms, attr:min{req, int},max{req, int}-->0
    </yellowLightTime>
    <beginTime opt="1, 2, 3">
      <!--ro, req, enum, recording start time, subType:int, attr:opt{req, string}, desc:recording start time-->1
    </beginTime>
    <leftNotYieldStraight>
      <!--ro, opt, object, detection of left turn not yield to straight-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable detection of left turn not yield to straight driving, attr:opt{req, string}-->true
      </enable>
      <capNo opt="3">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->3
      </capNo>
    </leftNotYieldStraight>
    <rightNotYieldLeft>
      <!--ro, opt, object, detection of right turn not yield to left turn-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable detection of right turn not yield to left turn, attr:opt{req, string}-->true
      </enable>
      <capNo opt="3">
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      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </rightNotYieldLeft>
  <uTurnNotYieldGoingStaight>
    <!--ro, opt, object, detection of U-turn not yield to straight driving-->
    <enable opt="true,false">
      <!--ro, req, bool, whether to enable detection of U-turn not yield to straight driving, attr:opt{req, string}-->true
    </enable>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </uTurnNotYieldGoingStaight>
  <leftSideDrivingAfterLeftTurn>
    <!--ro, opt, object, turn left in short turn radius-->
    <enable opt="true,false">
      <!--ro, req, bool, whether to enable detection of left turn not yield to straight driving, attr:opt{req, string}-->true
    </enable>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </leftSideDrivingAfterLeftTurn>
  <failYieldPed>
    <!--ro, opt, object, failing to yield for pedestrians-->
    <enable opt="true,false">
      <!--ro, req, bool, whether to enable detection of left turn not yield to straight driving, attr:opt{req, string}-->true
    </enable>
    <vehicleFromDirection opt="Front,Back">
      <!--ro, opt, enum, driving direction of the vehicle which does not yield to pedestrian, subType:string, attr:opt{req, string}, desc:"Front", "Back"--
>Front
    </vehicleFromDirection>
    <cameraPosition opt="crossing,roadSection">
      <!--ro, req, enum, camera position, subType:string, attr:opt{req, string}, desc:Yellow Time-->crossing
    </cameraPosition>
    <noDisplacementPedEnabled opt="true,false">
      <!--ro, req, bool, whether to enable no displacement capture of pedestrian, attr:opt{req, string}-->true
    </noDisplacementPedEnabled>
    <failYieldPedLineList>
      <!--ro, opt, array, list of failing to yield for pedestrians lines, subType:object-->
      <failYieldPedLine>
        <!--ro, opt, object, failing to yield for pedestrians line-->
        <lineId opt="1,2">
          <!--ro, req, int, line ID, range:[1,2], attr:opt{req, string}-->0
        </lineId>
        <lineName>
          <!--ro, req, string, line name, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList size="2">
          <!--ro, opt, array, subType:object, range:[0,2], attr:size{opt, int}-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </failYieldPedLine>
    </failYieldPedLineList>
    <notification>
      <!--ro, opt, string, linked capture, desc:pedestrian linkage-->test
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
      </enable>
      <ip>
        <!--ro, opt, string, IPv4 address, range:[0,32]-->test
      </ip>
      <port>
        <!--ro, opt, int, port No., range:[0,65535]-->1
      </port>
      <username>
        <!--ro, opt, string, user name, range:[0,32], desc:user name of traffic camera-->test
      </username>
      <password>
        <!--ro, opt, string, password, range:[0,16], desc:password of traffic camera,when transmitting,HTTPS is required-->test
      </password>
      <sensitivity min="0" max="100">
        <!--ro, req, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
      </sensitivity>
      <RS485Port min="1" max="10">
        <!--ro, req, int, RS-485 port No. for the signal light to access the camera,which is between 1 and 10, range:[1,10], attr:min{req, int},max{req,
int}-->1
      </RS485Port>
    </notification>
    <failYieldPedSensitivity min="1" max="100">
      <!--ro, req, int, not yielding to pedestrian sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
    </failYieldPedSensitivity>
    <failYieldnonMotor opt="true,false">
      <!--ro, opt, bool, whether to enable capturing not yielding to pedestrian, attr:opt{req, string}-->true
    </failYieldnonMotor>
    <failYieldNum min="1" max="7">
      <!--ro, opt, int, number of pedestrian not yielded to, attr:min{req, int},max{req, int}-->1
    </failYieldNum>
```

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    </failYieldNum>
    <failYieldAreaType opt="rectangle,polygon">
      <!--ro, opt, enum, detection area of failing to yield for pedestrians, subType:string, attr:opt{req, string}, desc:"rectangle", "polygon"-->rectangle
    </failYieldAreaType>
    <failYieldDistanceMode opt="car,lane">
      <!--ro, opt, enum, distance mode of failing to yield for pedestrians, subType:string, attr:opt{req, string}, desc:"car" (distance from pedestrian to vehicle center), "lane" (distance from pedestrian to lane center)-->car
    </failYieldDistanceMode>
    <failYieldSnapPosition min="0" max="100">
      <!--ro, opt, int, distance range for capturing failing to yield for pedestrians, attr:min{req, int},max{req, int}-->20
    </failYieldSnapPosition>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </failYieldPed>
  <illegalParking>
    <!--ro, opt, object, illegal parking detection-->
    <enable opt="true,false">
      <!--ro, req, bool, whether to enable illegal parking detection, attr:opt{req, string}-->true
    </enable>
    <sensitivity min="0" max="100">
      <!--ro, req, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
    </sensitivity>
    <lastTime min="1" max="180">
      <!--ro, req, int, parking time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
    </lastTime>
    <illegalParkingLineList>
      <!--ro, opt, array, subType:object, range:[0,2]-->
      <illegalParkingLine>
        <!--ro, opt, object-->
        <lineId opt="1,2,3">
          <!--ro, req, int, attr:opt{req, string}-->0
        </lineId>
        <lineName>
          <!--ro, req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList num="2">
          <!--ro, opt, array, subType:object, range:[0,2], attr:num{opt, int}-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </illegalParkingLine>
    </illegalParkingLineList>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </illegalParking>
  <gasser>
    <!--ro, opt, object, queue jumping-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable queue jumping detection, attr:opt{req, string}-->true
    </enabled>
    <sensitivity min="0" max="100">
      <!--ro, req, int, detection sensitivity of queue jumping, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <congestionThreshold min="1" max="100">
      <!--ro, req, int, congestion threshold, range:[1,100], attr:min{req, int},max{req, int}-->1
    </congestionThreshold>
    <capInterval min="0" max="6000">
      <!--ro, req, int, second and third capture detection, range:[0,6000], unit:ms, attr:min{req, int},max{req, int}-->0
    </capInterval>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </gasser>
  <congestion>
    <!--ro, opt, object, congestion-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable congestion detection, attr:opt{req, string}-->true
    </enabled>
    <interval min="1" max="10">
      <!--ro, req, int, alarm interval, range:[1,10], unit:min, attr:min{req, int},max{req, int}-->1
    </interval>
    <sensitivity min="1" max="100">
      <!--ro, req, int, congestion sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->0
    </sensitivity>
    <lastTime min="1" max="60">
      <!--ro, req, int, congestion duration, range:[1,60], unit:s, attr:min{req, int},max{req, int}-->1
    </lastTime>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </congestion>
  <accident>
    <!--ro, opt, object, accident detection-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable accident detection, attr:opt{req, string}-->true
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    </enabled>
    <lastTime min="1" max="180">
      <!--ro, req, int, event time limit, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->1
    </lastTime>
    <interval min="1" max="60">
      <!--ro, req, int, repeated capture interval, range:[1,60], unit:min, attr:min{req, int},max{req, int}-->1
    </interval>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </accident>
  <nonMotorEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable non-motor vehicle detection, attr:opt{req, string}-->true
  </nonMotorEnabled>
  <runGreenLight opt="true,false">
    <!--ro, opt, bool, whether to enable green light running detection, attr:opt{req, string}-->true
  </runGreenLight>
  <runGreenLightParam>
    <!--ro, opt, object, parameters of green light running detection-->
    <detectEvent opt="runGreenLight,crossingStop">
      <!--ro, opt, enum, Detection Logic, subType:string, attr:opt{req, string}, desc:"runGreenLight" (running green light in congestion), "crossingStop"
(overstay at intersection)-->runGreenLight
    </detectEvent>
    <interval min="1" max="180">
      <!--ro, opt, int, capture interval, range:[1,180], unit:s, attr:min{req, int},max{req, int}, desc:capture interval-->15
    </interval>
    <secondLampState opt="0,1,2">
      <!--ro, opt, enum, traffic light status of the second capture times, subType:int, attr:opt{req, string}, desc:0 (red light), 1 (green light), 2
(unlimited)-->2
    </secondLampState>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </runGreenLightParam>
  <helmet>
    <!--ro, opt, bool, without helmet-->true
  </helmet>
  <helmetCapNo opt="1,2,3">
    <!--ro, opt, int, number of captured pictures in which the driver does not wear the helmet, range:[1,3], attr:opt{req, string}-->1
  </helmetCapNo>
  <nonMotorExist>
    <!--ro, opt, bool, non-motor vehicle on motor vehicle lane-->true
  </nonMotorExist>
  <nonMotorExistCapNo opt="2,3">
    <!--ro, opt, int, number of captured pictures in which the motor lane is occupied by non-motor vehicles, range:[2,3], attr:opt{req, string}-->2
  </nonMotorExistCapNo>
  <motorIllegalCarrying>
    <!--ro, opt, bool, illegal manned motor vehicle-->true
  </motorIllegalCarrying>
  <nomotorIllegalCarrying>
    <!--ro, opt, bool, illegal manned non-motor vehicle-->true
  </nomotorIllegalCarrying>
  <canopy>
    <!--ro, opt, bool, umbrella tent installation violation on non-motor vehicle-->true
  </canopy>
  <pedRedLight>
    <!--ro, opt, bool, pedestrian red light running-->true
  </pedRedLight>
  <pedOnNomotorLane>
    <!--ro, opt, bool, whether to detect pedestrian walking on non-motor vehicle lane-->true
  </pedOnNomotorLane>
  <pedOnMotorLane>
    <!--ro, opt, bool, whether to detect pedestrian walking on motor vehicle lane-->true
  </pedOnMotorLane>
  <capType opt="vehicle,nonMotor,human,face,mixObject,motor,motorWithHum,nonMotorWithHum,motorNonmotor">
    <!--ro, opt, enum, capture types, subType:string, attr:opt{req, string}, desc:"vehWithHum" (motor vehicle, non-motor vehicle, and pedestrian), "vehicle"
(motor vehicle), "nonmotor" (non-motor vehicle), "human" (pedestrian), "motorWithHum" (motor vehicle and pedestrian), "nonmotorWithHum" (non-motor vehicle
and pedestrian), "motorNonmotor" (motor vehicle and non-motor vehicle)-->mixObject
  </capType>
  <deferredFrmNum min="0" max="25">
    <!--ro, opt, int, delay a few seconds to trigger the mode, range:[0,25], attr:min{req, int},max{req, int}-->0
  </deferredFrmNum>
  <redLightCapNo opt="3,4">
    <!--ro, opt, int, captured picture amount in red light, range:[3,4], attr:opt{req, string}-->3
  </redLightCapNo>
  <directionCapNo opt="3,4">
    <!--ro, opt, int, number of captured pictures of driving against direction guidance, range:[3,4], attr:opt{req, string}-->3
  </directionCapNo>
  <firstShowTailVep opt="0,1,2,3">
    <!--ro, opt, enum, place of the first capture time, subType:int, attr:opt{req, string}, desc:0 (none), 1 (expose vehicle tail), 2 (expose vehicle tail
and LPR success), 3(parking over stop line)-->0
  </firstShowTailVep>
  <banSignType opt="truck,yellowPlate,standard,bus">
    <!--ro, opt, enum, capture type list, subType:string, attr:opt{req, string}, desc:"truck", "yellowPlate" (yellow license plate vehicle), "standard",
"bus"-->truck
  </banSignType>
  <greenLightStopSensitivity min="1" max="100">
    <!--ro, opt, int, stopping sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->0
  </greenLightStopSensitivity>
  <parkingOverStopLine>
    <!--ro, opt, object, Over Stop Line-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
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    </enabled>
  </parkingOverStopLine>
  <nonMotorShed>
    <!--ro, opt, object, Non-Motor Vehicle Umbrella Tent-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="1,2">
      <!--ro, req, int, the number of captured pictures, range:[1,2], attr:opt{req, string}-->0
    </capNo>
  </nonMotorShed>
  <banNonMotorSign>
    <!--ro, opt, object, non-motor vehicle prohibition violation-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="2,3">
      <!--ro, req, int, the number of captured pictures, range:[2,3], attr:opt{req, string}-->0
    </capNo>
  </banNonMotorSign>
  <nonMotorPassenger>
    <!--ro, opt, object, Manned Non-Motor Vehicle-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="1,2">
      <!--ro, req, int, the number of captured pictures, range:[1,2], attr:opt{req, string}-->0
    </capNo>
    <refineMannedIllegalCode opt="true,false">
      <!--ro, opt, bool, whether to enable detailed manned violation code, attr:opt{req, string}-->true
    </refineMannedIllegalCode>
  </nonMotorPassenger>
  <SuonaDet>
    <!--ro, opt, object, whistle detection-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="2,3">
      <!--ro, opt, int, the number of captured pictures, range:[2,3], attr:opt{req, string}-->2
    </capNo>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </SuonaDet>
  <RoarDet>
    <!--ro, opt, object, thumping sound of vehicle detection-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="2,3">
      <!--ro, opt, int, the number of captured pictures, range:[2,3], attr:opt{req, string}-->2
    </capNo>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </RoarDet>
  <speedDetectMode opt="none,radar">
    <!--ro, opt, enum, speed detection mode, subType:string, attr:opt{req, string}, desc:"none", "radar"-->none
  </speedDetectMode>
  <BanMotorDet>
    <!--ro, opt, object, motorcycle driving in prohibited areas-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->false
    </enabled>
    <capNo opt="1,2,3">
      <!--ro, opt, int, the number of captured pictures, range:[1,3], attr:opt{req, string, range:[1,3]}-->2
    </capNo>
    <sensitivity min="1" max="100">
      <!--ro, opt, int, sensitivity, range:[1,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
  </BanMotorDet>
  <Radar>
    <!--ro, opt, object, radar parameters-->
    <radarType opt="none,MultiLane,csr,IOBox,SSTK,custom" def="MultiLane">
      <!--ro, opt, string, radar type, attr:opt{req, string},def{req, string}, desc:"none", "MultiLane" (multi-lane radar), "csr" (TransMicrowave),"IOBox"
(radar connecting I/O expansion box), "SSTK" (Sensortech),"custom" (custom type), the default value is MultiLane-->MultiLane
    </radarType>
    <radarSensitivity min="0" max="65535" def="0">
      <!--ro, opt, int, radar detection sensitivity, range:[0,65535], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist, it
indicates the function is supported. The default value is 0-->0
    </radarSensitivity>
    <radarAngle min="0" max="89" def="0">
      <!--ro, opt, int, angle of radar and horizontal direction, range:[0,89], unit:°, attr:min{req, int},max{req, int},def{req, int}, desc:if this node
does not exist, it indicates the function is supported. The default value is 0-->0
    </radarAngle>
    <validRadarSpeedTime min="1" max="9000" def="2000">
      <!--ro, opt, int, radar speed valid time, range:[0,9000], unit:ms, attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist,
it indicates the function is supported. The default value is 2,000-->0
    </validRadarSpeedTime>
    <radarLinearCorrection min="0.0" max="2.0" def="1.0">
      <!--ro, opt, float, linear correction parameters, range:[0.0,2.0], attr:min{req, float},max{req, float},def{req, float}, desc:if this node does not
exist, it indicates the function is supported. The default value is 1.0, and the value should be accurate to 3 decimal places-->1.0
    </radarLinearCorrection>
    <radarConstantCorrection min="-100" max="100" def="0">
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      <!--ro, opt, int, constant correction parameter,, range:[-100,100], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist, it indicates the function is supported. The default value is 0-->0
    </radarConstantCorrection>
    <radarRS485 min="1" max="5" def="2">
      <!--ro, opt, int, 485 radar No., range:[1,5], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist, it indicates the function is supported. The default value is 2-->1
    </radarRS485>
  </Radar>
  <NotSlowZebraCrossingEvent>
    <!--ro, opt, object, not slowing down at zebra crossing-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo min="1" max="3">
      <!--ro, req, int, the number of captured pictures, range:[1,3], step:1, attr:min{req, int},max{req, int}-->1
    </capNo>
    <lowSpeed min="0" max="100">
      <!--ro, req, int, lowest speed limit (km/h), range:[0,100], step:1, attr:min{req, int},max{req, int}-->0
    </lowSpeed>
    <highSpeed min="0" max="100">
      <!--ro, req, int, highest speed limit (km/h), range:[0,100], step:1, attr:min{req, int},max{req, int}-->0
    </highSpeed>
    <LineInfoList size="3">
      <!--ro, opt, array, list of lane line information, subType:object, attr:size{req, int}, desc:lane list-->
      <LineInfo>
        <!--ro, opt, object, single lane line information-->
        <lineId min="0" max="3">
          <!--ro, req, int, index, range:[1,3], attr:min{req, int},max{req, int}-->0
        </lineId>
        <RegionCoordinatesList size="2">
          <!--ro, opt, array, triggering line coordinate list, subType:object, attr:size{req, int}-->
          <RegionCoordinates>
            <!--ro, opt, object, point coordinates-->
            <positionX min="0" max="1000">
              <!--ro, req, int, X-coordinate, range:[0,1000], attr:min{req, int},max{req, int}-->0
            </positionX>
            <positionY min="0" max="1000">
              <!--ro, req, int, Y-coordinate, attr:min{req, int},max{req, int}-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </LineInfo>
    </LineInfoList>
  </NotSlowZebraCrossingEvent>
  <TurnRightStop>
    <!--ro, opt, object, large-sized vehicle not stopping before right turn-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <capNo min="0" max="3">
      <!--ro, req, int, the number of captured pictures, range:[1,3], step:1, attr:min{req, int},max{req, int}-->3
    </capNo>
    <durationTime min="0" max="600">
      <!--ro, req, int, duration, range:[1,600], unit:s, attr:min{req, int},max{req, int}-->5
    </durationTime>
    <slowUp min="0" max="100">
      <!--ro, req, int, upper limit of deceleration, range:[0,100], step:1, unit:km/h, attr:min{req, int},max{req, int}-->40
    </slowUp>
    <slowStep min="0" max="50">
      <!--ro, req, int, deceleration range, range:[0,50], step:1, unit:km/h, attr:min{req, int},max{req, int}-->5
    </slowStep>
    <slowDown min="0" max="100">
      <!--ro, req, int, lower limit of deceleration, range:[0,100], step:1, unit:km/h, attr:min{req, int},max{req, int}-->10
    </slowDown>
    <detectMode opt="stop,slow">
      <!--ro, req, enum, deceleration mode, subType:string, attr:opt{req, string}, desc:"stop", "slow"-->stop
    </detectMode>
    <SlowLane>
      <!--ro, opt, object, deceleration line-->
      <RegionCoordinatesList size="2">
        <!--ro, opt, object, deceleration line coordinate list, attr:size{req, int}-->
        <RegionCoordinates>
          <!--ro, opt, object, point coordinates-->
          <positionX min="0" max="1000">
            <!--ro, req, int, X-coordinate, range:[0,1], attr:min{req, int},max{req, int}-->0.1
          </positionX>
          <positionY min="0" max="1000">
            <!--ro, req, int, Y-coordinate, range:[0,1], attr:min{req, int},max{req, int}-->0.1
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </SlowLane>
  </TurnRightStop>
  <NotYieldToRightRoad>
    <!--ro, opt, object, not yield to vehicles from right roads-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <firstCapturePosition min="0" max="100">
      <!--ro, req, int, the capture position of the first picture, range:[0,100], step:1, attr:min{req, int},max{req, int}-->20
    </firstCapturePosition>
    <lateralDistance min="20" max="350">
      <!--ro, req, int, lateral distance between two vehicles, range:[20,350], step:1, attr:min{req, int},max{req, int}-->150
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    </lateralDistance>
    <longitudinalDistance min="10" max="150">
      <!--ro, req, int, longitudinal distance between two vehicles, range:[10,150], step:1, attr:min{req, int},max{req, int}-->50
    </longitudinalDistance>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </NotYieldToRightRoad>
  <EnterNotYieldToInRoundabout>
    <!--ro, opt, object, not yield at roundabout intersection-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <firstCapturePosition min="0" max="100">
      <!--ro, req, int, the capture position of the first picture, range:[0,100], step:1, attr:min{req, int},max{req, int}-->20
    </firstCapturePosition>
    <lateralDistance min="20" max="350">
      <!--ro, req, int, lateral distance between two vehicles, range:[20,350], step:1, attr:min{req, int},max{req, int}-->150
    </lateralDistance>
    <longitudinalDistance min="10" max="150">
      <!--ro, req, int, longitudinal distance between two vehicles, range:[10,150], step:1, attr:min{req, int},max{req, int}-->50
    </longitudinalDistance>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </EnterNotYieldToInRoundabout>
  <RampNotYieldToMainRoad>
    <!--ro, opt, object, vehicle from ramp not yielding to vehicle on main road-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the detection, attr:opt{req, string}-->true
    </enabled>
    <lateralDistance min="10" max="90">
      <!--ro, req, int, lateral distance of a vehicle when it drives from a ramp to a main road., range:[10,90], step:1, attr:min{req, int},max{req, int}--
>60
    </lateralDistance>
    <capNo opt="3">
      <!--ro, req, int, number of captures, attr:opt{req, string}-->3
    </capNo>
  </RampNotYieldToMainRoad>
  <BanSignTypeList size="10">
    <!--ro, opt, array, capture type list, subType:object, attr:size{req, int}-->
    <banSignType opt="standard,truck,yellowPlate,bus,yellowBus,dangerCar,instructionCar,pickupTruck,concreteMixer,slagTruck">
      <!--ro, opt, enum, prohibition violation type, subType:string, attr:opt{req, string}, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus), "dangercar" (dangerous goods transport vehicle), "instructioncar" (driver-training vehicle)--
>standard
    </banSignType>
  </BanSignTypeList>
  <TurnSignalDet>
    <!--ro, opt, object, turning without signal detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->false
    </enabled>
    <turnFiliter opt="true,false">
      <!--ro, opt, bool, whether to filter left turn and right turn, attr:opt{req, string}-->false
    </turnFiliter>
    <detectEvent opt="turnLeft,turnRight,changeLane">
      <!--ro, opt, enum, turning signal detection event, subType:string, attr:opt{req, string}, desc:"turnLeft", "turnRight", "changeLane"-->changeLane
    </detectEvent>
    <detectEventList size="3">
      <!--ro, opt, array, subType:object, attr:size{req, int}-->
      <detectEvent opt="turnLeft,turnRight,changeLane">
        <!--ro, opt, enum, subType:string, attr:opt{req, string}-->changeLane
      </detectEvent>
    </detectEventList>
    <capNo opt="3">
      <!--ro, opt, int, range:[3,3], attr:opt{req, int}-->3
    </capNo>
  </TurnSignalDet>
  <ReflectiveTapeDet>
    <!--ro, opt, object, whether to enable reflective stripe detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->false
    </enabled>
    <capNo min="1" max="3">
      <!--ro, opt, int, the number of captured pictures, range:[1,3], attr:min{req, int},max{req, int}-->2
    </capNo>
  </ReflectiveTapeDet>
  <ZipperDriveDet>
    <!--ro, opt, object, pass alternatively detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->false
    </enabled>
    <carNum opt="1,2,3">
      <!--ro, opt, int, capture triggered number of vehicles, range:[1,3], attr:opt{req, string, range:[1,3]}-->2
    </carNum>
    <parallDistance min="0" max="200" def="0">
      <!--ro, opt, int, parallel distance, range:[0,200], attr:min{req, int},max{req, int},def{req, int}, desc:if this node does not exist, it indicates the
function is supported. The default value is 0-->0
    </parallDistance>
  </ZipperDriveDet>
  <NonMotorCrossing>
    <!--ro, opt, object, non-motor vehicle parking over stop line, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
    <enabled opt="true,false">
      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->false
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      <!--ro, opt, bool, whether to enable the detection, attr:opt{req, string}-->false
    </enabled>
    <lastTime min="10" max="180">
      <!--ro, req, int, parking time, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->10
    </lastTime>
    <sensitivity min="1" max="100">
      <!--ro, req, int, sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->1
    </sensitivity>
    <nonMotorStopLine>
      <!--ro, opt, object, non-motor vehicle parking over stop line-->
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX min="0" max="1000">
            <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->0
          </positionX>
          <positionY min="0" max="1000">
            <!--ro, req, int, range:[0,1000], attr:min{req, int},max{req, int}-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </nonMotorStopLine>
  </NonMotorCrossing>
  <lineType opt="unknown,white,singleYellow,doubleYellow,guardRail,noCross,fullDottedLine,dottedFullLine">
    <!--ro, opt, enum, line type, subType:string, attr:opt{req, string}, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle), "fullDottedLine" (double
yellow line (left: dotted, right: solid)), "dottedFullLine" (double yellow line (left: solid, right: dotted))-->unknown
  </lineType>
  <TruckRunARedLight>
    <!--ro, opt, object, red light running of large truck detection, desc:this node is used for linking evidence intersection violation cameras can link the
checkpoints to capture the front license plate of the large trucks-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable, attr:opt{req, string}-->false
    </enabled>
    <secondCam min="0" max="255">
      <!--ro, req, int, the unique identifier for the linked cameras, range:[0,255], attr:min{req, int},max{req, int}, desc:all linked cameras share the
same configuration, which is related to the uploaded value of secondCam in ANPR-->1
    </secondCam>
    <sceneMode opt="singleLane,multiLane">
      <!--ro, opt, enum, detection scene type, subType:string, attr:opt{req, string}, desc:"singleLane", "multiLane"-->singleLane
    </sceneMode>
    <laneID min="1" max="3">
      <!--ro, opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,singleLane}, attr:min{req,
int},max{req, int}, desc:this node is valid when the camera is applied in single lane scene-->1
    </laneID>
    <bluePlateCarFilterEnabled opt="true,false">
      <!--ro, opt, bool, whether to enable filtering blue plate vehicles, attr:opt{req, string}-->true
    </bluePlateCarFilterEnabled>
    <NotificationList size="3">
      <!--ro, opt, array, list of linked camera, max. 3 linked cameras, subType:object, range:[1,3], attr:size{req, int}-->
      <notification>
        <!--ro, opt, object, picture capture-->
        <enabled opt="true,false">
          <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
        </enabled>
        <ipv4 min="0" max="32">
          <!--ro, req, string, IPv4 address, range:[0,32], attr:min{req, int},max{req, int}-->test
        </ipv4>
        <port min="0" max="66535">
          <!--ro, req, int, port No., range:[0,66535], attr:min{req, int},max{req, int}-->0
        </port>
        <username min="0" max="32">
          <!--ro, opt, string, user name, range:[0,32], attr:min{req, int},max{req, int}, desc:transmitting via HTTP may cause security problems. HTTPS is
required-->test
        </username>
        <password min="0" max="16">
          <!--ro, req, string, password, range:[0,16], attr:min{req, int},max{req, int}, desc:transmitting via HTTP may cause security problems. HTTPS is
required-->test
        </password>
        <laneID min="1" max="3">
          <!--ro, opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,multiLane}, attr:min{req,
int},max{req, int}, desc:this node is valid when the camera is applied in multiple lanes scene-->1
        </laneID>
      </notification>
    </NotificationList>
    <linkSceneMode opt="oneLinkMultiple, oneLinkOne, multipleLinkOne">
      <!--ro, opt, enum, detection scene type, subType:string, attr:opt{req, string}, desc:"oneLinkMultiple"(one camera links with multiple cameras),
"oneLinkOne" (one camera links with one camera), "multipleLinkOne" (multiple cameras link with multiple cameras)-->oneLinkOne
    </linkSceneMode>
  </TruckRunARedLight>
  <bigCarSignSpeed min="0" max="255">
    <!--ro, opt, int, marked speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSignSpeed>
  <bigCarLowSpeedLimit min="0" max="255">
    <!--ro, opt, int, low speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarLowSpeedLimit>
  <lowSpeedLimit min="0" max="60">
    <!--ro, opt, int, low speed limit for small-sized vehicle, attr:min{req, int},max{req, int}-->0
  </lowSpeedLimit>
  <bigCarSpeedLimit min="0" max="60">
    <!--ro, opt, int, maximum speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
  </bigCarSpeedLimit>
  <TruckTurnRightStop>
```

```
      <!--ro, opt, object, large-sized vehicle not stopping before right turn-->
      <enabled opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->false
      </enabled>
      <capNo opt="3,5">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->3
      </capNo>
      <sensitivity min="1" max="100">
        <!--ro, opt, int, stopping sensitivity, range:[0,100], attr:min{req, int},max{req, int}-->50
      </sensitivity>
      <stopTime min="1" max="180">
        <!--ro, opt, int, Parking Duration, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->10
      </stopTime>
      <capInterval min="1" max="10000">
        <!--ro, opt, int, attr:min{req, int},max{req, int}-->1000
      </capInterval>
    </TruckTurnRightStop>
    <carHighSpeed min="0" max="255">
      <!--ro, opt, int, abnormal speeding of small-sized vehicle, attr:min{req, int},max{req, int}-->0
    </carHighSpeed>
    <bigCarHighSpeed min="0" max="255">
      <!--ro, opt, int, abnormal speeding of large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarHighSpeed>
    <greenLightStopCapNo opt="3">
      <!--ro, opt, int, captured picture amount in green light, attr:opt{req, string}-->3
    </greenLightStopCapNo>
    <illegalTurnCapNo opt="3">
      <!--ro, req, int, number of captured violation turing picture(s), attr:opt{req, string}-->3
    </illegalTurnCapNo>
    <uphone>
      <!--ro, opt, object, making a phone call-->
      <enable opt="true,false">
        <!--ro, req, bool, whether to enable, attr:opt{req, string}-->true
      </enable>
      <capNo opt="1,2,3">
        <!--ro, req, int, number of captures, attr:opt{req, string}-->2
      </capNo>
    </uphone>
    <noBanLeftTypeList size="5">
      <!--ro, opt, array, allowlist of left-turn vehicle types, subType:object, attr:size{req, int}-->
      <noBanLeftType opt="standard,truck,yellowPlate,bus,yellowBus">
        <!--ro, opt, enum, vehicle types which are allowed to turn left, subType:string, attr:opt{req, string}, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus", "yellowbus" (yellow license plate bus)-->standard
      </noBanLeftType>
    </noBanLeftTypeList>
    <noBanRightTypeList size="5">
      <!--ro, opt, array, allowlist of right-turn vehicle types, subType:object, attr:size{req, int}-->
      <noBanRightType opt="standard,truck,yellowPlate,bus,yellowBus">
        <!--ro, opt, enum, vehicle types which are allowed to left-turn, subType:string, attr:opt{req, string}, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus", "yellowbus" (yellow license plate bus)-->standard
      </noBanRightType>
    </noBanRightTypeList>
    <NoAlterTrafficDet>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, opt, bool, attr:opt{req, string}-->true
      </enabled>
      <capNo opt="3">
        <!--ro, req, int, attr:opt{req, string}-->3
      </capNo>
      <ConvergenceLineList size="1">
        <!--ro, opt, array, subType:object, range:[1,1], attr:size{req, int}-->
        <ConvergenceLine>
          <!--ro, opt, object-->
          <lineId opt="1">
            <!--ro, req, int, attr:opt{req, string}-->1
          </lineId>
          <lineName>
            <!--ro, req, string, range:[0,64]-->test
          </lineName>
          <RegionCoordinatesList size="2">
            <!--ro, opt, array, subType:object, range:[0,2], attr:size{req, int}-->
            <RegionCoordinates>
              <!--ro, opt, object-->
              <positionX>
                <!--ro, req, int, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--ro, req, int, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </ConvergenceLine>
      </ConvergenceLineList>
    </NoAlterTrafficDet>
    <dangerCarSignSpeed min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </dangerCarSignSpeed>
    <dangerCarSpeedLimit min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </dangerCarSpeedLimit>
    <dangerCarLowSpeedLimit min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </dangerCarLowSpeedLimit>
```

```xml
    </dangerCarLowSpeedLimit>
    <dangerCarHighSpeed min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </dangerCarHighSpeed>
    <nonMotorSignSpeed min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </nonMotorSignSpeed>
    <nonMotorSpeedLimit min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </nonMotorSpeedLimit>
    <nonMotorLowSpeedLimit min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </nonMotorLowSpeedLimit>
    <nonMotorHighSpeed min="0" max="255">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
    </nonMotorHighSpeed>
    <reversing>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <snapTime min="2" max="3">
        <!--ro, opt, int, range:[2,3], unit:s, attr:min{req, int},max{req, int}-->2
      </snapTime>
    </reversing>
    <freightManned>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <capNo opt="2,3">
        <!--ro, req, int, attr:opt{req, string}-->2
      </capNo>
      <sensitivity min="1" max="100">
        <!--ro, opt, int, range:[1,100], attr:min{req, int},max{req, int}-->50
      </sensitivity>
    </freightManned>
    <busCapDirectionType opt="left,straight,leftStraight,right,leftRight,rightStraight,leftRightStraight">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->leftRightStraight
    </busCapDirectionType>
    <reverseSensitivity min="0" max="100">
      <!--ro, opt, int, range:[0,100], attr:min{req, int},max{req, int}-->0
    </reverseSensitivity>
    <VirtualLaneList size="6">
      <!--ro, opt, object, attr:size{req, int}-->
      <VirtualLane>
        <!--ro, req, object-->
        <lineName opt="laneRightBoundaryLine,turnLeftLine,turnRightLine,topZebraLine,bottomZebraLine,triggerLine">
          <!--ro, req, enum, subType:string, attr:opt{req, string}-->laneRightBoundaryLine
        </lineName>
        <RegionCoordinatesList size="2">
          <!--ro, opt, object, attr:size{req, int}-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </VirtualLane>
    </VirtualLaneList>
    <vehicleRestrictions>
      <!--ro, opt, object-->
      <enabled opt="true,false">
        <!--ro, req, bool, attr:opt{req, string}-->true
      </enabled>
      <capNo opt="2,3">
        <!--ro, req, int, attr:opt{req, string}-->2
      </capNo>
      <restrictionType opt="region">
        <!--ro, req, enum, subType:string, attr:opt{req, string}-->region
      </restrictionType>
      <nonRestrictedRegionList size="64">
        <!--ro, opt, object, attr:size{req, int}-->
        <nonRestrictedRegion>
          <!--ro, opt, object-->
          <nonRestrictedProvinces opt="京,津,沪,渝,辽,吉,黑,冀,鲁,苏,浙,闽,豫,鄂,湘,皖,赣,粤,琼,川,黔,滇,陕,甘,青,台,晋,蒙,宁,新,桂,藏,港,澳">
            <!--ro, req, enum, subType:string, attr:opt{req, string}-->京
          </nonRestrictedProvinces>
          <nonRestrictedCityList size="64">
            <!--ro, opt, object, attr:size{req, int}-->
            <nonRestrictedCity>
              <!--ro, opt, object-->
              <nonRestrictedCityCode opt="A,B,C,D,E,F,G,H,I,J,K,L,M,N,O,P,Q,R,S,T,U,V,W,X,Y,Z">
                <!--ro, req, enum, subType:string, attr:opt{req, string}-->A
              </nonRestrictedCityCode>
            </nonRestrictedCity>
          </nonRestrictedCityList>
        </nonRestrictedRegion>
      </nonRestrictedRegionList>
      <timeSwitchList size="7">
```

```
      <!--ro, opt, object, attr:size{req, int}-->
      <timeSwitch>
        <!--ro, req, object-->
        <dayOfWeek opt="1,2,3,4,5,6,7">
          <!--ro, req, enum, subType:int, attr:opt{req, string}-->1
        </dayOfWeek>
        <timeRangeList size="4">
          <!--ro, req, object, attr:size{req, int}-->
          <timeRange>
            <!--ro, req, object-->
            <ID min="1" max="4">
              <!--ro, req, int, range:[1,4], attr:min{req, int},max{req, int}-->1
            </ID>
            <startHour min="0" max="23">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </startHour>
            <startMinute min="0" max="59">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </startMinute>
            <endHour min="0" max="23">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </endHour>
            <endMinute min="0" max="59">
              <!--ro, opt, int, attr:min{req, int},max{req, int}-->0
            </endMinute>
          </timeRange>
        </timeRangeList>
      </timeSwitch>
    </timeSwitchList>
  </vehicleRestrictions>
  <nonMotorOccupyZebraCrossing>
    <!--ro, opt, object-->
    <enabled opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enabled>
    <capNo opt="1,2,3">
      <!--ro, opt, int, attr:opt{req, string}-->3
    </capNo>
  </nonMotorOccupyZebraCrossing>
  <yellowGridParking>
    <!--ro, opt, object-->
    <enable opt="true,false">
      <!--ro, req, bool, attr:opt{req, string}-->true
    </enable>
    <lastTime min="1" max="180">
      <!--ro, req, int, range:[1,180], unit:s, attr:min{req, int},max{req, int}-->10
    </lastTime>
    <yellowGridParkingLineList>
      <!--ro, opt, array, subType:object, range:[0,2]-->
      <yellowGridParkingLine>
        <!--ro, opt, object-->
        <lineId opt="1,2">
          <!--ro, req, int, attr:opt{req, string}-->0
        </lineId>
        <lineName>
          <!--ro, req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList num="2">
          <!--ro, opt, array, subType:object, range:[0,2], attr:num{opt, int}-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </yellowGridParkingLine>
    </yellowGridParkingLineList>
    <capNo opt="3">
      <!--ro, req, int, attr:opt{req, string}-->3
    </capNo>
  </yellowGridParking>
  <cameraPosition opt="crossing,roadSection">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->crossing
  </cameraPosition>
</VideoEpolice>
```


11.5.1.36 Get the trigger mode parameters of video intersection violation system

Request URL

GET /ISAPI/ITC/TriggerMode/videoEpolice?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Camera ID (this optional parameter is supported when isSupportChannelVideoEpolice is "true" returned from /ISAPI/ITC/TriggerMode/capabilities).

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<VideoEpolice xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, the trigger mode parameters of video intersection violation system, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable intersection violation system-->true
  </enabled>
  <relatedLaneCount>
    <!--ro, req, int, number of linked lanes, range:[1,4]-->0
  </relatedLaneCount>
  <TrafficLight>
    <!--ro, req, object, traffic lights-->
    <source>
      <!--ro, req, enum, signal source, subType:string, desc:"IO, RS485, videoDetect, network"-->IO
    </source>
    <IOLightList>
      <!--ro, opt, array, list, subType:object-->
      <IOLight>
        <!--ro, opt, object, I/O access-->
        <lightId>
          <!--ro, req, int, traffic light No., range:[1,6]-->1
        </lightId>
        <lightType>
          <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:"left, straight, right"-->straight
        </lightType>
        <relatedIO>
          <!--ro, req, enum, linked I/O input, subType:string, desc:"NON_IO, IO1, IO2, IO3, IO4, IO5, IO6, IO7, IO8"-->IO7
        </relatedIO>
        <redLightStatus>
          <!--ro, req, enum, red light level status, subType:string, desc:"low" (low-level red light), "high" (high-level red light)-->low
        </redLightStatus>
      </IOLight>
    </IOLightList>
    <RS485LightList>
      <!--ro, opt, array, list, subType:object-->
      <RS485Light>
        <!--ro, opt, object, RS-485 access-->
        <lightId>
          <!--ro, req, int, traffic light No., range:[1,6]-->0
        </lightId>
        <lightType>
          <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:"left, right, straight"-->straight
        </lightType>
        <relatedLightChan>
          <!--ro, req, int, detector channel No. linked to red light, range:[0,16]-->0
        </relatedLightChan>
        <yellowLightChan>
          <!--ro, req, int, detector channel No. linked to yellow light, range:[0,16]-->0
        </yellowLightChan>
      </RS485Light>
    </RS485LightList>
    <NetworkLightList>
      <!--ro, opt, array, list, subType:object-->
      <NetworkLight>
        <!--ro, opt, object, network indicator light-->
        <lightId>
          <!--ro, req, int, traffic light No., range:[1,6]-->0
        </lightId>
        <lightType>
          <!--ro, req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
        </lightType>
        <relatedLightChan>
          <!--ro, req, int, detector channel No. linked to red light, range:[0,16]-->0
        </relatedLightChan>
        <yellowLightChan>
          <!--ro, req, int, detector channel No. linked to yellow light, range:[0,16]-->0
        </yellowLightChan>
      </NetworkLight>
    </NetworkLightList>
    <videoDetectLightList>
      <!--ro, opt, array, list, subType:object-->
      <videoDetectLight>
        <!--ro, opt, object, video detection access-->
        <videoDetectLightId>
          <!--ro, req, int, video detection signal No., range:[1,12]-->0
        </videoDetectLightId>
```

```
    <lightNum>
      <!--ro, req, int, number of traffic lights, range:[0,3]-->0
    </lightNum>
    <straightLight>
      <!--ro, req, bool, whether the straight driving light is on-->true
    </straightLight>
    <leftLight>
      <!--ro, req, bool, whether the left-turn light is on-->true
    </leftLight>
    <rightLight>
      <!--ro, req, bool, whether the right-turn light is on-->true
    </rightLight>
    <redLight>
      <!--ro, req, bool, whether the red light is on-->true
    </redLight>
    <greenLight>
      <!--ro, req, bool, whether the green light is on-->true
    </greenLight>
    <yellowLight>
      <!--ro, req, bool, whether the yellow light is on-->true
    </yellowLight>
    <yellowLightTime>
      <!--ro, req, int, yellow light duration, range:[0,10], unit:s-->0
    </yellowLightTime>
    <rectRegion>
      <!--ro, req, object, traffic light region, desc:the origin is the upper-left corner of the screen-->
      <positionLeft>
        <!--ro, req, int, left boundary, range:[0,65535]-->0
      </positionLeft>
      <positionTop>
        <!--ro, req, int, top boundary, range:[0,65535]-->0
      </positionTop>
      <positionRight>
        <!--ro, req, int, right boundary, range:[0,65535]-->0
      </positionRight>
      <positionBottom>
        <!--ro, req, int, bottom boundary, range:[0,65535]-->0
      </positionBottom>
    </rectRegion>
  </videoDetectLight>
</videoDetectLightList>
<autoDetectTrafficLight>
  <!--ro, opt, object, dep:and,{$.VideoEpolice.TrafficLight.source,eq,autoDetectTrafficLight}-->
  <detectRegionList>
    <!--ro, opt, array, subType:object, range:[0,8]-->
    <detectRegion>
      <!--ro, opt, object-->
      <height>
        <!--ro, req, float, range:[0,1]-->0.001
      </height>
      <width>
        <!--ro, req, float, range:[0,1]-->0.001
      </width>
      <x>
        <!--ro, req, float, range:[0,1]-->0.001
      </x>
      <y>
        <!--ro, req, float, range:[0,1]-->0.001
      </y>
    </detectRegion>
  </detectRegionList>
  <trafficLightPanelList>
    <!--ro, opt, array, subType:object, range:[0,8]-->
    <trafficLightPanel>
      <!--ro, opt, object-->
      <ruleID>
        <!--ro, req, int, range:[1,8]-->1
      </ruleID>
      <lampPanelDirection>
        <!--ro, opt, enum, subType:string-->level
      </lampPanelDirection>
      <lampPanelType>
        <!--ro, opt, enum, subType:string-->combination
      </lampPanelType>
      <lampSequence>
        <!--ro, opt, enum, subType:string-->redYellowGreen
      </lampSequence>
      <lampPanelRegion>
        <!--ro, opt, object-->
        <height>
          <!--ro, req, float, range:[0,1]-->0.001
        </height>
        <width>
          <!--ro, req, float, range:[0,1]-->0.001
        </width>
        <x>
          <!--ro, req, float, range:[0,1]-->0.001
        </x>
        <y>
          <!--ro, req, float, range:[0,1]-->0.001
        </y>
      </lampPanelRegion>
      <trafficLightList>
        <!--ro, opt, array, subType:object, range:[0,8]-->
```

```
            <!--ro, opt, array, subType:object, range:[0,8]-->
            <trafficLight>
              <!--ro, opt, object-->
              <trafficLightType>
                <!--ro, opt, enum, subType:string-->roundLight
              </trafficLightType>
              <lampPanelRegion>
                <!--ro, opt, object-->
                <height>
                  <!--ro, req, float, range:[0,1]-->0.001
                </height>
                <width>
                  <!--ro, req, float, range:[0,1]-->0.001
                </width>
                <x>
                  <!--ro, req, float, range:[0,1]-->0.001
                </x>
                <y>
                  <!--ro, req, float, range:[0,1]-->0.001
                </y>
              </lampPanelRegion>
            </trafficLight>
          </trafficLightList>
        </trafficLightPanel>
      </trafficLightPanelList>
    </autoDetectTrafficLight>
  </TrafficLight>
  <LaneParamList>
    <!--ro, req, array, lane parameter list, subType:object-->
    <LaneParam>
      <!--ro, req, object, lane parameters-->
      <laneId>
        <!--ro, req, int, lane No., range:[1,6]-->0
      </laneId>
      <laneDirectionType>
        <!--ro, req, enum, lane direction No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <relatedDriveWay>
        <!--ro, req, int, linked lane No., range:[1,99]-->0
      </relatedDriveWay>
      <recordEnable>
        <!--ro, req, bool, whether to enable recording when red light running is detected-->true
      </recordEnable>
      <recordType>
        <!--ro, req, enum, recording type, subType:string, desc:"preRecord, delayRecord"-->preRecord
      </recordType>
      <beginTime>
        <!--ro, req, enum, recording start time, subType:int, desc:"0, 1, 2, 3"-->0
      </beginTime>
      <preRecordTime>
        <!--ro, req, int, pre-record time, range:[0,100], unit:s-->0
      </preRecordTime>
      <recordDelayTime>
        <!--ro, req, int, post-record time, range:[0,100], unit:s-->0
      </recordDelayTime>
      <recordTimeOut>
        <!--ro, req, int, recording timeout, range:[0,100], unit:s-->0
      </recordTimeOut>
      <signSpeed>
        <!--ro, opt, int, marked speed limit, range:[0,255], unit:km/h-->0
      </signSpeed>
      <speedLimit>
        <!--ro, opt, int, speed limit, range:[0,255], unit:km/h-->0
      </speedLimit>
      <usageType>
        <!--ro, req, enum, lane usage,only one of the options can be supported at a time, subType:string, desc:"carriageWay" (normal lane),"bus" (fast
lane),"slow" (slow lane), "nonmotor" (non-motor lane), "reverse" (reverse lane), "emergency" (emergency lane)-->carriageWay
      </usageType>
      <LaneUsageList>
        <!--ro, opt, array, lane usage, one or multiple options can be supported at a time, subType:object, dep:and,
{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special}, desc:lane usage, one or multiple options can be supported at a time-->
        <laneUsage>
          <!--ro, req, enum, lane usage, subType:string, desc:"bus" (bus lane), "banTrucks" (trucks prohibited lane), "banLeft" (left-turn prohibited lane),
"banRight" (right-turn prohibited lane), "motor" (motorcycle lane)-->bus
        </laneUsage>
      </LaneUsageList>
      <banTrucksTimeSwitchList>
        <!--ro, req, array, list of large vehicle prohibited time periods, subType:object-->
        <banTrucksTimeSwitch>
          <!--ro, req, object, large vehicle prohibited time periods-->
          <timeId>
            <!--ro, req, int, time period No., range:[1,4]-->0
          </timeId>
          <startHour>
            <!--ro, opt, int, start time (hour), range:[0,23]-->0
          </startHour>
          <startMinute>
            <!--ro, opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--ro, opt, int, end time (hour), range:[0,23]-->0
          </endHour>
          <endMinute>
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      <!--ro, opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </banTrucksTimeSwitch>
</banTrucksTimeSwitchList>
<busWayTimeSwitchList>
  <!--ro, req, array, list, subType:object-->
  <busWayTimeSwitch>
    <!--ro, req, object, bus allowed passing time-->
    <timeId>
      <!--ro, req, int, time period No., range:[1,4]-->0
    </timeId>
    <startHour>
      <!--ro, opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--ro, opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--ro, opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--ro, opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </busWayTimeSwitch>
</busWayTimeSwitchList>
<busCapDirectionType>
  <!--ro, opt, enum, subType:string-->leftRightStraight
</busCapDirectionType>
<emergencyTimeSwitchList>
  <!--ro, opt, array, time period for emergency lanes, subType:object-->
  <emergencyTimeSwitch>
    <!--ro, req, object, Emergency Lane Time-->
    <timeId>
      <!--ro, req, int, time period No., range:[1,4]-->0
    </timeId>
    <startHour>
      <!--ro, opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--ro, opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--ro, opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--ro, opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </emergencyTimeSwitch>
</emergencyTimeSwitchList>
<banLeftTimeSwitchList>
  <!--ro, req, array, left-turn prohibited time period, subType:object-->
  <banLeftTimeSwitch>
    <!--ro, req, object, left-turn prohibited time-->
    <timeId>
      <!--ro, req, int, time period No., range:[1,4]-->0
    </timeId>
    <startHour>
      <!--ro, opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--ro, opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--ro, opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--ro, opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </banLeftTimeSwitch>
</banLeftTimeSwitchList>
<banRightTimeSwitchList>
  <!--ro, req, array, right-turn prohibited time period, subType:object-->
  <banRightTimeSwitch>
    <!--ro, req, object, right-turn prohibited time-->
    <timeId>
      <!--ro, req, int, time period No., range:[1,4]-->0
    </timeId>
    <startHour>
      <!--ro, opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--ro, opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--ro, opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--ro, opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </banRightTimeSwitch>
</banRightTimeSwitchList>
<banStraightTimeSwitchList>
  <!--ro, req, array, straight prohibited time period, subType:object-->
```

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        <banStraightTimeSwitch>
          <!--ro, req, object, straight prohibited time-->
          <timeId>
            <!--ro, req, int, index, range:[1,4]-->0
          </timeId>
          <startHour>
            <!--ro, opt, int, start time (hour), range:[0,23]-->0
          </startHour>
          <startMinute>
            <!--ro, opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--ro, opt, int, end time (hour), range:[0,23]-->0
          </endHour>
          <endMinute>
            <!--ro, opt, int, end time (minute), range:[0,59]-->0
          </endMinute>
        </banStraightTimeSwitch>
      </banStraightTimeSwitchList>
      <variableLaneTimeSwitchList>
        <!--ro, opt, array, time period for changeable lanes, subType:object, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special},
desc:time period for changeable lanes-->
        <variableLaneTimeSwitch>
          <!--ro, req, object, time for changeable lanes-->
          <timeId>
            <!--ro, req, int, index, range:[1,4]-->0
          </timeId>
          <startHour>
            <!--ro, opt, int, start time (hour), range:[0,23]-->0
          </startHour>
          <startMinute>
            <!--ro, opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--ro, opt, int, end time (hour), range:[0,23]-->0
          </endHour>
          <endMinute>
            <!--ro, opt, int, end time (minute), range:[0,59]-->0
          </endMinute>
          <laneDirection>
            <!--ro, req, enum, Driving Direction, subType:string, desc:"unknown", "left", "straight", "leftStraight", "right", "leftRight", "rightStraight",
"leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->unknown
          </laneDirection>
        </variableLaneTimeSwitch>
      </variableLaneTimeSwitchList>
      <variableLaneTimeSwitchType>
        <!--ro, opt, enum, subType:string, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special}-->week
      </variableLaneTimeSwitchType>
      <variableLaneWeekSwitchList>
        <!--ro, opt, array, subType:object, range:[0,7], dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.usageType,eq,special}-->
        <variableLaneWeekSwitch>
          <!--ro, req, object-->
          <dayOfWeek>
            <!--ro, req, enum, subType:int-->1
          </dayOfWeek>
          <timeRangeList>
            <!--ro, req, array, subType:object, range:[0,4]-->
            <timeRange>
              <!--ro, req, object-->
              <ID>
                <!--ro, req, int, range:[1,4]-->1
              </ID>
              <startHour>
                <!--ro, opt, int, range:[0,23]-->0
              </startHour>
              <startMinute>
                <!--ro, opt, int, range:[0,59]-->0
              </startMinute>
              <endHour>
                <!--ro, opt, int, range:[0,23]-->0
              </endHour>
              <endMinute>
                <!--ro, opt, int, range:[0,59]-->0
              </endMinute>
              <laneDirection>
                <!--ro, opt, enum, subType:string-->unknown
              </laneDirection>
            </timeRange>
          </timeRangeList>
        </variableLaneWeekSwitch>
      </variableLaneWeekSwitchList>
      <radarEnabled>
        <!--ro, opt, bool, whether to enable radar speed detection-->true
      </radarEnabled>
      <Radar>
        <!--ro, opt, object, radar parameters-->
        <radarType>
          <!--ro, req, enum, radar type, subType:string, desc:"none", "adr" (Andoray), "adr4Byte" (Andoray) (no radar controller), "olvia" (Olvia), "csr"
(TransMicrowave), "IOBox" (radar connecting I/O expansion box), "SSTK" (WHST), "custom" (custom type)-->none
        </radarType>
        <radarSensitivity>
          <!--ro, opt, int, radar sensitivity, range:[0,65535]-->0
        </radarSensitivity>
        <radarAngle>
```

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          <!--ro, opt, int, angle between the radar and the horizontal direction, range:[0,90]-->0
        </radarAngle>
        <validRadarSpeedTime>
          <!--ro, req, int, effective duration of radar speed, range:[0,2000]-->0
        </validRadarSpeedTime>
        <radarLinearCorrection>
          <!--ro, req, float, linear correction parameter, range:[0.0,2.0]-->0.000
        </radarLinearCorrection>
        <radarConstantCorrection>
          <!--ro, req, int, constant correction parameter, range:[-100,100]-->0
        </radarConstantCorrection>
        <radarRS485>
          <!--ro, req, int, radar linked RS-485 port No., range:[1,5]-->0
        </radarRS485>
      </Radar>
      <directionType>
        <!--ro, req, enum, lane direction, subType:string, desc:"unknown, left, straight, leftStraight, right, leftRight, rightStraight, leftRightStraight,
leftWait, straightWait, strightWaitRight"-->unknown
      </directionType>
      <stopLineDistance>
        <!--ro, req, int, distance between the vehicle and the stop line in the second captured picture of red light running violation, range:[0,300]-->0
      </stopLineDistance>
      <loopSensitivity>
        <!--ro, req, int, loop sensitivity, range:[1,100]-->1
      </loopSensitivity>
      <capPositionOffset>
        <!--ro, req, int, minimum offset between the capture positions of the second and the third picture, range:[0,1500]-->0
      </capPositionOffset>
      <detectSnapType>
        <!--ro, req, object, type of capture triggered by video detection-->
        <post>
          <!--ro, req, bool, checkpoint-->true
        </post>
        <postCapNo>
          <!--ro, req, int, checkpoint capture times, range:[1,3]-->0
        </postCapNo>
        <checkpointQuickTriggerEnabled>
          <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.post,eq,true}-->true
        </checkpointQuickTriggerEnabled>
        <overSpeed>
          <!--ro, req, bool, whether to enable speeding detection-->true
        </overSpeed>
        <overSpeedCapNo>
          <!--ro, req, int, number of captured pictures in which the speeding violation is detected, range:[2,3]-->0
        </overSpeedCapNo>
        <driveLine>
          <!--ro, req, bool, whether to enable driving on lane line detection-->true
        </driveLine>
        <driveLineCapNo>
          <!--ro, req, int, number of continuously captured pictures in which the driving on lane line happened, range:[2,3]-->0
        </driveLineCapNo>
        <driveLineSensitivity>
          <!--ro, req, int, sensitivity of driving on lane line detection, range:[0,100]-->0
        </driveLineSensitivity>
        <reverse>
          <!--ro, req, bool, whether to enable wrong-way driving detection-->true
        </reverse>
        <reverseCapNo>
          <!--ro, req, int, number of captured pictures in which the wrong-way driving is detected, range:[1,3]-->1
        </reverseCapNo>
        <redLight>
          <!--ro, req, bool, whether the red light is on-->true
        </redLight>
        <redLightMode>
          <!--ro, req, enum, red light running mode, subType:int, desc:0-by direction,1-by lane-->0
        </redLightMode>
        <nonMotorRedLightEnabled>
          <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.redLight,eq,true}-->true
        </nonMotorRedLightEnabled>
        <direction>
          <!--ro, req, bool, whether to enable detection of driving in wrong lane at intersection-->true
        </direction>
        <illegalPriority>
          <!--ro, req, enum, traffic violation priority, subType:int, desc:0-none, 1-red light running first-->0
        </illegalPriority>
        <intersectionCongest>
          <!--ro, req, bool, whether to enable intersection congestion detection-->true
        </intersectionCongest>
        <nonDriveWay>
          <!--ro, req, bool, whether to enable detection of motor vehicle on non-motor vehicle lane-->true
        </nonDriveWay>
        <nonDriveWayCapNo>
          <!--ro, req, int, number of captured pictures in which motor vehicle on non-motor vehicle lane happened, range:[2,3]-->0
        </nonDriveWayCapNo>
        <nonDriveWayTime>
          <!--ro, req, int, capture duration of motor vehicle on non-motor vehicle lane, range:[10,240], unit:s-->10
        </nonDriveWayTime>
        <changeLane>
          <!--ro, req, bool, whether to enable illegal lane change detection-->true
        </changeLane>
        <changeLaneCapNo>
          <!--ro, req, int, number of captured pictures in which illegal lane change is detected, range:[2,3]-->0
        </changeLaneCapNo>
```

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<changeLaneFirstImageReserveEnabled>
  <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.changeLane,eq,true}-->true
</changeLaneFirstImageReserveEnabled>
<banSign>
  <!--ro, req, bool, whether to enable ban breaking detection-->true
</banSign>
<banSignCapNo>
  <!--ro, req, int, number of continuously captured pictures in which the ban breaking happened, range:[2,3]-->0
</banSignCapNo>
<notCatchingReverseVehiclesEnabled>
  <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.banSign,eq,true}-->true
</notCatchingReverseVehiclesEnabled>
<intersectionStop>
  <!--ro, opt, object, whether to enable intersection parking detection-->
  <enable>
    <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
  </enable>
  <lastTime>
    <!--ro, req, int, congestion duration, range:[10,180], unit:s-->0
  </lastTime>
  <sensitivity>
    <!--ro, opt, int, detection sensitivity, range:[1,100]-->0
  </sensitivity>
  <bigCarRatio>
    <!--ro, opt, int, range:[1,100]-->0
  </bigCarRatio>
  <smallCarRatio>
    <!--ro, opt, int, range:[1,100]-->0
  </smallCarRatio>
  <capNo>
    <!--ro, opt, int, range:[3,3]-->3
  </capNo>
</intersectionStop>
<greenLightStop>
  <!--ro, req, bool, whether to enable detection of parking at green light-->true
</greenLightStop>
<illegalTurn>
  <!--ro, req, bool, illegal turning-->true
</illegalTurn>
<reverseLaneCheckpointFilteringEnabled>
  <!--ro, opt, bool, dep:and,{$.VideoEpolice.LaneParamList[*].LaneParam.detectSnapType.illegalTurn,eq,true}-->true
</reverseLaneCheckpointFilteringEnabled>
<gasser>
  <!--ro, opt, object, queue jumping-->
  <enabled>
    <!--ro, req, bool, whether to enable intersection violation system-->true
  </enabled>
  <sensitivity>
    <!--ro, req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <congestionThreshold>
    <!--ro, req, int, congestion threshold, range:[1,100]-->0
  </congestionThreshold>
  <capInterval>
    <!--ro, req, int, time interval of continuous capture (2 to 3 pictures), range:[0,6000], unit:ms-->0
  </capInterval>
  <capNo>
    <!--ro, opt, int, range:[3,3]-->3
  </capNo>
</gasser>
<congestion>
  <!--ro, opt, object, congestion-->
  <enabled>
    <!--ro, opt, bool, whether to enable intersection violation system-->true
  </enabled>
  <interval>
    <!--ro, req, int, alarm time interval, range:[1,10], unit:min-->0
  </interval>
  <sensitivity>
    <!--ro, req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <lastTime>
    <!--ro, req, int, congestion duration, range:[1,60], unit:s-->0
  </lastTime>
  <capNo>
    <!--ro, opt, int, range:[1,1]-->1
  </capNo>
</congestion>
<accident>
  <!--ro, opt, object, accident detection-->
  <enabled>
    <!--ro, req, bool, whether to enable intersection violation system-->true
  </enabled>
  <lastTime>
    <!--ro, req, int, congestion duration, range:[1,180], unit:s-->0
  </lastTime>
  <interval>
    <!--ro, req, int, alarm time interval, range:[1,60], unit:min-->0
  </interval>
  <capNo>
    <!--ro, opt, int, range:[1,1]-->1
  </capNo>
</accident>
<nonMotorEnabled>
```

```
    <nonMotorEnabled>
      <!--ro, opt, bool, whether to enable non-motor vehicle detection-->true
    </nonMotorEnabled>
    <runGreenLight>
      <!--ro, opt, bool, whether to enable green light running detection-->true
    </runGreenLight>
    <runGreenLightParam>
      <!--ro, opt, object, parameters of green light running detection-->
      <detectEvent>
        <!--ro, req, enum, Detection Logic, subType:string, desc:"runGreenLight" (running green light in congestion), "crossingStop" (overstay at intersection)-->runGreenLight
      </detectEvent>
      <interval>
        <!--ro, req, int, capture interval, range:[1,180], unit:s, desc:capture interval-->15
      </interval>
      <secondLampState>
        <!--ro, opt, enum, traffic light status of the second capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
      </secondLampState>
      <threeLampState>
        <!--ro, opt, enum, traffic light status of the third capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
      </threeLampState>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
    </runGreenLightParam>
    <intervalType>
      <!--ro, req, enum, Burst Interval Type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
    </intervalType>
    <IntervalList>
      <!--ro, opt, array, list, subType:object-->
      <Interval>
        <!--ro, req, object, interval information-->
        <intervalValue>
          <!--ro, req, int, burst interval, desc:unit: ms/dm; capture interval by time is between 70 and 2000; capture interval by distance is between 1 and 20-->0
        </intervalValue>
      </Interval>
    </IntervalList>
    <redLightCapNo>
      <!--ro, opt, int, number of captured violation picture(s), range:[3,4]-->0
    </redLightCapNo>
    <directionCapNo>
      <!--ro, opt, int, number of captured pictures of driving against direction guidance, range:[3,4]-->0
    </directionCapNo>
    <banSignType>
      <!--ro, opt, enum, capture type list, subType:string, desc:"truck", "yellowPlate" (yellow license plate vehicle), "standard", "bus"-->truck
    </banSignType>
    <greenLightStopSensitivity>
      <!--ro, opt, int, stopping sensitivity, range:[1,100]-->0
    </greenLightStopSensitivity>
    <banNonMotorSign>
      <!--ro, opt, object, non-motor vehicle prohibition violation-->
      <enabled>
        <!--ro, req, bool, whether to enable intersection violation system-->true
      </enabled>
      <capNo>
        <!--ro, req, int, the number of captured pictures, range:[2,3]-->0
      </capNo>
    </banNonMotorSign>
    <nonMotorPassenger>
      <!--ro, opt, object, Manned Non-Motor Vehicle-->
      <enabled>
        <!--ro, req, bool, whether to enable intersection violation system-->true
      </enabled>
      <capNo>
        <!--ro, req, int, the number of captured pictures, range:[1,2]-->0
      </capNo>
      <refineMannedIllegalCode>
        <!--ro, opt, bool, whether to enable detailed manned violation code-->true
      </refineMannedIllegalCode>
    </nonMotorPassenger>
    <SuonaDet>
      <!--ro, opt, object, whistle detection-->
      <enabled>
        <!--ro, opt, bool, whether to enable intersection violation system-->true
      </enabled>
      <capNo>
        <!--ro, opt, int, the number of captured pictures, range:[2,3]-->0
      </capNo>
      <sensitivity>
        <!--ro, opt, int, detection sensitivity, range:[1,100]-->0
      </sensitivity>
    </SuonaDet>
    <BanMotorDet>
      <!--ro, opt, object, motorcycle driving in prohibited areas-->
      <enabled>
        <!--ro, opt, bool, whether to enable intersection violation system-->false
      </enabled>
      <capNo>
        <!--ro, opt, int, the number of captured pictures, range:[1,3]-->2
      </capNo>
      <sensitivity>
        <!--ro, opt, int, detection sensitivity, range:[1,100]-->0
      </sensitivity>
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    </BanMotorDet>
    <BanSignTypeList>
      <!--ro, opt, array, capture type list, subType:object, desc:capture type list-->
      <banSignType>
        <!--ro, opt, enum, prohibition violation type, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus", "yellowbus" (yellow license plate bus), "dangercar" (dangerous goods transport vehicle), "instructioncar" (driver-training vehicle)-->standard
      </banSignType>
    </BanSignTypeList>
    <TurnSignalDet>
      <!--ro, opt, object, turning signal detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
      <enabled>
        <!--ro, opt, bool, whether to enable intersection violation system-->false
      </enabled>
      <turnFiliter>
        <!--ro, opt, bool, whether to filter left turn and right turn-->true
      </turnFiliter>
      <detectEvent>
        <!--ro, opt, enum, capture type, subType:string, desc:"turnLeft", "turnRight", "changeLane"-->changeLane
      </detectEvent>
      <detectEventList>
        <!--ro, opt, array, subType:object, range:[1,3]-->
        <detectEvent>
          <!--ro, opt, enum, subType:string-->changeLane
        </detectEvent>
      </detectEventList>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
    </TurnSignalDet>
    <ReflectiveTapeDet>
      <!--ro, opt, object, reflective stripe detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
      <enabled>
        <!--ro, opt, bool, whether to enable intersection violation system-->false
      </enabled>
      <capNo>
        <!--ro, opt, int, the number of captured pictures, range:[1,3]-->2
      </capNo>
    </ReflectiveTapeDet>
    <ZipperDriveDet>
      <!--ro, opt, object, pass alternatively detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
      <enabled>
        <!--ro, opt, bool, whether to enable intersection violation system-->true
      </enabled>
      <carNum>
        <!--ro, opt, int, capture triggered number of vehicles, range:[1,3]-->2
      </carNum>
      <parallDistance>
        <!--ro, opt, int, parallel distance, range:[0,200]-->0
      </parallDistance>
    </ZipperDriveDet>
    <reversing>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->true
      </enabled>
      <snapTime>
        <!--ro, opt, int, range:[2,3], unit:s-->2
      </snapTime>
    </reversing>
    <NotYieldToRightRoad>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->false
      </enabled>
      <firstCapturePosition>
        <!--ro, req, int, range:[0,100], step:1-->20
      </firstCapturePosition>
      <lateralDistance>
        <!--ro, req, int, range:[20,350], step:1-->150
      </lateralDistance>
      <longitudinalDistance>
        <!--ro, req, int, range:[10,150], step:1-->50
      </longitudinalDistance>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
    </NotYieldToRightRoad>
    <EnterNotYieldToInRoundabout>
      <!--ro, opt, object-->
      <enabled>
        <!--ro, req, bool-->false
      </enabled>
      <firstCapturePosition>
        <!--ro, req, int, range:[0,100], step:1-->20
      </firstCapturePosition>
      <lateralDistance>
        <!--ro, req, int, range:[20,350], step:1-->150
      </lateralDistance>
      <longitudinalDistance>
        <!--ro, req, int, range:[10,150], step:1-->50
      </longitudinalDistance>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
```

```
        </capNo>
      </EnterNotYieldToInRoundabout>
      <RampNotYieldToMainRoad>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, req, bool-->false
        </enabled>
        <lateralDistance>
          <!--ro, req, int, range:[10,90], step:1-->60
        </lateralDistance>
        <capNo>
          <!--ro, opt, int, range:[3,3]-->3
        </capNo>
      </RampNotYieldToMainRoad>
      <reverseSensitivity>
        <!--ro, opt, int, range:[0,100]-->0
      </reverseSensitivity>
    </detectSnapType>
    <ViolationDetectLineList>
      <!--ro, req, array, list, subType:object-->
      <ViolationDetectLine>
        <!--ro, req, object, draw violation detection line for a single lane-->
        <lineName>
          <!--ro, req, enum, detection line name, subType:string, desc:"laneLine, stopLine, redlightLine, cancelLine, waitLine"-->laneLine
        </lineName>
        <lineType>
          <!--ro, opt, enum, detection line type, subType:string, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line), 
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
        </lineType>
        <customName>
          <!--ro, opt, string, custom triggering line name-->test
        </customName>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ViolationDetectLine>
    </ViolationDetectLineList>
    <firstShowTailVep>
      <!--ro, opt, enum, place of the first capture time, subType:int, desc:0 (none), 1 (expose vehicle tail), 2 (expose vehicle tail and LPR success), 
3(parking over stop line)-->0
    </firstShowTailVep>
    <lowSpeedLimit>
      <!--ro, opt, int, low speed limit for small-sized vehicle, range:[0,255]-->0
    </lowSpeedLimit>
    <bigCarSignSpeed>
      <!--ro, opt, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
    </bigCarSignSpeed>
    <bigCarSpeedLimit>
      <!--ro, opt, int, maximum speed limit for large-sized vehicle, range:[0,255]-->0
    </bigCarSpeedLimit>
    <bigCarLowSpeedLimit>
      <!--ro, opt, int, low speed limit for large-sized vehicle, range:[0,255]-->0
    </bigCarLowSpeedLimit>
    <noBanLeftTypeList>
      <!--ro, opt, array, allowlist of left-turn vehicle types, subType:object-->
      <noBanLeftType>
        <!--ro, opt, enum, vehicle types which are allowed to turn left, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle), 
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
      </noBanLeftType>
    </noBanLeftTypeList>
    <noBanRightTypeList>
      <!--ro, opt, array, allowlist of right-turn vehicle types, subType:object-->
      <noBanRightType>
        <!--ro, opt, enum, vehicle types which are allowed to left-turn, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle), 
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
      </noBanRightType>
    </noBanRightTypeList>
    <dangerCarSignSpeed>
      <!--ro, opt, int, range:[0,255], unit:km/h-->0
    </dangerCarSignSpeed>
    <dangerCarSpeedLimit>
      <!--ro, opt, int, range:[0,255], unit:km/h-->0
    </dangerCarSpeedLimit>
    <dangerCarLowSpeedLimit>
      <!--ro, opt, int, range:[0,255], unit:km/h-->0
    </dangerCarLowSpeedLimit>
    <dangerCarHighSpeed>
      <!--ro, opt, int, range:[0,255], unit:km/h-->0
    </dangerCarHighSpeed>
    <carHighSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </carHighSpeed>
    <bigCarHighSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarHighSpeed>
```

```
        <nonMotorSignSpeed>
          <!--ro, opt, int, range:[0,255], unit:km/h-->0
        </nonMotorSignSpeed>
        <nonMotorSpeedLimit>
          <!--ro, opt, int, range:[0,255], unit:km/h-->0
        </nonMotorSpeedLimit>
        <nonMotorLowSpeedLimit>
          <!--ro, opt, int, range:[0,255], unit:km/h-->0
        </nonMotorLowSpeedLimit>
        <nonMotorHighSpeed>
          <!--ro, opt, int, range:[0,255], unit:km/h-->0
        </nonMotorHighSpeed>
      </LaneParam>
    </LaneParamList>
    <VirtualLaneList>
      <!--ro, req, array, list, subType:object-->
      <VirtualLane>
        <!--ro, req, object, virtual lane line-->
        <lineName>
          <!--ro, req, enum, detection line name, subType:string, desc:"laneLine, stopLine, redlightLine, cancelLine, waitLine"-->laneRightBoundaryLine
        </lineName>
        <lineType>
          <!--ro, opt, enum, detection line type, subType:string, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line),
"doubleYellow" (double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
        </lineType>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </VirtualLane>
    </VirtualLaneList>
    <Posting>
      <!--ro, opt, object, positioning frame-->
      <postingEnable>
        <!--ro, req, bool, whether to enable positioning frame-->true
      </postingEnable>
      <postingSize>
        <!--ro, req, enum, positioning frame size, subType:string, desc:"small" (300 pixels), "medium" (400 pixels), "large" (500 pixels)-->small
      </postingSize>
    </Posting>
    <leftNotYieldStraight>
      <!--ro, opt, object, left turn not yield to straight-->
      <enable>
        <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
    </leftNotYieldStraight>
    <rightNotYieldLeft>
      <!--ro, opt, object, right turn not yield to left turn-->
      <enable>
        <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
    </rightNotYieldLeft>
    <uTurnNotYieldGoingStaight>
      <!--ro, opt, object, U-Turn not Yield to Straight-->
      <enable>
        <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
    </uTurnNotYieldGoingStaight>
    <leftSideDrivingAfterLeftTurn>
      <!--ro, opt, object, turn left in short turn radius-->
      <enable>
        <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <capNo>
        <!--ro, opt, int, range:[3,3]-->3
      </capNo>
    </leftSideDrivingAfterLeftTurn>
    <failYieldPed>
      <!--ro, opt, object, failing to yield for pedestrians-->
      <enable>
        <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <vehicleFromDirection>
        <!--ro, req, enum, driving direction of the vehicle which does not yield to pedestrian, subType:string, desc:"Front, Back"-->Front
      </vehicleFromDirection>
      <cameraPosition>
```

```xml
      <!--ro, req, enum, camera position, subType:string, desc:"crossing, roadSection"-->crossing
    </cameraPosition>
    <noDisplacementPedEnabled>
      <!--ro, req, bool, whether to enable no displacement capture of pedestrian-->true
    </noDisplacementPedEnabled>
    <failYieldPedLineList>
      <!--ro, opt, array, list, subType:object-->
      <failYieldPedLine>
        <!--ro, opt, object, failing to yield for pedestrians line-->
        <lineId>
          <!--ro, req, int, line ID, range:[1,2]-->0
        </lineId>
        <lineName>
          <!--ro, req, string, detection line name, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </failYieldPedLine>
    </failYieldPedLineList>
    <notification>
      <!--ro, opt, object, capture linkage, desc:pedestrian linkage-->
      <enable>
        <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <ip>
        <!--ro, opt, string, IP address of traffic camera, range:[0,32]-->test
      </ip>
      <port>
        <!--ro, opt, int, port No. of traffic camera, range:[0,65535]-->1
      </port>
      <username>
        <!--ro, opt, string, user name of traffic camera, range:[0,32]-->test
      </username>
      <password>
        <!--ro, opt, string, password of traffic camera, range:[0,16]-->test
      </password>
      <sensitivity>
        <!--ro, req, int, detection sensitivity, range:[0,100]-->0
      </sensitivity>
      <RS485Port>
        <!--ro, req, int, RS-485 port No. for the traffic light to access the camera, range:[1,10]-->1
      </RS485Port>
    </notification>
    <failYieldPedSensitivity>
      <!--ro, req, int, Not Yielding to Pedestrian Sensitivity, range:[0,100]-->0
    </failYieldPedSensitivity>
    <failYieldnonMotor>
      <!--ro, opt, bool, whether to enable capturing not yielding to pedestrian-->true
    </failYieldnonMotor>
    <failYieldNum>
      <!--ro, opt, int, Number of Pedestrian Not Yielded to, range:[1,7]-->1
    </failYieldNum>
    <failYieldAreaType>
      <!--ro, opt, enum, detection area of failing to yield for pedestrians, subType:string, desc:"rectangle", "polygon"-->rectangle
    </failYieldAreaType>
    <failYieldDistanceMode>
      <!--ro, opt, enum, distance mode of failing to yield for pedestrians, subType:string, desc:"car" (distance from pedestrian to vehicle center), "lane"
(distance from pedestrian to lane center)-->car
    </failYieldDistanceMode>
    <failYieldPedSnapPosition>
      <!--ro, opt, int, distance range for capturing failing to yield for pedestrians, range:[0,100]-->20
    </failYieldPedSnapPosition>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </failYieldPed>
  <illegalParking>
    <!--ro, opt, object, illegal parking-->
    <enable>
      <!--ro, req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
    <sensitivity>
      <!--ro, req, int, detection sensitivity, range:[0,100]-->0
    </sensitivity>
    <lastTime>
      <!--ro, req, int, congestion duration, range:[1,180], unit:s-->0
    </lastTime>
    <illegalParkingLineList>
      <!--ro, opt, array, list of illegal parking lines, subType:object, range:[0,3]-->
      <illegalParkingLine>
        <!--ro, opt, object, illegal parking lines-->
        <lineId>
          <!--ro, req, int, line ID, range:[1,3]-->0
        </lineId>
```

```xml
        <lineName>
          <!--ro, req, string, detection line name, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--ro, opt, array, list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--ro, req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </illegalParkingLine>
    </illegalParkingLineList>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </illegalParking>
  <detectMode>
    <!--ro, opt, enum, detection mode, subType:string, desc:"video", "coilLoop" (video and coil)-->video
  </detectMode>
  <helmet>
    <!--ro, opt, bool, Without Helmet-->true
  </helmet>
  <helmetCapNo>
    <!--ro, opt, int, number of captured pictures in which the driver does not wear the helmet, range:[1,3]-->1
  </helmetCapNo>
  <nonMotorExist>
    <!--ro, opt, bool, Non-Motor Vehicle on Motor Vehicle Lane-->true
  </nonMotorExist>
  <nonMotorExistCapNo>
    <!--ro, opt, int, number of captured pictures in which the motor lane is occupied by non-motor vehicle, range:[2,3]-->2
  </nonMotorExistCapNo>
  <motorIllegalCarrying>
    <!--ro, opt, bool, Illegal Manned Motor Vehicle-->true
  </motorIllegalCarrying>
  <nomotorIllegalCarrying>
    <!--ro, opt, bool, Illegal Manned Non-Motor Vehicle-->true
  </nomotorIllegalCarrying>
  <canopy>
    <!--ro, opt, bool, Umbrella Tent Installation Violation on Non-Motor Vehicle-->true
  </canopy>
  <pedRedLight>
    <!--ro, opt, bool, Pedestrian Red Light Running-->true
  </pedRedLight>
  <pedOnNomotorLane>
    <!--ro, opt, bool, whether to detect pedestrian walking on non-motor vehicle lane-->true
  </pedOnNomotorLane>
  <pedOnMotorLane>
    <!--ro, opt, bool, whether to detect pedestrian walking on motor vehicle lane-->true
  </pedOnMotorLane>
  <capType>
    <!--ro, opt, enum, "vehicle,nonMotor,human,face", subType:string, desc:"vehicle, nonMotor, human, face"-->mixObject
  </capType>
  <deferredFrmNum>
    <!--ro, req, int, delay a few seconds to trigger the mode, range:[0,25]-->0
  </deferredFrmNum>
  <parkingOverStopLine>
    <!--ro, opt, object, Over Stop Line-->
    <enabled>
      <!--ro, req, bool, whether to enable intersection violation system-->true
    </enabled>
  </parkingOverStopLine>
  <speedDetectMode>
    <!--ro, opt, enum, speed detection mode, subType:string, desc:"none", "radar"-->none
  </speedDetectMode>
  <NotSlowZebraCrossingEvent>
    <!--ro, opt, object, not slowing down at zebra crossing-->
    <enabled>
      <!--ro, req, bool, whether to enable intersection violation system-->true
    </enabled>
    <capNo>
      <!--ro, req, int, the number of captured pictures, range:[1,3], step:1-->1
    </capNo>
    <lowSpeed>
      <!--ro, req, int, lowest speed limit (km/h), range:[0,100], step:1-->0
    </lowSpeed>
    <highSpeed>
      <!--ro, req, int, highest speed limit (km/h), range:[0,100], step:1-->0
    </highSpeed>
    <LineInfoList>
      <!--ro, opt, object, list of lane line information-->
      <LineInfo>
        <!--ro, opt, object, single lane line information-->
        <lineId>
          <!--ro, req, int, lane index, range:[1,3]-->0
        </lineId>
        <RegionCoordinatesList>
          <!--ro, opt, array, list, subType:object, range:[0,2]-->
          <RegionCoordinates>
```

```xml
          <!--ro, opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </LineInfo>
  </LineInfoList>
</NotSlowZebraCrossingEvent>
<TurnRightStop>
  <!--ro, opt, object, large-sized vehicle not stopping before right turn-->
  <enabled>
    <!--ro, req, bool, whether to enable intersection violation system-->true
  </enabled>
  <capNo>
    <!--ro, req, int, the number of captured pictures, range:[1,3], step:1-->3
  </capNo>
  <durationTime>
    <!--ro, req, int, duration, range:[1,600], unit:s-->5
  </durationTime>
  <slowUp>
    <!--ro, req, int, upper limit of deceleration, range:[0,100], step:1, unit:km/h-->40
  </slowUp>
  <slowStep>
    <!--ro, req, int, deceleration range, range:[0,50], step:1, unit:km/h-->5
  </slowStep>
  <slowDown>
    <!--ro, req, int, lower limit of deceleration, range:[0,100], step:1, unit:km/h-->10
  </slowDown>
  <detectMode>
    <!--ro, req, enum, deceleration mode, subType:string, desc:"stop", "slow"-->stop
  </detectMode>
  <SlowLane>
    <!--ro, opt, object, Deceleration Line-->
    <RegionCoordinatesList>
      <!--ro, opt, array, list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </SlowLane>
</TurnRightStop>
<TruckRunARedLight>
  <!--ro, opt, object, red light running of large truck detection, desc:this node is used for linking evidence intersection violation cameras can link the
checkpoints to capture the front license plate of the large trucks-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->false
  </enabled>
  <secondCam>
    <!--ro, req, int, the unique identifier for the linked cameras, range:[0,255], desc:all linked cameras share the same configuration, which is related
to the uploaded value of secondCam in ANPR-->1
  </secondCam>
  <sceneMode>
    <!--ro, opt, enum, detection scene type, subType:string, desc:"singleLane", "multiLane"-->singleLane
  </sceneMode>
  <laneID>
    <!--ro, opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,singleLane}, desc:this node is
valid when the camera is applied in single lane scene-->1
  </laneID>
  <bluePlateCarFilterEnabled>
    <!--ro, opt, bool, whether to enable filtering blue plate vehicles-->true
  </bluePlateCarFilterEnabled>
  <NotificationList>
    <!--ro, opt, array, list of linked camera, up to 3 linked cameras are supported, subType:object, range:[1,3]-->
    <notification>
      <!--ro, opt, object, picture capture-->
      <enabled>
        <!--ro, req, bool, whether to enable the function-->true
      </enabled>
      <ipv4>
        <!--ro, req, string, IPv4 address, range:[0,32]-->test
      </ipv4>
      <port>
        <!--ro, req, int, port No., range:[0,66535]-->0
      </port>
      <username>
        <!--ro, opt, string, user name, range:[0,32], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
      </username>
      <password>
        <!--ro, req, string, password, range:[0,16], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
      </password>
      <laneID>
        <!--ro, opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,multiLane}, desc:this node is
valid when the camera is applied in multiple lanes scene-->1
      </laneID>
```

```
      </laneID>
    </notification>
  </NotificationList>
  <linkSceneMode>
    <!--ro, opt, enum, detection scene type, subType:string, desc:"oneLinkMultiple"(one camera links with multiple cameras), "oneLinkOne" (one camera links with one camera), "multipleLinkOne" (multiple cameras link with multiple cameras)-->oneLinkOne
  </linkSceneMode>
</TruckRunARedLight>
<TruckTurnRightStop>
  <!--ro, opt, object, large-sized vehicle not stopping before right turn-->
  <enabled>
    <!--ro, req, bool, whether to enable-->true
  </enabled>
  <capNo>
    <!--ro, req, enum, number of captures, subType:int, desc:number of captures-->3
  </capNo>
  <sensitivity>
    <!--ro, req, int, stopping sensitivity, range:[1,100]-->50
  </sensitivity>
  <stopTime>
    <!--ro, opt, int, Parking Duration, range:[1,180], unit:s-->10
  </stopTime>
  <capInterval>
    <!--ro, opt, int, range:[1,10000], unit:ms-->1000
  </capInterval>
</TruckTurnRightStop>
<uphone>
  <!--ro, opt, object, making a phone call-->
  <enable>
    <!--ro, req, bool-->true
  </enable>
  <capNo>
    <!--ro, req, int, number of captured pictures, range:[1,3]-->2
  </capNo>
</uphone>
<NoAlterTrafficDet>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <capNo>
    <!--ro, req, int, range:[3,3], step:1-->3
  </capNo>
  <ConvergenceLineList>
    <!--ro, opt, array, subType:object, range:[1,1]-->
    <ConvergenceLine>
      <!--ro, opt, object-->
      <lineId>
        <!--ro, req, int, range:[1,1]-->1
      </lineId>
      <lineName>
        <!--ro, req, string, range:[0,64]-->test
      </lineName>
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </ConvergenceLine>
  </ConvergenceLineList>
</NoAlterTrafficDet>
<greenLightStopCapNo>
  <!--ro, opt, int, range:[3,3]-->3
</greenLightStopCapNo>
<illegalTurnCapNo>
  <!--ro, opt, int, range:[3,3]-->3
</illegalTurnCapNo>
<nonMotorShed>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, req, bool-->true
  </enabled>
  <capNo>
    <!--ro, req, int, range:[1,2]-->1
  </capNo>
</nonMotorShed>
<NonMotorCrossing>
  <!--ro, opt, object-->
  <enabled>
    <!--ro, opt, bool-->true
  </enabled>
  <lastTime>
    <!--ro, req, int, range:[10,180], unit:s-->0
  </lastTime>
  <sensitivity>
    <!--ro, opt, int, range:[1,100]-->0
  </sensitivity>
```

```
    <nonMotorStopLine>
      <!--ro, opt, object-->
      <RegionCoordinatesList>
        <!--ro, opt, array, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object-->
          <positionX>
            <!--ro, req, int, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </nonMotorStopLine>
  </NonMotorCrossing>
  <freightManned>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <capNo>
      <!--ro, req, int, range:[2,3], step:1-->2
    </capNo>
    <sensitivity>
      <!--ro, opt, int, range:[1,100]-->50
    </sensitivity>
  </freightManned>
  <vehicleRestrictions>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <capNo>
      <!--ro, opt, int, range:[2,3]-->2
    </capNo>
    <restrictionType>
      <!--ro, req, enum, subType:string-->region
    </restrictionType>
    <nonRestrictedRegionList>
      <!--ro, opt, array, subType:object, range:[0,64], dep:and,{$.VideoEpolice.vehicleRestrictions.restrictionType,eq,region}-->
      <nonRestrictedRegion>
        <!--ro, opt, object-->
        <nonRestrictedProvinces>
          <!--ro, req, enum, subType:string-->京
        </nonRestrictedProvinces>
        <nonRestrictedCityList>
          <!--ro, opt, array, subType:object, range:[0,64]-->
          <nonRestrictedCity>
            <!--ro, opt, object-->
            <nonRestrictedCityCode>
              <!--ro, req, enum, subType:string-->A
            </nonRestrictedCityCode>
          </nonRestrictedCity>
        </nonRestrictedCityList>
      </nonRestrictedRegion>
    </nonRestrictedRegionList>
    <timeSwitchList>
      <!--ro, opt, array, subType:object, range:[0,7]-->
      <timeSwitch>
        <!--ro, req, object-->
        <dayOfWeek>
          <!--ro, req, enum, subType:int-->1
        </dayOfWeek>
        <timeRangeList>
          <!--ro, req, array, subType:object, range:[0,4]-->
          <timeRange>
            <!--ro, req, object-->
            <ID>
              <!--ro, req, int, range:[1,4]-->1
            </ID>
            <startHour>
              <!--ro, opt, int, range:[0,23]-->0
            </startHour>
            <startMinute>
              <!--ro, opt, int, range:[0,59]-->0
            </startMinute>
            <endHour>
              <!--ro, opt, int, range:[0,23]-->0
            </endHour>
            <endMinute>
              <!--ro, opt, int, range:[0,59]-->0
            </endMinute>
          </timeRange>
        </timeRangeList>
      </timeSwitch>
    </timeSwitchList>
  </vehicleRestrictions>
  <nonMotorOccupyZebraCrossing>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool-->true
    </enabled>
    <capNo>
```

```
    <capNo>
      <!--ro, opt, int, range:[1,3]-->1
    </capNo>
  </nonMotorOccupyZebraCrossing>
  <yellowGridParking>
    <!--ro, opt, object-->
    <enable>
      <!--ro, req, bool-->true
    </enable>
    <lastTime>
      <!--ro, req, int, range:[1,180], unit:s-->0
    </lastTime>
    <yellowGridParkingLineList>
      <!--ro, opt, array, subType:object, range:[0,2]-->
      <yellowGridParkingLine>
        <!--ro, opt, object-->
        <lineId>
          <!--ro, req, int, range:[1,2]-->0
        </lineId>
        <lineName>
          <!--ro, req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--ro, opt, array, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--ro, opt, object-->
            <positionX>
              <!--ro, req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--ro, req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </yellowGridParkingLine>
    </yellowGridParkingLineList>
    <capNo>
      <!--ro, opt, int, range:[3,3]-->3
    </capNo>
  </yellowGridParking>
  <cameraPosition>
    <!--ro, opt, enum, subType:string-->crossing
  </cameraPosition>
</VideoEpolice>
```


11.5.1.37 Set the trigger mode parameters of video intersection violation system

Request URL

PUT /ISAPI/ITC/TriggerMode/videoEpolice?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoEpolice xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, trigger mode parameters of video intersection violation system, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable intersection violation system-->true
  </enabled>
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,4]-->0
  </relatedLaneCount>
  <TrafficLight>
    <!--req, object, traffic light-->
    <source>
      <!--req, enum, signal source, subType:string, desc:"IO", "RS485", "videoDetect" (video detection), "network"-->IO
    </source>
    <IOLightList>
      <!--opt, array, list, subType:object-->
      <IOLight>
        <!--opt, object, I/O access-->
        <lightId>
          <!--req, int, traffic light No., range:[1,3]-->1
        </lightId>
        <lightType>
          <!--req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
        </lightType>
        <relatedIO>
          <!--req, enum, linked I/O input, subType:string, desc:"IO1", "IO2", "IO3", "IO4", "IO5", "IO6", "IO7", "IO8"-->IO7
        </relatedIO>
        <redLightStatus>
```

```
          <redLightStatus>
            <!--req, enum, red light level status, subType:string, desc:"low" (low-level red light), "high" (high-level red light)-->low
          </redLightStatus>
        </IOLight>
      </IOLightList>
      <RS485LightList>
        <!--opt, array, list, subType:object-->
        <RS485Light>
          <!--opt, object, RS-485 access-->
          <lightId>
            <!--req, int, traffic light No., range:[1,3]-->0
          </lightId>
          <lightType>
            <!--req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
          </lightType>
          <relatedLightChan>
            <!--req, int, detector channel No. linked to red light, range:[0,16]-->0
          </relatedLightChan>
          <yellowLightChan>
            <!--req, int, detector channel No. linked to yellow light, range:[0,16]-->0
          </yellowLightChan>
        </RS485Light>
      </RS485LightList>
      <NetworkLightList>
        <!--opt, array, list of network indicator light, subType:object-->
        <NetworkLight>
          <!--opt, object, network indicator light-->
          <lightId>
            <!--req, int, traffic light No., range:[1,3]-->0
          </lightId>
          <lightType>
            <!--req, enum, guiding direction of traffic light, subType:string, desc:"left", "right", "straight"-->straight
          </lightType>
          <relatedLightChan>
            <!--req, int, detector channel No. linked to red light, range:[0,16]-->0
          </relatedLightChan>
          <yellowLightChan>
            <!--req, int, detector channel No. linked to yellow light, range:[0,16]-->0
          </yellowLightChan>
        </NetworkLight>
      </NetworkLightList>
      <videoDetectLightList>
        <!--opt, array, list of video detection traffic lights, subType:object-->
        <videoDetectLight>
          <!--opt, object, video detection traffic light-->
          <videoDetectLightId>
            <!--req, int, video detection signal No., range:[1,2]-->0
          </videoDetectLightId>
          <lightNum>
            <!--req, int, number of traffic lights, range:[1,3]-->0
          </lightNum>
          <straightLight>
            <!--req, bool, whether the straight driving light is on-->true
          </straightLight>
          <leftLight>
            <!--req, bool, whether the left-turn light is on-->true
          </leftLight>
          <rightLight>
            <!--req, bool, whether the right-turn light is on-->true
          </rightLight>
          <redLight>
            <!--req, bool, whether the red light is on-->true
          </redLight>
          <greenLight>
            <!--req, bool, whether the green light is on-->true
          </greenLight>
          <yellowLight>
            <!--req, bool, whether the yellow light is on-->true
          </yellowLight>
          <yellowLightTime>
            <!--req, int, yellow light duration, range:[0,10], unit:s-->0
          </yellowLightTime>
          <rectRegion>
            <!--req, object, traffic light region, desc:the origin is the upper-left corner of the screen-->
            <positionLeft>
              <!--req, int, x-coordinate of upper left corner, range:[0,65535]-->0
            </positionLeft>
            <positionTop>
              <!--req, int, y-coordinate of upper left corner, range:[0,65535]-->0
            </positionTop>
            <positionRight>
              <!--req, int, x-coordinate of lower right corner, range:[0,65535]-->0
            </positionRight>
            <positionBottom>
              <!--req, int, y-coordinate of lower right corner, range:[0,65535]-->0
            </positionBottom>
          </rectRegion>
        </videoDetectLight>
      </videoDetectLightList>
    </TrafficLight>
    <carHighSpeed>
      <!--req, int, abnormal speeding of small-sized vehicle, range:[0,255]-->0
    </carHighSpeed>
    <bigCarHighSpeed>
```

```
      <!--req, int, abnormal speeding of large-sized vehicle, range:[0,255]-->0
    </bigCarHighSpeed>
    <LaneParamList>
      <!--req, array, lane parameter list, subType:object-->
      <LaneParam>
        <!--req, object, lane parameters-->
        <laneId>
          <!--req, int, lane No., range:[1,6]-->0
        </laneId>
        <laneDirectionType>
          <!--req, enum, lane direction No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
        </laneDirectionType>
        <relatedDriveWay>
          <!--req, int, linked lane No., range:[1,99]-->0
        </relatedDriveWay>
        <recordEnable>
          <!--req, bool, whether to enable recording when red light running is detected-->true
        </recordEnable>
        <recordType>
          <!--req, enum, recording type, subType:string, desc:"preRecord", "delayRecord"-->preRecord
        </recordType>
        <beginTime>
          <!--req, enum, recording start time, subType:int, desc:recording start time,"1,2,3"-->0
        </beginTime>
        <preRecordTime>
          <!--req, int, pre-record time, range:[0,100], unit:s-->0
        </preRecordTime>
        <recordDelayTime>
          <!--req, int, post-record time, range:[0,100], unit:s-->0
        </recordDelayTime>
        <recordTimeOut>
          <!--req, int, recording timeout, range:[0,100], unit:s-->0
        </recordTimeOut>
        <signSpeed>
          <!--req, int, marked speed limit for small-sized vehicle, range:[0,255], unit:km/h-->0
        </signSpeed>
        <speedLimit>
          <!--req, int, speed limit for small-sized vehicle driving in high way, range:[0,255], unit:km/h-->0
        </speedLimit>
        <usageType>
          <!--req, enum, lane purpose, subType:string, desc:"carriageWay" (roadway), "bus" (bus lane), "slow" (slow lane), "nonmotor" (non-motor lane), "reverse" (contraflow lane), "emergency" (emergency lane), "banNonMotor" (non-motor vehicle vehicles prohibited lane), "ramp", "fast" (overtaking lane)-->carriageWay
        </usageType>
        <LaneUsageList>
          <!--opt, array, lane purpose list, subType:object-->
          <laneUsage>
            <!--req, enum, lane purpose, subType:string, desc:"bus" (bus lane), "banTrucks" (trucks prohibited lane), "banLeft" (left-turn prohibited lane), "banRight" (right-turn prohibited lane), "motor" (motorcycle lane), "banStraight" (straight prohibited lane), "variableLane" (changeable lane)-->bus
          </laneUsage>
        </LaneUsageList>
        <banTrucksTimeSwitchList>
          <!--req, array, large-sized vehicle prohibited time period, subType:object-->
          <banTrucksTimeSwitch>
            <!--req, object, large-sized vehicle prohibited time-->
            <timeId>
              <!--req, int, time period No., range:[1,4]-->0
            </timeId>
            <startHour>
              <!--opt, int, start time (hour), range:[0,23]-->0
            </startHour>
            <startMinute>
              <!--opt, int, start time (minute), range:[0,59]-->0
            </startMinute>
            <endHour>
              <!--opt, int, end time (hour), range:[0,23]-->0
            </endHour>
            <endMinute>
              <!--opt, int, end time (minute), range:[0,59]-->0
            </endMinute>
          </banTrucksTimeSwitch>
        </banTrucksTimeSwitchList>
        <busWayTimeSwitchList>
          <!--req, array, bus allowed passing time period, subType:object-->
          <busWayTimeSwitch>
            <!--req, object, bus allowed passing time-->
            <timeId>
              <!--req, int, time period No., range:[1,4]-->0
            </timeId>
            <startHour>
              <!--opt, int, start time (hour), range:[0,23]-->0
            </startHour>
            <startMinute>
              <!--opt, int, start time (minute), range:[0,59]-->0
            </startMinute>
            <endHour>
              <!--opt, int, end time (hour), range:[0,23]-->0
            </endHour>
            <endMinute>
              <!--opt, int, end time (minute), range:[0,59]-->0
            </endMinute>
          </busWayTimeSwitch>
        </busWayTimeSwitchList>
```

```
</buswayTimeSwitchList>
<emergencyTimeSwitchList>
  <!--opt, array, time period for emergency lane, subType:object-->
  <emergencyTimeSwitch>
    <!--req, object, emergency lane time-->
    <timeId>
      <!--req, int, index, range:[1,4]-->0
    </timeId>
    <startHour>
      <!--opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </emergencyTimeSwitch>
</emergencyTimeSwitchList>
<banLeftTimeSwitchList>
  <!--req, array, left-turn prohibited time period, subType:object-->
  <banLeftTimeSwitch>
    <!--req, object, left-turn prohibited time-->
    <timeId>
      <!--req, int, time period No., range:[1,4]-->0
    </timeId>
    <startHour>
      <!--opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </banLeftTimeSwitch>
</banLeftTimeSwitchList>
<banRightTimeSwitchList>
  <!--req, array, right-turn prohibited time period, subType:object-->
  <banRightTimeSwitch>
    <!--req, object, right-turn prohibited time-->
    <timeId>
      <!--req, int, time period No., range:[1,4]-->0
    </timeId>
    <startHour>
      <!--opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </banRightTimeSwitch>
</banRightTimeSwitchList>
<banStraightTimeSwitchList>
  <!--req, array, straight prohibited time period, subType:object-->
  <banStraightTimeSwitch>
    <!--req, object, straight prohibited time-->
    <timeId>
      <!--req, int, index, range:[1,4]-->0
    </timeId>
    <startHour>
      <!--opt, int, start time (hour), range:[0,23]-->0
    </startHour>
    <startMinute>
      <!--opt, int, start time (minute), range:[0,59]-->0
    </startMinute>
    <endHour>
      <!--opt, int, end time (hour), range:[0,23]-->0
    </endHour>
    <endMinute>
      <!--opt, int, end time (minute), range:[0,59]-->0
    </endMinute>
  </banStraightTimeSwitch>
</banStraightTimeSwitchList>
<variableLaneTimeSwitchList>
  <!--req, array, time period for changeable lanes, subType:object-->
  <variableLaneTimeSwitch>
    <!--req, object, time for changeable lanes-->
    <timeId>
      <!--req, int, index, range:[1,4]-->0
    </timeId>
    <startHour>
      <!--opt, int, start time (hour), range:[0,23]-->0
```

```
          </startHour>
          <startMinute>
            <!--opt, int, start time (minute), range:[0,59]-->0
          </startMinute>
          <endHour>
            <!--opt, int, end time (hour), range:[0,23]-->0
          </endHour>
          <endMinute>
            <!--opt, int, end time (minute), range:[0,59]-->0
          </endMinute>
          <laneDirection>
            <!--req, enum, driving direction, subType:string, desc:"unknown", "left", "straight", "leftStraight", "right", "leftRight", "rightStraight",
"leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->unknown
          </laneDirection>
        </variableLaneTimeSwitch>
      </variableLaneTimeSwitchList>
      <radarEnabled>
        <!--opt, bool, whether to enable radar speed detection-->true
      </radarEnabled>
      <Radar>
        <!--opt, object, radar parameters-->
        <radarType>
          <!--req, enum, radar type, subType:string, desc:"none", "MultiLane" (multi-lane radar), "csr" (TransMicrowave),"IOBox" (radar connecting I/O
expansion box), "SSTK" (Sensortech),"custom" (custom type), "adr" (Andoray), "adr4Byte" (Andoray (no radar controller)),"olvia" (Olvia)-->none
        </radarType>
        <radarSensitivity>
          <!--req, int, sensitivity, range:[0,65535]-->0
        </radarSensitivity>
        <radarAngle>
          <!--req, int, angle between the radar and the horizontal direction, range:[0,90]-->0
        </radarAngle>
        <validRadarSpeedTime>
          <!--req, int, effective duration of radar speed, which is between 0 and 2000,unit: millisecond, range:[0,2000]-->0
        </validRadarSpeedTime>
        <radarLinearCorrection>
          <!--req, float, linear correction parameter, range:[0.0,2.0]-->0.000
        </radarLinearCorrection>
        <radarConstantCorrection>
          <!--req, int, constant correction parameter,, range:[-100,100]-->0
        </radarConstantCorrection>
        <radarRS485>
          <!--req, int, radar linked RS-485 port No., range:[1,5]-->0
        </radarRS485>
      </Radar>
      <directionType>
        <!--req, enum, lane direction, subType:string, desc:"unknown", "left", "straight", "leftStraight", "right", "leftRight", "rightStraight",
"leftRightStraight", "leftWait", "straightWait", "strightWaitRight"-->unknown
      </directionType>
      <stopLineDistance>
        <!--req, int, distance between the vehicle and the stop line in the second captured picture of red light running violation, range:[0,300]-->0
      </stopLineDistance>
      <loopSensitivity>
        <!--req, int, coil sensitivity, range:[1,100]-->1
      </loopSensitivity>
      <capPositionOffset>
        <!--req, int, minimum offset between the capture positions of the second and the third picture, range:[0,1500]-->0
      </capPositionOffset>
      <detectSnapType>
        <!--req, object, type of capture triggered by video detection-->
        <post>
          <!--req, bool, whether to enable checkpoint detection-->true
        </post>
        <postCapNo>
          <!--req, int, captured pictures of checkpoint, range:[2,3]-->0
        </postCapNo>
        <overSpeed>
          <!--req, bool, whether to enable speeding detection-->true
        </overSpeed>
        <overSpeedCapNo>
          <!--req, int, number of captured pictures in which the speeding violation is detected, range:[2,3]-->0
        </overSpeedCapNo>
        <driveLine>
          <!--req, bool, whether to enable driving on lane line detection-->true
        </driveLine>
        <driveLineCapNo>
          <!--req, int, number of continuously captured pictures in which the driving on lane line happened, range:[2,3]-->0
        </driveLineCapNo>
        <driveLineSensitivity>
          <!--req, int, sensitivity of driving on lane line detection, range:[0,100]-->0
        </driveLineSensitivity>
        <reverse>
          <!--req, bool, whether to enable wrong-way driving detection-->true
        </reverse>
        <reverseCapNo>
          <!--req, int, number of captured pictures in which the wrong-way driving is detected,which is between 2 and 3 and the default value is 2, range:
[0,100]-->0
        </reverseCapNo>
        <redLight>
          <!--req, bool, whether the red light is on-->true
        </redLight>
        <redLightMode>
          <!--req, enum, red light running mode,0-by direction,1-by lane, subType:int, desc:red light running mode,0-by direction,1-by lane-->0
        </redLightMode>
```

```xml
<direction>
  <!--req, bool, whether to enable detection of driving in wrong lane at intersection-->true
</direction>
<illegalPriority>
  <!--req, enum, traffic violation priority, subType:int, desc:0 (by direction), 1 (by lane)-->0
</illegalPriority>
<intersectionCongest>
  <!--req, bool, whether to enable intersection congestion detection-->true
</intersectionCongest>
<nonDriveWay>
  <!--req, bool, whether to enable detection of motor vehicle on non-motor vehicle lane-->true
</nonDriveWay>
<nonDriveWayCapNo>
  <!--req, int, number of captured pictures in which motor vehicle on non-motor vehicle lane happened, range:[2,3]-->0
</nonDriveWayCapNo>
<nonDriveWayTime>
  <!--req, int, capture duration of motor vehicle on non-motor vehicle lane, range:[10,240], unit:s-->10
</nonDriveWayTime>
<changeLane>
  <!--req, bool, whether to enable illegal lane change detection-->true
</changeLane>
<changeLaneCapNo>
  <!--req, int, number of captured pictures in which illegal lane change is detected, range:[2,3]-->0
</changeLaneCapNo>
<banSign>
  <!--req, bool, whether to enable ban breaking detection-->true
</banSign>
<banSignCapNo>
  <!--req, int, number of continuously captured pictures in which the ban is broken, range:[2,3]-->0
</banSignCapNo>
<intersectionStop>
  <!--opt, object, whether to enable intersection parking detection-->
  <enable>
    <!--req, bool, whether to enable detection of left-turn without straight driving-->true
  </enable>
  <lastTime>
    <!--req, int, congestion duration, range:[10,180], unit:s-->0
  </lastTime>
  <sensitivity>
    <!--opt, int, detection sensitivity, range:[1,100]-->0
  </sensitivity>
</intersectionStop>
<greenLightStop>
  <!--req, bool, whether to enable detection of parking at green light-->true
</greenLightStop>
<illegalTurn>
  <!--req, bool, illegal turning-->true
</illegalTurn>
<gasser>
  <!--opt, object, queue jumping-->
  <enabled>
    <!--req, bool, whether to enable intersection violation system-->true
  </enabled>
  <sensitivity>
    <!--req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <congestionThreshold>
    <!--req, int, congestion threshold, range:[1,100]-->0
  </congestionThreshold>
  <capInterval>
    <!--req, int, time interval of continuous capture (2 to 3 pictures), range:[0,6000], unit:ms-->0
  </capInterval>
</gasser>
<congestion>
  <!--opt, object, congestion-->
  <enabled>
    <!--opt, bool, whether to enable intersection violation system-->true
  </enabled>
  <interval>
    <!--req, int, alarm time interval, range:[1,10], unit:min-->0
  </interval>
  <sensitivity>
    <!--req, int, detection sensitivity, range:[0,100]-->0
  </sensitivity>
  <lastTime>
    <!--req, int, congestion duration, range:[1,60], unit:s-->0
  </lastTime>
</congestion>
<accident>
  <!--opt, object, accident detection-->
  <enabled>
    <!--req, bool, whether to enable intersection violation system-->true
  </enabled>
  <lastTime>
    <!--req, int, congestion duration, unit: second, range:[1,180], unit:s-->0
  </lastTime>
  <interval>
    <!--req, int, alarm time interval, unit: minute, range:[1,60], unit:min-->0
  </interval>
</accident>
<nonMotorEnabled>
  <!--opt, bool, whether to enable non-motor vehicle detection-->true
</nonMotorEnabled>
<runGreenLight>
```

```
        <runGreenLight>
          <!--opt, bool, whether to enable green light running detection-->true
        </runGreenLight>
        <runGreenLightParam>
          <!--ro, opt, object, parameters of green light running detection-->
          <detectEvent>
            <!--req, enum, detection logic, subType:string, desc:"runGreenLight" (running green light in congestion), "crossingStop" (overstay at 
intersection)-->runGreenLight
          </detectEvent>
          <interval>
            <!--req, int, capture interval, range:[1,180], unit:s, desc:capture interval-->15
          </interval>
          <secondLampState>
            <!--opt, enum, traffic light status of the second capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
          </secondLampState>
          <threeLampState>
            <!--opt, enum, traffic light status of the third capture times, subType:int, desc:0 (red light), 1 (green light), 2 (unlimited)-->2
          </threeLampState>
        </runGreenLightParam>
        <intervalType>
          <!--req, enum, burst interval type, subType:string, desc:"time" (by time), "distance" (by distance)-->time
        </intervalType>
        <IntervalList>
          <!--opt, array, list of time intervals, subType:object-->
          <Interval>
            <!--req, object, interval information-->
            <intervalValue>
              <!--req, int, burst interval, desc:unit: ms/dm; capture interval by time is between 70 and 2000; capture interval by distance is between 1 and 
20-->0
            </intervalValue>
          </Interval>
        </IntervalList>
        <redLightCapNo>
          <!--opt, int, number of captured violation picture(s), range:[3,4]-->0
        </redLightCapNo>
        <directionCapNo>
          <!--opt, int, number of captured pictures of driving against direction guidance, range:[3,4]-->0
        </directionCapNo>
        <banSignType>
          <!--opt, enum, capture type list, subType:string, desc:"truck", "yellowPlate" (yellow license plate vehicle), "standard", "bus"-->truck
        </banSignType>
        <greenLightStopSensitivity>
          <!--opt, int, stopping sensitivity, range:[1,100]-->0
        </greenLightStopSensitivity>
        <nonMotorShed>
          <!--opt, object, non-motor vehicle umbrella tent-->
          <enabled>
            <!--req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--req, int, number of captures, range:[1,2]-->0
          </capNo>
        </nonMotorShed>
        <banNonMotorSign>
          <!--opt, object, non-motor vehicle prohibition violation-->
          <enabled>
            <!--req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--req, int, number of captures, range:[2,3]-->0
          </capNo>
        </banNonMotorSign>
        <nonMotorPassenger>
          <!--opt, object, manned non-motor vehicle-->
          <enabled>
            <!--req, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--req, int, number of captures, range:[1,2]-->0
          </capNo>
          <refineMannedIllegalCode>
            <!--opt, bool, whether to enable detailed manned violation code-->true
          </refineMannedIllegalCode>
        </nonMotorPassenger>
        <SuonaDet>
          <!--opt, object, whistle detection-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->true
          </enabled>
          <capNo>
            <!--opt, int, number of captures, range:[2,3]-->0
          </capNo>
          <sensitivity>
            <!--opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
        </SuonaDet>
        <BanMotorDet>
          <!--opt, object, motorcycle driving in prohibited areas-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->false
          </enabled>
          <capNo>
            <!--opt, int, number of captures, range:[1,3]-->2
          </capNo>
```

```
          <sensitivity>
            <!--opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
        </BanMotorDet>
        <BanSignTypeList>
          <!--opt, array, capture type list, subType:object-->
          <banSignType>
            <!--opt, enum, prohibition violation type, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle), "standard", "bus", 
"yellowbus" (yellow license plate bus), "dangercar" (dangerous goods transport vehicle), "instructioncar" (driver-training vehicle)-->standard
          </banSignType>
        </BanSignTypeList>
        <TurnSignalDet>
          <!--opt, object, turning without signal detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->false
          </enabled>
          <turnFiliter>
            <!--opt, bool, whether to filter left turn and right turn-->true
          </turnFiliter>
          <detectEvent>
            <!--opt, enum, capture type, subType:string, desc:"turnLeft", "turnRight", "changeLane"-->changeLane
          </detectEvent>
        </TurnSignalDet>
        <ReflectiveTapeDet>
          <!--opt, object, reflective stripe detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->false
          </enabled>
          <capNo>
            <!--opt, int, number of captures, range:[1,3]-->2
          </capNo>
        </ReflectiveTapeDet>
        <ZipperDriveDet>
          <!--opt, object, pass alternatively detection, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->true
          </enabled>
          <carNum>
            <!--opt, int, capture triggered number of vehicles, range:[1,3]-->2
          </carNum>
          <parallDistance>
            <!--opt, int, parallel distance, range:[0,200]-->0
          </parallDistance>
        </ZipperDriveDet>
        <NonMotorCrossing>
          <!--opt, object, non-motor vehicle parking over stop line, desc:if this node is enabled, it is related to "illegalInfo" of ANPR-->
          <enabled>
            <!--opt, bool, whether to enable intersection violation system-->true
          </enabled>
          <lastTime>
            <!--req, int, congestion duration, range:[10,180], unit:s-->0
          </lastTime>
          <sensitivity>
            <!--opt, int, detection sensitivity, range:[1,100]-->0
          </sensitivity>
          <nonMotorStopLine>
            <!--opt, object, non-motor vehicle parking over stop line-->
            <RegionCoordinatesList size="2">
              <!--opt, object, list, attr:size{req, int}-->
              <RegionCoordinates>
                <!--opt, object, point coordinates-->
                <positionX>
                  <!--req, int, x-coordinate, range:[0,1]-->0.1
                </positionX>
                <positionY>
                  <!--req, int, y-coordinate, range:[0,1]-->0.1
                </positionY>
              </RegionCoordinates>
            </RegionCoordinatesList>
          </nonMotorStopLine>
        </NonMotorCrossing>
      </detectSnapType>
      <ViolationDetectLineList>
        <!--req, array, list, subType:object-->
        <ViolationDetectLine>
          <!--req, object, draw violation detection line for a single lane-->
          <lineName>
            <!--req, enum, line name, subType:string, desc:detection line name: "laneLine,stopLine,redlightLine,cancelLine,waitLine"-->laneLine
          </lineName>
          <lineType>
            <!--opt, enum, detection line type, subType:string, desc:"unknown", "white” (white solid line), "singleYellow” (single yellow line), 
"doubleYellow” (double yellow line), "guardRail” (lane line with guardrail), "noCross” (lane line prohibited to cross by vehicle)-->unknown
          </lineType>
          <customName>
            <!--opt, string, custom triggering line name-->test
          </customName>
          <RegionCoordinatesList>
            <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
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              <positionY>
                <!--req, int, y-coordinate-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </ViolationDetectLine>
      </ViolationDetectLineList>
      <firstShowTailVep>
        <!--opt, enum, place of the first capture time, subType:int, desc:0 (none), 1 (expose vehicle tail), 2 (expose vehicle tail and LPR success),
3(parking over stop line)-->0
      </firstShowTailVep>
      <lowSpeedLimit>
        <!--req, int, low speed limit for small-sized vehicle, range:[0,255]-->0
      </lowSpeedLimit>
      <bigCarSignSpeed>
        <!--req, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSignSpeed>
      <bigCarSpeedLimit>
        <!--req, int, maximum speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSpeedLimit>
      <bigCarLowSpeedLimit>
        <!--req, int, low speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarLowSpeedLimit>
      <noBanLeftTypeList>
        <!--opt, array, allowlist of left-turn vehicle types, subType:object-->
        <noBanLeftType>
          <!--opt, enum, vehicle types which are allowed to turn left, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
        </noBanLeftType>
      </noBanLeftTypeList>
      <noBanRightTypeList>
        <!--opt, array, allowlist of right-turn vehicle types, subType:object-->
        <noBanRightType>
          <!--opt, enum, vehicle types which are allowed to left-turn, subType:string, desc:"truck", "yellowplate" (yellow license plate vehicle),
"standard", "bus", "yellowbus" (yellow license plate bus)-->standard
        </noBanRightType>
      </noBanRightTypeList>
      <dangerCarSignSpeed>
        <!--opt, int, range:[0,255], unit:km/h-->0
      </dangerCarSignSpeed>
      <dangerCarSpeedLimit>
        <!--opt, int, range:[0,255], unit:km/h-->0
      </dangerCarSpeedLimit>
      <dangerCarLowSpeedLimit>
        <!--opt, int, range:[0,255], unit:km/h-->0
      </dangerCarLowSpeedLimit>
    </LaneParam>
  </LaneParamList>
  <VirtualLaneList>
    <!--req, array, list of virtual line, subType:object-->
    <VirtualLane>
      <!--req, object, virtual lane line-->
      <lineName>
        <!--req, enum, line name, subType:string, desc:detection line name: "laneLine,stopLine,redlightLine,cancelLine,waitLine"-->laneRightBoundaryLine
      </lineName>
      <lineType>
        <!--opt, enum, detection line type, subType:string, desc:"unknown", "white" (white solid line), "singleYellow" (single yellow line), "doubleYellow"
(double yellow line), "guardRail" (lane line with guardrail), "noCross" (lane line prohibited to cross by vehicle)-->unknown
      </lineType>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, x-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, y-coordinate-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </VirtualLane>
  </VirtualLaneList>
  <Posting>
    <!--opt, object, positioning frame-->
    <postingEnable>
      <!--req, bool, whether to enable positioning frame-->true
    </postingEnable>
    <postingSize>
      <!--req, enum, positioning frame size, subType:string, desc:"small,middle,large"-->small
    </postingSize>
  </Posting>
  <leftNotYieldStraight>
    <!--opt, object, left turn not yield to straight-->
    <enable>
      <!--req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
  </leftNotYieldStraight>
  <rightNotYieldLeft>
    <!--opt, object, right turn not yield to left turn-->
    <enable>
      <!--req, bool, whether to enable detection of left-turn without straight driving-->true
    </enable>
  </rightNotYieldLeft>
```

```
    <uTurnNotYieldGoingStaight>
      <!--opt, object, U-turn not yield to straight-->
      <enable>
        <!--req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
    </uTurnNotYieldGoingStaight>
    <leftSideDrivingAfterLeftTurn>
      <!--opt, object, turn left in short turn radius-->
      <enable>
        <!--req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
    </leftSideDrivingAfterLeftTurn>
    <failYieldPed>
      <!--opt, object, failing to yield for pedestrians-->
      <enable>
        <!--req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <vehicleFromDirection>
        <!--req, enum, driving direction of the vehicle which does not yield to pedestrian, subType:string, desc:"Front", "Back"-->Front
      </vehicleFromDirection>
      <cameraPosition>
        <!--req, enum, camera position, subType:string, desc:"crossing" (crossroad intersection), "roadsection"-->crossing
      </cameraPosition>
      <noDisplacementPedEnabled>
        <!--req, bool, whether to enable no displacement capture of pedestrian-->true
      </noDisplacementPedEnabled>
      <failYieldPedLineList>
        <!--opt, array, list, subType:object-->
        <failYieldPedLine>
          <!--opt, object, failing to yield for pedestrians line-->
          <lineId>
            <!--req, int, line ID, range:[1,2]-->0
          </lineId>
          <lineName>
            <!--req, string, detection line name, range:[0,64]-->test
          </lineName>
          <RegionCoordinatesList>
            <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </failYieldPedLine>
      </failYieldPedLineList>
      <notification>
        <!--req, object, capture linkage, desc:capture linkage via HCNetSDK-->
        <enable>
          <!--req, bool, whether to enable detection of left-turn without straight driving-->true
        </enable>
        <ip>
          <!--opt, string, IPv4 address, range:[0,32]-->test
        </ip>
        <port>
          <!--opt, int, port No. of traffic camera, range:[0,65535]-->1
        </port>
        <username>
          <!--opt, string, user name of traffic camera, range:[0,32]-->test
        </username>
        <password>
          <!--opt, string, password of traffic camera, range:[0,16]-->test
        </password>
        <sensitivity>
          <!--req, int, detection sensitivity, range:[0,100]-->0
        </sensitivity>
        <RS485Port>
          <!--req, int, RS-485 port No. for the traffic light to access the camera, range:[1,10]-->1
        </RS485Port>
      </notification>
      <failYieldPedSensitivity>
        <!--req, int, not yielding to pedestrian sensitivity, range:[0,100]-->0
      </failYieldPedSensitivity>
      <failYieldnonMotor>
        <!--req, bool, whether to enable capturing non-motor vehicle-->true
      </failYieldnonMotor>
      <failYieldNum>
        <!--opt, int, number of pedestrian not yielded to, range:[1,7]-->1
      </failYieldNum>
      <failYieldAreaType>
        <!--opt, enum, detection area of failing to yield for pedestrians, subType:string, desc:"rectangle", "polygon"-->rectangle
      </failYieldAreaType>
      <failYieldDistanceMode>
        <!--opt, enum, distance mode of failing to yield for pedestrians, subType:string, desc:"car" (distance from pedestrian to vehicle center), "lane"
  (distance from pedestrian to lane center)-->car
      </failYieldDistanceMode>
      <failYieldPedSnapPosition>
        <!--opt, int, distance range for capturing failing to yield for pedestrians, range:[0,100]-->20
      </failYieldPedSnapPosition>
    </failYieldPed>
```

```xml
    </failYieldPed>
    <illegalParking>
      <!--opt, object, illegal parking-->
      <enable>
        <!--req, bool, whether to enable detection of left-turn without straight driving-->true
      </enable>
      <sensitivity>
        <!--req, int, detection sensitivity, range:[0,100]-->0
      </sensitivity>
      <lastTime>
        <!--req, int, parking time, range:[1,180], unit:s-->0
      </lastTime>
      <illegalParkingLineList>
        <!--opt, array, list of illegal parking lines, subType:object-->
        <illegalParkingLine>
          <!--opt, object, illegal parking lines-->
          <lineId>
            <!--req, int, line ID, range:[1,3]-->0
          </lineId>
          <lineName>
            <!--req, string, line name, range:[0,64]-->test
          </lineName>
          <RegionCoordinatesList>
            <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </illegalParkingLine>
      </illegalParkingLineList>
    </illegalParking>
    <detectMode>
      <!--req, enum, detection mode, subType:string, desc:"video", "coilLoop" (video and coil)-->video
    </detectMode>
    <helmet>
      <!--opt, bool, whether the driver is not wearing helmet-->true
    </helmet>
    <helmetCapNo>
      <!--opt, int, number of captured pictures in which the driver does not wear the helmet, range:[1,3]-->1
    </helmetCapNo>
    <nonMotorExist>
      <!--opt, bool, whether the non-motor vehicle is occupying non-motor vehicle lane-->true
    </nonMotorExist>
    <nonMotorExistCapNo>
      <!--opt, int, number of captured pictures in which the motor lane is occupied by non-motor vehicle, range:[2,3]-->2
    </nonMotorExistCapNo>
    <motorIllegalCarrying>
      <!--opt, bool, illegal manned motor vehicle-->true
    </motorIllegalCarrying>
    <nomotorIllegalCarrying>
      <!--opt, bool, illegal manned non-motor vehicle-->true
    </nomotorIllegalCarrying>
    <canopy>
      <!--opt, bool, umbrella tent installation violation on non-motor vehicle-->true
    </canopy>
    <pedRedLight>
      <!--opt, bool, pedestrian red light running-->true
    </pedRedLight>
    <pedOnNomotorLane>
      <!--opt, bool, whether to detect pedestrian walking on non-motor vehicle lane-->true
    </pedOnNomotorLane>
    <pedOnMotorLane>
      <!--opt, bool, whether to detect pedestrian walking on motor vehicle lane-->true
    </pedOnMotorLane>
    <capType>
      <!--opt, enum, Capture Type, subType:string, desc:"vehicle" (motor vehicle), "nonMotor" (non-motor vehicle), "human" (pedestrian), "face",-->mixObject
    </capType>
    <deferredFrmNum>
      <!--req, int, delay a few seconds to trigger the mode, range:[0,25]-->0
    </deferredFrmNum>
    <parkingOverStopLine>
      <!--opt, object, over stop line-->
      <enabled>
        <!--req, bool, whether to enable driving on the lane line trigger mode-->true
      </enabled>
    </parkingOverStopLine>
    <speedDetectMode>
      <!--opt, enum, speed detection mode, subType:string, desc:"none", "radar"-->none
    </speedDetectMode>
    <NotSlowZebraCrossingEvent>
      <!--opt, object, not slowing down at zebra crossing-->
      <enabled>
        <!--req, bool, whether to enable not slowing down at zebra crossing triggering mode-->true
      </enabled>
      <capNo>
        <!--req, int, number of captures, range:[1,3], step:1-->1
      </capNo>
      <lowSpeed>
```

```xml
        <!--req, int, lowest speed limit (km/h), range:[0,100], step:1-->0
      </lowSpeed>
      <highSpeed>
        <!--req, int, highest speed limit (km/h), range:[0,100], step:1-->0
      </highSpeed>
      <LineInfoList>
        <!--opt, object, list of lane line information-->
        <LineInfo>
          <!--opt, object, single lane line information-->
          <lineId>
            <!--req, int, index, range:[1,3]-->0
          </lineId>
          <RegionCoordinatesList>
            <!--opt, array, list, subType:object, range:[0,2]-->
            <RegionCoordinates>
              <!--opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
              <positionX>
                <!--req, int, x-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, y-coordinate-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </LineInfo>
      </LineInfoList>
    </NotSlowZebraCrossingEvent>
    <TurnRightStop>
      <!--opt, object, large-sized vehicle not stopping before right turn-->
      <enabled>
        <!--req, bool, whether to enable intersection violation system-->true
      </enabled>
      <capNo>
        <!--req, int, number of captures, range:[1,3], step:1-->3
      </capNo>
      <durationTime>
        <!--req, int, duration, range:[1,600], unit:s-->5
      </durationTime>
      <slowUp>
        <!--req, int, upper limit of deceleration, range:[0,100], step:1, unit:km/h-->40
      </slowUp>
      <slowStep>
        <!--req, int, deceleration range, range:[0,50], step:1, unit:km/h-->5
      </slowStep>
      <slowDown>
        <!--req, int, lower limit of deceleration, range:[0,100], step:1, unit:km/h-->10
      </slowDown>
      <detectMode>
        <!--req, enum, deceleration mode, subType:string, desc:"stop", "slow"-->stop
      </detectMode>
      <SlowLane>
        <!--opt, object, deceleration line-->
        <RegionCoordinatesList>
          <!--opt, array, list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object, point coordinates, desc:the origin is the upper-left corner of the screen-->
            <positionX>
              <!--req, int, x-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </SlowLane>
    </TurnRightStop>
    <NotYieldToRightRoad>
      <!--opt, object, not yield to vehicles from right roads-->
      <enabled>
        <!--req, bool, whether to enable not yield to vehicles from right roads triggering mode-->false
      </enabled>
      <firstCapturePosition>
        <!--req, int, capture position of first picture, range:[0,100], step:1-->20
      </firstCapturePosition>
      <lateralDistance>
        <!--req, int, lateral distance between two vehicles, range:[20,350], step:1-->150
      </lateralDistance>
      <longitudinalDistance>
        <!--req, int, longitudinal distance between two vehicles, range:[10,150], step:1-->50
      </longitudinalDistance>
    </NotYieldToRightRoad>
    <EnterNotYieldToInRoundabout>
      <!--opt, object, not yield at roundabout intersection-->
      <enabled>
        <!--req, bool, whether to enable not yield at roundabout intersection triggering mode-->false
      </enabled>
      <firstCapturePosition>
        <!--req, int, capture position of first picture, range:[0,100], step:1-->20
      </firstCapturePosition>
      <lateralDistance>
        <!--req, int, lateral distance between two vehicles, range:[20,350], step:1-->150
      </lateralDistance>
      <longitudinalDistance>
```

```xml
        <!--req, int, longitudinal distance between two vehicles, range:[10,150], step:1-->50
      </longitudinalDistance>
    </EnterNotYieldToInRoundabout>
    <RampNotYieldToMainRoad>
      <!--opt, object, vehicle from ramp not yielding to vehicle on main road-->
      <enabled>
        <!--req, bool, whether to enable vehicle from ramp not yielding to vehicle on main road triggering mode-->false
      </enabled>
      <lateralDistance>
        <!--req, int, lateral distance of a vehicle when it drives from a ramp to a main road., range:[10,90], step:1-->60
      </lateralDistance>
    </RampNotYieldToMainRoad>
    <TruckRunARedLight>
      <!--opt, object, red light running of large truck detection, desc:this node is used for linking evidence intersection violation cameras can link the
checkpoints to capture the front license plate of the large trucks-->
      <enabled>
        <!--req, bool, whether to enable-->false
      </enabled>
      <secondCam>
        <!--req, int, the unique identifier for the linked cameras, range:[0,255], desc:all linked cameras share the same configuration, which is related to
the uploaded value of secondCam in ANPR-->1
      </secondCam>
      <sceneMode>
        <!--opt, enum, detection scene type, subType:string, desc:"singleLane", "multiLane"-->singleLane
      </sceneMode>
      <laneID>
        <!--opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,singleLane}, desc:this node is valid
when the camera is applied in single lane scene-->1
      </laneID>
      <bluePlateCarFilterEnabled>
        <!--opt, bool, whether to enable filtering blue plate vehicles-->true
      </bluePlateCarFilterEnabled>
      <NotificationList>
        <!--opt, array, list of linked camera, max. 3 linked cameras, subType:object, range:[1,3]-->
        <notification>
          <!--opt, object, picture capture-->
          <enabled>
            <!--req, bool, whether to enable-->true
          </enabled>
          <ipv4>
            <!--req, string, IPv4 address, range:[0,32]-->test
          </ipv4>
          <port>
            <!--req, int, port No., range:[0,66535]-->0
          </port>
          <username>
            <!--opt, string, user name, range:[0,32], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
          </username>
          <password>
            <!--req, string, password, range:[0,16], desc:transmitting via HTTP may cause security problems. HTTPS is required-->test
          </password>
          <laneID>
            <!--opt, int, lane No. of the linked camera, range:[1,3], dep:and,{$.VideoEpolice.TruckRunARedLight.sceneMode,eq,multiLane}, desc:this node is
valid when the camera is applied in multiple lanes scene-->1
          </laneID>
        </notification>
      </NotificationList>
      <linkSceneMode>
        <!--ro, opt, enum, detection scene type, subType:string, desc:"oneLinkMultiple"(one camera links with multiple cameras), "oneLinkOne" (one camera
links with one camera), "multipleLinkOne" (multiple cameras link with multiple cameras)-->oneLinkOne
      </linkSceneMode>
    </TruckRunARedLight>
    <TruckTurnRightStop>
      <!--opt, object, large-sized vehicle not stopping before right turn-->
      <enabled>
        <!--req, bool, whether to enable-->true
      </enabled>
      <capNo>
        <!--req, int, number of captures, range:[2,3], step:1-->3
      </capNo>
      <sensitivity>
        <!--req, int, stopping sensitivity, range:[1,100]-->50
      </sensitivity>
      <stopTime>
        <!--opt, int, parking duration, range:[1,180], unit:s-->10
      </stopTime>
    </TruckTurnRightStop>
    <uphone>
      <!--ro, opt, object, making a phone call-->
      <enabled>
        <!--ro, req, bool, whether to enable phone call detection-->true
      </enabled>
      <capNo>
        <!--ro, req, int, number of captures, range:[1,3]-->2
      </capNo>
    </uphone>
    <NoAlterTrafficDet>
      <!--opt, object-->
      <enabled>
        <!--opt, bool-->true
      </enabled>
      <capNo>
        <!--req, int, range:[3,3], step:1-->3
      </capNo>
```

```
    <ConvergenceLineList>
      <!--opt, array, subType:object, range:[1,1]-->
      <ConvergenceLine>
        <!--opt, object-->
        <lineId>
          <!--req, int, range:[1,1]-->1
        </lineId>
        <lineName>
          <!--req, string, range:[0,64]-->test
        </lineName>
        <RegionCoordinatesList>
          <!--opt, array, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object-->
            <positionX>
              <!--req, int, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </ConvergenceLine>
    </ConvergenceLineList>
  </NoAlterTrafficDet>
  <dangerCarHighSpeed>
    <!--opt, int, range:[0,255], unit:km/h-->0
  </dangerCarHighSpeed>
</VideoEpolice>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.38 Get the multiple-frame recognition mode parameters of smart video security

Request URL

GET /ISAPI/ITC/TriggerMode/videoMonitor

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, trigger mode parameters of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, total number of linked lanes, range:[1,3]-->0
  </relatedLaneCount>
  <capType>
    <!--ro, req, enum, captured target type, subType:string, desc:"vehicle" (motor vehicle), "vehWithHum" (non-motor vehicle)-->vehicle
  </capType>
  <ViolationDetectLineList>
    <!--ro, req, array, list of violation detection line on the single lane, subType:object-->
    <ViolationDetectLine>
      <!--ro, opt, object, violation detection line on the single lane-->
      <lineName>
        <!--ro, req, enum, line name, subType:string, desc:"laneLine" (lane line)-->laneLine
      </lineName>
      <relatedDriveWay>
```

```
      <relatedDriveWay>
        <!--ro, req, int, linked lane No., range:[1,99]-->1
      </relatedDriveWay>
      <laneDirectionType>
        <!--ro, req, enum, lane direction type No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <carDriveDirect>
        <!--ro, opt, enum, subType:string-->up2down
      </carDriveDirect>
      <RegionCoordinatesList>
        <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--ro, req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <signSpeed>
        <!--ro, opt, int, marked speed limit, range:[0,255]-->0
      </signSpeed>
      <speedLimit>
        <!--ro, opt, int, speed limit, range:[0,255]-->0
      </speedLimit>
      <lowSpeedLimit>
        <!--ro, opt, int, low speed limit for small-sized vehicle, range:[0,255]-->0
      </lowSpeedLimit>
      <bigCarSignSpeed>
        <!--ro, opt, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSignSpeed>
      <bigCarSpeedLimit>
        <!--ro, opt, int, speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarSpeedLimit>
      <bigCarLowSpeedLimit>
        <!--ro, opt, int, low speed limit for large-sized vehicle, range:[0,255]-->0
      </bigCarLowSpeedLimit>
    </ViolationDetectLine>
  </ViolationDetectLineList>
  <VirtualLane>
    <!--ro, req, object, draw violation detection line on all lanes-->
    <lineName>
      <!--ro, req, enum, line name, subType:string, desc:line name-->laneRightBoundaryLine
    </lineName>
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </VirtualLane>
  <enabled>
    <!--ro, opt, bool, whether to enable smart video security, desc:if this node is not returned, smart video security of the device is enabled by default-->true
  </enabled>
  <TriggerLine>
    <!--ro, opt, object, triggering line of smart video security capture-->
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinates list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </TriggerLine>
  <capNo>
    <!--ro, opt, int, captured pictures of smart video security linked cameras, range:[1,2]-->1
  </capNo>
  <interval>
    <!--ro, opt, int, capture interval of smart video security, range:[1,3000], unit:ms-->1
  </interval>
  <carDriveDirect>
    <!--ro, opt, enum, vehicle detection type, subType:string, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
  </carDriveDirect>
  <sceneMode>
    <!--ro, opt, enum, scene mode, subType:string, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance" (entrance and exit scene), "other" (other scenes)-->standardScene
  </sceneMode>
  <overSpeed>
```

```
      <!--ro, opt, bool, whether to enable overspeed trigger mode, desc:whether to enable overspeed trigger mode-->true
    </overSpeed>
    <overSpeedCapNo>
      <!--ro, opt, int, capture times of overspeed, range:[1,3]-->1
    </overSpeedCapNo>
    <lowSpeed>
      <!--ro, opt, bool, whether to enable low speed trigger mode-->true
    </lowSpeed>
    <lowSpeedCapNo>
      <!--ro, opt, int, capture times of low speed, range:[1,3]-->1
    </lowSpeedCapNo>
    <carHighSpeed>
      <!--ro, opt, int, abnormal overspeed, range:[0,255]-->0
    </carHighSpeed>
    <carLowSpeed>
      <!--ro, opt, int, abnormal low speed, range:[0,255]-->0
    </carLowSpeed>
    <bigCarHighSpeed>
      <!--ro, opt, int, abnormal overspeed of large-sized vehicle, range:[0,255]-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed>
      <!--ro, opt, int, abnormal low speed of large-sized vehicle, range:[0,255]-->0
    </bigCarLowSpeed>
    <nonmotorOverSpeed>
      <!--ro, opt, bool, whether to enable capturing non-motor vehicle-->true
    </nonmotorOverSpeed>
    <nonmotorHighSpeed>
      <!--ro, opt, int, speed limit for non-motor vehicle (unit: km/h), range:[0,255], unit:km/h-->0
    </nonmotorHighSpeed>
  </VideoMonitor>
```

11.5.1.39 Set the multiple-frame recognition mode parameters of smart video security

Request URL

PUT /ISAPI/ITC/TriggerMode/videoMonitor

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, trigger mode parameters of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--req, int, number of supported lanes, range:[1,3]-->0
  </relatedLaneCount>
  <capType>
    <!--req, enum, captured target type, subType:string, desc:"vehicle" (motor vehicle), "vehWithHum" (non-motor vehicle)-->vehicle
  </capType>
  <ViolationDetectLineList>
    <!--req, array, list of violation detection line on the single lane, subType:object-->
    <ViolationDetectLine>
      <!--opt, object, violation detection line on the single lane-->
      <lineName>
        <!--req, enum, detection line name, subType:string, desc:"laneLine" (for a single lane only)-->laneLine
      </lineName>
      <relatedDriveWay>
        <!--req, int, linked lane No., range:[1,99]-->1
      </relatedDriveWay>
      <laneDirectionType>
        <!--req, enum, lane direction type No., subType:int, desc:0 (unknown),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from
north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
      </laneDirectionType>
      <carDriveDirect>
        <!--opt, enum, subType:string-->up2down
      </carDriveDirect>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, only two coordinates are required, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <signSpeed>
        <!--opt, int, marked speed limit, range:[0,255]-->0
      </signSpeed>
      <speedLimit>
        <!--opt, int, speed limit, range:[0,255]-->0
      </speedLimit>
      <lowSpeedLimit>
```

```
          <!--opt, int, low speed limit for small-sized vehicle, range:[0,255]-->0
        </lowSpeedLimit>
        <bigCarSignSpeed>
          <!--opt, int, marked speed limit for large-sized vehicle, range:[0,255]-->0
        </bigCarSignSpeed>
        <bigCarSpeedLimit>
          <!--opt, int, speed limit for large-sized vehicle, range:[0,255]-->0
        </bigCarSpeedLimit>
        <bigCarLowSpeedLimit>
          <!--opt, int, low speed limit for large-sized vehicle, range:[0,255]-->0
        </bigCarLowSpeedLimit>
      </ViolationDetectLine>
    </ViolationDetectLineList>
    <VirtualLane>
      <!--req, object, draw violation detection line on all lanes-->
      <lineName>
        <!--req, enum, detection line name, subType:string, desc:"laneRightBoundaryLine" (parking space boundary line)-->laneRightBoundaryLine
      </lineName>
      <RegionCoordinatesList>
        <!--opt, array, area coordinate list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </VirtualLane>
    <enabled>
      <!--opt, bool, whether to enable smart video security, desc:if this node is not returned, smart video security of the device is enabled by default--
>true
    </enabled>
    <TriggerLine>
      <!--opt, object, triggering line of smart video security capture-->
      <RegionCoordinatesList>
        <!--opt, array, area coordinates list, subType:object, range:[0,2]-->
        <RegionCoordinates>
          <!--opt, object, area coordinates, desc:the origin is the upper-left corner of the screen-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </TriggerLine>
    <capNo>
      <!--opt, int, captured pictures of smart video security linked cameras, range:[1,2]-->1
    </capNo>
    <interval>
      <!--opt, int, capture interval of smart video security, range:[1,3000], unit:ms-->1
    </interval>
    <carDriveDirect>
      <!--opt, enum, vehicle detection type, subType:string, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
    </carDriveDirect>
    <sceneMode>
      <!--opt, enum, scene mode, subType:string, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance" (entrance and exit scene),
"other" (other scenes)-->standardScene
    </sceneMode>
    <overSpeed>
      <!--opt, bool, whether to enable overspeed trigger mode, desc:whether to enable overspeed trigger mode-->true
    </overSpeed>
    <overSpeedCapNo>
      <!--opt, int, capture times of overspeed, range:[1,3]-->1
    </overSpeedCapNo>
    <lowSpeed>
      <!--opt, bool, whether to enable low speed trigger mode-->true
    </lowSpeed>
    <lowSpeedCapNo>
      <!--opt, int, capture times of low speed, range:[1,3]-->1
    </lowSpeedCapNo>
    <carHighSpeed>
      <!--opt, int, abnormal overspeed, range:[0,255]-->0
    </carHighSpeed>
    <carLowSpeed>
      <!--opt, int, abnormal low speed, range:[0,255]-->0
    </carLowSpeed>
    <bigCarHighSpeed>
      <!--opt, int, abnormal overspeed of large-sized vehicle, range:[0,255]-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed>
      <!--opt, int, abnormal low speed of large-sized vehicle, range:[0,255]-->0
    </bigCarLowSpeed>
    <nonmotorOverSpeed>
      <!--ro, opt, bool, whether to enable capturing non-motor vehicle-->true
    </nonmotorOverSpeed>
    <nonmotorHighSpeed>
      <!--ro, opt, int, speed limit for non-motor vehicle (unit: km/h), range:[0,255], unit:km/h-->0
    </nonmotorHighSpeed>
  </VideoMonitor>
```

```
</VideoMonitor>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

11.5.1.40 Get the multiple-frame recognition mode capability of smart video security

Request URL

GET /ISAPI/ITC/TriggerMode/videoMonitor/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, multiple-frame recognition mode capability of smart video security, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount min="1" max="16">
    <!--ro, req, int, total number of linked lanes, attr:min{req, int},max{req, int}-->0
  </relatedLaneCount>
  <relatedDriveWay min="1" max="99">
    <!--ro, req, int, linked lane No., attr:min{req, int},max{req, int}-->0
  </relatedDriveWay>
  <laneDirectionType opt="0,1,2,3,4,5,6,7,8">
    <!--ro, req, enum, lane direction, subType:int, attr:opt{req, string}, desc:0-unknown, 1-from east to west, 2-from west to east, 3-from south to north, 4-from north to south, 5-from southeast to northwest, 6-from northwest to southeast, 7-from northeast to southwest, 8-from southwest to northeast-->0
  </laneDirectionType>
  <capType opt="vehWithHum,vehicle,nonmotor,human,motorWithHum,nonmotorWithHum,motorWithNonmotor">
    <!--ro, req, enum, captured target type, subType:string, attr:opt{req, string}, desc:"vehWithHum" (motor vehicle, non-motor vehicle, and pedestrian), "vehicle" (motor vehicle), "nonmotor" (non-motor vehicle), "human" (pedestrian), "motorWithHum" (motor vehicle and pedestrian), "nonmotorWithHum" (non-motor vehicle and pedestrian), "motorNonmotor" (motor vehicle and non-motor vehicle)-->vehWithHum
  </capType>
  <enabled opt="true,false">
    <!--ro, opt, bool, whether to enable smart video security, attr:opt{req, string}-->true
  </enabled>
  <TriggerLine>
    <!--ro, opt, object, triggering line of smart video security capture-->
    <RegionCoordinatesList>
      <!--ro, opt, array, area coordinate list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </TriggerLine>
  <capNo opt="1,2">
    <!--ro, opt, int, number of captured pictures, attr:opt{req, string}-->0
  </capNo>
  <interval min="1" max="3000">
    <!--ro, opt, int, capture time, unit:ms, attr:min{req, int},max{req, int}-->0
```

```
    </interval>
    <carDriveDirect opt="up2down,down2up">
      <!--ro, req, enum, vehicle detection type, subType:string, attr:opt{req, string}, desc:"up2down" (up to down), "down2up" (down to up)-->up2down
    </carDriveDirect>
    <sceneMode opt="standardScene ,lowPoleScene">
      <!--ro, opt, enum, scene mode, subType:string, attr:opt{req, string}, desc:"standardScene" (standard scene), "lowPoleScene" (low pole scene), "entrance" (entrance and exit scene), "other" (other scenes)-->standardScene
    </sceneMode>
    <overSpeed>
      <!--ro, opt, bool, whether to enable overspeed trigger mode-->true
    </overSpeed>
    <overSpeedCapNo opt="1,2,3">
      <!--ro, opt, int, capture times of overspeed, attr:opt{req, string}-->0
    </overSpeedCapNo>
    <lowSpeed>
      <!--ro, opt, bool, whether to enable low speed trigger mode-->true
    </lowSpeed>
    <lowSpeedCapNo opt="1,2,3">
      <!--ro, opt, int, capture times of low speed, attr:opt{req, string}-->0
    </lowSpeedCapNo>
    <signSpeed min="0" max="255">
      <!--ro, opt, int, marked speed limit, attr:min{req, int},max{req, int}-->0
    </signSpeed>
    <speedLimit min="0" max="255">
      <!--ro, opt, int, speed limit, attr:min{req, int},max{req, int}-->0
    </speedLimit>
    <bigCarSignSpeed min="0" max="255">
      <!--ro, opt, int, marked speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarSignSpeed>
    <bigCarHighSpeedLimit min="0" max="255">
      <!--ro, opt, int, maximum speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarHighSpeedLimit>
    <bigCarLowSpeedLimit min="0" max="255">
      <!--ro, opt, int, low speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarLowSpeedLimit>
    <lowSpeedLimit min="0" max="60">
      <!--ro, opt, int, low speed limit for small-sized vehicle, attr:min{req, int},max{req, int}-->0
    </lowSpeedLimit>
    <bigCarSpeedLimit min="0" max="60">
      <!--ro, opt, int, speed limit for large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarSpeedLimit>
    <carHighSpeed min="0" max="255">
      <!--ro, opt, int, abnormal overspeed, attr:min{req, int},max{req, int}-->0
    </carHighSpeed>
    <carLowSpeed min="0" max="255">
      <!--ro, opt, int, abnormal low speed, attr:min{req, int},max{req, int}-->0
    </carLowSpeed>
    <bigCarHighSpeed min="0" max="255">
      <!--ro, opt, int, abnormal overspeed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed min="0" max="255">
      <!--ro, opt, int, abnormal low speed of large-sized vehicle, attr:min{req, int},max{req, int}-->0
    </bigCarLowSpeed>
    <nonmotorHighSpeedLimit min="0" max="255">
      <!--ro, opt, int, speed limit for non-motor vehicle, attr:min{req, int},max{req, int}-->0
    </nonmotorHighSpeedLimit>
  </VideoMonitor>
```


11.5.1.41 Get the recommended value of smart video security multi-frame recognition parameter

Request URL

GET /ISAPI/ITC/TriggerMode/videoMonitor/recommendation?channelID=<channelID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMonitor xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, smart video security multi-frame recognition parameter, attr:version{req, string, protocolVersion}-->
  <relatedLaneCount>
    <!--ro, req, int, total number of linked lanes, range:[1,3]-->0
  </relatedLaneCount>
  <capType>
    <!--ro, req, enum, detection type, subType:string, desc:detection type-->vehicle
  </capType>
  <ViolationDetectLineList>
```

```
  <!--ro, req, array, subType:object-->
  <ViolationDetectLine>
    <!--ro, opt, object-->
    <lineName>
      <!--ro, req, enum, line name, subType:string, desc:line name-->laneLine
    </lineName>
    <relatedDriveWay>
      <!--ro, req, int, lane No., range:[1,99]-->1
    </relatedDriveWay>
    <laneDirectionType>
      <!--ro, req, enum, lane direction, subType:int, desc:0-unknown, 1-from east to west, 2-from west to east, 3-from south to north, 4-from north to south, 5-from southeast to northwest, 6-from northwest to southeast, 7-from northeast to southwest, 8-from southwest to northeast-->1
    </laneDirectionType>
    <carDriveDirect>
      <!--ro, opt, enum, subType:string-->up2down
    </carDriveDirect>
    <RegionCoordinatesList>
      <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
      <RegionCoordinates>
        <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
        <positionX>
          <!--ro, req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
    <signSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </signSpeed>
    <speedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </speedLimit>
    <lowSpeedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </lowSpeedLimit>
    <bigCarSignSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarSignSpeed>
    <bigCarSpeedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarSpeedLimit>
    <bigCarLowSpeedLimit>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarLowSpeedLimit>
  </ViolationDetectLine>
</ViolationDetectLineList>
<VirtualLane>
  <!--ro, req, object-->
  <lineName>
    <!--ro, req, enum, line name, subType:string, desc:line name-->laneRightBoundaryLine
  </lineName>
  <RegionCoordinatesList>
    <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
    <RegionCoordinates>
      <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</VirtualLane>
<enabled>
  <!--ro, opt, bool-->true
</enabled>
<TriggerLine>
  <!--ro, opt, object-->
  <RegionCoordinatesList>
    <!--ro, opt, array, region coordinate list, subType:object, range:[0,2]-->
    <RegionCoordinates>
      <!--ro, opt, object, region coordinates, desc:the origin is the upper-left corner of the screen-->
      <positionX>
        <!--ro, req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--ro, req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</TriggerLine>
<capNo>
  <!--ro, opt, int, range:[1,2]-->1
</capNo>
<interval>
  <!--ro, opt, int, range:[1,3000], unit:ms-->1
</interval>
<sceneMode>
  <!--ro, opt, enum, subType:string-->standardScene
</sceneMode>
<overSpeed>
```

```
    <overSpeed>
      <!--ro, opt, bool-->true
    </overSpeed>
    <overSpeedCapNo>
      <!--ro, opt, int, range:[1,3]-->1
    </overSpeedCapNo>
    <lowSpeed>
      <!--ro, opt, bool-->true
    </lowSpeed>
    <lowSpeedCapNo>
      <!--ro, opt, int, range:[1,3]-->1
    </lowSpeedCapNo>
    <carHighSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </carHighSpeed>
    <carLowSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </carLowSpeed>
    <bigCarHighSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarHighSpeed>
    <bigCarLowSpeed>
      <!--ro, opt, int, range:[0,255]-->0
    </bigCarLowSpeed>
    <nonmotorOverSpeed>
      <!--ro, opt, bool-->true
    </nonmotorOverSpeed>
    <nonmotorHighSpeed>
      <!--ro, opt, int, range:[0,255], unit:km/h-->0
    </nonmotorHighSpeed>
    <channelId>
      <!--ro, opt, int, range:[1,5]-->1
    </channelId>
    <radarTrack>
      <!--ro, opt, bool-->true
    </radarTrack>
  </VideoMonitor>
```


11.5.1.42 Set the video mark parameters

Request URL

PUT /ISAPI/ITC/videoMark

Query Parameter

None

Request Message


```
<?xml version="1.0" encoding="UTF-8"?>

<VideoMarkInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, video mark information, attr:version{req, string, protocolVersion}-->
  <horizontalField>
    <!--req, int, horizontal FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </horizontalField>
  <verticalField>
    <!--req, int, vertical FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </verticalField>
  <autoGetGps>
    <!--req, bool, whether to get GPS information automatically, desc:the value of this node is false by default-->true
  </autoGetGps>
  <longitude>
    <!--req, string, longitude, range:[1,11], desc:longitude, which contains 11 digits-->W040D30M28S
  </longitude>
  <latitude>
    <!--req, string, latitude, range:[1,11], desc:latitude, which contains 11 digits-->N040D30M28S
  </latitude>
</VideoMarkInfo>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.43 Get the video mark parameters

Request URL

GET /ISAPI/ITC/videoMark

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<VideoMarkInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video mark information, attr:version{req, string, protocolVersion}-->
  <horizontalField>
    <!--ro, req, int, horizontal FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </horizontalField>
  <verticalField>
    <!--ro, req, int, vertical FOV (field of view), range:[0,360], desc:the default value is 0-->1
  </verticalField>
  <autoGetGps>
    <!--ro, req, bool, whether to get GPS information automatically: "true"-yes, "false"-no (default), desc:the value of this node is false by default--
>true
  </autoGetGps>
  <longitude>
    <!--ro, req, string, longitude, range:[1,11], desc:longitude, which contains 11 digits-->W040D30M28S
  </longitude>
  <latitude>
    <!--ro, req, string, latitude, range:[1,11], desc:latitude, which contains 11 digits-->N040D30M28S
  </latitude>
</VideoMarkInfo>
```


11.5.1.44 Get the video tag capability

Request URL

GET /ISAPI/ITC/videoMark/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<VideoMarkInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, video mark information, attr:version{req, string, protocolVersion}-->
  <horizontalField min="0" max="360">
    <!--ro, req, int, horizontal FOV (field of view), range:[0,360], attr:min{req, int},max{req, int}, desc:the default value is 0-->1
  </horizontalField>
  <verticalField min="0" max="360">
    <!--ro, req, int, vertical FOV (field of view), range:[0,360], attr:min{req, int},max{req, int}, desc:the default value is 0-->1
  </verticalField>
  <autoGetGps>
    <!--ro, req, bool, whether to get GPS information automatically: "true"-yes, "false"-no (default), desc:the value of this node is false by default--
>true
  </autoGetGps>
  <longitude min="1" max="11">
    <!--ro, req, string, longitude, range:[1,11], attr:min{req, int},max{req, int}, desc:longitude, which contains 11 digits-->W040D30M28S
  </longitude>
  <latitude min="1" max="11">
    <!--ro, req, string, latitude, range:[1,11], attr:min{req, int},max{req, int}, desc:latitude, which contains 11 digits-->N040D30M28S
  </latitude>
</VideoMarkInfo>
```

11.5.1.45 Set ANR parameters

Request URL

PUT /ISAPI/Traffic/ANR

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ANRControl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, ANR parameters, attr:version{req, string, protocolVersion}-->
  <uploadInterval>
    <!--req, int, upload interval, unit:ms-->0
  </uploadInterval>
  <uploadTimeout>
    <!--opt, int, uploading timed out, unit:ms-->0
  </uploadTimeout>
  <DeviceIpAddress>
    <!--opt, object, IP address of ANR arming server-->
    <addressingFormatType>
      <!--req, enum, address type, subType:string, desc:"ipaddress", "hostname"-->ipaddress
    </addressingFormatType>
    <hostName>
      <!--opt, string, host server name, range:[0,64], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,hostname}-->test
    </hostName>
    <ipAddress>
      <!--opt, string, IPv4 address, range:[0,32], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipAddress>
    <ipv6Address>
      <!--opt, string, IPv6 address, range:[0,128], dep:and,{$.ANRControl.DeviceIpAddress.addressingFormatType,eq,ipaddress}-->test
    </ipv6Address>
  </DeviceIpAddress>
</ANRControl>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device 
applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

11.5.1.46 Set parameters of traffic picture composition

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/imageMerge

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<ImageMerge xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, attr:version{req, string, protocolVersion}-->
  <isMerge>
    <!--req, int, whether to enable composition, range:[0,1]-->0
  </isMerge>
  <oneMergeType>
    <!--req, enum, subType:int-->101
  </oneMergeType>
  <twoMergeType>
    <!--req, enum, subType:int-->201
  </twoMergeType>
  <threeMergeType>
    <!--req, enum, subType:int-->301
  </threeMergeType>
  <fourMergeType>
    <!--req, enum, subType:int-->401
  </fourMergeType>
  <closeupIndex>
    <!--opt, int, range:[0,5]-->0
  </closeupIndex>
  <closeupScale>
    <!--opt, int, range:[0,100]-->0
  </closeupScale>
  <positionOffset>
    <!--opt, int, range:[50,2047]-->50
  </positionOffset>
</ImageMerge>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```


11.5.1.47 Get the picture composition parameters

Request URL

GET /ISAPI/Traffic/channels/<channelID>/imageMerge

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<ImageMerge xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <isMerge>
    <!--ro, req, int, range:[0,1]-->0
  </isMerge>
  <oneMergeType>
    <!--ro, req, enum, subType:string-->101
  </oneMergeType>
  <twoMergeType>
    <!--ro, req, enum, subType:string-->201
  </twoMergeType>
  <threeMergeType>
    <!--ro, req, enum, subType:string-->301
  </threeMergeType>
  <fourMergeType>
    <!--ro, req, enum, subType:string-->401
  </fourMergeType>
  <closeupIndex>
    <!--ro, opt, int, range:[0,5]-->0
  </closeupIndex>
  <closeupScale>
    <!--ro, opt, int, range:[0,100]-->0
  </closeupScale>
  <positionOffset>
    <!--ro, opt, int, range:[0,2159]-->50
  </positionOffset>
</ImageMerge>
```


11.5.1.48 Set the parameters of vehicle detection captured pictures

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, parameters of vehicle detection captured pictures, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--opt, object, picture parameters-->
    <mode>
      <!--req, enum, capture mode, subType:string, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality>
      <!--opt, int, picture quality, range:[1,100]-->50
    </pictureQuality>
    <pictureSize>
      <!--opt, int, picture size, unit:KB-->100
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--opt, object, picture overlay-->
    <enabled>
      <!--req, bool, whether to enable text overlay-->true
    </enabled>
    <OverlapItemList>
      <!--opt, array, overlay item list, subType:object-->
      <OverlapItem>
        <!--opt, object, overlay items-->
        <id>
          <!--req, int, index-->1
        </id>
        <item opt="positionNo">
          <!--req, string, overlay content, attr:opt{opt, enum, subType:string}-->test
        </item>
      </OverlapItem>
    </OverlapItemList>
    <fontColor>
      <!--opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--opt, object, Capture Overlay, desc:professional OSD configuration for picture capture, which is only supported by capture cameras now-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--opt, object, Composite Picture Overlay, desc:professional OSD configuration for picture composite, which is only supported by capture cameras now-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--opt, object, License Plate Enhancement-->
    <enabled>
      <!--opt, bool, whether to enable license plate enhancement on captured picture-->true
    </enabled>
    <level>
      <!--opt, int, set the level of license plate enhancement from 0 to 100, range:[0,100]-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList>
    <!--opt, array, the list of the types of captured pictures, subType:object, range:[1,3], desc:all types of picture shall be captured if this node is not returned-->
    <capturePicType>
      <!--opt, enum, types of captured pictures, subType:string, desc:"licensePlatePicture" (license plate picture), "vehiclePicture" (vehicle picture), "detectionPicture" (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution>
    <!--opt, string, picture resolution, desc:subject to the returned range-->1920*1080
  </resolution>
  <capturePicIntervalEnabled>
    <!--ro, opt, bool, whether to enable picture capture interval-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime>
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled>
    <!--opt, bool, whether to enable background picture of panoramic channel, desc:a scene can be monitored by both channels of the front-end dual-channel device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the panoramic-image channel-->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications, 
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.49 Get parameters of vehicle detection captured pictures

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of vehicle detection captured pictures, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--ro, opt, object, picture parameters-->
    <mode>
      <!--ro, req, enum, capture mode, subType:string, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality>
      <!--ro, opt, int, picture quality, range:[1,100]-->50
    </pictureQuality>
    <pictureSize>
      <!--ro, opt, int, picture size, unit:KB-->100
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--ro, opt, object, picture overlay-->
    <enabled>
      <!--ro, req, bool, whether to enable text overlay-->true
    </enabled>
    <OverlapItemList>
      <!--ro, opt, array, overlay item list, subType:object-->
      <OverlapItem>
        <!--ro, opt, object, overlay items-->
        <id>
          <!--ro, req, int, ID-->1
        </id>
        <item opt="positionNo">
          <!--ro, req, string, overlay content, attr:opt{opt, enum, subType:string}-->test
        </item>
      </OverlapItem>
    </OverlapItemList>
    <fontColor>
      <!--ro, opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--ro, opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--ro, opt, object, capture overlay, desc:professional OSD configuration for picture capture, which is only supported by capture cameras now-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--ro, opt, object, composite picture overlay, desc:professional OSD configuration for picture composite, which is only supported by capture cameras 
now-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--ro, opt, object, license plate enhancement-->
    <enabled>
      <!--ro, opt, bool, whether to enable license plate enhancement on captured picture-->true
    </enabled>
    <level>
      <!--ro, opt, int, set the level of license plate enhancement from 0 to 100, range:[0,100]-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList>
    <!--ro, opt, array, the list of the types of captured pictures, subType:object, range:[1,3], desc:all types of picture shall be captured if this node is 
not returned-->
    <capturePicType>
      <!--ro, opt, enum, types of captured pictures, subType:string, desc:"licensePlatePicture” (license plate picture), "vehiclePicture” (vehicle picture), 
"detectionPicture” (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution>
    <!--ro, opt, string, picture resolution, desc:subject to the returned range-->1920*1080
  </resolution>
  <capturePicIntervalEnabled>
    <!--ro, opt, bool, whether to enable picture capture interval-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime>
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled>
    <!--ro, opt, bool, whether to enable background picture of panoramic channel, desc:a scene can be monitored by both channels of the front-end dual-
channel device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the panoramic-image channel-
->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```


11.5.1.50 Get captured picture configuration capability

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<PicParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, picture capture configuration capability, attr:version{opt, string, protocolVersion}-->
  <PictureCfg>
    <!--ro, req, object, picture parameters-->
    <mode opt="quality,size">
      <!--ro, req, enum, capture mode, subType:string, attr:opt{opt, enum, subType:string}, desc:"quality" (by quality), "size" (by size)-->quality
    </mode>
    <pictureQuality min="1" max="100">
      <!--ro, opt, int, picture quality, range:[1,100], attr:min{opt, string},max{opt, string}-->1
    </pictureQuality>
    <pictureSize min="1" max="6553500000">
      <!--ro, opt, int, picture size, unit:KB, attr:min{req, int},max{req, int}-->1
    </pictureSize>
  </PictureCfg>
  <Overlap>
    <!--ro, opt, object, picture overlay-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable text overlay, attr:opt{req, string}-->true
    </enabled>
    <OverlapItem opt="positionNo,positionInfo,cameraNo,captureTime,plateNo,vehicleColor,sceneName,carType,vehicleLogo,sceneNo,GPS,vehicleSpeed">
      <!--ro, req, enum, overlay items, subType:string, attr:opt{opt, enum}, desc:"positionNo" (detection point No.), "positionInfo" (detection point),
"cameraNo" (camera No.), "captureTime" (capture time), "plateNo" (license plate No.), "vehicleColor" (vehicle color), "sceneName" (scene name), "carType"
(vehicle type), "vehicleLogo" (main vehicle brand) , "sceneNo" (scene No.), "GPS" (GPS device)-->positionNo
    </OverlapItem>
    <fontColor>
      <!--ro, opt, string, font color, desc:the previous version of protocol; hexBinary-->test
    </fontColor>
    <backColor>
      <!--ro, opt, string, background color, desc:the previous version of protocol; hexBinary-->test
    </backColor>
  </Overlap>
  <CapturePicOverlays>
    <!--ro, opt, object, professional OSD configuration for captured pictures,currently this configuration is only supported by capture cameras,
desc:professional OSD configuration for captured pictures,currently this configuration is only supported by capture cameras-->
  </CapturePicOverlays>
  <MergePicOverlays>
    <!--ro, opt, object, professional OSD configuration for combined pictures,currently this configuration is only supported by capture cameras,
desc:professional OSD configuration for combined pictures,currently this configuration is only supported by capture cameras-->
  </MergePicOverlays>
  <PlateEnhancement>
    <!--ro, opt, object, license plate enhancement on captured picture-->
    <enabled opt="true,false" def="false">
      <!--ro, opt, bool, whether to enable license plate enhancement on captured picture, attr:opt{opt, string},def{opt, string}-->true
    </enabled>
    <level min="0" max="100" def="50">
      <!--ro, opt, int, license plate enhancement level range, range:[0,100], attr:min{opt, string},max{opt, string},def{opt, string}-->1
    </level>
  </PlateEnhancement>
  <CapturePicTypeList size="3">
    <!--ro, opt, array, the list of the types of captured pictures, subType:object, attr:size{req, int}, desc:all types of picture shall be captured if this
node is not returned-->
    <capturePicType opt="licensePlatePicture,vehiclePicture,detectionPicture">
      <!--ro, opt, string, types of captured pictures, attr:opt{req, string}, desc:"licensePlatePicture” (license plate picture), "vehiclePicture” (vehicle
picture), "detectionPicture” (detection picture (background picture))-->licensePlatePicture
    </capturePicType>
  </CapturePicTypeList>
  <resolution opt="1920*1080,2688*1520,1280*720,704*576,352*288">
    <!--ro, opt, string, picture resolution, attr:opt{req, string}-->1920*1080
  </resolution>
  <capturePicIntervalEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable picture capture interval, attr:opt{req, string}-->true
  </capturePicIntervalEnabled>
  <capturePicIntervalTime min="1" max="30" def="1">
    <!--ro, opt, int, capture interval, range:[1,30], unit:s, dep:and,{$.PicParam.capturePicIntervalEnabled,eq,true}, attr:min{req, int},max{req,
int},def{req, int}-->1
  </capturePicIntervalTime>
  <syncCaptureBackPanoramaEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable background picture of panoramic channel, attr:opt{req, string}, desc:a scene can be monitored by both channels of
the front-end dual-channel device. The alarm from the detailed-image channel should be linked with the capture of panoramic background pictures by the
panoramic-image channel-->true
  </syncCaptureBackPanoramaEnabled>
</PicParam>
```


11.5.1.51 Get the capability of OSD overlay on a captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{opt, string, protocolVersion}-->
  <overlayInfoEnabled opt="true,false">
    <!--ro, req, bool, whether to enable text overlay, attr:opt{req, string}, desc:whether to enable text overlay-->true
  </overlayInfoEnabled>
  <OverlayInfoList size="50">
    <!--ro, req, array, subType:object, attr:size{opt, string}-->
    <OverlayInfo>
      <!--ro, req, object-->
      <itemType
opt="0,1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24,25,26,27,28,29,30,31,32,33,34,35,36,37,38,39,40,41,42,43,44,45,46,47,48,49,50,51,52,53
,54,55,56,57,58,59,60,61,62,63,64,65,66,67,68,69,70,71,72,73,74,75,76,77,78,79,80,81,82,83,84,85,86,87,88,89,90,91,92,93,94,95,96,97,98,99,100,101,102,103,1
04,105,106,107,108,109,110,111,112,113,114,115,116,117,118,119" min="1" max="88">
        <!--ro, req, enum, subType:int, attr:opt{opt, string},min{req, int},max{req, int}-->70
      </itemType>
      <editAble opt="true,false">
        <!--ro, req, bool, whether the overlaid information is editable, attr:opt{opt, string}-->true
      </editAble>
      <itemOverlayEnabled opt="true,false">
        <!--ro, req, bool, attr:opt{opt, string}-->true
      </itemOverlayEnabled>
      <customName min="0" max="32">
        <!--ro, req, string, self-defined name of the overlay information, attr:min{opt, string},max{opt, string}-->test
      </customName>
      <changeLineNum min="0" max="100">
        <!--ro, req, int, range:[0,100], attr:min{opt, string},max{opt, string}-->50
      </changeLineNum>
      <spaceNum min="0" max="255">
        <!--ro, req, int, range:[0,255], attr:min{opt, string},max{opt, string}-->1
      </spaceNum>
      <startPosEnable opt="true,false">
        <!--ro, opt, bool, whether to enable coordinate configuration, attr:opt{opt, string}-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point-->0
      </startPosLeft>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent min="0" max="100">
    <!--ro, req, int, percentage of overlaying lines, range:[0,100], attr:min{opt, string},max{opt, string}-->100
  </linePercent>
  <itemsStlye opt=" horizontal,vertical">
    <!--ro, req, enum, overlay mode, subType:string, attr:opt{opt, string}, desc:"horizontal" (default),"vertical"-->vertical
  </itemsStlye>
  <charStyle opt="song_type,wei_type">
    <!--ro, req, enum, font type, subType:string, attr:opt{opt, string}, desc:font type-->song_type
  </charStyle>
  <charSize min="0" max="5">
    <!--ro, req, enum, character size, subType:int, attr:min{opt, string},max{opt, string}, desc:0-16*16(Chinese)/8*16(English),1-
32*32(Chinese)/16*32(English),2-48*48,3-64*64(Chinese)/32*64(English)-->1
  </charSize>
  <charPosition min="0" max="2">
    <!--ro, req, enum, text position overlaid on the picture, subType:int, attr:min{opt, string},max{opt, string}, desc:0 (in the picture),1 (outside the
top edge of the picture), 2 (outside the bottom edge of the picture)-->1
  </charPosition>
  <charInterval min="0" max="16">
    <!--ro, req, int, character separation distance, range:[0,16], unit:px, attr:min{opt, string},max{opt, string}-->1
  </charInterval>
  <foreColor min="0" max=" 0xffffff">
    <!--ro, req, int, foreground color, which is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value
is 0xffffff (white), attr:min{opt, string},max{opt, string}-->1
  </foreColor>
  <backColor min="0" max=" 0xffffff">
    <!--ro, req, int, background color, which is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value
is 0x0 (black), attr:min{opt, string},max{opt, string}-->1
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption: 0-no,1-yes-->true
  </colorAdapt>
```

```xml
    <charColorReverseEnabled opt="true, false">
      <!--ro, opt, bool, attr:opt{req, string}-->true
    </charColorReverseEnabled>
    <platePicOverlay>
      <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
    </platePicOverlay>
    <platePicOverlayMultiplier min="1" max="3">
      <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
    </platePicOverlayMultiplier>
    <platePicPosTop>
      <!--ro, req, int, start top coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
    </platePicPosTop>
    <platePicPosLeft>
      <!--ro, req, int, start left coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
    </platePicPosLeft>
    <colOverDetModEnabled opt="true,false">
      <!--ro, opt, bool, attr:opt{opt, string}-->true
    </colOverDetModEnabled>
    <overlaidTextTransparency min="0" max="100">
      <!--ro, opt, int, range:[0,100], unit:%, attr:min{req, int},max{req, int}-->1
    </overlaidTextTransparency>
    <dateStyle opt="YYYY-MM-DD,MM-DD-YYYY,DD-MM-YYYY,CHR-YYYY-MM-DD,CHR-MM-DD-YYYY,CHR-DD-MM-YYYY,CHR-YYYY/MM/DD,CHR-MM/DD/YYYY,CHR-DD/MM/YYYY">
      <!--ro, opt, string, attr:opt{req, string}-->YYYY-MM-DD
    </dateStyle>
    <timeStyle opt="12hour,24hour">
      <!--ro, opt, string, attr:opt{req, string}-->12hour
    </timeStyle>
    <plateColorOverlayEnabled opt="true,false">
      <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,10}, attr:opt{req, string}-->true
    </plateColorOverlayEnabled>
    <plateProvinceOverlayEnabled opt="true,false">
      <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,10}, attr:opt{req, string}-->true
    </plateProvinceOverlayEnabled>
    <overSpeedPercentAccurateEnabled opt="true,false">
      <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,20}, attr:opt{req, string}-->true
    </overSpeedPercentAccurateEnabled>
    <nonMotoPassengerNumEnabled opt="true,false">
      <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,60}, attr:opt{req, string}-->true
    </nonMotoPassengerNumEnabled>
    <PictureInfoList>
      <!--ro, opt, array, subType:object-->
    </PictureInfoList>
  </CapturePicOverlays>
```

11.5.1.52 Set the configuration parameters for overlaying all OSD text information on a captured picture

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, see details in the message of XML_OverlayInfo, subType:object, attr:version{opt, string, protocolVersion}-->
  <OverlayInfo>
    <!--opt, object, overlay information-->
    <overlayInfoID>
      <!--req, int, No., range:[1,50]-->1
    </overlayInfoID>
    <itemType>
      <!--req, enum, the type of overlaid information on a picture, subType:int, desc:the type of overlaid information on a picture-->70
    </itemType>
    <itemOverlayEnabled>
      <!--req, bool-->true
    </itemOverlayEnabled>
    <customName>
      <!--req, string, self-defined name of the overlay information, range:[1,32]-->test
    </customName>
    <changeLineNum>
      <!--req, int, range:[0,100]-->2
    </changeLineNum>
    <spaceNum>
      <!--req, int, range:[0,255]-->1
    </spaceNum>
    <startPosEnable>
      <!--opt, bool, whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of
intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and
the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--opt, string, overlay character string, range:[1,32], desc:overlay character string-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.53 Get the configuration parameters of overlaying all OSD text information on a captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, list of overlay information, subType:object, attr:version{opt, string, protocolVersion}-->
  <OverlayInfo>
    <!--ro, opt, object, overlay information-->
    <overlayInfoID>
      <!--ro, opt, int, range:[1,50]-->1
    </overlayInfoID>
    <itemType>
      <!--ro, req, enum, subType:int-->70
    </itemType>
    <itemOverlayEnabled>
      <!--ro, req, bool-->true
    </itemOverlayEnabled>
    <customName>
      <!--ro, req, string, self-defined name of the overlay information, range:[1,32]-->test
    </customName>
    <changeLineNum>
      <!--ro, req, int, range:[0,100]-->2
    </changeLineNum>
    <spaceNum>
      <!--ro, req, int, range:[0,255]-->1
    </spaceNum>
    <startPosEnable>
      <!--ro, opt, bool, whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--ro, opt, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--ro, opt, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--ro, opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of
intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and
the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--ro, opt, string, overlay character string, range:[1,32], desc:overlay character string-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```

11.5.1.54 Get the configuration parameters of overlaying a piece of OSD text information on a captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, overlay information, attr:version{opt, string, protocolVersion}-->
  <overlayInfoID>
    <!--ro, opt, int, No., range:[1,50]-->1
  </overlayInfoID>
  <itemType>
    <!--ro, req, enum, overlayed information type, subType:int, desc:overlayed information type-->70
  </itemType>
  <itemOverlayEnabled>
    <!--ro, req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--ro, req, string, self-defined name of the overlay information, range:[1,32]-->test
  </customName>
  <changeLineNum>
    <!--ro, req, int, number of line feeds, range:[0,100]-->2
  </changeLineNum>
  <spaceNum>
    <!--ro, req, int, number of spaces, range:[0,255]-->1
  </spaceNum>
  <startPosEnable>
    <!--ro, opt, bool, whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--ro, opt, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--ro, opt, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture
width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--ro, opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of
intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and
the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--ro, opt, string, overlay character string, range:[1,32], desc:overlay character string-->test
  </overlayInfoText2>
</OverlayInfo>
```


11.5.1.55 Set the configuration parameters for overlaying a

piece of OSD text information on a captured picture **Request URL**

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays/info/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, overlay information, attr:version{opt, string, protocolVersion}-->
  <overlayInfoID>
    <!--req, int, overlayed information ID, range:[1,50]-->1
  </overlayInfoID>
  <itemType>
    <!--req, enum, overlayed information type, subType:int, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line,etc.), 18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30/31/32-reserved, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-congestion-->70
  </itemType>
  <itemOverlayEnabled>
    <!--req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--req, string, custom overlaying name, range:[1,32]-->test
  </customName>
  <changeLineNum>
    <!--req, int, number of line feeds, range:[0,100]-->2
  </changeLineNum>
  <spaceNum>
    <!--req, int, number of spaces, range:[0,255]-->1
  </spaceNum>
  <startPosEnable>
    <!--opt, bool, whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--req, int, start top coordinate, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--req, int, start left coordinate, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--opt, string, overlay character string, range:[0,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--opt, string, overlay character string for camera 2, range:[1,32], desc:overlay character string for camera 2-->test
  </overlayInfoText2>
</OverlayInfo>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.56 Get overlay parameters of captured traffic pictures by channel

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays?multiCharPosition=<multiCharPosition>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
multiCharPosition	enum	--

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>
<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, see details in the message of XML_OverlayInfoList, attr:version{opt, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable text overlay, desc:whether to enable text overlay-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--ro, opt, array, subType:object-->
    <OverlayInfo>
      <!--ro, opt, object-->
      <overlayInfoID>
        <!--ro, req, int-->1
      </overlayInfoID>
      <itemType>
        <!--ro, req, enum, overlay mode, subType:int, desc:overlay mode-->70
      </itemType>
      <itemOverlayEnabled>
        <!--ro, req, bool-->true
      </itemOverlayEnabled>
      <customName>
        <!--ro, req, string, self-defined name of the overlay information-->test
      </customName>
      <changeLineNum>
        <!--ro, req, int, range:[0,100]-->2
      </changeLineNum>
      <spaceNum>
        <!--ro, req, int, space number, range:[0,255]-->1
      </spaceNum>
      <startPosEnable>
        <!--ro, opt, bool, whether to enable coordinate configuration-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point-->0
      </startPosLeft>
      <overlayInfoText>
        <!--ro, opt, string, overlay character string, range:[0,128]-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--ro, opt, string, overlay character string, range:[0,32]-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--ro, req, int, percentage of overlaying lines, range:[0,100]-->1
  </linePercent>
  <itemsStlye>
    <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal", "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--ro, req, enum, font type, subType:string, desc:font type: 0-SimSun (default),1-Wei-->song_type
  </charStyle>
  <charSize>
    <!--ro, req, enum, character size, subType:int, desc:0 (8*16), 1 (16*32), 2 (48*48), 3 (32*64), 4 (64*128)-->0
  </charSize>
  <charPosition>
    <!--ro, req, enum, text position overlaid on the picture, subType:int, desc:0 (in the picture), 1 (outside the top edge of the picture), 2 (outside the
bottom edge of the picture)-->1
  </charPosition>
  <charInterval>
    <!--ro, req, int, character separation distance, range:[0,16]-->0
  </charInterval>
  <foreColor>
    <!--ro, req, int, foreground color, range:[0,16777215]-->1
  </foreColor>
  <backColorEnabled>
    <!--ro, opt, bool-->true
  </backColorEnabled>
  <backColor>
    <!--ro, opt, int, background color, range:[0,16777215], desc:background color-->1
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <zeroizeEnable>
```

```
    <!--ro, opt, bool, whether to enable zero filling for OSD overlay-->true
  </zeroizeEnable>
  <snapTimeType>
    <!--ro, opt, enum, subType:string-->generationTime
  </snapTimeType>
  <charColorReverseEnabled>
    <!--ro, opt, bool-->true
  </charColorReverseEnabled>
  <platePicOverlay>
    <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicOverlayMultiplier>
    <!--ro, opt, int-->1
  </platePicOverlayMultiplier>
  <platePicPosTop>
    <!--ro, req, int, start top coordinate-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--ro, req, int, start left coordinate-->0
  </platePicPosLeft>
  <overlaidTextTransparency>
    <!--ro, opt, int, range:[0,100], unit:%-->1
  </overlaidTextTransparency>
  <featurePicOSDEnable>
    <!--ro, opt, bool-->true
  </featurePicOSDEnable>
  <dateStyle>
    <!--ro, opt, enum, subType:string-->YYYY-MM-DD
  </dateStyle>
  <timeStyle>
    <!--ro, opt, enum, subType:string-->12hour
  </timeStyle>
  <plateProvinceOverlayEnabled>
    <!--ro, opt, bool, dep:and,{$.CapturePicOverlays.OverlayInfoList[*].OverlayInfo.itemType,eq,10}-->true
  </plateProvinceOverlayEnabled>
  <prefixLanguage>
    <!--ro, opt, enum, subType:string-->English
  </prefixLanguage>
  <PictureInfoList>
    <!--ro, opt, array, subType:object-->
  </PictureInfoList>
</CapturePicOverlays>
```

11.5.1.57 Set overlay parameters of captured traffic pictures by channel

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/capturePicOverlays?multiCharPosition=<multiCharPosition>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
multiCharPosition	enum	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<CapturePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, see details in the message of XML_OverlayInfoList, attr:version{opt, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--req, bool, whether to enable text overlay, desc:whether to enable text overlay-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--opt, array, subType:object-->
    <OverlayInfo>
      <!--opt, object-->
      <overlayInfoID>
        <!--req, int-->1
      </overlayInfoID>
      <itemType>
        <!--req, enum, subType:int-->70
      </itemType>
      <itemOverlayEnabled>
        <!--req, bool-->true
      </itemOverlayEnabled>
      <customName>
        <!--req, string, self-defined name of the overlay information-->test
      </customName>
      <changeLineNum>
        <!--req, int, range:[0,100]-->2
      </changeLineNum>
      <spaceNum>
        <!--req, int, range:[0,255]-->1
```

```xml
      </spaceNum>
      <startPosEnable>
        <!--opt, bool, whether to enable coordinate configuration-->true
      </startPosEnable>
      <startPosTop>
        <!--req, int, top coordinate of the start point-->0
      </startPosTop>
      <startPosLeft>
        <!--req, int, left coordinate of the start point-->0
      </startPosLeft>
      <overlayInfoText>
        <!--opt, string, overlay character string-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--opt, string, overlay character string-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--req, int, percentage of overlaying lines, range:[0,100]-->1
  </linePercent>
  <itemsStlye>
    <!--req, enum, overlay mode, subType:string, desc:"horizontal" (default),"vertical”-->horizontal
  </itemsStlye>
  <charStyle>
    <!--req, enum, font type, subType:string, desc:0-SimSun (default),1-Wei-->song_type
  </charStyle>
  <charSize>
    <!--req, enum, character size, subType:int, desc:0-16*16(Chinese)/8*16(English),1-32*32(Chinese)/16*32(English),2-48*48,3-64*64(Chinese)/32*64(English)-
->0
  </charSize>
  <charPosition>
    <!--req, enum, text position overlayed on the picture, subType:int, desc:0 (in the picture),1 (outside the top edge of the picture), 2 (outside the
bottom edge of the picture)-->1
  </charPosition>
  <charInterval>
    <!--req, int, character separation distance, range:[0,16]-->0
  </charInterval>
  <foreColor>
    <!--req, int, foreground color, which is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value is
0xffffff (white)-->1
  </foreColor>
  <backColor>
    <!--req, int, background color, which is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value is
0x0 (black)-->1
  </backColor>
  <colorAdapt>
    <!--opt, bool, whether to enable color self-adaption: 0-no,1-yes-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--opt, bool, whether to enable zero filling for OSD overlay, which is used to enable zero filling for vehicle speed, speed limit, overspeed ratio and
lane No. Zero filling is enabled by default-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicPosTop>
    <!--req, int, start top coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture height, and the
default value is 0-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--req, int, start left coordinate, which is only valid for overlaying within the picture, the value is between 0 and the actual picture width, and the
default value is 0-->0
  </platePicPosLeft>
  <overlaidTextTransparency>
    <!--opt, int, range:[0,100], unit:%-->1
  </overlaidTextTransparency>
  <PictureInfoList>
    <!--opt, array, subType:object-->
  </PictureInfoList>
</CapturePicOverlays>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
CapturePicOverlays	[Message content]	application/xml	--	--	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, 
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error reason description in detail, desc:error reason description in detail-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.58 Get the overlay capability in the composite captured pictures

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, composite picture overlay, attr:version{req, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable the function-->true
  </overlayInfoEnabled>
  <OverlayInfoList size="50">
    <!--ro, req, array, list of overlay information, subType:object, attr:size{req, int}-->
    <OverlayInfo>
      <!--ro, req, object, overlay information-->
      <itemType opt="0,1,2,3,…,82,83,85">
        <!--ro, req, enum, overlay information type, subType:string, attr:opt{req, string}, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-
direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle 
color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic 
violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over 
yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 
24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow 
label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making 
a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box 
detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed 
limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-
wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading 
rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann 
emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & 
exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license 
plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-
reflective stripe-->0
      </itemType>
      <editAble opt="true,false">
        <!--ro, req, bool, whether the overlaid information is editable, attr:opt{req, string}-->true
      </editAble>
      <itemOverlayEnabled opt="true,false">
        <!--ro, req, bool, whether to overlay the item, attr:opt{req, string}-->true
```

```
      </itemOverlayEnabled>
      <customName min="0" max="32">
        <!--ro, req, string, custom overlaying name, range:[0,32], attr:min{req, int},max{req, int}, desc:the maximum length is 32 (including '\0'). If this node is not configured, the default name will be overlaid-->test
      </customName>
      <changeLineNum min="0" max="100">
        <!--ro, req, int, number of line feeds, range:[0,100], attr:min{req, int},max{req, int}, desc:the default value is 0-->0
      </changeLineNum>
      <spaceNum min="0" max="255">
        <!--ro, req, int, number of spaces, range:[0,255], attr:min{req, int},max{req, int}, desc:the default value is 0-->0
      </spaceNum>
      <startPosEnable opt="true,false">
        <!--ro, opt, bool, whether to enable coordinate configuration, attr:opt{req, string}, desc:it is valid only for information overlay within the picture. After enabled, the function of line feed and space will be invalid. You are recommended to disable coordinate configuration-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
      </startPosLeft>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent min="0" max="100">
    <!--ro, req, int, percentage of overlaying lines, range:[0,100], attr:min{req, int},max{req, int}, desc:the default value is 100-->1
  </linePercent>
  <itemsStlye opt="horizontal,vertical">
    <!--ro, req, enum, overlay mode, subType:string, attr:opt{req, string}, desc:"horizontal" (default), "vertical"-->horizontal
  </itemsStlye>
  <charStyle opt="song_type,wei_type">
    <!--ro, req, enum, font type, subType:string, attr:opt{req, string}, desc:"song_type"-SimSun (default), "wei_type"-Wei-->song_type
  </charStyle>
  <charSize min="0" max="5">
    <!--ro, req, enum, character size, subType:string, attr:min{req, int},max{req, int}, desc:"0"-16*16(Chinese)/8*16(English), "1"-32*32(Chinese)/16*32(English), "2"-48*48, "3"-64*64(Chinese)/32*64(English), "4"-128*128(Chinese)/64*128(English), "5"-96*96(Chinese)/48*96(English)-->0
  </charSize>
  <charPosition min="0" max="3">
    <!--ro, req, enum, position of the text overlaid on the picture, subType:int, attr:min{req, int},max{req, int}, desc:0 (in the picture),1 (outside the top edge of the picture), 2 (outside the bottom edge of the picture)-->0
  </charPosition>
  <charInterval min="0" max="16">
    <!--ro, req, int, character pitch, range:[0,16], attr:min{req, int},max{req, int}, desc:character separation distance, which is between 0 and 16 and the default value is 0, unit: pixel-->0
  </charInterval>
  <foreColor min="0" max="0xffffff">
    <!--ro, req, int, foreground color, attr:min{req, int},max{req, int}, desc:it is the RGB value directly obtained via the palette and its default value is 0xffffff (white)-->0
  </foreColor>
  <backColor min="0" max="0xffffff">
    <!--ro, req, int, background color, attr:min{req, int},max{req, int}, desc:it is the RGB value directly obtained via the palette and its default value is 0x0 (black)-->0
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <colOverDetModEnabled opt="true,false">
    <!--ro, opt, bool, whether to enable vehicle color overlay in detailed mode, attr:opt{req, string}-->true
  </colOverDetModEnabled>
</MergePicOverlays>
```

11.5.1.59 Set the configuration parameters for overlaying all OSD text information on the composite captured picture

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, array, list of overlay information, subType:object, attr:version{req, string, protocolVersion}-->
  <OverlayInfo>
    <!--req, object, overlay information-->
    <overlayInfoID>
      <!--req, int, overlay information ID-->0
    </overlayInfoID>
    <itemType>
      <!--req, enum, overlay information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for parking space patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask-->0
    </itemType>
    <itemOverlayEnabled>
      <!--req, bool, whether to overlay the information-->true
    </itemOverlayEnabled>
    <customName>
      <!--req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
    </customName>
    <changeLineNum>
      <!--req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
    </changeLineNum>
    <spaceNum>
      <!--req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
    </spaceNum>
    <startPosEnable>
      <!--opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
    </startPosEnable>
    <startPosTop>
      <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```


11.5.1.60 Get parameters of overlaying characters on composite pictures

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfoList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, array, overlay information list, subType:object, attr:version{req, string, protocolVersion}-->
  <OverlayInfo>
    <!--ro, req, object, overlay information-->
    <overlayInfoID>
      <!--ro, req, int, overlay ID-->0
    </overlayInfoID>
    <itemType>
      <!--ro, req, enum, overlaid information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for parking space patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask-->0
    </itemType>
    <itemOverlayEnabled>
      <!--ro, req, bool, whether to overlay this item-->true
    </itemOverlayEnabled>
    <customName>
      <!--ro, req, string, custom overlaying name, range:[0,32], desc:customize name of the overlay information-->test
    </customName>
    <changeLineNum>
      <!--ro, req, int, the default value is 0, range:[0,100], desc:the default value is 0-->0
    </changeLineNum>
    <spaceNum>
      <!--ro, req, int, the number of spaces, range:[0,255], desc:the default value is 0-->0
    </spaceNum>
    <startPosEnable>
      <!--ro, opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
    </startPosEnable>
    <startPosTop>
      <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
    </startPosTop>
    <startPosLeft>
      <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
    </startPosLeft>
    <overlayInfoText>
      <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
    </overlayInfoText>
    <overlayInfoText2>
      <!--ro, opt, string, overlay characters, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
    </overlayInfoText2>
  </OverlayInfo>
</OverlayInfoList>
```


11.5.1.61 Set the configuration parameters for overlaying a piece of OSD text information on the composite captured picture

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, overlay information, attr:version{req, string, protocolVersion}-->
  <overlayInfoID>
    <!--req, int, overlayed information ID-->0
  </overlayInfoID>
  <itemType>
    <!--req, enum, overlay information type, subType:string, desc:0-unknown,1-place,2-intersection No.,3-device No.,4-direction No.,5-direction,6-lane No.,7-lane,8-capture time (without millisecond),9-capture time (with millisecond),10-license plate number,11-vehicle color,12-vehicle type,13-vehicle brand,14-vehicle speed,15-speed limit sign,16-vehicle length (between 1 and 99 meters),17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information,19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30/31/32-reserved, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection,35-vehicle sub-brand detection, 36-vehicle direction,37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-congestion-->0
  </itemType>
  <itemOverlayEnabled>
    <!--req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--req, string, custom overlaying name, range:[0,32], desc:self-defined name of the overlay information-->test
  </customName>
  <changeLineNum>
    <!--req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
  </changeLineNum>
  <spaceNum>
    <!--req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
  </spaceNum>
  <startPosEnable>
    <!--opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
  </overlayInfoText2>
</OverlayInfo>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, error code description, desc:error code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:the detailed information of custom error returned by device applications, which is used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

11.5.1.62 Get the configuration parameters of overlaying a piece of OSD text information on the composite captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays/info/<textID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
textID	string	--

Request Message

None

Response Message


```
<?xml version="1.0" encoding="UTF-8"?>

<OverlayInfo xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, overlay information, attr:version{req, string, protocolVersion}-->
  <overlayInfoID>
    <!--ro, req, int, overlayed information ID, which is between 1 and 50-->0
  </overlayInfoID>
  <itemType>
    <!--ro, req, enum, overlayed information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 
6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-
vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more 
useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera 
information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 
26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-
dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-
verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm 
detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-
segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-
person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating 
personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke 
detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking 
space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing 
mask-->0
  </itemType>
  <itemOverlayEnabled>
    <!--ro, req, bool, whether to overlay the item-->true
  </itemOverlayEnabled>
  <customName>
    <!--ro, req, string, custom overlaying name, range:[0,32], desc:self-defined name of the overlay information-->test
  </customName>
  <changeLineNum>
    <!--ro, req, int, number of line feeds, and the default value is 0, range:[0,100], desc:the default value is 0-->0
  </changeLineNum>
  <spaceNum>
    <!--ro, req, int, number of spaces, and the default value is 0, range:[0,255], desc:the default value is 0-->0
  </spaceNum>
  <startPosEnable>
    <!--ro, opt, bool, whether to enable coordinate configuration, desc:whether to enable coordinate configuration-->true
  </startPosEnable>
  <startPosTop>
    <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture 
height, and the default value is 0-->0
  </startPosTop>
  <startPosLeft>
    <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture 
width, and the default value is 0-->0
  </startPosLeft>
  <overlayInfoText>
    <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:
[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction 
description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, 
verification certificate No., calibration expiration date is 128 bytes-->test
  </overlayInfoText>
  <overlayInfoText2>
    <!--ro, opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string 
length of camera 2 is 32 bytes-->test
  </overlayInfoText2>
</OverlayInfo>
```


11.5.1.63 Set the overlay parameters in the composite captured pictures

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays?multiCharPosition=<multiCharPosition>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
multiCharPosition	enum	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, see details in the message of XML_OverlayInfoList, attr:version{req, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--req, bool, whether to enable the function-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--req, array, list of overlay information, subType:object-->
    <OverlayInfo>
      <!--req, object, overlay information-->
      <overlayInfoID>
        <!--req, int, overlay ID-->0
      </overlayInfoID>
      <itemType>
        <!--req, enum, overlay information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
      </itemType>
      <itemOverlayEnabled>
        <!--req, bool, whether to overlay the information-->true
      </itemOverlayEnabled>
      <customName>
        <!--req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
      </customName>
      <changeLineNum>
        <!--req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
      </changeLineNum>
      <spaceNum>
        <!--req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
      </spaceNum>
      <startPosEnable>
        <!--opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
      </startPosEnable>
      <startPosTop>
        <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
      </startPosTop>
      <startPosLeft>
        <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
      </startPosLeft>
      <overlayInfoText>
        <!--opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--req, int, percentage of overlaying lines, range:[0,100], desc:the default value is 100-->1
  </linePercent>
  <itemsStlye>
    <!--req, enum, overlay mode, subType:string, desc:"horizontal" (default), "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--req, enum, font type, subType:string, desc:font type: 0-SimSun (default),1-Wei-->song_type
  </charStyle>
  <charSize>
    <!--req, enum, character size, subType:string, desc:0 (8*16), 1 (16*32), 2 (48*48), 3 (32*64), 4 (64*128)-->0
  </charSize>
  <charPosition>
    <!--req, enum, position of the text overlaid in the picture, subType:int, desc:0 (in the picture),1 (outside the top edge of the picture), 2 (outside
```

```
the bottom edge of the picture)-->0
  </charPosition>
  <charInterval>
    <!--req, int, character pitch, range:[0,16], desc:the default value is 0, unit: pixel-->0
  </charInterval>
  <colorAdapt>
    <!--opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--opt, bool, whether to enable zero filling for OSD overlay, desc:it is used to enable zero filling for vehicle speed, speed limit, overspeed ratio,
and lane No. Zero filling is enabled by default-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicPosTop>
    <!--req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height,
and the default value is 0-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width,
and the default value is 0-->0
  </platePicPosLeft>
</MergePicOverlays>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”,
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, custom error information description, range:[0,1024], desc:detailed information of custom error returned by device applications,
used for fast debugging-->badXmlFormat
  </description>
</ResponseStatus>
```

11.5.1.64 Get the parameters of OSD overlay on the composite captured picture

Request URL

GET /ISAPI/Traffic/channels/<channelID>/picParam/mergePicOverlays?multiCharPosition=<multiCharPosition>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
multiCharPosition	enum	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<MergePicOverlays xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, see details in the message of XML_OverlayInfoList, attr:version{req, string, protocolVersion}-->
  <overlayInfoEnabled>
    <!--ro, req, bool, whether to enable the function-->true
  </overlayInfoEnabled>
  <OverlayInfoList>
    <!--ro, req, array, list of overlay information, subType:object-->
    <OverlayInfo>
      <!--ro, req, object, overlay information-->
      <overlayInfoID>
        <!--ro, req, int, overlay ID-->0
```

```xml
      </overlayInfoID>
      <itemType>
        <!--ro, req, enum, overlay information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
      </itemType>
      <itemOverlayEnabled>
        <!--ro, req, bool, whether to overlay the information-->true
      </itemOverlayEnabled>
      <customName>
        <!--ro, req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
      </customName>
      <changeLineNum>
        <!--ro, req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
      </changeLineNum>
      <spaceNum>
        <!--ro, req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
      </spaceNum>
      <startPosEnable>
        <!--ro, opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
      </startPosEnable>
      <startPosTop>
        <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
      </startPosTop>
      <startPosLeft>
        <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
      </startPosLeft>
      <overlayInfoText>
        <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
      </overlayInfoText>
      <overlayInfoText2>
        <!--ro, opt, string, overlay character string, range:[1,32], desc:for camera 2, which is valid when <itemType> is camera information; the maximum string length of camera 2 is 32 bytes-->test
      </overlayInfoText2>
    </OverlayInfo>
  </OverlayInfoList>
  <linePercent>
    <!--ro, req, int, percentage of overlaying lines, range:[0,100], desc:the default value is 100-->1
  </linePercent>
  <itemsStlye>
    <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal" (default), "vertical"-->horizontal
  </itemsStlye>
  <charStyle>
    <!--ro, req, enum, font type, subType:string, desc:“song_type”-SimSun (default), “wei_type”-Wei-->song_type
  </charStyle>
  <charSize>
    <!--ro, req, enum, font size, subType:string, desc:“0”-16*16(Chinese)/8*16(English), “1”-32*32(Chinese)/16*32(English), “2”-48*48, “3”-64*64(Chinese)/32*64(English)-->0
  </charSize>
  <charPosition>
    <!--ro, req, enum, position of the text overlaid in the picture, subType:int, desc:“0”-overlaid on the picture, "1”-overlaid outside the top edge of the picture, “2”-overlaid outside the bottom edge of the picture-->0
  </charPosition>
  <charInterval>
    <!--ro, req, int, character pitch, range:[0,16], desc:the default value is 0, unit: pixel-->0
  </charInterval>
  <foreColor>
    <!--ro, req, int, foreground color, desc:it is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value is 0xffffff (white)-->0
  </foreColor>
  <backColor>
    <!--ro, req, int, background color, desc:it is the RGB value directly obtained via the palette and its default value is 0x0 (black)-->0
  </backColor>
  <colorAdapt>
    <!--ro, opt, bool, whether to enable color self-adaption-->true
  </colorAdapt>
  <zeroizeEnable>
    <!--ro, opt, bool, whether to enable zero filling for OSD overlay, desc:it is used to enable zero filling for vehicle speed, speed limit, overspeed ratio, and lane No. Zero filling is enabled by default-->true
  </zeroizeEnable>
  <platePicOverlay>
    <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
  </platePicOverlay>
  <platePicPosTop>
```

```
    <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
  </platePicPosTop>
  <platePicPosLeft>
    <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
  </platePicPosLeft>
  <PictureInfoList>
    <!--ro, opt, array, overlay information list of other captured picture types, subType:object, desc:non-detection picture (background picture)-->
    <PictureInfo>
      <!--ro, opt, object, picture information-->
      <pictureType>
        <!--ro, req, enum, picture type, subType:string, desc:"illegalPicture" (violation capture, which corresponds to "compositePicture" returned by ANPR for violation occurrence); whether violation occurred can be indicated by whether "illegalInfo" is returned by ANPR-->illegalPicture
      </pictureType>
      <overlayInfoEnabled>
        <!--ro, req, bool, whether to enable the function-->true
      </overlayInfoEnabled>
      <OverlayInfoList>
        <!--ro, req, array, list of overlay information, subType:object-->
        <OverlayInfo>
          <!--ro, req, object, overlay information-->
          <overlayInfoID>
            <!--ro, req, int, overlay ID-->0
          </overlayInfoID>
          <itemType>
            <!--ro, req, enum, overlaid information type, subType:string, desc:0-unknown, 1-place, 2-intersection No., 3-device No., 4-direction No., 5-direction, 6-lane No., 7-lane, 8-capture time (without millisecond), 9-capture time (with millisecond), 10-license plate number, 11-vehicle color, 12-vehicle type, 13-vehicle brand, 14-vehicle speed, 15-speed limit sign, 16-vehicle length (between 1 and 99 meters), 17-violation code (traffic violation information is more useful than code, e.g., normal, low speed, overspeed, reverse driving, running the red light, occupying lane, driving over yellow lane line, etc.),18-camera information, 19-traffic violation, 20-overspeed ratio, 21-red light start time, 22-red light end time, 23-red light time, 24-security code, 25-capture No., 26-seatbelt, 27-reserved, 28-sun visor, 29-lane direction, 30-license plate color, 31-scene No., 32-scene name, 33-yellow label vehicle detection, 34-dangerous goods transport vehicle detection, 35-vehicle sub-brand detection, 36-vehicle direction, 37-window hangings, 38-making a call, 39-confidence, 40-verification unit, 41-verification certificate No., 42-calibration expiration date, 43-longitude and latitude, 44-tissue box detection, 45-baby in arm detection, 46-label detection, 47-decoration detection, 48-face score, 49-face No., 50-violation description, 51-marked speed limit, 52-segment speed, 53-segment distance, 54-segment overspeed ratio, 55-segment name, 56-segment ID, 57-traffic accident detection, 58-smoking, 59-wearing helmet, 60-manned, 61-person features, 62-using mobile phone, 63-umbrella tent, 66-vehicle head/tail, 67-vehicle door status, 68-vehicle loading rate, 69-number of operating personnel, 70-parking space patrol, 71-intelligent tracing (scheduled capture for Parking Space Patrol), 72-Ringelmann emittance (for black smoke detection), 73-vehicle purpose (bus, school bus, coach, taxi, ambulance, etc.), 74-vehicle bodywork features, 75-vehicle entry & exit status, 76-parking space No., 77-parking space status, 78-total parking spaces, 79-occupied parking spaces, 80-vacant parking spaces, 81-fake license plate, 82-whether wearing mask, 83-vehicle noise decible detection by device (in inverse proportion to the distance between device and vehicle), 85-reflective stripe-->0
          </itemType>
          <itemOverlayEnabled>
            <!--ro, req, bool, whether to overlay the information-->true
          </itemOverlayEnabled>
          <customName>
            <!--ro, req, string, custom overlaying name, range:[0,32], desc:max. 32 bytes; if this node is not returned or empty, the default name will be overlaid-->test
          </customName>
          <changeLineNum>
            <!--ro, req, int, number of line feeds, range:[0,100], desc:the default value is 0-->0
          </changeLineNum>
          <spaceNum>
            <!--ro, req, int, number of spaces, range:[0,255], desc:the default value is 0-->0
          </spaceNum>
          <startPosEnable>
            <!--ro, opt, bool, whether to enable coordinate configuration, desc:it is valid for overlay within the picture; line breaking and space function will be unavailable after it is enabled; it is recommended to be disabled-->true
          </startPosEnable>
          <startPosTop>
            <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
          </startPosTop>
          <startPosLeft>
            <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
          </startPosLeft>
          <overlayInfoText>
            <!--ro, opt, string, it is valid when itemType is set to place, device No., direction No., direction description, lane information, and camera, range:[1,128], desc:the maximum string length of place is 128 bytes, the maximum string length of intersection No., device No., direction No., direction description and lane information is 32 bytes., the maximum string length of camera 1 is 44 bytes, and the maximum string length of verification unit, verification certificate No., calibration expiration date is 128 bytes-->test
          </overlayInfoText>
          <overlayInfoText2>
            <!--ro, opt, string, overlay character string, range:[1,32], desc:for handling camera 2 when the overlay information type is camera information; the maximum length of camera 2 is 32 bytes-->test
          </overlayInfoText2>
        </OverlayInfo>
      </OverlayInfoList>
      <linePercent>
        <!--ro, req, int, overlay line percentage, range:[0,100], desc:the default value is 100-->1
      </linePercent>
      <itemsStlye>
        <!--ro, req, enum, overlay mode, subType:string, desc:"horizontal" (default), "vertical"-->horizontal
      </itemsStlye>
      <charStyle>
        <!--ro, req, enum, font type, subType:string, desc:“song_type”-SimSun (default), “wei_type”-Wei-->song_type
      </charStyle>
      <charSize>
        <!--ro, req, enum, font size, subType:string, desc:“0”-16*16(Chinese)/8*16(English), “1”-32*32(Chinese)/16*32(English), “2”-48*48, “3”-64*64(Chinese)/32*64(English)-->0
      </charSize>
```

```
        <charPosition>
          <!--ro, req, enum, position of the text overlaid in the picture, subType:int, desc:"0"-overlaid on the picture, "1"-overlaid outside the top edge of the picture, "2"-overlaid outside the bottom edge of the picture-->0
        </charPosition>
        <charInterval>
          <!--ro, req, int, character pitch, range:[0,16], desc:the default value is 0, unit: pixel-->0
        </charInterval>
        <foreColor>
          <!--ro, req, int, foreground color, desc:it is the RGB value directly obtained by the palette, the value is between 0 and 0xffffff and the default value is 0xffffff (white)-->0
        </foreColor>
        <backColor>
          <!--ro, req, int, background color, desc:it is the RGB value directly obtained via the palette and its default value is 0x0 (black)-->0
        </backColor>
        <colorAdapt>
          <!--ro, opt, bool, whether to enable color self-adaption-->true
        </colorAdapt>
        <zeroizeEnable>
          <!--ro, opt, bool, whether to enable zero filling for OSD overlay, desc:it is used to enable zero filling for vehicle speed, speed limit, overspeed ratio, and lane No. Zero filling is enabled by default-->true
        </zeroizeEnable>
        <platePicOverlay>
          <!--ro, opt, bool, whether to enable overlaying license plate thumbnail-->true
        </platePicOverlay>
        <platePicPosTop>
          <!--ro, req, int, top coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture height, and the default value is 0-->0
        </platePicPosTop>
        <platePicPosLeft>
          <!--ro, req, int, left coordinate of the start point, desc:it is valid for overlay within the picture, the value is between 0 and the actual picture width, and the default value is 0-->0
        </platePicPosLeft>
      </PictureInfo>
    </PictureInfoList>
  </MergePicOverlays>
```

11.5.1.65 Set vehicle detection parameters

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/vehicleDetect

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <nation>
    <!--req, enum, region, subType:string, desc:"ER"-CIS Region,"EU"-Europe Region,"ME"-Middle East,"AP"-Asia Pacific,"AfricaAndAmerica"-Africa and America,"All"-All Region-->All
  </nation>
  <bestDetectionSize>
    <!--opt, int, range:[0,1000]-->0
  </bestDetectionSize>
  <stateOrProvince>
    <!--opt, enum, subType:string-->云
  </stateOrProvince>
  <VehicleDetectSceneList size="10">
    <!--opt, array, subType:object, attr:size{req, int}-->
    <VehicleDetectScene>
      <!--opt, object, Vehicle Detection Scene-->
      <id>
        <!--req, int, ID, range:[1,10]-->0
      </id>
      <sceneName>
        <!--opt, string, scene name, range:[0,64]-->test
      </sceneName>
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <PlateRecogParam>
        <!--opt, object, LPR Parameters-->
        <PlateRecogRegionList>
          <!--opt, array, subType:object-->
          <PlateRecogRegion>
            <!--opt, object-->
            <id>
```

```xml
        <!--req, string, ID, range:[1,10]-->test
      </id>
      <RegionCoordinatesList>
        <!--opt, array, area coordinates list, subType:object-->
        <RegionCoordinates>
          <!--opt, object, area coordinates-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
    </PlateRecogRegion>
  </PlateRecogRegionList>
</PlateRecogParam>
<LaneConfig>
  <!--opt, object, lane line-->
  <LaneList size="10">
    <!--opt, object, lane list, attr:size{req, int}-->
    <Lane>
      <!--opt, object, lane-->
      <laneId>
        <!--req, int, lane No., range:[1,10]-->1
      </laneId>
      <RegionCoordinatesList>
        <!--opt, array, area coordinates list, subType:object-->
        <RegionCoordinates>
          <!--opt, object, area coordinates-->
          <positionX>
            <!--req, int, X-coordinate, range:[0,1000]-->0
          </positionX>
          <positionY>
            <!--req, int, Y-coordinate, range:[0,1000]-->0
          </positionY>
        </RegionCoordinates>
      </RegionCoordinatesList>
      <lineType>
        <!--req, enum, Lane Line Type, subType:string-->laneBoundaryLine
      </lineType>
      <carDriveDirect>
        <!--req, enum, subType:string-->down_to_up
      </carDriveDirect>
      <RightLaneLine>
        <!--opt, object, right lane line-->
        <RegionCoordinatesList size="2">
          <!--req, array, area coordinates list, subType:object, attr:size{req, int}-->
          <RegionCoordinates>
            <!--opt, object, area coordinates-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </RightLaneLine>
    </Lane>
  </LaneList>
</LaneConfig>
<AtRoadsideCalib>
  <!--opt, object-->
  <RegionCoordinatesList>
    <!--opt, array, area coordinates list, subType:object-->
    <RegionCoordinates>
      <!--opt, object, area coordinates-->
      <positionX>
        <!--req, int, X-coordinate, range:[0,1000]-->0
      </positionX>
      <positionY>
        <!--req, int, Y-coordinate, range:[0,1000]-->0
      </positionY>
    </RegionCoordinates>
  </RegionCoordinatesList>
</AtRoadsideCalib>
<ParkingDetection>
  <!--opt, object-->
  <enabled>
    <!--req, bool, whether to enable the function-->true
  </enabled>
  <duration>
    <!--req, int, range:[5,60], unit:s-->0
  </duration>
  <alarmIntervalTime>
    <!--req, int, range:[1,30], unit:min-->0
  </alarmIntervalTime>
</ParkingDetection>
<AboveRoadCalib>
  <!--opt, object-->
  <RegionCoordinatesList>
    <!--opt, array, area coordinates list, subType:object-->
    <RegionCoordinates>
```

```
            <!--opt, object, area coordinates-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </AboveRoadCalib>
      <RelatedLaneParam>
        <!--opt, object, linked lane parameters-->
        <RelatedLaneNoList>
          <!--req, array, linked lane number, subType:object-->
          <relatedLaneNo>
            <!--req, int, linked lane number, range:[1,6]-->0
          </relatedLaneNo>
        </RelatedLaneNoList>
      </RelatedLaneParam>
    </VehicleDetectScene>
  </VehicleDetectSceneList>
  <PlateDetectionRegion>
    <!--opt, object-->
    <PlateSize>
      <!--opt, object-->
      <minWidth>
        <!--opt, int, Min. Width, range:[0,1000]-->0
      </minWidth>
      <maxWidth>
        <!--opt, int, Max. Width, range:[0,1000]-->0
      </maxWidth>
    </PlateSize>
    <plateMode>
      <!--opt, enum, license plate mode, subType:string, desc:"small,large",License plate mode-->small
    </plateMode>
  </PlateDetectionRegion>
  <autoBuildRecogArea>
    <!--opt, enum, subType:string-->line
  </autoBuildRecogArea>
  <RodeType>
    <!--opt, object-->
    <type>
      <!--opt, enum, transmitter type, subType:string, desc:"entrance,city,custom,alarmInput"-->entrance
    </type>
    <Custom>
      <!--opt, object, custom-->
      <delayTime>
        <!--opt, int, Delay Time, range:[0,15000]-->0
      </delayTime>
      <delayTimeUnit>
        <!--opt, enum, "ms", subType:string, desc:"ms"-->ms
      </delayTimeUnit>
    </Custom>
    <gateType>
      <!--opt, enum, Entrance and Exit Type, subType:string-->entry
    </gateType>
  </RodeType>
  <enhancedMode>
    <!--opt, bool-->true
  </enhancedMode>
  <picRecognition>
    <!--opt, bool-->true
  </picRecognition>
  <countryIndex>
    <!--opt, int, country/region No., range:[1,255]-->0
  </countryIndex>
  <IntelligentMode>
    <!--opt, object-->
    <mountingType>
      <!--opt, enum, subType:int-->1
    </mountingType>
    <captureStrategy>
      <!--opt, enum, subType:int-->2
    </captureStrategy>
    <pictureSplicingFormat>
      <!--opt, enum, Picture Splicing Format, subType:string-->2
    </pictureSplicingFormat>
  </IntelligentMode>
  <CameralDirection>
    <!--opt, object-->
    <frontLensFacingDirection>
      <!--opt, enum, subType:int-->0
    </frontLensFacingDirection>
    <rearLensFacingDirection>
      <!--opt, enum, subType:int-->0
    </rearLensFacingDirection>
  </CameralDirection>
  <CRIndex>
    <!--req, enum, country/region, subType:int, desc:0 (Unsupported. It indicates that the algorithm library does not support recognizing license plates of this country), 1 (Czech Republic), 10 (Moldova), 100 (Reserved), 101 (Reserved), 102 (Reserved), 103 (Reserved), 104 (Egypt), 105 (Libya), 106 (Sudan), 107 (Tunisia), 108 (Algeria), 109 (Morocco), 11 (Russia), 110 (Ethiopia), 111 (Eritrea), 112 (Somalia Democratic), 113 (Djibouti), 114 (Kenya), 115 (Tanzania), 116 (Uganda), 117 (Rwanda), 118 (Burundi), 119 (Seychelles), 12 (Ukraine), 120 (Chad), 121 (Central African), 122 (Cameroon), 123 (Equatorial Guinea), 124 (Gabon), 125 (Republic of the Congo (Congo-Brazzaville)), 126 (Democratic Republic of the Congo (Congo-Kinshasa)), 127 (São Tomé and Príncipe), 128
```

```
(Mauritania), 129 (Western Sahara (Saharaui)), 13 (Belgium), 130 (Senegal), 131 (Gambia), 132 (Mali), 133 (Burkina Faso), 134 (Guinea), 135 (Guinea-Bissau), 136 (Cape Verde), 137 (Sierra Leone), 138 (Liberia), 139 (Ivory Coast), 14 (Bulgaria), 140 (Ghana), 141 (Togo), 142 (Benin), 143 (Niger), 144 (Zambia), 145 (Angola), 146 (Zimbabwe), 147 (Malawi), 148 (Mozambique), 149 (Botswana), 15 (Denmark), 150 (Namibia), 151 (South Africa), 152 (Swaziland), 153 (Lesotho), 154 (Madagascar), 155 (Comoros), 156 (Mauritius), 157 (Nigeria), 158 (South Sudan), 159 (Saint Helena), 16 (Finland), 160 (Mayotte), 161 (Reunion), 162 (Canary Islands), 163 (AZORES), 164 (Madeira), 165 (Reserved), 166 (Reserved), 167 (Reserved), 168 (Reserved), 169 (Canada), 17 (United Kingdom Great Britain), 170 (Greenland), 171 (St. Pierre and Miquelon), 172 (United State), 173 (Bermuda), 174 (Mexico), 175 (Guatemala), 176 (Belize), 177 (El Salvador), 178 (Honduras), 179 (Nicaragua), 18 (Greece), 180 (Costa Rica), 181 (Panama), 182 (Bahamas), 183 (Turks and  Caicos Islands), 184 (Cuba), 185 (Jamaica), 186 (Cayman Islands), 187 (Haiti), 188 (Dominican), 189 (Puerto Rico), 19 (Croatia), 190 (United States Virgin Islands), 191 (British Virgin Islands), 192 (Anguilla), 193 (Antigua and Barbuda), 194 (Collectivité de Saint-Martin), 195 (Autonomous country), 196 (Saint-Barthélemy), 197 (Saint Kitts and Nevis), 198 (Montserrat), 199 (Guadeloupe), 2 (France), 20 (Hungary), 200 (Dominica), 201 (Martinique), 202 (St. Lucia), 203 (Saint Vincent and the Grenadines), 204 (Grenada), 205 (Barbados), 206 (Trinidad and Tobago), 207 (Curaçao), 208 (Aruba), 209 (Netherlands Antilles), 21 (Israel), 210 (Colombia), 211 (Venezuela), 212 (Guyana), 213 (Suriname), 214 (French Guiana), 215 (Ecuador), 216 (Peru), 217 (Bolivia), 218 (Paraguay), 219 (Chile), 22 (Luxembourg), 220 (Brazil), 221 (Uruguay), 222 (Argentina), 223 (Reserved), 224 (Reserved), 225 (Reserved), 226 (Reserved), 227 (Australia), 228 (New Zealand ), 229 (Papua New Guinea), 23 (Macedonia, North Macedonia after 2018), 230 (Solomon Islands), 231 (Vanuatu), 232 (New Caledonia), 233 (Palau), 234 (Federated States of Micronesia), 235 (Marshall Island), 236 (Northern Mariana Islands), 237 (Guam), 238 (Nauru), 239 (Kiribati), 24 (Norway), 240 (Fiji Islands), 241 (Tonga), 242 (Tuvalu), 243 (Wallis et Futuna), 244 (Samoa), 245 (Eastern Samoa), 246 (Tokelau), 247 (Niue), 248 (Cook Islands), 249 (French Polynesia), 25 (Portugal), 250 (Pitcairn Islands), 251 (Hawaii State), 252 (Reserved), 253 (invalid), 254 (Unrecognized), 255 (All), 26 (Romania), 27 (Serbia), 28 (Azerbaijan), 29 (Georgia), 3 (Germany), 30 (Kazakhstan), 31 (Lithuania), 32 (Turkmenistan), 33 (Uzbekistan), 34 (Latvia), 35 (Estonia), 36 (Albania), 37 (Austria), 38 (Bosnia and Herzegovina), 39 (Ireland), 4 (Spain), 40 (Iceland), 41 (Vatican), 42 (Malta), 43 (Sweden), 44 (Switzerland), 45 (Cyprus), 46 (Turkey), 47 (Slovenia), 48 (Montenegro), 49 (Kosovo), 5 (Italy), 50 (Andorra), 51 (Armenia), 52 (Monaco), 53 (Liechtenstein), 54 (San Marino), 55 (Reserved), 56 (Reserved), 57 (Reserved), 58 (Reserved), 59 (China), 6 (Netherlands), 60 (Bahrain), 61 (South Korea), 62 (Lebanon), 63 (Nepal), 64 (Thailand), 65 (Pakistan), 66 (United Arab Emirates), 67 (Bhutan), 68 (Oman), 69 (North Korea), 7 (Poland), 70 (Philippines), 71 (Cambodia), 72 (Qatar), 73 (Kyrgyzstan), 74 (Maldives), 75 (Malaysia), 76 (Mongolia), 77 (Saudi Arabia), 78 (Brunei), 79 (Laos), 8 (Slovakia), 80 (Japan), 81 (Turkey), 82 (Palestinian), 83 (Tajikistan), 84 (Kuwait), 85 (Syria), 86 (India), 87 (Indonesia), 88 (Afghanistan), 89 (Sri Lanka), 9 (Belarus), 90 (Iraq), 91 (Vietnam), 92 (Iran), 93 (Yemen), 94 (Jordan), 95 (Burma/Myanmar), 96 (Sikkim), 97 (Bangladesh), 98 (Singapore), 99 (Democratic Republic of Timor-Leste)-->0
  </CRIndex>
  <detectionMode>
    <!--opt, enum, detection mode, subType:string-->vehiclePriority
  </detectionMode>
  <Filtration>
    <!--opt, object-->
    <enabled>
      <!--opt, bool-->true
    </enabled>
    <filtrationTime>
      <!--opt, int, filtering time, range:[1,5], unit:min-->1
    </filtrationTime>
  </Filtration>
  <relatedPlateColor>
    <!--opt, string-->other
  </relatedPlateColor>
  <relatedVehicleType>
    <!--opt, string-->vehicle
  </relatedVehicleType>
</VehicleDetectCfg>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.66 Get vehicle detection parameters

Request URL

GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <enabled>
    <!--ro, req, bool, whether to enable the function-->true
  </enabled>
</VehicleDetectCfg>
```

11.5.1.67 Set vehicle detection parameters at a specific scene

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/vehicleDetect/<SID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
SID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<VehicleDetectScene xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--req, object, vehicle detection scene, attr:version{req, string, protocolVersion}-->
  <id>
    <!--req, int, index, range:[1,10]-->0
  </id>
  <sceneName>
    <!--opt, string, scene name, range:[0,64]-->test
  </sceneName>
  <enabled>
    <!--req, bool, whether to enable-->true
  </enabled>
  <PlateRecogParam>
    <!--opt, object-->
    <PlateRecogRegionList>
      <!--opt, array, subType:object-->
      <PlateRecogRegion>
        <!--opt, object, ANPR region-->
        <id>
          <!--req, string, index, range:[1,10]-->test
        </id>
        <RegionCoordinatesList>
          <!--opt, array, area coordinates list, subType:object, range:[3,10]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
      </PlateRecogRegion>
    </PlateRecogRegionList>
  </PlateRecogParam>
  <LaneConfig>
    <!--opt, object, lane line-->
    <LaneList size="10">
      <!--opt, object, lane list, attr:size{req, int}-->
      <Lane>
        <!--opt, object, lane-->
        <laneId>
          <!--req, int, lane No., range:[1,10]-->1
        </laneId>
        <RegionCoordinatesList>
          <!--opt, array, area coordinates list, subType:object, range:[0,2]-->
          <RegionCoordinates>
            <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
            <positionX>
              <!--req, int, X-coordinate, range:[0,1000]-->0
            </positionX>
            <positionY>
              <!--req, int, Y-coordinate, range:[0,1000]-->0
            </positionY>
          </RegionCoordinates>
        </RegionCoordinatesList>
        <lineType>
          <!--req, enum, line type, subType:string, desc:laneBoundaryLine; aneLine;-->laneBoundaryLine
        </lineType>
```

```xml
        <carDriveDirect>
          <!--req, enum, vehicle driving direction, subType:string, desc:down2up: from down to up; up2down: from up to down;-->down_to_up
        </carDriveDirect>
        <RightLaneLine>
          <!--opt, object, right lane line, desc:right lane line coordinate. When this node is returned,it indicates that both left and right lane lines are required to be configured for each lane line-->
          <RegionCoordinatesList size="2">
            <!--req, array, area coordinates list, subType:object, range:[0,2], attr:size{req, int}-->
            <RegionCoordinates>
              <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
              <positionX>
                <!--req, int, X-coordinate, range:[0,1000]-->0
              </positionX>
              <positionY>
                <!--req, int, Y-coordinate, range:[0,1000]-->0
              </positionY>
            </RegionCoordinates>
          </RegionCoordinatesList>
        </RightLaneLine>
      </Lane>
    </LaneList>
  </LaneConfig>
  <AtRoadsideCalib>
    <!--opt, object-->
    <RegionCoordinatesList>
      <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:The origin is the lower-left corner of the screen.-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </AtRoadsideCalib>
  <ParkingDetection>
    <!--opt, object, vehicle parking detection-->
    <enabled>
      <!--req, bool, whether to enable-->true
    </enabled>
    <duration>
      <!--req, int, detection time interval, range:[5,60], unit:s-->0
    </duration>
    <alarmIntervalTime>
      <!--req, int, uploading interval, range:[1,30], unit:min-->0
    </alarmIntervalTime>
  </ParkingDetection>
  <AboveRoadCalib>
    <!--opt, object-->
    <RegionCoordinatesList>
      <!--opt, array, area coordinates list, subType:object, range:[0,4], desc:rectangle-->
      <RegionCoordinates>
        <!--opt, object, area coordinates, desc:the origin is the lower-left corner of the screen-->
        <positionX>
          <!--req, int, X-coordinate, range:[0,1000]-->0
        </positionX>
        <positionY>
          <!--req, int, Y-coordinate, range:[0,1000]-->0
        </positionY>
      </RegionCoordinates>
    </RegionCoordinatesList>
  </AboveRoadCalib>
  <RelatedLaneParam>
    <!--opt, object, linked lane parameters-->
    <RelatedLaneNoList>
      <!--req, array, linked lane No., subType:object-->
      <relatedLaneNo>
        <!--req, int, linked lane No., range:[1,6]-->0
      </relatedLaneNo>
    </RelatedLaneNoList>
  </RelatedLaneParam>
  <relatedPlateColor>
    <!--opt, string-->other
  </relatedPlateColor>
  <relatedVehicleType>
    <!--opt, string-->vehicle
  </relatedVehicleType>
</VehicleDetectScene>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.68 Get the capability of vehicle detection parameters

Request URL

GET /ISAPI/Traffic/channels/<channelID>/vehicleDetect/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<VehicleDetectCfg xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{req, string, protocolVersion}-->
  <enabled opt="true,false">
    <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
  </enabled>
</VehicleDetectCfg>
```


11.5.1.69 Set vehicle detection configuration parameters

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/vehicleDetect/config

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<Configuration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, vehicle detection parameters, attr:version{req, string, protocolVersion}-->
  <needAuth>
    <!--req, bool, whether to enable certification-->true
  </needAuth>
  <dir>
    <!--opt, string, path-->test
  </dir>
  <picTime>
    <!--opt, string-->test
  </picTime>
  <picDisplay>
    <!--req, bool-->true
  </picDisplay>
</Configuration>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.5.1.70 Get the recognition result of the license plate captured manually

Request URL

GET /ISAPI/Traffic/MNPR/channels/<channelID>?laneNo=<laneNo>&OSD=<OSDType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
laneNo	string	lane No.
OSDType	string	whether to enable OSD overlay for manual capture: enabled by default; not returned and not enabled when is 1

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, vehicle information, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, req, enum, protocol type, subType:string, desc:transport protocol type: HTTP,HTTPS,EHome-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->2004-05-03T17:30:08+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventType>
  <eventState>
    <!--ro, req, enum, continuous event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, enum, event description, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventDescription>
```

```
</eventDescription>
<channelName>
  <!--ro, opt, string, channel name, range:[1,64]-->test
</channelName>
<deviceID>
  <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
</deviceID>
<ANPR>
  <!--ro, opt, object, information about license plate recognition alarm-->
  <region>
    <!--ro, req, enum, region, subType:string, desc:region-->ER
  </region>
  <country>
    <!--ro, req, enum, country or region, subType:int, desc:country or region-->253
  </country>
  <area>
    <!--ro, req, enum, province or state, subType:string, desc:province or state-->FJR
  </area>
  <licensePlate>
    <!--ro, req, enum, license plate number, subType:string, desc:license plate number,e.g.,"123456"-->noPlate
  </licensePlate>
  <line>
    <!--ro, req, int, recognized lane No.-->1
  </line>
  <direction>
    <!--ro, req, enum, target direction, subType:string, desc:"reverse", "forward", "unknown"-->reverse
  </direction>
  <confidenceLevel>
    <!--ro, req, int, confidence level, range:[0,100]-->50
  </confidenceLevel>
  <plateType>
    <!--ro, req, enum, license plate type, subType:string, desc:license plate type: "unknown", "arm"-police vehicle, "embassy"-embassy vehicle,
"motorola"-motorcycle, "coach"-driver-training vehicle, "tempTravl"-vehicle with temporary license plate, "trailer"-trailer, "consulate"-consular vehicle,
"tempEntry"-temporary vehicle, "newEnergy"-new energy vehicle-->unknown
  </plateType>
  <plateColor>
    <!--ro, req, enum, license plate color, subType:string, desc:license plate color: "white","yellow","blue","black","green","newEnergyGreen"-new energy
green,"newEnergyYellowGreen"-new energy flavogreen,"other"-other color-->black
  </plateColor>
  <licenseBright>
    <!--ro, opt, int, license plate brightness, which ranges from 0 to 255, range:[0,255]-->50
  </licenseBright>
  <Rect>
    <!--ro, opt, object, coordinates of the license plate thumbnail in the matched picture, desc:coordinates of the license plate thumbnail in the matched
picture-->
    <height>
      <!--ro, req, float, height, range:[0.000,1.000]-->1.000
    </height>
    <width>
      <!--ro, req, float, width, range:[0.000,1.000]-->1.000
    </width>
    <x>
      <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
    </x>
    <y>
      <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
    </y>
  </Rect>
  <pilotsafebelt>
    <!--ro, opt, enum, whether the driver is wearing a safety belt, subType:string, desc:whether the driver is wearing safety belt: "unknown,yes,no"--
>unknown
  </pilotsafebelt>
  <vicepilotsafebelt>
    <!--ro, opt, string, whether the co-driver is wearing a safety belt-->unknown
  </vicepilotsafebelt>
  <pilotsunvisor>
    <!--ro, opt, string, whether the driver room's sun visor is open-->unknown
  </pilotsunvisor>
  <vicepilotsunvisor>
    <!--ro, opt, string, whether the co-driver room's sun visor is open-->unknown
  </vicepilotsunvisor>
  <envprosign>
    <!--ro, req, enum, whether it is a yellow-label vehicle, subType:string, desc:"unknown" (unknown), "green" (green label), "yellow" (yellow label)--
>unknown
  </envprosign>
  <dangmark>
    <!--ro, req, enum, whether it is a dangerous goods vehicle, subType:string, desc:"unknown", "yes", "no"-->unknown
  </dangmark>
  <uphone>
    <!--ro, req, enum, whether the driver is making a phone call, subType:string, desc:"unknown", "yes", "no"-->unknown
  </uphone>
  <pendant>
    <!--ro, req, enum, whether there are window hangings detected, subType:string, desc:"unknown", "yes", "no"-->unknown
  </pendant>
  <tissueBox>
    <!--ro, req, enum, whether there is a tissue box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
  </tissueBox>
  <frontChild>
    <!--ro, req, enum, whether the co-driver is with a baby in arm, subType:string, desc:"unknown", "yes", "no"-->unknown
  </frontChild>
  <label>
    <!--ro, req, enum, whether there are stickers detected, subType:string, desc:"unknown", "yes", "no"-->unknown
  </label>
  <decoration>
```

```
      <!--ro, req, enum, whether there are decorations detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </decoration>
    <smoking>
      <!--ro, req, enum, whether anyone is smoking, subType:string, desc:whether there is smoking detected: "unknown,yes,no"-->yes
    </smoking>
    <perfumeBox>
      <!--ro, opt, enum, whether there is perfume box detected, subType:string, desc:whether there is perfume box detected: "unknown,yes,no"-->unknown
    </perfumeBox>
    <pdvs>
      <!--ro, req, enum, whether there is a person sticking out of sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pdvs>
    <helmet>
      <!--ro, req, enum, whether the non-motor vehicle driver wears a helmet, subType:string, desc:"unknown", "yes", "no"-->no
    </helmet>
    <twoWheelVehicle>
      <!--ro, opt, enum, unknown, subType:string, desc:whether there is two-wheel detected: "unknown,yes,no"-->unknown
    </twoWheelVehicle>
    <threeWheelVehicle>
      <!--ro, opt, enum, whether there is three-wheel detected, subType:string, desc:whether there is three-wheel detected: "unknown,yes,no"-->unknown
    </threeWheelVehicle>
    <blackness>
      <!--ro, opt, int, Ringelmann emittance, desc:it is used for black smoke detection-->2
    </blackness>
    <plateCharBelieve>
      <!--ro, opt, string, confidence of each character in the recognized license plate, desc:confidence of each character in the recognized license plate-->test
    </plateCharBelieve>
    <speedLimit>
      <!--ro, opt, int, upper speed limit, desc:it is valid only when overspeeding occurred-->50
    </speedLimit>
    <illegalInfo>
      <!--ro, opt, object, violation information-->
      <illegalCode>
        <!--ro, req, string, violation code, range:[0,64]-->1101
      </illegalCode>
      <illegalName>
        <!--ro, req, string, violation name, range:[0,128]-->test
      </illegalName>
      <illegalDescription>
        <!--ro, opt, string, violation description, range:[0,256]-->test
      </illegalDescription>
    </illegalInfo>
    <vehicleType>
      <!--ro, req, enum, vehicle type, subType:string, desc:vehicle type: "unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickupTruck"-->nonmotorVehicle
    </vehicleType>
    <postPicFileName>
      <!--ro, opt, string, name of the picture selected as the checkpoint picture when illegal action occurs, desc:name of the picture selected as the checkpoint picture when illegal action occurs, "none" refers to not selecting any picture-->test
    </postPicFileName>
    <featurePicFileName>
      <!--ro, opt, string, red light running capture, desc:name of the picture selected as the close-up picture when running the red light in the intersection violation system is detected, "none" refers to not selecting any picture-->test
    </featurePicFileName>
    <detectDir>
      <!--ro, req, enum, detection direction, subType:int, desc:1 (upward), 2 (downward), 3 (bidirection), 4 (from east to west), 5 (from south to north), 6 (from west to east), 7 (from north to south), 8 (other)-->1
    </detectDir>
    <detectType>
      <!--ro, req, enum, detection type, subType:int, desc:1 (inductive loop trigger), 2 (video trigger), 3 (multiple-frame recognition), 4 (radar trigger)-->1
    </detectType>
    <barrierGateCtrlType>
      <!--ro, opt, enum, whether to enable elapsed time, subType:int, desc:0 (enabled), 1 (disabled)-->0
    </barrierGateCtrlType>
    <alarmDataType>
      <!--ro, opt, enum, alarm data type (real-time data or history data), subType:int, desc:1 (history data), 0 (real-time data)-->0
    </alarmDataType>
    <dwIllegalTime>
      <!--ro, opt, int, illegal action duration, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the first picture-->100
    </dwIllegalTime>
    <vehicleInfo>
      <!--ro, opt, object, vehicle information-->
      <index>
        <!--ro, req, int, vehicle No.-->1
      </index>
      <vehicleType>
        <!--ro, req, enum, vehicle type, subType:int, desc:vehicle type: "unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickupTruck"-->0
      </vehicleType>
      <colorDepth>
        <!--ro, req, enum, shade of the vehicle color, subType:int, desc:0 (deep color), 1 (light color), 2 (unknown)-->0
      </colorDepth>
      <color>
        <!--ro, req, enum, vehicle color, subType:string, desc:"green", "brown", "pink", "purple", "deepGray" (dark gray), "cyan", "orange", "white", "silver" (silvery), "gray", "blacks", "red", "deepBlue" (dark blue), "blue", "yellow"-->green
      </color>
      <speed>
        <!--ro, req, int, target speed, unit:km/h-->1
      </speed>
```

```
      <length>
        <!--ro, req, int, length of the former vehicle, unit:dm-->10
      </length>
      <vehicleLogoRecog>
        <!--ro, req, int, vehicle main brand-->1
      </vehicleLogoRecog>
      <vehileSubLogoRecog>
        <!--ro, opt, int, vehicle sub-brand-->1
      </vehileSubLogoRecog>
      <vehileModel>
        <!--ro, opt, int, model year of vehicle sub-brand-->1
      </vehileModel>
      <vehicleTypeByWeight>
        <!--ro, opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1-class one vehicle (buses with seven or less seats,trucks with
capacity of 2 tons or less), 2-class two vehicle (buses with 8 to 19 seats,trucks with capacity of 2 to 5 (included) tons), 3-class three vehicle (buses
with 20 to 39 seats,trucks with capacity of 5 to 10 (included) tons), 4-class four vehicle (buses with 40 or more seats,trucks with capacity of 10 to 15
(included) tons), 5-class five vehicle (trucks with capacity of more than 15 tons)-->4
      </vehicleTypeByWeight>
    </vehicleInfo>
    <EntranceInfo>
      <!--ro, opt, object, entrance and exit information-->
      <parkingID>
        <!--ro, opt, string, parking space No.-->test
      </parkingID>
      <gateID>
        <!--ro, opt, string, entrance and exit No.-->test
      </gateID>
      <direction>
        <!--ro, req, enum, target direction, subType:string, desc:"enter", "leave"-->enter
      </direction>
      <cardNo>
        <!--ro, opt, string, card No.-->test
      </cardNo>
      <parkType>
        <!--ro, req, enum, parking type, subType:string, desc:"temporary", "permanent"-->temporary
      </parkType>
    </EntranceInfo>
    <pictureInfoList>
      <!--ro, req, array, picture list, subType:object-->
      <pictureInfo>
        <!--ro, req, object, picture information-->
        <fileName>
          <!--ro, req, enum, picture name, subType:string, desc:it must correspond to the picture name transmitted with the alarm message.
"detectionPicture.jpg" (background picture), "licensePlatePicture.jpg" (license plate picture), "pilotPicture.jpg" (driver's picture matting),
"copilotPicture.jpg" (co-driver's picture matting), "compositePicture.jpg" (composite picture), "plateBinaryPicture.jpg" (license plate binary picture),
"nonMotorPicture.jpg" (non-motor vehicle picture matting), "pedestrianDetectionPicture.jpg" (pedestrian picture), "pedestrianPicture.jpg" (pedestrian's
picture matting), "vehiclePicture.jpg" (vehicle picture).-->detectionPicture.jpg
        </fileName>
        <type>
          <!--ro, req, enum, parking type, subType:string, desc:"detectionPicture" (background picture), "licensePlatePicture" (license plate picture),
"pilotPicture" (driver's picture matting), "copilotPicture" (co-driver's picture matting), "compositePicture" (composite picture), "plateBinaryPicture"
(license plate binary picture), "nonMotorPicture" (non-motor vehicle picture matting), "pedestrianDetectionPicture" (pedestrian picture),
"pedestrianPicture" (pedestrian's picture matting), "vehiclePicture" (vehicle picture)-->vehiclePicture
        </type>
        <dataType>
          <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
        </dataType>
        <picRecogMode>
          <!--ro, opt, enum, license plate recognition mode, subType:int, desc:0 (front license plate recognition), 1 (rear license plate recognition)-->0
        </picRecogMode>
        <redLightTime>
          <!--ro, opt, int, red light time elapsed, unit:s-->60
        </redLightTime>
        <vehicleHead>
          <!--ro, req, enum, license plate recognition direction, subType:string, desc:"unknown", "forward” (front license plate recognition), "back” (rear
license plate recognition)-->unknown
        </vehicleHead>
        <absTime>
          <!--ro, opt, string, absolute time, desc:format: yyyyMMddHHmmssxxx, e.g.: 20090810235959999. the last three number is time in millisecond--
>20090810235959999
        </absTime>
        <plateRect>
          <!--ro, opt, object, license plate area coordinates, desc:this node is valid only when type is "detectionPicture"-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </plateRect>
        <vehicelRect>
          <!--ro, opt, object, vehicle area coordinates, desc:the normalized value is the current image size in percentage multiplying 1000. This node is
valid only when type is "detectionPicture"-->
          <X>
            <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
          </X>
          <Y>
            <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
```

```
          </Y>
          <width>
            <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
          </height>
        </vehicelRect>
        <pictureURL>
          <!--ro, opt, string, picture URL, desc:it is valid only when dataType is "URL"-->test
        </pictureURL>
        <pId>
          <!--ro, opt, string, strlen.max=32, range:[1,32], desc:the recommended generation rule is the device's serial number + the time elapsed since the 
device was started + a random number-->test
        </pId>
      </pictureInfo>
    </pictureInfoList>
    <hasMoreData>
      <!--ro, opt, bool, whether there is more data, desc:this node is used to report the license plate information first, and then report XML message with 
picture data; the XM message with picture data and license plate information are linked by UUID-->true
    </hasMoreData>
    <listType>
      <!--ro, opt, enum, name list attribute, subType:string, desc:"white" (allowlist), "black" (blocklist), "temporary" (temporary list)-->white
    </listType>
    <originalLicensePlate>
      <!--ro, opt, string, original license plate number, desc:when the license plate number is a minor language, return the original license plate number--
>test
    </originalLicensePlate>
    <CRIndex>
      <!--ro, req, enum, country/region, subType:int, desc:country or region index-->258
    </CRIndex>
    <VehicelGPSInfo>
      <!--ro, opt, object, GPS information of the vehicle-->
      <longitudeType>
        <!--ro, req, enum, W, subType:string, desc:longitude,"E,W"-->E
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, N, subType:string, desc:latitude,"S,N"-->S
      </latitudeType>
      <Longitude>
        <!--ro, req, object, longitude-->
        <degree>
          <!--ro, req, int, degree-->60
        </degree>
        <minute>
          <!--ro, req, int, minute, range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second, range:[0,59.999999]-->59.000000
        </sec>
      </Longitude>
      <Latitude>
        <!--ro, req, object, latitude-->
        <degree>
          <!--ro, req, int, degree-->60
        </degree>
        <minute>
          <!--ro, req, int, minute, range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second, range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
        </sec>
      </Latitude>
    </VehicelGPSInfo>
    <vehiclePositionControl>
      <!--ro, opt, enum, vehicle arming, subType:string, desc:arming type: "vehicleMonitor"-intelligent arming of vehicle (PUT 
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/<taskID>/startTask),"manualVehicleMonitor"-manual arming of vehicle (PUT 
/ISAPI/Traffic/channels/<ID>/manualVehicleMonitor),"dailyVehicleMonitor"-daily arming of vehicle (you can check whether this arming type is supported via 
the node isSupportDailyVehicleMonitor in the capability message returned by /ISAPI/Traffic/channels/<ID>/vehicleDetect/capabilities; when daily arming of 
vehicle is enabled,both alarm of ANPR and intelligent arming of vehicle will be uploaded; if this node is not returned,it is normal vehicle detection--
>dailyVehicleMonitor
    </vehiclePositionControl>
    <vehicleMonitorTaskID>
      <!--ro, opt, string, task ID of intelligent arming of vehicle, range:[1,64], desc:this node is returned when the value of vehiclePositionControl is 
"vehicleMonitor"-->test
    </vehicleMonitorTaskID>
    <vehicleListName>
      <!--ro, opt, string, vehicle list name, range:[0,128], desc:name of the list that the vehicle belongs to,the maximum size is 128 bytes-->test
    </vehicleListName>
    <vehicleThermometryEnabled>
      <!--ro, opt, bool, whether to enable vehicle temperature measurement-->true
    </vehicleThermometryEnabled>
    <currTemperature>
      <!--ro, opt, float, temperature-->36.5
    </currTemperature>
    <thermometryUnit>
      <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
    </thermometryUnit>
  </ANPR>
  <UUID>
    <!--ro, opt, string, UUID, desc:if the vehicle is captured many times, the UUID is the same-->test
  </UUID>
  <picNum>
```

```xml
      <!--ro, opt, int, number of pictures-->2
    </picNum>
    <monitoringSiteID>
      <!--ro, opt, string, camera number-->test
    </monitoringSiteID>
    <ePlateUUID>
      <!--ro, opt, string, electronic license plate ID, desc:If this node is configured with a value, it indicates that an electronic license plate is linked-
  ->test
    </ePlateUUID>
    <isDataRetransmission>
      <!--ro, opt, bool, data retransmission mark-->true
    </isDataRetransmission>
    <SceneInfo>
      <!--ro, opt, object, scene information-->
      <scenesID>
        <!--ro, opt, string, scene ID which is between 1 and 16-->test
      </scenesID>
      <sceneName>
        <!--ro, opt, string, scene name, range:[0,32]-->test
      </sceneName>
      <PTZPos>
        <!--ro, opt, object, PTZ position-->
        <elevation>
          <!--ro, opt, int, tilting angle, range:[-900,2700]-->0
        </elevation>
        <azimuth>
          <!--ro, opt, int, panning angle, range:[0,3600]-->0
        </azimuth>
        <absoluteZoom>
          <!--ro, opt, int, zooming ratio, range:[0,1000]-->1
        </absoluteZoom>
      </PTZPos>
    </SceneInfo>
    <monitorDescription>
      <!--ro, opt, string, camera information, range:[0,256]-->test
    </monitorDescription>
    <DeviceGPSInfo>
      <!--ro, opt, object, device GPS information-->
      <longitudeType>
        <!--ro, req, enum, W, subType:string, desc:longitude,"E,W"-->E
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, N, subType:string, desc:latitude,"S,N"-->S
      </latitudeType>
      <Longitude>
        <!--ro, req, object, longitude-->
        <degree>
          <!--ro, req, int, degree-->60
        </degree>
        <minute>
          <!--ro, req, int, minute, range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second, range:[0,59.999999]-->59.000000
        </sec>
      </Longitude>
      <Latitude>
        <!--ro, req, object, latitude-->
        <degree>
          <!--ro, req, int, degree-->60
        </degree>
        <minute>
          <!--ro, req, int, minute, range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second, range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
        </sec>
      </Latitude>
    </DeviceGPSInfo>
  </EventNotificationAlert>
```


Parameter Name	**Parameter Value**	**Parameter Type(Content-Type)**	**Content-ID**	**File Name**	**Description**
EventNotificationAlert	[Message content]	application/xml	--	--	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

11.5.1.71 License plate recognition

EventType:ANPR

```
<?xml version="1.0" encoding="UTF-8"?>

<EventNotificationAlert xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, license plate recognition result, attr:version{opt, string, protocolVersion}-->
  <ipAddress>
    <!--ro, req, string, IPv4 address of the device that triggers the alarm-->172.6.64.7
  </ipAddress>
  <ipv6Address>
    <!--ro, opt, string, IPv6 address of the device that triggers the alarm-->1080:0:0:0:8:800:200C:417A
  </ipv6Address>
  <portNo>
    <!--ro, opt, int, communication port No. of the device that triggers the alarm-->80
  </portNo>
  <protocol>
    <!--ro, req, enum, protocol type, subType:string, desc:transmission communication protocol type: "HTTP", "HTTPS", "EHome". The value should be "HTTP"
when ISAPI protocol is transmitted via EZ protocol. The value should be "EHome" when ISAPI protocol is transmitted via ISUP-->HTTP
  </protocol>
  <macAddress>
    <!--ro, opt, string, MAC address-->01:17:24:45:D9:F4
  </macAddress>
  <dynChannelID>
    <!--ro, opt, string, digital channel No.-->test
  </dynChannelID>
  <channelID>
    <!--ro, opt, int, channel No. of the device that triggers the alarm, desc:video channel No. that triggers the alarm-->1
  </channelID>
  <releatedChannelList>
    <!--ro, opt, string, list of alarm related channels, which are of the same camera with channelID, desc:this parameter is used for live view or playback
on the platform; multiple channel No.s are separated by commas-->1,2,3
  </releatedChannelList>
  <dateTime>
    <!--ro, req, datetime, alarm trigger time-->1970-01-01T00:00:00+08:00
  </dateTime>
  <activePostCount>
    <!--ro, opt, int, times that the same alarm has been uploaded, desc:event triggering frequency-->1
  </activePostCount>
  <eventType>
    <!--ro, req, enum, event type, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventType>
  <eventState>
    <!--ro, req, enum, continuous event status, subType:string, desc:for durative event: active (valid), inactive (invalid)-->active
  </eventState>
  <eventDescription>
    <!--ro, req, enum, event description, subType:string, desc:"ANPR" (license plate recognition)-->ANPR
  </eventDescription>
  <channelName>
    <!--ro, opt, string, channel name, range:[1,128]-->test
  </channelName>
  <deviceID>
    <!--ro, opt, string, device ID, desc:it should be returned for ISUP alarms, e.g., test0123 (Ehome2.0, Ehome4.0, and ISUP5.0)-->12345
  </deviceID>
  <ANPR>
    <!--ro, opt, object, information about license plate recognition alarm-->
    <region>
      <!--ro, opt, enum, region, subType:string, desc:"ER" (Russian-speaking region), "EU" (Europe), "EUandCIS" (Europe and Russia), "ME" (Middle East),
"other" (other regions), "APAC" (Asia-Pacific region), "AFandAM" (Africa and America), "THAandLA" (Thailand and Laos), "HKandMO" (Hong Kong China and Macao
China), "All" (all regions),-->ER
    </region>
    <country>
      <!--ro, opt, enum, country or region, subType:int, desc:253-invalid, 70-Philippines, 17-United Kingdom (previously Great Britain), 23-Macedonia (North
Macedonia since 2018), 188-Dominican, 227-Australia, 60-Bahrain, 95-Burma/Myanmar, 228-New Zealand, 28-Azerbaijan, 131-Gambia, 132-Mali, 159-Saint Helena,
133-Burkina Faso, 134-Guinea, 135-Guinea-Bissau, 136-Cape Verde, 137-Sierra Leone, 138-Liberia, 139-Ivory Coast, 140-Ghana, 141-Togo, 142-Benin, 143-Niger,
144-Zambia, 145-Angola, 146-Zimbabwe, 147-Malawi, 148-Mozambique, 149-Botswana, 150-Namibia, 151-South Africa, 152-Swaziland, 153-Lesotho, 154-Madagasca,
155-Comoros, 156-Mauritius, 157-Nigeria, 158-South Sudan, 160-Mayotte, 161-Reunion, 162-Canary Islands, 163-AZORES, 164-Madeira, 165-reserved, 166-reserved,
167-reserved, 168-reserved, 169-Canada, 170-Greenland Nuuk, 171-Pierre and Miquelon, 172-United States, 173-Bermuda, 174-Mexico, 175-Guatemala, 176-Belize,
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177-El Salvador, 178-Honduras, 179-Nicaragua, 180-Costa Rica, 181-Panama, 182-Bahamas, 183-Turks and  Caicos Islands, 184-Cuba, 185-Jamaica, 186-Cayman Islands, 187-Haiti, 189-Puerto Rico, 190-United States Virgin Islands, 191-British Virgin Islands, 192-Anguilla, 193-Antigua and Barbuda, 194-Collectivit de Saint-Martin, 195-Autonomous country, 196-Saint-Barthlemy, 197-Saint Kitts and Nevis, 198-Montserrat, 199-Guadeloupe, 200-Dominica, 201-Martinique, 202-St. Lucia, 203-Saint Vincent and the Grenadines, 204-Grenada, 205-Barbados, 206-Trinidad and Tobago, 207-Cura?ao, 0-Unsupported, 1-Czech Republic, 2-France, 3-Germany, 4-Spain, 5-Italy, 6-Netherlands, 7-Poland, 8-Slovakia, 9-Belarus, 10-Moldova, 11-Russia, 12-Ukraine, 13-Belgium, 14-Bulgaria, 15-Denmark, 16-Finland, 18-Greece, 19-Croatia, 20-Hungary, 21-Israel(), 22-Luxembourg, 24-Norway, 25-Portuga, 26-Romania, 27-Serbia, 29-Georgia(), 30-Kazakhstan(), 31-Lithuania, 32-Turkmenistan(), 33-Uzbekistan(), 34-Latvia, 35-Estonia, 36-Albania, 37-Austria, 38-Bosnia and Herzegovina, 39-Ireland, 40-Iceland, 41-Vatican, 42-Malta, 43-Sweden, 44-Switzerland, 45-Cyprus, 46-Turkey, 47-Slovenia, 48-Montenegro, 49-Kosovo, 50-Andorra, 51-Armenia(), 52-Monaco, 53-Liechtenstein, 54-San Marino, 55-reserved, 56-reserved, 90-Iraq, 57-reserved, 58-reserved, 59-China, 91-Vietnam, 61-South Korea, 62-Lebanon, 63-Nepal, 64-Thailand, 65-Pakistan, 66-United Arab Emirates, 67-Bhutan, 68-Oman, 69-North Korea, 71-Cambodia, 72-Qatar, 73-Kyrgyzstan, 74-Maldives, 75-Malaysia, 76-Mongolia, 77-Saudi Arabia, 78-Brunei, 79-Laos, 80-Japan, 81-Turkey, 82-Palestinian, 83-Tajikistan, 84-Kuwait, 85-Syria, 86-India, 87-Indonesia, 88-Afghanistan, 89-Sri Lanka, 92-Iran, 93-Yemen, 94-Jordan, 96-Sikkim, 97-Bangladesh, 98-Singapore, 99-Democratic Republic of Timor-Leste, 100-reserved, 101-reserved, 102-reserved, 103-reserved, 104-Egypt, 105-Libya, 106-Sudan, 107-Tunisia, 108-Algeria, 109-Morocco, 110-Ethiopia, 111-Eritrea, 112-Somalia Democratic, 113-Djibouti, 114-Kenya, 115-Tanzania, 116-Uganda, 117-Rwanda, 118-Burundi, 119-Seychelles, 120-Chad, 121-Central African, 122-Cameroon, 123-Equatorial Guinea, 124-Gabon, 125-Congo, 126-Democratic Republic of the Congo, 127-Sao Tome and Principe, 128-Mauritania, 129-Western Sahara, 130-Senegal, 208-Aruba, 209-Netherlands Antilles, 210-Colombia, 211-Venezuela, 212-Guyana, 213-Suriname, 214-Guyane Francaise, 215-Ecuador, 216-Peru, 217-Bolivia, 218-Paraguay, 219-Chile, 220-Brazil, 221-Uruguay, 222-Argentina, 223-reserved, 224-reserved, 225-reserved, 226-reserved, 229-Papua New Guinea, 230-Salomonen, 231-Vanuatu, 232-New Caledonia, 233-Palau, 234-Federated States of Micronesia, 235-Marshall Island, 236-Northern Mariana Islands, 237-Guam, 238-Nauru, 239-Kiribati, 240-Fidschi, 241-Tonga, 242-Tuvalu, 243-Wallis et Futuna, 244-Samoa, 245-Eastern Samoa, 246-Tokelau, 247-Niue, 248-Cook Islands, 249-French Polynesia, 250-Pitcairn Islands, 251-Hawaii State, 252-reserved, 254-unrecognized, 255-all-->253
    </country>
    <area>
      <!--ro, opt, enum, United Arab Emirates, subType:string, desc:"FJR" (Fujairah), "AD" (Abu Dhabi), "unknown", "UMW" (Umm al-Qaiwain), "other", "AM" (Ajman), "RAK" (Ras al-Khaimah), "DB" (Dubai), "SJ" (Sharjah)-->FJR
    </area>
    <licensePlate>
      <!--ro, req, string, license plate number, range:[1,32], desc:noPlate (vehicle without license plate), unknown (no license plate recognized), XXXXXXXX (specific license plate number; color information required by Chinese license plate; for motor vehicles, it is a 16-byte string; for non-motor vehicles, it is a 48-byte string)-->A283KY77
    </licensePlate>
    <cameraNo>
      <!--ro, opt, string, device No., range:[1,48], desc:corresponds to <cameraNum> from /ISAPI/System/Video/inputs/channels/<ID>/cameraInfo (recommended for new devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
    </cameraNo>
    <line>
      <!--ro, req, int, recognized lane No.-->1
    </line>
    <direction>
      <!--ro, opt, enum, license plate recognition direction, subType:string, desc:"reverse", "forward", "unknown"-->reverse
    </direction>
    <confidenceLevel>
      <!--ro, req, int, confidence level, range:[0,100]-->50
    </confidenceLevel>
    <plateType>
      <!--ro, opt, enum, license plate type, subType:string, desc:"unknown", "arm" (police vehicle), "92TypeArm" (type 92 armed police vehicle), "02TypePersonalized" (type 02 custom vehicle),  "yellowTwoLine" (yellow two-line rear license plate), "embassy" (embassy vehicle), "oneLineArm"(new armed police vehicle, one-line), "twoLineArm" (new armed police vehicle, two-line), "yellow1225FarmVehicle" (agricultural vehicle, yellow 1225), "green1325FarmVehicle" (agricultural vehicle, green 1325), "yellow1325FarmVehicle" (agricultural vehicle, yellow 1325), "motorola" (motorcycle), "coach" (driver-training vehicle), "tempTravl" (vehicle with temporary license plate), "trailer" (trailer), "consulate" (consular vehicle), "hongKongMacao" (Hong Kong China/Macao China entrance vehicle), "tempEntry" (temporary vehicle), "civilAviation" (civil aviation license plate), "newEnergy" (new energy vehicle), "92FarmVehicle" (two-line license plate civil vehicle), "emergency" (emergency license plate), "oneLineArmHeadquarters" (armed police headquarter license plate, one-line), "twoLineArmHeadquarters" (armed police headquarter license plate, two-line), "twoWheelVehicle" (two wheeler)-->unknown
    </plateType>
    <plateColor>
      <!--ro, opt, enum, license plate color, subType:string, desc:"black", "blue","golden", "orange", "red", "yellow", "white", "unknown", "other", "newEnergyYellowGreen"-new energy yellow-green,"civilAviationBlack", "civilAviationGreen" (civil aviation green), "green", "mixedColor", "newEnergyGreen" (new energy green), "brown”-->black
    </plateColor>
    <licenseBright>
      <!--ro, opt, int, license plate brightness, which ranges from 0 to 255, range:[0,255]-->50
    </licenseBright>
    <Rect>
      <!--ro, opt, object, coordinates of the license plate thumbnail in the matched picture, desc:used by back-end DeepinMind; the origin is the upper-left corner of the screen-->
      <height>
        <!--ro, req, float, height, range:[0.000,1.000]-->1.000
      </height>
      <width>
        <!--ro, req, float, width, range:[0.000,1.000]-->1.000
      </width>
      <x>
        <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
      </x>
      <y>
        <!--ro, req, float, the reference origin is the upper left corner of image, range:[0.000,1.000]-->1.000
      </y>
    </Rect>
    <pilotsafebelt>
      <!--ro, opt, enum, whether the driver is wearing a safety belt, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pilotsafebelt>
    <vicepilotsafebelt>
      <!--ro, opt, enum, whether the co-driver is wearing a safety belt, subType:string, desc:"unknown", "yes", "no"-->unknown
    </vicepilotsafebelt>
    <pilotsunvisor>
      <!--ro, opt, enum, whether the driver room's sun visor is open, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pilotsunvisor>
    <vicepilotsunvisor>
      <!--ro, opt, enum, whether the co-driver room's sun visor is open, subType:string, desc:"unknown", "yes", "no"-->unknown
    </vicepilotsunvisor>
    <envprosign>
      <!--ro, opt, enum, whether it is a yellow-label vehicle, subType:string, desc:"unknown", "green" (green-label), "yellow" (yellow-lable), "yes", "no"-->unknown
    </envprosign>
    <dangmark>
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    <dangmark>
      <!--ro, opt, enum, whether it is a dangerous goods vehicle, subType:string, desc:"unknown", "yes", "no"-->unknown
    </dangmark>
    <uphone>
      <!--ro, opt, enum, whether the driver is making a phone call, subType:string, desc:"unknown", "yes", "no"-->unknown
    </uphone>
    <pendant>
      <!--ro, opt, enum, whether there are window hangings detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pendant>
    <tissueBox>
      <!--ro, opt, enum, whether there is a tissue box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </tissueBox>
    <frontChild>
      <!--ro, opt, enum, whether the co-driver is with a baby in arm, subType:string, desc:"unknown", "yes", "no"-->unknown
    </frontChild>
    <label>
      <!--ro, opt, enum, whether there are stickers detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </label>
    <decoration>
      <!--ro, opt, enum, whether there are decorations detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </decoration>
    <smoking>
      <!--ro, opt, enum, whether anyone is smoking, subType:string, desc:"unknown", "yes", "no"-->yes
    </smoking>
    <perfumeBox>
      <!--ro, opt, enum, whether there is perfume box detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </perfumeBox>
    <pdvs>
      <!--ro, opt, enum, whether there is a person sticking out of sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
    </pdvs>
    <helmet>
      <!--ro, opt, enum, whether there is helmet detected, subType:string, desc:"unknown", "yes", "no"-->no
    </helmet>
    <twoWheelVehicle>
      <!--ro, opt, enum, whether there is a two-wheel vehicle detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </twoWheelVehicle>
    <threeWheelVehicle>
      <!--ro, opt, enum, whether there is a three-wheel vehicle detected, subType:string, desc:"unknown", "yes", "no"-->unknown
    </threeWheelVehicle>
    <blackness>
      <!--ro, opt, int, Ringelmann emittance, desc:it is used for black smoke detection-->2
    </blackness>
    <plateCharBelieve>
      <!--ro, opt, string, confidence of each character in the recognized license plate, desc:if "A12345" is detected as a license plate number and the
confidence is 10, 20, 30, 40, 50, 60, it indicates that the accuracy of "A" is 20%-->test
    </plateCharBelieve>
    <speedLimit>
      <!--ro, opt, int, maximum speed limit, desc:it is valid only when overspeeding occurred-->50
    </speedLimit>
    <illegalInfo>
      <!--ro, opt, object, vehicle traffic violation information, desc:illegal action information-->
      <illegalCode>
        <!--ro, req, string, illegal action code, range:[0,64]-->1301
      </illegalCode>
      <illegalName>
        <!--ro, req, string, illegal action name, range:[0,128]-->逆行
      </illegalName>
      <illegalDescription>
        <!--ro, opt, string, illegal action description, range:[0,256]-->车辆逆行
      </illegalDescription>
    </illegalInfo>
    <vehicleType>
      <!--ro, req, enum, vehicle type, subType:string, desc:vehicle type:
"unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickup
Truck"-->nonmotorVehicle
    </vehicleType>
    <postPicFileName>
      <!--ro, opt, string, name of the picture selected as the checkpoint picture when illegal action occurs, desc:"none" (no picture selected)-->test
    </postPicFileName>
    <featurePicFileName>
      <!--ro, opt, string, name of the picture selected as the close-up picture when running the red light in the intersection violation system is detected,
desc:"none" (no picture selected)-->test
    </featurePicFileName>
    <detectDir>
      <!--ro, opt, enum, detection direction, subType:int, desc:1 (upward), 2 (downward), 3 (bidirectional), 4 (westward), 5 (northward), 6 (eastward), 7
(southward), 8 (other)-->1
    </detectDir>
    <detectType>
      <!--ro, opt, enum, detection type, subType:int, desc:0 (vehicle detection), 1 (inductive loop trigger), 2 (video trigger), 3 (multiple-frame
recognition), 4 (radar trigger), 5 (mixed-traffic detection)-->1
    </detectType>
    <barrierGateCtrlType>
      <!--ro, opt, enum, whether to enable elapsed time, subType:int, desc:0 (enabled), 1 (disabled)-->0
    </barrierGateCtrlType>
    <alarmDataType>
      <!--ro, opt, enum, alarm data type (real-time data or history data), subType:int, desc:0 (real-time data), 1 (history data)-->0
    </alarmDataType>
    <dwIllegalTime>
      <!--ro, opt, int, illegal action duration, unit:ms, desc:it is the difference between the capture time of the last picture and the capture time of the
first picture-->100
    </dwIllegalTime>
    <vehicleInfo>
      <!--ro, opt, object, vehicle information-->
      <index>
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        <!--ro, req, int, vehicle No.-->1
      </index>
      <vehicleType>
        <!--ro, opt, enum, vehicle type, subType:int, desc:vehicle type:
"unknown,largeBus,truck,vehicle,van,buggy,pedestrian,twoWheelVehicle,threeWheelVehicle,SUVMPV,mediumBus,motorVehicle,nonmotorVehicle,smallCar,miniCar,pickup
Truck"-->0
      </vehicleType>
      <colorDepth>
        <!--ro, req, enum, shade of the vehicle color, subType:int, desc:0 (deep color), 1 (light color), 2 (unknown)-->0
      </colorDepth>
      <color>
        <!--ro, req, enum, vehicle color, subType:string, desc:"unknown", "green", "brown", "pink", "purple", "deepGray" (dark gray), "cyan", "orange",
"white", "silver" (silvery), "gray", "blacks", "red", "deepBlue" (dark blue), "blue", "yellow"-->green
      </color>
      <speed>
        <!--ro, opt, int, vehicle speed, unit:km/h-->1
      </speed>
      <length>
        <!--ro, req, int, length of the former vehicle, unit:dm-->10
      </length>
      <vehicleLogoRecog>
        <!--ro, req, int, vehicle main brand-->1
      </vehicleLogoRecog>
      <vehileSubLogoRecog>
        <!--ro, opt, int, vehicle sub-brand-->1
      </vehileSubLogoRecog>
      <vehileModel>
        <!--ro, opt, int, model year of vehicle sub-brand-->1
      </vehileModel>
      <vehicleTypeByWeight>
        <!--ro, opt, enum, vehicle type according to the vehicle weight, subType:int, desc:1 (class 1: buses with seven or less seats, trucks with capacity
of 2 tons or less), 2 (class 2: buses with 8 to 19 seats, trucks with capacity of 2 to 5 (included) tons), 3 (class 3: buses with 20 to 39 seats, trucks
with capacity of 5 to 10 (included) tons),4 (class 4: buses with 40 or more seats, trucks with capacity of 10 to 15 (included) tons), 5 (class 5: trucks
with capacity of more than 15 tons), 6 (class 6: special vehicle such as ambulance, fire truck, and garbage truck)-->4
      </vehicleTypeByWeight>
      <tollRoadVehicleSeries>
        <!--ro, opt, enum, vehicles on tollway, subType:int, desc:N/A-->2
      </tollRoadVehicleSeries>
      <tollRoadVehicleType>
        <!--ro, opt, enum, vehicle types on tollway, subType:int, desc:N/A-->4
      </tollRoadVehicleType>
      <CarWindowFeature>
        <!--ro, opt, object, vehicle window feature, desc:configured via <CarWindowFeature> from /ISAPI/ITC/carFeatureParam-->
        <tempPlate>
          <!--ro, opt, enum, whether there is temporary license plate, subType:string, desc:detect whether there is temporary license plate on the window:
"unknown", "yes", "no"-->unknown
        </tempPlate>
        <passCard>
          <!--ro, opt, enum, whether there is entry & exit pass, subType:string, desc:detect whether there is entry & exit pass on the window: "unknown",
"yes", "no"-->unknown
        </passCard>
        <carCard>
          <!--ro, opt, enum, whether there are cards detected, subType:string, desc:whether there is any card (business cards, flyers, etc.) attached to the
window: "unknown", "yes", "no"-->unknown
        </carCard>
      </CarWindowFeature>
      <CarBodyFeature>
        <!--ro, opt, object, bodywork feature, desc:configured via <CarBodyFeature> from /ISAPI/ITC/carFeatureParam-->
        <sparetire>
          <!--ro, opt, enum, whether it is with a spare tire, subType:string, desc:"unknown", "yes", "no"-->unknown
        </sparetire>
        <rack>
          <!--ro, opt, enum, whether it is with a luggage rack, subType:string, desc:"unknown", "yes", "no"-->unknown
        </rack>
        <sunRoof>
          <!--ro, opt, enum, whether there is a sunroof, subType:string, desc:"unknown", "yes", "no"-->unknown
        </sunRoof>
        <words>
          <!--ro, opt, enum, whether the vehicle is painted, subType:string, desc:"unknown", "yes", "no"-->unknown
        </words>
        <slagTruckCoverPlate>
          <!--ro, opt, enum, whether the back cover of the dump truck is covered, subType:string, desc:"unknown", "yes", "no"-->unknown
        </slagTruckCoverPlate>
        <reflectiveTape>
          <!--ro, opt, enum, subType:string-->unknown
        </reflectiveTape>
        <closureWindow>
          <!--ro, opt, enum, subType:string-->unknown
        </closureWindow>
      </CarBodyFeature>
      <vehicleUseType>
        <!--ro, opt, enum, vehicle purpose type, subType:string, desc:configured via vehicleUseEnable from /ISAPI/ITC/carFeatureParam: "taxi", "ambulance",
"bus", "schoolbus", "coach", "unknown"-->taxi
      </vehicleUseType>
      <carWindowInfo>
        <!--ro, opt, object, vehicle window information, desc:configured via /ISAPI/ITC/TriggerMode/HOVLaneDetection?format=json-->
        <carWindowNum>
          <!--ro, opt, int, number of vehicle windows-->4
        </carWindowNum>
        <carWindowStatusList>
          <!--ro, opt, object, vehicle window status list-->
          <carWindowStatus>
            <!--ro, opt, object, vehicle window status-->
            <passengers>
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            <passengers>
              <!--ro, opt, int, number of persons behind the window, range:[0,10]-->1
            </passengers>
            <windowStatus>
              <!--ro, opt, enum, whether the vehicle windows are transparent, subType:string, desc:"unknown", "yes", "no"-->yes
            </windowStatus>
          </carWindowStatus>
        </carWindowStatusList>
      </carWindowInfo>
      <actualPassengers>
        <!--ro, opt, int, number of passengers in the vehicle, range:[0,100]-->1
      </actualPassengers>
      <vehiclePosition>
        <!--ro, opt, enum, detected vehicle body part, subType:string, desc:"leftSide", "rightSide", "headStock", "tailStock", "roof"-->leftSide
      </vehiclePosition>
      <containerInfo>
        <!--ro, opt, object, container information-->
        <containerNum>
          <!--ro, opt, int, number of transferred containers, range:[1,10]-->2
        </containerNum>
        <containerList>
          <!--ro, opt, array, list of container information, subType:object, range:[1,10]-->
          <container>
            <!--ro, opt, object, container information-->
            <containerMainNum>
              <!--ro, opt, string, major No. of captured container, range:[1,32]-->abcd1234
            </containerMainNum>
            <containerSubNum>
              <!--ro, opt, string, minor No. of captured container, range:[1,32]-->abcd1234
            </containerSubNum>
            <containerISONum>
              <!--ro, opt, string, ISO No. of captured container, range:[1,32]-->abcd1234
            </containerISONum>
            <containerNumConfidence>
              <!--ro, opt, int, confidence of captured container No., range:[0,100]-->0
            </containerNumConfidence>
          </container>
        </containerList>
      </containerInfo>
      <RFID>
        <!--ro, opt, string, link RFID, range:[1,64], dep:and,{$.EventNotificationAlert.ANPR.vehicleType,eq,nonmotorVehicle}-->12345
      </RFID>
      <vehicleCarryFeature>
        <!--ro, opt, object-->
        <isCarry>
          <!--ro, opt, enum, subType:string-->unknown
        </isCarry>
        <isTarpaulin>
          <!--ro, opt, enum, subType:string-->unknown
        </isTarpaulin>
        <isCoverplate>
          <!--ro, opt, enum, subType:string-->unknown
        </isCoverplate>
      </vehicleCarryFeature>
    </vehicleInfo>
    <EntranceInfo>
      <!--ro, opt, object, entrance and exit information-->
      <parkingID>
        <!--ro, opt, string, parking space No.-->test
      </parkingID>
      <gateID>
        <!--ro, opt, string, entrance and exit No.-->test
      </gateID>
      <direction>
        <!--ro, opt, enum, target direction, subType:string, desc:"enter", "leave"-->enter
      </direction>
      <cardNo>
        <!--ro, opt, string, card No.-->test
      </cardNo>
      <parkType>
        <!--ro, opt, enum, parking type, subType:string, desc:"temporary", "permanent"-->temporary
      </parkType>
    </EntranceInfo>
    <pictureInfoList>
      <!--ro, req, array, picture list, subType:object, range:[0,8]-->
      <pictureInfo>
        <!--ro, req, object, picture information-->
        <fileName>
          <!--ro, req, enum, picture name, subType:string, desc:"detectionPicture.jpg" (background picture), "licensePlatePicture.jpg" (license plate
picture), "pilotPicture.jpg" (driver's picture matting), "copilotPicture.jpg" (co-driver's picture matting), "compositePicture.jpg" (composite picture),
"plateBinaryPicture.jpg" (license plate binary picture), "nonMotorPicture.jpg" (non-motor vehicle picture matting), "pedestrianDetectionPicture.jpg"
(pedestrian picture), "pedestrianPicture.jpg" (pedestrian's picture matting), "vehiclePicture.jpg" (vehicle picture). Picture name, which must correspond to
the picture name transmitted with the alarm message-->detectionPicture.jpg
        </fileName>
        <type>
          <!--ro, req, enum, picture type, subType:string, desc:"detectionPicture" (background picture), "licensePlatePicture" (license plate picture),
"pilotPicture" (driver's picture matting), "copilotPicture" (co-driver's picture matting), "compositePicture" (composite picture), "plateBinaryPicture"
(license plate binary picture), "nonMotorPicture" (non-motor vehicle picture matting), "pedestrianDetectionPicture" (pedestrian picture),
"pedestrianPicture" (pedestrian's picture matting), "vehiclePicture" (vehicle picture)-->vehiclePicture
        </type>
        <dataType>
          <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
        </dataType>
        <picRecogMode>
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      <!--ro, opt, enum, license plate recognition mode, subType:int, desc:0 (front license plate recognition), 1 (rear license plate recognition)-->0
    </picRecogMode>
    <redLightTime>
      <!--ro, opt, int, red light time elapsed, unit:s-->60
    </redLightTime>
    <vehicleHead>
      <!--ro, opt, enum, license plate recognition direction, subType:string, desc:"unknown", "forward" (front license plate recognition), "back" (rear 
license plate recognition)-->unknown
    </vehicleHead>
    <absTime>
      <!--ro, opt, string, absolute time, desc:format: yyyyMMddHHmmssxxx, e.g.: 20090810235959999, the last three digits are time in millisecond--
>20090810235959999
    </absTime>
    <plateRect>
      <!--ro, opt, object, license plate area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value 
is the current image size in percentage multiplying 1000 and it is accurate to three decimal places. The origin is the upper-left corner of screen-->
      <X>
        <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
      </X>
      <Y>
        <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
      </Y>
      <width>
        <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
      </width>
      <height>
        <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
      </height>
    </plateRect>
    <vehicelRect>
      <!--ro, opt, object, the vehicle area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value is 
the current image size in percentage multiplying 1000. The origin is the upper-left corner of screen-->
      <X>
        <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
      </X>
      <Y>
        <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
      </Y>
      <width>
        <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
      </width>
      <height>
        <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
      </height>
    </vehicelRect>
    <pictureURL>
      <!--ro, opt, string, picture URL, desc:it is valid when the value of dataType is "URL"-->test
    </pictureURL>
    <pId>
      <!--ro, opt, string, strlen.max=32, range:[1,32], desc:strlen.max=32-->test
    </pId>
    <province>
      <!--ro, opt, enum, province index, subType:int, desc:N/A-->0
    </province>
    <PilotRect>
      <!--ro, opt, object-->
      <x>
        <!--ro, req, int, range:[0,1000]-->1000
      </x>
      <y>
        <!--ro, req, int, range:[0,1000]-->1000
      </y>
      <width>
        <!--ro, req, int, range:[0,1000]-->1000
      </width>
      <height>
        <!--ro, req, int, range:[0,1000]-->1000
      </height>
    </PilotRect>
    <VicepilotRect>
      <!--ro, opt, object-->
      <x>
        <!--ro, req, int, range:[0,1000]-->1000
      </x>
      <y>
        <!--ro, req, int, range:[0,1000]-->1000
      </y>
      <width>
        <!--ro, req, int, range:[0,1000]-->1000
      </width>
      <height>
        <!--ro, req, int, range:[0,1000]-->1000
      </height>
    </VicepilotRect>
    <VehicelWindowRect>
      <!--ro, opt, object-->
      <x>
        <!--ro, req, int, range:[0,1000]-->1000
      </x>
      <y>
        <!--ro, req, int, range:[0,1000]-->1000
      </y>
      <width>
```

```
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </VehicelWindowRect>
        <extraPlateRect>
          <!--ro, opt, object-->
          <x>
            <!--ro, req, int, range:[0,1000]-->1000
          </x>
          <y>
            <!--ro, req, int, range:[0,1000]-->1000
          </y>
          <width>
            <!--ro, req, int, range:[0,1000]-->1000
          </width>
          <height>
            <!--ro, req, int, range:[0,1000]-->1000
          </height>
        </extraPlateRect>
      </pictureInfo>
    </pictureInfoList>
    <hasMoreData>
      <!--ro, opt, bool, whether there is more data, desc:this field is used to report the license plate information first, and then report XML message with
picture data; the XM message with picture data and license plate information are linked by UUID-->true
    </hasMoreData>
    <listType>
      <!--ro, opt, enum, List Property, subType:string, desc:"white" (allowlist), "black" (blocklist), "temporary" (temporary list)-->white
    </listType>
    <originalLicensePlate>
      <!--ro, opt, string, original license plate number, desc:when the original license plate number contains non-English characters, the original license
plate number can be returned-->test
    </originalLicensePlate>
    <CRIndex>
      <!--ro, opt, enum, country or region index, subType:int, desc:country or region index-->258
    </CRIndex>
    <VehicelGPSInfo>
      <!--ro, opt, object, GPS information of the vehicle-->
      <longitudeType>
        <!--ro, req, enum, longitude, subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)-->E
      </longitudeType>
      <latitudeType>
        <!--ro, req, enum, latitude, subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)-->S
      </latitudeType>
      <Longitude>
        <!--ro, req, object, longitude information-->
        <degree>
          <!--ro, req, int, degree(s)-->60
        </degree>
        <minute>
          <!--ro, req, int, minute(s), range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second(s), range:[0,59.999999]-->59.000000
        </sec>
      </Longitude>
      <Latitude>
        <!--ro, req, object, latitude information-->
        <degree>
          <!--ro, req, int, degree(s)-->60
        </degree>
        <minute>
          <!--ro, req, int, minute(s), range:[0,59]-->59
        </minute>
        <sec>
          <!--ro, req, float, second(s), range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
        </sec>
      </Latitude>
    </VehicelGPSInfo>
    <vehiclePositionControl>
      <!--ro, opt, enum, vehicle arming type, subType:string, desc:"vehicleMonitor" (intelligent arming of vehicle, related URI: PUT
/ISAPI/Traffic/channels/<ID>/vehicleMonitor/<taskID>/startTask), "manualVehicleMonitor" (manual arming of vehicle, related URI: PUT
/ISAPI/Traffic/channels/<ID>/manualVehicleMonitor), "dailyVehicleMonitor" (daily arming of vehicle)-->dailyVehicleMonitor
    </vehiclePositionControl>
    <vehicleMonitorTaskID>
      <!--ro, opt, string, task ID of intelligent arming of vehicle, range:[1,64], desc:the task ID is sent to the device by the upper layer, which should
guarantee the uniqueness of ID. It's valid when the value of vehiclePositionControl is "vehicleMonitor"-->test
    </vehicleMonitorTaskID>
    <vehicleListName>
      <!--ro, opt, string, name of the list that the vehicle belongs to, range:[0,128], desc:name of the list that the vehicle belongs to,the maximum size
is 128 bytes-->test
    </vehicleListName>
    <vehicleThermometryEnabled>
      <!--ro, opt, bool, whether to enable vehicle temperature measurement-->true
    </vehicleThermometryEnabled>
    <currTemperature>
      <!--ro, opt, float, temperature-->36.5
    </currTemperature>
    <thermometryUnit>
      <!--ro, opt, enum, temperature unit, subType:string, desc:"celsius", "fahrenheit", "kelvin"-->celsius
    </thermometryUnit>
    <plateCategory>
      <!--ro, opt, string, license plate additional information, range:[0,8], desc:license plate additional information-->test
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```xml
      </plateCategory>
      <plateSize>
        <!--ro, opt, enum, license plate size, subType:int, desc:license plate size-->1
      </plateSize>
      <isNeedVerification>
        <!--ro, opt, bool, whether it requires verification, desc:false-do not verify-->true
      </isNeedVerification>
      <ISO-CR>
        <!--ro, opt, string, ISO 3166-1 standard country/region code, desc:refer to protocol dictionary-->CN
      </ISO-CR>
      <vehicelRectName>
        <!--ro, opt, string, country/region where the vehicle triggering events belongs, range:[0,64]-->11
      </vehicelRectName>
      <rodeType>
        <!--ro, opt, enum, road type, subType:string, desc:"entrance" (entrance and exit), "city", "custom", "alarmInput" (alarm input triggered by coil),
"post" (checkpoint), "publicSecurity" (general video security mode, in which the scene with mixed vehicles and without lane lines can be detected)--
>entrance
      </rodeType>
      <Custom>
        <!--ro, opt, object, custom parameters, dep:and,{$.EventNotificationAlert.ANPR.rodeType,eq,custom}-->
        <direction>
          <!--ro, opt, enum, target direction, subType:string, desc:"enter", "parkind", "leave"-->enter
        </direction>
      </Custom>
      <accelerationNoise>
        <!--ro, opt, object, vehicle revving behavior that causes huge noise-->
        <startTime>
          <!--ro, req, datetime, start time of the behavior-->1970-01-01T00:00:00+08:00
        </startTime>
        <endTime>
          <!--ro, req, datetime, end time of the behavior-->1970-01-01T00:00:00+08:00
        </endTime>
        <noiseDecibel>
          <!--ro, req, float, noise decibel-->36.5
        </noiseDecibel>
      </accelerationNoise>
      <ADRNo>
        <!--ro, opt, string, hazardous material ID, range:[1,8]-->231965
      </ADRNo>
      <pilotmask>
        <!--ro, opt, enum, subType:string-->unknown
      </pilotmask>
      <vicepilotMask>
        <!--ro, opt, enum, subType:string-->unknown
      </vicepilotMask>
      <nonMotorMask>
        <!--ro, opt, enum, subType:string-->unknown
      </nonMotorMask>
      <pedestrianMask>
        <!--ro, opt, enum, subType:string-->unknown
      </pedestrianMask>
      <vehicleTemperature>
        <!--ro, opt, object-->
        <vehicleHeadTemperature>
          <!--ro, opt, float, range:[-20.0,550.0]-->50.2
        </vehicleHeadTemperature>
        <vehicleBodyTemperature>
          <!--ro, opt, float, range:[-20.0,550.0]-->35.2
        </vehicleBodyTemperature>
        <leftWheelHubList>
          <!--ro, opt, array, subType:object, range:[0,6]-->
          <leftWheelHub>
            <!--ro, opt, object-->
            <ID>
              <!--ro, opt, int, range:[1,6]-->1
            </ID>
            <wheelHubTemperature>
              <!--ro, opt, float, range:[-20.0,550.0]-->38.2
            </wheelHubTemperature>
          </leftWheelHub>
        </leftWheelHubList>
        <rightWheelHubList>
          <!--ro, opt, array, subType:object, range:[0,6]-->
          <rightWheelHub>
            <!--ro, opt, object-->
            <ID>
              <!--ro, opt, int, range:[1,6]-->1
            </ID>
            <wheelHubTemperature>
              <!--ro, opt, float, range:[-20.0,550.0]-->38.2
            </wheelHubTemperature>
          </rightWheelHub>
        </rightWheelHubList>
      </vehicleTemperature>
      <deliveryVan>
        <!--ro, opt, enum, subType:string-->unknown
      </deliveryVan>
      <nonMotorShedUmbrella>
        <!--ro, opt, enum, subType:string-->unknown
      </nonMotorShedUmbrella>
      <nonMotorManned>
        <!--ro, opt, enum, subType:string-->unknown
      </nonMotorManned>
```

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  </ANPR>
  <UUID>
    <!--ro, opt, string, unique common ID, which is used to link the same capture across multiple servers, desc:if the vehicle is captured many times, the UUID is the same-->test
  </UUID>
  <picNum>
    <!--ro, opt, int, number of pictures-->2
  </picNum>
  <monitoringSiteID>
    <!--ro, opt, string, camera No., desc:corresponds to <positionNum> from /ISAPI/System/Video/inputs/channels/<channelID>/cameraInfo (recommended for new devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
  </monitoringSiteID>
  <ePlateUUID>
    <!--ro, opt, string, electronic license plate ID, desc:if this field has a value, it indicates that an electronic license plate is linked-->test
  </ePlateUUID>
  <isDataRetransmission>
    <!--ro, opt, bool, whether the data is replenished-->true
  </isDataRetransmission>
  <SceneInfo>
    <!--ro, opt, object, scene information-->
    <scenesID>
      <!--ro, opt, string, scene ID, which is between 1 and 16-->test
    </scenesID>
    <sceneName>
      <!--ro, opt, string, scene name, range:[0,128]-->test
    </sceneName>
    <PTZPos>
      <!--ro, opt, object, PTZ information-->
      <elevation>
        <!--ro, opt, int, tilting angle, range:[-900,2700]-->0
      </elevation>
      <azimuth>
        <!--ro, opt, int, panning angle, range:[0,3600]-->0
      </azimuth>
      <absoluteZoom>
        <!--ro, opt, int, zooming ratio, range:[0,1000]-->1
      </absoluteZoom>
    </PTZPos>
  </SceneInfo>
  <monitorDescription>
    <!--ro, opt, string, camera information, range:[0,256], desc:corresponds to <positionInfo> from /ISAPI/System/Video/inputs/channels/<ID>/cameraInfo (recommended for new devices) and /ISAPI/Traffic/channels/<channelID>/cameraInfo applied by some traffic monitoring device-->test
  </monitorDescription>
  <DeviceGPSInfo>
    <!--ro, opt, object, device GPS information-->
    <longitudeType>
      <!--ro, req, enum, longitude, subType:string, desc:"E" (Eastern Hemisphere), W" (West Hemisphere)-->E
    </longitudeType>
    <latitudeType>
      <!--ro, req, enum, latitude, subType:string, desc:"S" (Southern Hemisphere), "N" (North Hemisphere)-->S
    </latitudeType>
    <Longitude>
      <!--ro, req, object, longitude information-->
      <degree>
        <!--ro, req, int, degree(s)-->60
      </degree>
      <minute>
        <!--ro, req, int, minute(s), range:[0,59]-->59
      </minute>
      <sec>
        <!--ro, req, float, second(s), range:[0,59.999999]-->59.000000
      </sec>
    </Longitude>
    <Latitude>
      <!--ro, req, object, latitude information-->
      <degree>
        <!--ro, req, int, degree(s)-->60
      </degree>
      <minute>
        <!--ro, req, int, minute(s), range:[0,59]-->59
      </minute>
      <sec>
        <!--ro, req, float, second(s), range:[0,59.999999], desc:the value is accurate to six decimal places-->59.000000
      </sec>
    </Latitude>
  </DeviceGPSInfo>
  <pilotStandardSafebelt>
    <!--ro, opt, enum, whether the driver wears the seat belt properly, subType:string, desc:"unknown", "yes", "no"-->yes
  </pilotStandardSafebelt>
  <vicepilotStandardSafebelt>
    <!--ro, opt, enum, whether the front passenger wears the seat belt properly, subType:string, desc:"unknown", "yes", "no"-->yes
  </vicepilotStandardSafebelt>
  <trafficLightSnap>
    <!--ro, opt, enum, whether it is captured at the traffic lights, subType:string, desc:"yes", "no"-->yes
  </trafficLightSnap>
  <sequence>
    <!--ro, opt, int, No. of vehicle capture from network-triggered burst, range:[1,4294967295], desc:command applied by network-triggered burst control; related API: /ISAPI/Traffic/startRecognition; SDK interface: NET_DVR_ContinuousShoot-->0
  </sequence>
  <relaLaneDirectionType>
    <!--ro, opt, enum, linked lane direction, subType:int, desc:a camera can be linked with multiple lane directions, which is independent of the camera setup angle <detectDir>; 0 (others),1 (from east to west), 2 (from west to east), 3 (from south to north), 4 (from north to south), 5 (from southeast to northwest), 6 (from northwest to southeast), 7 (from northeast to southwest), 8 (from southwest to northeast)-->1
  </relaLaneDirectionType>
```

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  <carDirectionType>
    <!--ro, opt, enum, vehicle running direction on the lane, subType:int, desc:independent of the camera setup angle <detectDir>; 0-from up to down, 1-from down to up-->1
  </carDirectionType>
  <targetID>
    <!--ro, opt, string, target ID, range:[1,64], desc:UUID, of which the uniqueness is guaranteed by device, and corresponds to <dwMatchNo> in the private protocol-->test
  </targetID>
  <isSecondCamera>
    <!--ro, opt, bool, whether to enable capture by the second camera, desc:the second camera: e.g., the perspective camera from distant/close-view capture cameras, the back camera from front/back capture camera, etc.-->false
  </isSecondCamera>
  <dataAnalysisType>
    <!--ro, opt, enum, data analysis type, subType:int, desc:0-data not analyzed, 1-data analyzed-->1
  </dataAnalysisType>
  <RecordInfo>
    <!--ro, opt, object, vehicle video information-->
    <fileName>
      <!--ro, req, enum, video name, subType:string, desc:the video name should match <Content-ID> for binary transmission: "record.mp4" (vehicle video)-->record.mp4
    </fileName>
    <dataType>
      <!--ro, req, enum, data type, subType:int, desc:0 (binary), 1 (URL)-->0
    </dataType>
    <URL>
      <!--ro, opt, string, URL, desc:it is valid when the value of dataType is "URL"-->test
    </URL>
  </RecordInfo>
  <VehicleWeightInfo>
    <!--ro, opt, object, vehicle weighing information, desc:this node is currently only supported by weighing data management server-->
    <isOverWeight>
      <!--ro, opt, bool, whether it is overweight, desc:set overweight reporting or overrun reporting as priority via <illegalPriorityType> from /ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->false
    </isOverWeight>
    <axleNum>
      <!--ro, opt, int, number of axles, range:[1,10]-->4
    </axleNum>
    <axleModel>
      <!--ro, opt, enum, vehicle type by axle, subType:int, desc:0 (unknown), 512 (2-axle truck), 756 (3-axle centre-axle trailer combination), 766 (3-axle articulated vehicle), 767 (3-axle truck), 768 (3-axle truck), 1024 (type-125 4-axle centre-axle trailer combination), 1025 (type-152 4-axle centre-axle trailer combination), 1026 (4-axle articulated vehicle), 1027 (4-axle full trailer combination), 1028 (4-axle truck), 1280 (type-155 5-axle centre-axle trailer combination), 1281 (type-1125 5-axle centre-axle trailer combination), 1282 (type-155 5-axle articulated vehicle), 1283 (type-1125 5-axle articulated vehicle), 1284 (type-129 5-axle articulated vehicle), 1285 (type-1522 5-axle full trailer combination), 1286 (type-11222 5-axle full trailer combination), 1536 (type-159 6-axle centre-axle trailer combination), 1537 (type-159-2 6-axle centre-axle trailer combination), 1538 (type-1155-1 6-axle centre-axle trailer combination), 1539 (type-1155-2 6-axle centre-axle trailer combination), 1540 (type-159-3 6-axle articulated vehicle), 1541 (type-159-4 6-axle articulated vehicle), 1542 (type-1129 6-axle articulated vehicle), 1543 (type-11522-1 6-axle full trailer combination), 1544 (type-11522-2 6-axle full trailer combination)-->0
    </axleModel>
    <overWeight>
      <!--ro, opt, float, over-limit weight (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </overWeight>
    <weight>
      <!--ro, opt, float, vehicle weight (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </weight>
    <limitWeight>
      <!--ro, opt, float, weight limit (unit: ton), range:[0.000,100.000], unit:吨, desc:accurate to three decimal places-->4.502
    </limitWeight>
    <axleLen>
      <!--ro, opt, float, distance between tyres, range:[0.00,100.00], unit:吨, desc:accurate to two decimal places-->4.502
    </axleLen>
    <devDescInfo>
      <!--ro, opt, string, device description, range:[0,64]-->test
    </devDescInfo>
    <AxleInfoList>
      <!--ro, opt, object, axle information list, desc:<axleNum>, number of axles-->
      <AxleInfo>
        <!--ro, opt, object, information of a single axle-->
        <axleWeight>
          <!--ro, opt, float, axle weight, range:[0.00,10000.00], unit:kg-->100.00
        </axleWeight>
        <axleDistance>
          <!--ro, opt, int, distance between axles, range:[0,100000], unit:mm, desc:the distance between the current axle and the next axle-->30000
        </axleDistance>
        <wheelNum>
          <!--ro, opt, int, range:[0,8]-->2
        </wheelNum>
      </AxleInfo>
    </AxleInfoList>
    <length>
      <!--ro, opt, int, vehicle length, range:[1,1000000], unit:cm-->4000
    </length>
    <width>
      <!--ro, opt, int, vehicle width, range:[1,1000000], unit:cm-->4000
    </width>
    <height>
      <!--ro, opt, int, vehicle height, range:[1,1000000], unit:cm-->4000
    </height>
    <tollwayVehicleType>
      <!--ro, opt, enum, vehicle types on tollway, subType:int, desc:0 (unknown), 1 (type-1 minibus), 2 (type-1 small-sized bus), 3 (type-2 medium-sized bus), 4 (type-2 bus and passenger trailer combination), 5 (type-3 large-sized bus), 6 (type-4 large-sized bus), 7 (type-1 2-axle truck), 8 (type-2 2-axle truck), 9 (type-3 3-axle truck), 10 (type-4 4-axle truck), 11 (type-5 5-axle truck), 12 (type-6 6-axle truck), 13 (type-1 2-axle operation vehicle), 14 (type-2 2-axle operation vehicle), 15 (type-3 3-axle operation vehicle), 16 (type-4 4-axle operation vehicle), 17 (type-5 5-axle operation vehicle), 18 (type-6 operation vehicle with 6 or more axles)-->1
```

```xml
    </tollwayVehicleType>
    <tiresNum>
      <!--ro, opt, int, number of tyres, range:[1,20]-->4
    </tiresNum>
    <approvedPassengers>
      <!--ro, opt, int, authorized passenger capacity, range:[1,100]-->7
    </approvedPassengers>
    <limitLength>
      <!--ro, opt, int, vehicle length limit, range:[1,1000000], unit:cm, desc:configured via
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->4000
    </limitLength>
    <limitWidth>
      <!--ro, opt, int, vehicle width limit, range:[1,1000000], unit:cm, desc:configured via /ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?
format=json-->4000
    </limitWidth>
    <limitHeight>
      <!--ro, opt, int, vehicle height limit, range:[1,1000000], unit:cm, desc:configured via
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->4000
    </limitHeight>
    <overLength>
      <!--ro, opt, int, over-limit length, range:[1,1000000], unit:cm, desc:the over-limit part equals <length> minus <limitLength>-->4000
    </overLength>
    <overWidth>
      <!--ro, opt, int, over-limit width, range:[1,1000000], unit:cm, desc:the over-limit part equals <width> minus <limitWidth>-->4000
    </overWidth>
    <overHeight>
      <!--ro, opt, int, over-limit height, range:[1,1000000], unit:cm, desc:the over-limit part equals <height> minus <limitHeight>-->4000
    </overHeight>
    <isOverLimit>
      <!--ro, opt, bool, whether it is over-limit, desc:set overweight reporting or overrun reporting as priority via <illegalPriorityType> from
/ISAPI/Traffic/channels/<channelID>/vehicleWeight/capabilities?format=json-->false
    </isOverLimit>
    <axleNumConfidence>
      <!--ro, opt, int, confidence of captured number of axles, range:[0,100]-->0
    </axleNumConfidence>
    <standardAxleTypeInfo>
      <!--ro, opt, object-->
      <axleType>
        <!--ro, req, enum, subType:int-->12
      </axleType>
      <axleDescribeInfo>
        <!--ro, req, string, range:[0,64]-->2轴载货汽车
      </axleDescribeInfo>
    </standardAxleTypeInfo>
  </VehicleWeightInfo>
  <isNotSlowZebraCrossing>
    <!--ro, opt, bool, whether the vehicle did not slow down at zebra crossing-->false
  </isNotSlowZebraCrossing>
  <isTurnRightStop>
    <!--ro, opt, bool, whether the vehicle did not stop at a right turn-->false
  </isTurnRightStop>
  <PlateInfoList>
    <!--ro, opt, object, this node is returned when license plates from Mainland, Hongkong, and Maokao are all supported-->
    <PlateInfo>
      <!--ro, opt, object, information of a single license plate-->
      <plateRect>
        <!--ro, opt, object, license plate area coordinates, desc:this field is valid when the value of type is "detectionPicture". The normalized value is
the current image size in percentage multiplying 1000 and it is accurate to three decimal places. The origin is the upper-left corner of screen-->
        <X>
          <!--ro, req, int, X-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
        </X>
        <Y>
          <!--ro, req, int, Y-coordinate of the upper-left corner of the boundary frame, range:[0,1000]-->1000
        </Y>
        <width>
          <!--ro, req, int, width of the boundary frame, range:[0,1000]-->1000
        </width>
        <height>
          <!--ro, req, int, height of the boundary frame, range:[0,1000]-->1000
        </height>
      </plateRect>
      <plateColor>
        <!--ro, req, enum, license plate color, subType:string, desc:license plate color: "white","yellow","blue","black","green","newEnergyGreen"-new
energy green,"newEnergyYellowGreen"-new energy flavogreen,"other"-other color-->black
      </plateColor>
      <licensePlate>
        <!--ro, req, string, license plate number, range:[1,32], desc:license plate number,e.g.,"123456"-->A283KY77
      </licensePlate>
      <confidenceLevel>
        <!--ro, req, int, confidence level, range:[0,100]-->50
      </confidenceLevel>
      <province>
        <!--ro, opt, enum, province index, subType:int, desc:N/A-->0
      </province>
      <isExtraPlate>
        <!--ro, opt, bool-->true
      </isExtraPlate>
    </PlateInfo>
  </PlateInfoList>
  <deviceUUID>
    <!--ro, opt, string, device No., range:[1,32], desc:the device number is the device UUID which is the device's serial number and can be edited by the
node <deviceID> in the message of /ISAPI/System/deviceInfo-->12345
  </deviceUUID>
  <VehicleGATInfo>
```

```
  <VehicleGATInfo>
    <!--ro, opt, object, N/A, desc:N/A-->
    <palteTypeByGAT>
      <!--ro, opt, enum, N/A, subType:int, desc:N/A-->1
    </palteTypeByGAT>
    <plateColorByGAT>
      <!--ro, opt, enum, N/A, subType:int, desc:0-white, 1-yellow, 2-blue, 3-black, 4-green, 5-not recognized, 9, other colors-->0
    </plateColorByGAT>
    <vehicleTypeByGAT>
      <!--ro, opt, enum, N/A, subType:string, desc:GA/T 16.4 complied: "K10" (large-sized bus), "K20" (medium-sized bus), "K30" (small-sized bus), "K33" (small-sized car), "H10" (heavy truck), "H20" (medium-sized truck), "H30" (light truck), "M10" (three-wheel motorcycle), "M20" (two-wheel motorcycle), "X99" (others)-->K10
    </vehicleTypeByGAT>
    <colorByGAT>
      <!--ro, opt, enum, N/A, subType:string, desc:GA/T 16.8 complied: A-white, B-gray, C-yellow, D-pink, E-red, F-purple, G-green, H-blue, I-brown, J-black, K-not recognized, Z-other colors-->A
    </colorByGAT>
  </VehicleGATInfo>
  <EPlateInfo>
    <!--ro, opt, object, RFID detection result-->
    <EPlateChipNo>
      <!--ro, opt, string, chip No. of electronic license plate (detected by RFID), range:[0,16]-->E855000011F9AAA5
    </EPlateChipNo>
    <EPlateCardNo>
      <!--ro, opt, string, electronic license plate No. (detected by RFID), range:[0,12], desc:electronic license plate No. (12 digits; the first two represent the province code-->E855000011F9AAA5
    </EPlateCardNo>
    <EPlateNo>
      <!--ro, opt, string, license plate No. (detected by RFID), range:[0,32], desc:"noPlate" (vehicle without license plate detected), "unknown" (license plate not detected; temporary license plate detected)-->浙A12345
    </EPlateNo>
    <EPlateManufacturingDate>
      <!--ro, opt, string, production date (detected by RFID), range:[0,7], desc:format: yyyy-MM-->2014-01
    </EPlateManufacturingDate>
    <EPlateInspectionValidity>
      <!--ro, opt, string, inspection validity (detected by RFID), range:[0,7], desc:accurate to month; format: yyyy-MM-->2014-01
    </EPlateInspectionValidity>
    <EPlateMandaScrapPeriod>
      <!--ro, opt, int, compulsory scrapping period (detected by RFID), range:[0,100], unit:年, desc:from the start of inspection validity to the compulsory scrapping time (unit: year)-->10
    </EPlateMandaScrapPeriod>
    <EPlatePassengersCapacity>
      <!--ro, opt, int, authorized passenger capacity (detected by RFID), range:[0,100], desc:e.g. 1-->1
    </EPlatePassengersCapacity>
    <EPlateCarryingCapacity>
      <!--ro, opt, float, tractive tonnage, range:[0,100], unit:吨, desc:tractive tonnage (unit: ton)-->2.0
    </EPlateCarryingCapacity>
    <EPlateVehicleColor>
      <!--ro, opt, enum, vehicle color (detected by RFID), subType:string, desc:"unknown", "white", "silver", "gray", "black", "red", "deepBlue", "blue", "yellow", "green", "brown", "pink", "purple", "deepGray", "cyan", "orange"-->black
    </EPlateVehicleColor>
    <EPlateNonMotorType>
      <!--ro, opt, enum, non-motor vehicle type (detected by RFID), subType:string, desc:"unknown", "threeWheelVehicle" (tricycle), "twoWheelVehicle" (two wheeler)-->threeWheelVehicle
    </EPlateNonMotorType>
    <EPlateNonMotorPlateType>
      <!--ro, opt, enum, non-motor vehicle license plate type (detected by RFID), subType:string, desc:"unknown", "temporary", "normal"-->unknown
    </EPlateNonMotorPlateType>
    <EPlateColor>
      <!--ro, opt, enum, vehicle color (detected by RFID), subType:string, desc:"black", "blue","golden", "orange", "red", "yellow", "white", "unknown", "other", "newEnergyYellowGreen"-new energy yellow-green,"civilAviationBlack", "civilAviationGreen" (civil aviation green), "green", "mixedColor", "newEnergyGreen" (new energy green), "brown"-->black
    </EPlateColor>
    <EPlateUseType>
      <!--ro, opt, enum, non-motor vehicle purpose (detected by RFID), subType:string, desc:"unknown", "nonOperating" (non-operating), "takeOut" (take-out delivery), "express" (express delivery)-->nonOperating
    </EPlateUseType>
    <EPlateTriggerTime>
      <!--ro, req, datetime, device detection trigger time of electronic license plate (detected by RFID)-->1970-01-01T00:00:00+08:00
    </EPlateTriggerTime>
    <EPlateListenAuthCode>
      <!--ro, opt, string, licensing authority code of electronic license plate (detected by RFID)-->粤B
    </EPlateListenAuthCode>
  </EPlateInfo>
  <accelerationNoise>
    <!--ro, opt, object, vehicle revving behavior that causes huge noise-->
    <isExistsAccelerationNoise>
      <!--ro, req, bool, whether there is vehicle revving behavior that causes huge noise-->true
    </isExistsAccelerationNoise>
    <startTime>
      <!--ro, req, datetime, start time of the behavior, desc:accurate to millisecond-->1970-01-01T00:00:00.000+08:00
    </startTime>
    <endTime>
      <!--ro, req, datetime, end time of the behavior, desc:accurate to millisecond-->1970-01-01T00:00:00.000+08:00
    </endTime>
  </accelerationNoise>
  <noiseDecibel>
    <!--ro, opt, float, noise decibel of vehicle running, range:[0,300], unit:分贝, desc:accurate to one decimal place-->65.5
  </noiseDecibel>
  <exhaustPipeNum>
    <!--ro, opt, int, number of exhaust pipes, range:[0,4]-->2
  </exhaustPipeNum>
  <exhaustPipePositionType>
    <!--ro, opt, enum, exhaust pipe installation location, subType:string, desc:"unilateral", "bilateral", "middle"-->unilateral
```

```xml
  </exhaustPipePositionType>
  <openGateType>
    <!--ro, opt, enum, reasons for opening the barrier gate, subType:string, dep:and,{$.EventNotificationAlert.ANPR.barrierGateCtrlType,eq,0}, desc:"white" (auto-open barrier gate via device according to the allowlist), "manual" (manually open barrier gate via platform), "abnormal" (auto-open barrier gate via device for exceptional circumstances)-->white
  </openGateType>
  <plateStandardStatus>
    <!--ro, opt, enum, whether the license plate is valid, subType:string, desc:"yes", "no"-->yes
  </plateStandardStatus>
  <limitSurfaceSpeed>
    <!--ro, opt, int, cross-sectional device-detected speed limit (required for overspeed and violation), range:[0,255], unit:km/h-->1
  </limitSurfaceSpeed>
  <startCaptureSpeed>
    <!--ro, opt, int, device-detected capture triggered speed (required for overspeed and violation), range:[0,255], unit:km/h-->1
  </startCaptureSpeed>
  <redlightIllegalCode>
    <!--ro, opt, string, traffic lights and lane matching code for red light running, range:[0,4], desc:0-all lanes, 1-turn left, 2-go straight, 3-turn right, 4-go straight & turn left, 5-go straight & turn right-->1
  </redlightIllegalCode>
  <redlightSignalStartTime>
    <!--ro, opt, string, start time of the red light (required for red light running), desc:ISO 8601 format yyyy-MM-ddTHH:mm:ss+current zone, e.g., 2017-08-01T17:30:08+08:00-->"2004-05-03T17:30:08+08:00"
  </redlightSignalStartTime>
  <redlightSignalEndTime>
    <!--ro, opt, string, end time of the red light (required for red light running), desc:ISO 8601 format yyyy-MM-ddTHH:mm:ss+current zone, e.g., 2017-08-01T17:30:08+08:00-->"2004-05-03T17:30:08+08:00"
  </redlightSignalEndTime>
  <trafficControlTimeSpanList>
    <!--ro, opt, array, time period for traffic restriction, subType:object-->
    <TimeRange>
      <!--ro, req, object, time period-->
      <beginTime>
        <!--ro, req, time, start time-->10:00:00
      </beginTime>
      <endTime>
        <!--ro, req, time, end time-->10:00:00
      </endTime>
    </TimeRange>
  </trafficControlTimeSpanList>
  <detectionBackgroundImageResolution>
    <!--ro, opt, object, -ro, desc:corresponds to the picture represented by <pictureURL> and <pId> when <fileName> is "detectionPicture.jpg" and <type> is "detectionPicture" from <pictureInfoList> and <pictureInfo> in ANPR when the normalized coordinates of <Rect>, <pictureInfoList>, <pictureInfo>, <plateRect>, <vehicelRect>, <PlateInfoList>, <PlateInfo>, and <plateRect> are converted to the resolution coordinates, this node is required-->
    <height>
      <!--ro, req, int, resolution (height), range:[1,65535], unit:px-->1
    </height>
    <width>
      <!--ro, req, int, resolution (width), range:[1,65535], unit:px-->1
    </width>
  </detectionBackgroundImageResolution>
  <trafficSurveyVehicleType>
    <!--ro, opt, enum, subType:int-->1
  </trafficSurveyVehicleType>
  <doorTextInfoList>
    <!--ro, opt, array, subType:object, range:[0,2]-->
    <doorTextInfo>
      <!--ro, opt, object-->
      <doorID>
        <!--ro, opt, int, range:[0,2]-->1
      </doorID>
      <doorTextList>
        <!--ro, opt, array, subType:object, range:[0,4]-->
        <doorText>
          <!--ro, opt, string, range:[0,128]-->载荷50人
        </doorText>
      </doorTextList>
      <approvedPassengers>
        <!--ro, opt, int, range:[1,100]-->7
      </approvedPassengers>
      <doorTextConfidence>
        <!--ro, opt, int, range:[0,100]-->0
      </doorTextConfidence>
    </doorTextInfo>
  </doorTextInfoList>
  <splicingDetectInfo>
    <!--ro, opt, object-->
    <startSplicingTime>
      <!--ro, req, datetime-->1970-01-01T00:00:00 000+08:00
    </startSplicingTime>
    <endSplicingTime>
      <!--ro, req, datetime-->1970-01-01T00:00:00 000+08:00
    </endSplicingTime>
  </splicingDetectInfo>
</EventNotificationAlert>
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
anpr.xml	[Message content]	application/xml	--	--	--
.value	[Binary picture data]	image/jpeg	detectionPicture	detectionPicture.jpg	--
.value	[Binary picture data]	image/jpeg	licensePlatePicture	licensePlatePicture.jpg	--
.value	[Binary picture data]	image/jpeg	pilotPicture	pilotPicture.jpg	--
.value	[Binary picture data]	image/jpeg	copilotPicture	copilotPicture.jpg	--
.value	[Binary picture data]	image/jpeg	compositePicture	compositePicture.jpg	--
.value	[Binary picture data]	image/jpeg	plateBinaryPicture	plateBinaryPicture.jpg	--
.value	[Binary picture data]	image/jpeg	nonMotorPicture	nonMotorPicture.jpg	--
.value	[Binary picture data]	image/jpeg	pedestrianDetectionPicture	pedestrianDetectionPicture.jpg	--
.value	[Binary picture data]	image/jpeg	vehiclePicture	vehiclePicture.jpg	--
.value	[Binary picture data]	image/jpeg	featurePicture	featurePicture.jpg	--
.value	[Binary picture data]	image/jpeg	panoramaPicture	panoramaPicture.jpg	--
.value	[Binary picture data]	image/jpeg	extraLicensePlatePicture	extraLicensePlatePicture.jpg	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

11.6 出入口管控

11.6.1 票箱管理

11.6.1.1 Set the control parameters of audio announcement

Request URL

PUT /ISAPI/Parking/channels/<channelID>/TMEVoiceConfiguration

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<TMEVoiceConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, audio prompt control parameters, attr:version{req, string, protocolVersion}-->
  <voiceSpeed>
    <!--req, int, speed, range:[0,100]-->0
  </voiceSpeed>
  <voicePitch>
    <!--req, int, which is between 0 and 100, range:[0,100]-->0
  </voicePitch>
  <voiceVolum>
    <!--req, int, volume, range:[0,100]-->0
  </voiceVolum>
  <voiceRole>
    <!--req, int, speaker, desc:options: "3", "51", "52", "53", "54", "55"-->0
  </voiceRole>
  <voiceContent>
    <!--req, string, audio announcement content-->test
  </voiceContent>
  <voiceFileName>
    <!--req, string, audio file name-->test
  </voiceFileName>
  <voicePlate>
    <!--opt, bool, whether to enable audio prompt for license plates-->true
  </voicePlate>
</TMEVoiceConfiguration>
```


Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", 
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, which describes the error in details, desc:sub status code, which describes the error in details-->OK
  </subStatusCode>
</ResponseStatus>
```

11.6.1.2 Get total capability of entrance & exit station

Request URL

GET /ISAPI/Parking/TME/capabilities

Query Parameter

None

Request Message

None

Response Message


```xml
<?xml version="1.0" encoding="UTF-8"?>

<TMECap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the total capability of entrance & exit station, attr:version{req, string, protocolVersion}-->
  <isSupportTMEChargeRule>
    <!--ro, opt, bool, related URL: /ISAPI/Parking/channels/<ID>/TMEChargeRule/capabilities, desc:related URL: 
/ISAPI/Parking/channels/<ID>/TMEChargeRule/capabilities-->true
  </isSupportTMEChargeRule>
  <isSupportPaperCharge>
    <!--ro, opt, bool, whether the device supports configuring charging payment information, desc:related URL: 
/ISAPI/Parking/channels/<ID>/paperChargeInfo/capabilities-->true
  </isSupportPaperCharge>
  <isSupportParkingSpace>
    <!--ro, opt, bool, whether the device supports configuring the parking space information, desc:related URL: 
/ISAPI/Parking/channels/<ID>/parkingSpaceInfo/capabilities-->true
  </isSupportParkingSpace>
  <isSupportTMEOffline>
    <!--ro, opt, bool, whether the device supports configuring offline mode parameters, desc:related URL: 
/ISAPI/Parking/channels/<ID>/TMEOfflineConfiguration/capabilities-->true
  </isSupportTMEOffline>
  <isSupportTMEMultiCtrl>
    <!--ro, opt, bool, related URL: /ISAPI/Parking/channels/<ID>/TMEMultiCtrlConfiguration/capabilities, desc:related URL: 
/ISAPI/Parking/channels/<ID>/TMEMultiCtrlConfiguration/capabilities-->true
  </isSupportTMEMultiCtrl>
  <isSupportTMECFG>
    <!--ro, opt, bool, whether the device supports configuring entrance & exit station parameters, desc:related URL: 
/ISAPI/Parking/channels/<ID>/TMEConfiguration/capabilities-->true
  </isSupportTMECFG>
  <isSupportChargeAmount>
    <!--ro, opt, bool, whether the device supports charging account configurations, desc:related URL: 
/ISAPI/Parking/channels/<ID>/chargeAmount/capabilities-->true
  </isSupportChargeAmount>
  <isSupportVehicleCtrl>
    <!--ro, opt, bool, whether the device supports vehicle arming configurations, desc:related URL: 
/ISAPI/Parking/channels/<ID>/vehicleControl/capabilities-->true
  </isSupportVehicleCtrl>
  <isSupportBarrierGate>
    <!--ro, opt, bool, whether the device supports barrier gate control, desc:related URL: /ISAPI/Parking/channels/<ID>/barrierGate/capabilities-->true
  </isSupportBarrierGate>
  <isSupportTemporaryParkingCard>
    <!--ro, opt, bool, whether the device supports temporary cards-->true
  </isSupportTemporaryParkingCard>
  <isSupportLEDConfiguration>
    <!--ro, opt, bool, whether the device supports LED display parameter configuration, desc:related URL: 
/ISAPI/Parking/channels/<ID>/LEDConfiguration/capabilities-->true
  </isSupportLEDConfiguration>
  <isSupportVoiceBroadcastInfo>
    <!--ro, opt, bool, whether the device supports audio prompt control, desc:related URL: /ISAPI/Parking/channels/<ID>/voiceBroadcastInfo/capabilities--
>true
  </isSupportVoiceBroadcastInfo>
  <isSupportPaperPrintFormatInfo>
```

```
    <!--ro, opt, bool, whether the device supports configuring the ticket printing format, desc:related URL:
/ISAPI/Parking/channels/<ID>/paperPrintFormatInfo/capabilities-->true
  </isSupportPaperPrintFormatInfo>
  <isSupportLockGateConfig>
    <!--ro, opt, bool, whether the device supports intelligent barrier gate configurations, desc:related URL:
/ISAPI/Parking/channels/<ID>/lockGateConfiguration/capabilities-->true
  </isSupportLockGateConfig>
  <isSupportTMEVoiceConfiguration>
    <!--ro, opt, bool, whether the device supports configuring audio parameters, desc:related URL: /ISAPI/Parking/channels/<ID>/TMEVoiceConfiguration-->true
  </isSupportTMEVoiceConfiguration>
  <isSupportVoiceFileUpload>
    <!--ro, opt, bool, whether the device supports importing audio files, desc:related URL: /ISAPI/Parking/channels/<ID>/VoiceFileUpload/<NAME>-->true
  </isSupportVoiceFileUpload>
  <isSupportVoiceFileList>
    <!--ro, opt, bool, whether the device supports getting audio file list, desc:related URL: /ISAPI/Parking/channels/<ID>/VoiceFileList-->true
  </isSupportVoiceFileList>
  <isSupportRequestCardOpenBarrierOperationResult>
    <!--ro, opt, bool, whether the device supports searching for results of pass by swiping card, desc:POST
/ISAPI/Parking/TME/RequestCardOpenBarrierOperationResult?format=json-->true
  </isSupportRequestCardOpenBarrierOperationResult>
  <isSupportCardReaderTriggerParams>
    <!--ro, opt, bool, whether the device supports configuring card reader linkage parameters, desc:/ISAPI/Parking/TME/CardReaderTriggerParams?format=json--
>true
  </isSupportCardReaderTriggerParams>
  <isSupportSearchSwipeCardRecord>
    <!--ro, opt, bool, whether the device supports searching for card swiping records, desc:POST
/ISAPI/Parking/TME/SwipeCardRecordMgr/SearchSwipeCardRecord?format=json-->true
  </isSupportSearchSwipeCardRecord>
  <isSupportTMECardVehicleDelete>
    <!--ro, opt, bool-->true
  </isSupportTMECardVehicleDelete>
  <isSupportAddVehicleCardInfo>
    <!--ro, opt, bool, whether the device supports adding vehicle card information of entrance/exit, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/AddVehicleCardInfo?format=json-->true
  </isSupportAddVehicleCardInfo>
  <isSupportModifyVehicleCardInfo>
    <!--ro, opt, bool, whether the device supports editing vehicle card information of entrance/exit, desc:PUT
/ISAPI/Parking/TME/VehicleCardInfoMgr/ModifyVehicleCardInfo?format=json-->true
  </isSupportModifyVehicleCardInfo>
  <isSupportSearchCardInfoByVehicle>
    <!--ro, opt, bool, whether the device supports searching for vehicle card information by vehicle, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/SearchCardInfoByVehicle?format=json-->true
  </isSupportSearchCardInfoByVehicle>
  <isSupportSearchVehicleCardInfo>
    <!--ro, opt, bool, whether the device supports searching for vehicle card information, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/SearchVehicleCardInfo?format=json-->true
  </isSupportSearchVehicleCardInfo>
  <isSupportDeleteCardVehicleInfo>
    <!--ro, opt, bool, whether the device supports deleting vehicle card information of entrance/exit, desc:POST
/ISAPI/Parking/TME/VehicleCardInfoMgr/DeleteCardVehicleInfo?format=json-->true
  </isSupportDeleteCardVehicleInfo>
  <isSupportCreateNewVehicleCard>
    <!--ro, opt, bool, whether the device supports adding vehicle card of entrance/exit, desc:GET
/ISAPI/Parking/TME/VehicleCardInfoMgr/CreateNewVehicleCard?format=json-->true
  </isSupportCreateNewVehicleCard>
  <isSupportWriteEncryptCardCfg>
    <!--ro, opt, bool, whether the device supports encryption configuration for writing data to cards,
desc:/ISAPI/Parking/TME/VehicleCardInfoMgr/WriteEncryptCardCfg?format=json-->true
  </isSupportWriteEncryptCardCfg>
  <isSupportCustomPaperTicketPrintInfo>
    <!--ro, opt, bool, whether the device supports customizing ticket printing content, desc:/ISAPI/Parking/TME/CustomPaperTicketPrintInfo?format=json--
>true
  </isSupportCustomPaperTicketPrintInfo>
  <isSupportPaperPrintTest>
    <!--ro, opt, bool, whether the device supports ticket printing test, desc:POST /ISAPI/Parking/TME/PaperPrintTest?format=json-->true
  </isSupportPaperPrintTest>
</TMECap>
```

11.6.2 车辆出入显示屏幕

11.6.2.1 Get the configuration capability of LCD (Liquid Crystal Display) parameters

Request URL

GET /ISAPI/Parking/channels/<channelID>/LCD/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "LCDCap": {
    /*ro, req, object, LCD configuration capability*/
        "allowListDisplayEnabled": {
        /*ro, opt, object, whether to automatically display the custom content configured by the user whose vehicle is in the authorized list, desc:If
enabled, if the passing vehicle is in the authorized list, the device will automatically overwrite the custom content configured by the user. If not
enabled, the display content will not be changed.*/
            "@opt": [true, false],
            /*ro, req, array, range, subType:bool*/
            "@def":  false
            /*ro, opt, bool, default value*/
        },
        "blockListDisplayEnabled": {
        /*ro, opt, object, whether to automatically display the custom content configured by the user whose vehicle is in the unauthorized list, desc:If
enabled, if the passing vehicle is in the unauthorized list, the device will automatically overwrite the custom content configured by the user. If not
enabled, the display content will not be changed.*/
            "@opt": [true, false],
            /*ro, req, array, range, subType:bool*/
            "@def":  false
            /*ro, opt, bool, the default value*/
        },
        "temporaryListDisplayEnabled": {
        /*ro, opt, object, whether to automatically display the custom content configured by the user whose vehicle is a temporary vehicle, desc:If enabled,
if the passing vehicle is a temporary vehicle and is not in the unauthorized/authorized list, the device will automatically overwrite the custom content
configured by the user. If not enabled, the display content will not be changed.*/
            "@opt": [true, false],
            /*ro, req, array, range, subType:bool*/
            "@def":  false
            /*ro, opt, bool, the default value*/
        },
        "timeFormat": {
        /*ro, opt, object, time format*/
            "@opt": ["YYYY-MM-DD", "hh-mm-ss", "hh:mm:ss", "YYYY-MM-DD hh:mm:ss", "YYYY-MM-DD hh-mm-ss", "YYYY:MM:DD", "YYYY:MM:DD hh-mm-ss", "YYYY:MM:DD
hh:mm:ss"],
            /*ro, req, array, options, subType:string*/
            "#text":  "YYYY-MM-DD"
            /*ro, opt, string, example*/
        },
        "backgroundDisplayEnabled": {
        /*ro, opt, object, whether to display background picture, desc:If enabled, when a vehicle is passing, the passing vehicle information will be
displayed with the background picture.*/
            "@opt": [true, false]
            /*ro, req, array, options, subType:bool*/
        },
        "CustomContentList": {
        /*ro, opt, object, list of custom contents*/
            "maxSize":  8,
            /*ro, req, int, the maximum number of custom contents*/
            "CustomContent": {
            /*ro, opt, object, custom content*/
                "id": {
                /*ro, req, object, No., desc:starts from 1*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  8
                    /*ro, req, int, the maximum value*/
                },
                "content": {
                /*ro, opt, object, custom content*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  16
                    /*ro, req, int, the maximum value*/
                },
                "value": {
                /*ro, opt, object, options of display content*/
                    "@opt": ["plate", "parkingLotNum", "triggerTime", "vehicleType", "plateType", "vehicleCategory"],
                    /*ro, req, array, range, subType:string, desc:"plate", "parkingLotNum" (remaining parking space), "triggerTime" (vehicle passing time),
"vehicleType", "plateType", "vehicleCategory"*/
                    "#text":  "plate"
                    /*ro, opt, string, the sample value*/
                },
                "fontSizeLine": {
                /*ro, opt, object, font size*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  100
                    /*ro, req, int, the maximum value*/
                },
                "fontColorLine": {
                /*ro, opt, object, font color*/
                    "R": {
                    /*ro, req, object, R*/
                        "@min":  0,
                        /*ro, req, int, the minimum value*/
                        "@max":  255
                        /*ro, req, int, the maximum value*/
                    },
                    "G": {
                    /*ro, req, object, G*/
                        "@min":  0,
                        /*ro, req, int, the minimum value*/
```

```
                        /*ro, req, int, the minimum value*/
                        "@max":  255
                        /*ro, req, int, the maximum value*/
                    },
                    "B": {
                    /*ro, req, object, B*/
                        "@min":  0,
                        /*ro, req, int, the minimum value*/
                        "@max":  255
                        /*ro, req, int, the maximum value*/
                    }
                },
                "fontBoldLine": {
                /*ro, opt, object, whether to enable bold font*/
                    "@opt": [true, false],
                    /*ro, req, array, options, subType:bool*/
                    "@def":  false
                    /*ro, opt, bool, default value*/
                },
                "passVehicleDisplayEnabled": {
                /*ro, opt, object, whether to display the content when a vehicle is passing*/
                    "@opt": [true, false],
                    /*ro, req, array, options, subType:bool*/
                    "@def":  false
                    /*ro, opt, bool, default value*/
                }
            }
        },
        "fontSize": {
        /*ro, opt, object, font size*/
            "@min":  1,
            /*ro, req, int, the minimum value*/
            "@max":  100
            /*ro, req, int, the maximum value*/
        },
        "fontColor": {
        /*ro, opt, object, font color*/
            "R": {
            /*ro, req, object, R value*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  255
                /*ro, req, int, the maximum value*/
            },
            "G": {
            /*ro, req, object, G value*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  255
                /*ro, req, int, the maximum value*/
            },
            "B": {
            /*ro, req, object, B value*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  255
                /*ro, req, int, the maximum value*/
            }
        },
        "fontMovementSpeed": {
        /*ro, opt, object, moving speed of font*/
            "@min":  0,
            /*ro, req, int, the minimum value*/
            "@max":  30,
            /*ro, req, int, the maximum value*/
            "@def":  20
            /*ro, req, int, the default value*/
        },
        "MediaDataInfo": {
        /*ro, opt, object, settings of playing the media data*/
            "cycleIntervalTime": {
            /*ro, req, object, cycle display interval,unit: second, desc:unit: second*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  2
                /*ro, req, int, the maximum value*/
            },
            "cyclePlayOrder": {
            /*ro, req, object, playing order of cycle display*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  2
                /*ro, req, int, the maximum value*/
            }
        },
        "ctrlMode": {
        /*ro, opt, object, control mode,
        desc:It setting the mode is not supported, by default it is valid for both camera and platform.
        Camera (automatic control by device): The device recognizes vehicles in the block and authorized list and temporary license plates by itself, and
automatically displays the vehicle passing information according to the strategy configured in displayPassingVehicleInfoEnabled, allowListDisplayEnabled,
blockListDisplayEnabled, and temporaryListDisplayEnabled.
         Platform (manual control): Display the content in CustomContentList applied by the user. the mode, The automatic control strategy configured in
displayPassingVehicleInfoEnabled, allowListDisplayEnabled, blockListDisplayEnabled, and temporaryListDisplayEnabled will not take effect.
        Camera and platform: the above two control modes take effect at the same time. When CustomContentList is applied, its content is displayed
```

```
immediately; when the CustomContentList is not applied, it will automatically display the default content on the device.*/
            "@opt": ["camera", "platform", "cameraAndplatform"],
            /*ro, req, array, options, subType:string, desc:"camera", "platform", "cameraAndplatform" (takes effect in camera and platform mode)*/
            "#text":  "camera"
            /*ro, opt, string, example*/
        },
        "fanSpeedLevel": {
        /*ro, opt, object, fan speed level*/
            "@opt": ["low", "middle", "high", "close"],
            /*ro, req, array, option, subType:string*/
            "#text":  "low"
            /*ro, opt, string, example*/
        },
        "screenWidth":  1,
        /*ro, opt, int, screen width*/
        "screenHeight":  1,
        /*ro, opt, int, screen height*/
        "softwareVersion":  "test",
        /*ro, opt, string, Android version of the LCD*/
        "BackBright": {
        /*ro, opt, object, backlight brightness management*/
            "enabled": {
            /*ro, opt, object, whether to enable*/
                "@opt": [true, false]
                /*ro, req, array, option, subType:bool*/
            },
            "value": {
            /*ro, opt, object, brightness*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  100
                /*ro, req, int, the maximum value*/
            },
            "timeCtrlEnabled": {
            /*ro, opt, object, whether to enable adjustment by time period*/
                "@opt": [true, false]
                /*ro, req, array, option, subType:bool*/
            },
            "TimeCtrlList": {
            /*ro, opt, object, time period settings*/
                "maxSize":  8,
                /*ro, req, int, the maximum number of custom contents*/
                "TimeCtrl": {
                /*ro, opt, object, time period*/
                    "startTime":  "17:30:08",
                    /*ro, req, string, start time*/
                    "endTime":  "18:30:08",
                    /*ro, req, string, end time*/
                    "value": {
                    /*ro, opt, object, Brightness*/
                        "@min":  0,
                        /*ro, req, int, the minimum value*/
                        "@max":  100
                        /*ro, req, int, the maximum value*/
                    },
                }
            }
        },
        "displayTime": {
        /*ro, opt, object, display duration*/
            "@min":  1,
            /*ro, opt, int, the minimum value*/
            "@max":  300
            /*ro, opt, int, the maximum value*/
        },
        "MediaDataImport": {
        /*ro, opt, object, settings for importing media data*/
            "maxPictureNum":  10,
            /*ro, req, int, maximum number of pictures that can be imported*/
            "maxVideoNum":  10,
            /*ro, req, int, maximum number of videos that can be imported*/
            "pictureType": {
            /*ro, opt, object, picture type*/
                "@opt": ["backgroundPicture", "contentPicture", "logo"],
                /*ro, req, array, options, subType:string*/
                "#text":  "backgroundPicture"
                /*ro, opt, string, example*/
            }
        },
        "HintContentList": {
        /*ro, opt, object, list of prompt contents*/
            "maxSize":  2,
            /*ro, req, int, the maximum number of custom contents*/
            "HintContent": {
            /*ro, opt, object, prompt content*/
                "id": {
                /*ro, req, object, No.*/
                    "@min":  1,
                    /*ro, req, int, the minimum value*/
                    "@max":  2
                    /*ro, req, int, the maximum value*/
                },
                "content": {
```

```
                    /*ro, opt, object, prompt content*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  16
                        /*ro, opt, int, the maximum value*/
                    },
                    "value": {
                    /*ro, opt, object, options of display content*/
                        "@opt": ["plate", "parkingLotNum", "triggerTime", "vehicleType", "plateType", "vehicleCategory"],
                        /*ro, req, array, range, subType:string, desc:"plate", "parkingLotNum" (remaining parking space), "triggerTime" (vehicle passing time),
"vehicleType", "plateType", "vehicleCategory"*/
                        "#text":  "plate"
                        /*ro, opt, string, the sample value*/
                    },
                    "contentType": {
                    /*ro, opt, object, content type*/
                        "@opt": ["free", "passingVehicle"],
                        /*ro, req, array, option, subType:string*/
                        "#text":  "free"
                        /*ro, opt, string, example*/
                    },
                    "fontSizeLine": {
                    /*ro, opt, object, font size*/
                        "@min":  1,
                        /*ro, req, int, the minimum value*/
                        "@max":  100
                        /*ro, req, int, the maximum value*/
                    },
                    "fontColorLine": {
                    /*ro, opt, object, font color*/
                        "R": {
                        /*ro, req, object, R*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  255
                            /*ro, req, int, the maximum value*/
                        },
                        "G": {
                        /*ro, req, object, G*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  255
                            /*ro, req, int, the maximum value*/
                        },
                        "B": {
                        /*ro, req, object, B*/
                            "@min":  0,
                            /*ro, req, int, the minimum value*/
                            "@max":  255
                            /*ro, req, int, the maximum value*/
                        }
                    },
                    "fontBoldLine": {
                    /*ro, opt, object, whether to enable bold font*/
                        "@opt": [true, false],
                        /*ro, req, array, options, subType:bool*/
                        "@def":  true
                        /*ro, opt, bool, default value*/
                    }
                }
            },
            "parkingLotDisplayEnable": {
            /*ro, opt, object, whether to display vacant parking spaces*/
                "@opt": [true, false]
                /*ro, req, array, option, subType:bool*/
            },
            "displayPlateTypeEnable": {
            /*ro, opt, object, whether to display the license plate type*/
                "@opt": [true, false]
                /*ro, req, array, option, subType:bool*/
            },
            "powerOffSaveEnabled": {
            /*ro, opt, object, Whether that the parameters can be saved after power failure is supported in the URL,
            desc:Whether the parameters are saved after the power is off;
            Since the parameters need to be written to flash after power failure, if this node is not applied, the parameters will not be saved in flash; if the
value is false, the LCD will be restored to the default parameters after the power is off and be restarted.
            For example: /ISAPI/Parking/channels/<channelID>/LCD?format=json&powerOffSaveEnabled=true*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, opt, bool, default value*/
            },
            "mediaOccupyConfigListCap": {
            /*ro, opt, object, configure the display partition of passing vehicle information and the free scene of LCD*/
                "@size":  2,
                /*ro, opt, int, display the maximum number supported in the array of display partition*/
                "mediaOccupyMode": {
                /*ro, opt, object, the display mode of LCD media content and custom content*/
                    "@opt": ["mediaOccupy", "customOccupy", "mediaRatioOccupy"],
                    /*ro, req, array, options, subType:string, desc:"mediaOccupy" (only display media information which can be configured via
ISAPI/Parking/channels/<channelID>/LCD/mediaData/import), "customOccupy" (only display custom information configured in the CustomContentList),
"mediaRatioOccupy" (display according to the percentage)*/
                    "@def":  "mediaOccupy"
                    /*ro, opt, string, default value*/
```

```
                /*ro, opt, string, default value*/
            },
            "mediaOccupyRatio": {
            /*ro, opt, object, percentage of area occupied by the media content of the LCD screen*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  100
                /*ro, req, int, the maximum value*/
            },
            "contentType": {
            /*ro, opt, object, content type*/
                "@opt": ["free", "passingVehicle"],
                /*ro, req, array, options, subType:string*/
                "@def":  "free"
                /*ro, opt, string, default value*/
            }
        },
        "logoDisplayEnable": {
        /*ro, opt, object, whether to display the logo which can be configured via ISAPI/Parking/channels/<channelID>/LCD/mediaData/import*/
            "@opt": [true, false]
            /*ro, req, array, options, subType:bool*/
        },
    }
}
```


11.6.2.2 Get a piece of LCD (Liquid Crystal Display) media data

Request URL

GET /ISAPI/Parking/channels/<channelID>/LCD/mediaData/<FileID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
FileID	string	--

Request Message

None

Response Message


```
{
    "MediaData": {
    /*ro, req, object, media data*/
        "id":  1,
        /*ro, req, int, No., desc:it starts from 1 and corresponds to the ID returned after the media data is imported*/
        "dataType":  "picture",
        /*ro, req, enum, type of the imported data, subType:string, desc:"picture", "video"*/
        "name":  "test",
        /*ro, req, string, file name, range:[1,64]*/
        "pictureType":  "backgroundPicture",
        /*ro, opt, enum, picture type, subType:string, desc:"backgroundPicture" (background picture), "contentPicture" (content picture)*/
        "size":  1
        /*ro, opt, int, file size, range:[1,1024000], desc:unit: KB*/
    }
}
```


11.6.2.3 Delete a piece of LCD (Liquid Crystal Display) media data

Request URL

DELETE /ISAPI/Parking/channels/<channelID>/LCD/mediaData/<FileID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
FileID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it corresponds to subStatusCode when statusCode is not 1*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

11.6.2.4 Import the LCD (Liquid Crystal Display) media data

Request URL

POST /ISAPI/Parking/channels/<channelID>/LCD/mediaData/import?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MediaDataInfo": {
    /*req, object*/
        "dataType":  "picture",
        /*req, enum, type of the imported data: "picture", "video", subType:string, desc:"picture", "video"*/
        "pictureType":  "backgroundPicture"
        /*opt, enum, picture type, subType:string, desc:"backgroundPicture" (background picture), "contentPicture" (content picture), "logo" (logo
picture)*/
    }
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
MediaDataInfo	[Message content]	application/json	--	--	--
picture	[Binary picture data]	image/jpeg	picture	图片	--
video	[Stream data]	application/octet-stream		视频	--

Note： The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

Response Message

```
{
    "id":  1
    /*ro, req, int, media data ID*/
}
```

11.6.2.5 Get all LCD (Liquid Crystal Display) media data

Request URL

GET /ISAPI/Parking/channels/<channelID>/LCD/mediaData?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "MediaDataList": [
    /*ro, req, array, media data list, subType:object*/
        {
            "MediaData": {
            /*ro, opt, object, media data*/
                "id":  1,
                /*ro, req, int, No., desc:it starts from 1 and corresponds to the ID returned after the media data is imported*/
                "dataType":  "picture",
                /*ro, req, enum, type of the imported data: "picture", "video", subType:string, desc:"picture", "video"*/
                "name":  "test",
                /*ro, req, string, file name, range:[1,64]*/
                "pictureType":  "backgroundPicture",
                /*ro, opt, enum, picture type, subType:string, desc:"backgroundPicture" (background picture), "contentPicture" (content picture)*/
                "size":  1
                /*ro, opt, int, data size, range:[1,1024000], desc:unit: KB*/
            }
        }
    ]
}
```

11.6.2.6 Delete all LCD (Liquid Crystal Display) media data

Request URL

DELETE /ISAPI/Parking/channels/<channelID>/LCD/mediaData?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this field is required when the value of statusCode is not 1*/
}
```

11.6.2.7 Set the LCD (Liquid Crystal Display) parameters.

Request URL

PUT /ISAPI/Parking/channels/<channelID>/LCD?format=json&powerOffSaveEnabled=<powerOffSaveEnabled>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
powerOffSaveEnabled	string	Whether the parameters are saved after the power is off;Since the parameters need to be written to flash after power failure, if this node is not applied, the parameters will not be saved in flash; if the value is false, the LCD will be restored to the default parameters after the power is off and be restarted. For example: /ISAPI/Parking/channels//LCD?format=json&powerOffSaveEnabled=true

Request Message

```
{
    "LCD": {
    /*req, object, LCD parameters*/
        "displayPassingVehicleInfoEnabled":  true,
        /*opt, bool, whether to enable the display of passing vehicle information*/
        "allowListDisplayEnabled":  false,
        /*opt, bool, whether to automatically display the custom content configured by the user whose vehicle is in the authorized list, desc:If enabled, if the passing vehicle is in the authorized list, the device will automatically overwrite the custom content configured by the user. If not enabled, the display content will not be changed.*/
        "blockListDisplayEnabled":  false,
        /*opt, bool, whether to automatically display the custom content configured by the user whose vehicle is in the unauthorized list, desc:If enabled, if the passing vehicle is in the unauthorized list, the device will automatically overwrite the custom content configured by the user. If not enabled, the display content will not be changed.*/
        "temporaryListDisplayEnabled":  false,
        /*opt, bool, whether to automatically display the custom content configured by the user whose vehicle is a temporary vehicle, desc:If enabled, if the passing vehicle is a temporary vehicle and is not in the unauthorized/authorized list, the device will automatically overwrite the custom content configured by the user. If not enabled, the display content will not be changed.*/
        "displayPlateEnable":  true,
        /*opt, bool, whether to display the license plate*/
        "displayTimeEnable":  true,
        /*opt, bool, whether to enable time display*/
        "timeFormat":  "YYYY-MM-DD",
        /*opt, enum, time display format, subType:string, desc:"YYYY-MM-DD", "hh-mm-ss"*/
        "backgroundDisplayEnabled":  true,
        /*opt, bool, whether to display background picture, desc:If enabled, when a vehicle is passing, the passing vehicle information will be displayed with the background picture*/
        "backgroundDisplayTime":  1,
        /*opt, int, display duration of the background picture, range:[1,180], unit:s, desc:If another vehicle is passing, the duration will be refreshed*/
        "CustomContentList": [
        /*opt, array, custom content list, subType:object, range:[1,8]*/
            {
                "CustomContent": {
                /*opt, object, custom content*/
                    "id":  1,
                    /*req, int, No., range:[1,8], desc:it increases from 1*/
                    "content":  "test",
                    /*opt, string, custom content, range:[1,16]*/
                    "value":  "plate",
                    /*opt, enum, options of display content, subType:string, desc:“plate”, “parkingLotNum” (remaining parking space), “triggerTime” (vehicle passing time), “vehicleType”, “plateType”, “vehicleCategory”*/
                    "contentType":  "advertisement",
                    /*opt, enum, content type, subType:string, desc:“advertisement","redLight","greenLight"*/
                    "fontSizeLine":  1,
                    /*opt, int, font size, range:[1,100]*/
                    "fontColorLine": {
                    /*opt, object, font color*/
                        "R":  1,
                        /*req, int, R, range:[0,255]*/
                        "G":  1,
                        /*req, int, G, range:[0,255]*/
                        "B":  1
                        /*req, int, B, range:[0,255]*/
                    },
                    "fontTypeLine":  "Microsoft YaHei",
                    /*opt, enum, font type, subType:string, desc:“Microsoft YaHei”, “SimSun”, “SimHei”, “FangSong”, “KaiTi”*/
                    "fontBoldLine":  false,
                    /*opt, bool, whether to enable bold font*/
                    "passVehicleDisplayEnabled":  false
                    /*opt, bool, whether to display the content when a vehicle is passing, desc:true (display the content when a vehicle is passing), false (display the content for free time), it is false by default.*/
                }
            }
        ],
        "fontSize":  1,
        /*opt, int, font size, range:[1,100]*/
        "fontColor": {
        /*opt, object, font color*/
            "R":  1,
            /*req, int, R, range:[0,255]*/
            "G":  1,
            /*req, int, G, range:[0,255]*/
```

```
            "B":  1
            /*req, int, B, range:[0,255]*/
        },
        "fontMovementSpeed":  20,
        /*opt, int, moving speed threshold of font, range:[0,30], desc:default value: 20*/
        "MediaDataInfo": {
        /*opt, object, settings of playing the media data*/
            "cycleIntervalTime":  1,
            /*req, int, cycle display interval, unit:s*/
            "cyclePlayOrder": [1, 3, 4]
            /*req, array, playing order of cycle display, subType:int*/
        },
        "ctrlMode":  "cameraAndplatform",
        /*opt, enum, control mode, subType:string,
        desc:"camera", "platform", "cameraAndplatform" (takes effect in camera and platform mode); If setting the mode is not supported, by default it is
valid for both camera and platform.
        Camera (automatic control by device): The device recognizes vehicles in the block and authorized list and temporary license plates by itself, and
automatically displays the vehicle passing information according to the strategy configured in displayPassingVehicleInfoEnabled, allowListDisplayEnabled,
blockListDisplayEnabled, and temporaryListDisplayEnabled.
         Platform (manual control): Display the content in CustomContentList applied by the user. the mode, The automatic control trategy configured in
displayPassingVehicleInfoEnabled, allowListDisplayEnabled, blockListDisplayEnabled, and temporaryListDisplayEnabled will not take effect.
        Camera and platform: the above two control modes take effect at the same time. When CustomContentList is spplied, its content is displayed
immediately; when the CustomContentList is not applied, it will automatically display the default content on the device.*/
        "fanSpeedLevel":  "low",
        /*opt, enum, fan speed, subType:string, desc:"low", "middle", "high", "close"*/
        "temperatureThreshold":  1,
        /*opt, int, screen temperature threshold, range:[0,100], unit:℃*/
        "screenWidth":  1,
        /*ro, opt, int, screen width*/
        "screenHeight":  1,
        /*ro, opt, int, screen height*/
        "softwareVersion":  "test",
        /*ro, opt, string, Android version of the LCD*/
        "BackBright": {
        /*opt, object, backlight brightness management*/
            "enabled":  true,
            /*opt, bool, whether to enable the function*/
            "value":  1,
            /*opt, int, brightness*/
            "timeCtrlEnabled":  true,
            /*opt, bool, whether to enable adjustment by time period*/
            "TimeCtrlList": [
            /*opt, array, time period configuration, subType:object, range:[1,8]*/
                {
                    "TimeCtrl": {
                    /*opt, object, time period*/
                        "startTime":  "17:30:08",
                        /*req, time, start time*/
                        "endTime":  "18:30:08",
                        /*req, time, end time*/
                        "value":  1,
                        /*req, int, brightness, range:[0,100]*/
                        "vehiclePassingValue":  1
                        /*opt, int, passing vehicle brightness, range:[0,100], desc:When a vehicle passes, the brightness is the value of this node; if no
vehicle passes, the brightness is the value of the node value. If configuring this node is not supported, the brightness is the value of the node value*/
                    }
                }
            ]
        },
        "SceneUIList": [
        /*opt, array, UI template management of multiple scenes, subType:object*/
            {
                "SceneUI": {
                /*opt, object, ID of scene UI template*/
                    "scenemode":  "entranceWithPlate",
                    /*req, enum, scene type, subType:string, desc:"entranceWithPlate" (UI template type for entered vehicles with license plate),
"entranceWithoutPlate" (UI template type for entered vehicles without license plate), "exitingWithoutCharge" (UI template type for exited vehicles with
unpaid fee), "exitingWithCharge" (UI template type for exited vehicles without unpaid fee), "free" (free UI template type)*/
                    "UIID":  1
                    /*req, int, ID of scene UI template*/
                }
            }
        ],
        "displayTime":  1,
        /*opt, int, display duration, range:[1,300]*/
        "HintContentList": [
        /*opt, array, list of prompt contents, subType:object, range:[1,2]*/
            {
                "HintContent": {
                /*opt, object, prompt content*/
                    "id":  1,
                    /*req, int, No., range:[1,2], desc:it increases from 1*/
                    "content":  "test",
                    /*opt, string, custom content, range:[1,16]*/
                    "value":  "plate",
                    /*opt, enum, options of display content, subType:string, desc:"plate", "parkingLotNum" (remaining parking space), "triggerTime" (vehicle
passing time), "vehicleType", "plateType", "vehicleCategory"*/
                    "contentType":  "free",
                    /*opt, enum, content type, subType:string, desc:"free", "passingVehicle"*/
                    "fontSizeLine":  1,
                    /*opt, int, font size, range:[1,100]*/
                    "fontColorLine": {
                    /*opt, object, font color*/
```

```
                        "R":  1,
                        /*req, int, R, range:[0,255]*/
                        "G":  1,
                        /*req, int, G, range:[0,255]*/
                        "B":  1
                        /*req, int, B, range:[0,255]*/
                    },
                    "fontTypeLine":  "Microsoft YaHei",
                    /*opt, enum, font type, subType:string, desc:“Microsoft YaHei”, “SimSun”, “SimHei”, “FangSong”, “KaiTi”*/
                    "fontBoldLine":  true
                    /*opt, bool, whether to enable bold font*/
                }
            }
        ],
        "parkingLotDisplayEnable":  true,
        /*opt, bool, whether to enable the display of vacant parking spaces*/
        "sceneMode":  "entryScene",
        /*opt, enum, scene mode, subType:string, desc:"entryScene", "exitScene", "normalScene"*/
        "displayPlateTypeEnable":  true,
        /*opt, bool, whether to enable the display of the license plate type*/
        "displayValidityPeriodEnable":  true,
        /*opt, bool, whether to enable the display of validity period*/
        "logoDisplayEnable":  true,
        /*opt, bool, whether to display the logo which can be configured via ISAPI/Parking/channels/<channelID>/LCD/mediaData/import*/
        "mediaOccupyConfigList": [
        /*ro, opt, array, parameters of configuring the area for displaying the content when a vehicle is displaying and when it is free time on the LCD screen, subType:object*/
            {
                "mediaOccupyMode":  "mediaOccupy",
                /*ro, opt, enum, display mode of media information and custom information on LCD screen, subType:string, desc:“mediaOccupy” (only display media information), “customOccupy” (only display custom information configured in the CustomContentList), ”mediaRatioOccupy” (display according to the percentage); “mediaOccupy” (only display media information which can be configured via ISAPI/Parking/channels/<channelID>/LCD/mediaData/import), “customOccupy” (only display custom information configured in the CustomContentList), “mediaRatioOccupy” (display according to the percentage)*/
                "mediaOccupyRatio":  0,
                /*ro, opt, int, percentage of area occupied by the media content of the LCD screen, range:[0,100]*/
                "contentType":  "free"
                /*ro, opt, enum, content type, subType:string, desc:“free”, “passingVehicle”*/
            }
        ],
        "SceneUI": {
        /*opt, object, UI template of multiple scenes*/
            "id":  1,
            /*req, int, valid template ID, desc:1, 2, 3*/
            "RegionList": [
            /*opt, array, template area list, subType:object*/
                {
                    "Region": {
                    /*opt, object, area*/
                        "id":  1,
                        /*req, int, No., desc:it increases from 1*/
                        "trafficLightEnabled":  true,
                        /*opt, bool, whether to switch the traffic light*/
                        "contentEnabled":  true,
                        /*opt, bool, whether to enable the advertisement text*/
                        "pictureEnabled":  true,
                        /*opt, bool, whether to enable the advertisement picture*/
                        "videoEnabled":  true,
                        /*opt, bool, whether to enable the advertisement video*/
                        "time":  1,
                        /*opt, int, display interval, unit:s*/
                        "previewEnabled":  true,
                        /*opt, bool, live view*/
                        "localTimeEnabled":  true
                        /*opt, bool, whether to display the local time*/
                    }
                }
            ]
        },
        "characterMovementMode":  "scrolling"
        /*opt, enum, subType:string*/
    }
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this node is required when the value of statusCode is not 1*/
}
```

11.6.2.8 Get the LCD (Liquid Crystal Display) parameters

Request URL

GET /ISAPI/Parking/channels/<channelID>/LCD?format=json&powerOffSaveEnabled=<powerOffSaveEnabled>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
powerOffSaveEnabled	string	Whether the parameters are saved after the power is off;Since the parameters need to be written to flash after power failure, if this node is not applied, the parameters will not be saved in flash; if the value is false, the LCD will be restored to the default parameters after the power is off and be restarted. For example: /ISAPI/Parking/channels//LCD?format=json&powerOffSaveEnabled=true

Request Message

None

Response Message

```
{
    "LCD": {
    /*ro, req, object, LCD*/
        "displayPassingVehicleInfoEnabled":  true,
        /*ro, opt, bool, whether to enable the display of passing vehicle information*/
        "allowListDisplayEnabled":  false,
        /*ro, opt, bool, whether to automatically display the custom content configured by the user whose vehicle is in the authorized list, desc:If enabled, if the passing vehicle is in the authorized list, the device will automatically overwrite the custom content configured by the user. If not enabled, the display content will not be changed.*/
        "blockListDisplayEnabled":  false,
        /*ro, opt, bool, whether to automatically display the custom content configured by the user whose vehicle is in the unauthorized list, desc:If enabled, if the passing vehicle is in the unauthorized list, the device will automatically overwrite the custom content configured by the user. If not enabled, the display content will not be changed.*/
        "temporaryListDisplayEnabled":  false,
        /*ro, opt, bool, whether to automatically display the custom content configured by the user whose vehicle is a temporary vehicle, desc:If enabled, if the passing vehicle is a temporary vehicle and is not in the unauthorized/authorized list, the device will automatically overwrite the custom content configured by the user. If not enabled, the display content will not be changed.*/
        "displayPlateEnable":  true,
        /*ro, opt, bool, whether to display the license plate*/
        "displayTimeEnable":  true,
        /*ro, opt, bool, whether to enable time display*/
        "timeFormat":  "YYYY-MM-DD",
        /*ro, opt, enum, time display format, subType:string, desc:time format: "YYYY-MM-DD","hh-mm-ss"*/
        "backgroundDisplayEnabled":  true,
        /*ro, opt, bool, whether to display background picture, desc:If enabled, when a vehicle is passing, the passing vehicle information will be displayed with the background picture.*/
        "backgroundDisplayTime":  60,
        /*ro, opt, int, display duration of the background picture, range:[1,180], unit:s, desc:If another vehicle is passing, the duration will be refreshed.*/
        "CustomContentList": [
        /*ro, opt, array, custom content list, subType:object, range:[1,8]*/
            {
                "CustomContent": {
                /*ro, opt, object, custom content*/
                    "id":  1,
                    /*ro, req, int, No., range:[1,8], desc:it increases from 1*/
                    "content":  "test",
                    /*ro, opt, string, custom content, range:[1,16]*/
                    "value":  "plate",
                    /*ro, opt, enum, options of display content, subType:string, desc:“plate”, “parkingLotNum” (remaining parking space), “triggerTime” (vehicle passing time), “vehicleType”, “plateType”, “vehicleCategory”*/
                    "contentType":  "advertisement",
                    /*ro, opt, enum, content type, subType:string, desc:"advertisement", "redLight", "greenLight"*/
                    "fontSizeLine":  1,
                    /*ro, opt, int, font size, range:[1,100]*/
                    "fontColorLine": {
                    /*ro, opt, object, font color*/
                        "R":  1,
                        /*ro, req, int, R, range:[0,255]*/
                        "G":  1,
                        /*ro, req, int, G, range:[0,255]*/
                        "B":  1
                        /*ro, req, int, B, range:[0,255]*/
                    },
                    "fontTypeLine":  "Microsoft YaHei",
                    /*ro, opt, enum, font type, subType:string, desc:“Microsoft YaHei”, “SimSun”, “SimHei”, “FangSong”, “KaiTi”*/
                    "fontBoldLine":  false,
                    /*ro, opt, bool, whether to enable bold font*/
```

```
                    "passVehicleDisplayEnabled":  false
                    /*ro, opt, bool, whether to display the content when a vehicle is passing, desc:true (display the content when a vehicle is passing), false (display the content for free time), it is false by default.*/
                }
            }
        ],
        "fontSize":  1,
        /*ro, opt, int, font size, range:[1,100]*/
        "fontColor": {
        /*ro, opt, object, font color*/
            "R":  1,
            /*ro, req, int, R, range:[0,255]*/
            "G":  1,
            /*ro, req, int, G, range:[0,255]*/
            "B":  1
            /*ro, req, int, B, range:[0,255]*/
        },
        "fontMovementSpeed":  20,
        /*ro, opt, int, moving speed threshold of font, range:[0,30], desc:default value: 20*/
        "MediaDataInfo": {
        /*ro, opt, object, settings of playing the media data*/
            "cycleIntervalTime":  1,
            /*ro, req, int, cycle display interval, unit:s*/
            "cyclePlayOrder": [1, 3, 4]
            /*ro, req, array, playing order of cycle display, subType:int*/
        },
        "ctrlMode":  "cameraAndplatform",
        /*ro, opt, enum, control mode, subType:string,
        desc:“camera”, “platform”, “cameraAndplatform” (takes effect in camera and platform mode); If setting the mode is not supported, by default it is valid for both camera and platform.
        Camera (automatic control by device): The device recognizes vehicles in the block and authorized list and temporary license plates by itself, and automatically displays the vehicle passing information according to the strategy configured in displayPassingVehicleInfoEnabled, allowListDisplayEnabled, blockListDisplayEnabled, and temporaryListDisplayEnabled.
         Platform (manual control): Display the content in CustomContentList applied by the user. the mode, The automatic control trategy configured in displayPassingVehicleInfoEnabled, allowListDisplayEnabled, blockListDisplayEnabled, and temporaryListDisplayEnabled will not take effect.
        Camera and platform: the above two control modes take effect at the same time. When CustomContentList is spplied, its content is displayed immediately; when the CustomContentList is not applied, it will automatically display the default content on the device.*/
        "fanSpeedLevel":  "low",
        /*ro, opt, enum, fan speed gear, subType:string, desc:fan speed level: "low", "middle", "high", "close"*/
        "temperatureThreshold":  1,
        /*ro, opt, int, screen temperature threshold, range:[0,100], unit:℃*/
        "screenWidth":  1,
        /*ro, opt, int, screen width*/
        "screenHeight":  1,
        /*ro, opt, int, screen height*/
        "softwareVersion":  "test",
        /*ro, opt, string, Android version of the LCD*/
        "BackBright": {
        /*ro, opt, object, backlight brightness management*/
            "enabled":  true,
            /*ro, opt, bool, whether to enable the function*/
            "value":  1,
            /*ro, opt, int, brightness*/
            "timeCtrlEnabled":  true,
            /*ro, opt, bool, whether to enable adjustment by time period*/
            "TimeCtrlList": [
            /*ro, opt, array, Time Period Configuration, subType:object, range:[1,8]*/
                {
                    "TimeCtrl": {
                    /*ro, opt, object, time period*/
                        "startTime":  "17:30:08",
                        /*ro, req, time, start time*/
                        "endTime":  "18:30:08",
                        /*ro, req, time, end time*/
                        "value":  1,
                        /*ro, req, int, brightness, range:[0,100]*/
                        "vehiclePassingValue":  1
                        /*ro, opt, int, passing vehicle brightness, range:[0,100], desc:When a vehicle passes, the brightness is the value of this node; if no vehicle passes, the brightness is the value of the node value. If configuring this node is not supported, the brightness is the value of the node value*/
                    }
                }
            ]
        },
        "SceneUIList": [
        /*ro, opt, array, UI template management of multiple scenes, subType:object*/
            {
                "SceneUI": {
                /*ro, opt, object, ID of scene UI template*/
                    "scenemode":  "entranceWithPlate",
                    /*ro, req, enum, scene type, subType:string, desc:"entranceWithPlate" (UI template type for entered vehicles with license plate), "entranceWithoutPlate" (UI template type for entered vehicles without license plate), "exitingWithoutCharge" (UI template type for exited vehicles with unpaid fee), "exitingWithCharge" (UI template type for exited vehicles without unpaid fee), "free" (free UI template type)*/
                    "UIID":  1
                    /*ro, req, int, ID of scene UI template*/
                }
            }
        ],
        "displayTime":  1,
        /*ro, opt, int, display duration, range:[1,300]*/
        "HintContentList": [
        /*ro, opt, array, list of prompt contents, subType:object, range:[1,2]*/
            {
                "HintContent": {
```

```
                /*ro, opt, object, prompt content*/
                    "id":  1,
                    /*ro, req, int, No., range:[1,2], desc:it increases from 1*/
                    "content":  "test",
                    /*ro, opt, string, prompt content, range:[1,16]*/
                    "value":  "plate",
                    /*ro, opt, enum, options of display content, subType:string, desc:"plate", "parkingLotNum" (remaining parking space), "triggerTime"
(vehicle passing time), "vehicleType", "plateType", "vehicleCategory"*/
                    "contentType":  "free",
                    /*ro, opt, enum, content type, subType:string, desc:"advertisement", "redLight", "greenLight"*/
                    "fontSizeLine":  1,
                    /*ro, opt, int, font size, range:[1,100]*/
                    "fontColorLine": {
                    /*ro, opt, object, font color*/
                        "R":  1,
                        /*ro, req, int, R, range:[0,255]*/
                        "G":  1,
                        /*ro, req, int, G, range:[0,255]*/
                        "B":  1
                        /*ro, req, int, B, range:[0,255]*/
                    },
                    "fontTypeLine":  "Microsoft YaHei",
                    /*ro, opt, enum, font type, subType:string, desc:"Microsoft YaHei", "SimSun", "SimHei", "FangSong", "KaiTi"*/
                    "fontBoldLine":  true
                    /*ro, opt, bool, whether to enable bold font*/
                }
            }
        ],
        "parkingLotDisplayEnable":  true,
        /*ro, opt, bool, whether to enable the display of vacant parking spaces*/
        "sceneMode":  "entryScene",
        /*ro, opt, enum, scene mode, subType:string, desc:"entryScene","exitScene","normalScene"*/
        "displayPlateTypeEnable":  true,
        /*ro, opt, bool, whether to enable the display of the license plate type*/
        "displayValidityPeriodEnable":  true,
        /*ro, opt, bool, whether to enable the display of validity period*/
        "mediaOccupyConfigList": [
        /*ro, opt, array, parameters of configuring the area for displaying the content when a vehicle is displaying and when it is free time on the LCD
screen, subType:object*/
            {
                "mediaOccupyMode":  "mediaOccupy",
                /*ro, opt, enum, display mode of media information and custom information on LCD screen, subType:string, desc:"mediaOccupy" (only display
media information), "customOccupy" (only display custom information configured in the CustomContentList), "mediaRatioOccupy" (display according to the
percentage); "mediaOccupy" (only display media information which can be configured via ISAPI/Parking/channels/<channelID>/LCD/mediaData/import),
"customOccupy" (only display custom information configured in the CustomContentList), "mediaRatioOccupy" (display according to the percentage)*/
                "mediaOccupyRatio":  0,
                /*ro, opt, int, percentage of area occupied by the media content of the LCD screen, range:[0,100]*/
                "contentType":  "free"
                /*ro, opt, enum, content type, subType:string, desc:"free", "passingVehicle"*/
            }
        ],
        "logoDisplayEnable":  true,
        /*ro, opt, bool, whether to display the logo which can be configured via ISAPI/Parking/channels/<channelID>/LCD/mediaData/import*/
        "SceneUI": {
        /*ro, opt, object, object,UI template of multiple scenes*/
            "id":  1,
            /*ro, req, int, No. which starts from 1, value range: [1,4], desc:1, 2, 3*/
            "RegionList": [
            /*ro, opt, array, template area list, subType:object*/
                {
                    "Region": {
                    /*ro, opt, object, area*/
                        "id":  1,
                        /*ro, req, int, No., desc:it increases from 1*/
                        "trafficLightEnabled":  true,
                        /*ro, opt, bool, whether to switch the traffic light*/
                        "contentEnabled":  true,
                        /*ro, opt, bool, advertisement text*/
                        "pictureEnabled":  true,
                        /*ro, opt, bool, advertisement picture*/
                        "videoEnabled":  true,
                        /*ro, opt, bool, advertisement video*/
                        "time":  1,
                        /*ro, opt, int, display interval, unit:s*/
                        "previewEnabled":  true,
                        /*ro, opt, bool, live view*/
                        "localTimeEnabled":  true
                        /*ro, opt, bool, display local time*/
                    }
                }
            ]
        },
        "characterMovementMode":  "scrolling"
        /*ro, opt, enum, subType:string*/
    }
}
```


11.6.2.9 Get the configuration capability of LED screen display

Request URL

GET /ISAPI/Parking/channels/<channelID>/LEDConfiguration/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<LEDConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, the configuration capability of LED screen display, attr:version{req, string, protocolVersion}-->
  <information min="0" max="512">
    <!--ro, opt, string, LED screen display content, attr:min{req, int},max{req, int}-->test
  </information>
  <displayMode opt="left,right,immediate">
    <!--ro, opt, enum, display mode, subType:string, attr:opt{req, string}, desc:display mode: "left"-move left,"right"-move right,"immediate"-display
immediately-->left
  </displayMode>
  <speedType opt="fast,medium,slow">
    <!--ro, opt, enum, speed type, subType:string, attr:opt{req, string}, desc:"fast", "medium", "slow"-->fast
  </speedType>
  <showTime min="1" max="300">
    <!--ro, req, int, display duration, range:[1,300], unit:s, attr:min{req, int},max{req, int}-->1
  </showTime>
  <showPlate>
    <!--ro, opt, bool, whether to display temporary license plate-->true
  </showPlate>
  <ledCount min="1" max="10">
    <!--ro, opt, int, number of LED screens. An LED screen is a logic unit which can be controlled independently, attr:min{req, int},max{req, int}, desc:the
default value is 1-->1
  </ledCount>
  <fontSize min="1" max="10">
    <!--ro, opt, int, font size, attr:min{req, int},max{req, int}-->1
  </fontSize>
  <fontColor min="0" max="10">
    <!--ro, opt, int, RGB value of the font color, attr:min{req, int},max{req, int}-->1
  </fontColor>
  <laneID min="1" max="32">
    <!--ro, opt, string, range:[1,32], attr:min{req, int},max{req, int}-->1
  </laneID>
  <passingIconType opt="passing,noPassing">
    <!--ro, opt, enum, subType:string, attr:opt{req, string}-->passing
  </passingIconType>
  <QRCodeInfo>
    <!--ro, opt, object-->
    <QRCode min="1" max="128">
      <!--ro, req, string, range:[1,128], attr:min{req, int},max{req, int}-->test
    </QRCode>
    <QRSize min="12" max="256">
      <!--ro, opt, int, range:[12,256], attr:min{req, int},max{req, int}-->16
    </QRSize>
    <QRColor opt="red,green,yellow,blue,cyan,purple,white">
      <!--ro, opt, enum, subType:string, attr:opt{req, string}-->green
    </QRColor>
  </QRCodeInfo>
</LEDConfiguration>
```

11.6.2.10 Set the display parameters of the LED screen

Request URL

PUT /ISAPI/Parking/channels/<channelID>/LEDConfiguration?id=<netScreenID>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
netScreenID	string	--

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<LEDConfiguration xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <id>
    <!--opt, int-->1
  </id>
  <information>
    <!--req, string-->test
  </information>
  <displayMode>
    <!--req, enum, subType:string-->left
  </displayMode>
  <speedType>
    <!--req, enum, subType:string-->fast
  </speedType>
  <showTime>
    <!--req, int, range:[1,60], unit:s-->1
  </showTime>
  <showPlate>
    <!--opt, bool-->true
  </showPlate>
  <fontSize>
    <!--opt, int-->1
  </fontSize>
  <fontColor>
    <!--opt, int-->1
  </fontColor>
</LEDConfiguration>
```

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, subType:int-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, subType:string-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string-->OK
  </subStatusCode>
</ResponseStatus>
```


11.6.2.11 Set the parameters for displaying a specified scene on the LED screen

Request URL

PUT /ISAPI/Parking/channels/<channelID>/LEDConfigurations/multiScene/<SID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--
SID	string	--

Request Message

```
{
    "SingleSceneLEDConfigurations": {
    /*opt, object*/
        "sid":  1,
        /*req, int*/
        "mode":  "passingVehicle",
        /*req, enum, subType:string*/
        "showFreeEnabled":  true,
        /*opt, bool, dep:and,{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
        "displayTime":  1,
        /*opt, int, range:[1,300], unit:s, dep:and,{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
        "LEDConfigurationList": [
        /*opt, array, subType:object*/
            {
                "LEDConfiguration": {
                /*opt, object*/
                    "id":  1,
                    /*req, int*/
                    "enabled":  true,
                    /*req, bool*/
                    "ShowInfoList": [
                    /*opt, array, subType:object*/
                        {
                            "ShowInfo": {
                            /*opt, object*/
                                "id":  1,
                                /*req, int*/
                                "fontSize":  16,
                                /*opt, enum, subType:int*/
                                "fontColor":  "red",
                                /*opt, enum, subType:string*/
                                "speedType":  "fast",
                                /*req, enum, subType:string*/
                                "displayMode":  "left",
                                /*req, enum, subType:string*/
                                "LineInfoList": [
                                /*opt, array, subType:object*/
                                    {
                                        "LineInfo": {
                                        /*opt, object*/
                                            "id":  1,
                                            /*req, int*/
                                            "value":  "plate",
                                            /*opt, enum, subType:string*/
                                            "customValue":  "欢迎光临"
                                            /*opt, string*/
                                        }
                                    }
                                ]
                            }
                        }
                    ]
                }
            }
        ],
    }
}
```


Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int*/
    "statusString":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "subStatusCode":  "ok",
    /*ro, opt, string, range:[1,64]*/
    "errorCode":  1,
    /*ro, opt, int*/
    "errorMsg":  "ok"
    /*ro, opt, string*/
}
```

11.6.2.12 Get the parameters for displaying a specified scene on the LED screen

Request URL

GET /ISAPI/Parking/channels/<channelID>/LEDConfigurations/multiScene/<SID>?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.
SID	string	scene No.

Request Message

None

Response Message

```
{
    "SingleSceneLEDConfigurations": {
    /*ro, opt, object, the parameters for displaying a specified scene on the LED screen*/
        "sid":  1,
        /*ro, req, int, scene ID*/
        "mode":  "passingVehicle",
        /*ro, req, enum, scene mode, subType:string, desc:“passingVehicle”, “noVehicle”*/
        "showFreeEnabled":  true,
        /*ro, opt, bool, whether to enable displaying in free time, dep:and,
{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
        "displayTime":  1,
        /*ro, opt, int, display duration, range:[1,300], unit:s, dep:and,
{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
        "vehicleDisplayEnabled":  true,
        /*ro, opt, bool, whether enable automatic displaying of passing vehicle information*/
        "allowListDisplayEnabled":  false,
        /*ro, opt, bool, whether to automatically display the custom content configured by the user whose vehicle is in the authorized list, desc:If
enabled, if the passing vehicle is in the authorized list, the device will automatically overwrite the custom content configured by the user. If not
enabled, the display content will not be changed.*/
        "blockListDisplayEnabled":  false,
        /*ro, opt, bool, whether to automatically display the custom content configured by the user whose vehicle is in the unauthorized list, desc:If
enabled, if the passing vehicle is in the unauthorized list, the device will automatically overwrite the custom content configured by the user. If not
enabled, the display content will not be changed.*/
        "temporaryListDisplayEnabled":  false,
        /*ro, opt, bool, whether to automatically display the custom content configured by the user whose vehicle is a temporary vehicle, desc:If enabled,
if the passing vehicle is a temporary vehicle and is not in the unauthorized/authorized list, the device will automatically overwrite the custom content
configured by the user. If not enabled, the display content will not be changed.*/
        "LEDConfigurationList": [
        /*ro, opt, array, the parameters list of LED screen, subType:object*/
            {
                "LEDConfiguration": {
                /*ro, opt, object, the parameters of LED screen*/
                    "id":  1,
                    /*ro, req, int, screen ID*/
                    "enabled":  true,
                    /*ro, req, bool, enable displaying multiple lines on LED screen*/
                    "ShowInfoList": [
                    /*ro, opt, array, display content, subType:object*/
                        {
                            "ShowInfo": {
                            /*ro, opt, object, display content*/
                                "id":  1,
                                /*ro, req, int, row No., desc:it increases from 1*/
                                "fontSize":  16,
                                /*ro, opt, enum, font size, subType:int, desc:16 (16 × 16 dot matrix), 32 (32 × 32 dot matrix)*/
                                "fontColor":  "red",
                                /*ro, opt, enum, font color, subType:string, desc:"red", "green", "yellow"*/
                                "speedType":  "fast",
                                /*ro, req, enum, speed type, subType:string, desc:"fast", "medium", "slow"*/
                                "displayMode":  "left",
                                /*ro, req, enum, display mode, subType:string, desc:"left", "right", "instant”*/
                                "LineInfoList": [
                                /*ro, opt, array, the list of combination of rows, subType:object*/
                                    {
                                        "LineInfo": {
                                        /*ro, opt, object, combination of rows, desc:value and customValue are mutually exclusive.*/
                                            "id":  1,
                                            /*ro, req, int, No., desc:it increases from 1*/
                                            "value":  "plate",
                                            /*ro, opt, enum, options of display content, subType:string, desc:“plate”, “YYYY-MM-DD”, “hh-mm-ss”*/
                                            "customValue":  "欢迎光临"
                                            /*ro, opt, string, custom display contents*/
                                        }
                                    }
                                ]
                            }
                        }
                    ]
                }
            }
        ],
        "LEDGeneralConfiguration": {
        /*ro, opt, object, the general parameters of LED screen, desc:The parameters of LEDConfigurationList will take effect if LEDConfigurationList is
configured.*/
            "FontSize":  16,
            /*ro, opt, enum, font size, subType:int, desc:16 (16 × 16 dot matrix), 32 (32 × 32 dot matrix)*/
            "FontColor":  "red",
```

```
            /*ro, opt, enum, font color, subType:string, desc:"red", "green", "yellow"*/
            "Frequency":  5,
            /*ro, req, int, the frequency of screen refresh, desc:it increases from 1*/
            "FreeTimeShowInfo": {
            /*ro, opt, object, display in free time*/
                "CustomContentList": [
                /*ro, opt, array, content list displayed in free time, subType:object*/
                    {
                        "CustomContent": {
                        /*ro, opt, object, display content in free time*/
                            "id":  1,
                            /*ro, req, int, No., range:[1,5], desc:it increases from 1*/
                            "content":  "test"
                            /*ro, opt, string, display content, desc:the maximum size is 40*/
                        }
                    }
                ],
                "TimeBlockList": [
                /*ro, opt, array, parameters for time control configuration, subType:object*/
                    {
                        "TimeBlock": {
                        /*ro, opt, object, time period*/
                            "id":  1,
                            /*ro, opt, int, No.*/
                            "beginTime":  "00:00",
                            /*ro, opt, string, start time, desc:control in different times during 24 hours, accurate to the minute, format: hh:MM*/
                            "endTime":  "24:00"
                            /*ro, opt, string, end time, desc:control in different times during 24 hours, accurate to the minute, format: hh:MM*/
                        }
                    }
                ],
                "showTime":  15
                /*ro, req, int, display duration, range:[1,30]*/
            },
            "IllegalShowInfo": {
            /*ro, opt, object, violation information for display*/
                "IllegalList": [
                /*ro, opt, array, violation information list for display, subType:object*/
                    {
                        "IllegalContent": {
                        /*ro, opt, object, violation information for display*/
                            "id":  1,
                            /*ro, req, int, No., range:[1,5], desc:it increases from 1*/
                            "content":  "test",
                            /*ro, opt, string, display content, desc:the maximum size is 40*/
                            "contentType":  "nonHelmet"
                            /*ro, opt, enum, content type, subType:string, desc:“nonHelmet”, “bycleManned” (carrying passengers on non-motor vehicle),
“reverse”*/
                        }
                    }
                ],
                "TimeBlockList": [
                /*ro, opt, array, whether to enable parameters of configuring time control, subType:object*/
                    {
                        "TimeBlock": {
                        /*ro, opt, object, time period*/
                            "id":  1,
                            /*ro, opt, int, No.*/
                            "beginTime":  "00:00",
                            /*ro, opt, string, start time, desc:control in different times during 24 hours, accurate to the minute, format: hh:MM*/
                            "endTime":  "24:00"
                            /*ro, opt, string, end time, desc:control in different times during 24 hours, accurate to the minute, format: hh:MM*/
                        }
                    }
                ],
                "showTime":  15
                /*ro, req, int, display times, range:[1,30]*/
            }
        },
        "speedMode":  "slowMode"
        /*ro, opt, enum, speed mode, subType:string, desc:“fastMode”, “slowMode”*/
    }
}
```


11.6.2.13 Get the capability of configuring the parameters for displaying a specified scene on the LED screen

Request URL

GET /ISAPI/Parking/channels/<channelID>/LEDConfigurations/multiScene/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	channel No.

Request Message

None

Response Message

```
{
    "MultiSceneLEDConfigurationsCap": {
    /*ro, opt, object, the capability of configuring the parameters for displaying multiple scenes on the LED screen*/
        "SingleSceneLEDConfigurations": {
        /*ro, opt, object, configuring the parameters for displaying a specified scene on the LED screen*/
            "sid": {
            /*ro, req, object, scene ID*/
                "@maxSize":  2
                /*ro, req, int, the maximum number of scenes*/
            },
            "mode": {
            /*ro, req, object, scene mode*/
                "@opt": ["passingVehicle", "noVehicle"],
                /*ro, req, array, options, subType:string, desc:"passingVehicle", "noVehicle”*/
                "#text":  "passingVehicle"
                /*ro, opt, string, the sample value*/
            },
            "ctrlMode": {
            /*ro, opt, object, control mode,
            desc:It setting the mode is not supported, by default it is valid for both camera and platform.
            Camera (automatic control by device): The device recognizes vehicles in the block and authorized list and temporary license plates by itself,
and automatically displays the vehicle passing information according to the strategy configured in vehicleDisplayEnabled, allowListDisplayEnabled,
blockListDisplayEnabled, temporaryListDisplayEnabled.
                  Platform (manual control): Display the content in CustomContentList applied by the user. the mode, The automatic control trategy
configured in vehicleDisplayEnabled, allowListDisplayEnabled, blockListDisplayEnabled, and temporaryListDisplayEnabled will not take effect.
            Camera and platform: the above two control modes take effect at the same time. When CustomContentList is spplied, its content is displayed
immediately; when the CustomContentList is not applied, it will automatically display the default content on the device.
                 Camera and platform: the above two control modes take effect at the same time. When CustomContentList is spplied, its content is displayed
immediately; when the CustomContentList is not applied, it will automatically display the default content on the device.*/
                "@opt": ["camera", "platform", "cameraAndplatform"],
                /*ro, req, array, options, subType:string, desc:“camera”, “platform”, “cameraAndplatform” (takes effect in camera and platform mode)*/
                "#text":  "camera"
                /*ro, opt, string, example*/
            },
            "showFreeEnabled": {
            /*ro, opt, object, whether to enable vacancy display mode, desc:it is valid when the value of mode is "passingVehicle" (passing-vehicle scene)*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
            "displayTime": {
            /*ro, opt, object, display duration, desc:it is valid when the value of mode is "passingVehicle"*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  300
                /*ro, req, int, the maximum value*/
            },
            "vehicleDisplayEnabled": {
            /*ro, opt, object, enable automatically displaying content when a vehicle is passing*/
                "@opt": [true, false]
                /*ro, req, array, options, subType:bool*/
            },
            "allowListDisplayEnabled": {
            /*ro, opt, object, whether to automatically display the custom content configured by the user whose vehicle is in the authorized list, desc:If
enabled, if the passing vehicle is in the authorized list, the device will automatically overwrite the custom content configured by the user. If not
enabled, the display content will not be changed.*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, opt, bool, the default value*/
            },
            "blockListDisplayEnabled": {
            /*ro, opt, object, whether to automatically display the custom content configured by the user whose vehicle is in the unauthorized list, desc:If
enabled, if the passing vehicle is in the unauthorized list, the device will automatically overwrite the custom content configured by the user. If not
enabled, the display content will not be changed.*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, opt, bool, the default value*/
            },
            "temporaryListDisplayEnabled": {
            /*ro, opt, object, whether to automatically display the custom content configured by the user whose vehicle is a temporary vehicle, desc:If
enabled, if the passing vehicle is a temporary vehicle and is not in the unauthorized/authorized list, the device will automatically overwrite the custom
content configured by the user. If not enabled, the display content will not be changed.*/
                "@opt": [true, false],
                /*ro, req, array, options, subType:bool*/
                "@def":  false
                /*ro, opt, bool, the default value*/
            },
            "LEDConfigurationList": {
            /*ro, opt, object, the list of LED configuration parameters*/
                "LEDConfiguration": {
                /*ro, opt, object, LED configuration parameters*/
                    "id": {
                    /*ro, req, object, display screen ID, desc:it increases from 1*/
                        "@maxSize":  1
                        /*ro, req, int, the maximum number of scenes*/
                    },
                    "enabled": {
```

```
                    "enabled": {
                    /*ro, req, object, whether to enable the LED screen to display multiple lines*/
                        "@opt": [true, false]
                        /*ro, req, array, options, subType:bool*/
                    },
                    "ShowInfoList": {
                    /*ro, opt, object, list of display information*/
                        "ShowInfo": {
                        /*ro, opt, object, display information*/
                            "id": {
                            /*ro, req, object, it increases from 1*/
                                "@maxSize":  4
                                /*ro, req, int, the maximum number of rows*/
                            },
                            "fontSize": {
                            /*ro, opt, object, font size*/
                                "@opt": ["16", "32"]
                                /*ro, opt, array, options, subType:string, desc:16 (16 × 16 dot matrix), 32 (32 × 32 dot matrix)*/
                            },
                            "fontColor": {
                            /*ro, opt, object, font color*/
                                "@opt": ["red", "green", "yellow"],
                                /*ro, req, array, options, subType:string, desc:"red", "green", "yellow", "off"*/
                                "#text":  "red"
                                /*ro, opt, string, the sample value*/
                            },
                            "speedType": {
                            /*ro, req, object, speed type*/
                                "@opt": ["fast", "medium", "slow"],
                                /*ro, req, array, options, subType:string, desc:"fast", "medium", "slow"*/
                                "#text":  "fast"
                                /*ro, opt, string, the sample value*/
                            },
                            "displayMode": {
                            /*ro, req, object, display mode*/
                                "@opt": ["left", "right", "instant"],
                                /*ro, req, array, options, subType:string, desc:"left", "right", "instant"*/
                                "#text":  "left"
                                /*ro, opt, string, the sample value*/
                            },
                            "LineInfoList": {
                            /*ro, opt, object, the list of line information*/
                                "LineInfo": {
                                /*ro, opt, object, line information, desc:value and customValue cannot be configured simultaneously, and one ID can only be configured with one of these two parameters*/
                                    "id": {
                                    /*ro, req, object, index, desc:it increases from 1*/
                                        "@maxSize":  4
                                        /*ro, req, int, the maximum number of rows*/
                                    },
                                    "value": {
                                    /*ro, opt, object, display information*/
                                        "@opt": ["plate", "YYYY-MM-DD", "hh-mm-ss"],
                                        /*ro, req, array, options, subType:string, desc:"plate", "YYYY-MM-DD", "hh-mm-ss"*/
                                        "#text":  "plate"
                                        /*ro, opt, string, the sample value*/
                                    },
                                    "customValue": {
                                    /*ro, opt, object, custom display information*/
                                        "@min":  1,
                                        /*ro, req, int, the minimum value*/
                                        "@max":  100
                                        /*ro, req, int, the maximum value*/
                                    }
                                }
                            }
                        }
                    }
                }
            },
        },
    }
}
```

11.6.2.14 Get picture data management parameters of the LED screen

Request URL

GET /ISAPI/Parking/channels/<channelID>/LEDConfigurations/multiScene/pictureData?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "PictureDataList": [
    /*ro, opt, array, subType:object*/
        {
            "PictureData": {
            /*ro, opt, object*/
                "pID":  1,
                /*ro, opt, int, range:[1,64]*/
                "name":  "test"
                /*ro, opt, string, range:[1,128]*/
            }
        }
    ]
}
```

11.6.2.15 Set the parameters of displaying multiple scenes on the LED screen

Request URL

PUT /ISAPI/Parking/channels/<channelID>/LEDConfigurations/multiScene?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "MultiSceneLEDConfigurations": [
    /*opt, array, the parameters of displaying multiple scenes on the LED screen, subType:object*/
        {
            "SingleSceneLEDConfigurations": {
            /*opt, object, the parameters of displaying a single scene on the LED screen*/
                "sid":  1,
                /*req, int, scene ID*/
                "mode":  "passingVehicle",
                /*req, enum, scene mode, subType:string, desc:read-only,the types of scene modes: "passingVehicle" (passing-vehicle scene),"noVehicle" (no-
passing-vehicle scene)*/
                "showFreeEnabled":  true,
                /*opt, bool, whether to enable vacancy display mode, dep:and,
{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
                "displayTime":  1,
                /*opt, int, display duration, range:[1,300], unit:s, dep:and,
{$.MultiSceneLEDConfigurations[*].SingleSceneLEDConfigurations.mode,eq,passingVehicle}*/
                "ctrlMode":  "cameraAndplatform",
                /*ro, opt, enum, subType:string*/
                "vehicleDisplayEnabled":  true,
                /*opt, bool*/
                "allowListDisplayEnabled":  false,
                /*opt, bool*/
                "blockListDisplayEnabled":  false,
                /*opt, bool*/
                "temporaryListDisplayEnabled":  false,
                /*opt, bool*/
                "LEDConfigurationList": [
                /*opt, array, the list of LED configuration parameters, subType:object*/
                    {
                        "LEDConfiguration": {
                        /*opt, object, LED configuration parameters*/
                            "id":  1,
                            /*req, int, display screen ID*/
                            "enabled":  true,
                            /*req, bool, enable the LED screen to display multiple lines*/
                            "ShowInfoList": [
                            /*opt, array, list of display information, subType:object*/
                                {
                                    "ShowInfo": {
                                    /*opt, object, display information*/
                                        "id":  1,
                                        /*req, int, read-only,display screen ID, desc:read-only,display screen ID*/
                                        "fontSize":  16,
                                        /*opt, enum, font size, subType:int, desc:16 (16 × 16 dot matrix), 32 (32 × 32 dot matrix)*/
                                        "fontColor":  "red",
                                        /*opt, enum, font color, subType:string, desc:"red", "green", "yellow"*/
                                        "speedType":  "fast",
                                        /*req, enum, speed type, subType:string, desc:"fast", "medium", "slow"*/
                                        "displayMode":  "left",
                                        /*req, enum, display mode, subType:string, desc:"left", "right", "instant”*/
                                        "LineInfoList": [
```

```
                                        /*opt, array, the list of line information, subType:object*/
                                            {
                                                "LineInfo": {
                                                /*opt, object, line information, desc:value and customValue cannot be configured simultaneously, one ID can only be configured with one of these two parameters*/
                                                    "id":  1,
                                                    /*req, int, No., desc:it increases from 1*/
                                                    "value":  "plate",
                                                    /*opt, enum, display information, subType:string, desc:"plate" (license plate number), "YYYY-MM-DD" (date displayed in this format), "hh-mm-ss" (time displayed in this format)*/
                                                    "customValue":  "欢迎光临"
                                                    /*opt, string, custom display information*/
                                                }
                                            }
                                        ]
                                    }
                                }
                            ]
                        }
                    }
                ],
                "LEDGeneralConfiguration": {
                /*opt, object*/
                    "FontSize":  16,
                    /*opt, enum, font size, subType:int, desc:16 (16 × 16 dot matrix), 32 (32 × 32 dot matrix)*/
                    "FontColor":  "red",
                    /*opt, enum, font color, subType:string, desc:"red", "green", "yellow"*/
                    "Frequency":  5,
                    /*req, int, desc:it increases from 1*/
                    "FreeTimeShowInfo": {
                    /*ro, opt, object*/
                        "CustomContentList": [
                        /*ro, opt, array, subType:object*/
                            {
                                "CustomContent": {
                                /*ro, opt, object*/
                                    "id":  1,
                                    /*ro, req, int, No., range:[1,5], desc:it increases from 1*/
                                    "content":  "test"
                                    /*ro, opt, string, display information*/
                                }
                            }
                        ],
                        "TimeBlockList": [
                        /*ro, opt, array, subType:object*/
                            {
                                "TimeBlock": {
                                /*ro, opt, object, time period*/
                                    "id":  1,
                                    /*ro, opt, int, No.*/
                                    "beginTime":  "00:00",
                                    /*ro, opt, string, start time*/
                                    "endTime":  "24:00"
                                    /*ro, opt, string*/
                                }
                            }
                        ],
                        "showTime":  15
                        /*ro, req, int, display interval*/
                    },
                    "IllegalShowInfo": {
                    /*ro, opt, object*/
                        "IllegalList": [
                        /*ro, opt, array, subType:object*/
                            {
                                "IllegalContent": {
                                /*ro, opt, object*/
                                    "id":  1,
                                    /*ro, req, int, No., range:[1,5], desc:it increases from 1*/
                                    "content":  "test",
                                    /*ro, opt, string, display information*/
                                    "contentType":  "nonHelmet"
                                    /*ro, opt, enum, content type, subType:string*/
                                }
                            }
                        ],
                        "TimeBlockList": [
                        /*ro, opt, array, subType:object*/
                            {
                                "TimeBlock": {
                                /*ro, opt, object, time period*/
                                    "id":  1,
                                    /*ro, opt, int, No.*/
                                    "beginTime":  "00:00",
                                    /*ro, opt, string, start time*/
                                    "endTime":  "24:00"
                                    /*ro, opt, string*/
                                }
                            }
                        ],
                        "showTime":  15
                        /*ro, req, int*/
                    }
                }
```

```
            }
        }
    ]
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "ok",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error description, desc:this node is required when the value of statusCode is not 1*/
}
```

11.6.2.16 Create screen parameters

Request URL

POST /ISAPI/Parking/server/deviceManagement/led?format=json

Query Parameter

None

Request Message

```
{
    "LedInfo": {
    /*opt, object, screen information*/
        "deviceID":  "test",
        /*opt, string, unique device ID*/
        "deviceType":  "led",
        /*opt, string, device type*/
        "deviceName":  "显示屏",
        /*opt, string, device name*/
        "displayType":  "enterLocation",
        /*opt, enum, display type, subType:string, desc:"enterLocation" (screen for entrance location), "enterPrompt" (screen for prompt at entrance), "exitCharge" (screen for charge at exit)*/
        "communicateMode":  "serial",
        /*req, enum, communication mode, subType:string, desc:"serial", "network"*/
        "SerialCtrlInfo": {
        /*opt, object, serial port control information*/
            "comNo":  1,
            /*req, int, serial port No.*/
            "ctrlCode":  1
            /*req, int, control code*/
        },
        "NetworkCtrlInfo": {
        /*opt, object, network control information*/
            "ipaddr":  "10.10.112.250",
            /*req, string, IP address*/
            "portNo":  1
            /*opt, int, port No., range:[0,65535], step:1*/
        },
        "Params": {
        /*opt, object, parameters*/
            "lcdType":  "monochrome",
            /*req, enum, screen type, subType:string, desc:"monochrome", "twocolor", "colorful"*/
            "ScreenSize": {
            /*opt, object, screen size*/
                "height":  1,
                /*req, int, height*/
                "width":  1
                /*req, int, width*/
            },
            "color":  "red",
            /*req, enum, color of prompt content, subType:string, desc:"red", "green", "yellow", "blue", "pink", "white", "cyan"*/
            "scrolltype":  "none",
            /*req, enum, scroll type, subType:string, desc:"none" (not scroll), "leftScroll" (scroll left), "rightScroll" (scroll right), "upScroll" (scroll up), "downScroll" (scroll down)*/
            "isfreeShowlLeft":  true,
            /*req, bool, display vacant parking spaces in free time*/
            "textSize":  1,
            /*req, int, font size*/
            "FixedContent": {
            /*opt, object, fixed content*/
                "enabled":  true,
                /*req, bool, whether to enable displaying fixed content*/
                "istimeShow":  true,
                /*req, bool, whether to enable displaying the time*/
                "content":  "test"
                /*req, string, fixed display content*/
            }
        },
        "NoticeInfo": [
        /*opt, array, prompt content, subType:object*/
            {
                "line":  1,
                /*req, int, number of lines of content*/
                "textType":  "plateNum",
                /*req, string, prompt content, desc:supports selecting multiple content types, "plateNum" (plate number), "cardNum" (card number), "vehicleType" (vehicle type), "parkingType" (parking type), "timeYearMonthDay" (year-month-day), "timeHourMinSec" (hour-min-sec), "leftParkSpace" (vacant parking spaces), "validTime" (validity)*/
                "textColor":  "red",
                /*req, enum, color of prompt content, subType:string, desc:"red", "green", "yellow", "blue", "pink", "white", "cyan"*/
                "textSize":  "test",
                /*req, string, font size of prompt content*/
                "scrolltype":  "none"
                /*req, enum, scroll type, subType:string, desc:"none", "leftScroll", "rightScroll", "upScroll", "downScroll"*/
            }
        ],
        "fullControl":  "none"
        /*opt, enum, full-screen control, subType:string, desc:"none" (no control), "forceShowFull" (force showing no more vehicles allowed)*/
    }
}
```


Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

11.6.2.17 Configure screen parameters

Request URL

PUT /ISAPI/Parking/server/deviceManagement/led?format=json&id=<deviceID>

Query Parameter

Parameter Name	Parameter Type	Description
deviceID	string	unique device ID

Request Message

```
{
    "LedInfo": {
    /*opt, object, screen information*/
        "deviceID":  "test",
        /*opt, string, unique device ID*/
        "deviceType":  "led",
        /*opt, string, device type*/
        "deviceName":  "显示屏",
        /*opt, string, device name*/
        "displayType":  "enterLocation",
        /*opt, enum, display type, subType:string, desc:“enterLocation” (screen for entrance location), “enterPrompt” (screen for prompt at entrance), “exitCharge” (screen for charge at exit)*/
        "communicateMode":  "serial",
        /*req, enum, communication mode, subType:string, desc:“serial”, “network”*/
        "SerialCtrlInfo": {
        /*opt, object, serial port control information*/
            "comNo":  1,
            /*req, int, serial port No.*/
            "ctrlCode":  1
            /*req, int, control code*/
        },
        "networkCtrlInfo": {
        /*opt, object, network control information*/
            "ipaddr":  "10.10.112.250",
            /*req, string, IP address*/
            "portNo":  1
            /*opt, int, port No., range:[0,65535], step:1*/
        },
        "Params": {
        /*opt, object, parameters*/
            "lcdType":  "monochrome",
            /*req, enum, screen type, subType:string, desc:“monochrome”, “twocolor”, “colorful”*/
            "ScreenSize": {
            /*opt, object, screen size*/
                "height":  1,
                /*req, int, height*/
                "width":  1
                /*req, int, width*/
            },
            "color":  "red",
            /*req, enum, color of prompt content, subType:string, desc:“red”, “green”, “yellow”, “blue”, “pink”, “white”, “cyan”*/
            "scrolltype":  "none",
            /*req, enum, scroll type, subType:string, desc:“none”, “leftScroll”, “rightScroll”, “upScroll”, “downScroll”*/
            "isfreeShowlLeft":  true,
            /*req, bool, display vacant parking spaces in free time*/
            "textSize":  1,
            /*req, int, font size*/
            "FixedContent": {
            /*opt, object, fixed content*/
                "enabled":  true,
                /*req, bool, whether to enable displaying fixed content*/
                "istimeShow":  true,
                /*req, bool, whether to enable displaying the time*/
                "content":  "test"
                /*req, string, fixed display content*/
            }
        },
        "NoticeInfo": [
        /*opt, array, prompt content, subType:object*/
            {
                "line":  1,
                /*req, int, number of lines of content*/
                "textType":  "plateNum",
                /*req, string, prompt content, desc:supports selecting multiple content types, “plateNum” (plate number), “cardNum” (card number), “vehicleType” (vehicle type), “parkingType” (parking type), “timeYearMonthDay” (year-month-day), “timeHourMinSec” (hour-min-sec), “leftParkSpace” (vacant parking spaces), “validTime” (validity)*/
                "textColor":  "red",
                /*req, enum, color of prompt content, subType:string, desc:“red”, “green”, “yellow”, “blue”, “pink”, “white”, “cyan”*/
                "textSize":  "test",
                /*req, string, font size of prompt content*/
                "scrolltype":  "none"
                /*req, enum, scroll type, subType:string, desc:“none”, “leftScroll”, “rightScroll”, “upScroll”, “downScroll”*/
            }
        ],
        "fullControl":  "none"
        /*opt, enum, full-screen control, subType:string, desc:“none” (no control), “forceShowFull” (force showing no more vehicles allowed)*/
    }
}
```


Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```

11.6.2.18 Get screen parameters

Request URL

GET /ISAPI/Parking/server/deviceManagement/led?format=json&id=<deviceID>

Query Parameter

Parameter Name	Parameter Type	Description
deviceID	string	Unique device ID

Request Message

None

Response Message

```
{
    "LedInfo": {
    /*ro, opt, object, screen information*/
        "deviceID":  "test",
        /*ro, opt, string, unique device ID*/
        "deviceType":  "led",
        /*ro, opt, string, device type*/
        "deviceName":  "显示屏",
        /*ro, opt, string, device name*/
        "displayType":  "enterLocation",
        /*ro, opt, enum, display type, subType:string, desc:"enterLocation" (screen for entrance location), "enterPrompt" (screen for prompt at entrance),
"exitCharge" (screen for charge at exit)*/
        "communicateMode":  "serial",
        /*ro, req, enum, communication mode, subType:string, desc:"serial", "network"*/
        "SerialCtrlInfo": {
        /*ro, opt, object, serial port control information*/
            "comNo":  1,
            /*ro, req, int, serial port No.*/
            "ctrlCode":  1
            /*ro, req, int, control code*/
        },
        "networkCtrlInfo": {
        /*ro, opt, object, network control information*/
            "ipaddr":  "10.10.112.250",
            /*ro, req, string, IP address*/
            "portNo":  1
            /*ro, opt, int, port No., range:[0,65535], step:1*/
        },
        "Params": {
        /*ro, opt, object, parameters*/
            "lcdType":  "monochrome",
            /*ro, req, enum, screen type, subType:string, desc:"monochrome", "twocolor", "colorful"*/
            "ScreenSize": {
            /*ro, opt, object, screen size*/
                "height":  1,
                /*ro, req, int, height*/
                "width":  1
                /*ro, req, int, Width.*/
            },
            "color":  "red",
            /*ro, req, enum, color of prompt content, subType:string, desc:"red", "green", "yellow", "blue", "pink", "white", "cyan"*/
            "scrolltype":  "none",
            /*ro, req, enum, scroll type, subType:string, desc:"none", "leftScroll", "rightScroll", "upScroll", "downScroll"*/
            "isfreeShowlLeft":  true,
            /*ro, req, bool, display vacant parking spaces in free time*/
            "textSize":  1,
            /*ro, req, int, font size*/
            "FixedContent": {
            /*ro, opt, object, fixed content*/
                "enabled":  true,
                /*ro, req, bool, whether to enable displaying fixed content*/
                "istimeShow":  true,
                /*ro, req, bool, whether to display the time*/
                "content":  "test"
                /*ro, req, string, fixed display content*/
            }
        },
        "NoticeInfo": [
        /*ro, opt, array, prompt content, subType:object*/
            {
                "line":  1,
                /*ro, req, int, number of lines of content*/
                "textType":  "plateNum",
                /*ro, req, string, prompt content, desc:supports selecting multiple content types, "plateNum", "cardNum", "vehicleType", "parkingType",
"timeYearMonthDay", "timeHourMinSec", "leftParkSpace" (vacant parking spaces), "validTime"*/
                "textColor":  "red",
                /*ro, req, enum, color of prompt content, subType:string, desc:"red", "green", "yellow", "blue", "pink", "white", "cyan"*/
                "textSize":  "test",
                /*ro, req, string, font size of prompt content*/
                "scrolltype":  "none"
                /*ro, req, enum, scroll type, subType:string, desc:"none", "leftScroll", "rightScroll", "upScroll", "downScroll"*/
            }
        ],
        "fullControl":  "none"
        /*ro, opt, enum, full-screen control, subType:string, desc:"none" (no control), "forceShowFull" (force showing no more vehicles allowed)*/
    }
}
```


11.6.2.19 Get the capability set of displaying information on the LCD screen

Request URL

GET /ISAPI/System/LCDScreen/capabilities?format=json

Query Parameter

None

Request Message

None

Response Message


```
{
    "LCDScreenCap": {
    /*ro, opt, object, capability set of displaying information on the LCD screen*/
        "DisplayInfoCap": {
        /*ro, opt, object, capability of the LCD screen*/
            "QRCodeImg": {
            /*ro, opt, object, QR code information*/
                "QRCodeNum":  "1",
                /*ro, opt, string, the maximum number of QR code pictures that can be displayed*/
                "QRCodeImgFormat": {
                /*ro, opt, object, payment method and picture type of the QR code defined by the file name and the extension name*/
                    "@opt":  "weixin.jpg,zhifubao.bmp"
                    /*ro, req, string, optional values*/
                },
                "paymentType": {
                /*ro, opt, object, payment method (multiple methods can be separated by commas)*/
                    "@opt":  "WeChatPay,Alipay"
                    /*ro, req, string, optional values*/
                },
                "preDisPlayEnabled": {
                /*ro, opt, object, whether to enable pre-display, desc:true (device stores the QR code and displays it in advance when a corresponding scene detected, allowing users to scan for registration), false or field not exist (device displays the QR code instantly and does not store the QR code)*/
                    "@opt": [true, false],
                    /*ro, req, array, options, subType:bool*/
                    "@def":  false
                    /*ro, opt, bool, default value*/
                }
            },
            "contentUrl": {
            /*ro, opt, object, picture URL, desc:the device downloads the picture via URL to display it*/
                "@min":  0,
                /*ro, req, int, minimum string length*/
                "@max":  256
                /*ro, req, int, maximum string length*/
            },
            "backgroundPicture": {
            /*ro, opt, object, background picture content, desc:a binary picture string encoded by Base64*/
                "@min":  0,
                /*ro, req, int, minimum string length*/
                "@max":  40960
                /*ro, req, int, maximum string length*/
            },
            "sence": {
            /*ro, opt, object, scene*/
                "@opt":  "1,2,3,4,5,6,7,8,9,10,11,12,13,14,15"
                /*ro, req, string, optional values, desc:"1" (entered vehicles with license plate), "2" (entered vehicles without license plate), "3" (exited vehicles with license plate and without unpaid fee), "4" (exited vehicles with license plate and unpaid fee), "5" (exited vehicles without license plate)*/
            },
            "amounts":  true,
            /*ro, opt, bool, whether it supports displaying the account*/
            "licenseLen":  64,
            /*ro, opt, int, the maximum string length of the license plate characters, desc:If this node exists, it indicates that displaying the license plate information is supported*/
            "phoneNumberLen":  20,
            /*ro, opt, int, the maximum string length of the phone No., desc:If this node exists, it indicates that displaying the contact information is supported*/
            "noticeLen":  256,
            /*ro, opt, int, the maximum string length of the notice information*/
            "enterTime":  true,
            /*ro, opt, bool, whether it supports displaying the entering time*/
            "leaveTime":  true,
            /*ro, opt, bool, whether it supports displaying the exiting time*/
            "customInfoLen":  128,
            /*ro, opt, int, the maximum string length of the displayed custom information, desc:If this node exists, it indicates that displaying the custom information is supported*/
            "customInfoList": {
            /*ro, opt, object*/
                "@size":  4,
                /*ro, req, int*/
                "customInfo": {
                /*ro, opt, object*/
                    "@min":  0,
                    /*ro, req, int*/
                    "@max":  64
                    /*ro, req, int*/
                },
                "fontSize": {
                /*ro, opt, object*/
                    "@min":  1,
                    /*ro, opt, int*/
                    "@max":  100
                    /*ro, opt, int*/
                },
                "fontColor": {
                /*ro, opt, object*/
```

```
                /*ro, opt, object*/
                    "R": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  255
                        /*ro, opt, int*/
                    },
                    "G": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  255
                        /*ro, opt, int*/
                    },
                    "B": {
                    /*ro, opt, object*/
                        "@min":  0,
                        /*ro, opt, int*/
                        "@max":  255
                        /*ro, opt, int*/
                    }
                }
            },
            "plateType": {
            /*ro, opt, object, license plate type*/
                "@opt":  "fixedPlate,temporaryPlate,blackPlate,payByPeriodPlate,noPlate"
                /*ro, req, string, optional values, desc:"fixedPlate" (registered vehicle), "temporaryPlate" (temperatory vehicle). "blackPlate" (vehicles
in the blocklist)*/
            },
            "validityPeriod": {
            /*ro, opt, object, validity period*/
                "@min":  0,
                /*ro, req, int, the minimum value*/
                "@max":  29
                /*ro, req, int, the maximum value*/
            },
            "remainingCharge":  "30",
            /*ro, opt, string, account balance*/
            "parkingLot":  2,
            /*ro, opt, int, number of vacant parking spaces*/
            "titleLen":  128,
            /*ro, opt, int, the maximum string length of the title, desc:if this node exists, it indicates that setting the title is supported*/
            "laneID": {
            /*ro, opt, object, lane No., desc:the traffic server connects cameras and link cameras to lane*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  32
                /*ro, req, int, the maximum value*/
            },
            "senceDisplayTime": {
            /*ro, opt, object, display duration of a scene, desc:the display duration of a scene before refresh. Unit: second*/
                "@min":  1,
                /*ro, req, int, the minimum value*/
                "@max":  600
                /*ro, req, int, the maximum value*/
            }
        }
    }
}
```

Parameter Name	Parameter Value	Parameter Type(Content-Type)	Content-ID	File Name	Description
	[Message content]	application/json	--	--	--

Note: The protocol is transmitted in form format. See Chapter 4.5.1.4 for form framework description, as shown in the instance below.

```
--<frontier>
Content-Disposition: form-data; name=Parameter Name;filename=File Name
Content-Type: Parameter Type
Content-Length: ****
Content-ID: Content ID
Parameter Value
```

- Parameter Name: the name property of Content-Disposition in the header of form unit; it refers to the form unit name.
- Parameter Type (Content-Type): the Content-Type property in the header of form unit.
- File Name (filename): the filename property of Content-Disposition of form unit Headers. It exists only when the transmitted data of form unit is file, and it refers to the file name of form unit body.
- Parameter Value: the body content of form unit.

11.6.2.20 Configure the information to be displayed on the screen.

Request URL

POST /ISAPI/System/LCDScreen/displayInfo?format=json

Query Parameter

None

Request Message

```
{
    "DisplayInfo": {
    /*opt, object, displayed information*/
        "QRCodeImg": [
        /*opt, array, array of object, list of QR codes, subType:object*/
            {
                "QRCodeBase64":  "abcdeadfadsf==",
                /*req, string, QR code encoded via Base64*/
                "QRCodeFileName":  "weixin.jpg",
                /*req, string, picture format of the QR code, desc:"weixin.jpg" (WeChat), "zhifubao.bmp" (Alipay)*/
                "paymentType":  "WeChatPay,Alipay",
                /*req, string, payment method, desc:"WeChatPay", "Alipay". Multiple methods can be separated by commas*/
                "QRContent":  "http://",
                /*opt, string, QR code content, desc:When the node QRCodeBase64 is not configured, the device will generate a QR code using this node*/
                "preDisPlayEnabled":  false
                /*opt, bool, whether to enable, desc:true (device stores the QR code and displays it in advance when a corresponding scene detected,
allowing users to scan for registration), false or field not exist (device displays the QR code instantly and does not store the QR code)*/
            }
        ],
        "contentUrl":  "http://xxx.jpg",
        /*opt, string, picture URL, range:[0,256], desc:the device downloads the picture via URL to display it*/
        "backgroundPicture":  "abcdeadfadsf=="
        /*opt, string, background picture content, range:[0,40960], desc:a binary picture string encoded by Base64*/
    },
    "sence":  1,
    /*req, enum, scene, subType:int, desc:1 (entered vehicles with license plate),2 (entered vehicles without license plate),3 (exited vehicles without
unpaid fee),4 (exited vehicles with unpaid fee),5 (exited vehicles without license plate),6 (the fee is paid),7 (exited vehicles without license plate and
unpaid fee),8 (exited vehicles without license plate and with unpaid fee),9 (paying failed),10 (free). This node is used to control the content and layout
displayed on the screen*/
    "amounts":  30.0,
    /*opt, float, float,amount, desc:float,amount*/
    "license":  "ABC12345",
    /*opt, string, license plate number*/
    "phoneNumber":  "13012345678",
    /*opt, string, phone number*/
    "notice":  "abcdefg",
    /*opt, string, notice information*/
    "enterTime":  "1970-01-01T00:00:00+08:00",
    /*opt, datetime, entering time*/
    "leaveTime":  "1970-01-01T00:00:00+08:00",
    /*opt, datetime, exiting time*/
    "customInfo":  "abcdefg",
    /*opt, string, displayed custom information*/
    "customInfoList": [
    /*opt, array, subType:object, range:[0,4]*/
        {
            "customInfo":  "abcdefg",
            /*req, string, range:[0,64]*/
            "fontSize":  1,
            /*opt, int, range:[1,100]*/
            "fontColor": {
            /*opt, object*/
                "R":  1,
                /*opt, int, range:[0,255]*/
                "G":  1,
                /*opt, int, range:[0,255]*/
                "B":  1
                /*opt, int, range:[0,255]*/
            }
        }
    ],
    "plateType":  "fixedPlate",
    /*opt, enum, license plate type, subType:string, desc:"fixedPlate" (registered vehicle), "temporaryPlate" (temporary vehicle), "blackPlate" (vehicles in
the blocklist)*/
    "validityPeriod":  "2020年11月12日-2020年12月11日",
    /*opt, string, validity period*/
    "remainingCharge":  "30",
    /*opt, string, the account balance, desc:the account balance*/
    "parkingLot":  2,
    /*opt, int, number of vacant parking spaces*/
    "tideLaneStatus":  "opening",
    /*opt, enum, reversible lane status, subType:string, desc:"opening", "closed"*/
    "title":  "abc",
    /*opt, string, displayed title*/
    "laneID":  "1",
    /*opt, string, lane No., range:[1,32], desc:the traffic server connects cameras and link cameras to lane*/
    "senceDisplayTime":  60
    /*opt, int, range:[1,600], unit:s*/
}
```


Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string, error codes categorized by functional modules, desc:all general error codes are in the range of this field*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string, error codes categorized by functional modules, desc:N/A*/
}
```

11.6.3 停车场闸机

11.6.3.1 Set the barrier gate control parameters supported by the device to open, close, lock or unlock the barrier gate

Request URL

PUT /ISAPI/Parking/channels/<channelID>/barrierGate

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BarrierGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, barrier gate control, attr:version{req, string, protocolVersion}-->
  <ctrlMode opt="open,close,lock,unlock">
    <!--req, enum, barrier gate control mode, subType:string, attr:opt{req, string}, desc:"open" (open the barrier gate), "close" (close the barrier gate), "lock" (lock the barrier gate), "unlock" (unlock the barrier gate)-->open
  </ctrlMode>
  <cardNo>
    <!--opt, string, range:[1,64]-->1234567890
  </cardNo>
</BarrierGate>
```


Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, opt, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 (Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format", "Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code description-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <MErrCode>
    <!--ro, opt, string-->0x00000000
  </MErrCode>
  <MErrDevSelfEx>
    <!--ro, opt, string-->0x00000000
  </MErrDevSelfEx>
</ResponseStatus>
```

11.6.3.2 Get the entrance/exit barrier gate status

Request URL

GET /ISAPI/Parking/channels/<channelID>/barrierGate/barrierGateStatus

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<BarrierGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, entrance/exit barrier gate, attr:version{req, string, protocolVersion}-->
  <barrierGateStatus>
    <!--ro, opt, enum, barrier gate status, subType:int, desc:0 (no signal), 1 (closed), 2 (open)-->1
  </barrierGateStatus>
</BarrierGate>
```

11.6.3.3 Get the capability of controlling the barrier gate

Request URL

GET /ISAPI/Parking/channels/<channelID>/barrierGate/capabilities

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<BarrierGate xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <ctrlMode opt="open,close,lock,unlock">
    <!--ro, req, enum, control mode, subType:string, attr:opt{req, string}, desc:control mode-->open
  </ctrlMode>
  <isSupportBarrierGateStatus>
    <!--ro, opt, bool, whether it supports getting the barrier gate status-->true
  </isSupportBarrierGateStatus>
  <cardNo min="1" max="64">
    <!--ro, opt, string, range:[1,64], attr:min{opt, string},max{opt, string}-->123456789
  </cardNo>
</BarrierGate>
```

11.6.4 车辆出入

11.6.4.1 Set parameters to open/close and lock/unlock barrier gate

Request URL

PUT /ISAPI/ITC/Entrance/barrierGateCtrl

Query Parameter

None

Request Message

```
<?xml version="1.0" encoding="UTF-8"?>

<BarrierGateCtrl xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, object, attr:version{req, string, protocolVersion}-->
  <barrietGateNum>
    <!--opt, int, barrier gate No.-->1
  </barrietGateNum>
  <BarrierGateCtrlList>
    <!--opt, array, subType:object-->
    <barrietGateOper>
      <!--opt, enum, barrier gate control type, subType:string, desc:"off"-fall, "on"-rise, "stop"-stop at a certain position, "locked"-lock;-->on
    </barrietGateOper>
  </BarrierGateCtrlList>
</BarrierGateCtrl>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status information, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.6.4.2 Get the entrance and exit capability

Request URL

GET /ISAPI/ITC/Entrance/capabilities

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EntranceCap xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, attr:version{req, string, protocolVersion}-->
  <supportEntrance>
    <!--ro, opt, bool, whether it supports entrance and exit control-->true
  </supportEntrance>
  <supportBarrierGateNum>
    <!--ro, opt, int-->1
  </supportBarrierGateNum>
  <supportAlarmINNum>
    <!--ro, opt, int-->1
  </supportAlarmINNum>
  <supportRelayNum>
    <!--ro, opt, int-->1
  </supportRelayNum>
  <isSupportWizardDeviceInfo>
    <!--ro, req, bool, whether to support configuring device information in wizard-->true
  </isSupportWizardDeviceInfo>
  <isSupportWizardFirstLogin>
    <!--ro, req, bool, whether to support judging if it is login for the first time in wizard-->true
  </isSupportWizardFirstLogin>
  <supportEntranceParam>
    <!--ro, opt, int-->1
  </supportEntranceParam>
</EntranceCap>
```

11.6.4.3 Get the entrance and exit parameters of traffic cameras

Request URL

GET /ISAPI/ITC/Entrance/entranceParam

Query Parameter

None

Request Message

None

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<EntranceParamList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EntranceParam>
    <!--ro, opt, object-->
    <laneNum>
      <!--ro, req, int, barrier gate No.-->1
    </laneNum>
    <bEnable>
      <!--ro, req, bool, whether to enable the parameters-->true
    </bEnable>
    <ctrlMode>
      <!--ro, opt, enum, subType:int-->1
    </ctrlMode>
    <relateTriggerMode>
      <!--ro, req, enum, linked trigger mode, subType:int, desc:0 (checkpoint single I/O trigger), 1 (checkpoint vehicle detector), 2 (MPR)-->1
    </relateTriggerMode>
    <vehControlMeasure>
      <!--ro, opt, object, vehicle management and control method-->
      <plateNumOnlyEnable>
        <!--ro, req, bool, only match license plate number-->true
      </plateNumOnlyEnable>
      <plateNumColorEnable>
        <!--ro, req, bool, match license plate number and license plate color simultaneously-->true
      </plateNumColorEnable>
    </vehControlMeasure>
    <vehInfoManagList>
      <!--ro, req, array, subType:object-->
      <vehInfoManag>
        <!--ro, req, object, vehicle information management-->
        <vehInfoManagNum>
          <!--ro, req, enum, vehicle information management, subType:int, desc:0 (temporary vehicle configuration), 1 (blocklist vehicle configuration), 2 
(allowlist vehicle configuration)-->1
        </vehInfoManagNum>
        <barrierGateOper>
          <!--ro, req, enum, barrier gate operation, subType:int, desc:0 (no operation), 1 (open barrier gate)-->1
        </barrierGateOper>
        <relayOutAlarmEnable>
          <!--ro, req, bool, whether to enable relay output alarm-->true
        </relayOutAlarmEnable>
        <upAlarmEnable>
          <!--ro, req, bool, upload arming alarm information-->true
        </upAlarmEnable>
        <hostUpAlarmEnable>
          <!--ro, req, bool, alarm host uploads alarm information-->true
        </hostUpAlarmEnable>
      </vehInfoManag>
    </vehInfoManagList>
    <relayList>
      <!--ro, req, array, relay list, subType:object-->
      <relay>
        <!--ro, req, object, relay-->
        <relayNum>
          <!--ro, req, int, relay ID-->1
        </relayNum>
        <relayFunction>
          <!--ro, req, enum, relay function, subType:int, desc:0-none,1-open barrier gate, 2-close barrier gate, 3-stop barrier gate, 4-alarm signal, 5-
solid light-->1
        </relayFunction>
      </relay>
    </relayList>
    <IOAlarmList>
      <!--ro, req, array, subType:object-->
      <IOAlarm>
        <!--ro, req, object-->
        <IOAlarmNum>
          <!--ro, req, int, alarm port No.-->1
        </IOAlarmNum>
        <IOAlarmType>
          <!--ro, req, enum, alarm type, subType:int, desc:alarm type: 0-none,1-fire alarm-->1
        </IOAlarmType>
      </IOAlarm>
    </IOAlarmList>
    <notCloseCarFollow>
      <!--ro, opt, bool, whether to enable the barrier gate not to close when the vehicle is followed by another vehicle-->true
    </notCloseCarFollow>
```

```xml
      <bigCarKeepOpen>
        <!--ro, opt, object, the barrier gate remains open when the large-sized vehicle passes-->
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <duration>
          <!--ro, req, int, duration of the barrier gate remaining open, unit:s-->1
        </duration>
      </bigCarKeepOpen>
      <ParkingDetection>
        <!--ro, opt, object-->
        <enabled>
          <!--ro, req, bool, whether to enable the function-->true
        </enabled>
        <judgeTime>
          <!--ro, req, int, unit:s-->1
        </judgeTime>
      </ParkingDetection>
    </EntranceParam>
  </EntranceParamList>
```

11.6.4.4 Get the entrance and exit parameter capability of traffic cameras

Request URL

GET /ISAPI/ITC/Entrance/entranceParam/capabilities

Query Parameter

None

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<entranceParam xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, opt, object, parameters of entrances and exits, attr:version{req, string, protocolVersion}-->
  <laneNum>
    <!--ro, opt, int, barrier gate No.-->1
  </laneNum>
  <isSupportBEnable>
    <!--ro, opt, bool-->true
  </isSupportBEnable>
  <ctrlMode opt="0,1,2">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </ctrlMode>
  <relateTriggerMode opt="0,1,2">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}-->1
  </relateTriggerMode>
  <isSupportPlateNumColor>
    <!--ro, opt, bool, match both license plate number and license plate color-->true
  </isSupportPlateNumColor>
  <isSupportPlateNumOnly>
    <!--ro, opt, bool, match license plate number only-->true
  </isSupportPlateNumOnly>
  <vehInfoManag opt="0,1,2">
    <!--ro, opt, enum, subType:int, attr:opt{req, string}, desc:0 (temporary vehicle configuration),1 (blocklist vehicle configuration), 2 (allowlist 
vehicle configuration)-->1
  </vehInfoManag>
  <barrierGateOper opt="0,1">
    <!--ro, opt, enum, barrier gate operation, subType:int, attr:opt{req, string}, desc:0 (no operation), 1 (open barrier gate)-->1
  </barrierGateOper>
  <isSupportRelayOutAlarm>
    <!--ro, opt, bool-->true
  </isSupportRelayOutAlarm>
  <isSupportUpAlarm>
    <!--ro, opt, bool-->true
  </isSupportUpAlarm>
  <isSupportHostUpAlarm>
    <!--ro, opt, bool-->true
  </isSupportHostUpAlarm>
  <relayNum min="1" max="5">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </relayNum>
  <relayFunction min="0" max="5">
    <!--ro, opt, int, attr:min{req, int},max{req, int}-->1
  </relayFunction>
  <relayFunctionIndex opt="0,1,2,3,4,5">
    <!--ro, opt, string, attr:opt{req, string}-->test
  </relayFunctionIndex>
  <IOAlarmNum opt="1,2">
    <!--ro, opt, int, attr:opt{req, string}-->1
  </IOAlarmNum>
  <IOAlarmType opt="0,1">
    <!--ro, opt, enum, alarm type, subType:int, attr:opt{req, string}-->1
  </IOAlarmType>
  <notCloseCarFollow>
    <!--ro, opt, bool, whether to enable the barrier gate not to close when the vehicle is followed by another vehicle-->true
  </notCloseCarFollow>
  <bigCarKeepOpen>
    <!--ro, opt, object, the barrier gate remains open when the large-sized vehicle passes-->
    <enabled opt="true,false">
      <!--ro, req, bool, whether to enable the function, attr:opt{req, string}-->true
    </enabled>
    <duration min="0" max="1000">
      <!--ro, req, int, duration of the barrier gate remaining open, unit:s, attr:min{req, int},max{req, int}-->1
    </duration>
  </bigCarKeepOpen>
  <ParkingDetection>
    <!--ro, opt, object-->
    <enabled>
      <!--ro, req, bool, whether to enable the function-->true
    </enabled>
    <judgeTime>
      <!--ro, req, int, unit:s-->1
    </judgeTime>
  </ParkingDetection>
</entranceParam>
```

11.6.4.5 Set the entrance and exit parameters

Request URL

PUT /ISAPI/ITC/Entrance/entranceParam

Query Parameter

None

Request Message

```xml
<?xml version="1.0" encoding="UTF-8"?>
<EntranceParamList xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--opt, array, subType:object, attr:version{req, string, protocolVersion}-->
  <EntranceParam>
    <!--opt, object-->
    <laneNum>
      <!--req, int, barrier gate No.-->1
    </laneNum>
    <bEnable>
      <!--req, bool, whether to enable this function-->true
    </bEnable>
    <ctrlMode>
      <!--opt, enum, subType:int-->1
    </ctrlMode>
    <relateTriggerMode>
      <!--req, enum, linked trigger mode, subType:int, desc:0 (checkpoint single I/O trigger), 1 (checkpoint vehicle detector), 2 (MPR)-->1
    </relateTriggerMode>
    <vehControlMeasure>
      <!--opt, object, vehicle management and control-->
      <plateNumOnlyEnable>
        <!--req, bool, match license plate number only-->true
      </plateNumOnlyEnable>
      <plateNumColorEnable>
        <!--req, bool, match both license plate number and license plate color-->true
      </plateNumColorEnable>
    </vehControlMeasure>
    <vehInfoManagList>
      <!--req, array, subType:object-->
      <vehInfoManag>
        <!--req, object-->
        <vehInfoManagNum>
          <!--req, enum, vehicle information management, subType:int, desc:0 (temporary vehicle configuration),1 (blocklist vehicle configuration), 2
(allowlist vehicle configuration)-->1
        </vehInfoManagNum>
        <barrierGateOper>
          <!--req, enum, barrier gate operation, subType:int, desc:0 (no operation), 1 (open barrier gate)-->1
        </barrierGateOper>
        <relayOutAlarmEnable>
          <!--req, bool, whether to enable relays to output alarms-->true
        </relayOutAlarmEnable>
        <upAlarmEnable>
          <!--req, bool, upload arming alarm information-->true
        </upAlarmEnable>
        <hostUpAlarmEnable>
          <!--req, bool, alarm host uploads alarm information-->true
        </hostUpAlarmEnable>
      </vehInfoManag>
    </vehInfoManagList>
    <relayList>
      <!--req, array, subType:object-->
      <relay>
        <!--req, object-->
        <relayNum>
          <!--req, int, relay ID-->1
        </relayNum>
        <relayFunction>
          <!--req, enum, relay function, subType:int, desc:0 (none), 1 (open barrier gate), 2 (close barrier gate), 3 (stop barrier gate), 4 (alarm signal),
5 (solid light)-->1
        </relayFunction>
      </relay>
    </relayList>
    <IOAlarmList>
      <!--req, array, subType:object-->
      <IOAlarm>
        <!--req, object-->
        <IOAlarmNum>
          <!--req, int, alarm port No.-->1
        </IOAlarmNum>
        <IOAlarmType>
          <!--req, enum, alarm type, subType:int, desc:0 (none), 1(fire alarm)-->1
        </IOAlarmType>
      </IOAlarm>
    </IOAlarmList>
    <notCloseCarFollow>
      <!--opt, bool, whether to enable the barrier gate not to close when the vehicle is followed by another vehicle-->true
    </notCloseCarFollow>
    <bigCarKeepOpen>
      <!--opt, object, the barrier gate remains open when the large-sized vehicle passes-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <duration>
        <!--req, int, duration of the barrier gate remaining open, unit:s-->1
      </duration>
    </bigCarKeepOpen>
    <ParkingDetection>
      <!--opt, object-->
      <enabled>
        <!--req, bool, whether to enable the function-->true
      </enabled>
      <judgeTime>
        <!--req, int, unit:s-->1
      </judgeTime>
```

```
    </ParkingDetection>
  </EntranceParam>
</EntranceParamList>
```

Response Message

```
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:"OK" (succeeded), "Device Busy", "Device Error", "Invalid Operation", "Invalid XML Format",
"Invalid XML Content", "Reboot" (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
</ResponseStatus>
```

11.6.5 车辆名单比对

11.6.5.1 Get the capability of deleting the license plate information in the blocklist or allowlist

Request URL

GET /ISAPI/Traffic/channels/<channelID>/DelLicensePlateAuditData/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

None

Response Message

```
{
    "id": {
    /*ro, req, object, ID list*/
        "@size":  10000
        /*ro, req, int, the maximum number of members in the array (the maximum number of IDs in the list)*/
    },
    "licensePlate": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int, the maximum number of members in the array (the maximum number of IDs in the list)*/
    },
    "cardNo": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int, the maximum number of members in the array (the maximum number of IDs in the list)*/
    },
    "deleteAllEnabled": {
    /*ro, opt, object*/
        "@opt": [true, false]
        /*ro, req, array, subType:bool*/
    },
    "plateColor": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "@opt": ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen",
"orange", "other", "red", "unknown", "white", "yellow"]
        /*ro, opt, array, options, subType:string*/
    },
    "plateType": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "@opt": ["02TypePersonalized", "04NewMilitay", "92FarmVehicle", "92TypeArm", "92TypeCivil", "arm", "civilAviation", "coach", "consulate", "embassy",
"emergency", "green1325FarmVehicle", "hongKongMacao", "leftRightMilitay", "motorola", "newEnergy", "oneLineArm", "oneLineArmHeadquarters", "tempEntry",
"tempTravl", "trailer", "twoLineArm", "twoLineArmHeadquarters", "unknown", "upDownMilitay", "yellow1225FarmVehicle", "yellow1325FarmVehicle",
"yellowTwoLine"]
        /*ro, opt, array, options, subType:string*/
    },
    "name": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int*/
    },
    "certificateType": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "@opt": ["officerID#军官证", "ID#身份证", "passportID#护照", "other#其他"]
        /*ro, opt, array, subType:string*/
    },
    "certificateNumber": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int*/
    },
    "virtualParkingNum": {
    /*ro, opt, object*/
        "@size":  1
        /*ro, req, int*/
    },
    "CompoundCond": {
    /*ro, opt, object*/
        "@size":  1,
        /*ro, req, int*/
        "licensePlate": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "plateColor": {
        /*ro, opt, object*/
            "@opt": ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen",
"orange", "other", "red", "unknown", "white", "yellow"]
            /*ro, opt, array, options, subType:string*/
        }
    }
}
```

11.6.5.2 Delete the license plate information in the blocklist or allowlist

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/DelLicensePlateAuditData?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	--

Request Message

```
{
    "id": ["1", "2", "3"],
    /*opt, array, subType:string*/
    "licensePlate": ["浙A12345", "浙A12346", "浙A12347"],
    /*opt, array, subType:string, range:[1,32]*/
    "cardNo": ["abcd1234", "abcd12345", "abcd123456"],
    /*opt, array, subType:string, range:[1,64]*/
    "deleteAllEnabled":  true,
    /*opt, bool*/
    "plateColor":  ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen", "orange", "other", "red", "unknown", "white", "yellow"],
    /*opt, enumarray, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow), "orange", "other" (other color), "red", "unknown", "white", "yellow"*/
    "plateType":  ["02TypePersonalized", "04NewMilitay", "92FarmVehicle", "92TypeArm", "92TypeCivil", "arm", "civilAviation", "coach", "consulate", "embassy", "emergency", "green1325FarmVehicle", "hongKongMacao", "leftRightMilitay", "motorola", "newEnergy", "oneLineArm", "oneLineArmHeadquarters", "tempEntry", "tempTravl", "trailer", "twoLineArm", "twoLineArmHeadquarters", "unknown", "upDownMilitay", "yellow1225FarmVehicle", "yellow1325FarmVehicle", "yellowTwoLine"],
    /*opt, enumarray, license plate type, subType:string, desc:"02TypePersonalized" (type 02 custom vehicle), "92FarmVehicle" (two-line license plate civil vehicle), "92TypeArm" (type 92 armed police vehicle), "92TypeCivil" (type 92 civil vehicle), "arm" (police vehicle), "civilAviation" (civil aviation license plate), "coach" (driver-training vehicle), "consulate" (consular vehicle), "embassy" (embassy vehicle), "emergency" (emergency license plate), "green1325FarmVehicle" (agricultural vehicle (green 1325)), "hongKongMacao" (Hong Kong/Macao entrance vehicle), "motorola" (motorcycle), "newEnergy" (new energy vehicle license plate), "oneLineArm" (new armed police vehicle (one-line)), "oneLineArmHeadquarters" (armed police headquarter license plate (one-line)), "tempEntry" (temporary entry car), "tempTravl" (temporary license plate car), "trailer", "twoLineArm" (new armed police vehicle (two-line)), "twoLineArmHeadquarters" (armed police headquarter license plate (two-line)), "unknown", "yellow1225FarmVehicle" (agricultural vehicle (yellow 1225)), "yellow1325FarmVehicle" (agricultural vehicle (yellow 1325)), "yellowTwoLine" (yellow two-line rear license plate)*/
    "name": ["张三", "李四"],
    /*opt, array, subType:string, range:[1,64]*/
    "certificateType":  ["officerID", "ID", "passportID", "other"],
    /*ro, opt, enumarray, subType:string*/
    "certificateNumber": ["1234", "5678"],
    /*ro, opt, array, subType:string*/
    "virtualParkingNum": ["123", "456"],
    /*ro, opt, array, subType:string*/
    "CompoundCond": [
    /*opt, array, subType:object*/
        {
            "licensePlate":  "浙A12345",
            /*opt, string, range:[1,32]*/
            "plateColor":  "black"
            /*opt, enum, license plate color, subType:string, desc:"black", "blue", "civilAviationBlack" (civil aviation black), "civilAviationGreen" (civil aviation green), "golden", "green", "mixedColor" (mixed color), "newEnergyGreen" (new energy green), "newEnergyYellowGreen" (new energy green and yellow), "orange", "other" (other color), "red", "unknown", "white", "yellow"*/
        }
    ]
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:it is required when the value of statusCode is not 1, and it corresponds to subStatusCode*/
    "errorMsg":  "ok"
    /*ro, opt, string, error information, desc:this field is required when the value of statusCode is not 1*/
}
```


11.6.5.3 Get the capability of adding license plate information to the blocklist or allowlist, or editing the license plate information in the blocklist or allowlist

Request URL

GET /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData/record/capabilities?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

None

Response Message


```
{
    "LicensePlateInfoCap": {
    /*ro, req, object, list of license plates to be added to the list*/
        "id": {
        /*ro, req, object, index, desc:the value of this node is the ID of the license plate information, and the current ID can be obtained by calling the URI /ISAPI/Traffic/channels/<channelID>/searchLPListAudit*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  16
            /*ro, opt, int*/
        },
        "listType": {
        /*ro, opt, object, list type, desc:"blockList", "allowList"*/
            "@opt": ["blockList", "allowList"]
            /*ro, opt, array, subType:string*/
        },
        "licensePlate": {
        /*ro, req, object, license plate number*/
            "@min":  1,
            /*ro, opt, int*/
            "@max":  32
            /*ro, opt, int*/
        },
        "cardID": {
        /*ro, opt, object, card ID (Wiegand protocol), desc:object,card ID (Wiegand protocol),which consists of 8 decimal digits*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  9
            /*ro, req, int*/
        },
        "createTime":  "1970-01-01T00:00:00+08:00",
        /*ro, opt, datetime, created time*/
        "effectiveTime":  "1970-01-01T00:00:00+08:00",
        /*ro, opt, datetime, effective time*/
        "effectiveStartDate":  "1970-01-01",
        /*ro, opt, date, start date of the validity period*/
        "cardNo": {
        /*ro, opt, object, card No.*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  64
            /*ro, req, int*/
        },
        "plateColor": {
        /*ro, req, object*/
            "@opt": ["black", "blue", "civilAviationBlack", "civilAviationGreen", "golden", "green", "mixedColor", "newEnergyGreen", "newEnergyYellowGreen", "orange", "other", "red", "unknown", "white", "yellow"]
            /*ro, opt, array, subType:string*/
        },
        "plateType": {
        /*ro, req, object*/
            "@opt": ["02TypePersonalized", "04NewMilitay", "92FarmVehicle", "92TypeArm", "92TypeCivil", "arm", "civilAviation", "coach", "consulate", "embassy", "emergency", "green1325FarmVehicle", "hongKongMacao", "leftRightMilitay", "motorola", "newEnergy", "oneLineArm", "oneLineArmHeadquarters", "tempEntry", "tempTravl", "trailer", "twoLineArm", "twoLineArmHeadquarters", "unknown", "upDownMilitay", "yellow1225FarmVehicle", "yellow1325FarmVehicle", "yellowTwoLine"]
            /*ro, opt, array, subType:string*/
        },
        "operationType": {
        /*ro, opt, object*/
            "@opt": ["add", "modify"]
            /*ro, opt, array, subType:string*/
        },
        "name": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  32
            /*ro, req, int*/
        },
        "certificateType": {
        /*ro, req, object*/
            "@opt": ["officerID#军官证", "ID#身份证", "passportID#护照", "other#其他"]
            /*ro, opt, array, subType:string*/
        },
        "certificateNumber": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  32
            /*ro, req, int*/
        },
        "virtualParkingNum": {
        /*ro, opt, object*/
            "@min":  1,
            /*ro, req, int*/
            "@max":  16
```

```
            "@max": 16
            /*ro, req, int*/
        },
        "plateDescription": {
        /*ro, opt, object*/
            "@min": 0,
            /*ro, req, int*/
            "@max": 128
            /*ro, req, int*/
        },
        "groupName": {
        /*ro, opt, object*/
            "@min": 1,
            /*ro, req, int*/
            "@max": 64
            /*ro, req, int*/
        }
    }
}
```

11.6.5.4 Add license plate information to the blocklist or allowlist, or edit the license plate information in the blocklist or allowlist

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData/record?format=json

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.

Request Message

```
{
    "LicensePlateInfoList": [
    /*req, array, subType:object*/
        {
            "id": "1",
            /*req, string, ID, range:[1,16], desc:index,the maximum string size is 16 bytes. The value of this node is the ID of the license plate
information,and the current ID can be obtained by calling the URI /ISAPI/Traffic/channels/<channelID>/searchLPListAudit*/
            "listType": "blockList",
            /*req, enum, list type, subType:string, desc:list type: "blockList","allowList"*/
            "LicensePlate": "test",
            /*req, string, license plate No., range:[1,32]*/
            "createTime": "1970-01-01T00:00:00+08:00",
            /*opt, datetime, creation time*/
            "effectiveTime": "1970-01-01T00:00:00+08:00",
            /*opt, datetime, end time of the validity period*/
            "cardID": "test",
            /*opt, string, card ID (Wiegand protocol), range:[1,9], desc:card ID by Wiegand rule,which is a number consisting of 8 digits in decimal format.
The maximum string size is 9 bytes*/
            "effectiveStartDate": "1970-01-01",
            /*opt, string, start date of the validity period*/
            "cardNo": "abcd1234",
            /*opt, string, card No., range:[1,64]*/
            "plateColor": "blue",
            /*req, enum, license plate color, subType:string, desc:license plate color*/
            "plateType": "92TypeCivil",
            /*req, enum, license plate type, subType:string, desc:license plate type*/
            "operationType": "add",
            /*opt, string*/
            "name": "张三",
            /*ro, opt, string, range:[0,96]*/
            "certificateType": "officerID",
            /*ro, opt, enum, subType:string*/
            "certificateNumber": "test",
            /*ro, opt, string, range:[0,32]*/
            "virtualParkingNum": "123",
            /*opt, string, range:[1,16]*/
            "plateDescription": "abc",
            /*opt, string, range:[0,128]*/
            "groupName": "group1"
            /*opt, string, range:[1,64]*/
        }
    ]
}
```

Response Message

```
{
    "statusCode":  1,
    /*ro, opt, int, status code, desc:1 (succeeded). It is required when an error occurred*/
    "statusString":  "OK",
    /*ro, opt, string, status description, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "subStatusCode":  "ok",
    /*ro, opt, string, sub status code, range:[1,64], desc:"ok" (succeeded). It is required when an error occurred*/
    "errorCode":  1,
    /*ro, opt, int, error code, desc:when the value of statusCode is not 1, it corresponds to subStatusCode*/
    "errorMsg":  "ok",
    /*ro, opt, string, error details, desc:this node is required when the value of statusCode is not 1*/
    "MErrCode":  "0x00000000",
    /*ro, opt, string*/
    "MErrDevSelfEx":  "0x00000000"
    /*ro, opt, string*/
}
```


11.6.5.5 Update the vehicle blocklist and allowlist file

Request URL

PUT /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData?fileType=<fileType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
fileType	enum	When "FileType" is xml, it is an XML file. If no "fileType" field or "fileType" is excel, it is an Excel file of XLS format. When "fileType" is csv, it is a Excel file of CSV format.

Request Message

None

Response Message

```xml
<?xml version="1.0" encoding="UTF-8"?>

<ResponseStatus xmlns="http://www.isapi.org/ver20/XMLSchema" version="2.0">
  <!--ro, req, object, response message, attr:version{ro, req, string, protocolVersion}-->
  <requestURL>
    <!--ro, req, string, request URL, range:[0,1024]-->null
  </requestURL>
  <statusCode>
    <!--ro, req, enum, status code, subType:int, desc:0 (OK), 1 (OK), 2 (Device Busy), 3 (Device Error), 4 (Invalid Operation), 5 (Invalid XML Format), 6 
(Invalid XML Content), 7 (Reboot Required)-->0
  </statusCode>
  <statusString>
    <!--ro, req, enum, status description, subType:string, desc:“OK” (succeeded), “Device Busy”, “Device Error”, “Invalid Operation”, “Invalid XML Format”, 
“Invalid XML Content”, “Reboot” (reboot device)-->OK
  </statusString>
  <subStatusCode>
    <!--ro, req, string, sub status code, desc:sub status code-->OK
  </subStatusCode>
  <description>
    <!--ro, opt, string, range:[0,1024]-->badXmlFormat
  </description>
  <AttachInfo>
    <!--ro, opt, object-->
    <StatusList>
      <!--ro, req, object-->
      <Status>
        <!--ro, req, object-->
        <index>
          <!--ro, req, int-->0
        </index>
        <statusCode>
          <!--ro, req, enum, subType:int-->0
        </statusCode>
        <statusString>
          <!--ro, req, enum, subType:string-->OK
        </statusString>
        <subStatusCode>
          <!--ro, req, string, range:[1,64]-->ok
        </subStatusCode>
        <description>
          <!--ro, opt, string, range:[0,1024]-->test
        </description>
        <plateNo>
          <!--ro, opt, string, range:[1,64]-->test
        </plateNo>
        <licensePlateListType>
          <!--ro, opt, enum, subType:string-->allowlist
        </licensePlateListType>
      </Status>
    </StatusList>
  </AttachInfo>
  <successNum>
    <!--ro, opt, int-->100
  </successNum>
</ResponseStatus>
```


11.6.5.6 Export the file of vehicle blocklist and allowlist data

Request URL

GET /ISAPI/Traffic/channels/<channelID>/licensePlateAuditData?fileType=<fileType>

Query Parameter

Parameter Name	Parameter Type	Description
channelID	string	Channel No.
fileType	enum	When "FileType" is xml, it is an XML file. If no "fileType" field or "fileType" is excel, it is an Excel file of XLS format. When "fileType" is csv, it is a Excel file of CSV format.

Request Message

None

Response Message

None

12 How-To Video Guidance

If you need access to corresponding video guidance for device integration, please register on https://tpp.hikvision.com and visit our Training Center: https://tpp.hikvision.com/tpp/Training. The Training Center is specifically designed to provide technical training and guidance resources for our partners. On this platform, you can find integration video tutorials for various devices, enabling better understanding and learning of the integration process. To offer more personalized service, our Training Center also supports filtering by integration protocols, devices, and applications.